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Circles, Spheres and Spherical Geometry

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Circles, Spheres and Spherical Geometry

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To Keiko and Angelika

Preface

Circles and spheres have been successfully studied in ancient Greece. The Greek geometers saw the circles and spheres as *perfect figures*. By a metaphysical reasoning, Aristotle (BC384-BC322) asserted that the orbits of planets are circles. Though the notions of ellipses and parabolas were known due to the famous book on conic sections by Apollonius of Perga (BC240-BC190), the opinion that the planet orbits must be circles continued to survive, partly supported by the famous name of Aristotle.

However, in the sixteenth century, Johannes Kepler (1571–1630) confirmed that the orbits of planets must be ellipses. Additionally, Kepler posed a famous problem on spheres: What is the densest packing of unit spheres (or solid unit balls) in three dimensions? The famous *Kepler conjecture* says that the face-centered cubic lattice packing yields this densest packing, and this was proved by Thomas C. Hales [371] in 2005.

Isaac Newton (1643–1727) and his friend David Gregory (1659–1708) argued about the maximum number of unit balls that can touch simultaneously a given unit ball. Newton claimed that 12 is the maximum number, while Gregory asserted 13. Which one is right? This problem is known as the *problem of thirteen balls*, or the Gregory-Newton problem. It was settled in favor of Newton in 1953 by K. Schütte and B. L. van der Waerden [716].

In the so-called *Japanese temple geometry*, geometers from the Edo-period (1600–1868) also studied circles and spheres. *Descartes' circle theorem* and its three-dimensional version, namely *Soddy's formula*, were found independently by them. The Japanese mathematicians dedicated a large variety of related results to temples and shrines as Sangaku (artistically painted wooden panels); see Hidetoshi Fukagawa and Dan Pedoe [284], Hidetoshi Fukagawa and John F. Rigby [285] as well as Hidetoshi Fukagawa and Tony Rothman [286].

Spherical geometry, or the *geometry of spherical space*, is the two-dimensional geometry of the surface of a ball, as non-Euclidean geometry having important applications in astronomy, geodesy, navigation, and further scientific fields, also in form of higher dimensional analogues (see [131], [809] and, also for historical background, [678]).

In this book we deal with the *geometry of circles and spheres*, and also with results in *spherical spaces*. As prerequisites for reading it, the authors recommend only Linear Algebra, Advanced Calculus, and (for Chap. 8) elementary Probability Theory. In addition, some results from Graph Theory are also used in this book. They are completely introduced when it becomes necessary.

We begin the first chapter with the planar *inversion on circles* and its three-dimensional analogue. Inversions are convenient transformations when one studies families of circles or spheres. For example, suppose that mutually tangent three big balls lie on the floor and touch the ceiling that is parallel to the floor. Imagine that in the room enclosed by these three big balls (and also between the floor and the ceiling), a number of small balls of various sizes are piled up in the following way: The first ball is tangent to the floor and the three big balls; the second ball is tangent to the first one and the three big balls; the next ball is tangent to the previous one and the three big balls; and so on. Then one question is: how many small balls could be piled up in such a way? Considering the inversion on a sphere with center at the contact point of any two big balls, we can see that exactly four small balls could be piled up in such a way. See also the literature on *Soddy's hexlet* in Chap. 1 of this book.

Bend formulas, as treated in Chap. 2, are related to the topics studied by Japanese mathematicians in the Edo-period. Relations among the radii of Steiner cycles, Soddy's hexlet, and a pair of linked four cycles of spheres are studied in Chap. 2, besides Descartes' circle theorem and Soddy's formula.

In Chap. 3, the *orthogonal-circle-representation theorem* (OCR theorem) is introduced and proved. This theorem characterizes a special type of planar graphs that can be represented by a family of circles satisfying certain conditions. From this theorem, the *coin-graph theorem* (also called *Koebe-Andreev-Thurston theorem*) and Steinitz's theorem concerning the graph of a convex 3-polytope follow easily.

Chapters 4, 5, and 7 refer to *spherical geometry*, where figures on a sphere, such as geodesic curves, spherical caps, spherical triangles, spherical polygons, spherical ellipses, etc., are studied. Many formulas, e.g., the spherical cosine law, the spherical sine law, Euler's formula for the spherical excess, etc., are derived by *Cesàro's method* (a method using stereographic projections).

In Chap. 6, the *problem of thirteen balls* is considered. This is done by using a result on *planar graphs*, together with a few results from spherical geometry.

In Chap. 8, problems from *geometric probability on the sphere* are studied, thus referring to random point arrangements and random polygons.

In Chap. 9, *intersection graphs of (also random) spherical caps* are studied.

Chapter 10 concerns *four-point sets (quartets) on a sphere* that determine the radius of the sphere.

Spheres and balls in high dimensions are considered in Chap. 11. Problems like the determination of the number of unit balls that can simultaneously touch a ball of small radius in n-dimensional space are considered there.

In Chap. 12, *Cayley-Menger matrices* are introduced, and *orthogonal-sphere systems* as well as *tangent-sphere systems* are investigated. Further on, *Ptolemy's theorem* is generalized.

Casey's theorem and a generalization of it are proved in Chap. 13. (Note that Casey's theorem is an extension of Ptolemy's theorem, where the points in Ptolemy's theorem are replaced by circles, and the distance between points is replaced by the common tangent distance between circles.)

At the end of each chapter, we present *exercises* of different levels of difficulty, and at the end of the book, solutions to exercises selected from this set are given.

Each chapter ends with *notes* that support further reading and, in some cases, bring the text close to the research front. Depending on the respective situation in the specialist literature, these notes are very heterogeneous in nature. In some chapters, they bring the text indeed close to the research front (reflecting also research papers), and in other cases they are relatively short, citing only related monographs, or monographs and surveys.

Topics that are obviously close to the contents of our book are, e.g., packing and covering problems for circles/spheres and related topological results (such as the Borsuk-Ulam theorem). However, we did not take them into consideration since they would have exceeded the scope of this book. (For packing and covering problems, we refer to the very recent extension [262] of L. Fejes Tóth's famous monograph, and the Borsuk-Ulam theorem is excellently discussed by J. Matoušek [561].)

We hope that our book finds interested readers who will share our enthusiasm for the subfield of geometry covered by it.

Okinawa, Japan — Hiroshi Maehara
Chemnitz, Germany — Horst Martini
2024

Contents

1 Inversion and Stereographic Projection 1
- 1.1 Inversion of the Plane 1
- 1.2 Euler's Triangle Theorem 5
- 1.3 Steiner Cycles 8
- 1.4 Inversions of 3-Space 9
- 1.5 Soddy's Hexlet 10
- 1.6 Stereographic Projection 11
- 1.7 Appendix: Poncelet's Porism 12
- 1.8 Exercises 15
- 1.9 Notes 15
  - 1.9.1 Notes on Inversion in the Plane and in 3-Space 15
  - 1.9.2 Notes on Euler's Triangle Theorem 17
  - 1.9.3 Notes on Steiner Chains and Steiner's Porism 18
  - 1.9.4 Notes on Soddy's Hexlet 20
  - 1.9.5 Notes on the Stereographic Projection 21
  - 1.9.6 Notes on Poncelet's Porism 26

2 Bend Formulas 31
- 2.1 Bend Formulas for Steiner Cycles 31
- 2.2 Bend Formula for Soddy's Hexlet 35
- 2.3 Descartes' Circle Theorem 37
- 2.4 Soddy's Formula 38
- 2.5 Linked 4-Cycle Pairs 40
- 2.6 Exercises 42
- 2.7 Notes 43
  - 2.7.1 Notes on Descartes' Circle Theorem and Soddy's Formula 43
  - 2.7.2 Notes on Apollonian Circle Packings 44
  - 2.7.3 Notes on Apollonius' Touching Problem 46

3 Graphs and Circle-Systems 49
3.1 Graphs and Planar Graphs 49
3.2 Quadrangulations 54
3.3 Orthogonal-Circle Representations 56
3.4 3-Connected Graphs and Radial Graphs 63
3.5 The Coin Graph Theorem 65
3.6 The Theorem of Steinitz 67
3.7 Exercises 68
3.8 Notes 69
3.8.1 Notes on the Coin Graph Theorem 69
3.8.2 Notes on the Theorem of Steinitz 71

4 Spherical Geometry I 75
4.1 Geodesic Segments 75
4.2 Cylindrical Projections 77
4.3 Spherical Polygons 80
4.4 The Inscribed Angle Theorem 83
4.5 The Polar Set 85
4.6 Exercises 88
4.7 Notes 88
4.7.1 Notes on Spherical Geometry 88
4.7.2 Notes on Cylindrical Projections and Related Topics 90
4.7.3 Notes on Spherical Polygons 94

5 Spherical Geometry II 99
5.1 The Cesàro Triangle 99
5.2 Edge-Lengths of Cesàro Triangles 101
5.3 Spherical Cosine Law and Sine Law 103
5.4 The Triangle Comparison Theorem for Spheres 105
5.5 Triangles with Two Fixed Edge-Lengths 109
5.6 The Isoperimetric Theorem for Quadrilaterals 111
5.7 Exercises 113
5.8 Notes 114
5.8.1 Notes on Spherical Trigonometry and the Respective Cosine and Sine Laws 114
5.8.2 Notes on Triangle Comparison Theorems 118

6 The Problem of Thirteen Balls 123
6.1 The Lemma on Proper Diagonals 123
6.2 Major Triangles 125
6.3 On a Triangulation of a Quadrilateral 126
6.4 The Problem 128
6.5 Solution 129
6.6 Exercises 135
6.7 Notes 135
6.7.1 Notes on the Problem of Thirteen Balls 135

7 Spherical Geometry III 143
7.1 Spherical Ellipses 143
7.2 Lexell's Theorem 145
7.3 Equilateral Spherical Triangles 146
7.4 Spherical Polygons Inscribed in a Cap 147
7.5 Regular Spherical Polygons 150
7.6 Appendix 151
7.6.1 Orthogonal Projections of a Spherical Ellipse 151
7.6.2 Proof of Lemma 7.1 153
7.7 Exercises 154
7.8 Notes 154
7.8.1 Notes on Spherical Ellipses 155
7.8.2 Notes on Lexell's Theorem 156

8 Geometric Probability on the Sphere 159
8.1 Random Points on a Sphere 159
8.2 Random Spherical Triangles 160
8.3 Random Quartets 162
8.4 Wendel's Theorem 164
8.5 Crofton's Formula 165
8.6 Random Points on a Hemisphere 167
8.7 Santaló's Chord Theorem 169
8.8 Exercises 170
8.9 Notes 170
8.9.1 Notes on Random Points and Triangles on the Sphere 170
8.9.2 Notes on Crofton's Formulae 175

9 Intersection Graphs of Spherical Caps 181
9.1 Connectivity 181
9.2 On the Intersection of a Few Equal Caps 183
9.3 Intersection Graphs of Equal Caps 186
9.4 Random Spherical Caps 190
9.5 Chebyshev's Inequality and a Lemma 192
9.6 Asymptotic Probability 193
9.7 Exercises 195
9.8 Notes 196
9.8.1 Notes on Intersection Graphs 196
9.8.2 Notes on Chebyshev's Inequality 198

10 Quartets on a Sphere 201
10.1 Metric Spaces 201
10.2 The Vertices of a Convex Quadrilateral 203
10.3 Unispherical Quartets 205
10.4 An Equation for the Radius 206
10.5 A Planar Quartet That Is also Spherical 210
10.6 Exercises 211

10.7 Notes 211
10.7.1 Notes on Four-Point Systems 212

11 Higher Dimensions 213
11.1 Figures in High Dimensions 213
11.2 Moser's Paradox 215
11.3 The Volume of a Ball 216
11.4 Two Lemmas 220
11.5 Kissing a Small Ball 223
11.6 Exercises 226
11.7 Notes 226
11.7.1 Notes on Cubes 226
11.7.2 Notes on Simplices 230
11.7.3 Notes on the Volume of (Unit) Balls 236

12 The Cayley-Menger Determinant 239
12.1 The Cayley-Menger Matrix 239
12.2 Odd Integral Distances 242
12.3 Orthogonal-Sphere Systems 243
12.4 Tangent-Sphere Systems 244
12.5 A Generalization of Ptolemy's Theorem 248
12.6 Exercises 250
12.7 Notes 250
12.7.1 Notes on Cayley-Menger Determinants 251

13 Casey's Theorem 261
13.1 Bicolored Sets of Circles 261
13.2 Inversion Invariants 263
13.3 Isometric Radii Change 266
13.4 Proof of Casey's Theorem 269
13.5 A Theorem on Five Circles 269
13.6 An Extension of Casey's Theorem 271
13.7 Exercises 272
13.8 Notes 273
13.8.1 Notes on Ptolemy's Theorem 273
13.8.2 Notes on Casey's Theorem 276

14 Solutions to the Selected Exercises 279

Bibliography 291

Index 325

Chapter 1
Inversion and Stereographic Projection



Mathematics is not about numbers, equations, computations, or algorithms: it is about understanding.

— William Paul Thurston

An inversion with respect to a circle in $\mathbb{R}^2$ (or with respect to a sphere in $\mathbb{R}^3$) is a one-to-one transformation of $\mathbb{R}^2$ (or $\mathbb{R}^3$) that interchanges the inside of the circle (sphere) and the outside of the circle (sphere). Inversions have nice properties, and they turn out to be convenient when we study arrangements of circles, or spheres. According to Coxeter [182], the inversion was invented by L. J. Magnus in 1831. Using inversions, Steiner's porism and Soddy's hexlet are introduced. The stereographic projection of a sphere is also obtained by restricting the domain of an inversion.

1.1 Inversion of the Plane

Let C be a circle in the plane $\mathbb{R}^2$ with radius r and center $\boldsymbol{p}$. Let

$$\varphi : \mathbb{R}^2 \setminus \{\boldsymbol{p}\} \to \mathbb{R}^2 \setminus \{\boldsymbol{p}\}$$

be the map that sends $\boldsymbol{x} \in \mathbb{R}^2 \setminus \{\boldsymbol{p}\}$ to a point $\boldsymbol{x}'$ on the ray $\overrightarrow{\boldsymbol{p}\boldsymbol{x}}$ satisfying

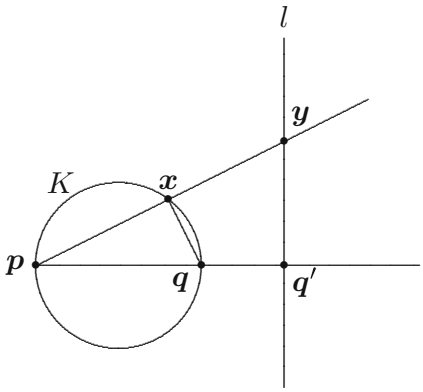
$$|\boldsymbol{p}\boldsymbol{x}| \cdot |\boldsymbol{p}\boldsymbol{x}'| = r^2,$$

where $|\boldsymbol{x}\,\boldsymbol{y}|$ denotes the length of the line segment connecting $\boldsymbol{x}$ to $\boldsymbol{y}$, namely, $|\boldsymbol{x}\,\boldsymbol{y}| = \|\boldsymbol{x} - \boldsymbol{y}\|$. This map φ is called the *inversion* of $\mathbb{R}^2$ with respect to the circle C. Note that $\varphi(\boldsymbol{p})$ is not defined. By the inversion with respect to a circle C, the inside and the outside of C are interchanged, and each point on C is fixed. The center $\boldsymbol{p}$ of C is called the *center of the inversion* φ and the squared radius of C is called the *power* of the inversion. If we apply φ again, then each point goes back to the original point,

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Fig. 1.1 $\triangle pqx \sim \triangle pyq'$



that is, $\varphi(\varphi(\boldsymbol{x})) = \boldsymbol{x}$ for every $\boldsymbol{x} \in \mathbb{R}^2 \setminus \{\boldsymbol{p}\}$. Thus any inversion is an involution and a one-to-one correspondence.

Theorem 1.1 (Circle-to-Circle Correspondence) *Via an inversion φ of $\mathbb{R}^2$ with center $\boldsymbol{p}$,*

(i) *every circle that passes through $\boldsymbol{p}$ goes to a line that does not pass through $\boldsymbol{p}$,*
(ii) *every circle that does not pass through $\boldsymbol{p}$ goes to a circle that does not pass through $\boldsymbol{p}$, and*
(iii) *every line that passes through $\boldsymbol{p}$ goes to itself.*

Proof Let r^2 be the power of φ. (i) Let K be a circle that passes through $\boldsymbol{p}$, and let $\boldsymbol{pq}$ be a diameter of K. Put $\boldsymbol{q}' = \varphi(\boldsymbol{q})$. Let us show that $\varphi(K)$ is a line l that passes through $\boldsymbol{q}'$ perpendicularly to the line $\boldsymbol{pq}'$, see Fig. 1.1. For a point $\boldsymbol{x}$ $(\neq \boldsymbol{p}, \boldsymbol{q})$ on K, let $\boldsymbol{y}$ be the point where the line $\boldsymbol{px}$ meets the line l. Since $\angle \boldsymbol{pxq}$ and $\angle \boldsymbol{pq}'\boldsymbol{y}$ are right angles, $\triangle \boldsymbol{pxq}$ and $\triangle \boldsymbol{pq}'\boldsymbol{y}$ are similar. Hence

$$|\boldsymbol{pq}| : |\boldsymbol{px}| = |\boldsymbol{py}| : |\boldsymbol{pq}'|,$$

from which we have $|\boldsymbol{px}| \cdot |\boldsymbol{py}| = |\boldsymbol{pq}| \cdot |\boldsymbol{pq}'| = r^2$, and we can see that $\varphi(\boldsymbol{x}) = \boldsymbol{y}$, namely $\varphi(K) = l$.

(ii) Now, let K be a circle that does not pass through $\boldsymbol{p}$. For a point $\boldsymbol{x}$ on K, let $\boldsymbol{y}$ be the other intersection point of the line $\boldsymbol{px}$ and K. By the power theorem, $|\boldsymbol{px}| \cdot |\boldsymbol{py}| = c$ (constant). If we put $\boldsymbol{x}' = \varphi(\boldsymbol{x})$, then $|\boldsymbol{px}| \cdot |\boldsymbol{px}'| = r^2$. Hence

$$r^2 = |\boldsymbol{px}| \cdot |\boldsymbol{px}'| = c\,\frac{|\boldsymbol{px}'|}{|\boldsymbol{py}|}.$$

Thus, $\boldsymbol{x}'$ is a point on the ray $\overrightarrow{\boldsymbol{py}}$ that satisfies $|\boldsymbol{px}'| = (r^2/c)|\boldsymbol{py}|$. This implies that if $\boldsymbol{y}$ moves along the circle K, then $\boldsymbol{x}'$ moves along the figure that is homothetic to K with center $\boldsymbol{p}$ and ratio r^2/c. Therefore, $\varphi(K)$ is a circle that does not pass through $\boldsymbol{p}$.

(iii) is obvious. □

From the above proof of (i), we have the following corollary.

Corollary 1.1 *Let φ be an inversion of the plane with center $\boldsymbol{p}$. If K is a circle that passes through $\boldsymbol{p}$, then $\varphi(K)$ is a line that is parallel to the tangent line of K at $\boldsymbol{p}$.*

A pair of circles (or a circle and a line) in the plane is said to *cross* each other if they share two different points. If they share exactly one point, then they are *tangent* to each other. By an inversion, a pair of mutually tangent circles are mapped to either a pair of tangent circles, or a circle and its tangent line, or a pair of parallel lines. For a pair of mutually crossing circles, the angle formed by the tangent lines of the circles at one of the intersection points is called the *crossing angle* of the pair of circles. Obviously, due to the axial symmetry of the whole figure the crossing angles at each of the two intersection points are equal. If the crossing angle of a pair of crossing circles K_1, K_2 is a right angle, then K_1, K_2 are said to be *orthogonally crossing* to each other, denoted by $K_1 \perp K_2$. If K_i has center $\boldsymbol{p}_i$ and radius r_i for $i = 1, 2$, then by the Pythagorean theorem we have

$$K_1 \perp K_2 \Leftrightarrow |\boldsymbol{p}_1\boldsymbol{p}_2|^2 = r_1^2 + r_2^2.$$

Theorem 1.2 *The crossing angle of a pair of mutually crossing circles K_1, K_2 is invariant under any inversion. In other words, for any inversion φ of the plane, the crossing angle of $\varphi(K_1), \varphi(K_2)$ is equal to the crossing angle of K_1, K_2.*

Proof Let $\boldsymbol{q}$ be an intersection point of K_1, K_2, and let l_i be the tangent line of K_i at $\boldsymbol{q}$ for $i = 1, 2$. Let $\boldsymbol{p}$ be the center of the inversion φ.

(i) The case $\boldsymbol{p} = \boldsymbol{q}$: By Corollary 1.1, $\varphi(K_i)$ is a line parallel to l_i. Hence the crossing angle of $\varphi(K_1)$ and $\varphi(K_2)$ is equal to the crossing angle of l_1 and l_2.

(ii) The case $\boldsymbol{p} \neq \boldsymbol{q}$: Since l_i and K_i share only one point $\boldsymbol{q}$, $\varphi(l_i)$ and $\varphi(K_i)$ share only one point, that is, they are tangent. Hence the crossing angle of $\varphi(l_1), \varphi(l_2)$ is equal to the crossing angle of $\varphi(K_1), \varphi(K_2)$.

(ii-a) Suppose that $\boldsymbol{p}$ lies on l_1. Then $\varphi(l_1) = l_1$, and $\varphi(l_2)$ is a circle whose tangent line at $\boldsymbol{p}$ is parallel to l_2. Hence the crossing angle of $\varphi(l_1)$ and $\varphi(l_2)$ is equal to the crossing angle of l_1 and l_2.

(ii-b) Suppose $\boldsymbol{p}$ lies neither on l_1 nor on l_2. Then $\varphi(l_i)$ is a circle passing through $\boldsymbol{p}$ and its tangent line at $\boldsymbol{p}$ is parallel to l_i. Hence the crossing angle of $\varphi(l_1), \varphi(l_2)$ is equal to the crossing angle of l_1, l_2.

□

Lemma 1.1 *Let K_1, K_2 be disjoint circles in $\mathbb{R}^2$ with centers $\boldsymbol{q}_1, \boldsymbol{q}_2$, respectively. If $\boldsymbol{q}_1 \neq \boldsymbol{q}_2$, then there is a circle Γ with center on the line $\boldsymbol{q}_1\boldsymbol{q}_2$ such that it orthogonally crosses both K_1, K_2.*

Proof Let r_i be the radius of K_i, and suppose that $r_1 \geq r_2$. We may put $\boldsymbol{q}_1 = (0, 0)$, $\boldsymbol{q}_2 = (a, 0), a > 0$. Since K_1, K_2 are disjoint, we have either $a > r_1 + r_2$ (when each circle lies outside the other), or $a < r_1 - r_2$ (when K_2 lies inside K_1),

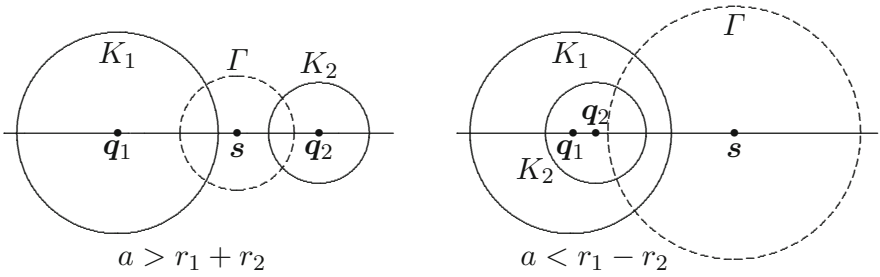


Fig. 1.2 $\Gamma \perp K_i$, $i = 1, 2$

see Fig. 1.2. Thus, $|a - r_1| > r_2$. Let Γ be a circle with center $\boldsymbol{s} = (b, 0)$ and radius r. The circle Γ orthogonally crosses both K_1, K_2 if and only if

$$\begin{cases} r^2 + r_1^2 = b^2 \\ r^2 + r_2^2 = (b - a)^2 \end{cases}. \tag{1.1}$$

Let us regard this as a simultaneous equation on b, r. From (1.1) we have $b^2 - r_1^2 = (b - a)^2 - r_2^2$, and hence $b = (a^2 + r_1^2 - r_2^2)/(2a)$. Since

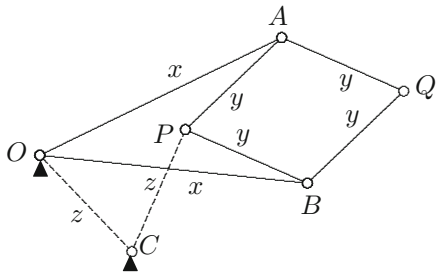
$$\begin{aligned} b^2 - r_1^2 &= \Big(\frac{a^2 + r_1^2 - r_2^2}{2a} - r_1\Big)\Big(\frac{a^2 + r_1^2 - r_2^2}{2a} + r_1\Big) \\ &= \Big(\frac{(a - r_1)^2 - r_2^2}{2a}\Big)\Big(\frac{(a + r_1)^2 - r_2^2}{2a}\Big) > 0, \end{aligned}$$

we have $r = \sqrt{b^2 - r_1^2}$. Thus, the simultaneous Eq. (1.1) has a unique solution $b, r > 0$. Hence there is a circle with center on the line $\boldsymbol{pq}$ which orthogonally crosses both K_1, K_2. □

Theorem 1.3 *For any pair of disjoint circles K_1, K_2 on the plane, there is a point $\boldsymbol{x}$ on the line connecting the centers of K_1, K_2, such that for any inversion φ with center $\boldsymbol{x}$, $\varphi(K_1)$ and $\varphi(K_2)$ are concentric circles.*

Proof Let $\boldsymbol{q}_i$ be the center of K_i. By Lemma 1.2, there is a circle Γ that orthogonally crosses both K_1, K_2, with center on the line $\boldsymbol{q}_1\boldsymbol{q}_2$. Let $\boldsymbol{x}$, $\boldsymbol{y}$ be the intersection points of Γ and the line $\boldsymbol{q}_1\boldsymbol{q}_2$. Let C be any circle with center $\boldsymbol{x}$, and φ be the inversion of the plane with respect to C. By this inversion φ, the line $\boldsymbol{q}_1\boldsymbol{q}_2$ goes to itself, and $\varphi(\Gamma)$ becomes a line l perpendicular to $\boldsymbol{q}_1\boldsymbol{q}_2$. By Theorem 1.2, each of the circles $\varphi(K_1)$, $\varphi(K_2)$ orthogonally crosses the lines $l = \varphi(\Gamma)$ and $\boldsymbol{q}_1\boldsymbol{q}_2$. In this case, the center of $\varphi(K_i)$ must be the intersection point of l and the line $\boldsymbol{q}_1\boldsymbol{q}_2$ for each $i = 1, 2$. Hence $\varphi(K_1)$, $\varphi(K_2)$ are concentric circles. □

Fig. 1.3 Peaucellier linkage



Example 1.1 (Peaucellier Linkage) The *Peaucellier linkage* is a planar straight line mechanism consisting of two rods (line segments) of length x and four rods of length $y(< x)$ connected as in Fig. 1.3. The joint O is fixed on the plane, and the angle formed by any two rods connected at a joint can vary freely.

In this apparatus, three points O, P, Q lie on the line that perpendicularly bisects AB, and since P, Q lies on the circle with center A and radius y, we have, by the power theorem, $|OP| \cdot |OQ| =$ constant. By considering the special case when P and Q coincide, we can see that $|OP| \cdot |OQ| = x^2 - y^2$. Thus, when P "draws" a figure F, then Q "draws" the figure $\varphi(F)$, where φ is the inversion of the plane with center O and power $x^2 - y^2$. If we connect P to a fixed point C on the plane, by a rod of length $z = |OC|$, then, since P is restricted on a circle with center C that passes through O, the point Q moves on a fixed line in the plane.

Concerning a planar linkage in which a vertex (link) draws a line, and a spatial linkage in which a link draws a plane, see H. Rademacher and O. Toeplitz [653, Chapter 16], and [397, Chapter 5] of D. Hilbert and S. Cohn-Vossen.

1.2 Euler's Triangle Theorem

The next theorem was discovered by the Romanian mathematician G. Ţiţeica in 1908, and rediscovered by R. A. Johnson [430, p.75] in 1916, see also [806, p.125] and, for a very recent contribution, [541].

Theorem 1.4 (The Ţiţeica-Johnson Theorem) *If three mutually crossing circles of the same radius r meet at a point, then the circle passing through the remaining three intersection points also has radius r, see Fig. 1.4.*

This theorem is also called the *three-circles theorem.*

Proof We may put $r = 1$. Suppose that the three unit circles meet at the origin O, and let $\boldsymbol{w}_1, \boldsymbol{w}_2, \boldsymbol{w}_3$ be the centers of these unit circles. Then $\|\boldsymbol{w}_i\| = 1$ $(i = 1, 2, 3)$, and the three intersection points $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z}$ other than O are given by

$$\boldsymbol{x} = \boldsymbol{w}_1 + \boldsymbol{w}_2, \ \boldsymbol{y} = \boldsymbol{w}_2 + \boldsymbol{w}_3, \ \boldsymbol{z} = \boldsymbol{w}_3 + \boldsymbol{w}_1.$$

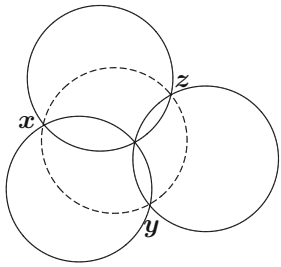


Fig. 1.4 The Ţiţeica-Johnson theorem

If we put $\boldsymbol{p} = \boldsymbol{w}_1 + \boldsymbol{w}_2 + \boldsymbol{w}_3$, then we have $\|\boldsymbol{p} - \boldsymbol{x}\| = \|\boldsymbol{p} - \boldsymbol{y}\| = \|\boldsymbol{p} - \boldsymbol{z}\| = 1$, and hence the unit circle with center $\boldsymbol{p}$ passes through $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z}$. □

This theorem is used by G. Pólya [641, Chapter 10] to show how an idea occurs to one who tries to solve a problem, and it is also listed in the book [199, Chapter 6 §3] by D. J. Davis and R. Hersh.

The three-circles theorem has many applications and extensions. A promising direction for future research is given by its *extension to normed planes*. It turns out that this theorem holds even in arbitrary such planes, where only strict convexity of the unit circle should be demanded. This generalization has immediate connections to *orthocenters* (following the important concept of *isosceles orthogonality*) and to *Feuerbach nine-point circles* (which, in general, become six-point circles) of triangles. These and many further related results can be found in [44], [551], and [556]. Also in strictly convex normed planes, the four occurring circle centers plus the four occuring intersection points give the 8 images of a parallel projection of the vertices of some 3-cube. To increase the number of participating circles, one can study analogous 2-projections of higher dimensional cubes, yielding planar *Clifford configurations of circles* (see [552]). These are derived again for arbitrary normed planes with strictly convex unit circles.

We come now to another theorem on circles, more precisely on the radii of in- and circumcircles of triangles.

Theorem 1.5 (Euler's Triangle Theorem) *Let R, r denote the radii of the circumscribed circle and the inscribed circle, respectively, of a triangle $\triangle \boldsymbol{x}\boldsymbol{y}\boldsymbol{z}$ in the plane, and let d be the distance between the centers of these circles. Then $d^2 = R(R - 2r)$ holds.*

Remark 1.1 If we replace the inscribed circle of $\triangle \boldsymbol{x}\boldsymbol{y}\boldsymbol{z}$ by an escribed circle, then $d^2 = R(R + 2r)$ holds instead.

Proof If $d = 0$, then $\triangle \boldsymbol{x}\boldsymbol{y}\boldsymbol{z}$ is an equilateral triangle as easily seen. In this case, we know that $R = 2r$. Hence $d^2 = R(R - 2r)$ holds. So in the following we assume that $d > 0$. Let C and K be the inscribed circle and the circumscribed circle of $\triangle \boldsymbol{x}\boldsymbol{y}\boldsymbol{z}$, and $\boldsymbol{p}, \boldsymbol{q}$ be the centers of C, K, respectively. Let $\boldsymbol{u}, \boldsymbol{v}$ be the intersection of the line $\boldsymbol{p}\boldsymbol{q}$ and the circumscribed circle K such that $\|\boldsymbol{u} - \boldsymbol{p}\| \leq \|\boldsymbol{v} - \boldsymbol{p}\|$, see Fig. 1.5.

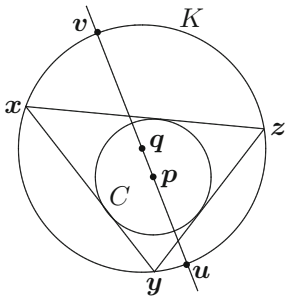


Fig. 1.5 $d^2 = R(R-2r)$

Let φ be the inversion with respect to the circle C. The images $\varphi(\boldsymbol{xy})$, $\varphi(\boldsymbol{yz})$, $\varphi(\boldsymbol{zx})$ of the lines $\boldsymbol{xy}$, $\boldsymbol{yz}$, $\boldsymbol{zx}$ are circles of the same *diameter* r and they meet at $\boldsymbol{p}$. Since $\varphi(K)$ is a circle that passes through $\varphi(\boldsymbol{x})$, $\varphi(\boldsymbol{y})$, $\varphi(\boldsymbol{z})$, the circle $\varphi(K)$ has also diameter r by Theorem 1.4. On the other hand, from

$$\|\boldsymbol{u}-\boldsymbol{p}\| = \|\boldsymbol{u}-\boldsymbol{q}\| - \|\boldsymbol{q}-\boldsymbol{p}\| = R-d,$$
$$\|\boldsymbol{v}-\boldsymbol{p}\| = \|\boldsymbol{v}-\boldsymbol{q}\| + \|\boldsymbol{q}-\boldsymbol{p}\| = R+d$$

we have $\|\varphi(\boldsymbol{u})-\boldsymbol{p}\| = r^2/(R-d)$, $\|\varphi(\boldsymbol{v})-\boldsymbol{p}\| = r^2/(R+d)$. Hence

$$\|\varphi(\boldsymbol{u})-\varphi(\boldsymbol{v})\| = \frac{r^2}{R-d} - \frac{r^2}{R+d} = \frac{2r^2R}{R^2-d^2}.$$

Since $\|\varphi(\boldsymbol{u})-\varphi(\boldsymbol{v})\|$ is equal to the diameter of $\varphi(K)$ (which is r), we have

$$r = \frac{2r^2R}{R^2-d^2},$$

from which $d^2 = R(R-2r)$ follows. □

To study *incircles and circumcircles of triangles in normed planes* is also still a promising topic. Many related results on their geometry can be found in the expository papers [557, Section 7] and [23].

Remark 1.2 For a tetrahedron T in $\mathbb{R}^3$, let r, R be the radii of the inscribed sphere of T and the circumscribed sphere of T, respectively, and d be the distance between the centers of these spheres. Then $d^2 \le (R+r)(R-3r)$ holds. This is known as *Grace-Danielsson inequality*, see, e.g., [314]. No elementary proof of this inequality is known so far.

The following theorem is a special case of *Poncelet's porism* (cf. our Section 1.7, and see also, e.g., [710] and [75].)

Theorem 1.6 *Let $\triangle \boldsymbol{abc}$ be a triangle in the plane, C be its inscribed circle, and K be its circumscribed circle. If a triangle $\triangle \boldsymbol{xyz}$ is circumscribed to C, and two vertices $\boldsymbol{x}$, $\boldsymbol{y}$ lie on the circle K, then the remaining vertex $\boldsymbol{z}$ also lies on K.*

Proof Let $\boldsymbol{p}$ be the center of C. We may suppose that the radius of C is 2. Let l, g, h be lines determined by the edges of the triangle $\triangle \boldsymbol{abc}$. By the inversion φ with respect to the circle C, $\varphi(l), \varphi(g), \varphi(h)$ become unit circles passing through $\boldsymbol{p}$. Since $\varphi(K)$ is a circle passing through the three intersection points of the unit circles $\varphi(l), \varphi(g), \varphi(h)$, it follows from Theorem 1.4 that $\varphi(K)$ is also a unit circle.

Now, since the lines $\boldsymbol{xy}$, $\boldsymbol{yz}$, $\boldsymbol{zx}$ are tangent to C, their images $\varphi(\boldsymbol{xy})$, $\varphi(\boldsymbol{yz})$, $\varphi(\boldsymbol{zx})$ are also unit circles, and $\varphi(\boldsymbol{xy})$ passes through the two points $\varphi(\boldsymbol{x})$, $\varphi(\boldsymbol{y})$. By Theorem 1.4, the circle that passes through $\varphi(\boldsymbol{x})$, $\varphi(\boldsymbol{y})$, $\varphi(\boldsymbol{z})$ is a unit circle. Note that $\varphi(K)$ is a unit circle passing through $\varphi(\boldsymbol{x})$, $\varphi(\boldsymbol{y})$, and it is different from the unit circle $\varphi(\boldsymbol{xy})$. Since there are only two unit circles that pass through $\varphi(\boldsymbol{x})$, $\varphi(\boldsymbol{y})$, the unit circle that passes through $\varphi(\boldsymbol{x})$, $\varphi(\boldsymbol{y})$, $\varphi(\boldsymbol{z})$ must coincide with $\varphi(K)$. Hence $\varphi(\boldsymbol{z})$ lies on $\varphi(K)$, and hence $\boldsymbol{z}$ lies on K. □

1.3 Steiner Cycles

Let α, β be mutually disjoint circles in the plane. A cyclic sequence of *mutually non-crossing* circles $\sigma_1, \sigma_2, \ldots, \sigma_n$ that satisfy the condition

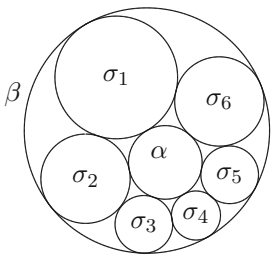
"each circle σ_i is tangent to four circles $\alpha, \beta, \sigma_{i-1}, \sigma_{i+1}$"

(where suffices are taken modulo n, that is, $\sigma_0 = \sigma_n, \sigma_{n+1} = \sigma_1$) is called a *Steiner cycle* or *Steiner n-cycle* for (α, β). Figure 1.6 shows a Steiner 6-cycle for (α, β).

Remark 1.3 Generally, the assumption "mutually noncrossing" is not necessary for defining a Steiner cycle, but to make our argument simple, we put such an assumption.

Theorem 1.7 (Steiner's Porism) *Let α, β be a pair of disjoint circles. If there is a Steiner cycle for (α, β), then for any circle σ that is tangent to both α, β, and that encloses none of α, β, there is a Steiner cycle that contains σ as one of its members; and every Steiner cycle for (α, β) consists of the same number of circles.*

Fig. 1.6 A Steiner 6-cycle



Proof In the case that α, β are concentric circles, the theorem is obvious. If α, β are not concentric, then there is an inversion φ of the plane that maps α, β to a pair of concentric circles. We may suppose that $\varphi(\alpha)$ lies inside $\varphi(\beta)$. If there is a Steiner cycle for (α, β), then it is sent by φ to a Steiner cycle for $(\varphi(\alpha), \varphi(\beta))$. Hence there is a Steiner cycle for $(\varphi(\alpha), \varphi(\beta))$, and $\varphi(\sigma)$ is tangent to both $\varphi(\alpha), \varphi(\beta)$, containing none of them in its interior. Hence there is a Steiner cycle containing $\varphi(\sigma)$ as a member, and the number of members in any Steiner cycle for $(\varphi(\alpha), \varphi(\beta))$ is the same. Now, applying the inversion φ again, the theorem follows. □

1.4 Inversions of 3-Space

An inversion of $\mathbb{R}^3$ is defined similarly to the planar case. Let S be a sphere in $\mathbb{R}^3$ with center $\boldsymbol{p}$ and radius r. The map $\varphi : \mathbb{R}^3 \setminus \{\boldsymbol{p}\} \to \mathbb{R}^3 \setminus \{\boldsymbol{p}\}$ that sends a point $\boldsymbol{x} \neq \boldsymbol{p}$ to a point $\boldsymbol{x}'$ on the ray $\overrightarrow{\boldsymbol{px}}$ such that

$$|\boldsymbol{px}| \cdot |\boldsymbol{px}'| = r^2$$

is called the inversion of $\mathbb{R}^3$ with respect to the sphere S. The point $\boldsymbol{p}$ is called the center of the inversion φ, and r^2 is called the power of φ. By the inversion φ, the interior and the exterior of S are interchanged, and every point of S is fixed. The inversion φ is a one-to-one correspondence.

Theorem 1.8 *Under an inversion of $\mathbb{R}^3$ with center $\boldsymbol{p}$,*

(i) *every sphere that passes through $\boldsymbol{p}$ goes to a plane that does not pass through $\boldsymbol{p}$,*
(ii) *every sphere that does not pass through $\boldsymbol{p}$ goes to a sphere that does not pass through $\boldsymbol{p}$, and*
(iii) *every plane that passes through $\boldsymbol{p}$ goes to itself.*

The proof is analogous to the proof of Theorem 1.1.

Similarly to Theorem 1.3, the following holds.

Theorem 1.9 *For any pair of disjoint spheres σ_1, σ_2 in $\mathbb{R}^3$, there is a point $\boldsymbol{x}$ on the line connecting the centers of σ_1, σ_2 such that for any inversion φ with center $\boldsymbol{x}$, $\varphi(\sigma_1)$ and $\varphi(\sigma_2)$ are concentric spheres.*

Proof Note that the restriction of an inversion of $\mathbb{R}^3$ with center $\boldsymbol{p}$ to a plane that passes through $\boldsymbol{p}$ is an inversion of the plane with center $\boldsymbol{p}$.

Now let H be a plane that contains the line l connecting the centers of $\sigma_i, i = 1, 2$. Then $H \cap \sigma_i, i = 1, 2$ are disjoint circles in the plane H. By Theorem 1.3, there is a point $\boldsymbol{x}$ on the line l such that for the inversion φ of $\mathbb{R}^3$ with center $\boldsymbol{x}$, the circles $\varphi(\sigma_i) \cap H, i = 1, 2$, are concentric circles on H. Note also that the point $\boldsymbol{x}$ on l does not depend on the choice of the plane H containing l. Hence $\varphi(\sigma_1), \varphi(\sigma_2)$ are concentric spheres. □

1.5 Soddy's Hexlet

A cyclic sequence $\sigma_1, \sigma_2, \ldots, \sigma_n$ of n spheres which are mutually noncrossing and satisfy that

$$\sigma_i \text{ and } \sigma_j \text{ are tangent } \Leftrightarrow i - j \equiv \pm 1 \pmod{n},$$

is called an *n-cycle of spheres*. For a 3-cycle τ_1, τ_2, τ_3 of spheres, a cycle of 6 spheres

$$\sigma_1, \sigma_2, \ldots, \sigma_6$$

is called a *hexlet* (or *Soddy's hexlet*) for τ_1, τ_2, τ_3 if each σ_i is tangent to τ_i, $i = 1, 2, 3$.

Theorem 1.10 (Soddy's Hexlet) *Let τ_1, τ_2, τ_3 be a 3-cycle of spheres. For any sphere S that is tangent to these three spheres, there is a hexlet for τ_1, τ_2, τ_3 that contains S as a member.*

Proof Let $\boldsymbol{p}$ be the contact point of τ_1, τ_2, and let φ be an inversion of $\mathbb{R}^3$ with center $\boldsymbol{p}$. Then $\varphi(\tau_1), \varphi(\tau_2)$ form a pair of parallel planes, and $\varphi(\tau_3), \varphi(S)$ are mutually tangent spheres, both tangent to the pair of parallel planes. We may suppose that $\varphi(\tau_1), \varphi(\tau_2)$ are the planes $z = -1, z = 1$ in $\mathbb{R}^3$, respectively, and that $\varphi(\tau_3), \varphi(S)$ are unit spheres with center $O = (0, 0, 0)$, $\boldsymbol{q}_1 = (2, 0, 0)$, respectively. Let $\boldsymbol{q}_j = (2\cos\frac{j\pi}{3}, 2\sin\frac{j\pi}{3}, 0)$ for $j = 1, 2, 3, 4, 5$, see Fig. 1.7. Then five unit spheres with centers $\boldsymbol{q}_j$, $j = 1, 2, 3, 4, 5$, and $\varphi(S)$ form together a 6-cycle of unit spheres all tangent to $\varphi(\tau_i), i = 1, 2, 3$. By applying φ to these five spheres and $\varphi(S)$, we get a hexlet for τ_1, τ_2, τ_3 that contains S as one of its members. □

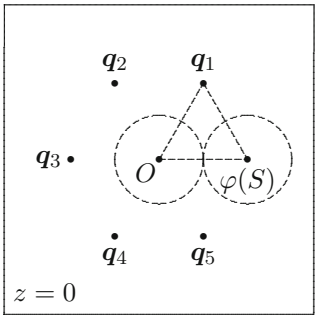


Fig. 1.7 Centers $\boldsymbol{q}_j$ of 5 unit spheres on the plane $z = 0$

1.6 Stereographic Projection

Let $\mathbb{S}$ be the sphere of radius r centered at the origin O in $\mathbb{R}^3$, and let $\boldsymbol{p} \in \mathbb{S}$, and $\boldsymbol{p}^*$ be its antipodal point (that is, the line segment $\boldsymbol{p}\boldsymbol{p}^*$ is a diameter of $\mathbb{S}$). Let H be the plane in $\mathbb{R}^3$ that is tangent to $\mathbb{S}$ at $\boldsymbol{p}^*$. For $\boldsymbol{x} \in \mathbb{S} \setminus \{\boldsymbol{p}\}$, let $\bar{\boldsymbol{x}}$ be the intersection point of the line $\boldsymbol{p}\boldsymbol{x}$ and the plane H. Then the map

$$\varphi : \mathbb{S} \setminus \{\boldsymbol{p}\} \to H;\ \boldsymbol{x} \mapsto \bar{\boldsymbol{x}}$$

is called the *stereographic projection* of $\mathbb{S}$ from $\boldsymbol{p}$, see Fig. 1.8. The stereographic projection φ is a bijection from $\mathbb{S} \setminus \{\boldsymbol{p}\}$ to H. As observed from Fig. 1.8, $\triangle \boldsymbol{p}\boldsymbol{x}\boldsymbol{p}^* \sim \triangle \boldsymbol{p}\boldsymbol{p}^*\bar{\boldsymbol{x}}$ (similar). Hence $|\boldsymbol{p}\boldsymbol{x}| : |\boldsymbol{p}\boldsymbol{p}^*| = |\boldsymbol{p}\boldsymbol{p}^*| : |\boldsymbol{p}\bar{\boldsymbol{x}}|$, and

$$|\boldsymbol{p}\boldsymbol{x}| \cdot |\boldsymbol{p}\bar{\boldsymbol{x}}| = |\boldsymbol{p}\boldsymbol{p}^*|^2 = (2r)^2. \tag{1.2}$$

Thus, the stereographic projection φ is the restriction of the inversion of $\mathbb{R}^3$ with center $\boldsymbol{p}$ and power $(2r)^2$ to $\mathbb{S} \setminus \{\boldsymbol{p}\}$.

Since a circle on $\mathbb{S}$ is the intersection of $\mathbb{S}$ and a sphere (or a plane) in $\mathbb{R}^3$, it is mapped by the inversion to the intersection of H and a sphere (or a plane) by Theorem 1.8. Hence the next theorem follows.

Theorem 1.11 (Circle-to-Circle Correspondence) *The stereographic projection $\varphi : \mathbb{S} \setminus \{\boldsymbol{p}\} \to H$ is a bijection. If K is a circle on $\mathbb{S}$ that passes through $\boldsymbol{p}$, then $\varphi(K)$ is a line on H. If K is a circle that does not pass through $\boldsymbol{p}$, then $\varphi(K)$ is a circle on H.* □

By a tangent line l of a circle K at a point $\boldsymbol{x} \in K$, we mean a tangent line of K at x in the plane containing K. If K lies on the sphere $\mathbb{S}$, then this tangent line l and $\mathbb{S}$ intersect in one point, and hence l is also a tangent line of $\mathbb{S}$ at $\boldsymbol{x}$. If a circle K on $\mathbb{S}$ passes through $\boldsymbol{p}^*$ and $p \notin K$, then the tangent line of K at $\boldsymbol{p}^*$ is contained in the

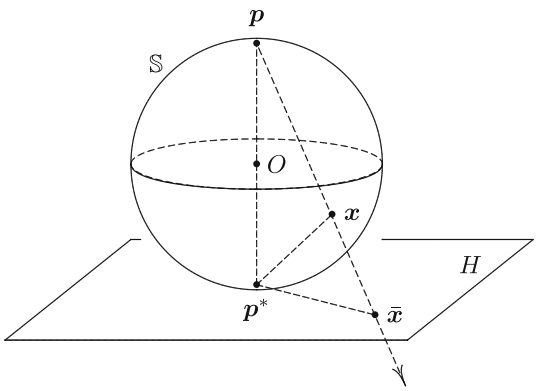


Fig. 1.8 Stereographic projection

plane H, and since l and $\varphi(K)$ share only one point p^*, l is also a tangent line of $\varphi(K)$ at $\bar{\boldsymbol{p}}^* = \boldsymbol{p}^*$.

For a pair of intersecting circles in the plane, the angle formed by the tangent lines of the two circles at an intersection point is called the *crossing angle* of the circles. Similarly, for a pair of circles on a sphere $\mathbb{S}$, their crossing angle is defined as the angle formed by the two tangent lines at an intersection point. If two circles are tangent, then their crossing angle is 0. When two circles cross each other, their crossing angle at one intersection point and the crossing angle at the other intersection point are equal.

Theorem 1.12 (Invariance of Angles) *The stereographic projection $\varphi : \mathbb{S} \setminus \{p\} \to H$ preserves the size of angles; that is, the crossing angle of two intersecting circles on $\mathbb{S}$ is equal to the crossing angle of their images on H.*

Proof Let K_1, K_2 be two circles on $\mathbb{S}$ intersecting at $\boldsymbol{q} \in \mathbb{S} \setminus \{\boldsymbol{p}\}$. Note that for each $i = 1, 2$, there is a circle K_i' that passes through $\boldsymbol{p}$ and is tangent to K_i at $\boldsymbol{q}$. Then the crossing angle of K_1, K_2 is equal to the crossing angle of K_1', K_2'. Note also that $\varphi(K_i)$ and $\varphi(K_i')$ are tangent at $\varphi(\boldsymbol{q})$. Hence, the crossing angle of $\varphi(K_1), \varphi(K_2)$ is equal to the crossing angle of $\varphi(K_1'), \varphi(K_2')$. Therefore, it is enough to show that for two intersecting lines on H, the angle formed by them is equal to the crossing angle formed by their correponding circles (the inverse images of the lines) on $\mathbb{S}$. Let l_1, l_2 be two crossing lines on H, and let K_1, K_2 be the corresponding circles on $\mathbb{S}$, respectively. Then, K_i is the intersection of $\mathbb{S}$ and the plane determined by $\boldsymbol{p}$ and l_i. Let l_i' be the line through $\boldsymbol{p}$ and parallel to l_i. Then l_i' is the tangent line of K_i. Hence the crossing angle of K_1, K_2 is equal to the angle formed by l_1', l_2', which is equal to the angle formed by l_1, l_2. □

1.7 Appendix: Poncelet's Porism

The following theorem is called Poncelet's porism for circles.

Theorem 1.13 *Let C, K be disjoint circles in the plane such that C lies inside K. Suppose that there is an n-gon Γ satisfying that*

(i) *all its edges are tangent to C and*
(ii) *all its vertices lie on K.*

Then, for any point P_1 on K, there is an n-gon that contains P_1 as a vertex and also satisfies (i) *and* (ii)*.*

Proof The following proof is based on [710]. Let R, r be the radii of K, C, and O, Q be their centers, respectively. Let A be the intersection of the ray $\overrightarrow{QO}$ and K, see Fig. 1.9. Let us use polar coordinates with pole O and polar axis $\overrightarrow{OA}$. For a point $X = (R, \theta) \in K$, let $Y = (R, \tau) \in K$ be the point such that $\theta < \tau < \theta + \pi$ and XY is tangent to C. Then the argument τ of Y is a continuous function of θ: $\tau = f(\theta)$.

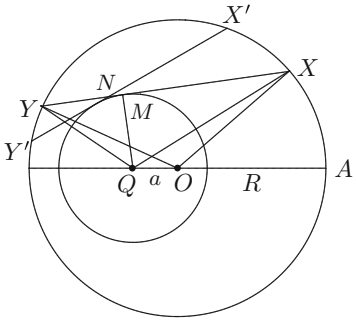


Fig. 1.9
$\theta = \angle AOX$, $\tau = \angle AOY$

First, let us find $\frac{d\tau}{d\theta} = f'(\theta)$. Let $X' = (R, \theta + \Delta\theta)$ and $Y' = (R, \tau + \Delta\tau)$ be a point such that $X'Y'$ is tangent to C. Let M be the contact point of XY and C, and N be the intersection point of XY and $X'Y'$. Then $\triangle XNX'$ and $\triangle Y'NY$ are similar. Hence $\frac{|YY'|}{|X'X|} = \frac{|NY|}{|NX'|}$. Since $\widehat{XX'} = R\Delta\theta$, $\widehat{YY'} = R\Delta\tau$, we have

$$\frac{d\tau}{d\theta} = \lim_{\Delta\theta\to 0}\frac{\Delta\tau}{\Delta\theta} = \lim_{\Delta\theta\to 0}\frac{\widehat{YY'}}{\widehat{XX'}} = \lim_{\Delta\theta\to 0}\frac{|YY'|}{|X'X|} = \lim_{\Delta\theta\to 0}\frac{|NY|}{|NX'|} = \frac{|MY|}{|MX|}.$$

Thus, $f'(\theta) = \frac{d\tau}{d\theta} = \frac{|MY|}{|MX|}$. Let us compute $|MX|, |MY|$. Put $a = |OQ|$. Since $\angle XOQ = \pi - \theta$, we have

$$|QX|^2 = R^2 + a^2 - 2Ra\cos(\pi - \theta) = R^2 + a^2 + 2Ra\cos\theta.$$

Similarly, we have $|QY|^2 = R^2 + a^2 + 2Ra\cos\tau$. Hence

$$|NX|^2 = |QX|^2 - r^2 = R^2 - r^2 + a^2 + 2Ra\cos\theta$$
$$|NY|^2 = |QY|^2 - r^2 = R^2 - r^2 + a^2 + 2Ra\cos\tau.$$

Put $G(t) = 1\Big/\sqrt{1 + \frac{2Ra\cos t}{R^2 - r^2 + a^2}}$. Note here that $G(t)$ is a positive periodic function with period 2π defined in $-\infty < t < \infty$. Using this function, we get

$$|NX| = \sqrt{R^2 - r^2 + a^2}/G(\theta), \quad |NY| = \sqrt{R^2 - r^2 + a^2}/G(\tau).$$

Hence we have $f'(\theta) = \frac{|NY|}{|NX|} = \frac{G(\theta)}{G(\tau)}$. Since $\tau = f(\theta)$, we can write this in the following way:

$$G(f(\theta))f'(\theta) = G(\theta). \tag{1.3}$$

Define $J(\theta)$ by

$$J(\theta) = \int_{\theta}^{f(\theta)} G(t)dt.$$

If $F(t)$ is the primitive function of $G(t)$, then $J(\theta) = F(f(\theta)) - F(\theta)$. Hence

$$\frac{dJ(\theta)}{d\theta} = F'(f(\theta))\, f'(\theta) - F'(\theta) = G(f(\theta))f'(\theta) - G(\theta) = 0$$

by (1.3), and hence $J(\theta) = \omega$, a constant independent from θ. Thus,

for any θ, the line segment connecting the points (R, θ) and $(R, f(\theta))$ is tangent to C, and $J(\theta) = \int_{\theta}^{f(\theta)} G(t)dt = \omega$.

Now, let $X_1, X_2, \ldots, X_n$ be the vertices of the n-gon Γ in anti-clockwise order, and let $X_i = (R, \theta_i)$, $i = 1, 2, \ldots, n$. Then, $f(\theta_i) = \theta_{i+1}$, $i = 1, 2, \ldots, n-1$. Since $\angle X_1 O X_2 + \angle X_2 O X_3 + \cdots + \angle X_n O X_1 = 2\pi$, we have

$$(\theta_2 - \theta_1) + (\theta_3 - \theta_2) + \cdots + (\theta_n - \theta_{n-1}) + (f(\theta_n) - \theta_n) = 2\pi.$$

Hence $f(\theta_n) = \theta_1 + 2\pi$, and therefore

$$\int_{\theta_1}^{\theta_1 + 2\pi} G(t)dt = J(\theta_1) + J(\theta_2) + \cdots + J(\theta_n) = n\omega.$$

Since $G(t)$ is a periodic function with period 2π,

$$\int_{\theta}^{\theta + 2\pi} G(t)dt = n\omega \quad \text{for every } \theta. \tag{1.4}$$

Let $P_1 = (R, \eta)$, and denote the iterated composition $\overbrace{f \circ f \circ \cdots \circ f}^{k}$ of the function f by f^k. Then

$$\int_{\eta}^{f^n(\eta)} G(t)dt = J(\eta) + J(f(\eta)) + J(f^2(\eta)) + \cdots + J(f^{n-1}(\eta)) = n\omega.$$

Since $G(t)$ is a positive function, (1.4) implies that $f^n(\eta) = \eta + 2\pi$. Therefore, the point $(R, f^n(\eta))$ coincides with $P_1 = (R, \eta)$, and hence the n-gon with vertices

$$P_1 = (R, \eta), P_2 = (R, f(\eta)), P_3 = (R, f^2(\eta)), \ldots, P_n = (R, f^{n-1}(\eta))$$

satisfies (i) and (ii). □

1.8 Exercises

1.1 Let φ be an inversion of the plane $\mathbb{R}^2$ with center O. For a point $\boldsymbol{q} \neq O$, let C be a circle that passes through $\boldsymbol{q}$ and $\varphi(\boldsymbol{q})$. Prove that $\varphi(C) = C$.

1.2 Prove that by an inversion of the plane, a pair of mutually tangent circles is sent to either (i) a pair of mutually tangent circles, or (ii) a circle and its tangent line, or (iii) a pair of parallel lines.

1.3 Let A, B be two circles in the plane. Prove that there is an inversion of the plane that sends A, B to circles of the same size. (Consider mutually tangent two circles that are also tangent to both A and B.)

1.4* Suppose that A, B are disjoint circles in the plane, and the circle B lies in the interior of the circle A. Prove that the locus of the centers of such circles that are internally tangent to A and externally tangent to B is an ellipse.

1.5 Let $\sigma_1, \sigma_2, \sigma_3, \sigma_4$ be mutually tangent four spheres in $\mathbb{R}^3$, and suppose that their contact points are all different. Prove that the number of spheres (including planes) that are tangent to all these four spheres is exactly two.

1.6 Prove that for each of the following conditions (1), (2), (3), there is a hexlet for a cycle of three spheres that satisfies the condition.

(1) Just one element in the hexlet is a plane.
(2) Two elements in the hexlet are planes.
(3) One element (sphere) in the hexlet encloses all other five elements in the hexlet.

1.7* Is it possible to draw more than five circles in the plane so that each circle is tangent to exactly five other circles?

1.9 Notes

Due to the existing literature, the following notes are very extensive. We report on main themes from Chap. 1, which are: inversion in the plane and in 3-space, Euler's triangle theorem, Steiner chains and Steiner's porism, Soddy's famous hexlet, the stereographic projection and, as an appendix, also Poncelet's porism.

1.9.1 Notes on Inversion in the Plane and in 3-Space

As a non-linear transformation in planar elementary geometry, the *inversion in circles* yields many elegant applications and insights, and the study of the inversive plane and higher dimensional analogues can create demanding geometric exercises. Regarding the history (at least with the background of non-Euclidean geometries), we refer to Rosenfeld's monograph [676]: e.g., the contributions of Apollonius are

given there on pp. 114–116, and various further contributors are discussed in other book parts.

We give here a selected family of basic books and booklets that deal completely, or in form of whole chapters, in a comprehensive way with this interesting mapping and its properties and applications. These references reflect different viewpoints, motivations, approaches, and applications, namely referring to *axiomatic and elementary geometry* (giving aid for solving related problems from circle geometry, like the Apollonian one, and presenting theorems important in *foundations of geometry*), *inversive planes and spaces*, theory of *circular tufts and conformal geometry, projections* (e.g., the stereographic one), investigations around *Möbius transforms, complex numbers*, (models of various) *non-Euclidean geometries* and, last not least, *teaching aspects*. As common features of all these selected references, one obviously sees a broad and geometric introduction of the inversion itself, and that at least one of the mentioned aspects and application fields is discussed there. Clearly, following all these criteria, the selection given now cannot be complete, but we try our best.

Books and Book Chapters

Some older references (mainly from the first half of the twentieth century) are Hadamard [353, Chapters V, VI and VII of the Complements to Book III], the first chapter of Coolidge's basic book [177], Johnson's widely cited monograph [430, pp. 43–57 and pp. 96–109], the famous book [179, Part II of Chapter 3] of Courant and Robbins, and the booklet [698] of Schmidt.

Altshiller-Court contributed with two related books: in Chapter 9 of [28] he introduces inversion, and there are also interesting applications; and in [29, Chapter VIII] he discusses comprehensively the three-dimensional version. We mention here the paper [92] in which problems around inversion are extended to 3-space, see also Section 5.8 in Borceux [103]. Later the following related references came to be: Yaglom [822, Chapters 1 to 4 and the supplement], Pedoe's booklet [627, Chapter I], the monograph [369, §39 and §40] due to Hajós, Coxeter's classical book [182, Chapter 6], Schwerdtfeger's monograph [721, §2 and Chapter II], the chapters on inversion and projection in Perfect's book [630], Bakelman's booklet [52], Guggenheimer's monograph [344, Chapters 8 and 9], Coxeter and Greitzer [186, Chapter 5], especially (and not only) Chapters II and VI of Pedoe's comprehensive book [625], Greenberg's excursion [322, Chapters 7 and 9] into non-Euclidean geometries, and Ewald's basic monograph [251, Chapter 6].

Further on, we should cite Giering's book [299, Chapters 18, 19, and 20], Section 9.5 and great parts of Chapter 10 in Berger's comprehensive double monograph [75] (see also Chapter II of his more recent book [78] where also inversion in space is explicitly discussed), Chapter 3 (from 3.6 on) of Eves [250], Chapter 5 on inversive geometry in [124], Hartshorne's basic book [382, Section 37] and applications in the sections following there, and the book [91] of Blair in which also Liouville's theorem on conformal maps and the question of when a convex body

remains convex under a sphere inversion are comprehensively discussed (see also [92]). Furthermore, [103, Sections 5.7 to 5.11] of Borceux should be mentioned, particularly Section 5.8 where spatial inversion is studied. Chapter 4 as well as further parts with applications in Aumann's book [48] should be added; this book broadly discusses circle geometry, like also the book [644], where inversion and applications of it yield the topic of Chap. 7.

In addition we refer to the broad representations [822], [823], and [820] of Yaglom from the encyclopedic collection [17]. Also the surveys [811] of Wilker and [665] of Reventós Tarrida treating inversive geometry should be mentioned here.

A Few Applications

Finally we want to mention two interesting applications of inversion: *mechanical linkages and fractals*.

Peaucellier's inversor, a mechanical instrument invented in 1873 to draw the inverse of a given curve, is explained in many important books; we mention here Johnson's monograph [430, Sections 67 to 72], the famous book [397, §40] of Hilbert and Cohn-Vossen, Schmidt's booklet [698, Chapter VI], Coxeter and Greitzer [186, Section 5.3], the end of Section 38 in the book [250] of Eves, and Aumann [48, Section 4.4]. The mechanical models described in the classical book [193] include two- and three-dimensional linkages, among them also the Peaucellier linkage. The spherical analogue of Peaucellier's inversor was studied by Goldberg [307], and Reed [661] used the linkage to prove that certain oblique sections of a torus are circular. In some of the references above also the related *inversor of Hart* is presented.

It is well-known that inversion in circles is also important for the construction of *fractals*, see Mandelbrot's basic monograph [543]. We mention some further results in this direction. Clancy and Frame [165] study fractal limit sets of groups of inversions in the plane, and Gdawiec [296] extends, in view of related infinite circle inversion fractals, circles to star-shaped sets. Similarly, Ramírez et al. [654] extend the used circle inversions to inversions in unit circles of certain normed planes and apply this extended transformation to famous fractals, such as the Sierpinski triangle and the Fibonacci fractal. Boreland and Kunze [104] present mathematical foundations to fractal constructions based on circle inversions, showing also a strong connection to the theory of iterated function systems.

1.9.2 Notes on Euler's Triangle Theorem

Euler's triangle theorem says that for any triangle the radius R of the circumcircle, the radius r of the incircle and the distance d between the centers of these two circles satisfy the equality $d^2 = R(R - 2r)$. This theorem implies the *Euler inequality* $R \geq 2r$, with equality iff the considered triangle is equilateral. Euler's theorem

was obtained already by Chapple in 1746 (see [156, page 123]) and is certainly much older. There are many books and articles where Euler's triangle theorem and/or his inequality are discussed or applied to get further inequalities. As examples we mention Johnson's book [430, p. 186], Altshiller-Court [29, Chapter 3, §152], as well as Alsina and Nelsen [24, p. 56]. In Chapter 6 of the monograph [723] by H. Sedrakyan and N. Sedrakyan problems and inequalities are selected which have to do with circle radii. Among them one can also find a discussion of the circum-inradius inequality for triangles. And in the glossary of the comprehensive book [228], where almost 50 terms, theorems, formulas, functions, equations, and techniques are collected, today commonly associated with Euler, also the theorem discussed here is listed (see page 300 there).

For all triangles inscribed in a circle, Pambuccian and Schacht [613] give a synthetic proof of Euler's inequality for Hilbert's absolute geometry, characterizing equilateral triangles by the equality case. And Carlitz [141] collects several inequalities which are particularly related to Euler's inequality. An additional characterization of equilateral triangles is given by their area.

And in [764] a proof of Euler's inequality in the *hyperbolic and spherical case* is given.

1.9.3 Notes on Steiner Chains and Steiner's Porism

A finite sequence of circles each being tangent to two fixed and non-intersecting circles and to two further circles of that sequence is called a *Steiner chain of circles* or a *Steiner cycle*. The so-called *Steiner's porism* can also be formulated as follows: if two such given circles allow such a chain, then they even admit an infinite number, and each of these chains contains the same number of circles. In addition, any circle tangent to the two given circles and surrounding either none or both of them is a member of such a chain.

Books and Book Chapters

Often in combination with an interesting variety of related results on circles and also on spheres in higher dimensions (e.g., the corresponding use of inversion, the arbelos, and Soddy's hexlet), theorems about Steiner chains and, in particular, Steiner's porism can be found in the books by Coolidge [177, pp. 31–37], Johnson [430, Chapter VI], Coxeter and Greitzer [186, pp. 123–126, 175–176, 180], Pedoe [625, Section 25], Eves [250, pp. 134–135], Honsberger [401, Chapter 6], Ogilvy [601, pp. 51–54], Wells [806, pp. 244–245], Barnes [60, Chapter 7], and Aumann [48, Chapter 9]. Also at the end of Chapter 1 of Yaglom's book [822] Steiner chains and related configurations are discussed. In addition we mention here the book [644]; in its Chapter 7 Steiner chains and the Apollonius problem are nicely presented.

Steiner chains and Soddy's hexlet belong to the famous objects that were also found by the *Japanese Wasan masters*, predating their discovery in the West. More precisely, Steiner chains were studied already in the Japanese temple geometry (Wasan) by Ajima Naonobu (1732–1798), i.e., earlier than by Jakob Steiner (1796–1863). Ozone [608] studies Steiner chains as given in Wasan, and he discusses elegant ways to study and prove them. Regarding the occurence of these two notions in the Japanese temple geometry, we also cite here Fukagawa and Pedoe [284, Chapters 1 and 9]. The interested reader might also look at Akopyan's nice booklet [13, Section 6.8] "without words, but figures".

Some Research Results

Besides these representations, the following articles might be additionally quoted. Generalizing former results, Iwata [424] gives an analytic condition for getting *a configuration of hyperspheres in n-space presenting an analogue of Steiner chains*. In the two papers [669] and [137] three-dimensional analogues of Steiner's porism are given. In the first paper, a computer algebra system is used, and in the second contribution the analogue of Steiner's idea is slightly generalized, showing also a surprising connection to Platonic solids. In the expository paper [665] of Reventós Tarrida Steiner's porism is elegantly derived via inversion.

Yiu [827] studies *rational Steiner chains* in which R and r (the radius of the larger resp. smaller given circle) as well as d (the distance between the midpoints of both circles) are given by rational numbers. It is easy to see that rational Steiner m-chains exist only for $m = 3, 4, 6$. Using previous results, the author presents a parametrization for all triples (R, r, d) with $R = 1$ corresponding to a rational Steiner chain. Protasov [649] gives a nice discussion of Steiner's porism, and Baralić et al. [56] extend the notion of Steiner chain beginning with more than two circles. Also for this situation they find an analogue to the Steiner porism, and they pose a couple of interesting open questions.

In [524] Maehara presents interesting bend formulas for Steiner chains of circles and extends this also to sphere systems in higher dimensions.

Musin [591] studies configurations of hyperspheres in n-dimensional space, getting far-reaching high-dimensional extensions of classical results concerning Steiner's porism (in the plane) and Soddy's hexlet (in 3-space). To obtain these results, he uses kissing arrangements of hyperspheres, corresponding packings and spherical codes.

For an odd prime power q, Hungerbühler and Kusejko [417] study Steiner's porism in *finite Miquelian Möbius planes* constructed over the pair of finite fields GF(q) and GF(q^2). The authors derive properties of common tangent circles for two given concentric circles and obtain a finite version of Steiner's porism for concentric circles. Using the quadratic residue theorem in GF(q), conditions on the length of a Steiner chain are derived. Also extensions to arbitrary pairs of non-intersecting circles are given.

The paper [719] of Schwartz and Tabachnikov deals with the 1-parameter family of Steiner chains of cardinality k, say, as they are described by Steiner's porism for two fixed given circles. E.g., they ask: what do the members of such a Steiner chain of cardinality k have in common? They show that the first $k-1$ moments of their curvatures remain constant within a 1-parameter family (the case $k=3$ is established by the circle theorem of Descartes). Using also *spherical designs*, first a more general result is proved, and from this the main results on Steiner chains are derived. In addition, the authors succeed in extending one of their main results on Steiner chains to spherical and hyperbolic geometries.

Motivated by invesitigations around *Poncelet's porism*, Heidari and Honari [390] derive an extension of Steiner's porism to n-dimensional space. Steiner chains for n given spheres are introduced, and by the three-inversion theorem it is shown that there exists a 1-parameter subgroup of Möbius transformations which preserves these n spheres. The authors obtain a corresponding 1-parameter family of Steiner chains by applying the elements of this subgroup to a Steiner chain.

1.9.4 Notes on Soddy's Hexlet

The famous *hexlet of Soddy* is a special three-dimensional analogue of the Steiner porism, and it was perhaps first observed in the Japanese temple geometry (see [284, Chapter 9] (Sangaku tablets from 1822 in the Kanagawa prefecture). Soddy, who received the Nobel prize in 1921 for discovering isotopes, rediscovered it (cf. [747]).

Books and Book Chapters

As classical geometric configuration of spheres, Soddy's hexlet is more detailed mentioned or studied in various geometry books, namely those of Coxeter [182, Section 1.59], Pedoe [625, Section 40.3], Honsberger [401, Chapter 6], Ogilvy [601, Sections 5.3 and 5.4], Eves [250, p. 152], Barnes [60, Chapter 7], as well as Fukagawa and Pedoe [284, Chapter 9]. In the geometry dictionary [806], Soddys hexlet is discussed on page 232.

Some Research Results

The envelope of a varying sphere touching three fixed spheres is a *Dupin cyclide*. Coxeter [181] studies two families of spheres which touch any three members of the other family and two from its own family, thus forming rings as it is the case with Soddy's hexlet. He considers the case when the cyclide is of such a shape that there is a closed ring of m spheres, every of them touching the next. Each sphere of the same family belongs to such a ring if there is one ring. And if this one goes round the

cyclide d times, then it is called a p-ring with $p = m/d$. Using stereographic projections of inspheres of cells of certain 4-polytopes, Coxeter proves the following nice result of Steiner: if one considered family has p-rings, then the other has q-rings, with the equality $1/p + 1/q = 1/2$. Soddy's hexlet is the subcase $p = 3, q = 6$.

In [811] Wilker developed an elegant method to represent n-balls in n-dimensional inversive space. He applies this method in [812] to study Soddy's hexlet, e.g. numerical relations between curvatures of participating spheres. He shows, for example, that the sum of the curvatures of opposite spheres of the hexlet is equal to twice the sum of the curvatures of the original three tangent spheres. Suitably defining *curvature* for the spherical and hyperbolic 3-space, Wilker succeeds in extending his results also to these non-Euclidean cases. The paper [669] of Roanes-Lozano was discussed already above, when referring to Steiner chains of circles. Variants related to Soddy's hexlet are studied. And also Musin's paper [591] was already presented there, due to his successful use of spheical codes when studying analogues of Steiner's porism and Soddy's hexlet.

Maehara and Oshiro [542] define an *m-cycle of balls* to be a cyclic sequence of m three-dimensional balls, non-overlapping, where every two consecutive balls are tangent to each other, and an (m, p)-link be a pair of an m-cycle and a p-cycle that form a non-splittable link with no two balls overlapping. It is shown that a $(3, p)$-link exists only when $p \geq 6$, and in any $(3, 6)$-link every ball from the 3-cycle is tangent to all balls of the 6-cycle. Further on, a $(4, 4)$-cycle exists, and in any such cycle every ball is tangent to all balls in the other 4-cycle, too. And also in the paper [524] variants of Steiner chains, Soddy's hexlet and further sphere systems are investigated.

1.9.5 Notes on the Stereographic Projection

In the classical sense, the *stereographic projection* is a perspective projection of the 2-sphere in 3-space, whose center of projection is the north pole of sphere, onto the tangent plane of it through the south pole. This projection is conformal (i.e., angles between curves are preserved; e.g., circles become circles), but it is not isometric or equiareal.

Some History and Applications

Its history goes back to ancient Greek astronomers who projected down the celestial sphere for studying movements of stars and planets in the plane. Important names occuring here are Hipparchus, Apollonius, Archimedes, and Ptolemy, but still many things about this early step are controversial. In form of the planispheric astrolabe the Byzantine and Arabic astronomers used and developed the related ideas further, and their knowledge reached Western Europe in the eleventh and twelfth century. Contributions and applications due to Mercator, Harriot, and Halley followed; the

latter gave the first proof for the conformality. The current name "stereographic projection" is due to d'Aguilon (1613), see [835]. Further material about the history is collected by Rosenfeld in [676, pp. 121–130, 242, 342]. Classifying the cartographic mapping procedures, Schröder [712] gives also a lot of historical information, particularly also around stereographic projection. In the second edition of [79] Berggren gives a more thorough exposition on using the stereographic projection in the astrolabe, an important step.

Today stereographic projection is famous as an elegant tool to visualize elegantly mathematical phenomena, often astonishing. In mathematics, this refers, e.g., to *models of non-Euclidean geometries, compactification in topology, complex analysis and the Riemann sphere, projective geometry, differential geometry, arithmetic geometry, polytope theory (Schlegel diagrams), simplifications of integrals involving trigonometric functions* and so on. In neighbouring disciplines we have it in cartography, geodesy, crystallography, photography and optics (e.g., fisheye lenses and mappings of spherical panoramas), geology, planetary science and so on.

Books and Book Chapters

Let us start with book chapters where stereographic projection is comprehensively introduced and applications are shown, first within mathematics and then in a few other fields.

Classical geometry books containing such comprehensive sections or chapters are Klein [455, pp. 50–51 and pp. 294–300], Johnson [430, pp. 106–109], Hilbert and Cohn-Vossen [397, §36], Altshiller-Court [28, pp. 261–262], Coxeter [182, Section 16.2], see also the survey Rosenfeld [674, Section 4.1]. And also books mainly devoted to other themes (like inversions) contain a lot of material on stereographic projection; see, e.g., Schmidt's book [698].

In the second part of Borsuk's monograph [105] on n-dimensional analytical geometry, stereographic projection is used to study projective and Möbius n-dimensional space. Similar in nature is Blaschke's book [94], where in Chap. 2 the natural way from stereographic projection to inversions and linkages is shown.

Yaglom [822, pp. 74–78] introduces and uses stereographic projection for proving that any circular transformation of the plane can be realized by an inversion followed by a similarity.

Milman and Parker [575, pp. 301–305] introduce the stereographic projection for studying mainly the *Klein and Poincaré disk models*. Similarly, in a classical sense and using it for models of non-Euclidean geometry, Prasolov and Tikhomirov [646, pp. 93–96] introduce the projection type. The latter books are nicely written also for teaching aspects on the undergraduate level. Even more elementary is the known book [186] of Coxeter and Greitzer, successfully integrating it into a network of related notions.

Stereographic projection yields also an elegant way to get the formulas for *spherical trigonometry* from those of planar geometry. The crystallographer and mineralogist G. Cesàro developed this way; we refer to Donnay's book [220] for

more details. For this theme we refer also to Chapter 8 of the more recent, nice book [131] of van Brummelen.

In the long Chapter 18 of his monograph [299] Giering introduces the stereographic projection more generally, for *quadrics and in the projective setting*, also with coordinate representations. The projection of spheres he treats as a subcase. (The same repeats with inversions, in the next chapter.) On an analogous level, Pedoe [625, pp. 387–390] studies stereographic projection, too. Here we mention also Schwerdtfeger's book [721] on Möbius transformations, complex numbers, and models of non-Euclidean geometries; in Part (3) of Chap. 1 there is a nice introduction to stereographic projection, with interesting examples.

Bigalke [87] introduces in Section 5.5 the stereographic projection, giving various properties, and in Section 9.1 he comprehensively discusses the construction of so-called pole figures of crystals generated by their facet normals, and how stereographic projection and its conformality is used in that framework. Basic constructions are carefully presented, like also the related Wulff net, and finally *crystallographic results* are represented, like the corresponding point groups. Further applications in crystallography are given.

Berger discusses in [75, Part II], more precisely in the subsections 18.1.4, 18.1.8.6, and 18.10.2, various sides of stereographic projection, e.g.: definitions, basic properties, cartography, and the Möbius group. In his more recent book [78] he discusses at the beginning of Section II.5 and in Section III.2 the conformal group and applications of the *Möbius group*.

Section 6.3 of the textbook [458] of Knörrer is one of the most careful introductions and representations of stereographical projections. It uses *group theory and Riemann's number ball*, and in Section 6.7 the author uses a four-dimensional stereographic projection to study the fibers of the Hopf map.

Borceux [103, Section 5.9], introduces the main properties of stereographic projection, and also Aumann [48, Section 7.4] gives a nice introduction to it, enumerating main properties and supporting this by wonderful figures. A nice presentation of properties of stereographic projection is also given in Section 5.4 of Glaeser's book [302].

Some Research Results: Differential Geometry

The following references are given in thematic order, and within these themes chronologically. We give first a few related publications from *differential geometry*.

We begin with Euler who studies in [246] properties of angle-preserving maps; in particular, he develops the differential geometry of the stereographic projection.

In his nice book [815], Wilson starts with Euclidean geometry, switching then to spherical and hyperbolic geometry and taking several versions of the *Gauss-Bonnet theorem* as guideline; finally this theorem is also discussed for closed surfaces. At the end of Chap. 2 he presents Möbius geometry based on stereographic projection.

The science of *invisibility* combines Einstein's general relativity and Maxwell's principles of electromagnetism; Leonhardt and Philbin have a part in its trans-

formation from fiction to science, e.g. using analogies between mechanics and optics, and the geometry of curved spaces. Their book [501] gives the mathematical foundations, mainly differential geometry, and shows nicely how highly theoretical concepts can become useful in electrical and optical engineering. Due to Maxwell's role in this book, in the "optical part" also his fisheye lens is discussed, and how it is deduced from the stereographic projection.

Cecil and Ryan present in [149] a thorough treatment of the differential geometry of hypersurfaces in real, complex, and quaternionic space forms. They lay special emphasis on isoparametric and Dupin hypersurfaces in real space forms and Hopf hypersurfaces in complex space forms. The book shows at several places how important the stereographic projection in this framework is. After its introduction in Chap. 2, this projection type is used in Chap. 4 regarding *Möbius geometry, isoparametric and Dupin hypersurfaces*, and in Chap. 5 with respect to the classification of taut hypersurfaces in Euclidean space. The first two chapters have much overlap with the book [148] of the same authors, where stereographic projection is also needed.

Cartography and Geodesy

Not being far from differential geometry, we continue with references related to *cartography* and *geodesy*.

In the booklet [697] of Scheffers the basic knowledge from *differential geometry* is presented. More precisely, the analytic derivations of projections of the sphere onto the plane are treated, and this refers to the mercator, stereographic and equal-area projections.

A main contribution of the classic book [466] of König and Weise is the successful use of general complex vector coordinates for the spheroid, to apply then a direct cartographic transformation of this interesting object such that methods of *complex function theory* come into the game. In Chapter VII the stereographic projection, the conical projection, and the general arc projection of the spheroid are comprehensively discussed.

Although mentioned already above (regarding history), we cite here once more Schröder's nice little book [712]. Stereographic projection is also historically discussed.

Feeman exposes in [255] the mathematical ideas involved in creating and analyzing (geographical) maps. His booklet seems to be the first more recent book in which the famous problem of mapping (portions of) the Earth in a very understandable and, at the same time, comprehensive style. Chapter 9 deals with conformal mappings and, of course, particularly with the stereographic projection.

Classical Geometry, Foundations of Geometry

We continue with references in the spirit of *classical geometry* (also foundations of geometry) and *non-Euclidean geometries*.

Hartl [380] gives an elementary proof of the n-dimensional generalization of the classical theorem that two complex quadrics, having a conic section in common, have a second conic section in common. He applies this result to classical results on the n-dimensional stereographic projection.

Benz [73] develops *Möbius sphere geometry* for an arbitrary real inner product space X (i.e., a real vector space X of dimension at least 2 and not necessarily finite-dimensional, with a fixed scalar product, positive definite). Extending X suitably, he also introduces (in section 9) stereographic projection.

The book [667] of Richter-Gebert, wonderfully emphasizing pearls from *classical projective geometry* and bringing many results from various non-Euclidean geometries and related topics, is interesting for students, mathematicians, computer scientists, and physicists. Many of the presented results are nicely illustrated. In Part III also Möbius transformations and the stereographic projection are treated.

Popko's nice book [643] treats a large variety of topics based on the sphere in Euclidean 3-space. (Platonic) polyhedra, spherical polygons, packing problems and kissing spheres are examples. In Appendix B also stereographic projection and its basic properties are treated.

Swart and Torrence [766] describe advances in *photographic technology* (e.g., Google Maps street view) and their mathematical background on the level of geometric projections. All that we see is imprinted on a sphere, where we are in its center. To get an ordinary (plane) picture, suitable projections are used. One of them is stereographic projection, on which the article is (not only, but) mainly based.

In the papers [770] and [734] two classical results are reproved by using in a refreshing way stereographic projection: in the first case it is the famous *Apollonius problem* (on circles touching three given ones), and in the second *Lexell's theorem* (on equiareal spherical triangles with a fixed given side). The reader should note that the used stereographic projection refers to Möbius and Laguerre planes.

Ionascu [419] deals with the analogue of the *circle of Apollonius in spherical geometry* which is the pre-image via stereographic projection of an algebraic curve of degree three. And this curve has two connected components, where each is the reflection of the other through the center of the sphere.

Further Topics

The final four references are from different areas.

Wilker [813] uses stereographic projection for presenting the classical *Hopf fibration*. In his approach the participating 3-sphere is clearly the union of two solid tori with common boundary, that the fibers are pairwise linked circles.

In his nice, expository paper [468], Kollár presents the most important results, problems, tools and ideas from *birational geometry*. So he surveys also results on

rational and rigid hypersurfaces, and he shows the connection with the classification of varieties developed by Enriques and others. Explaining the rationality of quadric hypersurfaces, he starts with the historically essential stereographic projection.

In order to calculate lasing modes and optical power flow in *multisection lasers* with open boundaries, O'Connor et al. [599] introduce a formalism which is based on a projection of the complex-valued electric field and its spatial derivative onto a suitably extended complex plane. In single-section lasers, a laser mode is a loxodrome on the extended complex plane, and in a multisection laser, loxodromes for individual sections of the laser are derived. A natural visualization of such constructions is given by the stereographic projection of the extended complex plane onto the Riemann sphere.

The book [298] of Geiges on the importance of geometry in celestial mechanics deals mainly with geometric interpretations of the *Kepler problem*. In Chap. 8 the author treats the needed differential geometry and spherical geometry, and the role of stereographic projection is clarified, too.

1.9.6 Notes on Poncelet's Porism

Poncelet published in his *Traité des propriétés projectives des figures* (1822) the following important theorem from projective geometry, which is called Poncelet's porism: *If for two smooth (real) conics in the plane there exists an m-sided polygon which is inscribed in one conic and circumscribed about the other, then there are infinitely many such polygons, and each point of the first conic is vertex of one of them*. The interscribed m-gon is often called *Poncelet polygon*. It seems that the first solution to Poncelet's porism was proposed by W. Chapple [156], before it became known as Poncelet's theorem. Giving here an overview to what happened mathematically around this theorem, we are in a comfortable situation: there exist various excellent comprehensive surveys and even books around this theorem such that in great parts it suffices to cite them.

Some History

The history of the porism, of variants, modifications, generalizations, and applications of it can be read in the comprehensive surveys [113, Lecture 8] by Bos, [115] due to Bos, Kers, Oort, and Raven (this article contains also new results), and the two papers [208] and [209] by Del Centina. Also the interesting paper [718] (with a discovery referring to Poncelet's prisoner-of-war time in Russia) should be mentioned here. And we add also Martini's survey [546]; in its Section 2.4 Poncelet's porism is (based on older sources) updated till 1993.

Books and Book Chapters

There are even two books completely dedicated to Poncelet's porism and related topics. The first one [271] by Flatto gives two short proofs of Poncelet's and Cayley's theorem on polygons inscribed in a conic and circumscribed about another one, and this is done in the *complex projective plane* as well as the *real affine plane*. The needed tools are clarified, i.e., a short course on analytic geometry in the projective plane, Riemann surfaces, tori, and elliptic curves and functions is given. The real case for two ellipses does not need these preparations. After the proofs, relations to billiards and other topics are presented. A supplement of Tabachnikov is added, interestingly offering another look at Poncelet's porism and billiards. The second book [225], due to Dragović and Radnović, is based on the possibility to prove Poncelet's theorem with the help of *elliptic curves* depending on the representation of an elliptic curve as the double cover of the outer participating conic, with four ramification points. (An analogous covering regarding the inner conic is possible, too.) Based on this, in a comprehensive way the authors want to show what current research directions are related to the framework around the Poncelet porism (namely, dynamics of integrable billiards, algebraic geometry of hyperelliptic Jacobians, and classical projective geometry of pencils of quadrics). But a historical view is also given, underlining basic ideas and the way to the recent state of the art.

Of course, Poncelet's porism can also be found in classical geometry books, partially with proofs. Examples here are the book II of Berger [75, Section 16.6], that of Prasolov and Tikhomirov [646, pp. 178–184], and Berger's more recent book [78, Section IV.8]. Here we also mention Schoenberg's book [710] whose 14th chapter is dedicated to Poncelet's porism and Steiner cycles.

Expository Chapters and Articles

But Poncelet's porism is also very popular and suitable for collections (in books or expository papers) of really nice mathematical results from different fields of mathematics or geometry; see, particularly, e.g. [300] and [737], or the final chapter of the popular book [283] of Fuchs and Tabachnikov. And in Chapter 4 of the collection [787] of lectures (meant together with its first part), Ueno et al. present three lectures about Poncelet's theorem. An appendix, consisting of two parts, adds a proof using elementary plane geometry, whereas the second lecture contains an analytic proof. Posamentier and Geretschläger [644] treat the porism in Chap. 3.

Some Research Results

We continue now with selected research results which are not mentioned in the wide surveys cited above.

Ronda and Valdés [673] study the relation between Kruppa equations and Poncelet's porism to show via some suitable parametrization how the classical theory of projective conics yields new insights and results on the 3D reconstruction from two images taken with uncalibrated cameras. Depending on the relation between Schwarzenberger bundles and Poncelet curves, Vallès [794] gives a nice and short proof of Poncelet's porism. Mirman [576] considers the subcase of two given circles and uses a recurrent procedure based on a biquadratic equation to find out whether there is an m-sided Poncelet polygon interscribed between these circles. This procedure has interesting applications for Hankel determinants. Extensions to given ellipses and hyperbolas are also presented.

The next results cited here are closely related to *billiards*. Considering a Poncelet m-gon P, Levi and Tabachnikov [502] study the system of extensions of the sides of P to lines, thus getting an arrangement of lines which has $m(m+1)/2$ intersection points. Schwartz showed that these points interestingly lie on a family of conics, and the authors continue by relating Schwartz' arguments to billiards inside ellipses. Dragović and Radnović [224] study higher-dimensional analogues of Poncelet-like problems by synthetic methods. Observing the billiard nature of some classical constructions and configurations, the billiard algebra is developed. This is a group structure on the set of lines simultaneously tangent to $n-1$ quadrics from a given confocal family in Euclidean n-space. Successfully using this tool, the authors obtain various results. Mozgawa [587] defines bar billiards for two nested conics C_1, C_2 having the Poncelet porism property and proves that for any oval C, in a suitably small neighborhood to the inner conic C_2, there exist ovals C_{in} and C_{out} inside and outside of C such that the pairings (C, C_{in}) and (C, C_{out}) have the Poncelet property. Continuing these studies, Cieślak, Martini and Mozgawa [164] study the rotation index of bar billiards and show that it is rational iff the two circles (considered here instead of the conics above) have the Poncelet porism property. *Billiard knots* are knots describable as non-self-intersecting periodic orbits in a three-dimensional billiard. Pecker [623] proves that for any elliptic right cylinder D every type of knot can be realized as the trajectory of a ball in D. To show this, Jacobi's proof of Poncelet's theorem by means of elliptic functions is used.

Halbeisen and Hungerbühler [370] consider chains of conics $C_0, \ldots, C_{m-1}$ such that for each $0 \le i \le m-2$ there is a *Poncelet triangle* T_j with vertices on C_j whose sides are tangent to C_{j+1}. They show that each chain can be closed by adding an appropriate conic C_{m+1} and confirm the existence of particular closed chains where the contact points of T_i with C_{i+1} serve as vertices of the subsequent triangle T_{i+1}. Further related interesting results are elegantly derived.

Kasner's theorem concerns what he termed the inscribed and diagonal pentagons constructed from a given pentagon. The extension of this theorem to Poncelet polygons was repeatedly investigated. Tabachnikov [769] discusses connections between Kasner's and Poncelet's work, showing also a link to dynamical systems via the pentagram map. It is shown that also nowadays there are various related research activities.

Garcia et al. [293] study loci of triangle centers over variants of two-well known triangle porisms: the bicentric and confocal families. In a natural way this leads

to the general version of Poncelet's porism. It is confirmed that the loci of certain triangle centers and associated points remain conics and/or circles. And in [292] the authors study one-dimensional families of Poncelet triangles "between" a circle C (outside) and an ellipse E, calling them *Brocard porisms*. The corresponding Brocard angle is invariant, and the Brocard points are stationary at the foci of E. There is also a Brocard circle passing through the two foci of E and the midpoint of C, and there is a second Brocard triangle whose vertices are obtained as intersections of the symmedians with the Brocard circle. The corresponding one-dimensional new family of triangles is inscribed to the Brocard circle and circumscribed about another ellipse. Continuing this generation procedure, an infinite, converging sequence of porisms created by certain nested objects is obtained. The authors derive many geometric results about this continuous family of porisms.

Chapter 2
Bend Formulas



It is clear that the chief end of mathematical study must be to make students think.

— John Wesley Young

If a Steiner cycle $\sigma_1, \ldots, \sigma_n$ for (α, β) is given, the following question occurs: what can we say about the relation between the radii of the circles? Such problems were considered in Japanese mathematics (Wasan) in the Edo-period (1600–1868), see the booklet of H. Fukagawa and D. Pedoe [284]. In this chapter, we discuss such relations on radii (or their reciprocals) for Steiner cycles, Soddy's hexlet, mutually tangent four circles in the plane (Descartes' circle theorem), mutually tangent five spheres in $\mathbb{R}^3$ (Soddy's formula), and others.

2.1 Bend Formulas for Steiner Cycles

A family of circles in the plane is called a *normal family* if for every circle in the family, all other circles lie in the same side (inside or outside) of the circle. For example, in Fig. 2.1, (a) is a normal family, but (b) and (c) are not. Note that the family of circles that forms a Steiner cycle for a pair of circles is a normal family since we imposed in the definition of Steiner cycle the condition that the circles are mutually non-crossing.

Denote the radius of a circle σ by $r(\sigma)$. For a normal family $\mathcal{F}$ of circles, the *bend* $b(\sigma)$ of a circle $\sigma \in \mathcal{F}$ is defined as follows:

$$b(\sigma) = \begin{cases} -1/r(\sigma) & \text{if all other circles lie inside } \sigma \\ 1/r(\sigma) & \text{otherwise.} \end{cases}$$

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H. Maehara, H. Martini, *Circles, Spheres and Spherical Geometry*,
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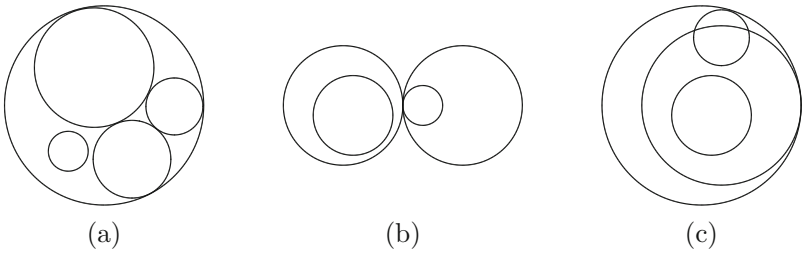


Fig. 2.1 Three families of circles

Theorem 2.1 *Let $\sigma_1, \sigma_2, \ldots, \sigma_n$ be a Steiner cycle for (α, β). If $n = km$ for some $k \geq 2$, then for every j, $1 \leq j \leq m$,*

$$\sum_{i=0}^{k-1} b(\sigma_{j+im}) = \frac{k}{2} \cot^2(\pi/n)\big(b(\alpha) + b(\beta)\big)$$

holds.

Example 2.1 In the Steiner 6-cycle shown in Fig. 1.6, we have $\cot^2(\pi/6) = 3$ and $6 = 2 \cdot 3$. Hence, by Theorem 2.1,

$$\begin{aligned}
b(\sigma_1) + b(\sigma_4) &= 3(b(\alpha) + b(\beta)) \\
b(\sigma_2) + b(\sigma_5) &= 3(b(\alpha) + b(\beta)) \\
b(\sigma_3) + b(\sigma_6) &= 3(b(\alpha) + b(\beta)) \\
b(\sigma_1) + b(\sigma_3) + b(\sigma_5) &= \tfrac{9}{2}(b(\alpha) + b(\beta)) \\
b(\sigma_2) + b(\sigma_4) + b(\sigma_6) &= \tfrac{9}{2}(b(\alpha) + b(\beta)).
\end{aligned}$$

From this simultaneous equation it follows that the radii of σ_i, $i = 1, 2, 3, 4$, can be determined by the radii of $\alpha, \beta, \sigma_5, \sigma_6$.

We prepare the proof of Theorem 2.1 by two lemmas. Let us denote by $f_{\boldsymbol{p}}$ the inversion of $\mathbb{R}^2$ with respect to the unit circle centered at $\boldsymbol{p} \in \mathbb{R}^2$. So, for a circle σ, $f_{\boldsymbol{p}}(\sigma)$ is its image. Note that if $\mathcal{F}$ is a normal family of circles and no circle in $\mathcal{F}$ passes through $\boldsymbol{p}$, then the family $\{f_{\boldsymbol{p}}(\sigma) : \sigma \in \mathcal{F}\}$ is also a normal family of circles.

Lemma 2.1 *Let $\{\sigma_1, \sigma_2, \ldots, \sigma_n\}$ be a normal family of n unit circles, and suppose that the centers $\boldsymbol{x}_i$ of σ_i satisfy*

$$\begin{cases}
\boldsymbol{x}_1 + \boldsymbol{x}_2 + \cdots + \boldsymbol{x}_n = O \text{ (the origin)}, \\
\|\boldsymbol{x}_1\| = \|\boldsymbol{x}_2\| = \cdots = \|\boldsymbol{x}_n\| = \delta \geq 1.
\end{cases}$$

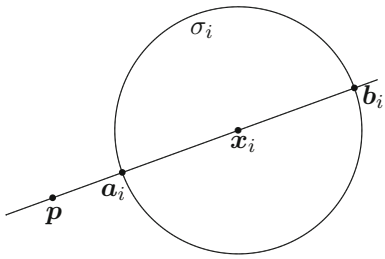


Fig. 2.2 σ_i has the diameter $\|\boldsymbol{b}_i - \boldsymbol{a}_i\|$

Then, for a point $\boldsymbol{p}$ *that lies on none of* σ_i*, we have*

$$b(f_{\boldsymbol{p}}(\sigma_i)) = \|\boldsymbol{x}_i - \boldsymbol{p}\|^2 - 1 \ (i = 1, 2, \ldots, n) \text{ and}$$

$$\sum_{i=1}^{n} b(f_{\boldsymbol{p}}(\sigma_i)) = n(\|\boldsymbol{p}\|^2 + \delta^2 - 1).$$

Proof Let us denote $f_{\boldsymbol{p}}(\sigma_i)$ by σ_i^*. Let $\boldsymbol{a}_i, \boldsymbol{b}_i$ be the intersection points of the line $\boldsymbol{p}\boldsymbol{x}_i$ and the circle σ_i such that $\|\boldsymbol{a}_i - \boldsymbol{p}\| < \|\boldsymbol{b}_i - \boldsymbol{p}\|$, see Fig. 2.2. Then the diameter of σ_i^* is equal to $\|f(\boldsymbol{b}_i) - f(\boldsymbol{a}_i)\|$.

(i) The case that $\boldsymbol{p}$ lies in the exterior of σ_i: Since

$$\|\boldsymbol{a}_i - \boldsymbol{p}\| = \|\boldsymbol{x}_i - \boldsymbol{p}\| - 1, \quad \|f_{\boldsymbol{p}}(\boldsymbol{a}_i) - \boldsymbol{p}\| \cdot \|\boldsymbol{a}_i - \boldsymbol{p}\| = 1,$$

we have $\|f_{\boldsymbol{p}}(\boldsymbol{a}_i) - \boldsymbol{p}\| = 1/(\|\boldsymbol{x}_i - \boldsymbol{p}\| - 1)$, $\|f_{\boldsymbol{p}}(\boldsymbol{b}_i) - \boldsymbol{p})\| = 1/(\|\boldsymbol{x}_i - \boldsymbol{p}\| + 1)$. Hence

$$\begin{aligned} 2r(\sigma_i^*) &= \|f_{\boldsymbol{p}}(\boldsymbol{a}_i) - f_{\boldsymbol{p}}(\boldsymbol{b}_i)\| = \|f_{\boldsymbol{p}}(\boldsymbol{a}_i) - \boldsymbol{p}\| - \|f_{\boldsymbol{p}}(\boldsymbol{b}_i) - \boldsymbol{p}\| \\ &= \frac{1}{\|\boldsymbol{x}_i - \boldsymbol{p}\| - 1} - \frac{1}{\|\boldsymbol{x}_i - \boldsymbol{p}\| + 1} = \frac{2}{\|\boldsymbol{x}_i - \boldsymbol{p}\|^2 - 1}, \end{aligned}$$

from which $b(\sigma_i^*) = \|\boldsymbol{x}_i - \boldsymbol{p}\|^2 - 1$ follows.

(ii) The case that $\boldsymbol{p}$ lies in the interior of σ_i: Since

$$\|\boldsymbol{a}_i - \boldsymbol{p}\| = 1 - \|\boldsymbol{x}_i - \boldsymbol{p}\|, \quad \|f_{\boldsymbol{p}}(\boldsymbol{a}_i) - \boldsymbol{p}\| \cdot \|\boldsymbol{a}_i - \boldsymbol{p}\| = 1,$$

we have $\|f_{\boldsymbol{p}}(\boldsymbol{a}_i) - \boldsymbol{p}\| = 1/(1 - \|\boldsymbol{x}_i - \boldsymbol{p}\|) = -1/(\|\boldsymbol{x}_i - \boldsymbol{p}\| - 1)$. Similarly, $\|f_{\boldsymbol{p}}(\boldsymbol{b}_i) - \boldsymbol{p}\| = 1/(\|\boldsymbol{x}_i - \boldsymbol{p}\| + 1)$. Hence

$$\begin{aligned} 2r(\sigma_i^*) &= \|f_{\boldsymbol{p}}(\boldsymbol{a}_i) - f_{\boldsymbol{p}}(\boldsymbol{b}_i)\| = \|f_{\boldsymbol{p}}(\boldsymbol{a}_i) - \boldsymbol{p}\| + \|f_{\boldsymbol{p}}(\boldsymbol{b}_i) - \boldsymbol{p}\| \\ &= \frac{-1}{\|\boldsymbol{x}_i - \boldsymbol{p}\| - 1} + \frac{1}{\|\boldsymbol{x}_i - \boldsymbol{p}\| + 1} = \frac{-2}{\|\boldsymbol{x}_i - \boldsymbol{p}\|^2 - 1}. \end{aligned}$$

Noting that, in this case, σ_i^* encloses all other circles σ_j^*, we have $b(\sigma_i^*) = -1/r(\sigma_i^*) = \|\boldsymbol{x}_i - \boldsymbol{p}\|^2 - 1$ by the definition of "bend".

From (i) and (ii), we always have $b(\sigma_i^*) = \|\boldsymbol{x}_i - \boldsymbol{p}\|^2 - 1$. Thus,

$$\sum_{i=1}^{n} b(\sigma_i^*) = \sum_{i=1}^{n} \|\boldsymbol{x}_i - \boldsymbol{p}\|^2 - n = n\delta^2 + n\|\boldsymbol{p}\|^2 - n - 2\sum_{i=1}^{n} \boldsymbol{x}_i \cdot \boldsymbol{p}$$

$$= n(\|\boldsymbol{p}\|^2 + \delta^2 - 1) - 2\Big(\sum_{i=1}^{n} \boldsymbol{x}_i\Big) \cdot \boldsymbol{p} = n(\|\boldsymbol{p}\|^2 + \delta^2 - 1).$$

□

Lemma 2.2 *Let τ_1, τ_2 be a pair of concentric circles in the plane such that $r(\tau_1) = \delta + 1$, $r(\tau_2) = \delta - 1$ ($\delta > 1$). Let $\boldsymbol{p}$ be a point that lies on neither of them. Then*

$$b(f_{\boldsymbol{p}}(\tau_1)) + b(f_{\boldsymbol{p}}(\tau_2)) = \frac{2}{\delta^2 - 1}(\|\boldsymbol{p}\|^2 + \delta^2 - 1).$$

Proof We may suppose that the center of the concentric circles is the origin O. To make our argument clear, let us suppose that $\boldsymbol{p}$ lies inside τ_2. (In other cases, proofs follow almost similarly.) Then the diameter of $f_{\boldsymbol{p}}(\tau_2)$ is computed as

$$\frac{1}{\delta - 1 - \|\boldsymbol{p}\|} + \frac{1}{\delta - 1 + \|\boldsymbol{p}\|} = \frac{2(\delta - 1)}{(\delta - 1)^2 - \|\boldsymbol{p}\|^2}.$$

Similarly, the diameter of $f_{\boldsymbol{p}}(\tau_1)$ is computed as

$$\frac{2(\delta + 1)}{(\delta + 1)^2 - \|\boldsymbol{p}\|^2}.$$

Since $f_{\boldsymbol{p}}(\tau_2)$ encloses the other circle, we have

$$b(f_{\boldsymbol{p}}(\tau_1)) + b(f_{\boldsymbol{p}}(\tau_2)) = \frac{(\delta + 1)^2 - \|\boldsymbol{p}\|^2}{\delta + 1} - \frac{(\delta - 1)^2 - \|\boldsymbol{p}\|^2}{\delta - 1}$$

$$= (\delta + 1) - (\delta - 1) + \|\boldsymbol{p}\|^2\Big(\frac{1}{\delta - 1} - \frac{1}{\delta + 1}\Big)$$

$$= \frac{2}{\delta^2 - 1}(\|\boldsymbol{p}\|^2 + \delta^2 - 1).$$

□

Proof of Theorem 2.1 By Theorem 1.3, there is a point $\boldsymbol{p} \in \mathbb{R}^2$ such that $f_{\boldsymbol{p}}(\alpha)$, $f_{\boldsymbol{p}}(\beta)$ are concentric circles, $r(f_{\boldsymbol{p}}(\alpha)) < r(f_{\boldsymbol{p}}(\beta))$. Then all $f_{\boldsymbol{p}}(\sigma_i)$, $i = 1, \ldots, n$, have the same radius. We may suppose that the center of the concentric

circles $f_{\boldsymbol{p}}(\alpha)$, $f_{\boldsymbol{p}}(\beta)$ is the origin O. Since the formula of the theorem is independent on the unit of length, we may suppose that all circles $f_{\boldsymbol{p}}(\sigma_i)$, $i = 1, \ldots, n$, are unit circles. Let δ be the distance between O and the center $\boldsymbol{x}_i$ of $f_{\boldsymbol{p}}(\sigma_i)$. Since the points $\boldsymbol{x}_i$, $i = 1, 2, \ldots, n$, form the vertices of a regular n-gon, we have

$$\sin(\pi/n) = 1/\delta,$$

and the radii of $f_{\boldsymbol{p}}(\alpha)$, $f_{\boldsymbol{p}}(\beta)$ are $\delta - 1$, $\delta + 1$, respectively. Note also that for every j $(1 \le j \le m)$ the k points $\boldsymbol{x}_{j+im}$, $i = 0, 1, \ldots, k-1$, are the vertices of a regular k-gon, and we have

$$\sum_{i=0}^{k-1} \boldsymbol{x}_{j+im} = O.$$

Since $f_{\boldsymbol{p}}\big(f_{\boldsymbol{p}}(\sigma_i)\big) = \sigma_i$, $f_{\boldsymbol{p}}\big(f_{\boldsymbol{p}}(\alpha)\big) = \alpha$, $f_{\boldsymbol{p}}\big(f_{\boldsymbol{p}}(\beta)\big) = \beta$, we have

$$\sum_{i=0}^{k-1} b(\sigma_{j+im}) = k(\|\boldsymbol{p}\|^2 + \delta^2 - 1),$$

$$b(\alpha) + b(\beta) = \frac{2}{\delta^2 - 1}(\|\boldsymbol{p}\|^2 + \delta^2 - 1),$$

by Lemmas 2.1 and 2.2. Now, since $\delta = 1/\sin(\pi/n)$ implies $\delta^2 - 1 = \cot^2(\pi/n)$, the equality of the theorem follows. □

2.2 Bend Formula for Soddy's Hexlet

Similarly to a circle-case, a family $\mathcal{F}$ of spheres is called a *normal family* if for any $\sigma \in \mathcal{F}$ all other spheres lie in the same side (inside or outside) of σ. The radius of σ is denoted by $r(\sigma)$. The *bend* $b(\sigma)$ of a sphere σ in a normal family $\mathcal{F}$ is defined (similarly to the circle-case) in the following way:

$$b(\sigma) = \begin{cases} -1/r(\sigma) & \text{if all other spheres lie inside } \sigma \\ 1/r(\sigma) & \text{otherwise.} \end{cases}$$

Theorem 2.2 *Let $\sigma_1, \sigma_2, \ldots, \sigma_6$ be a hexlet for a 3-cycle τ_1, τ_2, τ_3. Then we have*

$$b(\sigma_1) + b(\sigma_4) = 2\big(b(\tau_1) + b(\tau_2) + b(\tau_3)\big)$$
$$b(\sigma_2) + b(\sigma_5) = 2\big(b(\tau_1) + b(\tau_2) + b(\tau_3)\big)$$

$$b(\sigma_3)+b(\sigma_6)=2\big(b(\tau_1)+b(\tau_2)+b(\tau_3)\big)$$
$$b(\sigma_1)+b(\sigma_3)+b(\sigma_5)=3\big(b(\tau_1)+b(\tau_2)+b(\tau_3)\big)$$
$$b(\sigma_2)+b(\sigma_4)+b(\sigma_6)=3\big(b(\tau_1)+b(\tau_2)+b(\tau_3)\big).$$

Remark 2.1 From this it follows that if $b(\tau_1), b(\tau_2), b(\tau_3)$ and one of $b(\sigma_i)$ are rational numbers, then all bends of the hexlet are rational numbers.

Let us state here the sphere-version of Lemma 2.1, whose proof follows from the proof of Lemma 2.1 by replacing "circle" with "sphere".

Lemma 2.3 *Let $\{\sigma_1, \sigma_2, \ldots, \sigma_n\}$ be a normal family of spheres in $\mathbb{R}^3$, and suppose that the centers $\boldsymbol{x}_i$ of σ_i satisfy*

$$\begin{cases} \boldsymbol{x}_1+\boldsymbol{x}_2+\cdots+\boldsymbol{x}_n=O \ (\textit{the origin}), \\ \|\boldsymbol{x}_1\|=\|\boldsymbol{x}_2\|=\cdots=\|\boldsymbol{x}_n\|=\delta\geq 1. \end{cases}$$

Then for the inversion $f_{\boldsymbol{p}}$ of $\mathbb{R}^3$ with respect to a unit sphere with center $\boldsymbol{p}$ lying on none of σ_i, we have

$$b(f_{\boldsymbol{p}}(\sigma_i))=\|\boldsymbol{x}_i-\boldsymbol{p}\|^2-1 \ (i=1,2,\ldots,n), \ \textit{and}$$
$$\sum_{i=1}^{n} b(f_{\boldsymbol{p}}(\sigma_i))=n(\|\boldsymbol{p}\|^2+\delta^2-1).$$

□

Proof of Theorem 2.2 Let $\boldsymbol{p}\in\mathbb{R}^3$ be the contact point of τ_1 and τ_2, and let $f_{\boldsymbol{p}}$ be the inversion of $\mathbb{R}^3$ with respect to the unit sphere centered at $\boldsymbol{p}$. Put $\xi=f_{\boldsymbol{p}}(\tau_1), \eta=f_{\boldsymbol{p}}(\tau_2), \tau^*=f_{\boldsymbol{p}}(\tau_3)$. Then ξ, η are a pair of parallel planes, and τ^* is a sphere that is tangent to these two planes. Put $\sigma_i^*=f_{\boldsymbol{p}}(\sigma_i), i=1,2,\ldots,6$. Note that $\sigma_1^*,\ldots,\sigma_6^*$ are spheres, each being tangent to the planes ξ, η and to the sphere τ^*. The point $\boldsymbol{p}$ lies between the planes ξ, η and ouside of τ^*, σ_i^*. Since the equality of Theorem 2.2 that we are going to prove is independent of the choice of the unit of length, we may suppose that each σ_i^* and τ^* are unit spheres. We may further suppose that the center of τ^* is the origin O, and the centers $\boldsymbol{x}_i$ of σ_i^* lie on the plane $z=0$. Then $\|\boldsymbol{x}_i\|=2$. So, by putting $\delta=2$ and applying Lemma 2.3, we have

$$\sum_{i=0}^{2} b(f_{\boldsymbol{p}}(\sigma_{j+2i}^*))=3(\|\boldsymbol{p}\|^2+3) \qquad (\text{in the case of } k=3),$$
$$\sum_{i=0}^{1} b(f_{\boldsymbol{p}}(\sigma_{j+2i}^*))=2(\|\boldsymbol{p}\|^2+3) \qquad (\text{in the case of } k=2).$$

If we put $\boldsymbol{p} = (x, y, z)$, then $-1 < z < 1$. We may suppose $z \geq 0$. Then, since

$$2r(f_{\boldsymbol{p}}(\tau^*)) = \frac{1}{\|\boldsymbol{p}\| - 1} - \frac{1}{\|\boldsymbol{p}\| + 1} = \frac{2}{\|\boldsymbol{p}\|^2 - 1}$$
$$2r(f_{\boldsymbol{p}}(\xi)) = \frac{1}{z - 1}$$
$$2r(f_{\boldsymbol{p}}(\eta)) = \frac{1}{z + 1},$$

we have $b(f_{\boldsymbol{p}}(\tau^*)) + b(f_{\boldsymbol{p}}(\xi)) + b(f_{\boldsymbol{p}}(\eta)) = \|\boldsymbol{p}\|^2 + 3$. Thus the equality of the theorem holds. □

Remark 2.2 Frederick Soddy found a hexlet in 1936 without using inversions, and he proved, concerning the bends $b_1, \ldots, b_6$ of the six spheres, the equality $b_1 + b_4 = b_2 + b_5 = b_3 + b_6$. However, this equality was found more than 100 years earlier in Japanese temple geometry, see the booklet of H. Fukagawa and D. Pedoe [284, Ch. 10, Problem 9.10].

2.3 Descartes' Circle Theorem

We continue with a further famous theorem on circle arrangements.

Theorem 2.3 (Descartes' Circle Theorem) *If $\{\alpha_1, \alpha_2, \alpha_3, \alpha_4\}$ is a normal family of mutually tangent circles in the plane, then*

$$\Big(\sum_{i=1}^{4} b(\alpha_i)\Big)^2 = 2\sum_{i=1}^{4} b(\alpha_i)^2$$

holds, where $b(\alpha_i)$ is the bend of the circle α_i.

Proof Let $\boldsymbol{p}$ be the contact point of α_3 and α_4. Then $f_{\boldsymbol{p}}(\alpha_3)$ and $f_{\boldsymbol{p}}(\alpha_4)$ are parallel lines, and $f_{\boldsymbol{p}}(\alpha_1)$ and $f_{\boldsymbol{p}}(\alpha_2)$ are mutually tangent circles, both tangent to the two parallel lines. Since the formula in the theorem is independent of the unit of length, we may suppose that $f_{\boldsymbol{p}}(\alpha_1)$, $f_{\boldsymbol{p}}(\alpha_2)$ are unit circles with centers $(-1, 0)$ and $(1, 0)$, respectively. Put $\boldsymbol{p} = (x, y)$. We may suppose $0 \leq y < 1$. Then, by Lemma 2.1,

$$b(\alpha_1) = (x + 1)^2 + y^2 - 1 = x^2 + 2x + y^2,$$
$$b(\alpha_2) = (x - 1)^2 + y^2 - 1 = x^2 - 2x + y^2.$$

In the pair of parallel lines $f_{\boldsymbol{p}}(\alpha_i), i = 3, 4$, suppose that $\boldsymbol{p}$ is nearer to $f_{\boldsymbol{p}}(\alpha_3)$. Then $2r(\alpha_3) = 1/(1 - y)$, $2r(\alpha_4) = 1/(1 + y)$, hence

$$b(\alpha_3) = 2(1 - y), \quad b(\alpha_4) = 2(1 + y).$$

Therefore, $\sum_{i=1}^{4} b(\alpha_i) = 2(x^2 + y^2) + 4$. On the other hand,

$$\begin{aligned}\sum_{i=1}^{4} b(\alpha_i)^2 &= 2(x^2 + y^2)^2 + 8x^2 + 8 + 8y^2 \\ &= 2(x^2 + y^2)^2 + 8(x^2 + y^2) + 8.\end{aligned}$$

Hence the formula in the theorem holds. □

2.4 Soddy's Formula

We start with a lemma.

Lemma 2.4 *Let $\boldsymbol{x}_i \in \mathbb{R}^2$, $i = 1, 2, 3$, be the vertices of an equilateral triangle inscribed in the circle with radius δ centered at the origin O. Then, for any point $\boldsymbol{p}_0 \neq O$ in the plane $\mathbb{R}^2$, we have*

$$\sum_{i=1}^{3} \|\boldsymbol{x}_i - \boldsymbol{p}_0\|^2 = 3(\delta^2 + \|\boldsymbol{p}_0\|^2) \text{ and}$$

$$\sum_{i=1}^{3} \|\boldsymbol{x}_i - \boldsymbol{p}_0\|^4 = 6\delta^2\|\boldsymbol{p}_0\|^2 + 3(\delta^2 + \|\boldsymbol{p}_0\|^2)^2.$$

Proof Let $\angle \boldsymbol{p}_0 O \boldsymbol{x}_i = \theta_i$ (oriented angle) for $i = 1, 2, 3$. Then the three points $(\cos\theta_i, \sin\theta_i)$, $i = 1, 2, 3$, are the vertices of an equilateral triangle. In this case, the three points $(\cos 2\theta_i, \sin 2\theta_i)$, $i = 1, 2, 3$, are also the vertices of an equilateral triangle. Hence $\sum_{i=1}^{3} \cos\theta_i = \sum_{i=1}^{3} \cos 2\theta_i = 0$.

(i) From $\|\boldsymbol{x}_i - \boldsymbol{p}_0\|^2 = \delta^2 + \|\boldsymbol{p}_0\|^2 - 2\delta\|\boldsymbol{p}_0\| \cos\theta_i$, we have

$$\sum_{i=1}^{3} \|\boldsymbol{x}_i - \boldsymbol{p}_0\|^2 = 3(\delta^2 + \|\boldsymbol{p}_0\|^2).$$

(ii) From

$$\begin{aligned}\|\boldsymbol{x}_i - \boldsymbol{p}_0\|^4 &= \left(\delta^2 + \|\boldsymbol{p}_0\|^2 - 2\delta\|\boldsymbol{p}_0\|\cos\theta_i\right)^2 \\ &= 4\delta^2\|\boldsymbol{p}_0\|^2\cos^2\theta_i - 4\delta\|\boldsymbol{p}_0\|(\delta^2 + \|\boldsymbol{p}_0\|^2)\cos\theta_i \\ &\quad + (\delta^2 + \|\boldsymbol{p}_0\|^2)^2,\end{aligned}$$

$\cos^2\theta_i = \frac{1}{2} + \frac{\cos 2\theta_i}{2}$, and $\cos 2\theta_1 + \cos 2\theta_2 + \cos 2\theta_3 = 0$, it follows that

$$\sum_{i=1}^{3}\|\boldsymbol{x}_i - \boldsymbol{p}_0\|^4 = 6\delta^2\|\boldsymbol{p}_0\|^2 + 3(\delta^2 + |\boldsymbol{p}_0|^2)^2.$$

□

Theorem 2.4 (Soddy's Formula) *If $\{\sigma_1, \sigma_2, \ldots, \sigma_5\}$ is a normal family of mutually tangent five spheres in $\mathbb{R}^3$, then we have*

$$\Big(\sum_{i=1}^{5} b(\sigma_i)\Big)^2 = 3\sum_{i=1}^{5} b(\sigma_i)^2.$$

Remark 2.3 This formula became well-known by the poem "The kiss precise" published in *Nature* in 1936 by Frederic Soddy. But also this formula was already known to Japanese mathematicians in the Edo-period, see [284].

Proof Let $\boldsymbol{p}$ be the contact point of σ_4, σ_5, and denote by $f_{\boldsymbol{p}}$ the inversion of $\mathbb{R}^3$ with respect to the unit sphere centered at $\boldsymbol{p}$. Then $f_{\boldsymbol{p}}(\sigma_4)$ and $f_{\boldsymbol{p}}(\sigma_5)$ are parallel planes, $f_{\boldsymbol{p}}(\sigma_i)$, $i = 1, 2, 3$, are mutually tangent spheres each tangent to these two planes. Since the formula in Theorem 2.4 is independent of the choice of the unit of length, we may suppose that $f_{\boldsymbol{p}}(\sigma_i)$, $i = 1, 2, 3$, are unit spheres. Hence the centers $\boldsymbol{x}_i$, $i = 1, 2, 3$, of these spheres are the vertices of an equilateral triangle with edge-length 2. We may suppose that these three points $\boldsymbol{x}_i$, $i = 1, 2, 3$, lie on the plane $z = 0$ in $\mathbb{R}^3$, and $\boldsymbol{x}_1 + \boldsymbol{x}_2 + \boldsymbol{x}_3 = O$ (the origin). Put $\boldsymbol{p} = (x, y, z)$, $0 \le z < 1$, and $\boldsymbol{p}_0 = (x, y, 0)$. Since $\delta := \|\boldsymbol{x}_1\| = \|\boldsymbol{x}_2\| = \|\boldsymbol{x}_3\| = 2/\sqrt{3}$, it follows from Lemma 2.4 that, for $i = 1, 2, 3$,

$$\begin{aligned}b(\sigma_i) &= \|\boldsymbol{x}_i - \boldsymbol{p}\|^2 - 1 = \|\boldsymbol{x}_i - \boldsymbol{p}_0\|^2 + z^2 - 1 \\ b(\sigma_i)^2 &= \|\boldsymbol{x}_i - \boldsymbol{p}_0\|^4 + 2\|\boldsymbol{x}_i - \boldsymbol{p}_0\|^2(z^2 - 1) + (z^2 - 1)^2.\end{aligned}$$

By Lemma 2.4,

$$\sum_{i=1}^{3} b(\sigma_i) = \sum_{i=1}^{3}\|\boldsymbol{x}_i - \boldsymbol{p}_0\|^2 + 3(z^2 - 1) = 3(\delta^2 + \|\boldsymbol{p}_0\|^2) + 3(z^2 - 1),$$

and

$$\sum_{i=1}^{3} b(\sigma_i)^2 = \sum_{i=1}^{3} \Big(\|\boldsymbol{x}_i - \boldsymbol{p}_0\|^4 + 2\|\boldsymbol{x}_i - \boldsymbol{p}_0\|^2 (z^2 - 1) \Big) + 3(z^2 - 1)^2$$
$$= 6\delta^2 \|\boldsymbol{p}_0\|^2 + 3(\delta^2 + \|\boldsymbol{p}_0\|^2)^2 + 6(z^2 - 1)(\delta^2 + \|\boldsymbol{p}_0\|^2) + 3(z^2 - 1)^2$$
$$= 3\Big((\delta^2 + \|\boldsymbol{p}_0\|^2) + (z^2 - 1) \Big)^2 + 6\delta^2 \|\boldsymbol{p}_0\|^2.$$

On the other hand, since $b(\sigma_4) = 2(1 - z),\ b(\sigma_5) = 2(1 + z)$,

$$\sum_{i=1}^{5} b(\sigma_i) = 3(\delta^2 + \|\boldsymbol{p}_0\|^2) + 3(z^2 - 1) + 4,$$

$$\sum_{i=1}^{5} b(\sigma_i)^2 = 3\Big((\delta^2 + \|\boldsymbol{p}_0\|^2) + (z^2 - 1) \Big)^2 + 6\delta^2 \|\boldsymbol{p}_0\|^2 + 8(z^2 + 1).$$

From these two equalities and $\delta^2 = 4/3$, we can verify that

$$3 \sum_{i=1}^{5} b(\sigma_i)^2 = \Big(\sum_{i=1}^{5} b(\sigma_i) \Big)^2.$$

□

2.5 Linked 4-Cycle Pairs

Lemma 2.5 *Let $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ be a 4-cycle of spheres, and $\beta_1, \beta_2, \ldots, \beta_n$ be an n-cycle of spheres, $n \geq 3$. If α_i and β_j are tangent for every $i = 1, 2, 3, 4,\ j = 1, 2, \ldots, n$, then $n = 4$.*

Proof Since α_1 and α_3 are disjoint spheres, there is a point $\boldsymbol{p}$ such that $f_{\boldsymbol{p}}(\alpha_1)$ and $f_{\boldsymbol{p}}(\alpha_3)$ are concentric spheres by Theorem 1.9. We may suppose that the center of these concentric spheres is the origin O of $\mathbb{R}^3$. Put $\alpha_i^* = f_{\boldsymbol{p}}(\alpha_i), i = 1, 2, 3, 4$, and $\beta_j^* = f_{\boldsymbol{p}}(\beta_j), j = 1, 2, \ldots, n$. The spheres α_2^*, α_4^* and $\beta_j^*, j = 1, 2, \ldots, n$, are all tangent to the concentric spheres, and hence they have the same radius, say r. See Fig. 2.3. Let $\boldsymbol{x}_i$ be the center of α_i^* for $i = 2, 4$, and $\boldsymbol{y}_j$ be the center of β_j^* for $j = 1, 2, \ldots, n$. Then $\|\boldsymbol{y}_j - \boldsymbol{x}_i\| = 2r$ for $i = 2, 4,\ j = 1, 2, \ldots, n$. Hence $\boldsymbol{y}_j,\ j = 1, 2, \ldots, n$, lie on the plane H that perpendicularly bisects the line segment $\boldsymbol{x}_2\boldsymbol{x}_4$. Since $\|\boldsymbol{x}_2\| = \|\boldsymbol{x}_4\|$, O also lies on the plane H. Let $\boldsymbol{q}$ be the midpoint of $\boldsymbol{x}_2\boldsymbol{x}_4$. Then $\|\boldsymbol{y}_1 - \boldsymbol{q}\| = \|\boldsymbol{y}_2 - \boldsymbol{q}\| = \cdots = \|\boldsymbol{y}_n - \boldsymbol{q}\|$. Since $n \geq 3$, we can deduce that $\boldsymbol{q} = O$. Therefore, $\|\boldsymbol{x}_i - O\| = \|\boldsymbol{y}_j - O\| = \sqrt{2}r$ for $i = 2, 4,\ j = 1, 2, \ldots, n$.

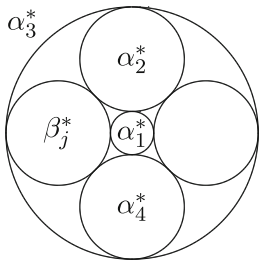


Fig. 2.3 α_1^*, α_3^* are concentric

This implies that $\angle \boldsymbol{y}_1 O \boldsymbol{y}_2 = \cdots = \angle \boldsymbol{y}_n O \boldsymbol{y}_1 = 90°$. Therefore, we must have $n = 4$. □

If a pair of 4-cycles $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ and $\beta_1, \beta_2, \beta_3, \beta_4$ satisfies that each α_i is tangent to all β_j, $j = 1, 2, 3, 4$, then the pair of 4-cycles is called a *linked 4-cycle pair*.

Example 2.2 Let α_1, and α_3 be concentric spheres in $\mathbb{R}^3$ centered at the origin O, and with radii $\sqrt{2}-1$, $\sqrt{2}+1$, respectively. Further on, let α_2, α_4 be the unit spheres with centers $(0, 0, \sqrt{2})$, $(0, 0, -\sqrt{2})$, respectively. Let β_j, $j = 1, 2, 3, 4$, be the unit spheres with centers $(\sqrt{2}, 0, 0)$, $(0, \sqrt{2}, 0)$, $(-\sqrt{2}, 0, 0)$, $(0, -\sqrt{2}, 0)$, respectively. Then the 4-cycles $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ and $\beta_1, \beta_2, \beta_3, \beta_4$ form a linked 4-cycle pair.

Theorem 2.5 (Bend Formula for a Linked 4-Cycle Pair) *Suppose that a pair of 4-cycles $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ and $\beta_1, \beta_2, \beta_3, \beta_4$ forms a linked 4-cycle pair. Then*

$$b(\alpha_1) + b(\alpha_3) = b(\alpha_2) + b(\alpha_4) = b(\beta_1) + b(\beta_3) = b(\beta_2) + b(\beta_4).$$

Proof By Theorem 1.9, there is a point $\boldsymbol{p}$ on the line connecting the centers of α_1, α_3 such that $f_{\boldsymbol{p}}(\alpha_1)$ and $f_{\boldsymbol{p}}(\alpha_3)$ are concentric spheres. We may suppose that their common center is the origin O of $\mathbb{R}^3$. Put $\alpha_i^* = f_{\boldsymbol{p}}(\alpha_i)$, $i = 1, 2, 3, 4$, and $\beta_j^* = f_{\boldsymbol{p}}(\beta_j)$, $j = 1, 2, 3, 4$. Note that these spheres have the same radius. We may suppose that these are all unit spheres. Let $\boldsymbol{x}_i$ be the center of α_i^* for $i = 2, 4$, and $\boldsymbol{y}_j$ be the center of β_j^* for $j = 1, 2, 3, 4$. Then

$$\|\boldsymbol{x}_2 - \boldsymbol{y}_j\| = \|\boldsymbol{x}_4 - \boldsymbol{y}_j\| = 2 \text{ for } j = 1, 2, 3, 4,$$

and

$$\|\boldsymbol{y}_1 - \boldsymbol{y}_2\| = \|\boldsymbol{y}_2 - \boldsymbol{y}_3\| = \|\boldsymbol{y}_3 - \boldsymbol{y}_4\| = \|\boldsymbol{y}_4 - \boldsymbol{y}_1\| = 2.$$

Hence

$$\boldsymbol{y}_1\boldsymbol{y}_2\boldsymbol{y}_3\boldsymbol{y}_4,\ \boldsymbol{x}_2\boldsymbol{y}_2\boldsymbol{x}_4\boldsymbol{y}_4,\ \boldsymbol{x}_2\boldsymbol{y}_1\boldsymbol{x}_4\boldsymbol{y}_3$$

are squares with side-length 2 centered at O. Thus, $\|\boldsymbol{x}_i\| = \|\boldsymbol{y}_j\| = \sqrt{2}$, and the radii of $f_{\boldsymbol{p}}(\alpha_1)$, $f_{\boldsymbol{p}}(\alpha_3)$ are $\sqrt{2} \pm 1$. By Lemma 2.1, we have

$$b(f_{\boldsymbol{p}}(\beta_1^*)) + b(f_{\boldsymbol{p}}(\beta_2^*)) + b(f_{\boldsymbol{p}}(\beta_3^*)) + b(f_{\boldsymbol{p}}(\beta_4^*)) = 4(\|\boldsymbol{p}\|^2 + 1)$$
$$b(f_{\boldsymbol{p}}(\alpha_2^*)) + b(f_{\boldsymbol{p}}(\beta_2^*)) + b(f_{\boldsymbol{p}}(\alpha_4^*)) + b(f_{\boldsymbol{p}}(\beta_4^*)) = 4(\|\boldsymbol{p}\|^2 + 1)$$
$$b(f_{\boldsymbol{p}}(\alpha_2^*)) + b(f_{\boldsymbol{p}}(\beta_1^*)) + b(f_{\boldsymbol{p}}(\alpha_4^*)) + b(f_{\boldsymbol{p}}(\beta_3^*)) = 4(\|\boldsymbol{p}\|^2 + 1),$$

from which we get

$$\begin{aligned} b(f_{\boldsymbol{p}}(\beta_1^*)) + b(f_{\boldsymbol{p}}(\beta_3^*)) &= b(f_{\boldsymbol{p}}(\beta_2^*)) + b(f_{\boldsymbol{p}}(\beta_4^*)) \\ &= b(f_{\boldsymbol{p}}(\alpha_2)) + b(f_{\boldsymbol{p}}(\alpha_4)) = 2(\|\boldsymbol{p}\|^2 + 1), \end{aligned}$$

that is,

$$b(\beta_1) + b(\beta_3) = b(\beta_2) + b(\beta_4) = b(\alpha_2) + b(\alpha_4) = 2(\|\boldsymbol{p}\|^2 + 1).$$

On the other hand, by Lemma 2.2,

$$b(f_{\boldsymbol{p}}(\alpha_1^*)) + b(f_{\boldsymbol{p}}(\alpha_3^*)) = 2(\|\boldsymbol{p}\|^2 + 1),$$

that is, $b(\alpha_1) + b(\alpha_3) = 2(\|\boldsymbol{p}\|^2 + 1)$. Hence the theorem follows. □

For bend-formulas in other sphere-systems and in higher dimensions see [524].

2.6 Exercises

2.1 Given a Steiner 6-cycle $\sigma_1, \ldots, \sigma_6$ for (α, β), prove that if the radii of $\alpha, \beta, \sigma_1, \sigma_2$ are rational numbers, then the radii of $\sigma_i, i = 3, 4, 5, 6$, are all rational numbers.

2.2 Let A, B, C be mutually externally tangent three circles with radii $1, 1, 2$, respectively. Find the radius of the circle that is internally tangent to all A, B, C.

2.3 Suppose that four balls lie on a table, and these four balls are mutually tangent. When three of the balls have unit radius, find the radius of the remaining ball.

2.4* Let $\sigma_1, \sigma_2, \sigma_3, \sigma_4, \tau_1, \tau_2$ be a system of six spheres in $\mathbb{R}^3$ such that, for $i = 1, 2$, the five spheres $\sigma_1, \sigma_2, \sigma_3, \sigma_4, \tau_i$ are mutually tangent. Prove that

$$b(\sigma_1) + b(\sigma_2) + b(\sigma_3) + b(\sigma_4) = b(\tau_1) + b(\tau_2).$$

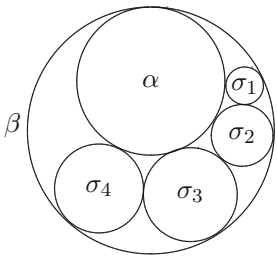


Fig. 2.4 Circles α and β are internally tangent

2.5* Consulting Fig. 2.4, let $\sigma_1, \sigma_2, \sigma_3, \sigma_4$ be four circles such that each is tangent to both α, β, and σ_i is externally tangent to σ_{i+1} for $i = 1, 2, 3$. Prove that $b(\sigma_1) + 3b(\sigma_3) = b(\sigma_4) + 3b(\sigma_2)$.

2.7 Notes

The notes completing Chap. 2 refer to the substantial themes of Descartes' circle theorem, Soddy's formula, Apollonian circle packings, and the problem of Apollonius on touching circles.

2.7.1 Notes on Descartes' Circle Theorem and Soddy's Formula

Given three mutually tangent circles, there are two circles touching all of them (and are also called Soddy circles). *Descartes' circle theorem* states that for each four mutually tangent circles, the radii of the circles satisfy a certain quadratic equation. It is closely related to the famous *problem of Apollonius* (on touching circles) and goes back to Descartes (namely to the year 1643, see the historical contributions [114] and [70] for more details). Independently, the Japanese mathematician Y. Nushizumi stated a form of Descartes' circle theorem in 1751.

Descartes' theorem occurs in not too many basic geometry books. It is, however, e.g. discussed in Coxeter's basic book [182, 1.56–1.59], in Pedoe's monograph [625, Section 40.3], and in the book [284, Section 1.7] on the Japanese temple geometry.

The papers [7], [624], [185], and [810] from the Sixties contain new proofs and nice generalizations of Descartes' theorem. These generalizations refer, for example, to the number of participating circles, and the paper [624] contains also a survey. Further straightforward and recent proofs of Descartes' theorem are given by Levrie [503] and Bradford [121].

Coxeter [185] finds a new approach to the n-dimensional version with $n + 2$ spheres (using inversions) and studies their spiral-shaped occurrence, where the

contact points of consecutive spheres belong to a "loxodromic curve". Various further interesting properties of these sphere arrangements are obtained, and Weiss [803] continues these investigations in view of Möbius transformations.

In the next two papers, relations between Steiner chains and Descartes' circle theorem are investigated. Schwartz and Tabachnikov [719] investigate curvatures in Steiner chains, getting consequences of Descartes' circle theorem as subcase. And Moritsugu [585] studies extensions of Descartes' theorem regarding the number of participating circles, and instead of inversions or related methods he uses Gröbner bases or resultants for the equations of in- and circumscribed circles.

Lagarias, Mallows, and Wilks [487] prove that the "curvature-centers" (curvature times center, where the center is considered as a complex number) satisfy Descartes' four-numbers relation. Northshield [598] presents an elegant, short proof of this result and gives an analogous version for spheres. Analogously, Tupan [784] gives a short proof of Descartes' theorem using only basic theorems from plane geometry and complex numbers.

Finally we cite the nice and elementary exposition [400] of Holshouser et al. discussing results around the problem of Apollonius and Descartes' theorem; in particular, Soddy's contribution like also Gosset's approach to the n-dimensional analogue for $n + 2$ spheres are discussed.

As already mentioned in Remark 2.3, *Soddy's formula for five spheres in 3-space* became known due to Soddy's poem "The kiss precise", but was already known in Japan during the Edo period. An interesting historical article to this formula is [729] by Sharygin and Shtogrin; they emphasize this older source from Japan (going back to the eighteenth century). In addition, the authors give a careful proof (based on determinants) of the n-dimensional analogue of Soddy's result, with $n+2$ mutually touching spheres. Kontorovich [474] studies sphere packings in 3-space whose bends (reciprocals of radii) are integers, calling them Soddy sphere packings. The quadratic form he uses is a natural analogue of that coming from Descartes' circle theorem. The paper has an interestingly rich collection of related references.

2.7.2 Notes on Apollonian Circle Packings

The configuration in Descartes' circle theorem can be seen as starting point for constructing *Apollonian circle packings*, coming from Apollonius' statement: Given three mutually tangent circles in the plane, there exist exactly two circles tangent to all three. (The given circle configuration presents a special case of the arbitrarily given configuration of three circles for Apollonius' famous touching problem, see below.)

An Apollonian circle packing arises by repeatedly filling the interstices between mutually tangent circles with further tangent circles; see, e.g., [78, II.9] and also [644, Chapter 4]. Lagarias et al. [487] present a new method for proving Descartes' theorem and various further results, such as analogues in higher dimensions, the

complex variant of Descartes' theorem, extensions to Apollonian circle packings, and extensions to spherical and hyperbolic geometry.

They continue, with two additional authors, their deep work on Apollonian circle packings by the trilogy [319], [320], and [321] on group theoretic and geometric aspects, which follows also the number-theoretic paper [318]. An Apollonian circle packing is called *integral* if every circle in the packing has an integer curvature. In [318] number-theoretic properties of the set of integer curvatures occurring in integral Apollonian circle packings are studied.

In the mentioned trilogy of articles, the authors observe that there are Apollonian packings which have strong integral properties; i.e., all circles in such packings have integer curvatures and rational centers. They start with the Descartes arrangement (four mutually tangent circles) and generate further circles via inversions in the four starting circles.

In Part I (i.e., in [319]) it is shown that there is a discrete group, called *Apollonian group* and acting on a parameter space of Descartes configurations, such that these configurations in a packing form an orbit under the action of this group. The authors show that there exist infinitely many types of integral Apollonian packings, where the integral structure is related to the integral nature of the Apollonian group, and they obtain many further related results.

In Part II (i.e., in [320]) the action of a larger discrete group, namely the *super-Apollonian group* also having an integral structure, is studied. The orbits describe the Descartes quadruples of a so-called super-packing. Its circles never cross but are nested to an arbitrary depth. It turns out that certain Apollonian packings and super-packings are strongly integral, and it is proved that there are exactly eight types of strongly integral super-packings, each containing a copy of every integral Apollonian circle packing (as before, up to scale). Further properties of the super-Apollonian group are shown.

In the third part [321], the authors extend the results of the other two parts to n-dimensional analogues of the Apollonian circle packings, having n-dimensional Desargues configurations consisting of $n + 2$ mutually touching spheres as starting point. They study n-dimensional analogues of the Apollonian group, its dual, and the super-Apollonian group, and they show that the Apollonian group and the dual Apollonian group are Coxeter groups. An Apollonian cluster ensemble is introduced, being any orbit under the Apollonian group; similar notions are developed for the other two types of groups. Among further results, it is shown in which dimensions there are rational Apollonian cluster ensembles (i.e., all curvatures are rational).

The curvatures of four mutually tangent circles with disjoint interiors clearly form a Descartes quadruple. The four least curvatures in an integral Apollonian circle packing form a so-called root Descartes quadruple, and it is a primitive root quadruple if the curvatures are relatively prime. Deriving a closed formula for the number of primitive root quadruples with minimum curvature, Northshield [597] confirms a conjecture of Mallows. The paper contains further interesting results in that direction.

It is known that if the original four circles of an Apollonian circle packing have integer curvatures, all of the circles in the packing will have this property. Bourgain and Fuchs [118] present a lower bound for the number $k(P, X)$ of integers up to X occurring as curvatures in a bounded integer Apollonian packing P. They confirm a positive-density conjecture of Graham et al., namely that the ratio $k(P, X)/X$ is larger than 0 for X tending to infinity. In other words, there exists a constant $c > 0$ depending on P such that $k(P, X) \geq cX$. There exist various further and similar interesting results on Apollonian circle packings with integer curvatures, and Fuchs [282] gives a nice survey on related results and methods. Another expository paper on recent results on Apollonian circle packings and the classical Apollonian problem in view of counting problems is [602] by Oh. The author discusses integral and primitive Apollonian packings, circles with prime curvatures in such structures, the importance of group-theoretical methods (leading to so-called thin subgroups) and various further aspects. Here we should also mention the older review [693] of Sarnak, who treats the construction of integral Apollonian circle packings emphasizing also a number of Diophantine problems, which come to be in the context of such packings.

2.7.3 Notes on Apollonius' Touching Problem

The famous *problem of Apollonius* on constructing all circles that touch three given ones in arbitrary positions has a long history and is, in particular, an excellent "exercise" in using inversion. Mostly in the literature, the given or constructed circles are allowed to be presented also by the borderline cases *lines* (radius infinity) or *points* (radius zero).

Book Chapters and Surveys

Ways of solution can be found in several book chapters; see, e.g., Johnson [430, Chapter VI], Schmidt [698, Chapter VII in Part A and Chapter II in Part B], Dörrie [219, pp. 154–160], Pedoe [625, Section 26.1], Berger's book [75, 10.11.1], Ogilvy [601, Section 4.3], Hartshorne's monograph [382, Section 38], Aumann [48, Chapter 8], as well as Posamentier and Geretschläger [644, Chapter 7]. Also the survey [820, Section 12.2] and Section 1.4 in [738] (on conics and the problem of Apollonius) should be mentioned here. Many of these references contain historical material on this problem. A paper dedicated as a whole to the history of the problem is [514] due to Lyuter.

Some Research Results

Gyarmathi [351] solves the n-dimensional Apollonius problem constructively by using the *cyclographic mapping*. Also Iwata [423] solves the n-dimensional version and extends the problem to *elliptic space*; he presents also Descartes' circle theorem.

Coxeter [185] investigates special cases of Apollonius' problem depending on the position of the given circles, and he applies inversion as well as hyperbolic geometry (via stereographic projection). It is clear that the type of the three given figures (if we allow also points and lines) and their position to each other determine the number of figures (points, lines or circles) that solve the problem. Pedoe [626] shows that the numbers 0, 1, 2, 3, 4, 5, 6, 8 are all possible, but that 7 cannot occur. Fitz-Gerald [270] completes this by showing that the case of one solution cannot exist if points are excluded from the definition of circles. Based on inversion, Bruen et al. [130] present a very detailed treatment of all possible cases in the real Möbius plane. Special care is given to Pedoe's result from [626].

The three-dimensional analogue (with four given spheres) is studied by Langlet [491]. He reduces the case that the sphere is externally or internally tangent to all given spheres to solving a simple quadratic equation for the radius and a system of three linear equations (for the three coordinates of the centers). This method can be extended to higher dimensions. Paluszny and Wilker [611] discuss the n-dimensional case referring to $(n-1)$-spheres touching $n+1$ given spheres. E.g., if no two of the given spheres intersect, and no three have the property that one separates the other two, the expected solution number is 2^{n+1}. The number of solutions depends then on the relative size and position of the given spheres. This is discussed in detail for $n=3$, regarding the six inversely invariant parameters which represent configurations of four given spheres. If the solution number is finite, it can be any integer from 0 to 16, and if it is infinite, it can be a one, two or threefold infinity depending on the number of one-parameter families of solutions.

The two papers [713] and [714] of Schroth are dedicated to the Apollonius problem in *Laguerre geometry*, and the problem is completely solved there for flat Laguerre planes and for 4-dimensional locally compact topological Laguerre planes, respectively.

Already Schoute contributed to an n-dimensional analogue of Apollonius' problem with given hyperplanes, and Drechsler and Sterz [226] continue these studies using modern *cohomological theory and enumerative geometry in projective space*.

Knight [457] investigates the Apollonius problem with the help of *Lie contact geometry*. He develops a related concept, called signature Lie triads, for the classification task. The established proof yielding the classification is relatively long.

With the help of inversion, Morita [584] studies the following configuration clearly related to Apollonius' problem: One starts with circles O_1, O_2, O_3 touching each other externally, and takes S as the circle touching all O_1, O_2, O_3 and having them in its interior. Now let I_1, I_2, I_3 be the three circles tangent to S, O_2, O_3, to S, O_1, O_3 and to S, O_1, O_2, respectively. Then let s_i and t_i be the two common

external tangents of O_i and I_i, $i = 1, 2, 3$. It follows that there is a circle tangent to all of s_1, t_1, s_2, t_2, s_3 and t_3. Furthermore, let $A_1A_2A_3$ be the triangle the edges of which lie on the outer common external tangents of O_1, O_2 and O_3, and let $B_1B_2B_3$ be determined analogously by the outer common external tangents of I_1, I_2 and I_3. Let, finally, l_1, l_2 and l_3 be the lines joining opposite vertices of the triangles $A_1A_2A_3$ and $B_1B_2B_3$. Then the lines l_1, l_2, l_3 have a common point. The author also derives an analogous three-dimensional configuration.

Berger et al. [74] study a typical *location problem* based on the Apollonius task, namely to determine the optimal location of production facilities in the plane. On the base of a uniform representation of Möbius, Laguerre, and Minkowski geometry, Taherian and Mohseni [770] obtain a new approach to the Apollonius problem, where also stereographic projection is used.

Baragar and Kontorovich [57] present some history of the Apollonius problem, including the names Viète and Gergonne, and lay special emphasize on the efficiency (i.e., the number of moves) in constructing the circle which is tangent to three mutually touching circles. It is shown that seven moves are sufficient.

Chapter 3
Graphs and Circle-Systems



Mathematics compares the most diverse phenomena and discovers the secret analogies that unite them.

— Joseph Fourier

We will explain some fundamentals on graphs, including planar graphs and Euler's formula. As a representation of a graph by a circle-system, we consider first the orthogonal-circle representation (abbreviated as OCR) of a special type of graphs, called quadrangulations. An OCR of a graph is a circle-system consisting of circles each corresponding to a vertex of the graph, in which each pair of circles corresponding to a pair of adjacent vertices is orthogonally crossing, whereas each pair of circles corresponding to non-adjacent vertices do not cross each other. The OCR theorem gives a necessary and sufficient condition for a quadrangulation to have an OCR. From the OCR theorem, the coin graph theorem (also called Koebe's theorem, or the Koebe-Andreev-Thurston theorem) and Steinitz' theorem characterizing the polyhedral graphs, are derived easily.

3.1 Graphs and Planar Graphs

Suppose that for any two distinct elements of a set, it is somehow defined whether the two elements are *adjacent* or not. Then we say the set is equipped with the *adjacency relation*. A non-empty finite set V equipped with an adjacency relation is called a *graph*, and each element of V is called a *vertex* of the graph. For example:

- For a set consisting of a finite number of points in $\mathbb{R}^3$, two points are defined to be adjacent if and only if this pair of points is exactly unit distance apart. Then we have a graph. This graph is known as a *unit-distance graph* in $\mathbb{R}^3$.
- In the set of vertices of a convex polyhedron Π in $\mathbb{R}^3$, two vertices are defined to be adjacent if they are the end-points of an edge of the polyhedron Π. Then we have a graph. This graph is called the *graph of the polyhedron* Π.

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- In a set consisting of a finite number of mutually non-overlapping disks in the plane, two disks are defined to be adjacent if and only if they are tangent. Then we have a graph whose vertices are disks. This graph is called a *coin graph*.

In a graph, each pair (u, v) of mutually adjacent vertices is called an *edge* of the graph, and it is simply denoted by uv or vu. Since the adjacency-relation of a graph can be given by the set $E = \{uv, xy, \dots\}$ of all edges, we may regard a *graph* as a pair $G = (V, E)$ of the set V of all vertices and the set E of all edges.

Vertices of a graph $G = (V, E)$ are (tentatively) denoted by $u, v, w, \dots$ and its edges are denoted by $a, b, c, \dots$. If $a = uv \in E$, then a is called the edge connecting u and v; the vertices u, v are called the endpoints of the edge a, and u is said to be *adjacent* to v, or u is a *neighbor* of v. For a vertex $v \in V$, the set of all vertices that are adjacent to v is denoted by $N(v)$, and called the *neighborhood* of v. Thus

$$N(v) = \{x \in V : xv \in E\}.$$

The number of vertices in $N(v)$ is called the *degree* of v and denoted by $\deg v$. For a graph $G = (V, E)$, the minimum value and the maximum value of $\deg v$ for $v \in V$, are called the *minimum degree*, and the *maximum degree* of G, respectively. If, in a graph, the degree of every vertex is known, then the number of edges of the graph is derived from the following "hand-shaking lemma", the proof of which is easy and will be omitted.

Lemma 3.1 (Hand-Shaking Lemma) *In a graph $G = (V, E)$,*

$$\sum_{v \in V} \deg v = 2 \times (\text{the number of edges}).$$

Suppose that two graphs $G = (V, E)$ and $G' = (V', E')$ satisfy $V' \subset V$ and $E' \subset E$. Then G' is called a *subgraph* of G. If, further, $G \neq G'$ holds, then G' is called a *proper subgraph* of G. If a graph has n vertices, and every pair of distinct vertices is adjacent, then the graph is called the *complete graph* with n vertices, and it is usually denoted by K_n. Each vertex of K_n has degree $n - 1$, and the number of edges of K_n is $\binom{n}{2}$.

Every graph can be "realized" as a figure in $\mathbb{R}^3$. For a graph $G = (V, E)$, represent each vertex by a point in $\mathbb{R}^3$, and connect each pair of points corresponding to a pair of adjacent vertices with a simple curve, so that different curves never intersect except in their common endpoints. This is clearly possible in $\mathbb{R}^3$. The figure obtained in this way is called a *realization* of the graph G. In a realization of a graph G, simple curves corresponding to edges of G are also called edges of the realization, and points corresponding to the vertices of G are also called the vertices of the realization. Thus the realization of a graph is also regarded as a graph. For a convex polyhedron Π in $\mathbb{R}^3$, its 1-skeleton (that is, the set of vertices and edges of Π) is a realization of the graph of Π. Hereafter, by *the graph of the convex polyhedron* we mean the 1-skeleton of the convex polyhedron.

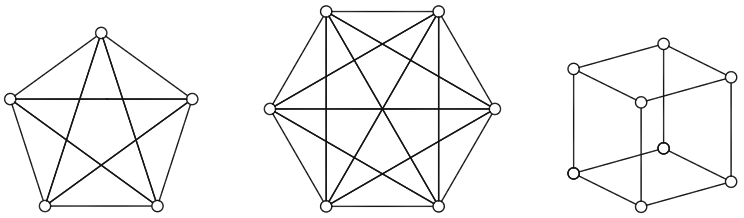


Fig. 3.1 The complete graphs K_5, K_6 and the graph of a cube

In many cases, it is convenient to "represent" graphs by figures in the plane. So, for a graph, we use

a suitable projection of one of its realization in $\mathbb{R}^3$ into a plane

as a *representation of the graph*. In such a figure, projections of edges of the realization (simple curves) could cross each other. So, to distinguish vertices of the graph from such crossing points of curves, vertices are usually denoted by $\circ$ or $\bullet$; see Fig. 3.1 which shows the complete graphs K_5, K_6 and the graph of a cube.

In a graph $G = (V, E)$, a sequence of edges for which the corresponding sequence of edges in its realization constitute a simple curve is called a *path* of G, and if u, v are vertices corresponding to the end-points of the simple curve, the path is said to connect u and v. A sequence of edges in G for which the corresponding sequence of edges in its realization constitutes a simple closed curve, is called a *cycle*. The number of edges contained in a path or cycle is called the *length* of the path or the cycle. A cycle of length k (that is, a cycle consisting of k edges) is called a k-cycle. A cycle of even length (resp. odd length) is simply called an *even-cycle* (resp. *odd-cycle*). For example, the graph of a cube contains 4-cycles, 6-cycles, 8-cycles, but contains no odd-cycle. If $n > 3$, then the complete graph K_n contains k-cycles for $k = 3, 4, \ldots, n$.

Lemma 3.2 *If each vertex of G has degree at least 2, then G contains a cycle.*

Proof Let us regard a realization of G as a road network. Starting from a vertex u we go along an edge to an adjacent vertex v. Since $\deg v \geq 2$, without returning the same edge, we can proceed along another edge to another vertex w. Since $\deg w \geq 2$, without returning the last edge, we can proceed further along an edge to another vertex. If we continue in this way, since the number of vertices is finite, sooner or later we come to a vertex that we already visited. Let z be the first vertex we visit twice. Then the vertices we visited after the first visit of z till the second visit of z are all different. Hence the sequence of edges during the first visit of z and the second visit of z form a cycle. □

A graph G is called *connected* if for any two vertices of G there is a path in G that connects the two vertices. A graph consisting of a single vertex is defined to be connected. A graph that is not connected is called *disconnected*. A disconnected graph can be regarded as the disjoint union of connected graphs. In this sense, each

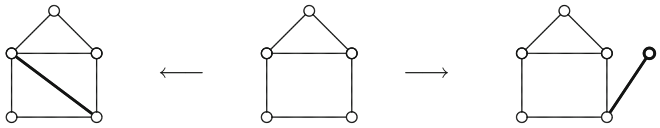


Fig. 3.2 Addition of a bypass (left) and a pointed-edge (right)

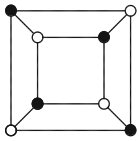


Fig. 3.3 A realization of the graph of a cube in the plane

connected graph is called a *connected component* (or simply a *component*) of the disconnected graph.

In a connected graph $G = (V, E)$, the operation to add a new edge $e \notin E$ that connects a pair of nonadjacent vertices in G is called the *addition of a bypass*. The operation to add a new vertex v and an edge that connects v to a vertex of G is called the *addition of a pointed-edge* (Fig. 3.2).

The following theorem is obvious.

Theorem 3.1 *Every connected graph with at least two vertices can be obtained from a single vertex graph by repeating addition of a pointed-edge and addition of a bypass.* □

If a graph G can be realized as a figure contained in a subset $X \subset \mathbb{R}^3$, then we say that G is *embeddable* in X. A graph that is embeddable in a plane is called a *planar graph*, whereas a graph realized in a plane is called a *plane graph*. For example, the graph of a cube (Fig. 3.1 right) is a planar graph, and Fig. 3.3 shows a plane graph obtained as a realization of the graph of a cube.

A graph "embedded" on a sphere (that is, realized on a sphere) can be mapped by a stereographic projection to a plane graph. And every plane graph can be mapped to a graph embedded on a sphere by the inverse mapping of some stereographic projection. Thus, we have the following theorem.

Theorem 3.2 *A graph G is planar if and only if G is embeddable on a sphere.* □

Let G be the graph of a convex polyhedron Π in $\mathbb{R}^3$, that is, the 1-skeleton of Π. Let P be an interior point of the convex polyhedron Π, and $\mathbb{S}$ be a sphere with center P. By the central projection $\mathbb{R}^3 \to \mathbb{S}$ from P, G can be embedded in $\mathbb{S}$.

Corollary 3.1 *The graph of a convex polyhedron is a planar graph.* □

A plane graph G divides the plane into several regions. Each region is called a *face* of G. Note that the outermost infinite region is also a face. For example, the plane graph in Fig. 3.3 has six faces. A plane graph consisting of a single vertex has one face. Faces of a plane graph are denoted by $\alpha, \beta, \ldots$. An edge of a plane graph G is called a *boundary edge* of a face α if one side (or both sides) of the edge

form the face α. A subgraph of a plane graph is called the *boundary* of a face α, if it consists of all boundary edges of α and the end-vertices of the boundary edges.

Let us note here that every connected plane graph with at least two vertices is obtained from a single vertex by repeating addition of a pointed-edge and addition of a bypass *in the plane*. For a plane graph G, let $v(G), e(G), f(G)$ denote the number of vertices, the number of edges, and the number of faces of G, respectively.

Theorem 3.3 (Euler's Formula) *For every connected plane graph G,*

$$v(G) - e(G) + f(G) = 2$$

holds.

For example, if G is the plane graph in Fig. 3.3, then $v(G) = 8, e(G) = 12, f(G) = 8$. Thus, $v(G) - e(G) + f(G) = 8 - 12 + 6 = 2$.

Proof For a graph G, denote $v(G) - e(G) + f(G)$ simply by $\psi(G)$. If we apply "addition of a pointed-edge in the plane" to a connected plane graph, then the number of vertices and the number of edges increase by 1, but the number of faces does not change. Hence, the value of ψ is invariant by addition of pointed-edge in the plane. If we apply "addition of a bypass in the plane" to a connected plane graph, then the number of vertices does not change, and the number of edges increases by 1. If u, v are the vertices connected by the bypass, then, since the original graph is connected, there is a path in the original graph that connects u and v. This path and the added bypass make together a simple closed curve in the plane, and this simple closed curve divides the plane into two regions, inside and outside of it. Since one side of the added bypass becomes the interior of this simple closed curve, and the other side becomes the exterior of this simple closed curve, the face where the bypass passes through is divided by the bypass into two regions. Hence, by addition of a bypass in the plane, the number of faces increases by 1. Therefore, the value of ψ does not change by addition of a bypass in the plane. Recall that every connected plane graph with at least two vertices is obtained from a single vertex graph by repeating addition of a pointed-edge and addition of a bypass in the plane. In a single vertex graph, $\psi = 1 - 0 + 1 = 2$, and the value of ψ is invariant under addition of a pointed-edge and a bypass in the plane, yielding $\psi(G) = 2$. □

Lemma 3.3 *If a planar graph G has n vertices, $n \geq 3$, then*

$$e(G) \leq 3n - 6.$$

Proof We may suppose that G is realized in the plane. If G is not connected, then by adding edges we can change it to a connected plane graph. So, it is enough to prove the lemma in the case that G is connected.

Now, imagine that for every edge of the plane graph G, we put one stone $\diamond$ to each side of the edge, see Fig. 3.4. The number of put stones is $2e(G)$. Since G has at least two edges, each face of G must contain at least three stones. Hence, we

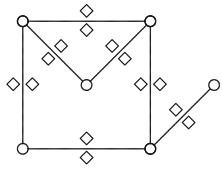


Fig. 3.4 One stone ⋄ to each side of every edge

have $3f(G) \leq 2e(G)$. From this and three times Euler's formula $3v(G) - 3e(G) + 3f(G) = 3 \cdot 2$, we have $e(G) \leq 3n - 6$. □

Remark 3.1 Suppose that for every edge of a plane graph we put one stone to each side of the edge. If a face α contains exactly three stones, then the boundary of the face α must be a 3-cycle. Such a face α is called a *triangular face*, or simply a *triangle*.

Example 3.1 In the complete graph K_5, we have $v(K_5) = 5$ and $e(K_5) = \binom{5}{2} = 10$. Since $10 \not\leq 9 = 3v(K_5) - 6$, K_5 is not planar.

A connected plane graph whose faces (including the outermost face) are all triangular faces is called a *maximal plane graph*. From the proof of Lemma 3.3, we can see that a maximal plane graph with n vertices has exactly $3n - 6$ edges and $2n - 4$ faces.

3.2 Quadrangulations

A connected plane graph with minimum degree ≥ 3 whose faces are all quadrilaterals (that is, the boundary of every face is a 4-cycle) is called a *quadrangulation* (of the plane). For example, the plane graph in Fig. 3.3 is a quadrangulation, but the plane graph in Fig. 3.5 is not a quadrangulation (though its faces are all quadrilaterals) since its minimum degree is 2.

Since the outer region of a quadrangulation is also a quadrilateral (i.e., its boundary is a 4-cycle), and its four vertices have degree at least three, it follows easily that there must be at least four more vertices. Thus, every quadrangulation has at least eight vertices.

Example 3.2 If G is a quadrangulation and $v(G) = n$, then $e(G) = 2n - 4$, $f(G) = n - 2$. This can be seen as follows: Since $2e(G) = 4f(G)$, from Euler's

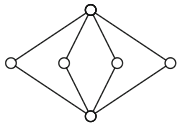


Fig. 3.5 The minimum degree is 2

formula we have $4 = 2n - 2e(G) + 2f(G) = 2n - e(G)$. Hence $e(G) = 2n - 4$ and $f(G) = n - 2$.

If we can divide the vertex set of a graph G into two groups so that the vertices in the same group become nonadjacent, then the graph is called a *bipartite graph*. For example, the graph of a cube (Fig. 3.3) is a bipartite graph. It is obvious that a bipartite graph contains no odd cycle.

Lemma 3.4 *Every quadrangulation is a bipartite graph.*

Proof First, we show that no odd cycle exists in a quadrangulation. Suppose that a quadrangulation $G = (V, E)$ has a $(2k + 1)$-cycle C. Since G is a quadrangulation, there must be quadrilateral faces inside the simple closed curve C, and outside C. Let $H = (V_H, E_H)$ be the graph obtained from G by removing the edges and vertices lying outside C. Let m be the number of quadrilateral faces inside C. Then

$$2|E_H| = 4m + 2k + 1,$$

where $|\cdot|$ denotes the number of elements of the set. This contradicts the fact that $|E_H|$ is an integer. Hence G has no odd cycle.

Next, let P, Q be two paths that connect a vertex x to another vertex y, and let E_P, E_Q denote the set of edges in P and in Q, respectively, see Fig. 3.6. We show that $|E_P| \equiv |E_Q| \pmod 2$. In order to do this, it is sufficient to show that $|E'_P| \equiv |E'_Q| \pmod 2$, where $E'_P = E_P \setminus (E_P \cap E_Q)$, $E'_Q = E_Q \setminus (E_P \cap E_Q)$. Note that $E'_P \cup E'_Q$ is divided into a several subsets, each of which consists of edges of a cycle. Then, since every cycle of G is an even cycle, we can deduce that $|E'_P \cup E'_Q| \equiv 0 \pmod 2$. Hence, $|E'_P| + |E'_Q| = |E'_P \cup E'_Q| \equiv 0 \pmod 2$, and $|E'_P| \equiv |E'_Q| \pmod 2$.

Therefore, if there is a path of even length that connects x and y, then every path connecting x and y has even length. Now, take a pair x, y of adjacent vertices of G, and put $w \in V \setminus \{x, y\}$ in the same group as x if x and w are connected by a path of even length. Otherwise, put w in the same group as y. Then no two vertices in the same group are adjacent. Hence G is a bipartite graph. □

A 4-cycle of a quadrangulation G is called a *separating* 4*-cycle* if each side of the simple closed curve formed by the 4-cycle contains a vertex of G, see Fig. 3.7. A quadrangulation is called *irreducible* if it has no separating 4-cycle. The graph of a cube in Fig. 3.3 is an irreducible quadrangulation.

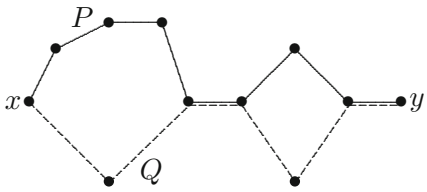


Fig. 3.6 Solid lines for P, dotted lines for Q

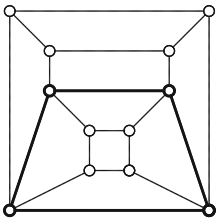


Fig. 3.7 A separating 4-cycle of a quadrangulation

Exercise 3.1 Let H be a plane graph with $n \geq 4$ vertices. Prove that if H is a bipartite graph and has a face that is not a quadrilateral, then $e(H) \leq 2n - 5$.

Solution If H is disconnected, then by adding the fewest necessary edges, we can get a connected plane graph that is still bipartite. Then the outermost region of this new graph is not a quadrilateral face. Hence, we may consider the case that H is connected. Since H has a face that is neither a triangle nor a quadrilateral, we can see, by putting one stone to each side of every edge, that $2 \times e(H) > 4 \times f(H)$. From this and Euler's formula we can deduce that $e(H) < 2n - 4$. □

3.3 Orthogonal-Circle Representations

As we said, two circles in the plane are said to be mutually crossing if they share two points. And if their tangent lines at a crossing point intersect orthogonally, then the two circles are said to be orthogonally crossing.

When two circles C_1 and C_2 cross orthogonally, then the line passing through the center of C_1 and one of the crossing points is a tangent line of C_2. Hence the center of C_1 lies outside C_2. When C_1 and C_2 cross orthogonally, we write $C_1 \perp C_2$. And $C_1 \perp C_2 \perp C_3 \perp C_4 \perp \ldots$ means that $C_1 \perp C_2$, $C_2 \perp C_3$, $C_3 \perp C_4$, etc.

Lemma 3.5 *Let C_1, C_2 be two circles with centers z_1, z_2 and $z_1 \neq z_2$, respectively. If a circle C is orthogonally crossing both C_1, C_2, then the center of C lies on a fixed line L that is perpendicular to the line z_1z_2. If C_1 and C_2 are disjoint, then L is disjoint from both C_1, C_2.*

The line L is called the *radical axis* of the circles C_1, C_2.

Proof Put $z_1 = (0, 0)$, $z_2 = (a, 0)$, and let r_i be the radius of C_i, for $i = 1, 2$. Let (x, y) be the center of C. Then, by the Pythagorean theorem,

$$r^2 = x^2 + y^2 - r_1^2$$
$$r^2 = (x - a)^2 + y^2 - r_2^2.$$

From this we have $x = \frac{1}{2a}(a^2 + r_1^2 - r_2^2)$. Since the value of x is constant, the center (x, y) of C lies on a line L that is perpendicular to the line z_1z_2. If C_1 and C_2 are

disjoint, then either $a > r_1 + r_2$ or $a < |r_1 - r_2|$ holds. If $a > r_1 + r_2$, then

$$x - r_1 = \frac{a^2 + r_1^2 - r_2^2}{2a} - r_1 = \frac{(a - r_1)^2 - r_2^2}{2a} > 0, \tag{3.1}$$

and hence L is disjoint from C_1. And, similarly, L is disjoint from C_2.

In the case $a < |r_1 - r_2|$, by assuming $r_1 > r_2$, we have $r_1 - a > r_2$. Hence, by (3.1), L is disjoint from C_1. (Since C_2 lies inside C_1, L is also disjoint from C_2.) □

Lemma 3.6 *If four circles C_i, $i = 1, 2, 3, 4$, are such that $C_1 \perp C_2 \perp C_3 \perp C_4 \perp C_1$, the following two assertions* (i) *and* (ii) *hold.*

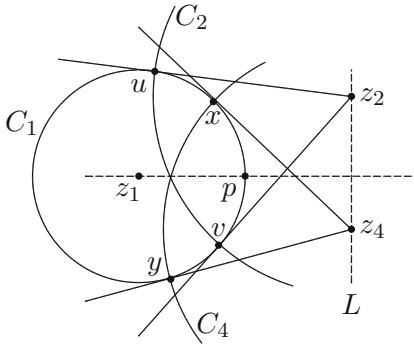
(i) *If C_1, C_3 are disjoint, then C_2, C_4 cross each other.*
(ii) *If C_1, C_3 do not cross and C_2, C_4 do not cross, then four circles meet at a point.*

Proof Let z_i be the center of C_i, $i = 1, 2, 3, 4$. By Lemma 3.5, the line z_1z_3 and the line z_2z_4 are perpendicular.

(i) Let u, v be the intersection points of C_1, C_2, and x, y be the intersection points of C_1, C_4. The lines z_2u, z_2v are tangent lines of C_1, and u, v are the contact points. Similarly, the lines z_4x, z_4y are tangent lines of C_1, and x, y are contact points, see Fig. 3.8. If C_1 and C_3 are disjoint, then by Lemma 3.5, the line z_2z_4 is did joint from both C_1, C_3. Hence, none of the minor arc $\widehat{uv}$ of C_1 and the minor arc $\widehat{xy}$ of C_1 contains the other. (If $\widehat{uv}$ contains $\widehat{xy}$, then z_4 lies in the angular region $\angle uz_2v$, the line z_2z_4 must intersect C_1.) Let p be the point on C_1 that is nearest to the line z_2z_4. Then, since both lines z_2p, z_4p cut C_1, $\widehat{uv}$ and $\widehat{xy}$ contain p. Hence the circle C_2 (which intercepts $\widehat{uv}$ from C_1) and the circle C_4 (which intercepts $\widehat{xy}$ from C_1) cross each other.

(ii) Suppose that C_1, C_3 do not cross, and C_2, C_4 do not cross. Since C_2, C_4 do not cross, C_1, C_3 are not disjoint by (i). And since C_1, C_3 do not cross, they must be tangent. Let w be their contact point. Then the line z_2z_4 is a common tangent line of C_1, C_3 passing through w. In this case, w must lie on C_2 and C_4. □

Fig. 3.8 $C_4 \perp C_1 \perp C_2$



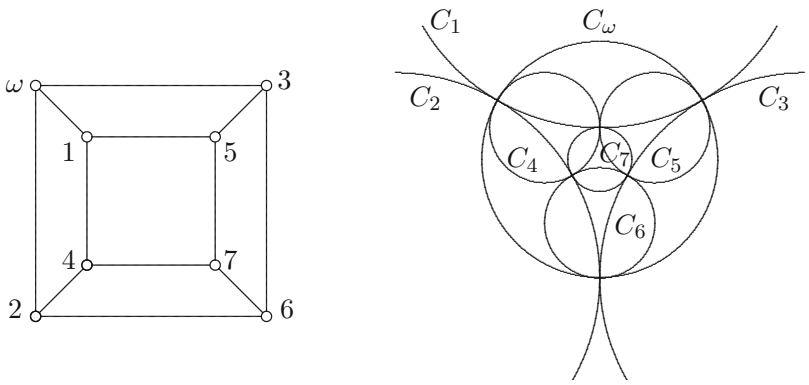


Fig. 3.9 C_ω is the special circle

For a quadrangulation $G = (V, E)$, a family $\mathcal{F} = \{C_v : v \in V\}$ of $|V|$ circles on the plane is called an *orthogonal-circle representation* (abbreviated *OCR*) of G if it satisfies the following conditions 1°, 2°.

1° C_u, C_v cross each other $\Leftrightarrow C_u \perp C_v \Leftrightarrow uv \in E$.
2° There is a special circle $C_\omega \in \mathcal{F}$ that encloses $C_v \in \mathcal{F} \setminus \{C_\omega\}$ unless $C_v \perp C_\omega$, and no other circle encloses another circle.

In any OCR of a quadrangulation $G = (V, E)$, the following 3° also holds, by Lemma 3.6(ii).

3° For each face (quadrilateral) $abcd$ of G, four circles C_a, C_b, C_c, C_d meet at a point and $C_a \perp C_b \perp C_c \perp C_d \perp C_a$. (Hence C_a and C_c are tangent, and also C_b and C_d are tangent.)

Figure 3.9 right shows an OCR of the graph of a cube (Fig. 3.9 left).

Theorem 3.4 (OCR Theorem) *A quadrangulation $G = (V, E)$ has an OCR if and only if G is irreducible.*

Proof Suppose that there is an OCR $\mathcal{F} = \{C_v : v \in V\}$ of G, and G is still not irreducible. Let $abcd$ be a separating 4-cycle. Let $abij$, $abkl$ be two faces of G sharing the edge ab. Let p, q be the intersection points of C_a and C_b. Then, by Lemma 3.6(ii), C_a and C_c are tangent at either p or q. Similarly, C_a and C_i, and also C_a and C_k are tangent at one of p, q. This implies that some three of C_a, C_c, C_i, C_k are tangent at the same point. In this case, one of the three circles encloses just one other circle. This contradicts the condition 2° of OCR. Thus, if G has an OCR, then G must be irreducible.

Next, suppose G is irreducible and let $n = |V|$. Then $n > 4$. By Example 3.2, $e(G) = 2n - 4$. Hence G has a vertex of degree 3. (If the minimum degree is greater than 3, then we have $2e(G) \geq 4n$ and $e(G) \geq 2n$.) Let ω be a vertex of degree 3, and denote other vertices by $1, 2, 3, \ldots, n-1$, where $N(\omega) = \{1, 2, 3\}$.

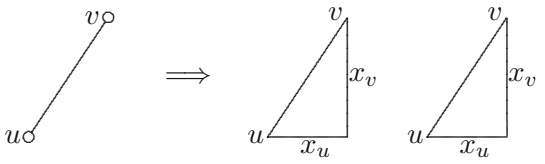


Fig. 3.10 A pair of congruent right triangles for each edge uv

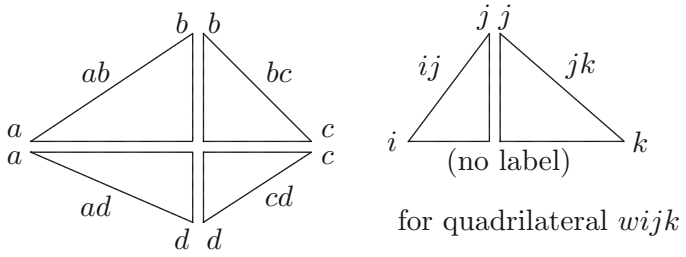


Fig. 3.11 Make a quadrangle and a (composite) triangle

In order to find an OCR for G, take $n-1$ positive reals

$$x_1, x_2, \ldots, x_{n-1}$$

and, corresponding to each edge uv ($u \neq \omega$, $v \neq \omega$), prepare a pair of congruent right triangles (made from cardboard) with arms of lengths x_u, x_v, see Fig. 3.10. The hypotenuses of such right triangles are labeled by uv.

For each quadrilateral face of G with vertices $a, b, c, d \in \{1, 2, \ldots n-1\}$, create a quadrangle $abcd$ by attaching four right triangles with hypotenuse ab, bc, cd, da, at the arms of the lengths x_a, x_b, x_c, x_d, see Fig. 3.11 left. For each quadrilateral face ωijk with ω as one of its vertices, create a (composite) triangle ijk by attaching right triangles with hypotenuses ij, jk at their arms of length x_j, see Fig. 3.11 right. Then, by Example 3.2, there arise three (composite) triangles and $n-5$ quadrangles. In the edges of these (composite) triangles and quadrangles, each "label" of edge of G other than 12, 23, 31 appears exactly twice as labels.

Try to attach these triangles and quadrangles along the edges of the same labels in the plane. Suppose that we succeed to fit them together without gaps and without overlappings such that the resulting figure becomes a triangle with vertices 1, 2, 3 in the plane. Then, for each vertex $v \neq \omega$, draw a circle C_v of radius x_v centered at the point with label v, and finally draw the inscribed circle of the triangle 123 as C_ω. Then these circles form an OCR of G. Figure 3.12 shows the case where G is the graph of a cube, see also Fig. 3.9.

Now we ask for what positive reals $x_1, \ldots, x_{n-1}$ it is possible to fit these triangles and quadrangles together into a (large) triangle 123 without gaps and overlappings?

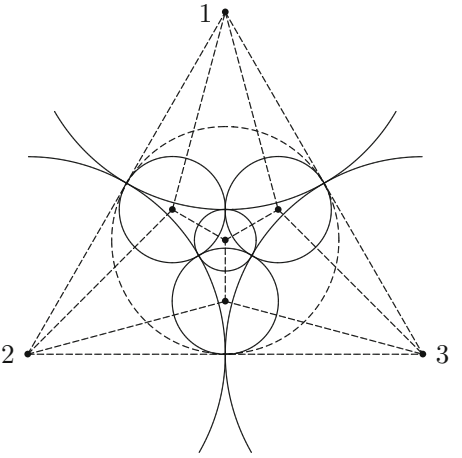


Fig. 3.12 Draw a circle with radius x_v and center v

Let $2\theta_i(x_1, x_2, \ldots, x_{n-1})$ denote the sum of the interior angles at vertex i for all right triangles with vertex i. Then

$$\theta_i(x_1, x_2, \ldots, x_{n-1}) = \sum_{j \in N(i)} \tan^{-1} \frac{x_j}{x_i},$$

where $N(i)$ is the neighborhood of the vertex i. If $2\theta_i(x_1, x_2, \ldots, x_{n-1}) = 2\pi$ holds for every $i \in V \setminus \{\omega, 1, 2, 3\}$, then it is possible to fit these (composite) triangles and quadrangles together into a triangle with three vertices $1, 2, 3$. Hence the theorem will follow from the next lemma. □

Lemma 3.7 *Let $G = (V, E)$ be an irreducible quadrangulation, and $\omega \in V$ be a vertex of degree 3. Let $1, 2, \ldots, n-1$ denote the vertices of V other than ω, and suppose that $\{1, 2, 3\}$ is the nighborhood of ω. Then there are positive reals $x_1, x_2, \ldots, x_{n-1}$ for which*

$$\sum_{j \in N(i)} \tan^{-1} \frac{x_j}{x_i} = \pi, \quad i = 4, 5, \ldots, n-1$$

hold, where $N(i)$ denotes the neighborhood of the vertex i.

Proof Let us regard an n-tuple of real numbers $(x_1, \ldots, x_n)$ as a point in n-dimensional space $\mathbb{R}^n$, and call each x_i its ith coordinate. The set of points in $\mathbb{R}^n$ whose coordinates are all positive is denoted by $\mathbb{R}^n_+$.

Since $V = \{1, 2, 3, \ldots, n-1, \omega\}$, from now on we denote the vertex ω of G by n. For a point $\boldsymbol{x} = (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n_+$, we put

$$\theta_i(\boldsymbol{x}) = \sum_{j \in N(i)} \tan^{-1} \frac{x_j}{x_i}.$$

To prove the lemma, it is enough to show that there is an $\boldsymbol{x} \in \mathbb{R}_+^n$ such that

$$(\theta_1(\boldsymbol{x}), \theta_2(\boldsymbol{x}), \ldots, \theta_n(\boldsymbol{x})) = (\frac{\pi}{3}, \frac{\pi}{3}, \frac{\pi}{3}, \pi, \ldots, \pi) \tag{3.2}$$

holds. Hereafter, we regard each coordinate x_i of $\boldsymbol{x}$ as a variable, and denote the right hand side of (3.2) by $(\eta_1, \eta_2, \ldots, \eta_n)$. From the identity

$$\tan^{-1} \frac{x_j}{x_i} + \tan^{-1} \frac{x_i}{x_j} = \frac{\pi}{2} \tag{3.3}$$

and $e(G) = 2n - 4$, we have

$$\sum_{i=1}^{n} \theta_i(\boldsymbol{x}) = \sum_{ij \in E} \left(\tan^{-1} \frac{x_j}{x_i} + \tan^{-1} \frac{x_i}{x_j} \right) = \frac{(2n-4)\pi}{2} = \sum_{i=1}^{n} \eta_i \tag{3.4}$$

for any $\boldsymbol{x} \in \mathbb{R}_+^n$. Since $\tan^{-1}(x_j/x_i)$ is monotonically decreasing in x_i and monotonically increasing in x_j, $\theta_i(\boldsymbol{x})$ is monotonically decreasing in x_i, and monotonically non-decreasing in the other variables. Since

$$\lim_{x_i \to \infty} \tan^{-1}(x_j/x_i) = 0$$

for a fixed x_j, for any $\boldsymbol{x} \in \mathbb{R}_+^n$ and $1 \leq i \leq n$, there is a smallest $\xi_i = \xi_i(\boldsymbol{x}) \geq 0$ such that

$$\theta_i(x_1, \ldots, x_{i-1}, x_i + \xi_i, x_{i+1}, \ldots, x_n) \leq \eta_i. \tag{3.5}$$

(If $\theta_i(\boldsymbol{x}) \leq \eta_i$, then $\xi_i = 0$, whereas, if $\theta_i(\boldsymbol{x}) > \eta_i$, then ξ_i is the unique solution of the equation $\theta_i(x_1, \ldots, x_{i-1}, x_i + \xi_i, x_{i+1}, \ldots, x_n) = \eta_i$.) Define a map $f : \mathbb{R}_+^n \to \mathbb{R}_+^n$ by

$$f(x_1, x_2, \ldots, x_n) = (x_1 + \xi_1, x_2 + \xi_2, \ldots, x_n + \xi_n),$$

where each $\xi_i \geq 0$ is the minimum number satisfying (3.5). When each coordinate of $\boldsymbol{x}$ varies in a very small range, then ξ_i varies also in a small range; so, $\xi_i = \xi_i(\boldsymbol{x})$ is a continuous function of $\boldsymbol{x} = (x_1, x_2, \ldots, x_n)$. If there is a point $\boldsymbol{x}_0 \in \mathbb{R}_+^n$ such that $f(\boldsymbol{x}_0) = \boldsymbol{x}_0$, then $\theta_i(\boldsymbol{x}_0) \leq \eta_i$ holds for every i. Hence, by (3.4), we must have $\theta_i(\boldsymbol{x}_0) = \eta_i$ for all i. Thus, to prove the lemma, it is sufficient to show that there is a point $\boldsymbol{x} \in \mathbb{R}_+^n$ for which $f(\boldsymbol{x}) = \boldsymbol{x}$ holds.

Now consider the sequence of points in $\mathbb{R}_+^n$

$$\boldsymbol{x}^{(1)} = (1, 1, \ldots, 1), \boldsymbol{x}^{(2)} = f(\boldsymbol{x}^{(1)}), \boldsymbol{x}^{(3)} = f(\boldsymbol{x}^{(2)}), \boldsymbol{x}^{(4)} = f(\boldsymbol{x}^{(3)}), \ldots.$$

By the definition of the map f, the sequence $x_i^{(1)}, x_i^{(2)}, x_i^{(3)}, \ldots$ of the ith coordinate of this sequence $\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}, \boldsymbol{x}^{(3)}, \ldots$ is a monotonically increasing sequence. Put

$$I = \{i : \lim_{\nu \to \infty} x_i^{(\nu)} = \infty\}.$$

If $I = \emptyset$, then the sequence $x_i^{(1)}, x_i^{(2)}, x_i^{(3)}, \ldots$ has a limit value for each i, and hence the limit point $\lim_{\nu \to \infty} \boldsymbol{x}^{(\nu)} = \boldsymbol{x}^{(\infty)}$ exists. Since $\xi_i(\boldsymbol{x})$ is a continuous function of $\boldsymbol{x}$, we have

$$\xi_i(\boldsymbol{x}^{(\infty)}) = \lim_{\nu \to \infty} \xi_i(\boldsymbol{x}^{(\nu)}) = \lim_{\nu \to \infty} \left(x_i^{(\nu+1)} - x_i^{(\nu)}\right) = 0.$$

Therefore, $f(\boldsymbol{x}^{(\infty)}) = \boldsymbol{x}^{(\infty)}$. Hence it is sufficient to show that $I = \emptyset$.

Suppose $I \neq \emptyset$. Then for every $i \in I$, there is a ν for which $\theta_i(\boldsymbol{x}^{(\nu)}) > \eta_i$. (Indeed, if $\theta_i(\boldsymbol{x}^{(\nu)}) \leq \eta_i$ for all $\nu = 1, 2, 3, \ldots$, then by the definition of the map f, we must have $x_i^{(1)} = x_i^{(2)} = x_i^{(3)} = \ldots$, contradicting $i \in I$.) Since $\theta_i(\boldsymbol{x})$ is monotonically non-decreasing in the variables other than x_i, $\theta_i(\boldsymbol{x}^{(\nu)}) > \eta_i$ for $\nu = \nu_0$ implies that $\theta_i(\boldsymbol{x}^{(\nu)}) \geq \eta$ for all $\nu > \nu_0$ by the definition of the map f. Thus, there is a ν_1 such that for $\nu \geq \nu_1$,

$$i \in I \Rightarrow \theta_i(\boldsymbol{x}^{(\nu)}) \geq \eta_i$$

holds. Therefore, $|I| < n$, and otherwise, (3.2) holds for sufficiently large ν by (3.4).

Put $E_I = \{ij \in E : i, j \in I\}$. In the computation of $\sum_{i \in I} \theta_i(\boldsymbol{x}^{(\nu)})$, $\pi/2$ will be added from each edge in E_I by (3.3), and $\tan^{-1}(x_j^{(\nu)}/x_i^{(\nu)})$ will be added from each edge ij, $i \in I$, $j \notin I$. However, since

$$\lim_{\nu \to \infty} \tan^{-1}(x_j^{(\nu)}/x_i^{(\nu)}) = 0$$

for $i \in I$, $j \notin I$, we need not consider the latter case. Therefore

$$\sum_{i \in I} \eta_i \leq \lim_{\nu \to \infty} \sum_{i \in I} \theta_i(\boldsymbol{x}^{(\nu)}) = |E_I|\pi/2.$$

Since we assumed $I \neq \emptyset$, the left hand side is positive, and hence $|E_I| > 0$, which implies $|I| \geq 2$. Since G is a quadrangulation, there is no edge connecting $1, 2, 3$. Hence I contains a vertex other than $1, 2, 3$.

Let us derive a contradiction in the following, by dividing our argument into two cases.

(i) The case $\{1, 2, 3\} \not\subset I$: Since $|I| \geq 2$, and I contains a vertex other than $1, 2, 3$, we have

$$\pi + \pi/3 \leq \sum_{i \in I} \eta_i \leq |E_I|\pi/2.$$

Hence $3 \leq |E_I|$, and hence $4 \leq |I|$, which means that I contains at least two vertices other than 1, 2, 3. Since the graph (I, E_I) is a bipartite graph, we have $|E_I| \leq 2|I| - 4$ for $|I| \geq 4$. Therefore,

$$(|I| - 2)\pi + 2\pi/3 \leq \sum_{i \in I} \eta_i \leq |E_I|\pi/2 \leq (2|I| - 4)\pi/2,$$

a contradiction.

(ii) The case $\{1, 2, 3\} \subset I$: In this case, we must have $n \in I$. (Otherwise, we have

$$\lim_{\nu \to \infty} \theta_n(\boldsymbol{x}^{(\nu)}) = \lim_{\nu \to \infty} \sum_{i=1}^{3} \tan^{-1}(x_i^{(\nu)}/x_n^{(\nu)}) = 3\pi/2 > \pi = \eta_n,$$

which contradicts the definition of the sequence $\boldsymbol{x}^{(\nu)}$.) Hence $|I| \geq 4$. The plane graph $H := (I, E_I)$ is clearly not a 4-cycle, and H has a face that is not a quadrilateral. (If every face of H is a quadrilateral, then the boundary of the outer face of H is a 4-cycle of G that contains a vertex of G in its interior. Since H is a proper subgraph of G, this contradicts the fact that G is irreducible.) Hence, by Exercise 3.1, we have $|E_I| \leq 2|I| - 5$. Then,

$$\pi(|I| - 3) + 3(\pi/3) = \sum_{i \in I} \eta_i \leq |E_I|\pi/2 \leq (2|I| - 5)\pi/2,$$

which is a contradiction. □

Remark 3.2 Theorem 3.4 is a reformulation of a result obtained by Brightwell and Scheinerman [128] as OCR theorem for a quadrangulation. Their proof is slightly improved in our OCR theorem.

3.4 3-Connected Graphs and Radial Graphs

For a graph $G = (V, E)$ and a subset $S \subsetneqq V$, $G - S$ denotes the graph obtained from G by removing the vertices in S and removing edges that are incident to vertices in S. If $G - S$ is disconnected, then S is called a *cut set* of G. In a complete graph, there is no cut set. For a connected graph, a cut set with minimum cardinality is called the *minimal cut set*. A connected graph with at least $k + 1$ vertices is called *k-connected* if the graph has no cut set consisting of $k - 1$ vertices. Thus, if G is a 3-connected graph, then for any two vertices x, y of G, $G - \{x, y\}$ is connected. A complete graph K_n is 3-connected for $n \geq 4$. The graph of a cube is also 3-connected as easily verified. The minimum degree of a 3-connected graph G is at least 3. (If a vertex v of G has degree at most 2, then its neighborhood $N(v)$ becomes a cut set with two vertices, and hence G is not 3-connected.)

Lemma 3.8 *Let S be a minimal cut set of a connected graph G, and let H be a connected component of $G - S$. Then every $x \in S$ is adjacent to some vertex in H.*

Proof Suppose that $x \in S$ is not adjacent to any vertex of H. Then H is still a connected component of $G - (S \setminus \{x\})$. This contradicts the fact that S is a minimal cut set. □

Theorem 3.5 *Let $G = (V, E)$ be a graph such that $|V| \geq 4$. If for every $x \in V$ there is a cycle of $G - \{x\}$ that contains $N(x)$, then G is 3-connected.*

Proof Suppose that G is not 3-connected, and let S be a minimal cut set, $|S| \leq 2$. Let x be a vertex in S. Then $N(x)$ is contained in a cycle C of $G - \{x\}$. Let C' be the graph obtained from C by removing a vertex in S other than x (if it exists), and edges adjacent to the vertices in S. Since a cycle cannot be made disconnected by removing a vertex, C' is connected. Hence C' is contained in a connected component H of $G - S$. Since $N(x)$ is contained in C, a connected component of $G - S$ other than H has no vertex that is adjacent to x. This contradicts Lemma 3.8. □

Exercise 3.2 Prove that a maximal plane graph with at least four vertices is 3-connected.

Solution Let G be a maximal plane graph with at least 4 vertices. For a vertex x of G, let $v_1, v_2, \ldots, v_k$ be the vertices (in anti-clockwise order around x) that are adjacent to x. Since each face of G is a triangle, $v_1v_2, v_2v_3, \ldots, v_kv_1$ are all edges of G. Hence these k vertices and these k edges form a cycle in $G - \{x\}$. Since $N(x)$ is contained in this cycle, G is 3-connected by Theorem 3.5. □

Lemma 3.9 *In a 3-connected plane graph G, the following properties* (1) *and* (2) *hold.* (1) *For every edge $e = xy$ of G, its both sides are distinct faces of G.* (2) *The boundaries of two different faces of G cannot share a pair of non-adjacent vertices.*

Proof

(1) Suppose that both sides of an edge $e = xy$ are one and the same face α of G. Then there is a simple closed curve Γ that crosses the edge e at a point, and except that point, the curve lies in the interior of the face α. Then Γ meets G only at the edge e, and never meets $G - \{x\}$. Since $\deg x \geq 3$, there is a vertex $z \neq y$ of G that is adjacent to x. Then y and z do not lie in the same side of Γ. Hence, $G - \{x\}$ is disconnected, contradicting the fact that G is 3-connected.

(2) Suppose that the boundaries of two different faces α, β of G share a pair of non-adjacent vertices x, y. Then there is a simple curve connecting x and y and passing through the interior of α except the endpoints x, y. Similarly, there is a simple curve connecting x, y passing through the interior of the face β. Attaching there two simple curves end to end, we obtain a simple closed curve Ω, that meets G only at x, y. Then each side of Ω contains a vertex of $G - \{x, y\}$, and hence $G - \{x, y\}$ is disconnected. This contradicts the fact that G is 3-connected. □

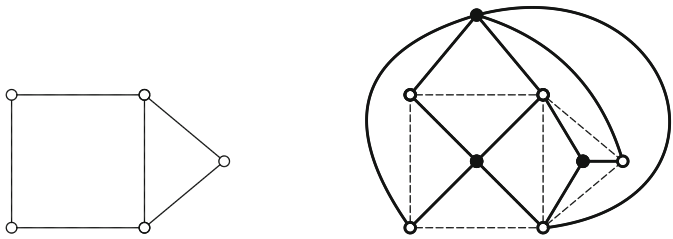


Fig. 3.13 A plane graph and its radial graph

For each face of a plane graph G, put one point inside the face. Such a point is called a *face-point*. The face-point inside the face α is denoted by α^*. For each face of α, connect the face-point α^* and the vertices on the boundary of α by edges (simple curves) so that these edges do not meet in the midway. These edges are called *radial edges*. After that, remove the original edges of G. Then we have a plane graph whose edges are all radial edges. This graph is called the *radial graph* of G and denoted by $R(G)$. The vertices of the radial graph $R(G)$ are divided into two groups, the vertices of G, and the face-points. The radial graph $R(G)$ is clearly a *bipartite* plane graph. Figure 3.13 shows a plane graph and its radial graph.

Lemma 3.10 *The radial graph of a* 3*-connected plane graph* G *is an irreducible quadrangulation.*

Proof Since G is 3-connected, both sides of an edge are distinct faces. Hence, each face of $R(G)$ is a quadrilateral whose one diagonal is an edge of G. Since the degree of each vertex of G is at least 3, each vertex of G is a common point of the boundaries of at least three faces of G. Hence the degree of each vertex of $R(G)$ is at least 3, and hence $R(G)$ is a quadrangulation. Suppose that $R(G)$ is not irreducible. Then $R(G)$ has a separating 4-cycle $u\alpha^* v\beta^*$, where u, v are vertices of G and α^*, β^* are face-points. Since there are vertices of $R(G)$ inside this 4-cycle, and outside this 4-cycle, there are quadrilaterals inside this 4-cycle and outside this 4-cycle. This implies that $G - \{u, v\}$ is disconnected, contradicting that G is 3-connected. Thus, $R(G)$ is an irreducible quadrangulation. □

3.5 The Coin Graph Theorem

For a given graph $G = (V, E)$, a family of mutually nonoverlapping disks $\mathcal{D} = \{D_v : v \in V\}$ in the plane that satisfies the condition

$$uv \in E \Leftrightarrow D_u \text{ and } D_v \text{ are tangent}$$

is called the *coin representation* of G. A graph for which a coin representation exists is called a *coin graph*.

Let $G = (V, E)$ be a coin graph, and $\mathcal{D} = \{D_v : v \in V\}$ be its coin representation. Then, by representing each vertex v by the center of D_v, and representing an edge $uv \in E$ by a line segment connecting the centers of D_u, D_v, we have a realization of G in the plane. Thus, we have the following theorem.

Theorem 3.6 *Every coin graph is a planar graph.* □

The converse is also true.

Theorem 3.7 (Coin Graph Theorem) *Every planar graph is a coin graph.*

Proof A graph with at most 3 vertices is obviously a coin graph. Let G be a plane graph with at least four vertices. If G is not connected, then by taking a face point in the outer face, and adding some edges that connect the face point to a vertex of each component of G, we can get a connected plane graph. If this connected plane graph has a coin representation, then, by removing the disks corresponding to the face point, we can get a coin representation of G. Thus, we may suppose that G is connected. If G has a face that is not a triangle, then take a face-point inside the face, and divide the face into triangular faces by adding *radial edges*. If this new graph has a coin representation $\mathcal{D}$, then by removing from $\mathcal{D}$ the disks corresponding to the added face-points, we have a coin representation of the original plane graph. Therefore, to prove the theorem, it is enough to show that every maximal plane graph G with at least 4 vertices has a coin representation. Since G is 3-connected by Exercise 3.2, its radial graph $R(G)$ is an irreducible quadrangulation. Let ω be the face-point of the outermost face of G. Then the degree of ω is 3. By the OCR theorem, there is an OCR $\mathcal{F}$ of $R(G)$ in which any circle other than the circle corresponding to ω does not enclose any other circle. By removing the circles corresponding to face-points and then replacing each circle by disks, we have from such a family $\mathcal{F}$ a coin representation of G. □

In a coin representation $\mathcal{D} = \{D_v : v \in V\}$ of a planar graph $G = (V, E)$, connect the centers of mutually tangent disks with line segments. Then the line segments never cross. Hence we have the following corollary.

Corollary 3.2 *Every planar graph can be realized in the plane such that each edge of it is a line segment.* □

Remark 3.3 The coin graph theorem was discovered by P. Koebe in 1936, and rediscovered again by E. Andreev in 1970 and by W. Thurston in 1985 (see [776]). This theorem is also called the circle-packing theorem of Koebe-Andreev-Thurston. In the book [610, Ch. 8] of J. Pach and P. K. Agarwal a direct proof of the coin graph theorem is presented.

3.6 The Theorem of Steinitz

We already noted in Corollary 3.1 that the graph of a convex polyhedron is a planar graph. Let Π be a convex polyhedron in $\mathbb{R}^3$. For a vertex v of Π, let $\alpha, \beta, \gamma, \ldots$ be the faces (facets) of Π (in clockwise order) around the vertex v. Then the boundary of the union of these faces becomes a cycle of the graph of Π, and the neighbors of v are vertices contained in this cycle. Hence, by Theorem 3.5, the graph of Π is 3-connected.

Theorem 3.8 *The graph of a convex polyhedron is a* 3*-connected planar graph.* □

The converse is also true, known as Steinitz' theorem.

Theorem 3.9 (Steinitz) *Every* 3*-connected planar graph can be realized as the* 1*-skeleton of a convex polyhedron in* $\mathbb{R}^3$.

Proof Let $G = (V, E)$ be a 3-connected *plane* graph. Then its radial graph $R(G)$ is an irreducible quadrangulation, and hence there is an OCR $\mathcal{F}$ of $R(G)$. Roughly speaking, the proof goes along the following line. By applying the inverse map of the stereographic projection, this OCR $\mathcal{F}$ is mapped to a system of circles on a sphere. Then the planes determined by the circles on the sphere corresponding to face-points of $R(G)$ enclose a convex polyhedron whose 1-skeleton is a realization of G.

Now the details. Suppose that an OCR $\mathcal{F}$ of $R(G)$ is given in the xy-plane in $\mathbb{R}^3$, and the special circle C_ω (which encloses all circles that do not cross C_ω) is centered at the origin $O = (0, 0, 0)$ with radius r. Put $P = (0, 0, r)$ and let $\mathbb{S}$ be the sphere with diameter OP. The sphere $\mathbb{S}$ is tangent to the xy-plane at O. Let φ denote the inverse mapping of the stereographic projection $\mathbb{S} \setminus \{P\} \to \{xy\text{-plane}\}$ from the point P. Then $\varphi(C_\omega)$ is the great circle (the equator) of $\mathbb{S}$, and for other circles $C \in \mathcal{F}$, its image $\varphi(C)$ is a small circle of $\mathbb{S}$. Indeed, if C is enclosed by C_ω, then $\varphi(C)$ is a small circle lying in the "southern hemisphere" of $\mathbb{S}$, and if C crosses C_ω orthogonally, then the center O lies outside of C, and hence $\varphi(C)$ is also a small circle on $\mathbb{S}$. Thus, $\{\varphi(C) : C \in \mathcal{F}, C \neq C_\omega\}$ is a family of small circles on $\mathbb{S}$, and for each $C \in \mathcal{F}$, no circle lies in the interior of the "smaller" cap bounded by C.

For each face-point α^* of $R(G)$, let π_α denote the plane determined by $\varphi(C_\alpha)$, where $C_\alpha \in \mathcal{F}$ is a circle corresponding to α^*. If the boundaries of the faces α, β have an edge uv in common, then, in $R(G)$, $\alpha^* u \beta^* v$ is a quadrilateral face, and hence the circles $\varphi(C_\alpha)$ and $\varphi(C_\beta)$ are tangent to each other. The intersection of the planes π_α and π_β is a common tangent line of the circles $\varphi(C_\alpha)$ and $\varphi(C_\beta)$, and also a tangent line of $\mathbb{S}$. Let $\alpha, \beta, \gamma, \ldots, \delta$ be the faces of G around the vertex v in clockwise order. Then the intersection lines of the pairs of planes (π_α, π_β), $(\pi_\beta, \pi_\gamma), \ldots, (\pi_\delta, \pi_\alpha)$ are all tangent lines of $\mathbb{S}$ orthogonally crossing the circle $\varphi(C_v)$. In this case, these tangent lines of $\mathbb{S}$ intersect at one and the same point $\hat{v}$, see Fig. 3.14. Therefore, the planes π_α for all faces α of G enclose the

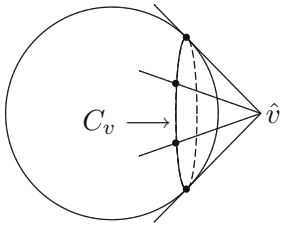


Fig. 3.14 Tangent lines of $\mathbb{S}$ orthogonal to a small circle

convex polyhedron with vertices $\{\hat{v} : v \in V\}$, and its 1-skeleton is a realization of G. □

Remark 3.4 Steinitz' theorem was obtained by E. Steinitz in 1922. B. Grünbaum [340] presented an improved proof of this theorem, but it is long and not easy. A comparatively easy proof is given by G. M. Ziegler [834], and it is also mentioned there that Steinitz' theorem can be proved using the OCR theorem. An elementary simple proof of the special case of Steinitz' theorem that states "every maximal planar graph with more than 3 vertices can be represented by a 1-skeleton of a convex polyhedron" was obtained by H. Maehara [518]. D. Barnette [61] gave an intuitive explanation of the following fact: Every 3-connected planar graph with maximal degree 3 can be realized by the 1-skeleton of a convex polyhedron.

3.7 Exercises

3.1 Prove the hand-shaking lemma.

3.2 A sphere is divided into several regions by $n \geq 3$ great circles. Assuming that no three great circles intersect at the same point, find the number of regions.

3.3 Prove that if a connected graph is not a complete graph then it has a cut set.

3.4 Prove that if a graph with $n(\geq 3)$ vertices has more than $\binom{n-1}{2}$ edges, then the graph is connected.

3.5* Prove that every quadrangulation has at least eight vertices.

3.6 Find an irreducible quadrangulation that has an odd number of vertices.

3.7* Let G be a 3-connected plane graph. For every edge e of G, we can draw a simple curve e^* that connects the face points in the both sides of e and crosses the edge e at a point. This curve e^* is called a *dual edge* of e. It is also possible that no two dual edges cross each other in the midway. The plane graph whose vertices are the face points of G and whose edges are the dual edges of the edges of G is called the *dual graph* of G, and it is denoted by G^*. Now, a question. Is it possible to re-draw G and G^* on the plane so that the following (i) and (ii) hold?

(i) All edges of G and G^* are line-segments or half lines.
(ii) Every edge of G and its dual edge cross each other orthogonally.

(We regard that the face-point w^* of the outermost face of G is lying at infinity, and the edges of G^* incident to w^* are half-lines.)

3.8 Notes

The notes to Chap. 3 here restrict to the following two important themes: results around the coin graph theorem and the famous theorem of Steinitz on polyhedral graphs.

3.8.1 Notes on the Coin Graph Theorem

A *unit disk graph* is the intersection graph of a family of unit disks in the plane. If the disks form a packing (i.e., they do not overlap), it is called a *unit coin graph* (or *penny graph*, or sometimes also *tangency graph*). An important observation in this direction goes back to Koebe [461] and is named today the *Koebe-Andreev-Thurston theorem* or *coin graph theorem* (see Koebe [461] and Thurston [776]). Namely, the vertices of any *planar graph* can be represented as midpoints of nonoverlapping circular disks in the plane such that any two of them are tangent to each other if the corresponding vertices are joined by an edge in the graph. And for triangulated planar graphs this representation is unique, up to conformal transformations of the plane (there are various further cases of uniqueness). Thus, connecting the centers of the disks via segments in the touching case, a straight-line drawing of the considered graph is established.

Book Parts and Surveys

Various problems and results on unit coin graphs can be found in several problem books or collections, see, e.g., Hartsfield and Ringel [381, Problems 8.4.6, 8.4.7 and 8.4.8] or Brass, Moser and Pach [125, Sections 5.7 and 9.2]. In connection also with Steinitz's theorem, discussions and applications of the Koebe-Andreev-Thurston theorem and its extension due to Brightwell and Scheinerman [128] are given in monographs by Pach and Agarwal [610, Chapter 13], Ziegler [834, Notes to Lecture 4], Matoušek [560, Section 5.3], Grünbaum [340, Section 13.7], Felsner [263, Section 2.7], Gruber [336, Sections 15 and 34], and Lovász [512, Chapter 5].

A closely related little survey is due to Sachs, see [683]. He discusses several known generalizations of the Koebe-Andreev-Thurston theorem and related results as well as interesting interconnections between them. Pisanski and Randić [637] survey (in a beautifully illustrated way) interesting relations between graph theory and geometry. In Sections 1 and 2 there, properties of and relations between coin graphs and polyhedral graphs (in view of Steinitz's theorem) are presented, like

also Koebe's result. And in the survey [399], Hlinený and Kratochvíl give an overview to recognition-complexity results for disk graphs and coin graphs. Koebe's theorem and its consequences are one main theme. They also show several results on coin graphs in higher dimensions. Among other things, it is proved here that the recognition of unit-ball coin graphs is NP-hard in dimensions 3, 4, 8, and 24. The paper contains interesting constructions and a helpful bibliography.

Some Research Papers

The Koebe-Andreev-Thurston theorem, unit coin graphs and extensions thereof have various further properties and applications, e.g. with respect to distance problems in finite point sets, geometric graphs and polytope theory. We give here some selected papers presenting such investigations.

Schramm [711] proves that any 3-connected planar graph even represents a 3-polytope whose edges are all tangent to the unit sphere such that the origin is the barycenter of the contact points. Sachs [683] presents several known generalizations of the Koebe-Andreev-Thurston theorem. Also it is clarified that Schramm's result remains true if the unit sphere is replaced by the boundary of any smooth, strictly convex body.

It is known that to find a *maximum independent set* (i.e., an independent set that is not a subset of any other independent set) in a unit disk graph is NP-hard. In [150], Cerioli et al. extend this result to penny graphs. Additionally they prove that to find a minimum clique partition in a penny graph is NP-hard, too. They also offer linear-time approximation algorithms for the computation of clique partitions, one for unit disk graphs and one for penny graphs.

Eppstein [244] studies *triangle free penny graphs*. Squaregraphs are the graphs that can be embedded in the plane such that each bounded face is a 4-cycle, and every vertex not on the unbounded face has degree at least 4. The author first discusses results about these graphs and their relations to other families of planar graphs, and he derives then deeper results on them. These refer to degeneracy, diameter, and upper bounds on edge numbers.

Colin de Verdière introduces in [173] a new *spectral invariant of graphs* which is shown to be monotonic regarding edge removal and an operation having homeomorphic reduction as a subcase. It is essential for planarity of graphs. Particularly, it turns out that this invariant is at most 3 iff the respective graph is planar. Alfakih [21] studies this invariant for penny graphs. He proves lower bounds on the graph invariant introduced in [173] when the complement of the considered graph is a penny graph. And Mitchell [577] extends results of Alfakih [21] getting insights that apply to complements of contact graphs of unit spheres in arbitrary dimensions.

3.8.2 Notes on the Theorem of Steinitz

Steinitz's theorem belongs to the interesting little field of polytopal graphs, and there particularly to the three-dimensional case. For this whole field we refer to the following general references: the nice book [61] of Barnette and the excellent survey [339] of Grünbaum. The *Steinitz problem*, namely to find necessary and sufficient conditions for a cell-composed sphere to be isomorphic to the boundary complex of a (higher dimensional) polytope (or to find a complete intrinsic characterization for boundary complexes of convex polytopes, or to give a characterization of the polytopal $(n-1)$-spheres among all $(n-1)$-spheres) is in general open. But for convex three-dimensional polytopes (shortly called *3-polytopes*), Steinitz' theorem holds. It says that *a given graph is the graph of a 3-polytope if and only if it is simple, planar, and 3-connected*. Due to this result, the graphs with these properties are usually called *polyhedral graphs*.

Book Chapters

For the history of his famous theorem we refer to Steinitz himself (see [758] and [759]), Sachs [683], Ziegler [834, Lecture 4 and its notes], and Grünbaum [340, Chapter 13]; Section 13.7 from [340] should be particularly mentioned since there different proofs, extensions, and analogues until 2003 are cited.

Extensive representations (with full proofs or explanations of proof methods and related questions) are given by Grünbaum [340, Chapter 13], Ziegler [834, Lecture 4 and its notes], Richter-Gebert [666], Matoušek [560, Section 5.3], Felsner [263, Chapters 1 and 2], Gruber [336, Chapter 15], and Lovász [512, Chapters 2 and 3]. Also in the book [251, Chapter III] of Ewald, strongly related to algebraic geometry, the Steinitz problem is taken care of.

In all these references further topics and notions are introduced or applied, since they efficiently help in proofs or for extending the field. Because we do not go into details here, we mention only some of them. All of them are well-studied mathematical tools or notions (and therefore easy to find). We mean here Schlegel diagrams, the stereographic projection, Möbius transformations, semialgebraic sets, realization spaces, the Koebe-Andreev-Thurston circle-packing theorem and the Koebe-Brightwell-Scheinerman theorem (the latter used, e.g., by Gruber for his proof of the Steinitz theorem given in [336, Section 15.3]).

Surveys

Besides in books, Steintz's theorem is also discussed in several surveys. In his exposition [339] on *polytopal graphs* (but still mainly treating the three-dimensional case), Grünbaum excellently presents typical proof methods, types of related results, and open questions around polyhedral and polytopal graphs. A large variety of

problems related to the combinatorial boundary structure of polytopes is surveyed in the excellent exposition [252] of Ewald et al.; in its second section partial results regarding the Steinitz problem for $n \geq 4$ (e.g., for simple complexes or combinatorial spheres with few vertices) are presented. In Section 5.2 of the handbook article [68], Bayer and Lee present, among other things, Balinski's result (that the graph of any n-polytope is still n-connected) and discuss also other connectivity criteria. And in Sect. 6.3, numbers of combinatorial types of 3-polytopes are studied, also with the help of Steinitz' theorem. Bokowski's handbook article [99] gives a good survey of the results related to the Steinitz problem from the viewpoint of *oriented matroids*. In the first part of his handbook survey [437], treating 3-polytopes, Kalai discusses Steinitz's theorem together with the classical theorems of Kotzig, Eberhard, and Motzkin.

Some Research Results and Directions

One interesting research direction directly coming from Steinitz's theorem can be described as follows: Steinitz's theorem gives a completely combinatorial characterization of the face lattices of convex 3-polytopes, and already Steinitz himself observed that his approach implies that for any 3-polytope the set of all its realizations is a trivial topological set (i.e., *realization spaces* of 3-polytopes are contractible and therefore connected). Further results in this direction and for dimensions $n \geq 4$ showed that realization spaces of n-polytopes with $n+4$ vertices can have arbitrary homotopy type. Richter-Gebert investigates this and presents and uses also related basic results of Mnëv [580] in his book [666]. He studies the structure of realization spaces of polytopes in fixed dimension and proves a *universality theorem* for 4-polytopes. This theorem says that for any primary basic *semialgebraic set* V there is a 4-polytope whose realization space is stably equivalent to V. (Recall that a basic semialgebraic set is a set defined by polynomial equalities and polynomial inequalities, and that a *semialgebraic set* is a finite union of basic semialgebraic sets.)

There are many deep and surprising results around such problems. E.g., Barnette [62] constructs a certain complex of six triangles that, when embedded in a 3-sphere, prevents the polyhedrality of this sphere. (A triangulated 3-sphere is called *polyhedral* if it is isomorphic to the boundary complex of a 4-polytope.) Improving former bounds, Onn and Sturmfels [604] show that any 3-polytope with m vertices can be realized in Euclidean 3-space with all vertex coordinates being integers whose absolute value does not exceed m^{169m^3}. Ziegler [833] poses research problems related to Steinitz's theorem and treating 4-polytopes. He discusses known results, methods, constructions, and examples - the latter with detailed proofs.

Ewald and Schulz [253] present $(n-1)$-spheres which cannot be embedded in n-space as starshaped sets. Namely, they verify that for any $n \geq 4$ *non-starshaped* simplicial $(n-1)$-spheres in n-space exist that have $n+8$ vertices. E.g., they also ask whether this number $n+8$ is best possible and conjecture that every $(n-1)$-sphere

with $n + 4$ vertices has a starshaped or even convex embedding in n-space. They show the relations of their investigations to algebraic geometry (more precisely, to toric varieties).

Searching analogues and extensions of the classical Steinitz theorem, Takayama and Hibi [771] try to find combinatorial characterizations of graphs referring to *Cohen-Macaulay simplicial complexes*. They derive a necessary and sufficient condition for a finite simple graph to be the graph of a two-dimensional simplicial complex that is Cohen-Macaulay over an arbitrary field. They also present the example of a graph which cannot be the graph of any Cohen-Macaulay simplicial complex over a field of characteristic 2, but is the graph of a two-dimensional Cohen-Macaulay simplicial complex over a field of characteristic $\neq 2$. Almohammad, Lángi and Naszódi [22] extend Steinitz's theorem to ball polytopes in 3-space, i.e., to intersections of finitely many unit balls.

Belotti et al. [71] treat realization spaces of polytopes, in particular realizations of 3-polytopes whose edges are tangent to the unit sphere (such realization spaces, in terms of the vertex coordinates or the facets, are semialgebraic sets). They consider Koebe realizations (with edges tangent to the sphere) and Springborn realizations (Koebe realizations whose edge barycenter is the origin), see Koebe [461] and Springborn [753]. Based on the fact that Koebe and Springborn realizations admit descriptions as semialgebraic sets defined by *rational polynomials*, the authors study the minimal degrees of such polynomials.

Chapter 4
Spherical Geometry I



If I were again beginning my studies, I would follow the advice of Plato and start with mathematics.

— Galileo Galilei

In this chapter, we show that geodesic curves on the sphere are great circular arcs, and we verify that the cylindrical projection preserves the areas of figures. Girard's formula for the area of a spherical triangle is also proved. From the area formula for spherical polygons obtained by applying Girard's formula, Legendre's proof of Euler's polyhedral formula is derived. The theorem on inscribed angles is presented, and the notion of polar set is introduced.

4.1 Geodesic Segments

Let us denote by $\mathbb{S}(r)$, or simply by $\mathbb{S}$, the sphere of radius r centered at the origin $O = (0, 0, 0)$ in the 3-dimensional Euclidean space $\mathbb{R}^3$. Every circle on the sphere $\mathbb{S}$ is obtained as the section of $\mathbb{S}$ by a plane in $\mathbb{R}^3$. The section of $\mathbb{S}$ by a plane that passes through the origin O is a *great circle* on $\mathbb{S}$. A circle on $\mathbb{S}$ that is not a great circle is called a *small circle* on $\mathbb{S}$. Every circle on $\mathbb{S}$ divides $\mathbb{S}$ into two regions, and each such region is called a *spherical cap* or simply a *cap*. If the circle is a great circle, each region is called a *hemisphere* instead of a cap. For a point $X \in \mathbb{S}$, X^* denotes its *antipodal point*, that is, the intersection point of the line XO and $\mathbb{S}$ other than X. Clearly, $(X^*)^* = X$. For any two distinct points A, B on $\mathbb{S}$, there is a great circle passing through A, B. (If $A^* \neq B$, then there is unique plane that passes through A, B, O, and the great circle that passes through A, B is unique. On the other hand, if $A^* = B$, then there are infinitely many great circles that pass through A, B.) When this great circle is divided by A, B, the shorter arc (minor arc or semicircle) is simply denoted by AB. (Since the line segment connecting A, B is also denoted by AB, we need to judge from the context which one AB represents.)

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Birkhäuser Advanced Texts Basler Lehrbücher,
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Among the curves on $\mathbb{S}$ that connect A and B, the great circular arc AB is the shortest curve.

In general, among the curves on a smooth surface M in $\mathbb{R}^3$ that connect two given points $P, Q \in M$, the shortest one is called a *geodesic segment* connecting P and Q.

Similarly to a point on the surface of the Earth, a point on $\mathbb{S}$ can be represented by its longitude τ, $-\pi \le \tau \le \pi$ (where the meridian passing through $(r, 0, 0)$ has longitude 0; east longitude has $+$ sign, west longitude has $-$ sign) and its latitude θ, $-\pi/2 \le \theta \le \pi/2$ (where north latitude has $+$ sign, south latitude has $-$ sign). Thus, (x, y, z)-coordinates and (τ, θ) of a point on $\mathbb{S}$ are related by

$$x = r\cos\tau\cos\theta,\ \ y = r\sin\tau\cos\theta,\ \ z = r\sin\theta.$$

Let $A, B \in \mathbb{S}$ be two points, $A^* \neq B$. Then A, B determine a great circle uniquely. Let us show that among the curves on $\mathbb{S}$ that connect A and B, the great circular arc AB has the minimum length. We may suppose, by rotating $\mathbb{S}$ if necessary, that A, B have the same longitude. Let $\gamma(t), a \le t \le b$, be a curve on $\mathbb{S}$ that connects A and B:

$$\gamma(t) = (x(t), y(t), z(t))\ (a \le t \le b),\ \ \gamma(a) = A, \gamma(b) = B.$$

The length of the curve $\gamma(t)$ is given by

$$\int_a^b \sqrt{\dot{x}(t)^2 + \dot{y}(t)^2 + \dot{z}(t)^2}\, dt,$$

where $(\dot{\ })$ denotes the derivative by t. (Regarding t as the time parameter, $\dot{x}(t), \dot{y}(t), \dot{z}(t)$ are the velocity of the point $\gamma(t)$ in the direction of x, y, z-axes, respectively, and hence $\sqrt{\dot{x}(t)^2 + \dot{y}(t)^2 + \dot{z}(t)^2}$ is the magnitude of the velocity vector of $\gamma(t)$. Hence, by integrating it from $t = a$ to $t = b$, we have the distance that the point $\gamma(t)$ has moved, namely, the length of the curve $\gamma(t)$.) Let $\theta(t), \tau(t)$ be the latitude and the longitude of $\gamma(t)$. Then

$$\begin{aligned}
\dot{x}(t) &= -r\dot{\tau}(t)\sin\tau(t)\cos\theta(t) - r\dot{\theta}(t)\cos\tau(t)\sin\theta(t)\\
\dot{y}(t) &= r\dot{\tau}(t)\cos\tau(t)\cos\theta(t) - r\dot{\theta}(t)\sin\tau(t)\sin\theta(t)\\
\dot{z}(t) &= r\dot{\theta}(t)\cos\theta(t),
\end{aligned}$$

from which we get $\dot{x}(t)^2 + \dot{y}(t)^2 + \dot{z}(t)^2 = r^2\dot{\tau}(t)^2\cos^2\theta(t) + r^2\dot{\theta}(t)^2$. Therefore,

$$\begin{aligned}
\text{the length of } \gamma &= \int_a^b \sqrt{\dot{x}(t)^2 + \dot{y}(t)^2 + \dot{z}(t)^2}\, dt \ge r\int_a^b |\dot{\theta}(t)| dt\\
&\ge r\left|\int_a^b \dot{\theta}(t) dt\right| = r|\theta(b) - \theta(a)|.
\end{aligned}$$

Since we assumed that A and B have the same longitude, $r|\theta(b) - \theta(a)|$ is equal to the length of the great circular arc AB. Thus, the great circular arc AB has the minimum length among the curves on $\mathbb{S}$ connecting A and B.

Let us state this as a theorem.

Theorem 4.1 *For two points A, B on a sphere $\mathbb{S}(r)$, the great circular arc AB is a geodesic segment connecting A and B.* □

4.2 Cylindrical Projections

In 3-dimensional space $\mathbb{R}^3$, a plane or a line that is parallel to the xy-plane is simply called a *horizontal plane* or a *horizontal line*. Let Γ be the cylinder (the surface of revolution) obtained by rotating the line segment connecting $(r, 0, r)$ and $(r, 0, -r)$ around the z-axis. The sphere $\mathbb{S}(r)$ is inscribed in this cylinder Γ and tangent to it at the equator. Put

$$\check{\mathbb{S}} = \mathbb{S}(r) \setminus \{(0, 0, r), (0, 0, -r)\}.$$

For $X = (x, y, z) \in \check{\mathbb{S}}$, let $\varphi(X) \in \Gamma$ denote the intersection of the horizontal ray $\overrightarrow{Z_X X}$ and the cylinder Γ, where $Z_X = (0, 0, z)$. Two points X and $\varphi(X)$ have the same z-coordinate, and if X lies on the equator of $\mathbb{S}$, then $\varphi(X) = X$, see Fig. 4.1. The map

$$\check{\mathbb{S}} \ni X \mapsto \varphi(X) \in \Gamma$$

is called the *cylindrical projection*.

Remark 4.1 A point on $\check{\mathbb{S}}$ is represented by (τ, θ) using its longitude $\tau \in [-\pi, \pi]$ and its latitude $\theta \in [-\pi/2, \pi/2]$, and a point on Γ can be also represented by (τ, h) using its longitude τ and its height $h \in [-r, r]$. Then

$$\varphi(\tau, \theta) = (\tau, r \sin\theta) \in \Gamma.$$

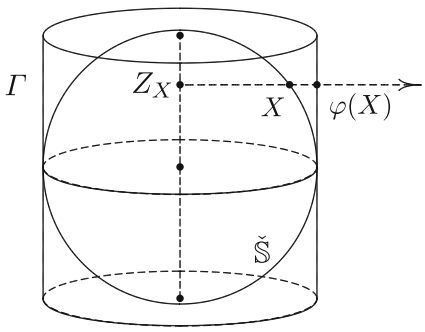


Fig. 4.1 The cylindrical projection

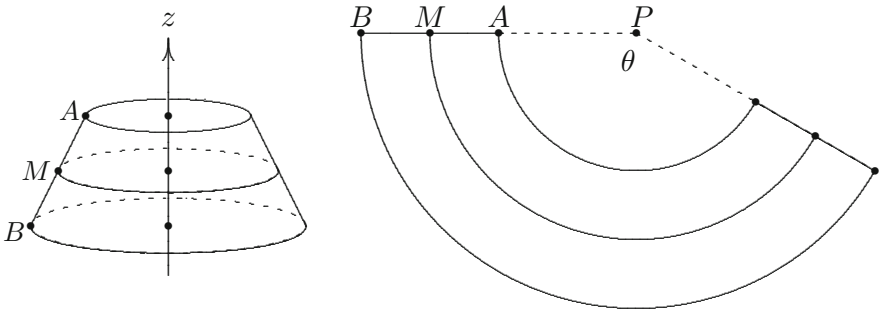


Fig. 4.2 The development of the surface of revolution

Theorem 4.2 (Archimedes-Lambert) *The cylindrical projection preserves the area, that is, for any region W on $\check{\mathbb{S}}$, W and $\varphi(W)$ have the same area.*

Remark 4.2 Archimedes proved that the area of $\mathbb{S}$ is equal to the area of Γ. J. H. Lambert generalized this as in Theorem 4.2. See, e.g., [255, 569].

To prove Theorem 4.2, we start with a lemma.

Lemma 4.1 *Let AB be a line segment of length d in the xz-plane which does not intersect the z-axis. By rotating AB around the z-axis, we have a surface of revolution (frustum of right cone), see Fig. 4.2 left. The area of this surface of revolution is equal to $d \times l$, where l is the length of the circle obtained as the locus of the midpoint M of AB.*

Proof If the line segment AB is parallel to the z-axis, then the surface of revolution is a circular cylinder, and the lemma is obvious in this case. Suppose that the ray $\overrightarrow{BA}$ intersects the z-axis at a point P. By cutting the surface of revolution along the line segment AB and flattening it into the plane, we have the right figure in Fig. 4.2. Let θ be the angle at the pivot P of the sector as shown in Fig. 4.2 right, and put $\rho = |PA|$, the length of the line segment PA. Since $|PM| = \rho + \frac{d}{2}$, we have $l = (\rho + \frac{d}{2})\theta$. Since $PB = \rho + d$, the area of this is

$$\left(\pi(\rho + d)^2 - \pi\rho^2)\right) \cdot \tfrac{\theta}{2\pi} = d(\rho + \tfrac{d}{2})\theta = d \times l.$$

□

Proof of Theorem 4.2 For a small $h > 0$ and θ with $|\theta| < \pi/2$, $|r \sin\theta| < r - h/2$, consider the zonal region U of $\check{\mathbb{S}}$ lying between the horizontal planes $z = r\sin\theta \pm \frac{h}{2}$. If h is sufficiently small, then U can be approximated by the lateral face of a frustum of a right cone. Hence, by Lemma 4.1, the area of U is approximated by the product of the length of the "center curve" and the width of the zone. The length l of the center curve is $2\pi r\cos\theta$. Consulting Fig. 4.3, we can see that the width d of the zone U satisfies $h/d = \cos\theta$. Hence $d = h/\cos\theta$, and therefore the area of U is

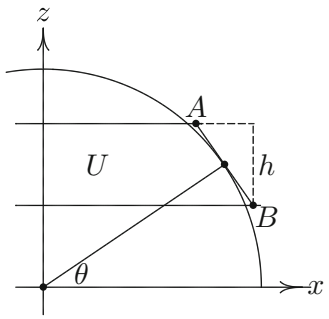


Fig. 4.3 A zone U and the lateral face of the frustum of a cone

approximately $d \times l = 2\pi rh$. On the other hand, the area of $\varphi(U)$ is clearly $2\pi rh$. Thus, the areas of U and of $\varphi(U)$ are (approximately) equal.

Now choose a large integer n, and divide $\check{\mathbb{S}}$ by the planes $z = \pm kr/n\,(k = 0, 1, 2, \ldots, n)$ into $2n$ zones. Furthermore, take $2n$ meridians with longitude $\pm\pi k/n, k = 0, 1, 2, \ldots n$, and divide each zone into $2n$ equal pieces. Then $\check{\mathbb{S}}$ is divided into $2n \times 2n$ pieces. The images of $2n$ meridians by φ are $2n$ generating lines of the cylinder Γ that divide Γ into $2n$ equal regions. Hence, each of the $4n^2$ pieces of $\check{\mathbb{S}}$ and its image by φ have the same area. For a region W of $\check{\mathbb{S}}$, its area can be approximated by the sum of the areas of small pieces of $\check{\mathbb{S}}$ that intersect W. And by increasing n, we can increase the accuracy of the approximation. Thus, we can deduce that the area of W is equal to the area of $\varphi(W)$. □

Corollary 4.1 (Archimedes) *The area of* $\mathbb{S}$ *is equal to* $4\pi r^2$. □

For a point P on $\mathbb{S}(r)$, the set

$$\{X \in \mathbb{S} : \angle POX \leq \theta\}$$

is called a *cap of angular radius* (or *spherical radius*) θ with *center* P. A cap with angular radius $\pi/2$ is a hemisphere. The center of a hemisphere is also called its *pole*.

Example 4.1 The area of a cap with angular radius θ on $\mathbb{S}(r)$ is equal to $2\pi r^2(1 - \cos\theta)$.

This can be seen as follows: We may suppose that the center of the cap is the north-pole $(0, 0, r)$. Then the cap is

$$\{(x, y, z) \in \mathbb{S} : z \geq r - r\cos\theta\}.$$

By the cylindrical projection φ, this cap is mapped to the part $z \geq r - r\cos\theta$ of Γ, which has area $2\pi r^2(1 - \cos\theta)$.

4.3 Spherical Polygons

A *simple polygonal curve* $A_0A_1A_2\ldots A_n$ on $\mathbb{S}(r)$ is defined to be a curve consisting of geodesic segments A_iA_{i+1}, $i = 0, 1, 2, \ldots, n-1$, each of length less than πr, such that

(i) A_i, $i = 1, 2, \ldots, n$, are all different,
(ii) two distinct segments are disjoint unless they share a common endpoint,
(iii) no two consecutive segments lie on the same great circle.

Each segment A_iA_{i+1} is called an *edge*, and each A_i is called a *vertex* of the polygonal curve. If $A_0 = A_n$, then the simple polygonal curve is called a simple *closed* polygonal curve. A *spherical polygon* is a region on $\mathbb{S}(r)$ bounded by a simple closed polygonal curve. The vertices and edges of the bounding polygonal curve are called the vertices and edges of the spherical polygon. Since a simple closed polygonal curve divides $\mathbb{S}(r)$ into two regions we need, for specifying a spherical polygon bounded by the polygonal curve, to choose one of the regions as the spherical polygon. If a simple closed polygonal curve lies in a hemisphere, then the region in the hemisphere is chosen as the spherical polygon bounded by the polygonal curve. When a spherical polygon has n edges, then it is called a *spherical n-gon*. If $n = 3$, then it is called a *spherical triangle.* Since a simple polygonal curve with three vertices always lies in a hemisphere by the condition (iii) of a simple polygonal curve, a spherical triangle always lies in a hemisphere. A spherical triangle with three vertices A, B, C is denoted by $\breve{\triangle}ABC$. The interior angle of a spherical polygon at a vertex is defined to be the plane angle formed by two tangent lines of the edges emanating from the vertex. The interior angle of $\breve{\triangle}ABC$ at A is denoted by $\angle CAB$ or $\angle A$.

A subset X of $\mathbb{S}$ is called *convex* if X lies in a hemisphere, and for any two points in X, the geodesic segment connecting the two points always lies in X. A convex set contains no pair of mutually antipodal points. By the condition (iii) of the polygonal curve, the vertices of a spherical triangle all lie on a small circle, and a spherical triangle is convex. Note that if a spherical polygon is convex, then for every vertex, the interior angle of the spherical polygon at the vertex is less than π.

If the intersection of a finite number of hemispheres has positive area (that is, it has interior points) and contains no pair of mutually antipodal points, then the intersection is a region bounded by a closed polygonal curve. Hence, it is a spherical polygon, and it is convex.

Lemma 4.2 *Let Γ be a convex spherical n-gon on $\mathbb{S}(r)$, $n \geq 3$. For an edge ϵ of Γ, let $G(\epsilon)$ denote the great circle determined by the edge ϵ. Then Γ lies in one side of $G(\epsilon)$.*

Proof Let $\epsilon = AB$, where A, B are vertices of Γ. Suppose that Γ contains two points X, Y that lie in the interiors of different sides of $G(\epsilon)$. Then, since Γ is convex, $\breve{\triangle}ABX$ and $\breve{\triangle}ABY$ are both contained in Γ. Hence the edge $\epsilon = AB$ does not lie on the boundary of Γ, a contradiction. □

Thus, every edge ϵ_i of Γ determines a hemisphere $H(\epsilon_i)$ that is bounded by $G(\epsilon_i)$ and contains Γ. For an exterior point P of a convex spherical polygon Γ, there is a geodesic segment PQ connecting P to an interior point Q of Γ. Then PQ crosses some edge ϵ_j of Γ, hence P and Q lie in different sides of the great circle $G(\epsilon_j)$, and hence $P \notin H(\epsilon_j)$. Thus, we have the next corollary.

Corollary 4.2 *Let Γ be a convex spherical n-gon with edges $\epsilon_1, \epsilon_2, \ldots, \epsilon_n$, and let $H_i := H(\epsilon_i)$ be the hemisphere bounded by the great circle determined by the edge ϵ_i and containing Γ. Then $\Gamma = H_1 \cap H_2 \cap \cdots \cap H_n$.* □

The intersection of two hemispheres is called a *lune*. The two great semicircles in the boundary of a lune are called the edges of the lune, and their (mutually antipodal) end-points are called the vertices of the lune.

Remark 4.3 The tangent line of an edge emanating from a vertex P of a spherical polygon on $\mathbb{S}(r)$ lies on the tangent plane of $\mathbb{S}(r)$ at P. So, if several spherical polygons with a vertex P in common tessellate together a neighborhood of P without overlapping, then the sum of the interior angles of these spherical polygons at P is equal to 2π.

In a lune L, the two interior angles at its vertices are equal. The area of a lune with interior angle θ is equal to $\frac{\theta}{2\pi} \times$(the area of the sphere). Thus, the area of a lune with interior angle θ is equal to $2\theta r^2$.

In the following, we take the radius of the sphere as the unit of distance, and regard the sphere $\mathbb{S}$ as a *unit sphere*.

Theorem 4.3 (Girard's Formula) *Let α, β, γ denote the interior angle of a spherical triangle $\breve{\triangle}ABC$ on a unit sphere $\mathbb{S}$. Then the area of $\breve{\triangle}ABC$ is equal to*

$$\alpha + \beta + \gamma - \pi.$$

Remark 4.4 $\alpha + \beta + \gamma - \pi$ is called the *spherical excess* of $\breve{\triangle}ABC$, and it is denoted by 2ε:

$$2\varepsilon = \alpha + \beta + \gamma - \pi.$$

Proof of Theorem 4.3 Denote the areas of $\breve{\triangle}ABC$, $\breve{\triangle}CBA^*$, $\breve{\triangle}CA^*B^*$, $\breve{\triangle}CB^*A$ by Δ, x, y, z, respectively (see Fig. 4.4). Since $\breve{\triangle}CA^*B^*$ and $\breve{\triangle}C^*AB$ are symmetric with respect to the center of $\mathbb{S}$, $\Delta + y$ is equal to the area of the lune CAC^*B. Hence $\Delta + y = 2\gamma$. Since $\Delta + x = 2\alpha$, $\Delta + z = 2\beta$, we have

$$(\Delta + x) + (\Delta + y) + (\Delta + z) = 2(\alpha + \beta + \gamma).$$

Since the left-hand side is $2\Delta + (\Delta + x + y + z) = 2\Delta + 2\pi$, we have $2\Delta = 2(\alpha + \beta + \gamma) - 2\pi$, and the formula of the theorem follows. □

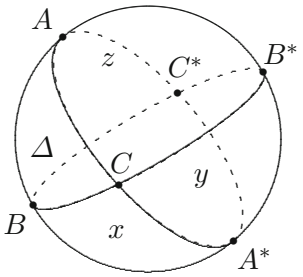


Fig. 4.4 The areas of four spherical triangles Δ, x, y, z

Since a convex spherical n-gon can be partitioned into $n-2$ spherical triangles by $n-3$ diagonals, we have the following corollary.

Corollary 4.3 *For $n > 3$, the area of a convex spherical n-gon on the unit sphere $\mathbb{S}$ is given by*

$$[\textit{the sum of the interior angles}] - (n-2)\pi.$$

□

Let us present here Legendre's proof of Euler's polyhedral formula. A polyhedron in $\mathbb{R}^3$ is called *convex* if for any two points of the polyhedron, the line segment connecting the two points belongs to the polyhedron.

Theorem 4.4 (Euler's Polyhedral Formula) *Let v, e, f denote the numbers of vertices, edges and faces, respectively, of a convex polyhedron Π in $\mathbb{R}^3$. Then $v - e + f = 2$ holds.*

Proof Take a point O in the interior of the convex polyhedron Π, and let $\mathbb{S}$ be the unit sphere centered at O. Consider the central projection of $\mathbb{R}^3 \setminus \{O\}$ from O onto $\mathbb{S}$ (i.e., the map that sends $O \neq X \in \mathbb{R}^3$ to the intersection point of the ray $\overrightarrow{OX}$ and $\mathbb{S}$). The images of the vertices, edges and faces of Π by this central projection constitute together a partition of $\mathbb{S}$ into f convex spherical polygons $\Gamma_1, \Gamma_2, \ldots, \Gamma_f$. Let $a(\Gamma_i)$ denote the area of Γ_i and let e_i denote the number of edges of Γ_i. Then, by Corollary 4.3, we have

$$a(\Gamma_i) = [\text{the sum of interior angles of } \Gamma_i] - (e_i - 2)\pi.$$

By Remark 4.3, we have $\sum_{i=1}^{f} [\text{the sum of interior angles of } \Gamma_i] = 2\pi v$. Note also that $\sum_{i=1}^{f} (e_i - 2)\pi = (2e - 2f)\pi$. Therefore

$$4\pi = \sum_{i=1}^{f} a(\Gamma_i) = 2\pi v - (2e - 2f)\pi,$$

from which $v - e + f = 2$ follows. □

Remark 4.5 There are many different proofs of Euler's polyhedral formula. (We already presented a proof in Chap. 3.) According to [131, Ch. 7, p.117], Legendre's proof seems to be the first one among the published rigorous proofs of Euler's polyhedral formula.

4.4 The Inscribed Angle Theorem

For a spherical triangle $\breve{\triangle}ABC$, three vertices lie on a small circle. The spherical cap bounded by this small circle and containing $\breve{\triangle}ABC$ is called the *circumscribed cap* of $\breve{\triangle}ABC$, and it is denoted by $cap(ABC)$.

Theorem 4.5 (The Inscribed Angle Theorem on the Sphere) *Let P be the center of the circumscribed cap of $\breve{\triangle}ABC$. Then*

$$\angle C - (\angle A + \angle B) = \pm 2\angle PBA,$$

where the sign $\pm$ takes $-$ if $\overset{\frown}{ACB}$ is a major arc, and $+$ if $\overset{\frown}{ACB}$ is a minor arc. (If $\overset{\frown}{ACB}$ is a semicircle, then $\angle BAP = 0$.) Therefore, for every point $X(\neq A, B)$ on $\overset{\frown}{ACB}$, we have

$$\angle AXB - (\angle XAB + \angle XBA) = constant.$$

Proof We consider the case (which looks most complicated) that $\overset{\frown}{ACB}$ is a major arc, and yet P does not lie in $\breve{\triangle}ABC$. Consulting Fig. 4.5, we have $\angle PAB = \angle PBA$, $\angle PBC = \angle PCB$, $\angle PAC = \angle PCA$, by symmetry. Hence

$$\begin{aligned}
\angle C &= \angle PCB - \angle PCA \\
&= \angle PBC - \angle PAC \\
\angle A &= \angle PAB - \angle PAC \\
&= \angle PBA - \angle PAC \\
\angle B &= \angle PBA + \angle PBC
\end{aligned}$$

From this it follows that $\angle C - (\angle A + \angle B) = -2\angle PBA$. □

Corollary 4.4 *In a spherical triangle $\breve{\triangle}ABC$,*

$\angle C < \angle A + \angle B \Leftrightarrow \overset{\frown}{ACB}$ *is a major arc,*
$\angle C = \angle A + \angle B \Leftrightarrow \overset{\frown}{ACB}$ *is a semicircle,*
$\angle C > \angle A + \angle B \Leftrightarrow \overset{\frown}{ACB}$ *is a minor arc.*

Since $\angle A + \angle B + \angle C > \pi$, $\angle C \leq \pi/2$ means that $\overset{\frown}{ACB}$ is a major arc. □

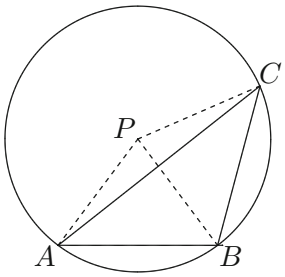


Fig. 4.5 The arc ACB is a major arc and $P \notin \breve{\triangle}ABC$

Lemma 4.3 *For a spherical triangle $\breve{\triangle}ABC$ on the unit sphere $\mathbb{S}$, we have*

$$\text{the area of } \breve{\triangle}ABC = \pi \;\Leftrightarrow\; \overset{\frown}{A^*CB^*} \text{ is a semicircle.}$$

Proof Let α, β, γ be the interior angles of $\breve{\triangle}ABC$ at A, B, C, respectively. Then the interior angles of $\breve{\triangle}A^*B^*C$ at A^*, B^*, C are $\pi - \alpha, \pi - \beta, \gamma$, respectively, and

$$\begin{aligned} \text{the area of } \breve{\triangle}ABC = \pi &\Leftrightarrow \alpha + \beta + \gamma - \pi = \pi \\ &\Leftrightarrow \gamma = (\pi - \alpha) + (\pi - \beta) \\ &\Leftrightarrow \overset{\frown}{A^*CB^*} \text{ is a semicircle.} \end{aligned}$$

□

Two spherical triangles on the unit sphere $\mathbb{S}$ are called (directly) *congruent* if one is obtained from the other by applying rotations around the center of $\mathbb{S}$.

Applying Lemma 4.3 we prove the following theorem, see also [129].

Theorem 4.6 *If the area of $\breve{\triangle}ABC$ is π, then the unit sphere $\mathbb{S}$ can be partitioned into four congruent copies of $\breve{\triangle}ABC$.*

Proof By Lemma 4.3, A^*B^* is a diameter of the circle A^*CB^*. Let D be a point on this circle A^*CB^* such that $A^*B^* = CD$. Then $\breve{\triangle}ABC$ and $\breve{\triangle}BAD$ are congruent. (Indeed, by the 180° rotation around the line through the midpoint M of AB and the midpoint M^* of A^*B^*, $\breve{\triangle}ABC$ becomes $\breve{\triangle}BAD$.) Similarly, since the area of $\breve{\triangle}ACB$ is π, A^*C^* is a diameter of the circle A^*BC^*, and BD is also a diameter of this circle. Hence, by 180° rotation around the line through the midpoint of AC and the midpoint of A^*C^*, $\breve{\triangle}ABC$ becomes $\breve{\triangle}CDA$. Thus four spherical triangles

$$\breve{\triangle}ABC,\ \breve{\triangle}ABD,\ \breve{\triangle}ACD,\ \breve{\triangle}BCD$$

are mutually congruent, and $\mathbb{S}$ is partitioned into these four spherical triangles. □

4.5 The Polar Set

For a point $P \in \mathbb{S}$, let $H(P)$ denote the hemisphere with pole P, that is,

$$H(P) = \{Y \in \mathbb{S} : \angle YOP \leq \pi/2\},$$

where O is the center of $\mathbb{S}$. For a nonempty subset $\mathcal{X}$ of $\mathbb{S}$, the set

$$\mathcal{X}^\circ = \bigcap_{X \in \mathcal{X}} H(X)$$

is called the *polar set* of $\mathcal{X}$. For example, $\{A, B, C\}^\circ = H(A) \cap H(B) \cap H(C)$. The polar set $(\mathcal{X}^\circ)^\circ$ of $\mathcal{X}^\circ$ is simply denoted by $\mathcal{X}^{\circ\circ}$.

Lemma 4.4 *For a geodesic segment PQ of length less than π,*

$$(PQ)^\circ = H(P) \cap H(Q).$$

Proof Obviously, $(PQ)^\circ \subset \{P, Q\}^\circ = H(P) \cap H(Q)$. If $X \in H(P) \cap H(Q)$, then $XP \leq \pi/2$, $XQ \leq \pi/2$. Among the points on PQ, P or Q is farthest from X. Hence for every point $Y \in PQ$ we have $XY \leq \pi/2$, and hence $X \in H(Y)$. Thus, $X \in H(P) \cap H(Q)$ implies that $X \in H(Y)$ for every $Y \in PQ$, and hence $X \in (PQ)^\circ$. Therefore, $H(P) \cap H(Q) \subset (PQ)^\circ$. □

Theorem 4.7 *The operation to take the polar set $(\ )^\circ$ satisfies the following four conditions:*

(i) *For any family $\{\mathcal{X}_i : i \in J\}$ of subsets of $\mathbb{S}$,*

$$\Big(\bigcup_{i \in J} \mathcal{X}_i\Big)^\circ = \bigcap_{i \in J} \mathcal{X}_i^\circ.$$

(ii) $\mathcal{X} \subset \mathcal{Y} \Rightarrow \mathcal{X}^\circ \supset \mathcal{Y}^\circ$.
(iii) $\mathcal{X} \subset \mathcal{X}^{\circ\circ}$.
(iv) $\mathcal{X}^\circ = \mathcal{X}^{\circ\circ\circ}$.

Proof

(i) is obvious from the definition of the operation $(\ )^\circ$.
(ii) Since $\mathcal{Y} = \mathcal{X} \cup (\mathcal{Y} \setminus \mathcal{X})$, we have $\mathcal{Y}^\circ = \mathcal{X}^\circ \cap (\mathcal{Y} \setminus \mathcal{X})^\circ$ by (i). Hence $\mathcal{Y}^\circ \subset \mathcal{X}^\circ$.
(iii) $P \in \mathcal{X} \Rightarrow \mathcal{X}^\circ \subset H(P) \Rightarrow$ for every $X \in \mathcal{X}^\circ$, $XP \leq \pi/2$. Therefore,

$$P \in \bigcap_{X \in \mathcal{X}^\circ} H(X) = \mathcal{X}^{\circ\circ}.$$

(iv) By (iii), we have $\mathcal{X}^\circ \subset (\mathcal{X}^\circ)^{\circ\circ}$. From $\mathcal{X} \subset \mathcal{X}^{\circ\circ}$ and (ii), we have $\mathcal{X}^{\circ\circ\circ} \subset \mathcal{X}^\circ$. □

Lemma 4.5 *Let $\Lambda(P,\theta) = \{X \in \mathbb{S} : \angle XOP \le \theta\}$, the cap of angular radius θ and center P. If $0 < \theta < \pi/2$, then $\Lambda(P,\theta)^\circ = \Lambda(P, \pi/2 - \theta)$.*

Proof If $X \in \Lambda(P,\theta)$, $Y \in \Lambda(P, \pi/2-\theta)$, then

$$XY \le XP + YP \le \pi/2.$$

Hence, $\Lambda(P, \pi/2-\theta) \subset H(X)$, and hence $\Lambda(P, \pi/2-\theta) \subset \Lambda(P,\theta)^\circ$. On the other hand, if $Z \in \mathbb{S}$ is not contained in $\Lambda(P, \pi/2-\theta)$, then $ZP > \pi/2-\theta$. Hence $ZP + \theta > \pi/2$. Then there is a point $X \in \Lambda(P,\theta)$ that lies on the extension of ZP beyond P and satisfies $ZX > \pi/2$. This implies that $Z \notin H(X)$, and hence $Z \notin \Lambda(P,\theta)^\circ$. Thus, if $Z \in \mathbb{S}$ is not contained in $\Lambda(P, \pi/2-\theta)$, then $Z \notin \Lambda(P,\theta)^\circ$. Therefore, $\Lambda(P,\theta)^\circ = \Lambda(P, \pi/2-\theta)$. □

Theorem 4.8 *Every convex spherical n-gon $\Gamma = P_1P_2\dots P_n$, $n \ge 3$, on $\mathbb{S}$ has the following three properties:*

(1) $\Gamma^{\circ\circ} = \Gamma$,
(2) $\Gamma^\circ = H(P_1) \cap H(P_2) \cap \cdots \cap H(P_n)$,
(3) Γ° *is a convex spherical n-gon.*

Proof

(1) By Corollary 4.2, Γ is the intersection of n hemispheres. Let $\mathcal{X}$ denote the set of poles of these n hemispheres. Then $\Gamma = \mathcal{X}^\circ$. Hence $\Gamma^{\circ\circ} = \mathcal{X}^{\circ\circ\circ} = \mathcal{X}^\circ = \Gamma$.
(2) Denote the boundary of Γ by $\partial\Gamma$. Since Γ is convex, it can be represented as

$$\Gamma = \bigcup_{X \in \partial\Gamma} (P_1X).$$

Then, by Theorem 4.7 (i) and Lemma 4.5, we have

$$\Gamma^\circ = \bigcap_{X\in\partial\Gamma} (P_1X)^\circ = \bigcap_{X\in\partial\Gamma} (H(P_1) \cap H(X)) = \bigcap_{X\in\partial\Gamma} H(X) = (\partial\Gamma)^\circ.$$

Since $\partial\Gamma = (P_1P_2) \cup (P_2P_3) \cup \cdots \cup (P_nP_1)$,

$$\Gamma^\circ = (\partial\Gamma)^\circ = (P_1P_2)^\circ \cap (P_2P_3)^\circ \cap \cdots \cap (P_nP_1)^\circ = H(P_1) \cap \cdots \cap H(P_n).$$

(3) Since Γ is contained in the interior of some hemisphere, there is a cap of angular radius $\pi/2 - \varepsilon$ (with sufficiently small $\varepsilon > 0$) that contains Γ. By Lemma 4.5, the polar set of a cap with angular radius $\pi/2-\varepsilon$ is a cap of angular radius ε, and we can deduce from Theorem 4.7(ii) that Γ° contains a cap of angular radius ε. Hence Γ° has positive area, and hence Γ° is a convex spherical polygon. If Γ° is a convex m-gon, then $m \le n$ by (2). Applying the same argument to $\Gamma = (\Gamma^\circ)^\circ$, we have $n \le m$. Hence $m = n$. □

By Theorem 4.8 (2), the polar set of a spherical triangle $\breve{\triangle}ABC$ is a spherical triangle $\breve{\triangle}A'B'C'$, where A' is a vertex not lying on the boundary of $H(A)$, B' is a vertex not lying on the boundary of $H(B)$, etc. This $\breve{\triangle}A'B'C'$ is called the *polar triangle* of $\breve{\triangle}ABC$. In other words, if $A', B', C' \in \mathbb{S}$ satisfy

$$(*)\begin{cases} AA' < \pi/2, & BA' = CA' = \pi/2, \\ BB' < \pi/2, & AB' = CB' = \pi/2, \\ CC' < \pi/2, & AC' = BC' = \pi/2, \end{cases}$$

then $\breve{\triangle}A'B'C'$ is called the polar triangle of $\breve{\triangle}ABC$. In this case, $\breve{\triangle}ABC$ is the polar triangle of $\breve{\triangle}A'B'C'$.

Theorem 4.9 *Suppose that $\breve{\triangle}ABC$ and $\breve{\triangle}A'B'C'$ are polar triangles of each other. Let a', b', c' be the lengths of the edges of $\breve{\triangle}A'B'C'$ opposite to the vertices A', B', C', respectively. Then we have*

$$\angle A + a' = \pi, \angle B + b' = \pi, \angle C + c' = \pi.$$

Proof Consulting Fig. 4.6, we see that $\angle B'AC = \pi/2 = \angle BAC'$. Hence

$$\angle B'AC' + \angle BAC = \angle B'AC + \angle BAC' = \pi.$$

Since $B'A = C'A = \pi/2$, we have $\angle B'AC' = B'C' = a'$. Therefore, $a' + \angle A = \pi$. Similarly we have $b' + \angle B = \pi$, $c' + \angle C = \pi$. □

From Theorem 4.9 and Girard's formula, we have the following corollary.

Corollary 4.5 *The sum of the area of a spherical triangle and the perimeter of its polar triangle is equal to 2π.* □

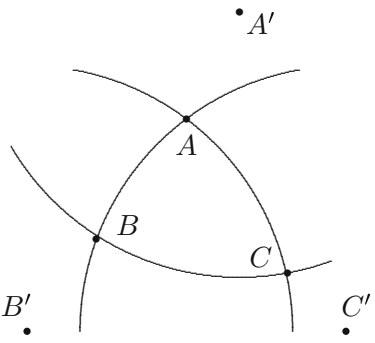


Fig. 4.6 $\breve{\triangle}ABC$ and its polar triangle $\breve{\triangle}A'B'C'$

4.6 Exercises

4.1 Let A, B be two points on the unit sphere $\mathbb{S}$ such that A lies at longitude 0, latitude 0, and B lies at longitude $\pi/4$, latitude $\pi/4$. Find the spherical distance AB.

4.2 The state of Alaska in US is approximated by a rectangular region with longitude between $140°W$ and $165°W$, latitude between $60°N$ and $70°N$. Find its area. (The radius of the Earth is approximately 6370 km.)

4.3 Let A, B, C, D be points on a small circle on $\mathbb{S}$ lying in this cyclic order. Prove that, in the spherical quadrilateral $ABCD$, the equality $\angle A + \angle C = \angle B + \angle D$ holds.

4.4* Prove that for a spherical triangle $\breve{\triangle}ABC$ on the unit sphere, the following two statements (i) and (ii) are equivalent:

(i) The area of a spherical triangle $\breve{\triangle}ABC$ is greater than π.

(ii) $\overset{\frown}{A^*CB^*}$ is a minor arc.

4.5* For $X \subset \mathbb{S}$ the intersection of all hemispheres that contain X is called the *convex hull* of X, and it is denoted by $conv(X)$. If there is no hemisphere that contains X, then we put $conv(X) = \mathbb{S}$. Prove that if $conv(X) \neq \mathbb{S}$, then

$$conv(X) = \bigcap_{P \in X^\circ} H(P) = X^{\circ\circ}.$$

4.6* (Rusty Compass) Using a rusty compass that can draw only circles of a fixed radius ρ, draw a circle on a big sphere. Then the area of the spherical cap bounded by the circle and containing the needle-point on the sphere is constant and independent of the radius of the big sphere. Prove this fact.

4.7 Notes

We start these notes with some thoughts about spherical geometry in general, also referring to its history. Then we treat, in a somewhat longer part, mathematical cartography and come, in particular, to the cylindrical projection. Finally, also in a longer part, we discuss the geometry of spherical polygons and triangles.

4.7.1 Notes on Spherical Geometry

In this subsection we will cite some general references on classical spherical geometry, meaning the *geometry of spherical spaces* in the spirit of non-Euclidean geometries. The reason for having chosen also this theme (spherical geometry) for

one notes part is the main headline of our Chaps. 4, 5, and 7. As the text and the reference list will show, the literature on classical spherical geometry is relatively scattered.

Busemann's more general investigations of types of geodesic metric spaces (yielding also characterizations of spherical spaces) will not be discussed in our book. For such concepts and their importance nowadays we refer to the monographs [136] and [617] of Busemann and Papadopoulos, respectively, and to the very recent exposition [620].

We start with the *history* of the field. Rosenfeld dedicates the whole first chapter of his monograph [676] to spherical geometry, clarifying the contributions of Menelaus, Ptolemy, the Arabic period, Regiomontanus, Vièta, Copernicus, Girard, Euler, and many others.

Also mainly historical in nature are the comprehensive papers [616] and [618] of Papadopoulos, in which the *contributions of Euler* and his academic students to spherical geometry are presented. These refer to spherical trigonometry, various spherical analogues of and differences to Euclidean results, and mathematical cartography (for the latter see below).

Regarding *applications* we should mention here the detailed monograph [743] of Smart from 1931, treating *spherical astronomy* and containing all needed tools and notions from spherical geometry.

In Part III of his classical book [96], Blumenthal discusses the *distance geometry* (also) in spherical n-space with the help of determinants which generalize those of Cayley-Menger. Some spherical Helly-type theorems are studied, like also results on congruence in hemispheres and small caps.

Also the book [428] of Jennings should be cited here. Its second chapter refers completely to the geometry of spherical spaces. The author treats *geodesics* as arcs of great circles and as distance minimizing curves. Fundamental results of *spherical trigonometry and the laws of sine and cosine for sides and angles* are treated. The relation between area and the *angular defect* is deduced, and interesting applications of all this and more material are given. These refer to navigation, map making and a discussion of conformal mappings.

Also Hartshorne [382, pp. 318 and 458–459] discusses topics from spherical geometry and, in particular, properties of *spherical polygons*. Prasolov and Tikhomirov discuss spherical geometry in their monograph [646, Sections 5.1 and 5.3], also in view of Riemannian geometry.

Similarly, the second chapter of Wilson's monograph [815] is dedicated to spherical spaces, giving *spherical sine and cosine formulas*, some related curve theory, finite groups of isometries, the Gauss-Bonnet theorem, and an extension to Möbius geometry.

Borceux treats *spherical triangles and spherical trigonometry* in Section 4.12 of his trilogy book [103].

The monograph [809] of Whittlesey is completely dedicated to the geometry of spherical spaces, introduced in an *axiomatic way* (several proofs are given) and showing also different approaches (one of them based on quaternions) and interesting applications. These applications refer e.g. to the surface of the earth,

the study of our planet system, three- and four-dimensional polyhedra, and also crystallography.

Melzak gives a thorough treatment of *spherical trigonometry* in Chapter 7 of his book [569], and Gowers [313] presents parts of spherical geometry and hyperbolic geometry (using a disc model) to give useful information on the parallel postulate.

Berger [75, Chapter 18] collects a lot of material on spherical geometry. He presents a comprehensive investigation of *spherical triangles and (convex) spherical polygons*, the *intrinsic metric of the sphere*, its *isometry group, spherical trigonometry*, and further topics. Also in his more recent book [78] he studies spherical geometry (e.g., its metric and spherical trigonometry) in Section III.1.

In Chapter 3 of his book [227], Dunajski presents (among other non-Euclidean geometries) basics of spherical geometry, e.g. studying mapmaking, Mercator's projection, and a derivation of the area formula for spherical triangles.

The book [131] of van Brummelen is the most comprehensive, recent contribution to *spherical trigonometry*. Starting with the representation of the contributions from the fruitful ancient, Arabic, Indian, medieval, and modern periods, the author gives a fascinating overview to needed or related tools, notions and applications (e.g., astronomy, regular polyhedra, stereographic projection, crystallography and the Cèsaro method). Also the book [643] of Popko and Kitrick should be mentioned here. It is not directly referring to spherical space, but contains a lot of related material (like trigonometric formulae, geodesics and loxodromes, applied topics like climate models etc.).

Lassak's survey [492] gives an overview on results in the spirit of *convex geometry*, but holding for spherical spaces and mainly depending on the *width function* of convex sets. They are related to quantities like *minimal width* and *diameter*, and therefore also *bodies of constant width, diametrically complete sets* and *reduced bodies* in such spaces are discussed. Also the papers [581], [290], [345], and [346] refer to typical convexity results in spherical spaces; see also page 242 in the book [558] of Martini, Montejano, and Oliversos.

4.7.2 Notes on Cylindrical Projections and Related Topics

In these notes we reflect results and references from the field of *mathematical cartography*, i.e., we treat *map projections* (which are not always real projections) to obtain planar images from parts of the earth's surface. Our interest here, in these notes, is mainly (but not only) directed to *cylindrical projections*.

The cylindrical projection introduced in our Sect. 4.2 is, more precisely, the so-called *cylindrical equal-area projection of Archimedes and Lambert*. Here it is our aim to give a short overview to references which contain informations to this and to related (mainly cylindrical and conical) cartographical maps. The topic of such "map projections" is classical, and therefore most of the references are of older date, but have sometimes also recent new editions. There are complete books treating only this area, or also book chapters and survey-like publications. The case of book

chapters refers often to parts of books on *differential geometry*, on *non-Euclidean (spherical) geometry* or on *classical geometry*; see below.

Complete Books

The classical books [398] of Hinks and [568] of Melluish present systematic studies of the whole field of map projections and have modern editions. In [398] numerous map projections are compared, where also informations on the relationships between used methods of projection and the qualities and disadvantages of each method are given. Since it is the author's intention to present a source for "map makers" and "map users", the practical point of view is central. Thus, too extensive mathematical theory is avoided, and many nice figures are presented. This book is interesting for readers specialized in *geography, the history of cartography* and the relations between different map projections. The book [568] addresses more the mathematical theories underlying the map constructions. It will be a good source of knowledge for eveybody interested in the *mathematical foundations of map projections*.

König and Weise [466] introduce complex vector coordinates of the spheroid such that *complex function theory* can be applied. These coordinates are the complex longitude, latitude and meridian arc length, and they lead to known projections (Mercator, Gauss-Krüger etc.). Then the pattern of *conformal projections* (including various cylindrical and conical ones) can be obtained by selected complex function transformations. These projections are presented for the geoid and for the sphere. Many further interesting methods and results are presented.

Also the comprehensive book [316] of Grafarend and Krumm goes much deeper into the *mathematical basics* needed for investigating a *broad variety of map projections*. The first four chapters treat (without detailed proofs) the basic facts from *differential geometry*. E.g., these are: mappings from Riemann manifolds to Riemann manifolds (mainly of dimension 2), the needed coordinates (or parameters), and surfaces of Gaussian curvature zero which are necessary for cartography. Concrete mappings follow, for instance sphere and ellipsoid-of-revolution to tangential planes, to cylinders, or to cones. For studying geodesic mappings, the Riemann, Soldner, and Fermi coordinates on the ellipsoid-of-revolution are presented, and various further related topics are given. The appendices offer, e.g., elliptic functions and elliptic integrals, for geodesics the geodesic curvature and torsion, mixed cylindric map projections, and Gauss surface normal coordinates. The authors give also interesting historical comments, and the bibliography contains 1387 references!

Clearly, *spherical harmonics* and related topics are close to the geometry of the sphere, but would open here several further doors which are not so close to the aims of this book. However, the monograph [277] of Freeden and Schreiner (giving a unified, impressive overview to the theory of spherical functions) should be considered as an exception (i.e., should be mentioned here), due to its closeness to

mathematical geosciences and mathematical geography. E.g., the behavior of such functions to be used in data analysis and geo-applications is taken care of.

Feeman's monograph [255] is an excellent and modern book on map projections. Topics presented (after giving basic facts on the earth) are: *basic spherical geometry*, the reasons why one cannot make a perfect flat map of the planet, different properties that maps can have, facts on how to design maps that have such desired properties, conformal and equal-area maps, and so on. The author quantitatively analyzes distortions that arise when making world maps, and he indicates how to generalize methods to produce maps of arbitrary surfaces of revolution. There is also a chapter that shows how to use Maple add-on software to get maps from geographic data points. The book would excellently complete and enrich courses in linear algebra, differential geometry, or calculus. On the other hand, it would excellently function as fundament of an undergraduate course in mathematics or a geography course, especially appealing to the teacher who might search exciting visual applications even for the classroom.

There are further books completely dedicated to map projections (and containing particular representations of cylindrical map projections). For example, we mention here Wagner [799], Urmaev [789], Fiala [265], Hoschek [404], and Schröder [712]. E.g., in [789] the author collects special applications of the differential equations of mathematical cartography together with some approximate and numerical methods of integration, and Hoschek's book also collects several "optimal drafts" (like that of Winkel etc.).

Book Chapters

Books in *differential geometry* with own chapters on map projections, in particular treating also cylindrical ones, are due to Gdowski [297], Klotzek [456, Section 3.10, Appendix], Wünsch [818, Section 7.4], and Trapp [781]. E.g., Gdowski clearly presents the fundaments of differential geometry needed in the fields of higher geodesy and cartography, and Trapp studies mathematical cartography based on *conformal and equiareal functions*.

Chapter 18.1 of Berger's monograph [75] presents various needed properties of the sphere and related basic projections (like the stereographic one), to show then a certain variety of different cartographical projections. A very detailed discussion of map projections is presented by Bigalke [87, Chapter 5], including also the already mentioned *draft of Winkel* (1913), which is one of the most popular map designs generated by a "mixture" of different methods.

Agricola and Friedrich wrote the nice book [10] on *elementary geometry*. Its last chapter is dedicated to spherical geometry, and for teaching purposes this book contains a large collection of interesting exercises and problems. Since also being close to elementary geometry, we mention here the book [462] of Koecher and Krieg; Chap. 4 presents a (today rarely occurring) collection of theorems on triangles and *their circles*.

Survey-Like and Historical Articles

Proskurowski [648] refers to cylindrical projections and Mercator's contributions; the author gives also an interesting overview regarding the Dutch cartographers of the sixteenth century.

Also very *famous mathematicians* contributed to mathematical cartography and related fields. For example, *Euler* published in 1777 three memoirs on mathematical cartography, namely on geographical maps in view of three criteria (one being conformality, and the others prescribing properties of images of meridians), and he also rediscovered Lambert's cylindrical equal-area projection (originally introduced in 1772). All this, including also valuable work of *Lagrange*, is nicely presented by Charitos [157]. Also Papadopoulos [619] deals with Euler's contributions to cartography, stimulated by the French geographer Delisle who introduced a useful method of drawing geographical maps. This method motivated Euler to write one of the mentioned memoirs.

The work of *Gauss* on *geodesy, topography and cartography*, distributed over different periods of his life, is discussed by Kautzleben in [444]. Via land measurement and triangulation, yet in the environment of Braunschweig, Gauss reduced data successfully via the method of *least squares*, and his interests in cartography were clearly influenced by his interests in *differential geometry*, and certainly vice versa.

Also *Chebyshev* contributed to mathematical cartography. Inspired by *Lagrange*, who developed differential-geometric formulas for *angle-preserving maps of the sphere to the plane*, he studied the following problem: for a region C given on the sphere, find the angle-presering map which minimizes the variation of distance-dilatation within that region. The solution, that he found himself, is a map for which the distance-dilatation is constant along the boundary of C. Chebyshev obtained also interesting approximation results regarding distances in that framework.

Greinke's paper [325] is remarkable: it contains an elementary approach to various conformal and area-preserving maps of the sphere onto certain surfaces, including also the plane. In its elementary style (that also Schröder's book [712] has) it is practically a complete teaching module for high schools.

Finally we mention that there are several "picture galleries of map projections" which can be visited via internet. Here is an excellent example, created by H. Havlicek (TU Vienna) and his research group:
https://www.geometrie.tuwien.ac.at/karto/

Here is the right place to mention also the recent Proceedings [138] edited by Caddeo and Papadopoulos on mathematical geography of the nineteenth century. They contain many articles (partially already mentioned here) with deep insights on many findings (mainly of Euler, Lagrange and Lambert) from this fruitful century that determine our world view still today.

4.7.3 Notes on Spherical Polygons

It turns out that the theory of *spherical polygons* (i.e., the knowledge about polygons defined for spherical spaces or as tiles of interesting spherical tilings) yields a particularly rich geometric topic. We select here references treating this notion, on this way also showing the large variety of related inspiring further problems.

Books, Book Parts, and Surveys

Strongly related to his two other books on highly symmetric polyhedra, Wenninger [808] nicely describes constructions of paper models of *spherical tilings* (i.e., spherical polygons play a natural role) connected with the symmetries of Platonic and Archimedean solids, with geodesic domes, honeycombs and so on. The author was strongly inspired by Coxeter.

The chapter on spherical geometry in Fenn's book [264] contains a detailed section on *geodesic triangles*, and another one on the celestial sphere.

Wilson [815] studies spherical geometry in Chap. 2, particularly also *sine and cosine formulae*, interesting related properties of curves and the Gauss-Bonnet theorem (including its Euclidean borderline case). Shortly we mention here also the book [809] of Wittlesey with an *axiomatic approach to spherical geometry*, see the notes above. And also Bigalke's book [87] contains interesting knowledge about spherical triangles and polygons.

In the monograph [643] of Popko and Kitrick several topics based on the geometry of the sphere are collected, among them *spherical subdivisions* (e.g., by great circles), packing problems and interesting point distributions on the sphere, *spherical polygons*, and various other concepts. Interesting application fields are presented, such as architecture ("Bucky's dome"), celestial catalogs, climate models, fishpharming, polyhedra with regularity properties, golf ball design and so on.

In the booklet [166] of Clemens on two-dimensional geometries a *problem-solving approach to planar geometries of constant curvature* is used. Part III treats *elementary spherical geometry*. E.g., the area of spherical triangles is studied, and a synthetic treatment of spherical geometry through the *angle-excess formula* is given. Part VI refers again to spherical geometry, but now from a more advanced point of view. *Spherical coordinates, shortest paths* between any two points, and also formulas for *spherical polygon areas* and the *sum of their interior angles* are discussed. Various further related topics can be found.

Berger [75, Chapter 18] discusses spherical triangles and (convex) spherical polygons in a broader way, and his later book [78, Chapter III] contains also various results on spherical geometry. Furthermore, we refer here to Hartshorne [382, Section 45 and its exercises]. Van Brummelen devotes the whole book [131] to spherical trigonometry.

Regarding *survey articles*, we mention Rosenfeld [674, §2 and §4] who mainly studies various properties of *spherical triangles*, like area, congruence, cosine theorem for sides and for angles, and also cotangent formulae. And we shortly mention the survey [492], regarding the type of studied questions mainly dedicated to convex geometry, but delivering various results also for polygons (see below for a better description).

Some Research Contributions

Here we present various research results on spherical polygons in some thematic order.

1. Spherical Tilings by Polygons

Grünbaum and Shephard [341] consider *spherical tilings with strong symmetry properties*. Namely, by central projection from the centroid the boundary structures of regular and uniform 3-polytopes can be projected onto a centered 2-sphere, to form there a spherical tiling whose transitivity properties come from the symmetry group of the projected polytopes. The authors classify all tiling types whose symmetry groups are transitive on the tiles (called isohedral tilings), on the edges (isotoxal tilings), or on the vertices (isogonal tilings). It should be noticed that these authors published various papers on the closely related notion of spherical patterns.

L. Fejes Tóth [259] discusses problems on spherical polygons in *spherical tilings* created by great circles, making also use of isoperimetric properties of regular spherical polygons. Florian [272] studies, for $A > 0$ and $P > 0$ with certain natural restrictions, the class of all compact, spherically convex sets in the 2-sphere having area at least A and perimeter at most P. He describes spherically convex sets closest to circumscribed [resp. inscribed] convex spherical m-gons; the distance considered here is measured in terms of area and perimeter deviation.

Types of spherical triangles which tile the sphere edge-to-edge are well-known, but relaxing the edge-to-edge demand, one gets a larger variety. As a first step in this direction, Dawson [200] characterizes all *isosceles triangles that tile the sphere*. He finds one infinite family and three sporadic tiles that tile only edge-to-edge. The same author together with Doyle published a series of papers on *right spherical triangles which can tile the sphere*. In the third and provisionally last part [201] they study the so-called asymptotically obtuse families of triangles, and they establish that six triangle types within such families can actually tile the sphere. The corresponding tilings are classified.

Gao et al. [291] study *edge-to-edge tilings* of the 2-sphere by twelve geometrically *congruent pentagonal tiles*. They first look at combinatorially congruent tiles, to investigate then all cases where the tiles have even the same edge length combination and arrangement, doing then the same with the cases where all the tiles have the same angle combination and arrangement. The classification of the spherical tilings in five classes, which require the edges of the tiles to be great arcs, is derived.

A *folding tiling* of the sphere is an edge-to-edge covering of it by spherical geodesic polygons without gaps or overlaps. Avelino and dos Santos [49] study dihedral folding tilings of the sphere by kites and isosceles triangles. Assuming that each tiling has at least two non-congruent tiles in some case of adjacency, they give geometric conditions on the tiles with consequences to local configurations. Finally presented in a table, this yields a characterization of a discrete family of all possible tilings. The authors continued these investigations, and Part IV of a series of related papers is [49].

Sakano and Akama [684] investigate edge-to-edge tilings of the 2-sphere by mutually *congruent polygons*. They first characterize all anisohedral spherical triangles (admitting tilings of the sphere such that the symmetry group of the tiling does not act transitively on the tiles). Further on, they give a complete classification of all tilings by kites (convex quadrangles whose cyclic list of edge-lengths is $aabb$ for $a \neq b$), darts (non-convex quadrangles whose cyclic list of edge-lengths is also $aabb$, $a \neq b$), and rhombi. These results are based on classification results on tilings of the 2-sphere by congruent triangles obtained by Ueno and Agaoka [786].

Breda and dos Santos [127] demonstrate how to generate new families of specific spherical tilings with the help of the free software *GeoGebra*. For this purpose some spherical geometry capabilities of *GeoGebra* had to be extended.

Lee et al. [498] present a *new sphere subdivision method* which generates a large number of spherical pentagons and is based on a successive subdivision of a module of a spherical dodecahedron. Using special design parameters, this method controls the shapes of generated spherical pentagons. The authors are able to divide a sphere into spherical pentagons of equal area and to minimize the number of different arc lengths used in the pentagonal structure. The efficiency of the new method is shown by various examples, and taking additional constraints on the optimization problem into consideration may deliver sphere subdivisions of interesting specific types.

Akama et al. [12] continue with a series of results and classify tilings of the sphere by congruent equilateral pentagons. They show that there are eight edge-to-edge tilings of the sphere by congruent equilateral pentagons, and that the number of participating tiles is between 12 and 60.

2. Isoperimetric-Type Results

Wimmer [816] confirms the following theorem also for the spherical plane: Among all quadrilaterals with given side-lengths there is one of greatest area, and for this one the sum of two opposite angles is the same as the sum of the two remaining angles. These quadrilaterals are *cyclic* and do not exceed the half-sphere (they consist of Eulerian triangles, i.e., triangles whose angles and edges are smaller than π).

Liu and Chang [511] prove various results of *isoperimetric type* for spherical m-gons, $m \geq 3$. E.g., for all spherical convex k-gons of fixed diameter $\pi/2$ with $3 \leq k \leq m$ and odd m, the regular spherical triangle has maximal perimeter, and the regular spherical m-gon has maximal area. For the same family, but of fixed minimal width $\pi/2$, the regular spherical m-gon attains the minimum perimeter,

and the regular triangle has minimal area. Some further results (e.g., also on reduced spherical polygons) are obtained.

Also Maehara and Martini [537] derive several inequalities for spherical polygons of isoperimetric type, and they use stereographic projection. E.g., they show that among spherical polygons with fixed side-lengths and perimeter less than 2π, a cyclic polygon has the largest area, and that among spherical k-gons inscribed in a fixed spherical cap, spherical regular k-gons have the largest perimeter. Also some of the following results on cyclic or regular spherical polygons are of isoperimetric type.

3. Cyclic and Regular Spherical Polygons

We say that K, a compact set in the n-dimensional sphere ($n \geq 2$), has the *Pompeiu property* if only the zero function has zero surface integral over every rotated copy of K. There exist regular spherical polygons bounded by arcs of great circles and not having the Pompeiu property. In [40], by Armitage the number of sides of such polygons is studied for the 2-sphere. It is proved that if $m \geq 4$, then there are m-sided regular spherical polygons which are bounded by arcs of great circles and do not have the Pompeiu property.

Maehara [520] shows that among the spherical polygons obtained by deforming a convex polygon on the unit 2-sphere, without changing the edge lengths, a *cyclic polygon* has maximum area. He also reproves that the unit 2-sphere in 3-space cannot be touched simultaneously by 13 further unit spheres. This famous result is due to Schütte and van der Waerden [716] and has an interesting history, see the reports [635] and [145] as well as the very recent contribution [304] due to Glazyrin.

Vu [798] studies the symplectic volume of the moduli space of closed spherical polygons and proves that *regular m-gons* maximize the volume inside the class of m-gons with fixed perimeter, and that under certain assumptions the regular m-gons are the unique m-gons maximizing the volume.

Guo and Sönmez [349] show that any formula on side-lengths and diagonal-lengths and the circumradius of cyclic polygons in *Euclidean, hyperbolic and spherical geometry*, which holds in one of these three geometries, is (under certain additional conditions) also true in the other two ones. The authors give also a useful little survey on *cyclic polygons* in these geometries.

Kakulashvili [436] collects calculations for polygons in spherical geometry. E.g., spherical cyclic quadrilaterals are investigated, and explicit calculations of *metric quantities of spherical polygons* are presented.

Kamiyama [440] studies the configuration space $M_m(a)$ of *regular spherical m-gons* of side-length $a \in (0, \pi)$ to determine the corresponding Euler characteristic. Here he computes this characteristic when m is even by showing that $M_m(a)$ may be obtained by Morse surgeries on the configuration space of equilateral m-gons in the plane. In [441] the author completes these studies.

It is known that for general *cyclic m-gons* in the Euclidean setting there is no area formula for $m \geq 5$ in terms of its side-lengths based only on the four arithmetic

operations and k-th roots. By Komori et al. [469] it is shown that a similar result holds in spherical and hyperbolic geometry.

4. Convexity-Related Results

Close to sets of constant width, Goldberg (see [305] and [306]) presents a kinematic construction for *spherical rotors*, and in the second paper he constructs two types of "basic" rotors for polygons in a kinematic way.

From the viewpoint of *computational geometry*, Ha and Shin [352] present a linear-time algorithm for computing the *intersection of a pair of spherical convex polygons*.

As already announced, we mention once more the excellent survey [492] of Lassak on spherical convex bodies, polygons, and polytopes. This survey is mainly related to typical questions and notions of *convex geometry*. The author collects results related to width and minimal width of these bodies. Important topics collected here in connection with spherical convex bodies (the polytope case always contained) are: *lunes and strips* containing the studied bodies, *minimal width* (= thickness) and *bodies of constant width* and of *constant diameter*, *diametrically complete sets* and *reduced bodies*. (Note that reduced bodies are convex bodies with the property that each proper convex subset of them has smaller minimal width; and diametrically complete sets have the property that each superset of them has a larger diameter. In Euclidean space, bodies of constant width share both these properties.) Some of the quantities and notions under particular consideration in that survey are *width functions* (in particular, *minimal width* and *diameter*), *perimeter, area* and *extreme points* of spherical convex bodies and polygons.

The *perimeter deviation* of two convex bodies K_1 and K_2 in the Euclidean plane is defined as the perimeter of $K_1 \cup K_2$ minus the perimeter of $K_1 \cap K_2$. It is known that a convex polygon with at most m vertices minimizing the perimeter deviation from K_1 is always inscribed in K_1. Fodor [273] shows that the analogous result does not hold on the sphere.

Chapter 5
Spherical Geometry II



Mathematics consists of proving the most obvious thing in the least obvious way.

— George Pólya

Using a known method of Cesàro, we prove the spherical cosine law, the spherical sine law, and Euler's formula for the spherical excess. The spherical cosine law is applied to prove a triangle comparison theorem for spheres. Euler's formula for spherical excess is applied to prove a theorem for the area of a spherical triangle with two fixed edges and one variable edge. We also derive an isoperimetric theorem for spherical quadrilaterals.

5.1 The Cesàro Triangle

Let us take the radius of the sphere $\mathbb{S}$ as unit length, and regard $\mathbb{S}$ as the *unit sphere*. For a non-antipodal pair of points $A, B \in \mathbb{S}$, the shortest geodesic segment connecting A and B and its length are denoted by AB. (The linear distance between A and B is denoted by $|AB|$ or $\|A-B\|$.) Recall also that $\breve{\triangle}ABC$ denotes a spherical triangle, whereas $\triangle ABC$ denotes a planar triangle. The antipodal point of $X \in \mathbb{S}$ is denoted by X^*.

For a spherical triangle $\breve{\triangle}ABC$ on $\mathbb{S}$, let Π be the plane that is tangent to $\mathbb{S}$ at A, and let $\varphi : \mathbb{S} \setminus \{A^*\} \to \Pi$ be the stereographic projection of $\mathbb{S}$ from A^*. Denote the image of $X \in \mathbb{S}$ by $\bar{X}$:

$$\bar{X} = \varphi(X).$$

Then, the planar triangle $\triangle \bar{A}\bar{B}\bar{C}$ is called the *Cesàro triangle* of $\breve{\triangle}ABC$ with respect to A. Planar triangles $\triangle \bar{A}\bar{B}^*\bar{C}$ and $\triangle \bar{A}\bar{B}\bar{C}^*$ are called (Cesáro's) *derived triangles*, see Fig. 5.1.

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H. Maehara, H. Martini, *Circles, Spheres and Spherical Geometry*,
Birkhäuser Advanced Texts Basler Lehrbücher,
https://doi.org/10.1007/978-3-031-62776-7_5

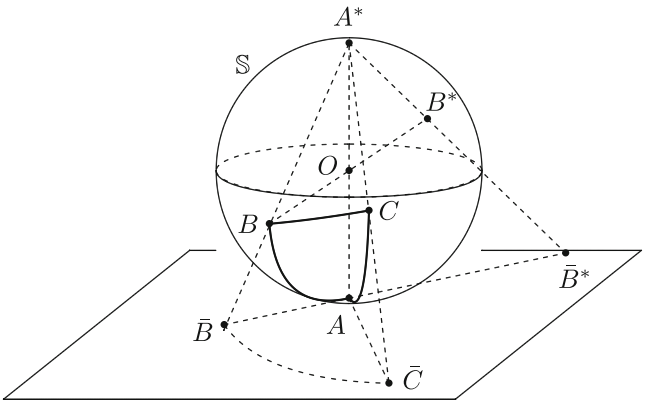


Fig. 5.1 Vertices $\bar{A}(=A)$, $\bar{B}$, $\bar{C}$ of a Cesàro triangle, and $\bar{B}^*$

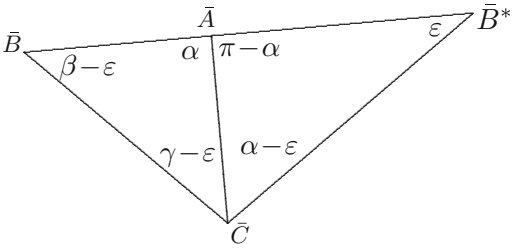


Fig. 5.2 The Cesàro triangle $\triangle\bar{A}\bar{B}\bar{C}$, and the derived triangle $\triangle\bar{A}\bar{B}^*\bar{C}$

Theorem 5.1 *The interior angles of the Cesáro triangle* $\triangle\bar{A}\bar{B}\bar{C}$ *of* $\breve{\triangle}ABC$ *with respect to A are given by*

$$\angle\bar{A} = \alpha,\ \angle\bar{B} = \beta - \varepsilon,\ \angle\bar{C} = \gamma - \varepsilon,$$

and the interior angles of the derived triangle $\triangle\bar{A}\bar{B}^*\bar{C}$ *are given by*

$$\angle\bar{A} = \pi - \alpha,\ \angle\bar{B}^* = \varepsilon,\ \angle\bar{C} = \alpha - \varepsilon$$

(see Fig. 5.2), where α, β, γ *are the interior angles of* $\breve{\triangle}ABC$ *at* A, B, C*, respectively, and* 2ε *is the spherical excess of* $\breve{\triangle}ABC$*, that is,* $2\varepsilon = \alpha + \beta + \gamma - \pi$.

Proof By the stereographic projection from A^*, every circle passing through A^* goes to a line on the tangent plane of $\mathbb{S}$ at A, and every circle that does not pass through A^* goes to a circle on the tangent plane. Hence the image of any arc of a great circle that passes through A goes to a line segment, and since the great circle BC does not pass through A^*, it goes to a circle Γ on the tangent plane of $\mathbb{S}$ at A. Hence we can see that the image of $\breve{\triangle}ABC$ under the stereographic projection is the curvilinear triangle enclosed by the line segments $\bar{A}\bar{B}$, $\bar{A}\bar{C}$ and the circular arc $\overset{\frown}{\bar{B}\bar{C}}$

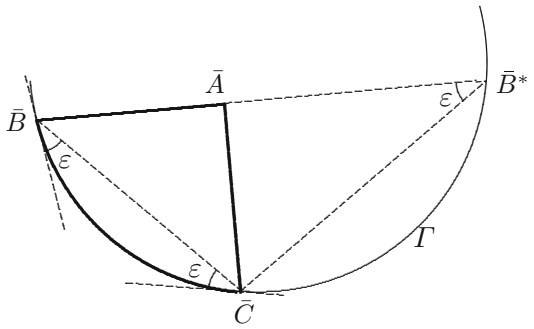


Fig. 5.3 Curvilinear triangle $\bar{A}\bar{B}\bar{C}$

of Γ not containing B^*, see Fig. 5.3. Since the sizes of angles are preserved by the stereographic projection, the interior angles of this "curvilinear" triangle at $\bar{A}$, $\bar{B}$, $\bar{C}$ become α, β, γ, respectively. Since the spherical excess 2ε of $\breve{\triangle}ABC$ is equal to the sum of three interior angles of this curvilinear triangle minus π, we can see that 2ε is equal to the sum of three interior angles of the curvilinear triangle minus the sum of interior angles of $\triangle\bar{A}\bar{B}\bar{C}$. Therefore, the angle formed by Γ and its chord $\bar{B}\bar{C}$ (the part not containing $\bar{A}$) is equal to ε. Thus, $\angle\bar{A}\bar{B}\bar{C} = \beta - \varepsilon$, $\angle\bar{A}\bar{C}\bar{B} = \gamma - \varepsilon$.

In $\triangle\bar{A}\bar{B}^*\bar{C}$, we have clearly $\angle\bar{B}^*\bar{A}\bar{C} = \pi - \alpha$, and by the alternate segment theorem, we have $\angle\bar{A}\bar{B}^*\bar{C} = \varepsilon$. And $\angle\bar{A}\bar{C}\bar{B}^* = \alpha - \varepsilon$.

□

Remark 5.1 Concerning Cesàro triangles, we refer to J. D. H. Donnay [220], van Brummelen [131], and H. Maehara and H. Martini [537].

Remark 5.2 In a planar triangle, "*the angle opposite to the greater edge is greater*" (see Euclid's Elements). The same assertion is also true for spherical triangles. This can be seen as follows. Suppose that $AC > AB$ in a spherical triangle $\breve{\triangle}ABC$ on $\mathbb{S}$. Then B lies in the interior of the spherical cap with center A and radius AC. By stereographic projection from A^*, this spherical cap is mapped to a disk with center $\bar{A}$ and radius $|\bar{A}\bar{C}|$, and $\bar{B}$ lies in the interior of this disk. Hence we have $|\bar{A}\bar{C}| > |\bar{A}\bar{B}|$ in the planar triangle $\triangle\bar{A}\bar{B}\bar{C}$. Hence $\angle\bar{B} > \angle\bar{C}$ by Proposition 18 of Euclid's Elements. Since $\angle\bar{B} = \beta - \varepsilon$, $\angle\bar{C} = \gamma - \varepsilon$, we have $\beta > \gamma$.

5.2 Edge-Lengths of Cesàro Triangles

Theorem 5.2 *The edge-lengths of the Cesàro triangle $\triangle\bar{A}\bar{B}\bar{C}$ of $\breve{\triangle}ABC$ with respect to A and the edge-lengths of the derived triangle $\triangle\bar{A}\bar{B}^*\bar{C}$ are given as follows:*

$$|\bar{A}\bar{B}| = 2\tan\tfrac{c}{2},\ |\bar{B}\bar{C}| = 2\sin\tfrac{a}{2}/(\cos\tfrac{b}{2}\cos\tfrac{c}{2}),\ |\bar{A}\bar{C}| = 2\tan\tfrac{b}{2},$$
$$|\bar{B}^*\bar{C}| = 2\cos\tfrac{a}{2}/(\sin\tfrac{b}{2}\sin\tfrac{c}{2}),\ |\bar{A}\bar{B}^*| = 2\cot\tfrac{c}{2},$$

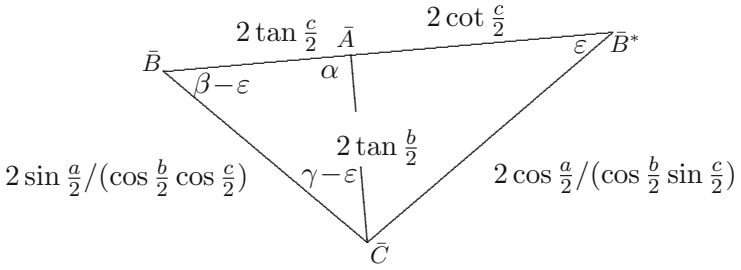


Fig. 5.4 Edge-lengths of a Cesàro triangle

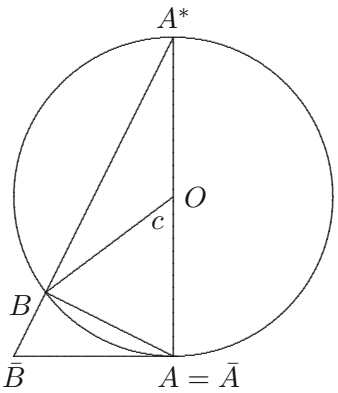


Fig. 5.5 Section by the plane A^*AB

see Fig. 5.4, where a, b, c denote the lengths of the edges of $\breve{\triangle}ABC$ opposite to A, B, C.

Proof

(i) Edge-lengths of $\triangle\bar{A}\bar{B}\bar{C}$: Fig. 5.5 shows the section of $\mathbb{S}$ and its tangent plane at A by the plane ABO (O is the center of $\mathbb{S}$). Since $\angle AOB = c$, we have $\angle AA^*B = c/2$, and since $|AA^*| = 2$, we have $|AB| = 2\sin\frac{c}{2}$, $|\bar{A}\bar{B}| = 2\tan\frac{c}{2}$. Similarly we have $|\bar{A}\bar{C}| = 2\tan\frac{b}{2}$. From $\triangle A^*BA \sim \triangle A^*A\bar{B}$ (similar), we have $|A^*B| \cdot |A^*\bar{B}| = |A^*A|^2$. Similarly, from $\triangle A^*CA \sim \triangle A^*A\bar{C}$, we have $|A^*C| \cdot |A^*\bar{C}| = |A^*A|^2$. Hence $|A^*B| \cdot |A^*\bar{B}| = |A^*C| \cdot |A^*\bar{C}|$, and therefore $|A^*B| : |A^*C| = |A^*\bar{C}| : |A^*\bar{B}|$. Since $\triangle A^*BC$ and $\triangle A^*\bar{C}\bar{B}$ have the common angle $\angle A^*$, $|A^*B| : |A^*C| = |A^*\bar{C}| : |A^*\bar{B}|$ implies that $\triangle A^*BC \sim \triangle A^*\bar{C}\bar{B}$. Therefore,$|BC| : |\bar{C}\bar{B}| = |A^*B| : |A^*\bar{C}|$, and hence

$$|\bar{C}\bar{B}| = \frac{|BC| \cdot |A^*\bar{C}|}{|A^*B|}.$$

Similarly to $|AB| = 2\sin\frac{c}{2}$, we can obtain $|BC| = 2\sin\frac{a}{2}$. From $|A^*B| = |A^*A| \cdot \cos\frac{c}{2}$ and $\cos\frac{b}{2} = \frac{|A^*\bar{A}|}{|A^*\bar{C}|} = \frac{2}{|A^*\bar{C}|}$, we have $|A^*\bar{C}| = 2/\cos\frac{b}{2}$, and hence

$$|\bar{C}\bar{B}| = \frac{2\sin\frac{a}{2} \cdot (2/\cos\frac{b}{2})}{2\cos\frac{c}{2}} = \frac{2\sin\frac{a}{2}}{\cos\frac{c}{2}\cos\frac{b}{2}}.$$

(ii) Edge-lengths of $\triangle\bar{A}\bar{B}^*\bar{C}$: Let a', b', c' be the edge-length of $\breve{\triangle}AB^*C$ opposite to A, B^*, C, respectively. By (i), the edge-lengths of the Cesàro triangle $\triangle\bar{A}\bar{B}^*\bar{C}$ of $\breve{\triangle}AB^*C$ with respect to A are $|\bar{A}\bar{B}^*| = 2\tan\frac{c'}{2}$, $|\bar{A}\bar{C}| = 2\tan\frac{b'}{2}$, $|\bar{B}^*\bar{C}| = 2\sin\frac{a'}{2}/(\cos\frac{b'}{2}\cos\frac{c'}{2})$. Since $a' = \pi - a, b' = b, c' = \pi - c$, we have

$$|\bar{B}^*\bar{C}| = 2\cos\tfrac{a}{2}/(\cos\tfrac{b}{2}\sin\tfrac{c}{2}),\ |\bar{A}\bar{C}| = 2\tan\tfrac{b}{2},\ |\bar{A}\bar{B}^*| = 2\cot\tfrac{c}{2}.$$

□

5.3 Spherical Cosine Law and Sine Law

Theorem 5.3 (Spherical Cosine Law) *For $\breve{\triangle}ABC$ on $\mathbb{S}$, the following holds:*

$$\cos a = \cos b\cos c + \sin b\sin c\cos\alpha. \tag{5.1}$$

Proof Figure 5.6 shows the enlarged Cesàro triangle $\triangle\bar{A}\bar{B}\bar{C}$ of Fig. 5.4 with ratio $\frac{1}{2}\cos\frac{b}{2}\cos\frac{c}{2}$. Applying the (planar) cosine law to this enlarged triangle, we have

$$\sin^2\tfrac{a}{2} = \sin^2\tfrac{b}{2}\cos^2\tfrac{c}{2} + \sin^2\tfrac{c}{2}\cos^2\tfrac{b}{2} - 2\sin\tfrac{b}{2}\cos\tfrac{c}{2}\sin\tfrac{c}{2}\cos\tfrac{b}{2}\cos\alpha.$$

Rewriting this by using the identities

$$2\sin^2\tfrac{x}{2} = 1 - \cos x,\ 2\cos^2\tfrac{x}{2} = 1 + \cos x,\ 2\sin\tfrac{x}{2}\cos\tfrac{x}{2} = \sin x,$$

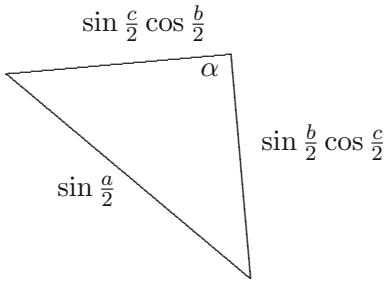


Fig. 5.6 Enlarged triangle $\triangle\bar{A}\bar{B}\bar{C}$, with ratio $\frac{1}{2}\cos\frac{b}{2}\cos\frac{c}{2}$

we have

$$1-\cos a=\tfrac{1}{2}(1-\cos b)(1+\cos c)+\tfrac{1}{2}(1-\cos c)(1+\cos b)-\sin b\sin c\cos\alpha,$$

from which (5.1) follows. □

Applying the spherical cosine law to the polar triangle of $\breve{\triangle}ABC$ we have the following corollary.

Corollary 5.1 *In $\breve{\triangle}ABC$, we have*

$$\cos\angle A=\cos\angle B\cos\angle C-\sin\angle B\sin\angle C\cos a.$$

□

Remark 5.3 By G. A. Jennings [428] and L. Fejes Tóth [257], the spherical cosine law is proved using the vector product.

Theorem 5.4 (Spherical Sine Law) *In $\breve{\triangle}ABC$ on $\mathbb{S}$, we have*

$$\frac{\sin a}{\sin\alpha}=\frac{\sin b}{\sin\beta}=\frac{\sin c}{\sin\gamma}. \tag{5.2}$$

Proof This time we use the enlarged triangle of the derived triangle $\triangle\bar{A}\bar{B}^*\bar{C}$ of $\breve{\triangle}ABC$ (Fig. 5.4) with ratio $\frac{1}{2}\cos\frac{b}{2}\sin\frac{c}{2}$, see Fig. 5.7. Applying the (planar) sine law to this enlarged triangle, we have

$$\frac{\cos\frac{a}{2}}{\sin(\pi-\alpha)}=\frac{\sin\frac{b}{2}\sin\frac{c}{2}}{\sin\varepsilon}.$$

Multiplying $2\sin\frac{a}{2}$ to both sides, and using $2\sin\frac{a}{2}\cos\frac{a}{2}=\sin a$, we have

$$\frac{\sin a}{\sin\alpha}=\frac{2\sin\frac{a}{2}\sin\frac{b}{2}\sin\frac{c}{2}}{\sin\varepsilon}.$$

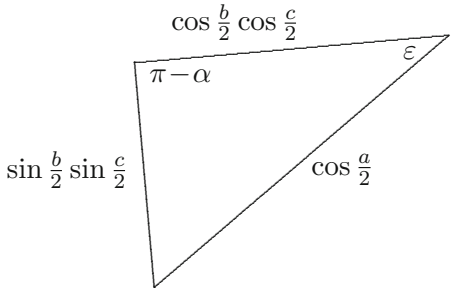


Fig. 5.7 Enlarged triangle $\triangle\bar{A}\bar{B}^*\bar{C}$, with ratio $\frac{1}{2}\cos\frac{b}{2}\sin\frac{c}{2}$

Since the right hand side is independent of the choice of an interior angle (and its opposite edge), (5.2) follows.

□

Theorem 5.5 (Euler's Formula for the Spherical Excess) *For* $\breve{\triangle}ABC$,

$$\cos\varepsilon = \frac{1+\cos a+\cos b+\cos c}{4\cos\frac{a}{2}\cos\frac{b}{2}\cos\frac{c}{2}} \tag{5.3}$$

holds, where a, b, c *are lengths of the edges opposite to* A, B, C *respectively, and* 2ε *is the spherical excess of* $\breve{\triangle}ABC$.

Proof Applying the (planar) cosine law to the enlarged triangle of Fig. 5.7, we have

$$\begin{aligned}\sin^2\tfrac{b}{2}\sin^2\tfrac{c}{2} = {} & \cos^2\tfrac{a}{2}+\cos^2\tfrac{b}{2}\cos^2\tfrac{c}{2}\\ & -2\cos\tfrac{a}{2}\cos\tfrac{b}{2}\cos\tfrac{c}{2}\cos\varepsilon.\end{aligned}$$

From this we get

$$\cos\varepsilon = \frac{\cos^2\frac{a}{2}+\cos^2\frac{b}{2}\cos^2\frac{c}{2}-\sin^2\frac{b}{2}\sin^2\frac{c}{2}}{2\cos\frac{a}{2}\cos\frac{b}{2}\cos\frac{c}{2}}.$$

The numerator of the right hand side fraction is equal to

$$\cos^2\tfrac{a}{2}+\big(\cos\tfrac{b}{2}\cos\tfrac{c}{2}+\sin\tfrac{b}{2}\sin\tfrac{c}{2}\big)\big(\cos\tfrac{b}{2}\cos\tfrac{c}{2}-\sin\tfrac{b}{2}\sin\tfrac{c}{2}\big).$$

Rewriting this by using the identities

$$\begin{aligned}2\cos^2\tfrac{x}{2} &= 1+\cos x,\\ 2\cos x\cos y &= \cos(x+y)+\cos(x-y),\end{aligned}$$

we have (5.3). □

5.4 The Triangle Comparison Theorem for Spheres

Applying the spherical cosine law, we prove the following result in an elementary way, without invoking Riemannian geometry.

Theorem 5.6 *For a spherical triangle* T *on the unit sphere, draw a spherical triangle* T' *on a sphere of radius* $R > 1$ *with the same edge-lengths as* T. *Then each interior angle of* T' *is smaller than the corresponding interior angle of* T.

Since a triangle in the plane is regarded as the limit case $R \to \infty$ of a spherical triangle on a sphere of radius R, we have the following corollary.

Corollary 5.2 *For a spherical triangle T on the unit sphere, draw a triangle on the plane with the same edge-lengths as T. Then every interior angle of the planar triangle is smaller than the corresponding interior angle of T.* □

Theorem 5.6 is a consequence of Toponogov's triangle comparison theorem, see [614] and [780, Theorem 3.91, p. 189]. We present here a proof of Theorem 5.6 given in [530].

First, we prepare two lemmas.

Lemma 5.1 *For $0 < \lambda < 1$, $\sin \lambda x / \sin x$ is monotonically increasing in $0 < x < \pi$, and*

$$\lim_{x \to 0} \frac{\sin \lambda x}{\sin x} = \lambda.$$

Proof Put $f(x) = \sin \lambda x / \sin x$. Then

$$f'(x) = (\lambda \cos \lambda x \sin x - \sin \lambda x \cos x) / \sin^2 x.$$

Denote the numerator by $g(x)$, that is,

$$g(x) = \lambda \cos \lambda x \sin x - \sin \lambda x \cos x.$$

Then $g(0) = 0$ and

$$\begin{aligned} g'(x) &= -\lambda^2 \sin \lambda x \sin x + \lambda \cos \lambda x \cos x - \lambda \cos \lambda x \cos \lambda x + \sin \lambda x \sin x \\ &= (1 - \lambda^2) \sin \lambda x \sin x > 0. \end{aligned}$$

Hence $g(x) > 0$ in $0 < x < \pi$, and hence $f'(x) > 0$ in this range. Now, using L'Hopital's Rule, we have

$$\lim_{x \to 0} \frac{\sin \lambda x}{\sin x} = \lim_{x \to 0} \frac{\lambda \cos \lambda x}{\cos x} = \lambda.$$

□

Lemma 5.2 *For fixed reals b, c, λ; $0 < b \leq c < \pi$, $0 < \lambda < 1$, put*

$$\varphi(x) = \frac{\cos \lambda x - \cos \lambda b \cos \lambda c}{\sin \lambda b \sin \lambda c} - \frac{\cos x - \cos b \cos c}{\sin b \sin c}.$$

Then, $\varphi(x) > 0$ for $c - b < x < \min\{b + c, 2\pi - (b + c)\}$.

Proof Put $d = \min\{b + c, 2\pi - (b + c)\}$. Since

$$\begin{aligned}\varphi(x) &= \frac{\cos\lambda x - \cos\lambda b\cos\lambda c}{\sin\lambda b\sin\lambda c} - 1 - \frac{\cos x - \cos b\cos c}{\sin b\sin c} + 1\\ &= \frac{\cos\lambda x - \cos(\lambda b + \lambda c)}{\sin\lambda b\sin\lambda c} - \frac{\cos x - \cos(b + c)}{\sin b\sin c},\end{aligned}$$

we have $\varphi(c - b) = \varphi(d) = 0$. Now

$$\begin{aligned}\varphi'(x) &= \frac{-\lambda\sin\lambda x}{\sin\lambda b\sin\lambda c} + \frac{\sin x}{\sin b\sin c}\\ \varphi''(x) &= \frac{-\lambda^2\cos\lambda x}{\sin\lambda b\sin\lambda c} + \frac{\cos x}{\sin b\sin c}\\ \varphi'''(x) &= \frac{\lambda^3\sin\lambda x}{\sin\lambda b\sin\lambda c} - \frac{\sin x}{\sin b\sin c}.\end{aligned}$$

If $0 < x \le c$, then

$$\frac{\varphi'(x)\sin\lambda b\sin\lambda c}{\sin x} = -\frac{\lambda\sin\lambda x}{\sin x} + \frac{\sin\lambda b\sin\lambda c}{\sin b\sin c}.$$

Since $\dfrac{\sin\lambda b}{\sin b} > \lambda$ and $\dfrac{\sin\lambda c}{\sin\lambda c} \ge \dfrac{\sin\lambda x}{\sin x}$ by Lemma 5.1, we have

$$\varphi'(x) > 0 \text{ in } 0 < x \le c. \tag{5.4}$$

Hence $\varphi(c - b) = 0$ implies

$$\varphi(x) > 0 \text{ in } c - b < x \le c. \tag{5.5}$$

Next, we show that

$$\varphi''(x) \text{ is monotonically decreasing in } 0 < x < \pi/2, \tag{5.6}$$

$$\varphi''(x) < 0 \text{ in } \pi/2 \le x < d. \tag{5.7}$$

If $0 < x < \pi/2$, then $\dfrac{\sin\lambda x}{\sin x} < \dfrac{\sin(\lambda\pi/2)}{\sin(\pi/2)} < 1$, and

$$\begin{aligned}\lambda\sin\lambda x - \sin x &= \sin x\left(\frac{\lambda\sin\lambda x}{\sin x} - 1\right)\\ &< \sin x\left(\frac{\lambda\sin(\lambda\pi/2)}{\sin(\pi/2)} - 1\right) < (\sin x)(\lambda - 1) < 0.\end{aligned}$$

Since

$$(\sin b \sin c)\varphi'''(x) = \frac{\sin b \sin c}{\sin \lambda b \sin \lambda c}\lambda^3 \sin \lambda x - \sin x$$

and $\dfrac{\sin b \sin c}{\sin \lambda b \sin \lambda c} < 1/\lambda^2$ by Lemma 5.1, we have

$$(\sin b \sin c)\varphi'''(x) < \lambda \sin \lambda x - \sin x < 0.$$

Therefore, $\varphi'''(x) < 0$ in $0 < x < \pi/2$, and hence (5.6) holds.

If $\pi/2 \le x < \pi$, then $\cos x \le 0$, and $\dfrac{\sin \lambda b \sin \lambda c}{\sin b \sin c} > \lambda^2$ by Lemma 5.1. Hence

$$(\sin \lambda b \sin \lambda c)\varphi''(x) = -\lambda^2 \cos \lambda x + \frac{\sin \lambda b \sin \lambda c}{\sin b \sin c} \cos x < \lambda^2(\cos x - \cos \lambda x).$$

Since $\cos x$ is monotonically decreasing in $0 < x < \pi$, we have $\cos x - \cos \lambda x < 0$ in $0 < x < \pi$. Since $\sin \lambda b \sin \lambda c > 0$, we have $\varphi''(x) < 0$ in $\pi/2 \le x < d$. Thus (5.7) holds.

Now, since

$$(\sin \lambda b \sin \lambda c)\varphi''(0) = -\lambda^2 + \frac{\sin \lambda b \sin \lambda c}{\sin b \sin c} > 0,$$

we have $\varphi''(0) > 0$. Since $\varphi''(\pi/2) < 0$ by (5.7), and since $\varphi''(x)$ is (strictly) monotonically decreasing in $0 < x < \pi/2$ by (5.6), there must be a unique $x_0 \in (0, \pi/2)$ such that $\varphi''(x_0) = 0$. Then $\varphi'(x)$ is monotonically increasing in $0 < x < x_0$, and monotonically decreasing in $x_0 < x < \pi/2$. Since $\varphi'(x)$ is also monotonically decreasing in $\pi/2 < x < d$ by (5.7), we can deduce that $\varphi'(x)$ is monotonically decreasing in $x_0 < x < d$. Therefore,

$$\text{there is just one } x \in (0, d) \text{ such that } \varphi'(x) = 0. \tag{5.8}$$

Now, if the lemma is not true, then there must be $x_1 \in (c, d)$ such that $\varphi(x_1) = 0$. Then, since $\varphi(c-b) = \varphi(x_1) = \varphi(d) = 0$, $\varphi'(x)$ must be 0 at some $x \in (c-b, x_1)$, and at some $x \in (x_1, d)$. This contradicts (5.8). Therefore, the lemma holds. □

Proof of Theorem 5.6 Let a, b, c denote the edge-lengths of T. Then $a + b + c < 2\pi$. We assume that $b \le c$. Then

$$c - b < a < \min\{b + c, 2\pi - (b + c)\} \le \pi.$$

Let α be the interior angle of T at the vertex opposite to the edge of length a, and α' be the corresponding interior angle of the spherical triangle T' drawn on the sphere of radius $R > 1$ with the same edge-lengths as T. To prove the theorem, it is enough

to show that $\alpha' < \alpha$. By the spherical cosine law, we have

$$\cos\alpha = \frac{\cos a - \cos b \cos c}{\sin b \sin c}, \quad \cos\alpha' = \frac{\cos \lambda a - \cos \lambda b \cos \lambda c}{\sin \lambda b \sin \lambda c},$$

where $\lambda = 1/R$. Now, by Lemma 5.2, we have $\cos\alpha' - \cos\alpha = \varphi(a) > 0$. Since $\cos x$ is monotonically decreasing in $0 < x < \pi$, we can deduce that $\alpha' < \alpha$. □

5.5 Triangles with Two Fixed Edge-Lengths

For fixed positive numbers b, c and a variable x satisfying $|b - c| < x < |b + c|$, the area $S(x)$ of a "planar" triangle with three edge-lengths b, c, x becomes the maximum when $x = x_0 := \sqrt{b^2 + c^2}$. And $S(x)$ is monotonically increasing in $x < x_0$, and monotonically decreasing in $x > x_0$. Let us show a similar result for spherical triangles.

Lemma 5.3 *Let a, b, c denote the lengths of the edges of $\breve{\triangle}ABC$ opposite to A, B, C, respectively. Put $a_0 = 2\sin^{-1}\sqrt{\sin^2(b/2) + \sin^2(c/2)}$. Then the following statements hold:*

(i) $\overset{\frown}{BAC}$ *is a semicircle* $\Leftrightarrow a = a_0$,

(ii) $\overset{\frown}{BAC}$ *is a major arc* $\Leftrightarrow a < a_0$,

(iii) $\overset{\frown}{BAC}$ *is a minor arc* $\Leftrightarrow a > a_0$.

Proof Note that if x is the length of the geodesic segment PQ on $\mathbb{S}$, then the length of the *line segment* PQ is $2\sin(x/2)$. When $\overset{\frown}{BAC}$ is a semicircle, the interior angle of the planar triangle $\triangle ABC$ at A is the right angle. Hence we have

$$2\sin(a/2) = \sqrt{4\sin^2(b/2) + 4\sin^2(c/2)},$$

from which (i) follows. The other two equivalences follow similarly. □

Remark 5.4 Note that the length of a semicircle is at most π. Hence, if $b + c \geq \pi$, then $\overset{\frown}{BAC}$ is always a major arc, and hence $a < a_0$.

Denote by $\breve{\triangle}(x, b, c)$ a spherical triangle with edge-lengths x, b, c, where b, c are fixed positive numbers and x is a variable satisfying $|b - c| < x < b + c$. Since $\cos t$ is monotonically decreasing in $0 < t < \pi$, from the spherical cosine law (Theorem 5.3) we have the following lemma.

Lemma 5.4 *Let $y(x)$ denote the interior angle of $\breve{\triangle}(x, b, c)$ opposite to the edge of length x. Then $y = y(x)$ is a monotonically increasing function of x.* □

Theorem 5.7 *Let $f(x)$ denote the area of $\breve{\triangle}(x,b,c)$. Then the following two implications hold.*

(1) *If $b+c \geq \pi$, then $f(x)$ is monotonically increasing in $x > 0$.*
(2) *If $b+c < \pi$, then $f(x)$ is monotonically increasing in $x < x_0$, and monotonically decreasing in $x > x_0$, where*

$$x_0 = 2\sin^{-1}\sqrt{\sin^2(b/2)+\sin^2(c/2)}.$$

Thus the maximum value of $f(x)$ is $f(x_0)$.

Proof By Euler's formula for the spherical excess, we have

$$\cos\varepsilon = \frac{1+\cos x+\cos b+\cos c}{4\cos(x/2)\cos(b/2)\cos(c/2)} = \frac{2\cos^2(x/2)+\cos b+\cos c}{4\cos(x/2)\cos(b/2)\cos(c/2)},$$

where $f(x) = 2\varepsilon$. Put $t = x/2$ and $g(t) = \cos(b/2)\cos(c/2)\cos\varepsilon$. Since

$$g(t) = \tfrac{1}{2}\cos t + \frac{\cos b+\cos c}{4\cos t},$$

we have

$$\begin{aligned} g'(t) &= -\tfrac{1}{2}\sin t + \left(\frac{\cos b+\cos c}{4}\right)\frac{\sin t}{\cos^2 t} \\ &= \frac{\sin t}{4\cos^2 t}(\cos b+\cos c-2\cos^2 t). \end{aligned}$$

Hence

$$\begin{aligned} g'(t) \gtreqless 0 &\Leftrightarrow \cos b+\cos c \gtreqless 2\cos^2 t \\ &\Leftrightarrow 2-2\sin^2(b/2)-2\sin^2(c/2) \gtreqless 2\cos^2 t \\ &\Leftrightarrow \sin^2 t \lesseqgtr \sin^2(b/2)+\sin^2(c/2) \\ &\Leftrightarrow t \lesseqgtr \sin^{-1}\sqrt{\sin^2(b/2)+\sin^2(c/2)}, \end{aligned}$$

where the triple-symbol $(\lesseqgtr)$ is meant in the same order. Since $\cos(b/2)\cos(c/2) > 0$, and $\cos t$ is monotonically decreasing in $0 < t < \pi$, we have

$$f'(x) \gtreqless 0 \Leftrightarrow x \lesseqgtr \sin^{-1}\sqrt{\sin^2(b/2)+\sin^2(c/2)}.$$

Now, from Lemma 5.2 and Remark 5.4 the theorem follows. □

For a figure F on $\mathbb{S}$, its area is denoted by $[F]$. For example, the area of a spherical triangle ABC is denoted by $[ABC]$, the area of $\breve{\triangle}(a,b,c)$ by $[\breve{\triangle}(a,b,c)]$, the area of a spherical quadrilateral $ABCD$ by $[ABCD]$.

From Theorem 5.7 (1), we have the following corollary.

Corollary 5.3 *If the length of the shortest edge of a spherical triangle $\breve{\triangle}ABC$ is at least $\pi/2$, then $[\breve{\triangle}ABC] \geq [\breve{\triangle}(\pi/2, \pi/2, \pi/2)]$*

Corollary 5.4 *In the range $x_1 \leq x \leq x_2$, the area of $\breve{\triangle}(x,b,c)$ becomes minimum either at $x = x_1$ or at $x = x_2$.* □

5.6 The Isoperimetric Theorem for Quadrilaterals

The next lemma is proved in [520], and it is also listed as a problem in "Curve on a Sphere" [817, p. 52].

Lemma 5.5 *If a simple closed curve on $\mathbb{S}$ has length smaller than 2π, then it is contained in some hemisphere.*

Proof Let Γ be a simple closed curve on the unit sphere $\mathbb{S}$ whose length is less than 2π. Let P, Q be two points on Γ that bisect the length of Γ. Then, since P, Q are connected by a curve of length less than π, the length of the geodesic segment PQ is less than π. Let M be the midpoint of PQ. Let $H(M)$ the hemisphere with pole M. Since $PQ < \pi$, P, Q lie in the interior of $H(M)$.

We show that Γ is contained in the interior of $H(M)$. Suppose that Γ is not contained in the interior of $H(M)$. Then Γ meets the boundary of $H(M)$ at some point R. Since M is the midpoint of PQ, by the rotation through 180° around M, P goes to Q, and R goes to R^* (the antipodal point of R that also lies on the boundary of $H(M)$). Hence $PR = QR^*$. Let s be the length of the partial curve $P \to R \to Q$ along Γ (hence $2s$ is the length of Γ). Then we have

$$s \geq PR + RQ = QR^* + RQ \geq RR^* = \pi,$$

which implies that the length $2s$ of Γ is greater than or equal to 2π, a contradiction. □

A spherical polygon on $\mathbb{S}$ is called *cyclic* if its vertices are concyclic, that is, all vertices lie on the same circle.

Theorem 5.8 *Let a, b, c be positive numbers satisfying that $a+b+c < \pi$. Among the spherical quadrilaterals $ABCD$ on $\mathbb{S}$ such that $AB = a$, $BC = b$, $CD = c$, a cyclic one in which $\overset{\frown}{ABCD}$ is a semicircle has the maximum area.*

Proof Since the perimeter of $ABCD$ is less than 2π, $ABCD$ lies in a hemisphere. Note that we always have $[ABCD] \leq [ABC] + [ACD]$. If the diagonal segment AC is not contained in the spherical quadrilateral $ABCD$, then, replacing the part

$\breve{\triangle}ABC$ by its "mirror image" with respect to the plane ACO, we have one for which $[ABCD] = [ABC] + [ACD]$ holds. Now, put $t = AC$. Then $[ABCD] = [\breve{\triangle}(a, b, t)]+[ACD]$. For every $t \leq a+b$, the maximum value of $[ACD]$ is attained when $\overset{\frown}{ACD}$ is a semicircle by Theorem 5.7, and hence the maximum value is given by a continuous function $g(t)$ of t. Thus, the maximum value $[\breve{\triangle}(a, b, t)] + g(t)$ of $[ABCD]$ for $AC = t$ is also a continuous function of t. Since the range of t is a closed interval $a - b \leq t \leq a + b$, there is a t where the maximum value is attained. Therefore, there is a spherical quadrilateral $ABCD$ that has the maximum area. In this $ABCD$, $\overset{\frown}{ACD}$ is a semicircle. The quadrilateral $ABCD$ also contains the diagonal BD. (For otherwise, replacing the part $\breve{\triangle}ABD$ by its "mirror image" with respect to the plane BDO, we could have a quadrilateral with larger area.) Then, by a similar argument as it was used above, we see that $\overset{\frown}{ABD}$ is also a semicircle. Then, since AC is contained in $ABCD$, B must lie on the subarc $\overset{\frown}{AC}$ of the semicircle $\overset{\frown}{ACD}$. Therefore, $ABCD$ is cyclic and $\overset{\frown}{ABCD}$ is a semicircle. □

Lemma 5.6 *Every spherical quadrilateral with perimeter less than 2π can be continuously deformed, with keeping its four edge-lengths, to a cyclic one.*

Proof Let $ABCD$ be a non-cyclic spherical quadrilateral with perimeter less than 2π. We may suppose that D lies in the interior of the circumscribed cap $cap(ABC)$ of $\breve{\triangle}ABC$. We also assume that $BC+CD \geq DA+AB$. If we increase the geodesic distance between D and B (with keeping the four edge-lengths fixed) so that it becomes nearly $DA + AB$ (in other words, $\angle DAB$ becomes nearly 180°), then D goes to the exterior of $cap(ABC)$. Therefore, there is a moment that D lies on the boundary of $cap(ABC)$. At that moment, $ABCD$ becomes cyclic. □

A spherical polygon is said to be *inscribed* in a cap if the spherical polygon is contained in the cap and all vertices of the spherical polygon lie on the boundary circle of the cap. The cap is also called the *circumscribed cap* of the spherical polygon. Thus a spherical polygon inscribed in a cap is a cyclic one.

Theorem 5.9 (Isoperimetric Theorem for Spherical Quadrilaterals) *Among the spherical quadrilaterals on $\mathbb{S}$ with fixed four edge-lengths and perimeter less than 2π, a cyclic one has the maximum area.*

Proof Let $ABCD$ be a non-cyclic quadrilateral on $\mathbb{S}$ with given fixed edge-lengths. We may suppose that CD is the longest edge, and $AD \geq BC$. Then, since $AD + DC \geq AB + BC$, D is not contained in $\breve{\triangle}ABC$. If B lies in the interior of $\breve{\triangle}ACD$, then replace B by its "mirror image" with respect to the plane ACO. By such a replacement $[ABCD]$ becomes larger, and B goes outside $\breve{\triangle}ACD$. By Lemma 5.4, $ABCD$ can be deformed with fixing four edge-lengths, and fixing two vertices C, D, to a cyclic quadrilateral $A'B'CD$. We may suppose that A' and A lie in the same side of the great circle CD. We show that $[ABCD] < [A'B'CD]$. Let $A'P$ be a (spherical) diameter of the circumscribed spherical cap of $A'B'CD$, see Fig. 5.8.

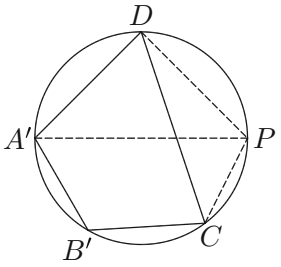


Fig. 5.8 $A'P$ is a diameter of the spherical cap

Since CD is the longest edge and $A'D \geq B'C$, it follows that $A'P$ and CD intersect. (It is possible that $P = C$, but we assume $P \neq C$ in the following.) We have

$$[A'B'CD] = [A'B'CP] + [A'DP] - [CPD].$$

Let us divide our argument into two cases.

(i) The case that AP and CD intersect: Since $B \notin \breve{\triangle}ACD$, $D \notin \breve{\triangle}ABC$, AP never intersects BC. Hence we have $[ABCD] = [ABCP] + [ADP] - [CDP]$. Since $[ADP] < [A'DP]$, $[ABCP] < [A'B'CP]$ by Theorems 5.7 and 5.8,

$$\begin{aligned}[ABCD] &\leq [ABCP] + [ADP] - [CPD] \\ &< [A'B'CP] + [A'DP] - [CPD] = [A'B'CD].\end{aligned}$$

(ii) The case that AP and CD do not intersect: In this case, the spherical quadrilateral $ABCP$ contains D in its interior. Hence we have $[ABCD] = [ABCP] - [ADP] - [CDP]$. Since $[ABCP] < [A'B'CP]$ by Theorem 5.8,

$$\begin{aligned}[ABCD] &\leq [ABCP] - [ADP] - [CPD] \\ &< [A'B'CP] + [A'DP] - [CPD] = [A'B'CD].\end{aligned}$$

□

5.7 Exercises

5.1* Let A, B be two points on the unit sphere with the same latitude $\pi/4$ and the difference of their longitudes $\pi/2$. Find the spherical distance between A and B.

5.2* In an equilateral spherical triangle on the unit sphere, let d be its edge-length and θ be its interior angle. Prove that

$$\cos d = \frac{\cos\theta}{1-\cos\theta} \text{ and } \cos\theta = \frac{\cos d}{1+\cos d}.$$

5.3* Prove that in $\breve{\triangle}ABC$ the following implications hold.

(i) $AC > AB \Rightarrow \angle B > \angle C$.
(ii) $AC = AB \Rightarrow \angle B = \angle C$.
(iii) $\angle B > \angle C \Rightarrow AC > AB$.

5.4* Let $\breve{\triangle}ABC$ be a *non-obtuse* triangle (that is, none of the interior angles is greater than $\pi/2$). Prove that if $X, Y \in \breve{\triangle}ABC$, then $XY \leq \max\{AB, BC, CA\}$.

5.5 Prove that it is impossible to take four points A, B, C, D on a sphere (of any radius) such that

$$AB = BC = CA = 1, AD = BD = CD = 1/\sqrt{3}.$$

5.6* From Theorem 5.7 (2), derive the following result: Among the convex quadrilaterals with edge-lengths a, a, b, b on the unit sphere $\mathbb{S}$, one that is inscribed in a spherical cap has the maximum area.

5.8 Notes

In these notes we will concentrate on the following topics: spherical trigonometry (particularly, also the corresponding sine and cosine theorems) and triangle comparison theorems (due to Toponogov).

5.8.1 Notes on Spherical Trigonometry and the Respective Cosine and Sine Laws

Spherical trigonometry, and particularly the *spherical sine and cosine theorems* (see our Sect. 5.3), represent an old, well-researched mathematical topic. Therefore the following part cannot be dominated by recent research papers and is more dedicated to books with interesting representations of the little field. However, the presented references contain nevertheless interesting sides and new applications. Most of them refer to classical spherical geometry, i.e., to the geometry of the 2-sphere. Extensions to spherical n-dimensional spaces are normally identified as such.

Some History

Highlights from the *history of spherical trigonometry* are connected with names like Menelaus, several Indian and Arabic mathematicians from the Middle Ages, Regiomontanus, Pitiscus (a Silesian mathematician from the sixteenth century), Napier, Euler and various further contributors.

In Björnbo's long paper [90] (JFM 33.0062.01) *Menelaus'* essential achievements in spherical trigonometry are summarized and honored, and further existing material is ordered.

The contributions of *Islamic scientists* during the early Middle Ages are nicely discussed and summarized in [204], [657] and [658], showing also once more the impact of Menelaus' findings. Although various efforts had their origins in problems from astronomy, the Arabic mathematicians also discussed notions from spherical geometry (e.g., sines and chords) in more abstract terms.

In view of spherical geometry and trigonometry in *India* (Middle Ages), e.g. also going back to works of Brahmagupta (seventh century), we refer to Datta and Singh [198], i.e., mainly to the last part of [198], which is dedicated to spherical geometry.

The role of *Regiomontanus* (fifteenth century) as well as *Clavius and Pitiscus* (sixteenth century) in spherical trigonometry is studied by Olivares [603]. Pitiscus introduced the name "trigonometry" and presented possibly the first proper proof of the spherical cosine theorem (in 1595). The paper [603] shows the evolution of the cosine theorem over all these centuries, and also its proof due to Pitiscus is presented.

Dietrich and Girstmair [213] study *Napier's achievements* in spherical trigonometry (namely Napier's rules, his aim to use his logarithmic calculus mainly in trigonometry and related topics). One sees also that the right notation in this little field was not really optimal (a level that Euler reached a little later). This is also the right place to cite Havil's book [385] on the life and mathematical work of Napier, including of course also spherical geometry. A further essential contribution to Napier's work on spherical trigonometry is [732] due to Silverberg.

The deplorable condition that spherical trigonometry disappeared from the curriculum of general elementary mathematics (like also its connections to navigation and astronomy) led Plofker to write the very nice historical article [640] on this little field, representing contributions from the respective Hellenistic and Islamic period to the corresponding Indian and medieval European time. Later the obtained insights were exceeded by the Europeans to the framework of non-Euclidean geometries, to which spherical geometry clearly belongs. Especially Euler's achievements are recognized in detail. E.g., it is shown in [640] how Euler established the foundations of spherical trigonometry via the *method of maxima and minima*, and how he derived the spherical cosine theorem by an argument using minima. Euler's contributions to spherical geometry and trigonometry are also nicely summarized in [618] by Papadopoulos, and an informative historical survey of the whole field is presented there. The author gives also a *mathematical overview* of Euler's results in spherical geometry, particularly including spherical projections and cartography. An outline

of the scientific connection between Euler and Lagrange is given, too. We mention here also once more the Proceedings [138] on mathematical geography.

It can be seen by the collection [439] of Kambly that, at the beginning of the twentieth century, the level of the material on spherical geometry covered in the mathematical Abitur was high. This example refers to Prussia. A similar example is the book [188] of Crawley. It is a selection of one thousand exercises (with solutions) in plane and spherical trigonometry, which appeared 1914 in Pennsylvania.

Two essential overviews from the first half of the twentieth century which contain much material on the history and on the (in those days) recent state in spherical geometry and trigonometry are those due to Zacharias [828, Parts 18 and 19] and Tropfke [782, Part II]. And, e.g., Hessenberg [395], Macaulay [516, Chapter XIII] as well as Kells et al. [445] were *popular related reference books* in this period.

We finish this little part on historical aspects by the relatively recent and excellent book [131] of Van Brummelen. The third chapter discusses the basic ancient contributions of Hipparchus of Rhodes and Menelaus of Alexandria (the author of *Sphaerica*—the spherical version of Menelaus' theorem plays a central role here), and in Chap. 4 the achievements of Indian and Islamic mathematicians are appreciated. The latter had a special motivation to apply spherical geometry: their praying in the direction to Mecca. Further on, the contribution of Napier, Euler and others are comprehensively discussed. In the very final part, applications like modern celestial navigation are presented.

More Recent Representations

In the middle of the twentieth century, the Hilbert doctorands Lietzmann and Dörrie wrote the two nice and comprehensive monographs [510] and [217], respectively. The first one clearly and systematically provides all the needed tools from school geometry and presents numerical and constructive methods needed in spherical geometry; applications in geodesy and astronomy are discussed, too. The second book [217], extremely rich in content and dedicated to planar as well as spherical trigonometry, shows many applications far from each other and therefore presenting a broad view on the richness of the topic. E.g., extensions of Ceva's theorem on the sphere, location of ships, sundials, and Gauss's method from orbital mechanics (a subfield of celestial mechanics) are presented.

Not far from this, the three books [315] by Graf, [731] by Sigl and [757] by Steinert have a strong reference to cartographic, geodesic and astronomic applications of spherical geometry and trigonometry in common. Here we can add Hammond's monograph [377] on spherical trigonometry and applications, e.g. emphasizing Napier's rules and navigation, and showing also a balance in its analytical and geometric interpretations.

A very nice book chapter devoted to *spherical trigonometry and respective sine and cosine theorems* is [100, § 6] by Boltyanski and Yaglom (where a detailed study of spherical trigonometry, based on vector calculus, is given). An analogous way to the cosine theorem of spherical trigonometry is presented in [45], see also

the related classroom note [396]). Further on we refer to [675, § 4] by Rosenfeld, where a complete derivation of these theorems and of relatives is given.

Also the important geometry book [75] of Berger contains a longer discussion of spherical triangles and trigonometry, namely the whole Section 18.6 (extensions to *spherical polygons* follow in Section 18.7). Due to the close relations between *spherical and Riemann geometry*, also the exposition [677, § 2] should at least be mentioned here.

Bigalke's comprehensive monograph [87] is generally seen as a more modern continuation of "classical books" (such as [510] and [217]) in the field considered here. In Chapter 2 of [87] a *whole theory of spherical triangles* is given, and one can find as basic theorems also the spherical sine and cosine theorems and various interesting applications.

Jennings dedicates the whole Chapter 2 of his monograph [428] to spherical geometry. He derives basic results from spherical trigonometry using cross products, shows the relation between *angular defect and area*, and derives the laws of sines and cosines for sides and angles. Discussed applications come from *navigation and cartography*.

Section 5.1 of the geometry book [646] of Prasolov and Tikhomirov contains only a short part treating spherical geometry, but the laws of sines and cosines are carefully derived there. From the viewpoint of *differential geometry*, Wilson [815] studies spherical geometry in his Chapter 2. He presents properties of curves on the sphere using the spherical distance metric, and he treats the spherical laws of cosines and sines.

The fifth chapter of the book [605] of Ostermann and Wanner comprehensively presents material on (spherical) trigonometry and its applications, and particularly Sects. 5.6 and 5.7 study also sine and cosine laws with nice applications. And we mention the monograph [103] of Borceux whose Section 14.2 contains also related results. Of course, also here the basic book [131] of van Brummelen should be cited once more, especially Chaps. 2 and 5.

There are various books in the field of *celestial mechanics*. In their monograph [596], Neutsch and Scherer introduce modern methods from this field as mathematical ones, to present also *different aspects of current research* there. The first two parts present historical facts and needed mathematical notions and concepts, among them also spherical trigonometry.

Regarding *applications*, we mention the book [135] of Burkholder. He introduces a *global spatial data system* which is able to use different types of geoinformation frameworks; it is a collection of mathematical concepts (e.g., certain equations from geometrical geodesy and solid geometry are used) that can handle spatial data locally as well as globally. In Chap. 3 mathematical fundaments are given, among them also basics from spherical trigonometry not commonly covered in elementary mathematical courses.

Also the book [416] of Hull should be mentioned here. It shows the relations between *Origami* and various mathematical concepts; spherical trigonometry is one of them.

A Few Related Journal Articles

Yang and Zhang [824] establish a higher dimensional extension of the *Neuberg-Pedoe inequality* yielding pairs of directly similar simplices. They obtain also an application of this result in spherical trigonometry and conjecture, that it holds also in n-dimensional elliptic geometry.

With the help of *stereographic projection*, Jeger [427] elegantly derives sine and cosine formulas for spherical triangles. McCleary [562] succeeds in proving simultaneously various basic trigonometric relations for spherical and hyperbolic geometry, among them also the laws of sines and cosines. The list of references in this article is very interesting.

Supporting the thesis that there are close relations between the theory of *integrability of (discrete) dynamical systems* and fundamental geometric structures, Petrera and Suris [634] verify that the cosine laws for simplices in spherical space define in various ways integrable systems. For this purpose, e.g. the derivation of the spherical cosine and sine laws in terms of Gram matrices is recalled, and then nice examples of integrable dynamical systems are obtained.

Slutskiy [742] describes an interesting, alternative approach (150 years old) of the Belgian scientist de Tilly to hyperbolic geometry, and also his studies of spherical geometry. In his original contributions, de Tilly also deduced the laws of sines and cosines in spherical space (see particularly Section 4 in [742]).

5.8.2 Notes on Triangle Comparison Theorems

Roughly speaking, the *triangle comparison theorem* (often called *Toponogov's theorem*) compares the angles of two geodesic triangles with the same side lengths lying on surfaces with different curvatures; see our Sect. 5.4 above. By the following overview, one sees the importance of this theorem in differential geometry, particularly in Riemannian geometry.

Some Historical Remarks

More precisely, the names *Aleksandrov Comparison Theorem for the Angles of a Triangle* (see Toponogov [780, Theorem 3.9.1]) for two dimensions, *Aleksandrov-Toponogov theorem* (cf. Chavel's book [158, p. 400] and [15]) and *Cartan-Aleksandrov-Toponogov theorem* (see Burago et al. [134, p. 240] in higher dimensions are common, and this theorem has two variants. In the first one, *lengths are compared*: given two geodesic triangles with two sides of the correspondingly same length and the angle between them of the same measure, lying in two surfaces with Gaussian curvature of the first everywhere larger than or equal to the maximal Gaussian curvature of the second, then the third triangle side is smaller than or equal to the third triangle side. The second variant *compares angles*: if, under the same

conditions for the two surfaces, the triangles have correspondingly congruent sides, then the angles are respectively larger or equal to their corresponding counterparts. The importance of these statements is nicely expressed by Petersen [633, p. 333]: "Toponogov's theorem is a very useful refinement of Gauss' early realization that curvature and angle excess of triangles are related."

Pambuccian and Zamfirescu [614] correct statements around the *history of the theorem* in the following way: The two-dimensional case is usually presented as result due to A. D. Aleksandrov (see [16]), and the n-dimensional Riemannian case as result of Toponogov [779]. But as the authors of [614] underline, the two-dimensional case was first derived by Pizzetti [639], and they present a biographical sketch and a short outline of Pizzetti's proof of the triangle comparison theorem. In [614] it is also explained how the names of Cartan and also of Darboux came into this framework, yielding finally the name of the related *CAT(k) spaces* by the initials of Cartan, Aleksandrov, and Toponogov.

It turned out that the proofs of Pizzetti, Aleksandrov, and Toponogov cannot be conveniently compared, since the definitions of the used terms are very different. In this direction, the interesting role of Rauch [660] was evaluated by Chavel [158, p. 399] (saying "by triangle comparison theorems we mean global forms of the *Rauch comparison theorem*") and Grove and Petersen [335, p. ix]. Further interesting remarks on the history of triangle comparison theorems can be found in [443], [77, pp. 52 and 6] and [632, p. 325].

Maehara [530] gave very recently an elementary proof of one classical form of Toponogov's theorem (which is essential for our book here), namely: if two spherical triangles on two different spheres have the same side-lengths, then the corresponding angles of that triangle which lies on the sphere with larger radius are smaller; see the beginning of our Sect. 5.4. Replacing the two spheres in this version by suitable surfaces or manifolds, one gets Toponogov's theorem and more general statements, very important in Riemannian geometry and further settings. As a helpful tool it found entrance into various further mathematical disciplines. In the following we present a suitable selection of related works.

The Comparison Theorem in Several Books

Toponogov himself published, late in his life, a book on differential geometry (see [780]). Its third chapter treats topics like *geodesics, shortest paths, and the intrinsic geometry of surfaces* (Gauss-Bonnet and Hopf-Rinow theorem). At the end, a proof of Aleksandrov's version of the comparison theorem is given.

Going chronologically back, we start with the lecture notes [333] of Gromoll, Klingenberg, and Meyer. In Chap. 6 the authors collect *various comparison theorems*, among them the angle comparison theorem of Toponogov.

Cheeger and Ebin present with [159] a comprehensive, elegantly written book on *global Riemannian geometry*; they are particularly interested in geometric and topological consequences following from assumptions on curvature. They create techniques for *comparing the geometry of a general manifold with that of a simply*

connected model space of constant curvature. E.g., these are based on comparison theorems, like also that of Toponogov.

Chavel's monograph [158] gives a comprehensive and broad approach to Riemannian geometry, having unexpectedly extensive parts of notes and exercises at the end of each chapter and, following from this, a large bibliography, thus opening enough space for further ideas and extensions. Discretization of Riemann manifolds, Hamiltonian and Lagrangian mechanics, and manifolds without conjugate points (including Hopf's theorem) are interesting topics. In the last chapter, entitled "Comparison and finiteness theorems", the author studies *Rauch's comparison theorem and applications* and also Aleksandrov-Toponogov triangle comparison theorems.

Also in the book [583] of Morgan and Tian on Ricci flow and the Poincaré conjecture one can find Toponogov's result as a helpful tool. In the second chapter the theory of *manifolds of non-negative sectional curvature* is discussed, and it contains the Toponogov theory on the geometry of triangles in this setting.

Research Papers

We continue with a selection of *research papers*, mainly in chronological order, thus showing also some of the historical milestones.

Elerath [241] extends Toponogov's comparison theorem by studying the curvature along *geodesics* with the curvature along geodesics in a suitable comparison space S. The usual comparison theorems take S as space of constant curvature, but the author uses instead a "flattening" surface of revolution in which the curvature decreases with the distance from the vertex. Thus, first he has to establish certain crucial properties of *geodesics and cut points* in such a surface. These ideas with geodesics are continued by Itokawa et al. in [420] and [421] regarding manifolds with restricted radial curvature. (Recall that the *radial curvature* of a Riemannian manifold S is the curvature of S along and tangent to the geodesics emanating from a fixed base point.)

Harris [379] derives an analogue of Toponogov's triangle comparison theorem for *Lorentz manifolds*. He obtains first two Rauch comparison theorems using *timelike geodesics*, and using a new concept of sectional curvature he finally gives his triangle comparison theorem for triangles which have all three sides consisting of timelike geodesic segments.

In [1], Abresch studies the *topology of asymptotically non-negatively curve manifolds*. He considers an extension of Toponogov's triangle comparison theorem to the situation that the comparison manifold is not simply connected and of constant curvature, but a surface with only rotational symmetries. In [2] Abresch continues [1] improving Gromov's "Betti number theorem" from [334] by extending it to non-compact asymptotically nonnegatively curved manifolds. Important expositions, containing survey parts, around Riemannian comparison theorems and related topics on manifolds of nonnegative sectional curvature are those of Karcher [443] and of Kobayashi (the pages 140–169 before [443]). A series of essential results in this area

is selected and presented with detailed proofs. Also interesting technical details are emphasized, and a comprehensive survey on essential developments in the studied field is given. In particular, a new proof of Toponogov's theorem is given, with applications to critical points of distance functions, the cut locus and the sphere theorems. Karcher's exposition is very interesting even for experts in the field, since in many cases his arguments differ essentially from presentations in standard textbooks. Marenich [544] proves a *characterization theorem of hyperspheres as special Riemannian manifolds* with positive sectional curvature containing an isometric embedded neighbourhood of an equatorial Euclidean sphere. The proofs are based on an extremal case of Toponogov's triangle comparison theorem.

Dai and Wei [194] establish a triangle comparison of Toponogov type for Riemannian manifolds *without sectional curvature bound*. They also give an application to the topological type of complete manifolds with non-negative Ricci curvature and small volume growth.

Colding [171] investigates *n-dimensional closed Riemannian manifolds with positive Ricci curvature* regarding their Gromov-Hausdorff distance to an n-sphere, and for the proof he develops a suitable version of Toponogov's triangle comparison theorem for positive Ricci curvature. The same author presents in [172] a new integral estimate of distances and angles on manifolds having a lower bound for their Ricci curvature. He thus obtains an *integral version of Toponogov's theorem for Ricci curvature*.

In [479], Kürzel proves a version of the classical Toponogov triangle comparison theorem, where the model space is a surface (possibly singular) of revolution having non-constant curvature.

For a given *Kähler manifold* S, Adachi [6] studies a nontrivial Kähler magnetic field (i.e., a closed two-form being a constant multiple of S) and finds a comparison theorem analogous to Toponogov's theorem in Riemannian geometry. Toponogov's triangle is here replaced by a bow-shape which consists of a trajectory segment and a kind of geodesic segment on manifolds with sectional curvature bounded from above.

Kondo and Tanaka [470] prove the *finiteness of the topological type of a complete open Riemannian manifold* S with base point $x \in S$ whose radial curvature at x is bounded below by a non-compact model surface of revolution S' with finite total curvature and has no pair of cut points in a sector based at $x' \in S'$. To prove this, they use a new comparison theorem of Toponogov type, and they apply the obtained theorem to get a partial answer to a conjecture of Milnor that complete open Riemannian manifolds with nonnegative Ricci curvature have finitely generated fundamental group.

To get similar results, Kondo and Tanaka derive in [471] and [472] a *Toponogov comparison theorem for open triangles* standing on the boundary of complete connected n-dimensional Riemannian manifolds with smooth convex boundary whose radial curvature is bounded below by certain model surfaces.

Lang and Schroeder present the expository note [488] with a transparent proof of *Toponogov's theorem for Aleksandrov spaces in the general case* (not assuming local compactness of the underlying metric space). In other words, they show that

with certain local assumptions in geodesic metric spaces the comparisons under consideration hold also in the large.

Kondo et al. [473] study the relationship between the topology and the curvature of *Finsler manifolds*. They prove a finiteness theorem of topological type and a diffeomorphism theorem as applications of a Toponogov-type triangle comparison theorem of Finsler geometry.

Hebda and Ikeda [388] prove the following result: let S be a complete Riemannian manifold with a base point x, and S' be a complete, simply connected surface, rotationally symmetric about the vertex x' such that the cut locus of every point x' lies in the opposite meridian of x'. Now suppose that (S', x') has weaker radial attraction than (S, x). Then, for every suitable geodesic triangle in S, there is a corresponding geodesic triangle in S' whose respective sides are equal and that satisfies the angle comparison and Aleksandrov convexity in Toponogov's Theorem. In addition, it is shown that rigidity fails in this system. Hebda and Ikeda [389] prove an *analogue of Toponogov's triangle theorem for surfaces of revolution as model spaces*. Namely, for a model surface and a Riemannian manifold with a fixed base point, the authors give necessary and sufficient conditions under which each geodesic triangle in the manifold, having a vertex at the base point, has a corresponding Aleksandrov triangle in the model.

Hu et al. [406] present an *elegant new proof* of Toponogov's Theorem for triangles in complete Aleksandrov spaces. This proof does not depend on further important, basic approaches to that theorem for Aleksandrov spaces in the literature; see, e.g., Burago, Gromov and Perel'man [133] and Shiohama [730]. The new proof in [406] yields a new idea useful in the framework of Aleksandrov spaces, namely that of using *variations of geodesics* to detect curvature.

Chapter 6
The Problem of Thirteen Balls



Millions saw the apple fall, but Newton asked why.

— Bernard Baruch

How many unit balls can touch one and the same unit ball simultaneously? (We assume that unit balls cannot overlap each other.) This problem is known as the problem of thirteen balls, or the *Gregory-Newton problem*. Applying spherical geometry and a lemma on planar graphs, we answer this problem.

6.1 The Lemma on Proper Diagonals

We start with a geometric statement.

Lemma 6.1 *Let $\breve{\triangle}ABC$ be a spherical triangle with edge-lengths $c = AB, b = AC$, $c \leq b$, in which $\angle B > \pi/2$. Suppose that A, C are fixed and B moves on $\mathbb{S}$ with keeping the length $c = AB$ fixed. Then the following three statements are equivalent:*

(i) *the length of BC decreases,*
(ii) *$\angle BAC$ decreases,*
(iii) *$\angle BCA$ decreases.*

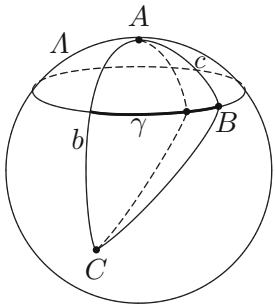
And any of (i), (ii), *and* (iii) *implies that the area $[ABC]$ decreases.*

Proof The equivalence of (i) and (ii) follows from the spherical cosine law. Let Λ be the spherical cap with center A, radius $c = AB$, and let γ the arc of the boundary circle of Λ intercepted by the interior angle $\angle A$ of $\breve{\triangle}ABC$, see Fig. 6.1. Since $c = AB \leq b = AC$ and $\angle ABC > \pi/2$, the arc γ is contained in $\breve{\triangle}ABC$. Hence (ii) is equivalent to the property that B becomes an interior point of the arc γ, which is equivalent to (iii), and also implies that the area $[ABC]$ of $\breve{\triangle}ABC$ decreases. □

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H. Maehara, H. Martini, *Circles, Spheres and Spherical Geometry*,
Birkhäuser Advanced Texts Basler Lehrbücher,
https://doi.org/10.1007/978-3-031-62776-7_6

Fig. 6.1 Lemma 6.1



Remark 6.1 Suppose that $AB = AC$ in $\breve{\triangle}ABC$. Then

$$\angle ABC \lessgtr \pi/2 \Longleftrightarrow AB \lessgtr \pi/2 \quad \text{(the triple symbol } \lessgtr \text{ in the same order)}$$

holds.

Let $\mathcal{Q} = ABCD$ be a spherical quadrilateral. Note that $A^* \neq C$. For otherwise, $\overset{\frown}{ABC}$ is a geodesic segment, and B cannot be a vertex. Suppose that the diagonal AC is contained in $\mathcal{Q}$. Then $\mathcal{Q}$ is divided by AC into two spherical triangles $\breve{\triangle}ABC$ and $\breve{\triangle}ACD$. If D is not an interior point of the cap circumscribed to $\breve{\triangle}ABC$, then AC is called a *proper diagonal* of the spherical quadrilateral $\mathcal{Q}$. In this case, B is not an interior point of $cap(ACD)$. If D lies in the interior of $cap(ABC)$, then the diagonal BD is also contained in $\mathcal{Q}$, and $\mathcal{Q}$ is divided by BD into two spherical triangles $\breve{\triangle}ABD$, $\breve{\triangle}BCD$. In this case, C is not an interior point of $cap(ABD)$, and hence BD is a proper diagonal of $\mathcal{Q}$. If $\mathcal{Q}$ is a cyclic quadrilateral, then both AC, BD are proper diagonals of $\mathcal{Q}$. Thus, a spherical quadrilateral that can be divided into two spherical triangles always has a proper diagonal.

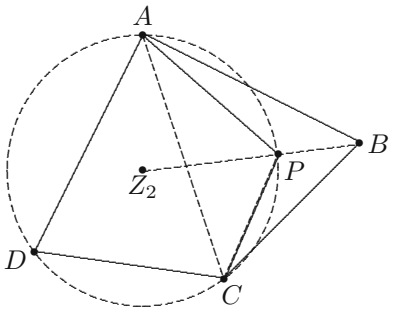
Lemma 6.2 (The Lemma on Proper Diagonals) *Let $\mathcal{Q}$ be a spherical quadrilateral that has a proper diagonal. If we deform $\mathcal{Q}$ with keeping its four edge-lengths so that a proper diagonal becomes shorter, then the area of $\mathcal{Q}$ decreases.*

Proof Let $\mathcal{Q} = ABCD$ and suppose that AC is a proper diagonal of $\mathcal{Q}$. If B lies on the boundary of $cap(ADC)$, then $\mathcal{Q}$ is cyclic. In this case, if we decrease AC with keeping the length of the edges of $\mathcal{Q}$, then $\mathcal{Q}$ becomes not cyclic, and hence its area decreases. So, we may suppose that B lies outside $cap(ADC)$.

(1) First, consider the case that one of $\angle BAC$, $\angle BCA$ is greater than $\pi/2$, and one of $\angle DAC$, $\angle DCA$ is greater than $\pi/2$. If we shorten AC with keeping the lengths of AB, BC, CD, DA, then both $[ABC], [ACD]$ decrease by Lemma 6.1, and hence $[\mathcal{Q}]$ decreases.

(2) Next, suppose $\angle BAC \leq \pi/2$ and $\angle BCA \leq \pi/2$. Let Z_1, Z_2 be the centers of $cap(ABC), cap(ACD)$, respectively. Since $\angle BAC \leq \pi/2$, the arc $\overset{\frown}{BAC}$ is a major arc by Corollary 4.4. Hence Z_1 and A lie in the same side of BC.

Fig. 6.2 Proper diagonal AC



Similarly, since $\angle BCA \leq \pi/2$, Z_1 and C lie in the same side of BA. Therefore, Z_1 is contained in the lune determined by the angle $\angle ABC$ of $\breve{\triangle}ABC$. Let M be the midpoint of the arc $\overset{\frown}{ABC}$. Then Z_1 lies between M and Z_2 and the great circle MZ_2 perpendicularly bisects AC. If Z_1, Z_2, M lie in the same side of AC, then Z_2 is an interior point of $\breve{\triangle}ABC$. If $Z_2 \notin \breve{\triangle}ABC$, then since $\angle Z_2AC < \pi/2, \angle Z_2CA < \pi/2$ and $\angle B^*AC \geq \pi/2, \angle B^*CA \geq \pi/2$, Z_2 is an interior point of $\breve{\triangle}ACB^*$, and hence Z_2 is an interior point of the lune determined by the angle $\angle ABC$ of $\breve{\triangle}ABC$. Let P be the intersection of BZ_2 and the boundary circle of $cap(ACD)$. Since Z_2 is an interior point of the lune with angle $\angle ABC$, P is an interior point of $\breve{\triangle}ABC$, see Fig. 6.2.

Since $\angle APZ_2 < \pi/2, \angle CPZ_2 < \pi/2$, we have $\angle APB > \pi/2$ and $\angle CPB > \pi/2$. Suppose that AC is shortened with keeping the lengths of AB, BC, CD, DA and AP, PC.

(i) Since the cyclic quadrilateral $ADCP$ becomes not cyclic, the area $[ADCP]$ decreases.
(ii) Since $\angle ABC$ becomes smaller, one of $\angle ABP$, $\angle CBP$, say $\angle ABP$, becomes smaller. Then, since $\angle APB > \pi/2$, $\angle BAP$ also becomes smaller by Lemma 6.1, and hence PB becomes shorter, which implies that $[ABP]$ and $[BCP]$ decrease. Therefore $[ABCD] = [ADCP] + [ABP] + [ACP]$ decreases. □

6.2 Major Triangles

A spherical triangle is called a *major triangle* if it contains the center of the circumscribed cap.

Example 6.1

(i) For every $x, y, z \in [\pi/3, \pi/2]$, $\breve{\triangle}(x, y, z)$ is a major triangle.
(ii) For every $x, y \in [\pi/3, 2\pi/3]$, $\breve{\triangle}(x, y, \pi/2)$ is a major triangle.

To see (i), let $\breve{\triangle}ABC$ be a spherical triangle with edge-lengths x, y, z. Then the edge-lengths of the planar triangle $\triangle ABC$ lie between $2\sin(\pi/6) = 1$ and $2\sin(\pi/4) = \sqrt{2}$. Thus the planar triangle $\triangle ABC$ is not an obtuse triangle, and hence it contains the center of the circumscribed circle. Hence $\breve{\triangle}ABC$ also contains the center of circumscribed cap, and it is a major triangle.

(ii) is seen as follows. The planar triangle formed by the three vertices of $\breve{\triangle}(x, y, \pi/2)$ has one edge of length $\sqrt{2}$, and the lengths of the other two edges lie between 1 and $2\sin(\pi/3) = \sqrt{3}$. Hence it is not an obtuse triangle, and it contains the center of the circumscribed circle. Hence $\breve{\triangle}(x, y, \pi/2)$ also contains the center of its circumscribed cap.

Lemma 6.3 (Major Triangle Lemma) *In a major triangle $\breve{\triangle}ABC$, if we decrease the length of AC with keeping the lengths of edges AB, BC, then the area $[ABC]$ decreases.*

Proof For a major triangle $\breve{\triangle}ABC$, the arc $\overset{\frown}{ABC}$ is a major arc, and the lemma follows from Lemma 5.1 and Theorem 5.7. □

6.3 On a Triangulation of a Quadrilateral

To solve the problem of thirteen balls, let us prepare a lemma concerning plane graphs. By a *triangulation of a quadrilateral*, we mean a plane graph such that the boundary of the unbounded outer face is a 4-cycle, and other faces are all triangles. If the number of triangles in a triangulation G of a quadrilateral is N, then G is called a triangulation of a quadrilateral into N triangles. A 3-cycle of a plane graph is called a *non-facial 3-cycle* if it is not the boundary of a face of G.

For example, Fig. 6.3 shows a triangulation of a quadrilateral into 20 triangles. It has 13 vertices and has no non-facial 3-cycle. The maximum degree of this graph is 6.

Lemma 6.4 *Let G be a triangulation of a quadrilateral into* 20 *triangles. If* (i) *G has* 13 *vertices and* (ii) *G has no non-facial* 3*-cycle, then the maximum degree of G is greater than* 5.

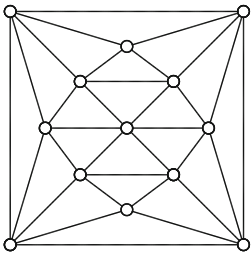


Fig. 6.3 A triangulation of a quadrilateral into 20 triangles

Proof The proof works by contradiction. Suppose that there is a triangulation $G = (V, E)$ of a quadrilateral into 20 triangles such that (i) it has 13 vertices, (ii) it has no non-facial 3-cycle, and (iii) the maximum degree is at most 5.

Since the number of edges is $(4+3\times 20)/2 = 32$, we have $\sum_{v\in V} \deg v = 64$ by the hand-shaking lemma. Since $5 \times 13 = 65$, G must have one vertex of degree 4, and 12 vertices of degree 5. Let $ABCD$ be the boundary 4-cycle of the unbounded face of G. In the following we are going to derive contradictions by dividing our argument into two cases.

Case (1): The degrees of the vertices A, B, C, D are all 5:

From each of A, B, C, D, three edges go into the interior of the "quadrilateral" $ABCD$. Since there is no edge that connects A and C, or B and D by condition (ii), the terminal vertices of these edges emanating from A, B, C, D are different from A, B, C, D. Noting that each of the edges AB, BC, CD, DA of the quadrilateral $ABCD$ must be an edge of a triangular face of G, we may put $v_1, v_2, v_3, \ldots, v_7, v_8$ as the terminal vertices of these edges emanating from A, B, C, D, as shown in Fig. 6.4 left. (At present, we do not know whether these 8 vertices $v_1, v_2, \ldots, v_8$ are all different or not. But we know, for example, that v_2 and v_7 are different by condition (ii).) Among these 12 edges emanating from A, B, C, D into the interior of the quadrilateral $ABCD$, any pair of "neighboring edges" emanating from the same vertex must be two edges of one and the same triangular face. Hence v_1 and v_2 must be adjacent. Similarly, v_2 and v_3, $\ldots$, v_7 and v_8, v_8 and v_1 must be adjacent, respectively. See Fig. 6.4 right. Then, since the maximum degree of each vertex of G is at most 5, we can deduce that the vertices $v_1, v_2, v_3, \ldots, v_8$ are all different. Since the graph G has just one vertex of degree 4, we may suppose that v_7 or v_8 is the vertex of degree 4. Then, from each of the vertices v_2, v_4, v_6, one more edge emanates. Let x, y, z be the terminal vertices of these three edges emanating from v_4, v_6, v_8. (At present, we do not know whether x, y, z are different or not.) By condition (ii), $x \neq v_4, v_8$. Since no other edge emanates from v_2, the edges v_2v_1 and v_2x are neighboring edges. Hence v_1 and x are adjacent. Similarly v_3 and x are adjacent.

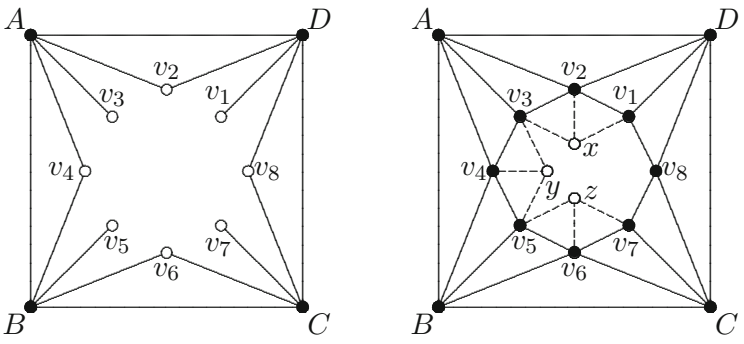


Fig. 6.4 No edge of G lies outside the quadrilateral $ABCD$

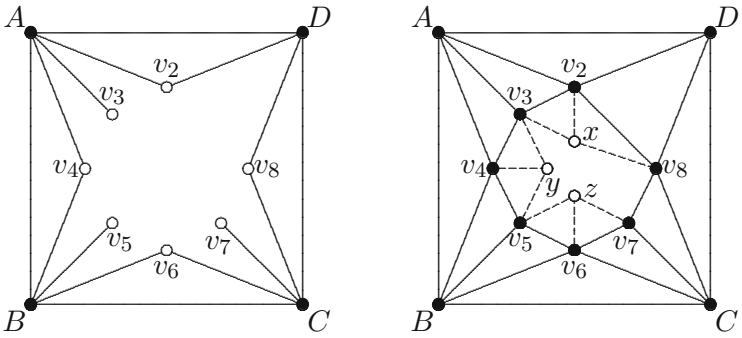


Fig. 6.5 $\deg D = 4$

And similarly, y is adjacent to v_3 and v_5; z is adjacent to v_5 and v_7. Note that from v_3 two more edges must go out other than the edges v_3A, v_3v_2, v_3v_4. Hence x and y must be different by condition (ii). By a similar reasoning, y and z must be different. Since the degree of every vertex is at most 5, we can now deduce that $A, B, C, D, v_1, v_2, v_3, v_4, v_5, v_6, v_7, x, y, z$ are all different. This contradicts condition (i).

Case (2): The vertex D has degree 4 (see Fig. 6.5).

In this case, v_2 is adjacent to v_3, v_8. Similarly to the case (1), from each of v_2, v_4, v_6, one more edge emanates. Let x, y, z be the terminal vertices of theses edges emanating from v_2, v_4, v_6. Then we can see that 14 vertices $A, B, C, D, v_2, v_3, \ldots, v_8, x, y, z$ are all different, which contradicts the assumption (i) of the theorem.

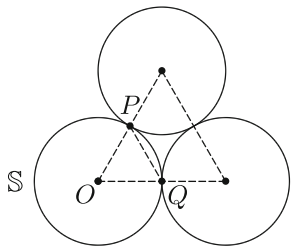
□

6.4 The Problem

How many unit balls can touch one and the same unit ball simultaneously? (We assume unit balls cannot overlap each other.) When two unit balls touch the unit sphere $\mathbb{S}$, the spherical distance between the two contact points is clearly greater than or equal to $\pi/3$, see Fig. 6.6. Conversely, if P, Q are points on the unit sphere $\mathbb{S}$ such that the spherical distance between them is at least $\pi/3$, then mutually non-overlapping two unit balls can touch $\mathbb{S}$ at P and Q. So, the problem is equivalent to the following one: How many points can be put on $\mathbb{S}$ so that the spherical distance between any pair is at least $\pi/3$?

Consider a regular icosahedron inscribed in $\mathbb{S}$. By projecting the edges of the icosahedron onto $\mathbb{S}$ by the central projection from the center of $\mathbb{S}$, the sphere $\mathbb{S}$ is divided into 20 equilateral spherical triangles with area $4\pi/20 \approx 0.628$. The area of

Fig. 6.6 $\angle POQ \geq \pi/3$



an equilateral spherical triangle with edge-length $\pi/3$ is computed as follows: By Euler's formula for the spherical excess, we have

$$\cos\varepsilon = \frac{1+3/2}{4\cdot 3\sqrt{3}/8} = \frac{5}{3\sqrt{3}},$$

and $\varepsilon = \cos^{-1}\frac{5}{3\sqrt{3}} \approx 0.2756$. Hence

$$\text{the area of } \breve{\triangle}(\pi/3, \pi/3, \pi/3) = 2\varepsilon \approx 0.551.$$

Since $4\pi/20 \approx 0.628 > 0.551$, we see that the spherical distance between any two vertices of the regular icosahedron is greater than $\pi/3$. Since an icosahedron has 12 vertices, we can deduce that at least 12 unit balls can simultaneously touch one and the same unit ball.

6.5 Solution

Theorem 6.1 *The maximum number of unit balls that can simultaneously touch one and the same unit ball is twelve.*

This section is dedicated to the proof of this theorem. Let n be the maximum number of unit balls that can simultaneously touch one and the same unit ball. As shown in the previous section, we have $n \geq 12$.

Suppose that n unit balls simultaneously touch a unit ball with boundary $\mathbb{S}$. Then the spherical distance between any two contact points on $\mathbb{S}$ is at least

$$a := \pi/3 \approx 1.047.$$

(Hereafter, the symbols a, b, c are used to stand for special constants.) Let $\mathcal{X}$ denote the set of n contact points on $\mathbb{S}$. Then $\mathcal{X}$ does not lie in a hemisphere. For otherwise, another unit ball can touch $\mathbb{S}$. Hence, the convex hull $conv(\mathcal{X})$ of $\mathcal{X}$ is a convex polyhedron containing the center of $\mathbb{S}$ in its interior. For any facet F of $conv(\mathcal{X})$, the circumscribed spherical cap of F has spherical radius less than $\pi/3$. For otherwise,

another unit ball can touch at the center of the spherical cap of F. Triangulate those facets of $conv(\mathcal{X})$ that have more than 3 edges by adding suitable diagonals. Then it follows easily from Euler's formula that the number of triangles is equal to $2n-4$. Project the triangulation of the surface of $conv(\mathcal{X})$ onto $\mathbb{S}$ by the central projection from the center of $\mathbb{S}$. Then $\mathbb{S}$ is triangulated into $2n-4$ spherical triangles. In the following, we are going to show that if $n \geq 13$, then

[A] at most one edge of the above triangulation of $\mathbb{S}$ has length $\geq \cos^{-1}\frac{1}{7}$,
[B] the planar graph obtained from the triangulation of $\mathbb{S}$ by eliminating the edges of length $\geq \cos^{-1}(1/7)$ has maximum degree ≤ 5, but
[C] no such planar graph exists.

Remark 6.2 The lower bound $\cos^{-1}(1/7)$ in [B] is chosen to guarantee the degree condition.

Cut $\mathbb{S}$ along the edges of the spherical triangles in the triangulation of $\mathbb{S}$ into $2n-4$ pieces. Let $\mathcal{D}$ denote the collection of these $2n-4$ pieces (spherical triangles). Note that for any triangle in $\mathcal{D}$,

($*$) the length of the shortest edge is at least $a = \pi/3$, and the radius of the circumscribed spherical cap is less than a.

Lemma 6.5 *The length of the second longest edge of each triangle in $\mathcal{D}$ is less than*

$$c := \cos^{-1}\left(\frac{1-\sqrt{6}}{4}\right) \approx 1.942.$$

Proof Among the spherical triangles satisfying ($*$), the supremum s of the length of the second longest edge is attained by an isosceles triangle with edge lengths $s, s, \pi/3$ inscribed in a spherical cap of radius $\pi/3$, see Fig. 6.7.

Let φ be the interior angle of an equilateral triangle with edge length $a = \pi/3$. By the spherical cosine law, we have

$$\cos\varphi = \frac{\cos a - \cos^2 a}{\sin^2 a} = \frac{1}{3}.$$

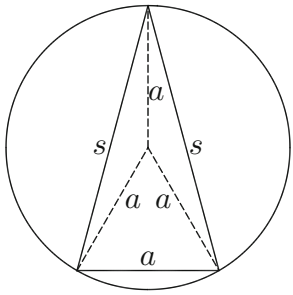


Fig. 6.7 An isosceles triangle with base $a = \pi/3$

Hence

$$\cos s = \cos^2 a + \sin^2 a \cos(\pi - \varphi/2),$$

from which we have $s = \cos^{-1}(\frac{1-\sqrt{6}}{4})$. □

Put

$$b = \cos^{-1}(1/7) \approx 1.427.$$

By the spherical cosine law, the base angle of the isosceles triangle with edge lengths a, a, b is computed as $\pi/3$.

Lemma 6.6 *Let $ABCD$ be a convex quadrilateral each of whose edges has length at least $a = \pi/3$. Suppose that $AD = b$ and $b < BD < c$. Then $AC > b$.*

Proof If the lengths of three edges AB, BC, CD are not all equal to $a = \pi/3$, then, with keeping the lengths of CD, DA, BD, shorten the edges AB, BC to the length a. Then, by the spherical cosine law, the angles $\angle ADB, \angle BDC$ decrease, and hence $\angle ADC$ decreases. Therefore AC becomes shorter. Next, with keeping $BC = a$ and the lengths of BD, shorten CD to length a. Then $\angle DBC$ (and hence $\angle ABC$) decreases, and AC decreases. Hence it is sufficient to prove the lemma in the case $AB = BC = CD = a$. In a spherical triangle $\breve{\Delta}(x, y, z)$, denote by $f(x, y, z)$ the cosine of the angle opposite to the edge of length z, that is,

$$f(x, y, z) = \frac{\cos z - \cos x \cos y}{\sin x \sin y}.$$

Then $f_z(x, y, z) < 0$ and

$$f_x(x, y, z) = \frac{\cos y - \cos x \cos z}{\sin^2 x \sin y}.$$

Put $BD = x$. Then $\cos \angle CBD = f(x, a, a)$, and $\cos \angle ABD = f(x, a, b)$. Since $f_x(x, a, a) > 0$, $f_x(x, a, b) > 0$, we have $f(x, a, a) < f(c, a, a) \approx 0.8439$ and $f(x, a, b) < f(c, a, b) \approx 0.4016$. Then

$$\angle ABC = \angle ABD + \angle CBD > \cos^{-1} f(c, a, b) + \cos^{-1} f(c, a, a) \approx 1.7238,$$

and hence

$$\cos AC < \frac{1}{4} + \frac{3}{4} \cos \angle ABC \approx 0.1357.$$

Therefore, $AC > \cos^{-1} 0.1357 \approx 1.437 > b$. □

Now we return to the collection $\mathcal{D}$ of cut-out spherical triangles. If there are edges longer than b in the edges of spherical triangles in $\mathcal{D}$, then eliminate them by repeating the following procedure 1, 2, 3.

1. Find the longest edge e and take out from $\mathcal{D}$ the pair of spherical triangles that have the edge e in common.
2. Make a spherical quadrilateral by attaching these two spherical triangles along the edge e.
3. Deform the quadrilateral with keeping its edge-lengths so that the length of a proper diagonal becomes b. Then cut the quadrilateral along the diagonal of length b into two spherical triangles, and paint these triangles red, and return them to $\mathcal{D}$.

Comments

(i) If neither of a pair taken out from $\mathcal{D}$ is red, then e itself is the proper diagonal of the quadrilateral obtained by attaching the pair of triangles. (This can be seen as follows: If a triangle is not red, then its three vertices must have been the vertices of a facet of $conv(\mathcal{X})$. Since the interior of the circumscribed spherical cap of a facet of $conv(\mathcal{X})$ cannot contain vertices of $conv(\mathcal{X})$, e itself must be a proper diagonal of the quadrilateral.) Thus, in this case, the proper diagonal e is shortened to length b.

(ii) In every red triangle, all edges are shorter than c by Lemma 6.5. Hence, if one of a pair of triangles taken out from $\mathcal{D}$ is red, then e is shorter than c, and the proper diagonal of the quadrilateral (maybe not e) is longer than b by Lemma 6.6. The proper diagonal is also shortened then.

(iii) Thus, proper diagonals are always shortened by the above procedure, and the sum of the areas of the triangles in $\mathcal{D}$ decreases by the lemma of proper diagonals.

(iv) The number of edges with length $\geq b$ is preserved.

If all edge-lengths of the triangles in $\mathcal{D}$ become less than or equal to b, then shorten those edges with length shorter than b to the length a. By the major triangle lemma, the sum of areas of the triangles in $\mathcal{D}$ also decreases in this new change. Thus, finally every triangle in $\mathcal{D}$ falls into one of the four types of triangles with edge-lengths

$$(a, a, a),\ (a, a, b),\ (a, b, b),\ (b, b, b),$$

respectively. Let

$$\delta,\ \alpha,\ \beta,\ \gamma$$

denote the area of these four types of triangles, respectively. Applying Euler's formula for spherical excess, we can compute their values as

$$\delta \approx 0.5513,\ \alpha \approx 0.667,\ \beta \approx 0.8923,\ \gamma \approx 1.1948.$$

Since the sum of these $2n-4$ triangles in $\mathcal{D}$ is less than the surface area 4π of $\mathbb{S}$, we have

$$2n-4 < 4\pi/\delta \approx 22.8.$$

Hence $n \leq 13$.

Suppose $n = 13$. Then there are 22 spherical triangles of the above four types in $\mathcal{D}$. Let $n_\delta, n_\alpha, n_\beta, n_\gamma$ denote the number of triangles in the types (a,a,a), (a,a,b), (a,b,b), (b,b,b), respectively. Then

$$\begin{aligned} 12.566 \approx 4\pi &> (22 - n_\alpha - n_\beta - n_\gamma)\delta + n_\alpha\,\alpha + n_\beta\,\beta + n_\gamma\,\gamma \\ &= 22\delta + n_\alpha(\alpha-\delta) + n_\beta(\beta-\delta) + n_\gamma(\gamma-\delta) \\ &\approx 0.5513 \times 22 + 0.1157\,n_\alpha + 0.341\,n_\beta + 0.6435\,n_\gamma, \end{aligned}$$

namely,

$$0.438 > 0.1157\,n_\alpha + 0.341\,n_\beta + 0.6435\,n_\gamma.$$

Therefore $n_\gamma = 0$. If there is a triangle $\breve{\triangle}$ of area β in $\mathcal{D}$, then there must be at least two other triangles each of which shares an edge of length b with $\breve{\triangle}$. Then the value of the right-hand side in the above inequality becomes at least $0.1157 \times 2 + 0.314 = 0.5724$, which is impossible. Hence $n_\beta = 0$. Since the number of edges with length b is even, n_α is also even. Hence $n_\alpha = 0$ or 2. Therefore, only the following two possibilities remain: (1) $\mathcal{D}$ consists of 22 triangles of area δ, or (2) $\mathcal{D}$ consists of 20 triangles of area δ and 2 triangles of area α. This implies that in the triangulation of $\mathbb{S}$ with vertex set $\mathcal{X}$, at most one edge has length greater than of equal to b. This proves [A].

Let G be the graph obtained from the triangulation of $\mathbb{S}$ by removing the edge of length $\geq b$. By the next lemma, for every vertex of G, the angle determined by two neighboring edges emanating from the vertex is greater than $\pi/3$. Hence the maximum degree of G is at most 5, which proves [B].

Lemma 6.7 *If a spherical triangle $\breve{\triangle}ABC$ on $\mathbb{S}$ satisfies*

$$a \leq AB \leq BC < b,\ a \leq AC,$$

then $\angle B > \pi/3$.

Proof Let $x = AB$, $y = BC$, $z = CA$ and $f(x,y,z) = \cos\angle B$. By the spherical cosine law $\cos z = \cos x \cos y + \sin x \sin y \cos\angle B$, we have

$$f(x,y,z) = \cos\angle B = \frac{\cos z - \cos x \cos y}{\sin x \sin y}.$$

Since

$$f_z(x, y, z) = \frac{-\sin z}{\sin x \sin y} < 0,$$

we have $f(x, y, z) \le f(x, y, a)$ for $a \le x \le y \le b, a \le z < \pi$. And since

$$f_y(x, y, a) = \frac{\cos x - \cos y \cos a}{\sin^2 y \sin x} > 0$$

for $a < x \le y < b$, we have $f(x, y, a) < f(x, b, a)$. Finally, from

$$f_x(x, b, a) = \frac{-2 + 7\cos x}{8\sqrt{3}\sin^2 x}$$

$$f_{xx}(x, b, a) = \frac{-4\cos x + 7(1 + \cos^2 x)}{3\sqrt{2}\sin^3 x},$$

we have $f_{xx}(x, b, a) > 0$ in $a \le x \le b$, and hence $f(x, b, a)$ takes its maximum value either at $x = a$ or at $x = b$. Since $f(a, b, a) = 1/2 > 47/96 = f(b, b, a)$, we have $f(x, y, z) < f(a, b, a) = 1/2$. Thus, $\cos \angle B < 1/2$, that is, $\angle B > \pi/3$.

□

If there is no edge of length $\ge b$ in the triangulation of $\mathbb{S}$, then G has $(3\times 22)/2 = 33$ edges. Hence the average degree of the vertices is $66/13$, which is greater than 5, a contradiction. So there must be an edge of length $\ge b$ in the triangulation of $\mathbb{S}$. Then G has 20 triangular faces and one quadrilateral face. If XYZ be a 3-cycle of G, then since the lengths of XY, YZ, ZX are all less than b, the area $[XYZ]$ is less than γ. If $\breve{\triangle}XYZ$ contains a vertex of $\mathcal{X}$ in its interior, then $\breve{\triangle}XYZ$ contains at least three triangles in the triangulation of $\mathbb{S}$, and hence the area $[XYZ]$ is at least $3\delta \approx 1.6539$. This is greater than $\gamma \approx 1.1948$, and impossible. Thus, every 3-cycle of G is a boundary cycle of a triangular face.

Thus, we derived that if $n = 13$, then there is a graph G on the sphere such that it has 13 vertices, one quadrilateral face, and 20 triangular face, every 3-cycle is the boundary of a triangular face, and maximum degree is at most 5.

Let $P \in \mathbb{S}$ be a point inside the quadrilateral face of G, Then, by the stereographic projection φ from P, G is mapped to a plane graph $\varphi(G)$ on the plane tangent to $\mathbb{S}$ at P^*, the antipodal point of P. This graph $\varphi(G)$ is a triangulation of a quadrilateral into 20 triangles, with 13 vertices and having no non-facial 3-cycle. And the maximum degree of $\varphi(\breve{G})$ is less than 6, which contradicts Lemma 6.4. This proves [C].

Remark 6.3 The first rigorous proof of the fact that the maximum number of unit spheres that can touch one and the same unit sphere is 12 was obtained by K. Schütte and B. L. van der Waerden [716]. There appeared many proofs of this problem, e.g., in [33, 106, 145, 499, 523, 588]. The above proof is based on [520].

6.6 Exercises

6.1 Prove that if the shortest edge of a spherical triangle T on the unit sphere $\mathbb{S}$ has length at least $\pi/2$, then T is a major triangle.

6.2* Prove that there is no quadrangulation with nine vertices.

6.3 Prove that the maximum number of balls of radius 1.1 that can simultaneously touch one and the same unit ball is twelve.

6.4* Let T be a spherical triangle on the unit sphere $\mathbb{S}$ with edge lengths a, b, c ($a \leq b \leq c$). If the spherical radius of the circumscribed cap of T is less than a, then the area of T is at least the area of $\breve{\triangle}(a, a, a)$. This result is known as Fejes Tóth's Lemma. (For a proof of this lemma, see Exercise 7.3*.) Using this lemma prove the following: The maximum number of balls of radius 2 that can simultaneously touch one and the same unit ball is six.

6.5 (i) Check that the distance between any two points in the eight points

$$(\tfrac{\sqrt{7}}{5}, 0, \pm\tfrac{3\sqrt{2}}{5}), (\tfrac{\sqrt{7}}{5}, \pm\tfrac{3\sqrt{2}}{5}, 0), (\tfrac{-\sqrt{7}}{5}, \pm\tfrac{3}{5}, \pm\tfrac{3}{5})$$

on the unit sphere is at least $6/5$.

(ii) Using Fejes Tóth's Lemma prove that the maximum number of balls of radius $3/2$ that can simultaneously touch one and the same unit ball is eight.

6.6 (i) Check that the distance between any two points in the nine points

$$(0, 0, 1), (\pm\tfrac{\sqrt{83}}{7\sqrt{3}}, \pm\tfrac{4}{7}, \tfrac{4}{7\sqrt{3}}), (\pm\tfrac{\sqrt{83}}{7\sqrt{3}}, 0, -\tfrac{8}{7\sqrt{3}}), (0, \pm\tfrac{\sqrt{3}}{2}, -\tfrac{1}{2})$$

on the unit sphere $\mathbb{S}$ is greater than $8/7$.

(ii) Using Fejes Tóth's Lemma, prove that the maximum number of balls of radius $4/3$ that can simultaneously touch one and the same unit ball is nine.

6.7 Notes

We restrict these notes to only one point, namely the problem of thirteen balls itself. The reason is simple: this problem created sufficiently many activities and interesting further discussions.

6.7.1 Notes on the Problem of Thirteen Balls

We want to show parts of the network that the 13-spheres problem was able to create. As already its discussion in the basic monograph [262] shows, this structure

of notions and results clearly belongs to the field of *discrete geometry*. Due to the large variety of related papers that exist, we shall lay emphasize mainly (but not only) on survey-like papers.

The Problem Itself and Its Proofs

In 1694, Newton and Gregory discussed the *problem of thirteen balls*. The first complete, rigorous solution to this problem is due to Schütte and van der Waerden [716]. They confirm Newton's opinion that no more than twelve unit balls can do this, and they use an interesting mixture of trigonometric and topological ideas. Shortly after this success, Leech [499] presented an elegant new proof of that result. His approach is based on the confirmation of the non-existence of a polyhedron with 13 vertices (one of valence 4, and twelve of valence 5) and 21 facets (20 triangular ones and 1 quadrangle), and this proof is usually chosen when the result and its proofs are presented somewhere (meanwhile there are several proofs of the result). E.g., in the books [11] of Aigner and Ziegler as well as [839] due to Zong one can find variants of Leech's proof. And Maehara [520] nicely reproves this result with the help of the *isoperimetric theorem for spherical polygons* (which is also elegantly derived).

Another approach (using *spherical Delaunay triangulations*) is due to K. Böröczky [106], see also Sect. 4.4 of the book [107]. And also Hsiang [405, Chapter 2] gives a new proof of Newton's opinion. Anstreicher [33] presents a proof based on *linear programming bounds* and (in another way than above) on properties of spherical Delaunay triangulations which is not (as in former proofs) directly depending on the dimension. Following the arguments of Leech, but using also a tool from spherical geometry, Maehara [523] succeeded in giving an elementary proof, suitable even for *undergraduate students*. And a very short proof, using *linear programming methods*, is also given by Glazyrin [304]. Chapter 1 of Zong's booklet [839] gives a concise and elegantly written overview to the whole subject.

First Generalizations: Related Book Chapters and Surveys

We continue with the concept of optimal packings, where the following steps of generalizations are already included: to replace Euclidean spheres by convex bodies, to consider the global situation, and to look also at higher dimensions. Due to the richness of this framework, we discuss mainly *surveys*, sometimes also given in form of *book chapters*.

Coxeter [184] gives a possible upper bound for the number of congruent non-overlapping $(n-1)$-spheres in n-space which can touch a given one of the same radius. He comprehensively discusses also the history of the problem (from the 17th till to the middle of the twentieth century, with an impressive list of references). The contributions of famous mathematicians like Kepler, Newton and Gregory, Schläfli

(who was the first to bring spherical trigonometry up to n dimensions), Poincaré, Sommerville, Guinand, and Minkowski are comprehensively presented.

In view of generalizations, we define now Newton numbers and related concepts referring to convex bodies. For each n-dimensional *convex body* K (i.e., for each compact, convex set with interior points in n-space), the maximum number of neighbors of a member of any packing with congruent copies of K is called the *Newton number* $N(K)$ *of* K. Also the name *kissing number* is often used. Thus, the dispute between Newton and Gregory refers to the subcase that K is a three-dimensional ball in 3-space. If "congruent" is replaced by "translative", we speak about the *Hadwiger number* $H(K)$ *of* K (or translative kissing number). The latter notion can be conveniently used also for *normed* or even *gauge spaces*, identifying K with the respective unit ball (in the non-symmetric case, i.e., for gauge spaces, inducing even an asymmetric distance function). E.g., from this viewpoint Hadwiger numbers are discussed by Swanepoel in the survey [765].

Surveys on Newton and Hadwiger numbers of convex bodies are Chapters 13.1 and 13.2 of the monograph [262], Section 5.3 of the handbook article [261], and Chapter 4 of Zong's book [837]. Also K. Bezdek surveys in [85] Hadwiger and Newton numbers of convex bodies, but also variants of them, called *one-sided Hadwiger and kissing numbers*.

Sloane [741] gives a nice exposition on the historical development of *sphere packing problems* and related methods. He excellently discusses kissing numbers and considers related *lattice and non-lattice packings*. Connections to other fields are shown, too. The article [741] appeared shortly before the first announcement of Hales regarding the Kepler conjecture, and the reader is referred there to the forthcoming third edition of the book [176] of Conway and Sloane, containing (with respect to the second edition) 800 new referencces. This important book contains much material on problems treating sphere packings. In particular, also kissing numbers are discussed (see in [175] Section 2.1 of Chapter 1 for the 13-spheres problem, Section 2.2 of Chapter 1 for other dimensions, and Chap. 13 for bounds and numerical results). Various results where lattice arrangements are optimal can be found there, too. Interesting connections to related fields (like *finite groups, quadratic forms, error-correcting codes, number theory* etc.) are discussed and successfully applied, and also applications outside of mathematics (chemistry, superstring theory and so on) are studied.

Zong [838] gives a nice survey on *Hadwiger numbers* of n-dimensional convex bodies K, and the subcase where the translates touching K are taken from a lattice packing (thus yielding *lattice kissing numbers*) is also taken care of. In some cases also proofs are presented. The same author surveys results on the relations between *kissing numbers, covering numbers* (i.e., the smallest numbers of translates of the interior of K covering K) and the relatively new related concept of blocking numbers in [842]. (Note that the *blocking number* of a convex body K is the smallest number of non-overlapping translates of K all of which touch K at its boundary such that their union makes it impossible that any further non-overlapping translate of K can touch K.) It turns out that blocking numbers form a "bridge" between the other

two notions, and such relations like also further results are excellently presented in [842].

Coming back to the *Newton-Gregory problem*, we refer to the surveys of Casselman [145] as well as Pfender and Ziegler [635]. They present related historical facts, from the classical beginnings till to extended problems, e.g. also referring to 4-space. Based on a method initiated by Delsarte in the early seventies (concerning inequalities for the distance distributions of kissing configurations; see the fundamental work [211] of Delsarte et al. on spherical codes and designs), interesting results regarding dimensions eight and twenty-four are discussed. We note that Delsarte's method strongly influenced further developments; see again [635]. So the authors discuss also Musin's excellent solution of the four-dimensional case (see [589]), based on a modification of Delsarte's method, and the paper [169] of Cohn and Elkies, opening new possibilities to deal with optimal sphere packings more efficiently. Various further results are offered, also treating optimal *lattice* packings of spheres. Basic ideas and tools are nicely explained.

In [337] Gruber gives an interesting overview regarding *lattice packing and covering problems*, related mainly to the geometry of numbers and referring to n-dimensional balls and convex bodies. He offers more recent results on uniqueness of densest lattice packings and the thinnest lattice coverings of a generic convex body. Estimates and bounds for the kissing number of densest lattice packings of generic convex bodies are characterized.

The concept of spherical codes (cf. again Delsarte et al. [211]) is more general than that of kissing numbers. (Recall that a *spherical code* with parameters (n, m, t) is a set of m points on the unit hypersphere in n dimensions for which the dot product of unit vectors from the origin to any two points is less than or equal to t. E.g., the kissing number problem and the Tammes problem can be formulated as subcases in terms of this concept.) Keeping this more general level of spherical codes, Boyvalenkov et al. [120] give a comprehensive survey on results and methods referring to kissing numbers.

Liberti [507] gives a relatively short review on bounds for kissing numbers of n-dimensional unit balls based on *mathematical programming*, forming also a cornerstone for the theory of spherical codes. This might be the right place to mention also a few important research papers, in which (motivated also by Delsarte's approach) methods from *semidefinite programming* are used to get upper bounds for kissing numbers in higher dimensions (different to the known cases $n = 1, 2, 3, 4, 8, 24$). Bachoc and Vallentin [51] consider $A(n, \alpha)$, i.e., the maximum number of points on the unit $(n-1)$-sphere with minimal angular distance α; the *kissing number problem* is then equivalent to finding $A(n, \pi/3)$. A "semidefinite program" is created which yields, for some dimensions n, better upper bounds for $A(n, \alpha)$ than known ones. Better software, faster computers and optimized use of computation time enable Mittelmann and Vallentin [579] to get further improvements in several cases. The results of Kuklin [480] go into the same direction. And results on *lattice kissing numbers* (where the packed spheres are centered at the points of an n-dimensional lattice and have midpoints at minimal distance from the center of the kissed sphere) are obtained by Skoruppa [740].

Cohn [167] gives, based on five lectures (from 2014), an excellent overview to several aspects of sphere packings, e.g. including *lattices, linear programming bounds, kissing numbers* and *spherical codes*. Unfortunately, the later spectacular results of Viazovska on sphere packings in dimension 8 (see [795]) and of Viazovska with colleagues (cf. [170]) in dimension 24 are not contained in this very readable material. (Note that Viazovska got the Fields medal for her results in dimension 8.) However, Cohn's later survey [168] contains these deep results and is highly recommendable. This overview is completed by Viazovska's survey [796], where the so-called *sharp sphere packings* are discussed, whose density attains the linear programming upper bound. In [796] she mainly concentrates on dimensions 8 and 24, where her work is located; in these dimensions no unit ball packing has larger density than the 8-dimensional lattice packing and the Leech lattice packing, respectively. A new lower bound on the density of packings of congruent spheres is given by Campos et al. in [140].

There are also *problem books* discussing the issue of 13 spheres and related topics. The first one is [190] by Croft, Falconer and Guy, where problems closely related to our discussion here (i.e., the Kepler conjecture and kissing numbers) are summarized in Sections D10 and D12. And the monograph [125] of Brass, Moser and Pach treats such questions in Sections 1.8, 1.9, and 2.4.

Further Extensions and Variations

We start with a few older (but refreshing) extensions and modifications, and we continue then with more recent contributions.

L. Fejes Tóth and Heppes consider the following *extension of the problem of thirteen spheres* in n-space: Let A be a unit ball in n-space, let B be a family of unit balls all of them touching A, and C be a family of unit balls, each of them touching a member of B. Let $Q = B \cup C$, and denote by q_n the maximal cardinality that such a set Q can have. The authors prove that $q_2 = 18, 56 \leq q_3 \leq 63$, and $168 \leq q_4 \leq 232$.

Inspired by the Gregory-Newton problem, Propp [647] discusses the following problem: what *permutations* can be obtained if one marks the 12 touching spheres and lets them change their positions at will? Using analytic and group theoretic methods, interesting results (in some sense related to Rubik's famous cube) are obtained. E.g., the arrangement of 12 such balls, with midpoints forming the vertex set of a regular icosahedron, can be deformed in an arrangement forming the vertex set of an Archimedean cuboctahedron. Also it is conjectured that each general position of 12 equal balls around a congruent one can be generated in such a way from any other arrangement. Grünbaum proves in [338] that the *Hadwiger number* of a planar convex body is realized by a lattice packing which is not true for 9-dimensional balls. Zong [836] constructs a counterexample for any dimension $n \geq 3$. It is some suitable truncation of the cube. G. Kuperberg and Schramm [481] consider *finite packings* P of p *non-congruent* balls in 3-space. For m giving the number of contact points of the given balls, the authors call $k(P) = 2m/p$

the *average kissing number* of P. If k denotes the supremum of all numbers $k(P)$ over all such finite packings, the authors show that $12.566 \leq k \leq 14.928$, which is particularly interesting regarding the lower bound (since for congruent balls this number is 12). Using *semidefinite programming*, in [223] by Dostert et al. new upper bounds on the average kissing number for certain dimensions are derived. And in [538] *kissing numbers of non-congruent balls* are investigated, namely regarding the smallest number of non-overlapping unit balls which can touch a ball of radius $r > 0$ in n-space.

Dostert and Kolpakov [222] find new upper and lower bounds for kissing numbers of congruent spheres in spherical and hyperbolic n-space.

Clearly, the 13-spheres problem is essential for the local situation in the sphere packing problem, asking for the densest packing of infinitely many congruent spheres in 3-space. The *Kepler conjecture* (or the so-called *sphere packing problem*) says that the *densest packing of congruent spheres* in 3-space is reached by the *face-centered cubic lattice packing* with density $\pi/\sqrt{18}$. With the help of Ferguson, Hales published (in a series of papers) a proof in 2005 and 2006. A revision followed in 2010, and all related papers, including this revision and further supporting material, can be found in the book [374] of Hales and Ferguson. Around 2003 Hales started with a project named FLYSPECK to create an automated (computerized) formal verification of his proof. In the strongly related book publication [372] Hales deals with a proof of the Kepler conjecture written in a formal logical system and being logically independent of the former proof. Presenting, more precisely, a blueprint version of this formal proof in [372], this book brings the above mentioned project closer to the reader. Not requiring knowledge on details from parts of the previous approaches, it is self-contained; see also the review [485] of Lagarias on it. The officially published completion of the FLYSPECK project is confirmed in [376] (2017). For comprehensive overviews to the Kepler conjecture and its history we refer to the book [768] of Szpiro and to Sections 7.1 and 7.2 as well as subsection 8.7.1 of the monograph [262] of L. and G. Fejes Tóth and W. Kuperberg. Two further nice expositions on the Kepler conjecture and its history are due to Henk and Ziegler [392] as well as to Joswig [434].

Related Problems

Böröczky and Szabó [111] study the strongly related *Tammes problem* of arranging k points on the unit 2-sphere so that the minimum distance between any two of them is maximized; see also the classical book [257] of L. Fejes Tóth and its strong extension [262, Section 8.5] for related references. Denoting this maximum by a_k, in [111] it is shown that $a_{13} < 1.02746114$. This implies the following: If $B_1, \ldots, B_{14}$ are 14 different unit balls in 3-space, forming a packing so that each of the balls $B_2, \ldots, B_{13}$ touches B_1, then the distance between the centres of B_1 and B_{14} is at least $2.20527\ldots$. Also Musin and Tarasov [592] treat the Tammes problem (placing 13 points suitably on the unit sphere) and its importance for the thirteen-spheres problem. Musin and Tarasov solve this problem by showing that

a certain arrangement of 13 points is best possible, namely at the minimal angular distance of ca. 57.1367 (an equivalent formulation of the thirteen-spheres problem asks whether the respective distance is smaller than 60). Depending on this approach to the Tammes problem, Böröczky and Szabó [112] present a geometric proof of the related Fejes Tóth conjecture that the balls in any 12-neighbour packing of unit balls in 3-space are positioned in hexagonal layers, as in a densest lattice packing. The Tammes problem and similar problems around *circle packing types* are nicely discussed in Berger's refreshing survey [76] which treats also various related things.

The *dodecahedral conjecture*, going back to L. Fejes Tóth [256], refers to the Voronoi cells of all ball centers from packings of unit balls in 3-space. It says that in any such packing every Voronoi cell has volume at least that of a Platonic dodecahedron having a unit sphere as insphere, where equality holds exactly when the Voronoi cell is such a Platonic solid. It was positively solved in 1998, and the paper [375] by Hales and McLaughlin contains a strongly revised version of the original proof which, however, is essentially the same as the original one. The paper [375] contains all basic references to the problem. The dodecahedral conjecture is not directly related to the Kepler conjecture, but the proofs of both share various ideas and methods.

And in the paper [373] Hales gives outlines of proofs of two further problems on packings of unit spheres in 3-space: the *strong dodecahedral conjecture*, and the conjecture of L. Fejes Tóth on *sphere packings with kissing number 12*. The *strong dodecahedral conjecture* is due to K. Bezdek [84] and says that in such a sphere packing the surface area of any Voronoi cell is at least as large as the surface area of the Platonic dodecahedron with inradius 1. The other, already mentioned conjecture goes back to the paper [258] of L. Fejes Tóth saying that if each sphere in a packing of unit spheres in 3-space is touched precisely by twelve other spheres, then this packing consists of hexagonal layers. Hales [373] carefully explains the basic geometric ideas and methods as well as their combinations with the computer calculation parts, to give a clear picture of the computer assisted proofs of both conjectures.

Musin presents in [591] five essays on topics close to L. Fejes Tóth, treating kissing numbers and related topics. Among them are the Tammes problem and his bound on circle packings, the maximum kissing number of packings on the sphere, and one-sided kissing numbers.

Chapter 7
Spherical Geometry III



The art of doing mathematics consists in finding that special case which contains all the germs of generality.

— David Hilbert

A spherical ellipse is similarly defined as an ellipse in the plane. It is the locus of a point X on $\mathbb{S}$ for which the sum of the distances from two fixed points to X is constant. A regular spherical polygon is a cyclic polygon whose edge-lengths are all equal. Using spherical ellipses in this chapter, we show that among cyclic n-gons inscribed in a cap of fixed spherical radius, regular n-gons have the maximum perimeter, and among the n-gons on $\mathbb{S}$ with fixed perimeter, regular n-gons have the maximum area.

7.1 Spherical Ellipses

Let $\mathbb{S}$ be the unit sphere centered at the origin O of $\mathbb{R}^3$. Let $A, B \in \mathbb{S}$ be a pair of points such that $AB = 2\phi < \pi$. For a given s, $\phi < s < \pi - \phi$, the set

$$\mathcal{E}(A, B : s) = \{X \in \mathbb{S} : AX + BX = 2s\}$$

is called a *spherical ellipse* with *foci* A, B and *parameter* s.

First, let us observe that a spherical ellipse on $\mathbb{S}$ is a simple closed curve on $\mathbb{S}$. Suppose that the midpoint M of AB is the south-pole $(0, 0, -1)$ of $\mathbb{S}$. Then, each meridian (a semicircle connecting the north-pole to the south-pole) intersects $\mathcal{E}(A, B : s)$ exactly in one point. (Since $AM + BM = 2\phi < 2s$, and $AM^* + BM^* = 2\pi - 2\phi > 2s$, each meridian intersects $\mathcal{E}(A, B : s)$. If two distinct points $X, X' \in \mathcal{E}(A, B : s)$ lie on the same meridian, then one of $\breve{\triangle}ABX$, $\breve{\triangle}ABX'$ contains the other. Then $AX + BX \neq AX' + BX'$, a contradiction.) Similarly, for two distinct points $X, Y \in \mathcal{E}(A, B : s)$, none of $\breve{\triangle}ABX$, $\breve{\triangle}ABY$ contains the other.

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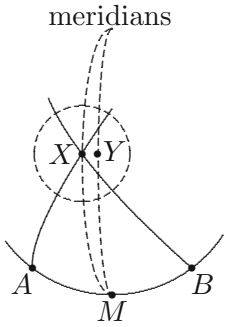


Fig. 7.1 None of $\breve{\triangle}ABX$, $\breve{\triangle}ABY$ contains the other

This implies that Y lies in the angular region (lune) of an exterior angle of $\breve{\triangle}ABX$ at the vertex X, see Fig. 7.1. Therefore, for any given ϵ-neighborhood of X, if the longitude of Y is sufficiently close to the longitude of X, then Y is contained in the ϵ-neighborhood, and the difference of the latitudes of X and Y becomes less than ϵ. This implies that the latitude of X is a continuous function of the longitude of X. Thus, $\mathcal{E}(A, B : s)$ is a simple closed curve on $\mathbb{S}$.

A spherical ellipse $\mathcal{E}(A, B : s)$ divides $\mathbb{S}$ into two regions

$$\{X \in \mathbb{S} : AX + BY < 2s\} \text{ and } \{X \in \mathbb{S} : AX + BX > 2s\}.$$

The former is called the *interior* of $\mathcal{E}(A, B : s)$, and the latter is called the *exterior* of $\mathcal{E}(A, B : s)$. The spherical ellipse with foci A, B and passing through C is also denoted by $\mathcal{E}(A, B : C)$. If $AC + BC = 2s$, then $\mathcal{E}(A, B : C) = \mathcal{E}(A, B : s)$.

Remark 7.1 (i) The foci A, B of $\mathcal{E}(A, B : s)$ lie in the interior of the spherical ellipse, and their antipodals A^*, B^* lie in the exterior of the spherical ellipse. (ii) If $X \in \mathcal{E}(A, B : s)$, then $A^*X + B^*X = 2\pi - (AX + BX) = 2\pi - 2s$. Hence the spherical ellipses $\mathcal{E}(A^*, B^* : \pi - s)$ and $\mathcal{E}(A, B : s)$ are identical curves, and hence $\mathcal{E}(A^*, B^* : C) = \mathcal{E}(A, B : C)$ as figures. Note that, however, the interior of $\mathcal{E}(A^*, B^* : C)$ is the exterior of $\mathcal{E}(A, B : C)$.

Lemma 7.1 *For a spherical triangle ABC with $AC \neq BC$, let D be the point on $\overset{\frown}{ACB}$ such that $AD = BC$, $BD = AC$. Then the spherical ellipse $\mathcal{E}(A, B : C)$ and the circular arc $\overset{\frown}{ACB}$ intersect only at C, D, with crossing each other at C, D, and the subarc $\overset{\frown}{CD}$ of the circular arc $\overset{\frown}{ACB}$ lies, except for the two endpoints, in the exterior of $\mathcal{E}(A, B : C)$.*

The proof of this lemma will be given in Sect. 7.6, Appendix.

Corollary 7.1 *Let $\breve{\triangle}ABC$ be a spherical triangle on $\mathbb{S}$ such that $AC < BC$, and let $\overset{\frown}{ACB}$ be a point such that $AC < AE$, $AC < BE$. Then $AE + BE > AC + BC$.* □

7.2 Lexell's Theorem

For a geodesic segment AB and a circular arc $\overset{\frown}{AXY}$ on the unit sphere $\mathbb{S}$, the angle between the tangent lines of AB and $\overset{\frown}{AXY}$ at A is denoted by $\angle\overset{\frown}{AXY}$.

Theorem 7.1 *Let 2ε be the area of $\breve{\triangle}ABC$ on the unit sphere $\mathbb{S}$. Then $\angle(B^*A, \overset{\frown}{B^*CA^*}) = \varepsilon$, see Fig. 7.2.*

Proof Let $\varphi : \mathbb{S} \to \Pi$ be the stereographic projection of $\mathbb{S}$ from A^* onto the plane Π tangent to $\mathbb{S}$ at A. For every $X \in \mathbb{S}$, $X \neq A^*$, $\varphi(X)$ is denoted by $\bar{X}$. By this stereographic projection, any circle that passes through A^* is mapped to a line on Π. So, the arc $\overset{\frown}{B^*CA^*}$ is mapped to a half line $\overrightarrow{\bar{B}^*\bar{C}}$. Therefore, the region enclosed by $\overset{\frown}{B^*CA^*}$ and the great circular arcs AB^*, AC is mapped to the planar triangle $\triangle\bar{A}\bar{B}^*\bar{C}$. Since this triangle is a derived triangle $\breve{\triangle}ABC$, (consulting Fig. 5.2 in Chap. 5) we have $\angle\bar{A}\bar{B}^*\bar{C} = \varepsilon$. Since the stereographic projection preserves the magnitude of angles, $\angle(B^*A, \overset{\frown}{B^*CA^*}) = \varepsilon$. □

For a spherical triangle $\breve{\triangle}ABC$, the arc $\overset{\frown}{B^*CA^*}$ is called the *Lexell arc* of $\breve{\triangle}ABC$ with base AB.

Theorem 7.2 (Lexell) *For a spherical triangle $\breve{\triangle}ABC$ on $\mathbb{S}$, let $\mathcal{H}$ denote the interior of the hemisphere bounded by the great circle AB and containing C. For a point $X \in \mathcal{H}$, $\breve{\triangle}ABX$ has the same area as $\breve{\triangle}ABC$ if and only if X lies on $\overset{\frown}{B^*CA^*}$.*

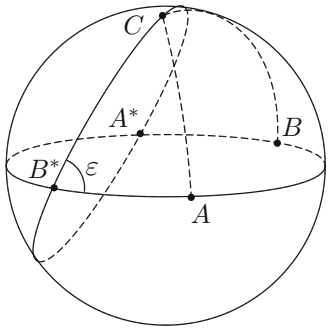


Fig. 7.2 Theorem 7.1

Proof By Theorem 7.1, the area of $\breve{\triangle}ABX$ is equal to $2 \times \angle(B^*A, \overset{\frown}{B^*XA^*})$. Now,

$$\angle(B^*A, \overset{\frown}{B^*XA^*}) = \angle(B^*A, \overset{\frown}{B^*CA^*}) \Leftrightarrow \overset{\frown}{B^*XA^*} = \overset{\frown}{B^*CA^*}$$
$$\Leftrightarrow X \in \overset{\frown}{B^*CA^*}.$$

□

Remark 7.2 There are many different proofs of this theorem, see, e.g., [257, Ch.1, §8] (cf. analogously [262]), [616, §5], and [520, 531, 537]. A proof using stereographic projection was first obtained by A. Simonič [734]. Recently, the authors presented a collection of seven proofs whose main ideas go back to famous mathematicians, see [539].

Lemma 7.2 *Let $\breve{\triangle}ABC$ be a spherical triangle on $\mathbb{S}$ such that $AC \neq BC$. Let $D \in \overset{\frown}{A^*CB^*}$ be a point that satisfies $AD = BC$, $BD = AC$. Then the subarc $\overset{\frown}{CD}$ of $\overset{\frown}{A^*CB^*}$ lies inside $\mathcal{E}(A, B : C)$.*

Proof By Remark 7.1(ii), $\mathcal{E}(A^*, B^* : C) = \mathcal{E}(A, B : C)$ as curves, and the interior of $\mathcal{E}(A^*, B^* : C)$ is the exterior of $\mathcal{E}(A, B : C)$. By Lemma 7.1, the subarc $\overset{\frown}{CD}$ of $\overset{\frown}{A^*CB^*}$ lies outside $\mathcal{E}(A^*, B^*, C)$. Hence $\overset{\frown}{CD}$ lies inside $\mathcal{E}(A, B : C)$. □

7.3 Equilateral Spherical Triangles

Theorem 7.3 *Among the spherical triangles inscribed in a fixed cap Λ on $\mathbb{S}$, equilateral triangles have the maximum perimeter.*

Proof Let Z be the center of the cap Λ. Let us show that the perimeter of the non-equilateral spherical triangle $\breve{\triangle}ABC$ inscribed in Λ is smaller than that of an equilateral spherical triangle inscribed in Λ. Let AC be the shortest edge, and BC be the longest edge of $\breve{\triangle}ABC$. We may consider the case that Z lies in $\breve{\triangle}ABC$. (If Z is not an interior point of $\breve{\triangle}ABC$, then let M be the midpoint of the arc $\overset{\frown}{ACB}$. Clearly, $AM > AC$, $BM > AC$. By Corollary 7.1, we have that $AM + BM > AC + BC$, the perimeter of $\breve{\triangle}ABM$ is greater than that of $\breve{\triangle}ABC$, and Z is an interior point of $\breve{\triangle}ABM$. Hence we may replace $\breve{\triangle}ABC$ by $\breve{\triangle}ABM$.)

Let D be the point on $\overset{\frown}{ACB}$ such that $BD = AC$. By Corollary 7.1, the subarc $\overset{\frown}{CD}$ of $\overset{\frown}{ACB}$ lies in the exterior of $\mathcal{E}(A, B : C)$ except for its two endpoints. Since Z is an interior point of $\breve{\triangle}ABC$, we have $\angle AZC < 2\pi/3 < \angle CZB$. Let E be a point on $\overset{\frown}{ACB}$ such that $\angle AZE = 2\pi/3$. This point E lies on the subarc $\overset{\frown}{CD}$,

and is different to C, D. Hence E is an exterior point of $\mathcal{E}(A, B : C)$, and we have $AE + BE > AC + BC$, which means that the perimeter of $\breve{\triangle}ABE$ is greater than the perimeter of $\breve{\triangle}ABC$. If $\breve{\triangle}AEB$ is equilateral, then we are done. If $\breve{\triangle}AEB$ is not equilateral, then $AB \neq EB$. Let F be the midpoint of the arc $\overset{\frown}{EBA}$. Then the spherical triangle $\breve{\triangle}EBA$ is an equilateral triangle whose perimeter is longer than that of $\breve{\triangle}ABC$. □

Theorem 7.4 *Let $\breve{\triangle}(a, b, c)$ be a spherical triangle with $a > b$. Then, for every e, $a > e > b$, the area of $\breve{\triangle}(e, a + b - e, c)$ is greater than the area of $\breve{\triangle}(a, b, c)$.*

Proof Let $\breve{\triangle}ABC$ be a spherical triangle in which the opposite edges of A, B, C have lengths a, b, c, respectively. Let $\mathcal{H}$ be the hemisphere bounded by the great circle AB and containing C. Let D be a point on $\overset{\frown}{ACB}$ such that $AD = BC$, $BD = AC$. Since $AC = b < a = BC = AD$ and $a > e > b$, there is a point E on the subarc $\overset{\frown}{CD}$ of the "half ellipse" $\mathcal{H} \cap \mathcal{E}(A, B : C)$ such that $AE = e$. Since $BE = a + b - e$, the perimeter of $\breve{\triangle}ABE$ is equal to $a + b + c$. Since C and D lie in the different sides of the semicircle AEA^* (which divides $\mathcal{H}$ into two regions), the subarc $\overset{\frown}{CD}$ of $\overset{\frown}{A^*CB^*}$ intersects the semicircle AEA^* at a point, say F. Then, since F lies on the Lexcell arc $\overset{\frown}{A^*CB^*}$, we have $[ABF] = [ABC]$. Since $\breve{\triangle}ABF$ is contained in $\breve{\triangle}ABE$, we have $[ABF] < [ABE]$. Therefore, $[ABE] > [ABC]$. Thus,

$$[\breve{\triangle}(e, a + b - e, c)] = [ABE] > [ABC] = [\breve{\triangle}(a, b, c)].$$

□

Theorem 7.5 *Among the spherical triangles with fixed perimeter p, equilateral triangles have the maximum area.*

Proof Let $\breve{\triangle}(a, b, c)$ be a spherical triangle with $a + b + c = p$, and suppose that a is the maximum and b is the minimum, $a > b$. Put $e = p/3$. Then $a > e > b$. By Theorem 7.4, $[\breve{\triangle}(e, a + b - e, c)] > [\breve{\triangle}(a, b, c)]$. If $a + b - e = c$, then $\breve{\triangle}(e, e, e) = \breve{\triangle}(e, a + b - e, c)$. If $a + b - e \neq c$, then by applying Theorem 7.4, we have $[\breve{\triangle}(e, e, e)] > [\breve{\triangle}(e, a + b - e, c)]$. Therefore, $[\breve{\triangle}(e, e, e)] > [\breve{\triangle}(a, b, c)]$. □

7.4 Spherical Polygons Inscribed in a Cap

In a planar polygon, a vertex x is called an *ear* of the polygon if the line segment connecting the two neighboring vertices of x lies in the interior of the polygon except for its two endpoints. For example, in a pentagon $ABCDE$, if the diagonal AD lies inside the polygon, then E is an ear of the polygon, see Fig. 7.3. The *two ears theorem* states that *if $n \geq 4$, then every n-gon contains two ears that are not*

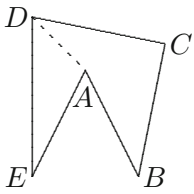


Fig. 7.3 The vertex E is an ear

adjacent. Similarly to the planar polygon case, an ear of a spherical polygon is a vertex x such that the geodesic segment between the two neighboring vertices of x lies in the interior of the spherical polygon, except for its two endpoints.

Lemma 7.3 *Every spherical n-gon, $n \geq 4$, lying in the interior of a hemisphere has at least two ears.*

Proof Let Γ be a spherical n-gon, $n \geq 4$, lying in the interior of the lower hemisphere $z \leq 0$. Project it from the center O of $\mathbb{S}$ into the plane that is tangent to $\mathbb{S}$ at $(0, 0, -1)$. Then we have a planar n-gon. It has at least two ears. By the inverse projection these ears are mapped to the ears of Γ. □

Lemma 7.4 *For every spherical polygon Γ with perimeter less than 2π, there is a cyclic one with the same edge-length sequence as Γ.*

Proof Let $a_1, a_2, \ldots, a_n$ be the edge-lengths of Γ, and suppose that a_n is the maximum. Let K be a circle on $\mathbb{S}$ whose spherical radius is nearly $\pi/2$. Take n points $P_1, P_2, \ldots, P_n$ on K so that $P_1P_2 = a_1, P_2P_3 = a_2, \ldots, P_{n-1}P_n = a_{n-1}$. If the spherical radius of K is sufficiently close to $\pi/2$, then we have $P_1P_n > a_n$. Squeeze the radius of K with keeping $P_1, P_2, \ldots, P_n$ on it and preserving the spherical distances $P_1P_2, P_2P_3, \ldots, P_{n-1}P_n$ until the length P_1P_n becomes a_n. □

Let us denote by $\Psi(a_1, a_2, \ldots, a_n)$ a cyclic spherical n-gon with edge-lengths $a_1, a_2, \ldots, a_n$ in this cyclic order.

Theorem 7.6 *Among the spherical n-gons with fixed edge-lengths*

$$a_1, a_2, \ldots, a_n, \text{ where } a_1 + a_2 + \cdots + a_n < 2\pi,$$

a cyclic n-gon has the maximum area.

Proof The proof works by induction on n. The case $n = 4$ follows from Theorem 5.8. Suppose that $n \geq 4$ and the theorem is true for n. Let $\mathcal{F}$ be the set of spherical $(n+1)$-gons $\langle X \rangle$ with vertices $X_0, X_1, \ldots, X_n$ and $X_iX_{i+1} = a_i$, $i = 0, 1, 2, \ldots, n$ (where the indices are taken modulo $n+1$). By Lemma 7.3, each $\langle X \rangle \in \mathcal{F}$ has at least two ears. Let

$$\mathcal{F}(i) = \{\langle X \rangle \in \mathcal{F} : X_i \text{ is an ear of } \langle X \rangle\}.$$

Then, for each $\langle X\rangle \in \mathcal{F}(i)$,

$$\text{the area of } \langle X\rangle = [X_{i-1}X_iX_{i+1}] + [X_0X_1\dots X_{i-1}X_{i+1}\dots X_n].$$

If we put $t = X_{i-1}X_{i+1}$, then by the inductive assumption we have

$$[X_0X_1\dots X_{i-1}X_{i+1}\dots X_n] \le \Psi(a_0, a_1, \dots, a_{i-2}, t, a_{i+1}, \dots, a_n).$$

Hence

$$\begin{aligned}\text{the area of } \langle X\rangle \le\ & [\breve{\Delta}(t, a_{i-1}, a_{i+1})] \\ & + [\Psi(a_0, a_1, \dots, a_{i-2}, t, a_{i+1}, \dots, a_n)],\end{aligned} \tag{7.1}$$

and equality holds only when the vertices other than X_i lie on the same circle. The right hand side of (7.1) is a continuous function of t, and the range of t is a closed interval

$$\max\{\delta, |a_{i-1} - a_i|\} \le t \le \min\{p - a_{i-1} - a_i, a_{i-1} + a_i\}$$

where δ is the infimum of x such that there exists a cyclic n-gon with edge-lengths $a_0, a_1, \dots, a_{i-2}, x, a_{i+1}, \dots, a_n$, and $p = a_0 + a_1 + \cdots + a_n$. Hence there is a maximum value $f(i)$ of the right hand side of (7.1). This maximum value is attained by an $\langle X\rangle \in \mathcal{F}(i)$ for which the vertices other than X_i lie on the same circle. Put $f(k) = \max\{f(i) : i = 0, 1, \dots, n\}$, and suppose that $\langle Y\rangle \in \mathcal{F}(k)$ has the area $f(k)$. Then the vertices of $\langle Y\rangle$ other than Y_k lie on the same circle. Since $\langle Y\rangle$ has another ear Y_j ($j \ne k$), we have also $\langle Y\rangle \in \mathcal{F}(j)$, and hence $f(k) \le f(j)$. Thus $f(k) = f(j)$, and hence the vertices of $\langle Y\rangle$ other than Y_j also lie on the same circle. Since $n + 1 \ge 5$, we can deduce that all vertices of $\langle Y\rangle$ lie on the same circle. Thus, $\langle Y\rangle \in \mathcal{F}(i)$, $i = 0, 1, \dots, n$, and $f(0) = f(1) = \cdots = f(n)$. Therefore the cyclic $(n + 1)$-gon has the maximum area among the $(n + 1)$-gons in $\mathcal{F}$. □

Lemma 7.5 *For every permutation σ of $\{1, 2, \dots, n\}$, two spherical n-gons*

$$\Psi(a_1, a_2, \dots, a_n) \text{ and } \Psi(a_{\sigma(1)}, a_{\sigma(2)}, \dots, a_{\sigma(n)}),$$

have the same area and the same perimeter.

Proof Let Z be the center of the cap circumscribed to $\Psi(a_1, a_2, \dots, a_n)$. Suppose that Z lies in the interior of $\Psi(a_1, a_2, \dots, a_n)$. Dividing this n-gon by geodesic segments connecting Z to the vertices of this n-gon into n spherical triangles, and then rearranging these spherical triangles, we can obtain $\Psi(a_{\sigma(1)}, a_{\sigma(2)}, \dots, a_{\sigma(n)})$. Hence they have the same area and the same perimeter.

If Z does not lie in the interior of $\Psi(a_1, a_2, \dots, a_n)$ we need to change the proof slightly (with introducing a "negative" spherical triangle), but it would be essentially the same. □

7.5 Regular Spherical Polygons

Theorem 7.7 *Among the spherical n-gons inscribed in a fixed cap Λ, regular n-gons have the maximum perimeter.*

Proof Let Z be the center of the cap Λ, and let $\Gamma = A_1A_2 \dots A_n$ be a non-regular n-gon inscribed in Γ. Let us show that a regular n-gon inscribed in Λ has a perimeter larger than the perimeter of Γ. By Lemma 7.5, we may suppose that A_2A_3 is the longest edge, and A_1A_2 is the shortest edge. Thus $A_1A_2 < A_2A_3$. We may consider the case that Z lies in the interior of Γ. (If Z is not contained in the interior of Γ, then, denoting by M the midpoint of $\overset{\frown}{A_1A_2A_3}$, we have $A_1M > A_1A_2$, $A_3M > A_1A_2$, and hence, by Corollary 7.1, we have $A_1M + MA_3 > A_1A_2 + A_2A_3$. Then the perimeter of the cyclic n-gon $A_1MA_3 \dots A_n$ is greater than that of Γ. So, we may replace Γ by $A_1MA_3 \dots A_n$.)

Now, since Γ contains Z in its interior,

$$\angle A_1ZA_2 < 2\pi/n < \angle A_2ZA_3.$$

Let E_2 be a point on $\overset{\frown}{A_1A_2A_3}$ satisfying $\angle A_1ZE_2 = 2\pi/n$. Then, $A_1E_2 > A_1A_2$. Since

$$\angle A_1ZA_2 + \angle A_2ZA_3 = \angle A_1ZE_2 + \angle E_2ZA_3 = 2\pi/n + \angle E_2ZA_3,$$

we have $\angle E_2ZA_3 = \angle A_1ZA_2 + \angle A_2ZA_3 - 2\pi/n > \angle A_1ZA_2$, and so we get $E_2A_3 > A_1A_2$. Hence by Corollary 7.1, $A_1E_2 + E_2A_3 > A_1A_2 + A_2A_3$. Therefore, the perimeter of the n-gon $\Gamma_1 := A_1E_2A_3 \dots A_n$ is greater than the perimeter of Γ. Notice that Γ_1 contains Z in its interior. If Γ_1 is not a regular n-gon, then, replacing the order of the edges, we can bring the longest edge and the shortest edge into consecutive position. For instance, suppose that E_2A_3 is the shortest edge and A_3A_4 is the longest edge. Let E_3 be a point on $\overset{\frown}{E_2A_3A_4}$ satisfying $\angle E_2ZE_3 = 2\pi/n$. Then the n-gon $\Gamma_2 := A_1E_2E_3A_4 \dots A_n$ is inscribed in Λ, and its perimeter is greater than that of Γ_1. Repeating a similar process, we can get a regular n-gon whose perimeter is greater than that of Γ. □

Theorem 7.8 *Among the spherical n-gons with fixed perimeter $p < 2\pi$, regular n-gons have the maximum area.*

Proof Let Γ be a spherical n-gon with perimeter p.

(i) By Lemma 7.4, there is a cyclic n-gon with the same edge-length as Γ. Then by Theorem 7.6, the area of this cyclic n-gon is greater than or equal to the area of Γ.

(ii) By Lemma 7.5, we can reorder the edge-lengths of this cyclic n-gon so that the longest edge (of length a) and the shortest edge (of length b) become consecutive. (Thus, it becomes a $\Psi(\dots, a, b, \dots)$.)
(iii) Replace a, b in $\Psi(\dots, a, b, \dots)$ with $e, a+b-e$ (where $e = p/n$) and then change the n-gon to a cyclic one. Then the area of the resulting n-gon is at least the area of $[\Gamma]$ by Theorem 7.4.

By repeating (i), (ii) and (iii) we can show that a regular n-gon with perimeter p has area like at least that of Γ. □

7.6 Appendix

7.6.1 *Orthogonal Projections of a Spherical Ellipse*

Suppose that A, B lie on the xz-plane, the midpoint of AB is the south-pole $(0, 0, -1)$, and

$$A = (-\sin\phi, 0, -\cos\phi),\ B = (\sin\phi, 0, -\cos\phi),$$

where $2\phi < \pi$. For $X = (x, y, z) \in \mathcal{E}(A, B : s)$, put $AX = s - t$, $BX = s + t$, where $-\phi < t < \phi$. Then, since $\angle AOX = AX = s - t$, $\angle BOX = BX = s + t$, we have

$$\cos(s-t) = \cos(AX) = \overrightarrow{OA} \cdot \overrightarrow{OX} = -x\sin\phi - z\cos\phi,$$
$$\cos(s+t) = \cos(BX) = \overrightarrow{OB} \cdot \overrightarrow{OX} = +x\sin\phi - z\cos\phi,$$

and therefore

$$\cos s\cos t + \sin s\sin t = -x\sin\phi - z\cos\phi,$$
$$\cos s\cos t - \sin s\sin t = +x\sin\phi - z\cos\phi,$$

from which we have

$$\begin{cases} \cos s\cos t &= -z\cos\phi \\ \sin s\sin t &= -x\sin\phi. \end{cases} \tag{7.2}$$

If $2s = \pi$, then we have $z = 0$, and $\mathcal{E}(A, B : s)$ is a great circle of $\mathbb{S}$. If $2s < \pi$, then $\mathcal{E}$ lies in the lower hemisphere, whereas, if $2s > \pi$, then $\mathcal{E}$ lies in the upper hemisphere. Suppose $2s \neq \pi$. From (7.2), we have

$$\left(\frac{\cos\phi}{\cos s}\right)^2 z^2 + \left(\frac{\sin\phi}{\sin s}\right)^2 x^2 = \cos^2 t + \sin^2 t = 1.$$

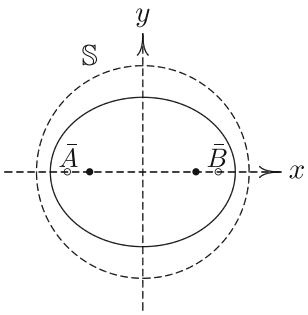


Fig. 7.4 The foci of the ellipse are denoted by •

Put $p = \cos\phi/\cos s, q = \sin\phi/\sin s$. Then, since $\phi < s < \pi - \phi$, we have $p^2 > 1 > q^2$ and

$$p^2z^2 + q^2x^2 = 1. \tag{7.3}$$

From this and $z^2 = 1 - x^2 - x^2$, we have

$$1 = p^2(1 - x^2 - y^2) + q^2x^2 = p^2 - (p^2 - q^2)x^2 - p^2y^2,$$

and hence

$$\left(\frac{p^2 - q^2}{p^2 - 1}\right)x^2 + \left(\frac{p^2}{p^2 - 1}\right)y^2 = 1. \tag{7.4}$$

This represents an ellipse in the xy-plane. Thus, the orthogonal projection of $\mathcal{E}$ into the xy-plane is an ellipse, which is contained in the unit circle centered at the origin. Figure 7.4 shows the ellipse (7.4) for the case $\phi = \pi/4$, $s = \pi/3$, and $\bar{A}$, $\bar{B}$ show the projections of A, B, respectively. Note that $\bar{A}$, $\bar{B}$ are not the foci of the ellipse (7.4).

The Eq. (7.3) also represents an ellipse in the xz-plane, but in this case the projection of $\mathcal{E}(A, B : s)$ is only a part of the ellipse (7.3).

Next, let us consider the figure obtained by the orthogonal projection of $\mathcal{E}(A, B : s)$ into the yz-plane. From (7.3) and $x^2 = 1 - y^2 - z^2$, we have

$$(p^2 - q^2)y^2 - q^2z^2 = 1 - q^2. \tag{7.5}$$

In the yz-plane, the Eq. (7.5) represents a hyperbola. See Fig. 7.5, which shows the hyperbola in the case $\phi = \pi/4, s = \pi/3$.

Thus, the projection of $\mathcal{E}$ into the yz-plane is a part of a hyperbola, see also [425, Theorem 3.1]. The hyperbola (7.5) divides the yz-plane into three regions, and the region that contains the y-axis is called the *exterior*. The other two regions are called the *interior* of the hyperbola. Denote by $\bar{\mathcal{E}}$ the orthogonal projection of $\mathcal{E}$ into the yz-plane, and by $\bar{X}$ the orthogonal projection of a point $X \in \mathbb{S}$. Note that if $2s \neq \pi$, then since $\mathcal{E}$ is connected, $\bar{\mathcal{E}}$ is contained in one of the connected components of

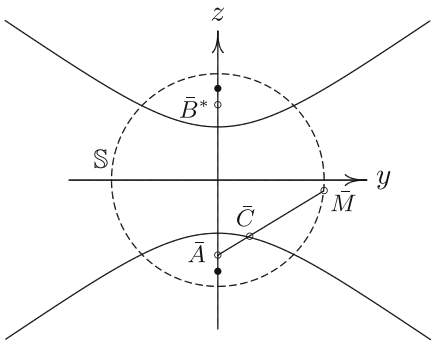


Fig. 7.5 The case $\phi = \pi/4, s = \pi/3$

the hyperbola (7.5). Since A, B, A^*, B^* lie on the xz-plane, the geodesic segments AB^*, AB are mapped into the z-axis by the orthogonal projection into the yz-plane. Since the geodesic segment AB^* crosses the ellipse at a point, and AB^* is mapped onto the line segment $\bar{A}\bar{B}^*$ bijectively, the line segment $\bar{A}\bar{B}^*$ must cross $\bar{\mathcal{E}}$, and since the hyperbola (7.5) is symmetric with respect to the y-axis, we can deduce that both $\bar{A}$, $\bar{B}^*$ lie in the interior of the hyperbola (7.5).

7.6.2 *Proof of Lemma 7.1*

Applying the orthogonal projection into the yz-plane, we prove Lemma 7.1.

Let $2s = AC + BC$. In the case $s = \pi/2$ (that is, when $\mathcal{E}(A, B : C)$ is a great circle), the lemma will be trivial. So, we assume that $s \neq \pi/2$. Then $\bar{\mathcal{E}}$ is contained in one of the components of the hyperbola (7.5). If $s < \pi/2$, then $\bar{\mathcal{E}}$ is contained in the $(z < 0)$-component of the hyperbola (7.5), whereas $s > \pi/2$ implies that $\bar{\mathcal{E}}$ is contained in the $(z > 0)$-component of the hyperbola (7.5). Recall that $\bar{A}$ and $\bar{B}^*$ lie in the interior of the hyperbola (7.5).

Since the line through A and B is parallel to the x-axis, we have $\bar{A} = \bar{B}$, and the plane that contains the arc $\overset{\frown}{ACB}$ is perpendicular to the yz-plane. Hence the orthogonal projection of $\overset{\frown}{ACB}$ into the yz-plane becomes a line segment connecting $\bar{A}$ and $\bar{M}$, where M is the midpoint of the arc $\overset{\frown}{ACB}$, see Fig. 7.5. Since $\bar{A}$ is a point in the interior of the hyperbola (7.5), the ray $\overrightarrow{\bar{A}\bar{M}}$ intersects $\bar{\mathcal{E}}$ only at the point $\bar{C}(= \bar{D})$. Since any line orthogonal to the yz-plane intersects $\mathbb{S}$ in at most two points, we have $\mathcal{E}(A, B : C) \cap \overset{\frown}{ACB} = \{C, D\}$. Since the line segment $\bar{A}\bar{M}$ crosses $\bar{\mathcal{E}}$ at $\bar{C}$, M must lie in the exterior of $\mathcal{E}$, and we can deduce that the subarc $\overset{\frown}{CD}$ lies, except its two endpoints, in the exterior of $\mathcal{E}$. □

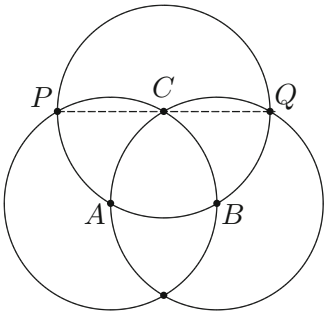


Fig. 7.6 Exercise 7.2 (see below)

7.7 Exercises

7.1 For a spherical triangle $\breve{\triangle}ABC$ on $\mathbb{S}$, let $X \in \mathbb{S}$, $X \neq A^*$, be a point such that the shortest geodesic AX intersects the Lexell arc $\overset{\frown}{A^*CB^*}$. Prove that $[ABX] \geq [ABC]$.

7.2* Let $\breve{\triangle}ABC$ be an equilateral spherical triangle with edge length $d < \pi/2$. Let K_A, K_B, K_C be circles on $\mathbb{S}$ with spherical radius d and centers A, B, C, respectively. Let P be the crossing points of K_A, K_C different to B, and Q be the crossing point of K_B, K_C different to A, as shown in Fig. 7.6. Prove that the arc $\overset{\frown}{PCQ}$ is a subarc of the Lexell arc $\overset{\frown}{A^*CB^*}$.

7.3* (Fejes Tóth's lemma) Let $\breve{\triangle}(a, b, c)$, $a \leq b \leq c$, be a spherical triangle on $\mathbb{S}$. Prove that if the spherical radius of the circumscribed cap of this triangle is smaller than a, then $[\breve{\triangle}(a, b, c)] \geq [\breve{\triangle}(a, a, a)]$. (If $a \geq \pi/2$, then $\breve{\triangle}(a, b, c)$ is a major triangle, and the result follows from the major triangle lemma.)

7.4 Represent the edge-length of an equilateral spherical triangle on $\mathbb{S}$ using its area.

7.5 By the orthogonal projection of the spherical ellipse $\mathcal{E}(A, B : s)$ with foci $A = (-\sin\phi, 0, -\cos\phi)$ and $B = (\sin\phi, 0, -\cos\phi)$ on $\mathbb{S}$ to the xy-plane, we get the ellipse $(5/3)x^2 + (3/2)y^2 = 1$. Find ϕ and s.

7.6* Among the spherical convex regions on $\mathbb{S}$ with a fixed perimeter, there exists one that has the maximal area. Assuming this is true, prove that one that has the maximum area is a spherical cap.

7.8 Notes

Since some of the themes and topics discussed in this chapter here occur already in other parts of the book, we focus in these notes on spherical ellipses and Lexell's theorem.

7.8.1 Notes on Spherical Ellipses

The authors were not able to locate many references on spherical ellipses and related topics, and only two (partially) corresponding surveys (namely [403] and [425]) could be found. Therefore all identified papers are listed below in chronological order.

Fabricius-Bjerre [254] presents a nicely unified and simplified derivation of geometric properties of *spherical conics*. Schilling [696] considers *spherical ellipses* in 3-space as intersections of a 2-sphere S and a suitable cone whose apex is the center of S. These curves can be seen as ellipses defined analogously to the usual concept with foci, and with the help of spherical distances. He shows that many of the elementary *focal properties* of usual ellipses can be carried over to these curves in S. This yields then corresponding theorems on ellipses in *elliptic geometry*. Pursuing the same goals, Namikawa [593] studies *spherical ellipses and hyperbolas*.

Similar to Schilling [696], Dirnböck [214] introduces spherical conics as intersection curves of the 2-sphere and certain cones, and he studies their *polar conics* and *focal properties* with the help of spherical evolutes. With analogous methods, spherical loxodromic curves and the spherical tractrix are studied, and their kinematic generation is presented.

A somehow special *topological result* is obtained by Chilakamarri et al. [163]. They complete the construction of *all possible types of spherical Venn diagrams on five curves* by a careful case analysis. Recall that a Venn diagram is *simple* if at most two curves intersect at a point, and it is *exposed* if each of its curves has an arc on the boundary of the unbounded exterior region. Further on, it is *convex* if it is isomorphic with a Venn diagram all of whose curves are convex. Completing a series of earlier papers dedicated to this subject, the authors use graph theory to enumerate all simple spherical Venn diagrams generated by five participating curves. E.g., it is shown that there are 20 nonisomorphic simple spherical Venn diagrams of 5 curves, 11 of which are convex. They establish which of these spherical Venn diagrams are convex, which are exposed, and which can be drawn with congruent ellipses. Kang and Lin [442] investigate *equilateral spherical drawings of planar Cayley graphs*, concentrated on the case that the underlying group is generated by two rotations. Due to this restriction, the set of such equilateral drawings can be parametrized by spherical ellipses on the unit sphere.

Schröcker [715] generates so-called *sphero-conics* on the sphere S as Schilling [696] (see above) does it with his spherical ellipses (as intersection of S and quadratic cones). Schröcker characterizes a sphero-conic as locus of points such that the absolute value of the sum or difference of tangent distances to two fixed circles is constant. From the two given proofs the first one is based on methods of descriptive and projective geometry, and the second one on algebraic methods. The proofs remain valid in case of purely imaginary tangent distances (i.e., when the sphero-conic is enclosed by both circles). Minor modifications of the algebraic proof yield also the hyperbolic case.

Besides other topics, Weber and Schröcker [800] study, for the elliptic plane, the analogue of the unique ellipse of minimal area enclosing a compact, convex set. They show that in this non-Euclidean situation the uniqueness of this *minimal enclosing ellipse* is guaranteed if the center or the axis of the conic is prescribed. With new methods they establish also sufficient conditions for the uniqueness of such ellipses. Finally, for ellipses of line sets corresponding uniqueness results are obtained. (This follows the duality principle between points and lines in the elliptic plane.) Finally, some open questions concerning extremal quadrics are formulated. For analogous results on minimal area ellipses in the hyperbolic case we refer to [801].

Horváth [403] presents an informative survey on *conics* and roulettes (generated by points that maintain a fixed position relative to a curve rolling on another one, without slipping) in non-Euclidean geometries, among them also *spherical geometry*.

Izmestiev [425] nicely surveys results on conics (also) in spherical geometry. After defining *conic sections and polarities in spherical geometry*, the author gives a corresponding classification and presents several theorems with proofs. E.g., he shows that a tangent to a spherical conic cuts from the lune formed by the focal lines a triangle of constant area. Also *Ivory's theorem* (the diagonals in a quadrilateral formed by four suitable confocal conics have equal lengths) is studied for the case of spherical geometry.

Maehara [529] shows that the following holds for spherical ellipses and triangles ABC: For ABC with $AC \neq BC$ the circular arc from A to B via C and the spherical ellipse with foci A and B passing through C intersect in precisely two points, and these two curves intersect each other at the intersection points. From this, *extremal properties of spherical equilateral triangles* are derived.

7.8.2 Notes on Lexell's Theorem

Lexell's theorem refers to spherical triangles of equal surface area, and so it has various applications and extensions in spherical geometry. An additional peculiarity of that theorem is the fact that many famous mathematicians proved it in different ways. By that reason, this subsection here is strongly historical in nature.

The Theorem Itself and Different Proofs

Lexell's theorem says that *each spherical triangle on the 2-sphere, with a fixed base AB and having the same surface area, has its third vertex on a small circle (called "Lexell's circle") which passes through each of the two points being antipodal to A and B*. Thus, the locus of the moving vertex of spherical triangles with fixed base and fixed spherical area is a small circle. Leonhard Euler and Anders Johan Lexell gave written proofs of this theorem at the same time, and Euler mentioned that he

was inspired by Lexell. But it turned out that the publications of the result followed in different years, namely in 1784 by Lexell [504] and in 1797 by Euler [247].

In [539], Maehara and Martini present seven proofs of Lexell's theorem, most of them coming from famous mathematicians (namely Leonhard Euler, Jakob Steiner, Carl Friedrich Gauss, Josef-Alfred Serret, and László Fejes Tóth) or being of particular elegance. The sequence of the following selection of very shortly described proofs of Lexell's theorem agrees with that in [539]. Lexell's own proof, using angle relations, is given and commented in [504] and [616]. Euler (see [247] and [248]) proved Lexell's theorem in a way similar to the proof of Proposition 37 in Euclid's Elements (based on the elementary statement that "triangles which are on the same base and between the same parallels are equal to one another"); see again [616]. And also Gauss took a similar way, cf. [46]; he inserted a proof into a letter to Schumacher in 1841, in response to a related proof sent from Clausen via Schumacher to him. Jacob Steiner (see [755], [144, p. 93], and [616]) used the inscribed angle theorem, and Maehara [519] found independently an analogous approach. According to Casey [144, p. 93], Josef-Alfred Serret proved Lexell's theorem by showing that the radius of the Lexell circle of the triangle ABC depends only on the length of AB and the area. Using polar triangles, also László Fejes Tóth gave an own proof of Lexell's theorem in [257, Section 1.8], see also [262, Section 1.8]. And Aleksander Simonić [734] presented an elegant proof via stereographic projection.

Besides these proofs reflected in [539] there are further ones due to famous mathematicians. Jacob Steiner, already mentioned above, continued in 1841 with a new proof of Lexell's theorem, see [756]. In addition, he mentions in this paper that the theorem was also known to Adrien-Marie Legendre who published a proof based on spherical trigonometry in [500, Note X, Problem III]. Later, in 1855, Henri Leon Lebesgue became involved, repeating in [496] one of Euler's approaches and mentioning that a related proof of Eugene Charles Catalan exists, which is indeed the case (see [146, Book VII, Problem VII]). Since in some of these publications Euler's related work is underestimated or even not mentioned, we refer once more to the excellent modern reference [616]. Related contributions of Nicholas Fuss (the secretary of Euler in Saint Petersburg) and other collaborators of Euler are nicely discussed in [46]. And we mention that further modern proofs and related discussions can be found in [519, 531], and [537].

Some History

The scientific life of the Swedish astronomer and mathematician Anders Johan Lexell (1740–1784) is one of the best examples for a strong combination of mathematics and astronomy. The first complete biography of him was written by Johan Stén [760] (here we refer also to Calinger's review [139]). Lexell was born in Turku (today located in Finland, but in those days belonging to the Swedish kingdom), and there he also began his mathematical studies on *differential and integral calculus*. For "mathematical tourists" it should be mentioned that there is a

monument of Lexell at the University of Turku (see [628]). In 1766 (2 years after his return to the Petersburg Academy) Leonhard Euler invited Lexell to work with him at the Kunstkammer there (see [248] and Coxeter's corresponding review [183]), and a fruitful period of collaboration began. Lexell created a *complete theory of comets*, and various famous astronomical findings are due to him. For instance, the discovery of the *first near-Earth object*, a Jupiter-family comet, resulted in naming this object after him (see [825] and [139]). A lunar crater is also named after him.

During a journey throughout Europe, Lexell was meeting several famous mathematicians, such as Johann III Bernoulli, Jean d'Alembert, Marie-Jean-Antoine-Nicolas de Caritat (Marquis de Concordet), and Pierre-Simon de Laplace. His mathematical work was mainly concentrated on *the theory of ordinary differential equations and spherical trigonometry*. In astronomy, he was the first "calculator". Namely, he could show that a new heavenly object (which was discovered by Herschel in 1781, called "King George's Star", and thought to be a comet) was in fact a planet, later named *Uranus*. Lexell was also involved in the determination of the solar parallax in connection with the Venus passage 1769, and he obtained a very good approximation of that parallax.

In 1783, Lexell became the *successor of Euler* at the Imperial Academy of Sciences in Saint Petersburg, but he filled that position only briefly, due to his premature death in 1784. Nevertheless, he became one of the most successful members of the Russian Academy of Sciences at that time. His mathematical work was praised by Leonhard Euler and Daniel Bernoulli. Further interesting historical articles on Lexell and his scientific activities are Atzema [46], Lysenko [513], and Zhukova [832], the first of which directly treats Lexell's theorem as a central topic. This theorem was also included into influential monographs of Lexell's time, e.g., into Legendre's book [500] from 1808. Later it was discussed in Hadamard's book [353, pp. 392–393], and more recent examples of monographs discussing it are Berger's book [75, 18.11.10] and Section 1.8 in the already mentioned classical book [257] of L. Fejes Tóth and its strongly expanded edition [262].

Extensions and Modifications

Regarding research aspects around Lexell's theorem, especially *non-Euclidean variants* are of interest. In these directions, various modifications and analogues of Lexell's theorem have been derived. For the *hyperbolic case* the reader should consult Papadopoulos and Su [621] as well as Frenkel and Su [281]. And regarding analogues in *absolute geometry* we refer to the papers [631] of Persson and [612] of Pambuccian.

Using so-called *Cesàro triangles* (recall that these are planar triangles obtained via stereographic projection from spherical ones), various results on spherical triangles (which are projected to get the Cesàro ones) are obtained by Maehara and Martini [537]. These concern the relation of sides of a spherical triangle and their opposite angles, Lexell's theorem, and a result on the area of spherical triangles with one variable side-length. The obtained results are applied to derive several *isoperimetric theorems for spherical polygons*.

Chapter 8
Geometric Probability on the Sphere



If we knew what it was we were doing, it would not be called research, would it?

— Albert Einstein

In this chapter, we study the probabilities concerning several random variables or events, determined by random points or random great circles on the sphere.

8.1 Random Points on a Sphere

Let $\mathbb{S}$ denote the unit sphere centered at the origin O in $\mathbb{R}^3$. A randomly chosen point on $\mathbb{S}$ is called a *random point* on $\mathbb{S}$. More precisely, a random point is a point uniformly distributed on $\mathbb{S}$, that is, a point $\boldsymbol{p} \in \mathbb{S}$ chosen in such a way that, for any subset $U \subset \mathbb{S}$, the probability that $\boldsymbol{p} \in U$ is equal to [the area of U]$/(4\pi)$.

Example 8.1 Let $\boldsymbol{u}$, $\boldsymbol{v}$ be independent random numbers, both uniformly distributed in the interval $[0, 1]$. Let $\boldsymbol{p}$ be a point on $\mathbb{S}$ with longitude $\tau = \pi(2\boldsymbol{u}-1)$ and latitude $\theta = \sin^{-1}(2\boldsymbol{v} - 1)$. Then $\boldsymbol{p}$ is a random point on $\mathbb{S}$.

This can be checked as follows. Put $\boldsymbol{p} = (\tau, \theta)$, and let $\varphi : \check{\mathbb{S}} \to \Gamma$ be the cylindrical projection. Consulting Remark 4.1 in Chap. 4, for any $U \subset \mathbb{S}$ we have

$$\boldsymbol{p} \in U \Leftrightarrow \varphi(\tau, \theta) \in \varphi(U) \Leftrightarrow (\pi(2\boldsymbol{u} - 1), 2\boldsymbol{v} - 1) \in \varphi(U).$$

Note that $(\pi(2\boldsymbol{u}-1), 2\boldsymbol{v}-1)$ is uniformly distributed on the quadrilateral $[-\pi, \pi] \times [-1, 1]$ obtained from the lateral face of the cylinder Γ by cutting along the line $\tau = 0$ and unfolding it. Hence the probability that $(\pi(2\boldsymbol{u} - 1), 2\boldsymbol{v} - 2) \in \varphi(U)$ is equal to [the area of $\varphi(U)$]$/(4\pi)$. Since the cylindrical projection φ preserves the area of a figure, it follows that $\Pr(\boldsymbol{p} \in U) =$ (the area of U)$/(4\pi)$. Thus, $\boldsymbol{p}$ is a random point on $\mathbb{S}$.

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H. Maehara, H. Martini, *Circles, Spheres and Spherical Geometry*,
Birkhäuser Advanced Texts Basler Lehrbücher,
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Example 8.2 If $\boldsymbol{p}$ is a random point of $\mathbb{S}$, then its antipodal point $\boldsymbol{p}^*$ is also a random point.

Indeed, if we denote by $U^* \subset \mathbb{S}$ the set which is symmetric to $U \subset \mathbb{S}$ with respect to the center O of $\mathbb{S}$, then "$\boldsymbol{p}^* \in U \Leftrightarrow \boldsymbol{p} \in U^*$" holds, and since the areas of U and U^* are equal, we have

$$\Pr(\boldsymbol{p}^* \in U) = \Pr(\boldsymbol{p} \in U^*) = \frac{\text{the area of } U^*}{4\pi} = \frac{\text{the area of } U}{4\pi}.$$

For a random point $\boldsymbol{p}$ on $\mathbb{S}$, the pair $\{\boldsymbol{p}, \boldsymbol{p}^*\}$ is called a *random diameter*. Thus, each endpoint of a random diameter is a random point. A great circle divides $\mathbb{S}$ into two hemispheres; the poles (centers) of the two hemispheres are called the poles of the great circle. Thus, every great circle has two poles, and they are mutually antipodal. Any point on $\mathbb{S}$ determines a great circle, being one of its poles. A *random great circle* is a great circle determined by taking a random point (or a random diameter) as its pole (or its poles).

Example 8.3 If $\boldsymbol{G}$ is a random great circle on $\mathbb{S}$, then each pole of $\boldsymbol{G}$ is a random point on $\mathbb{S}$.

Example 8.4 The probability that independently chosen three random great circles pass through the same point is 0.

This can be seen as follows. Let us take three random great circles independently in the order $\boldsymbol{G_1}, \boldsymbol{G_2}, \boldsymbol{G_3}$. Let x, x^* be the intersection points of $\boldsymbol{G}_1, \boldsymbol{G}_2$. Then $\boldsymbol{G}_3$ passes x whenever the pole (which is a random point) of $\boldsymbol{G}_3$ lies on the great circle with pole x. Since the area of a great circle is 0, the probability that independently taken three random great circles pass through the same point is 0.

8.2 Random Spherical Triangles

Let $\boldsymbol{x}, \boldsymbol{y}$ be independent random points on $\mathbb{S}$. Then the length of the geodesic segment $\theta = \boldsymbol{xy}$ is a random variable. Let us find the probability density function $f(\theta)$ of θ. When we consider the distribution of $\theta = \boldsymbol{xy}$, by the symmetry of $\mathbb{S}$, we may suppose that $\boldsymbol{x}$ is the north pole $(0, 0, 1)$ of $\mathbb{S}$. Then $\boldsymbol{xy} \leq \theta$ if and only if $\boldsymbol{y}$ lies in the spherical cap with center $\boldsymbol{x}$ and angular radius θ. Since this cap has area $2\pi(1 - \cos\theta)$, the distribution function $F(\theta) = \Pr(\boldsymbol{xy} \leq \theta)$ is given by

$$F(\theta) = \frac{2\pi(1 - \cos\theta)}{4\pi} = \frac{1 - \cos\theta}{2}.$$

Hence the probability density function $f(\theta) = \frac{dF(\theta)}{d\theta}$ is given by

$$f(\theta) = \frac{\sin\theta}{2}.$$

Therefore, the expected value $E(\theta)$ is

$$E(\theta) = \int_0^\pi \theta f(\theta) d\theta = \int_0^\pi \frac{\theta \sin\theta}{2} d\theta$$
$$= \frac{1}{2}\left[\sin\theta - \theta\cos\theta\right]_0^\pi = \frac{\pi}{2}.$$

Theorem 8.1 *For a spherical triangle determined by three random points chosen independently on $\mathbb{S}$, the expected value of its perimeter L is $3\pi/2$, and the expected value of its area A is $\pi/2$.*

Proof Let $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z}$ be three random points on $\mathbb{S}$. Then

$$E(L) = E(\boldsymbol{xy} + \boldsymbol{yz} + \boldsymbol{zx}) = 3\pi/2.$$

By Girard's formula, $A = \angle\boldsymbol{x} + \angle\boldsymbol{y} + \angle\boldsymbol{z} - \pi$. When we consider the size of $\angle\boldsymbol{x} = \angle\boldsymbol{yxz}$, we may regard $\boldsymbol{x}, \boldsymbol{y}$ as fixed points. Then $\Pr(\angle\boldsymbol{yxz} < t)$ is equal to the probability that $\boldsymbol{z}$ lies in the lune of angle $2t$ with bisector $\boldsymbol{xy}$, which is equal to t/π. Therefore, $\angle\boldsymbol{yxz}$ is uniformly distributed in the interval $[0, \pi]$. Hence $E(\angle\boldsymbol{x}) = \pi/2$. Thus $E(A) = 3\pi/2 - \pi = \pi/2$. □

For two random circles $\boldsymbol{G}_i, i = 1, 2$, chosen independently on $\mathbb{S}$, let $\varphi = \angle(\boldsymbol{G}_1, \boldsymbol{G}_2)$ be the non-obtuse angle formed by $\boldsymbol{G}_1, \boldsymbol{G}_2$ at their intersection. When we consider the distribution of φ, we may regard $\boldsymbol{G}_1$ as a fixed great circle. Let $\boldsymbol{p}$ be a pole of this $\boldsymbol{G}_1$, and let $H(\boldsymbol{p})$ be the hemisphere with center $\boldsymbol{p}$. Let $\boldsymbol{x}_2$ denote the pole of $\boldsymbol{G}_2$ contained in $H(\boldsymbol{p})$. Then $\boldsymbol{x}_2$ is uniformly distributed in $H(\boldsymbol{p})$. Note that $\varphi = \boldsymbol{px}_2$. Hence $\Pr(\varphi < t) = 2\pi(1 - \cos t)/(2\pi) = 1 - \cos t$. Thus, we have the following

Theorem 8.2 *For two independent random great circles $\boldsymbol{G}_1, \boldsymbol{G}_2$, the probability density function of the intersection angle $\varphi = \angle(\boldsymbol{G}_1, \boldsymbol{G}_2)$ is given by $f(\varphi) = \sin\varphi, 0 \le \varphi \le \pi/2$.* □

The expected value $E(\varphi)$ is computed as follows.

$$E(\varphi) = \int_0^{\pi/2} \varphi \sin\varphi d\varphi$$
$$= \int_0^{\pi/2} (\cos\varphi - (\varphi\cos\varphi)')d\varphi$$
$$= [\sin\varphi]_0^{\pi/2} - [\varphi\cos\varphi]_0^{\pi/2} = 1.$$

8.3 Random Quartets

For a subset $V \subset \mathbb{S}$, the intersection of all hemispheres that contain V is called the *convex hull of V in $\mathbb{S}$*. (If no hemisphere contains V, then the convex hull of V is $\mathbb{S}$.) For example, the convex hull of a pair of points $x, y \in \mathbb{S}$ lying in the interior of a hemisphere is the geodesic segment xy. If three points $x, y, z \in \mathbb{S}$ lying in the interior of a hemisphere do not lie on a great circle, then the convex hull of x, y, z is a spherical triangle.

A set of n (≥ 3) great circles is said to be *in general position*, if no three circles of them pass through the same point.

Lemma 8.1

(1) *If four great circles on $\mathbb{S}$ are in general position, then these four great circles divide the sphere $\mathbb{S}$ into 14 regions. Among them, eight are spherical triangles, and six are spherical quadrilaterals. See Fig. 8.1, with $\boldsymbol{G}_1, \boldsymbol{G}_2, \boldsymbol{G}_3, \boldsymbol{G}_4$ as four great circles in general position.*

(2) *If four great circles on $\mathbb{S}$ are not in general position, then among the regions divided by these great circles, no region is a quadrilateral.*

Proof

(1) (Though it is almost clear from Fig. 8.1, let us present a proof.) Two great circles meet at two points, and since no three great circles share the same point, there arise $2\binom{4}{2} = 12$ points as the intersections of great circles. Since each great circle is divided into 6 arcs by other three circles, there arise 24 arcs in total. Hence, we obtain a connected graph realized on $\mathbb{S}$ that has 12 vertices, and 24 edges. Then by Euler's formula $12 - 24 + f = 2$, and hence the number f of faces of this graph is 14. Since the number of great circles is four, each face is either a triangle or a (convex) quadrilateral. Let f_3, f_4 denote the numbers of triangles and quadrilaterals, respectively. Then $f_3 + f_4 = 14$ and $3f_3 + 4f_4 = 2 \times 24$, and we have $f_3 = 8,\ f_4 = 6$.

(2) Among the four great circles, suppose that three of them pass through the same point x. By these three circles, $\mathbb{S}$ is divided into six lunes. The remaining

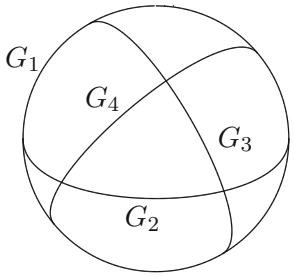


Fig. 8.1 Division of $\mathbb{S}$ by four great circles in general position

great circle divides each lune into at most two triangles, and there arises no quadrilateral.

□

Theorem 8.3 *When we choose four random points on* $\mathbb{S}$ *independently, then the probability that the convex hull of these four points in* $\mathbb{S}$ *becomes a spherical quadrilateral on* $\mathbb{S}$ *is* $3/8$.

Proof We take four random points on $\mathbb{S}$ in the following two-steps way. First choose four random great circles $\boldsymbol{G}_1, \boldsymbol{G}_2, \boldsymbol{G}_3, \boldsymbol{G}_4$ independently. Next, from the pair of poles of each great circle $\boldsymbol{G}_i$, choose either one with equal probability. In this two-steps way, we can take four independent random points $\boldsymbol{p}_1, \boldsymbol{p}_2, \boldsymbol{p}_3, \boldsymbol{p}_4$. Let $\boldsymbol{K}$ denote the convex hull of these four points in $\mathbb{S}$. If $\boldsymbol{K}$ is a spherical quadrilateral, then $\boldsymbol{p}_1, \boldsymbol{p}_2, \boldsymbol{p}_3, \boldsymbol{p}_4$ are the vertices of the quadrilateral. Now, by Theorem 4.7, we have

$$\begin{aligned}&\boldsymbol{K} \text{ is a quadrilateral}\\ &\quad\Leftrightarrow \boldsymbol{K}^\circ = H(\boldsymbol{p}_1) \cap H(\boldsymbol{p}_2) \cap H(\boldsymbol{p}_3) \cap H(\boldsymbol{p}_4) \text{ is a quadrilateral.}\end{aligned}$$

Four great circles $\boldsymbol{G}_1, \boldsymbol{G}_2, \boldsymbol{G}_3, \boldsymbol{G}_4$ divide $\mathbb{S}$ into several regions. Since to choose a pole of $\boldsymbol{G}_i$ corresponds to choose one side of $\boldsymbol{G}_i$, $\boldsymbol{K}^\circ = H(\boldsymbol{p}_1) \cap H(\boldsymbol{p}_2) \cap H(\boldsymbol{p}_3) \cap H(\boldsymbol{p}_4)$ is nonempty if and only if $\boldsymbol{K}^\circ$ is one of the regions in $\mathbb{S}$ divided by $\boldsymbol{G}_1, \ldots, \boldsymbol{G}_4$. By Lemma 8.1(2), in the case of four great circles that are not in general position, there arises no quadrilateral. Hence $\boldsymbol{K}^\circ$ is a spherical quadrilateral if and only if

(i) $\boldsymbol{G}_1, \boldsymbol{G}_2, \boldsymbol{G}_3, \boldsymbol{G}_4$ are in general position, and
(ii) for each $\boldsymbol{G}_i$, one side of it is chosen so that their intersection becomes a quadrilateral.

By Example 8.4, (i) occurs with probability 1. Under the condition (i), the number of ways to choose one side of each $\boldsymbol{G}_i$ for $i = 1, 2, 3, 4$ is $2^4 = 16$, and among these 16 ways, just 6 ways determine quadrilaterals. Namely, under the condition (i), the probability that (ii) occurs is $6/16 = 3/8$. Therefore, the probability that the convex hull of $\boldsymbol{p}_1, \boldsymbol{p}_2, \boldsymbol{p}_3, \boldsymbol{p}_4$ becomes a spherical quadrilateral is $3/8$. □

Remark 8.1 When we choose independently four random points on $\mathbb{S}$, the probability that the convex hull of these four points becomes a spherical triangle is $8/16 = 1/2$.

Remark 8.2 When we choose independently random four points, each uniformly distributed in a bounded region W in the plane, what is the probability $P(W)$ that the convex hull of these four points becomes a quadrilateral? This is known as *Sylvester's four-point-problem*. It is known that, for example,

$$P(\text{triangle}) = 2/3,\ P(\text{rectangle}) = 25/36,\ P(\text{circle}) = 1 - \frac{35}{12\pi^2}.$$

Theorem 8.3 is regarded as the sphere version of Sylvester's four-point-problem.

Corollary 8.1 *For four random points $\boldsymbol{x}_i$, $i = 1, 2, 3, 4$, taken independently on $\mathbb{S}$, the probability that the geodesic segments $\boldsymbol{x}_1\boldsymbol{x}_2$ and $\boldsymbol{x}_3\boldsymbol{x}_4$ cross each other is $1/8$.*

Proof The two segments $\boldsymbol{x}_1\boldsymbol{x}_2$ and $\boldsymbol{x}_3\boldsymbol{x}_4$ cross each other if and only if the convex hull of $V := \{\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3, \boldsymbol{x}_4\}$ is a spherical quadrilateral, and $\boldsymbol{x}_1\boldsymbol{x}_2$ is a diagonal of the quadrilateral. Under the condition that the convex hull of V is a spherical quadrilateral, each of the three segments $\boldsymbol{x}_1\boldsymbol{x}_2, \boldsymbol{x}_1\boldsymbol{x}_3, \boldsymbol{x}_1\boldsymbol{x}_4$ becomes a diagonal of the quadrilateral with equal probability. Hence $\boldsymbol{x}_1\boldsymbol{x}_2$ and $\boldsymbol{x}_3\boldsymbol{x}_4$ cross each other with probability $\frac{3}{8} \times \frac{1}{3} = \frac{1}{8}$. □

Remark 8.3 For more about geometric probability on the sphere, see H. Maehara and H. Martini [532]. On higher dimensional spherical versions of Sylvester's problem, see [534].

8.4 Wendel's Theorem

Lemma 8.2 *If n great circles on $\mathbb{S}$ are in general position, then they divide $\mathbb{S}$ into $n^2 - n + 2$ regions.*

Proof Since two great circles intersect at two points, we have $2\binom{n}{2} = n^2 - n$ points as the intersection of n great circles. Let us regard the figure obtained by these n great circles as a connected graph realized on $\mathbb{S}$. It has $n^2 - n$ vertices, and each vertex has degree 4. Hence due to the Hand-shaking lemma, it has $2(n^2 - n)$ edges. Now, by Euler's formula, the number of faces is equal to $n^2 - n + 2$. □

The next theorem is the 2-dimensional version of a theorem due to J. G. Wendel [807].

Theorem 8.4 (J .G .Wendel) *The probability that independently chosen n random points on $\mathbb{S}$ lie on a hemisphere is equal to $(n^2 - n + 2)/2^n$.*

Proof We take n random points on $\mathbb{S}$ in the following two-steps way. First choose n random great circles $\boldsymbol{G}_1, \boldsymbol{G}_2, \ldots, \boldsymbol{G}_n$ independently. Next, from the pair of poles of each great circle $\boldsymbol{G}_i$, choose either one with equal probability. In this two-steps way, we can take n independent random points $\boldsymbol{p}_1, \boldsymbol{p}_2, \ldots, \boldsymbol{p}_n$. Let $H(\boldsymbol{p}_i)$ be the hemisphere with center $\boldsymbol{p}_i$. If $\boldsymbol{p}_i$, $i = 1, 2, \ldots, n$, lie in a hemisphere with center $\boldsymbol{q}$, then $\boldsymbol{p}_i\boldsymbol{q} \leq \pi/2$ for every i, which implies that $\boldsymbol{q} \in H(\boldsymbol{p}_i)$ for $i = 1, 2, \ldots, n$. Thus,

$$\boldsymbol{p}_1, \boldsymbol{p}_2, \ldots, \boldsymbol{p}_n \text{ lie on a hemisphere } \Leftrightarrow H(\boldsymbol{p}_1) \cap H(\boldsymbol{p}_2) \cap \cdots \cap H(\boldsymbol{p}_n) \neq \emptyset.$$

Random great circles $\boldsymbol{G}_i, i = 1, 2, \ldots, n$, are in general position with probability 1. Hence they divide $\mathbb{S}$ into $n^2 - n + 2$ regions by Lemma 8.2. $H(\boldsymbol{p}_1) \cap H(\boldsymbol{p}_2) \cap \cdots \cap H(\boldsymbol{p}_n) \neq \emptyset$ if and only if $H(\boldsymbol{p}_1) \cap H(\boldsymbol{p}_2) \cap \cdots \cap H(\boldsymbol{p}_n)$ is one of $n^2 - n + 2$ regions. Once n random great circles are chosen, then there are 2^n ways to choose one pole for each great circle. Thus, once n random great circles are chosen, then the

probability that $\boldsymbol{p}_i$, $i = 1, 2, \ldots, n$, lie on a hemisphere is equal to $(n^2 - n + 2)/2^n$. Since this is constant and independent from the choice of n random great circles, the theorem follows. □

Let us prove here the circle version of the above theorem in a similar way.

Theorem 8.5 *The probability that n random points independently and uniformly distributed on a circle lie on a semicircle is equal to* $2n/2^n$.

Proof Let us take n random points on a unit circle in the following two-steps way: First, take n random "diameters" I_i, $i = 1, 2, \ldots, n$, of the circle independently, and then choose one endpoint $\boldsymbol{x}_i$ from each diameter I_i with equal probability. The n points $\boldsymbol{x}_i$, $i = 1, 2, \ldots, n$, are random independent points. For a point $\boldsymbol{p}$ on the circle, let us denote by $S(\boldsymbol{p})$ the semicircle with center (midpoint) $\boldsymbol{p}$. Then

$$\begin{aligned} \boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_n \in S(\boldsymbol{p}) &\Leftrightarrow \boldsymbol{p}\boldsymbol{x} \leq \pi/2, \text{ for } i = 1, 2, \ldots, n \\ &\Leftrightarrow \boldsymbol{p} \in S(\boldsymbol{x}_i), \text{ for } i = 1, 2, \ldots, n. \end{aligned}$$

Hence, $\boldsymbol{x}_i$, $i = 1, 2, \ldots, n$, lie on a semicircle if and only if

$$S(\boldsymbol{x}_1) \cap S(\boldsymbol{x}_2) \cap \cdots \cap S(\boldsymbol{x}_n) \neq \emptyset. \tag{8.1}$$

The $2n$ endpoints of $S(\boldsymbol{x}_i)$, $i = 1, 2, \ldots, n$, divide the circle into $2n$ arcs, and (8.1) holds whenever $S(\boldsymbol{x}_1) \cap \cdots \cap S(\boldsymbol{x}_n)$ is one of the $2n$ arcs. Once n random diameters are chosen, $\boldsymbol{x}_i$, $i = 1, 2, \ldots, n$, are chosen in 2^n ways, and (8.1) holds in $2n$ ways. Thus, the probability that $\boldsymbol{x}_i$, $i = 1, 2, \ldots, n$, lie on a semicircle under the condition that n diameters are given is equal to $2n/2^n$. Since this is constant, we can deduce that the probability that $\boldsymbol{x}_i$, $i = 1, 2, \ldots, n$, lie on a semicircle is $2n/2^n$. □

Example 8.5 For three random points $\boldsymbol{x}$, $\boldsymbol{y}$, $\boldsymbol{z}$ chosen independently on a circle, the probability that the planar triangle $\triangle \boldsymbol{x}\boldsymbol{y}\boldsymbol{z}$ contains the center of the circle is equal to $1 - 2 \cdot 3/2^3 = 1/4$.

8.5 Crofton's Formula

Let γ be a geodesic segment with length θ ($< \pi$), and $\boldsymbol{G}$ be a random great circle with poles $\boldsymbol{x}, \boldsymbol{x}^*$ on $\mathbb{S}$. Then $\boldsymbol{G}$ passes through a point $\boldsymbol{p}$ on γ if and only if $\boldsymbol{x}\boldsymbol{p} = \pi/2$. Hence, if we put

$$W = \{\boldsymbol{y} \in \mathbb{S} : \boldsymbol{y}\boldsymbol{p} = \pi/2 \text{ for some } \boldsymbol{p} \in \gamma\},$$

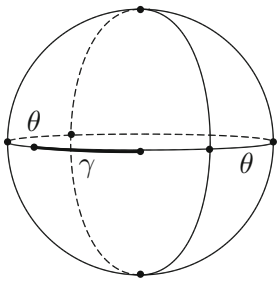


Fig. 8.2 A segment γ and two lunes of angle θ

then we have $\boldsymbol{G} \cap \gamma \neq \emptyset \Leftrightarrow \boldsymbol{x} \in W$. Since W is the sum of two lunes with angle θ (see Fig. 8.2), it follows that

$$\Pr(\boldsymbol{G} \cap \gamma \neq \emptyset) = \frac{2 \times 2\theta}{4\pi} = \frac{\theta}{\pi}.$$

Let Γ be a simple polygonal curve consisting of k segments γ_i of length $\theta_i < \pi$, $i = 1, 2, \ldots, k$. Let $\nu_i := \nu(\boldsymbol{G} \cap \gamma_i)$ denote the number of points in $\boldsymbol{G} \cap \gamma_i$. Then ν_i is a random variable with possible values 0, 1, ∞, and $\Pr(\nu_i = \infty) = 0$. Put

$$\nu(\boldsymbol{G} \cap \Gamma) = \nu_1 + \nu_2 + \cdots + \nu_k.$$

Since the expected value of ν_i is given by $E(\nu_i) = \theta_i/\pi$, the expected value $E(\nu(\boldsymbol{G} \cap \Gamma)$ is given by

$$\nu(\boldsymbol{G} \cap \Gamma) = \frac{\theta_1}{\pi} + \cdots + \frac{\theta_k}{\pi} = \frac{\theta_1 + \cdots + \theta_k}{\pi}.$$

By approximating a given curve on $\mathbb{S}$ by a polygonal curve on $\mathbb{S}$, the next theorem follows.

Theorem 8.6 *Let Γ be a curve of length L on $\mathbb{S}$. Then $E(\nu(\boldsymbol{G} \cap \Gamma)) = \frac{L}{\pi}$, where $\boldsymbol{G}$ is a random great circle on $\mathbb{S}$, and $\nu(\boldsymbol{G} \cap \Gamma)$ is the number of intersection points of $\boldsymbol{G}$ and Γ.* □

Let K be a convex region on $\mathbb{S}$ whose boundary ∂K has length L. The probability that a random great circle $\boldsymbol{G}$ is tangent to ∂K is 0. If $\boldsymbol{G}$ cuts K, then $\nu(\boldsymbol{G} \cap \partial K) = 2$. Thus, the value of $\nu(\boldsymbol{G} \cap \partial K)$ is either 0 or 2. Hence

$$\frac{L}{\pi} = E(\nu(\boldsymbol{G} \cap \partial K)) = 2 \times \Pr(\boldsymbol{G} \cap K \neq \emptyset).$$

Thus, we have the next theorem.

Theorem 8.7 (Crofton) *Let K be a convex region with perimeter L. Then*

$$\Pr(\boldsymbol{G} \cap K \neq \emptyset) = \frac{L}{2\pi}.$$

□

8.6 Random Points on a Hemisphere

Let H be a fixed hemisphere of $\mathbb{S}$. By a random point $\boldsymbol{x}$ on H we mean a point uniformly distributed on H, that is, for any $W \subset H$, we have $\Pr(\boldsymbol{x} \in W) = (area\, W)/(2\pi)$. Since the hemisphere H is not isotropic, computations of probabilities and expected values for random figures on H are generally more difficult than computations on $\mathbb{S}$. Applying Crofton's formula, let us calculate the expected value of the length of a geodesic arc connecting two random points chosen independently on H.

First, let us recall the *law of total probability*: Let T be a continuous random variable with probability density function $f(t)$. Then

$$\Pr(\cdots) = \int_{-\infty}^{\infty} \Pr(\cdots | T = t) f(t) dt,$$

where $\Pr(\cdots | T = t)$ is the conditional probability of the event "$\cdots$" under the condition $T = t$.

Theorem 8.8 *For two random points $\boldsymbol{x}$, $\boldsymbol{y}$ chosen independently on the hemisphere H, the expected value of the spherical distance between $\boldsymbol{x}$ and $\boldsymbol{y}$ is equal to $4/\pi$.*

Proof Let T be the length of the segment $\gamma = \boldsymbol{x}\boldsymbol{y}$, and let $f(t)$ be the probability density function of T. For a random great circle $\boldsymbol{G}$ on $\mathbb{S}$, we have

$$\Pr(\boldsymbol{G} \cap \gamma \neq \emptyset) = \int_0^{\pi} \Pr(\boldsymbol{G} \cap \gamma \neq \emptyset \,|\, T = t) f(t) dt,$$

which is, applying Crofton's formula, equal to

$$\int_0^{\pi} \frac{t}{\pi} f(t) dt = \frac{E(T)}{\pi}.$$

Hence we have $E(T) = \pi \Pr(\boldsymbol{G} \cap \gamma \neq \emptyset)$. The hemisphere H is divided by $\boldsymbol{G}$ into two regions, and "$\boldsymbol{G} \cap \gamma \neq \emptyset$" is equivalent to the property that $\boldsymbol{x}$ and $\boldsymbol{y}$ lie in different regions. Let $\boldsymbol{p}$ be the center (pole) of H, and $\boldsymbol{z}$ be a pole of $\boldsymbol{G}$. Let θ denote the spherical distance between $\boldsymbol{z}$ and $\boldsymbol{p}$. Then θ is a random variable, and its probability density function is given by $\frac{1}{2}\sin\theta$ as already seen in Sect. 8.2. The intersection angle $\angle(\boldsymbol{G}, \partial H)$ of $\boldsymbol{G}$ and ∂H (the boundary of hemisphere H) is equal to θ, see Fig. 8.3.

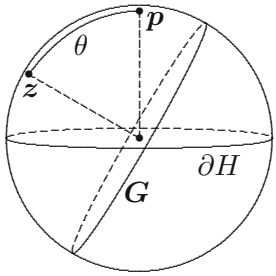


Fig. 8.3
$\angle(\boldsymbol{G}, \partial H) = \boldsymbol{zp} = \theta$

Now, $\boldsymbol{G}$ divides H into two lunes with angle θ and angle $\pi - \theta$. Note that $\boldsymbol{G} \cap \gamma = \emptyset$ if and only if both $\boldsymbol{x}$, $\boldsymbol{y}$ lie in the same lune (of angle θ or angle $\pi - \theta$). Hence

$$\Pr(\boldsymbol{G} \cap \gamma = \emptyset \,|\, \theta = t) = \left(\frac{t}{\pi}\right)^2 + \left(\frac{\pi - t}{\pi}\right)^2,$$

and hence

$$\begin{aligned}\Pr(\boldsymbol{G} \cap \gamma \neq \emptyset \,|\, \theta = t) &= 1 - \Pr(\boldsymbol{G} \cap \gamma = \emptyset \,|\, \theta = t) \\ &= 2t/\pi - 2t^2/\pi^2 = \frac{2t(\pi - t)}{\pi^2}.\end{aligned}$$

Thus,

$$\begin{aligned}\Pr(\boldsymbol{G} \cap \gamma \neq \emptyset) &= \int_0^\pi \Pr(\boldsymbol{G} \cap \gamma \neq \emptyset \,|\, \theta = t) \frac{1}{2} \sin t dt \\ &= \int_0^\pi \frac{2t(\pi - t)}{\pi^2} \frac{1}{2} \sin t dt.\end{aligned}$$

Since

$$\int t \sin t dt = t \sin t - t \cos t,$$

$$\int t^2 \sin t dt = 2t \sin t - (t^2 - 2) \cos t,$$

we have

$$\begin{aligned}\Pr(\boldsymbol{G} \cap \gamma \neq \emptyset) &= \frac{1}{\pi} [t \sin t - t \cos t]_0^\pi - \frac{1}{\pi^2} \left[t \sin t - (t^2 - 2) \cos t\right]_0^\pi \\ &= \frac{4}{\pi^2}.\end{aligned}$$

Therefore, $E(T)/\pi = 4/\pi^2$, and $E(T) = 4/\pi$. □

Corollary 8.2 *The expected value of the perimeter of a spherical triangle whose vertices are random points taken independently on a fixed hemisphere is equal to* $12/\pi$. □

8.7 Santaló's Chord Theorem

Since $\mathbb{S}$ is symmetric and isotropic, any point on $\mathbb{S}$ chosen randomly not depending on the already given subset of $\mathbb{S}$ can be regarded as a random point uniformly distributed on $\mathbb{S}$. In particular, each of the intersection points of two independent random great circles on $\mathbb{S}$ is a random point when we ignore the two random great circles. Let us state this as a lemma.

Lemma 8.3 *Each of the intersection points of two independent random great circles is a random point on* $\mathbb{S}$. □

Theorem 8.9 (Santaló's Chord Theorem) *Let K be a convex region on $\mathbb{S}$. For a random great circle $\boldsymbol{G}$ on $\mathbb{S}$, let φ denote the length of the chord $\boldsymbol{G} \cap K$, see Fig. 8.4. Then*

$$E(\varphi) = (\textit{the area of } K)/2.$$

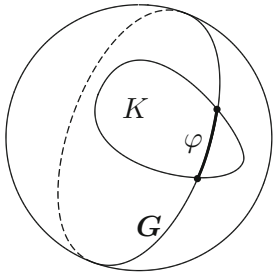
Proof Let $\boldsymbol{G}_i, i = 1, 2$, be two random great circles chosen independently on $\mathbb{S}$. We may regard each of the intersection points of $\boldsymbol{G}_1, \boldsymbol{G}_2$ as a random point. Hence $\Pr(\boldsymbol{G}_1 \cap \boldsymbol{G}_2 \cap K \neq \emptyset) = (\text{the area of } K)/(2\pi)$.

On the other hand, denoting the probability density function of φ by $f(\varphi)$, and applying Theorem 8.6, we can compute $\Pr(\boldsymbol{G}_1 \cap \boldsymbol{G}_2 \cap K \neq \emptyset)$ in the following way:

$$\int_0^\pi \Pr(\boldsymbol{G}_1 \cap (\boldsymbol{G}_2 \cap K)) \neq \emptyset) \mid \text{the length of } \boldsymbol{G}_2 \cap K = \varphi) f(\varphi)\, d\varphi$$
$$= \int_0^\pi \frac{\varphi}{\pi} f(\varphi)\, d\varphi = \frac{E(\varphi)}{\pi}.$$

Therefore, $E(\varphi) = (\text{the area of } K)/2$. □

Fig. 8.4 Theorem 8.9



The sphere-version of Crofton's formula and Santaló's chord theorem were obtained by L. A. Santaló in [687].

8.8 Exercises

8.1 When we choose five random great circles on $\mathbb{S}$, it is divided into 22 regions with probability 1. Find the numbers of triangles, quadrilaterals, and pentagons. (Study this by drawing figures.)

8.2* Prove that the probability that five random points on $\mathbb{S}$, chosen independently, form the vertices of a convex pentagon is $1/16$.

8.3 Prove that for four random points $\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3, \boldsymbol{x}_4$ on $\mathbb{S}$, chosen independently, the probability that the geodesic segments $\boldsymbol{x}_1\boldsymbol{x}_2$ and $\boldsymbol{x}_3\boldsymbol{x}_4$ cross each other is $1/8$.

8.4 What is the probability that a random great circle on $\mathbb{S}$ intersects a fixed spherical cap of angular radius $\rho < \pi/2$?

8.5 What is the probability that the geodesic segment connecting two random points on $\mathbb{S}$ chosen independently, intersects a fixed great circle on $\mathbb{S}$?

8.6* What is the probability that a random great circle on $\mathbb{S}$ intersects a given semicircle of a circle with radius $\rho < \pi/2$ in *two* points?

8.9 Notes

In this chapter the notes consist of the following two parts: First we discuss a longer part on random points and triangles on the sphere, and then a comprehensive report on Crofton's formula and related topics is given.

8.9.1 Notes on Random Points and Triangles on the Sphere

These announced results on configurations of random points and triangles (mainly) on the sphere have clearly also to do with random polygons, polytopes, and with famous problems (like Sylvester's four-point problem, see Sect. 8.3) from the field of *geometric probability*. In most cases, the studied objects are finite sets of independent, uniformly distributed and random points and their convex hulls. After listing survey-like publications, we present a selection of research papers; in both cases, mainly chronological order is chosen.

Book Parts and Surveys

Miles presents with [574] the first large survey on problems and results from *geometric probability on the sphere*. E.g., Section 3 of this expository paper refers to spherical caps that contain two, three or m uniform independent particles. Using integral geometry, such results are also extended to arbitrary domains (instead of caps). Section 5 treats *random spherical triangles*, getting the joint distributions of the lengths of the three sides, and of the sizes of the three angles (the sides being pairwise independent). In Section 6, *tessellations of the sphere* determined by n uniform random great circles are studied. For the polygons in such a tessellation a long list of first and second order moments of the area, perimeter, and number of angles is presented. Section 7 offers then related results on *Voronoi and Delaunay tessellations*, and finally the duality between points and great circles in this framework is discussed, like also extensions to higher dimensions.

Deák [203] gives an overview to the known (computerized) techniques for generating points uniformly inside an n-dimensional hypersphere and also uniformly on the surface of such a sphere. He presents modifications converting an interior algorithm into a surface algorithm and vice versa. An inductive method of projecting points on the surface of an $(n-1)$-dimensional sphere onto the surface of an n-dimensional sphere is discussed. Further on, Deák investigates the relationship between the inductive projection method and the polar transformation method for getting points within a hypersphere.

In Chapter 5 of their book [63], Barrett and Mackay discuss topics from the geometry on the sphere, including packing problems and *arranging points on spheres at random*. Tichy [777] discusses distribution properties of random points on the sphere and respective applications to *numerical integration* and to *numerical methods for solving related equation systems*.

Croft et al. [190] survey in their Section B5 several questions on *random polygons and polyhedra*, e.g. of the following type: Find bounds on $V(K,m)$, which denotes the *expected volume* of the convex hull of m random points in a convex body K (which can also be a ball/ellipsoid) of unit volume. For instance, Groemer [331] showed that the expected volume $V(K,n+1)$ is maximal when K is an ellipsoid. Also Sylvester's four-point problem (see Sect. 8.3) is discussed in Section B5 of [190].

Together with several coauthors Kendall obtained many results on a *theory of shape* in the plane, with statistical applications, e.g. to the asymptotic probability of given random sets being nearly circular. We refer here to his monograph [449]. And Kendall surveys in [447] analogous results on the 2-sphere, which mainly refer to *spherical triangles*.

In the monograph [706] Schneider and Weil present results from *spherical integral geometry* mainly in Sect. 6.5. Subjects like the local spherical Steiner formula, (generalized) support measures, spherical intrinsic volumes, and the spherical convex ring are discussed there. See also the dissertation [303] of Glasauer, where (among other results) local kinematic formulas for spherically convex bodies are studied.

Reitzner [662] gives a nice *survey on random polytopes*. In particular he underlines the distributional sides of functionals of random polytopes, such as estimates for higher moments, limit theorems, and large deviation inequalities. The floating body problem and visibility regions flank this part as geometric tools. And also Hug [409] presents a nice and comprehensive *survey on random polytopes*. In the first part he discusses topics from two main directions: results for random points in *smooth* convex bodies that have been extended to convex bodies with weaker smoothness assumptions, and results on expectations which have been supplemented by *distributional results*. In the second part, results on *intersecting random halfspaces, hyperplane tessellations*, and cells of *random mosaics* are studied. The author presents also an impressive collection of used geometric notions and methods as well as probabilistic tools (such as duality, stability of results, associated bodies, zonoids and further topics). This rich representation of methods and tools is one of the reasons for citing these surveys of Reitzner and Hug here, although their goal is not the study of *spherical* polytopes. The paper [410] of Hug and Schneider is dedicated to *tessellations of the Euclidean* $(n-1)$*-spheres by great* $(n-2)$*-subspheres*. These are clearly equivalent to tessellations of Euclidean n-space by hyperplanes passing through the origin, also called *conical tessellations*. The authors study expectations for general classes of geometric functionals of *random polyhedral cones*. E.g., these are: vertex number, total edge length, surface area and volume of the generated spherical polytopes. The complete covariance structure of these geometric functionals for an isotropic random Schläfli cone is determined, yielding the spherical analogue of a result due to Miles. This research paper contains also a comprehensive list of references showing various aspects of this little field between stochastic geometry and polytope theory; therefore it is cited here (also as survey).

The exposition [531] of Maehara and Martini collects many interesting results from *geometric probability on the 2-sphere in Euclidean 3-space*. First the authors present typical tools and notions from classical spherical geometry which enable them to offer the results in an intuitive, geometric way. Based on this, they study *random spherical polygons, expectations for perimeters*, and *areas of spherical triangles*, determined by three random points, and probabilities for the shape of the convex hull of four and five random points. The increasing difficulties for more points are explained. Then they derive Crofton's formula and the chord theorem of Santaló, yielding *probability results on spherical caps* (e.g., conditions for covering the whole 2-sphere when being of the same size). Also a *higher dimensional outlook* is given. The authors lay emphasize on elementary proofs and pose interesting open problems.

Hug and Thäle [411] study *splitting tessellations* of the Euclidean standard $(n-1)$-hypersphere. These are obtained when an existing tessellation cell is splitted into two cells by a random great $(n-2)$-hypersphere properly meeting that cell. The authors give a detailed description of such tessellations, e.g. treating the total cell curvatures, the geometry of a "typical cell" and various further geometric properties. The whole little field is comprehensively presented, such that we also decided to bring the paper here, although it is a research paper.

In Chapter 3 of the Lecture Notes [705], Schneider treats several results and notions from *spherical geometry*. These are integral-geometric identities, a spherical version of Blaschke's known selection theorem, and facts about the gnomonic projection. *Valuations on spherical polytopes* are discussed, yielding several interesting problems, and *inequalities of isoperimetric type* are discussed, too. Several formulas (like the Steiner formula or the kinematic formula) are studied in Chaps. 4, and 5 refers to arrangements of finitely many hyperplanes through the origin, which naturally generate tessellations of the space into polyhedral convex cones and can be conveniently presented with the help of spherical tessellations generated by great $(n-2)$-spheres. Different models of *random cones*, based on random hyperplane arrangements with certain distributions subject to mild assumptions, are investigated in Chap. 5, and this study of random cones is also continued in Chap. 6.

Some Selected Research Papers

Le (see [494] and [495]) investigates random point configurations and triangles in the 2-sphere, namely in view of evaluations of shape densities on the corresponding *shape space* (see the book [449] for general background). The author studies *statistical properties of occurring shapes* when the used points are generated by a random mechanism.

Arbeiter and Zähle [38] deal with *random mosaics on the* $(n-1)$*-sphere* whose cells are piecewise smooth. The authors choose the k-dimensional surface area measure and $k-1$ different curvature measures concentrated on the k-skeleton of the mosaic, $k = 0, ..., n-1$, and they derive relations between the respective densities. This yields a parameter system, based on which all mean values can be obtained. Also great hypersphere mosaics are studied.

Among various further results, the authors of [108] derive a precise asymptotic formula for the *volume of spherical regular simplices* as the dimension n tends to infinity.

Sugimoto and Tanemura [762] study the *random sequential coverings* of congruent spherical caps under the condition that none of them covers the center of another one. An algorithm for such type of coverings is given.

Gao et al. [290] prove that among all spherically convex bodies K of given volume, the total measure of great hyperspheres hitting K is minimal if K is a spherical cap. This isoperimetric type result gives a *spherical analogue of the usual Urysohn inequality* (volume against mean width), and it can be seen as *generalization of the Blasche-Santaló inequality*, since in this setting for prescribed volume spherical caps maximize the volume of the polar body. The authors present two approaches, each of them interesting for itself.

Maehara studies *intersection graphs of random spherical caps* in [522]. Let m be a positive integer, and $r := r(m)$ be a non-negative real number that is at most 1. Now choose m points independently and uniformly at random from the 2-sphere in 3-space to be centers $C_1, \ldots, C_m$ of the spherical caps of area $4r\pi$. We call a graph G_r the *intersection graph* of the family $C_1, \ldots, C_m$ when $C_1, \ldots, C_m$ form

its vertices, and the edges of G_r come to be whenever there is nonempty intersection between two caps. The author studies various interesting *asymptotic properties* of G_r with m going to infinity.

Pivovarov [638] derives threshold results for expectations of volume ratios referring to random Gaussian polytopes (being the convex hull of m independent identically distributed (i.i.d.) Gaussian random vectors) and *random polytopes whose vertex set is taken from the Euclidean sphere*. E.g., for such spherical polytopes and n tending to infinity, a super-exponential result is obtained, and in the dual sense corresponding results for *spherical polytopes generated by random facets* are obtained, too.

Maeda [517] classifies spherical triangles into seven types regarding sides and angles. He calculates for *random spherical triangles* the probabilities of occurrence of these types. As an application, the expectation of numbers of acute triangulations of a random spherical triangle in the 2-sphere can be estimated. For the spherical convex hulls P_m of m independent, uniformly distributed random points in a *halfsphere*, Bárány et al. [59] study the *asymptotic behaviour of the expectation of certain characteristics* when m tends to infinity. They succeed with *explicit integral expressions* for the expected facet number, surface area, and spherical mean width. Going on, they use for mean volume and vertex number more sophisticated methods. Finally, the results are compared with analogues for Euclidean space, and interesting problems are posed.

Results on *f-vectors* (which express the numbers of k-faces) of so-called Poisson zero n-polytopes Z^n of homogeneous and isotropic Poisson hyperplane tessellation in n-space are derived by Kabluchko [435]. More precisely, he gives an explicit combinatorial formula in terms of a special array of numbers satisfying a recurrence relation. A result on *intrinsic volumes* of Z^n has applications for a type of *random spherical polytopes*. Namely, the expected f-vector and also the expected volume of the spherical convex hull of random points chosen uniformly and independently (again) from the n-dimensional *halfsphere* can be computed, and an *extension of Sylvester's four-point problem* is obtained, too.

Also Brauchart et al. [126] study the convex hulls of m random points from the unit sphere of n-space, regarding *surface area measure* and asymptotics (as m goes to infinity) for the expected moments of the *radii of spherical caps* corresponding to the *facets* of the obtained polytopes. The authors pose conjectures for the asymptotic distribution of the scaled radii of these spherical caps, and also for the expected value of the largest of these radii (the covering radius). Numerical results, also given in this paper, support these conjectures.

Sylvester's four-point problem (asking for the probability that four random points in the plane form a re-entrant quadrilateral) has a long history, which is nicely described by Pfiefer [636] (see also our Sect. 8.3). Maehara and Martini [534] prove a *spherical extension of Sylvester's four-point problem*: Let X be a finite subset of the unit $(n-1)$-sphere in Euclidean n-space. This set is called *extremal* if for every $x \in X$ there is a hemisphere that has all points of X as interior points, except for x being on its boundary. The probability that a random sample of $n+1$ points, chosen uniformly from the unit $(n-1)$-sphere, forms an extremal set is $1-(n+2)/2^n$. In

important exposition dealing with further extensions of Sylvester's problem is due to Bárány [58].

Inspired by results on the asymptotic behaviour of the volume of Euclidean n-polytopes determined by i.i.d. points uniformly distributed in a convex body, Besau and Thäle [81] derive analogous properties of random polytopes also in *spherical spaces*, and in addition they look at the hyperbolic case and *Hilbert spaces* (using there the Busemann and Holmes-Thompson notions of volumes) and dual intrinsic volumes in Euclidean case. Among other tools, weighted floating bodies are used in the proofs. Somehow as counterpart to these results, Besau et al. [82] derive interesting *central limit theorems* (with n going to infinity) for random n-polytopes inscribed to the boundary of a smooth convex body. Besides in *spherical geometry*, these results also hold in hyperbolic and projective Finsler geometry. This way, a large family of well-known theorems is extended and unified.

And again we consider, for m random points distributed independently and uniformly on the unit $(n-1)$-sphere, the convex hull P_m. Inspired by high dimensional statistics, Bonnet and O'Reilly [102] study the n-polytopes P_m when *both numbers* m and n tend *simultaneously* to infinity. Under these assumptions, they study the asymptotic behaviour of the *facet number*, of the *facet shape* and the *facet distances from the origin*.

8.9.2 Notes on Crofton's Formulae

The contribution of M. W. Crofton (1826–1915) to *integral geometry* is fundamental. The famous *Crofton formula* (also called the *Cauchy-Crofton formula*, see our Sect. 8.5) in its classical form is a typical basic result in this field. It relates the length of a planar curve to the expected number of times that a "random" line intersects this curve (see [191]).

Some History

We refer also to the paper [497] of Lebesgue, the obituary [192] and the historical articles [489, 490, 682] and [727] on integral geometry and geometric probability, treating Crofton's formulae in various ways and with interesting connections to further geometric fields.

Later extended *Crofton-type formulae* became one cornerstone of the fast development of integral geometry that, roughly speaking, started with *Blaschke's school* (1935–1939), Santaló, Chern, as well as Hadwiger, the latter in the middle of the twentieth century. The corresponding development before this is documented in the books [212] of Deltheil, [95] of Blaschke, and [448] of Kendall and Moran. Examples of early publications in this direction, partially also combined with typical problems from *convex geometry*, are again Blaschke's book [95, §7] and his report [93], as well as Santaló's papers [686] (containing a corresponding new proof of

the isoperimetric inequality in the plane), [687] (treating spherical curves), and [688] (referring to surfaces of constant negative curvature with *Crofton's formula for chords*). This is completed by Santaló's exposition [689] on n-dimensional *spaces of constant curvature* containing needed generalizations of the classical Crofton formula. Also Chern's contributions [162, Part II] (see below) and [161, §3] have to be mentioned here, since the corresponding generalizations of Crofton formulas needed there are discussed in a very detailed way. In [162, Part II] also the developments around *applications*, combined with the names of Helgason and Gelfand, are represented. And Hadwiger's contributions can, in great parts, be found in his books [355, §34] and [356, Chapter 6], followed by Stoka's monograph [761].

Later Basic Books

After his book [690], Santaló presented with great success the standard monograph [691] on *integral geometry and geometric probability*, treating *Crofton's formulae for Euclidean n-space* in Chapter 14 (Section 3), and for *non-Euclidean n-spaces* in Chapter 18 (Section 1). A strongly related historical article is [692] of Santaló. For introducing the concept of *vectorial integral geometry*, extending common integral geometry to vector-valued functions, we refer to the paper [357] of Hadwiger and Schneider. Crofton type integral formulae have their natural analogues in this theory.

Since after the appearance of [691] students in China did not have the benefit of a similar book in Chinese, Ren collected material from related courses to write finally an excellent monograph. This follows, in its structure of contents, closely Santaló's book and contains also several new results; an English translation of this book is [663]. In Chapter 5 of his book [749], Solomon applies Crofton's mean value theorems for a detailed approach to Sylvester's four-points problem and its extension to 3-space. After clarifying the notion of valuation and characterizing, in Chapters 8 and 9 of the monograph [453], the various intrinsic volumes of convex bodies, Klain and Rota give in Chapter 10 a general Crofton formula for the intrinsic valuations in all dimensions.

In their basic monograph [706] on *stochastic and integral geometry*, Schneider and Weil present Crofton type formulas in Sections 5.1 and 6.4, and in the notes to Section 5.1 they also mention *differential-geometric versions of Crofton formulas* (see also Sulanke and Wintgen [763] and again Santaló [691]). Closer to convex geometry, in Section 6.4 of [706], *Crofton formulas for mixed measures* are derived, first in a translative and then in its kinematic form. These integral formulas for mixed measures and their global versions (i.e., mixed functionals) lead to kinematic formulas for certain *mixed volumes, projection functions* and *support functions of convex bodies*. And this yields, also presented in Section 6.4, Crofton type formulas for *projection functions*.

Also in books dedicated directly and mainly to *convex geometry* one can find various generalizations and specifications of Crofton-type formulas, forming a central topic which is connected with *projection and intersection formulas*. As

examples, we mention here Section 4.4 in Schneider's monograph [704] (Crofton's intersection formulae) and Chapter 5 of the book [412] of Hug and Weil.

In the second of the two volumes dedicated to selected papers and lectures of S.-S. Chern on differential and integral geometry (see [162, Part II] one finds eight lectures reflecting the two essential periods of integral geometry, namely the classical one with Blaschke and Santaló (see above), and the second one with Gelfand and Helgason, mainly referring to deep applications. Crofton's theorem is carefully introduced at the end of the first lecture, and in the second lecture essential corollaries of it, holding for closed convex curves, are studied. Inspired by this, further related results are presented, combined also with notions like *width* of such curves and *kinematic measures* (which are needed in further lectures there).

Surveys

The following surveys reflect results on and applications of Crofton-type formulae in the spirit of convex geometry. The handbook article [707] *on integral geometry*, due to Schneider and Wieacker, deals with measures of sets of geometric objects being invariant regarding an underlying geometric transformation group. The authors have chosen the direct measure-theoretic approach on the lines of Federer. Among other intersection formulae given in Section 2, *Crofton's intersection formula* and also a *translative Crofton-type formula* are treated there. And in the handbook article [802] of Weil and Wieacker, dedicated to *stochastic geometry*, versions of the *principal kinematic formula* and the *Crofton formula* are discussed. Further on, in Section 8 (treating *stereology*) Crofton formulae are applied to obtain an expectation formula for the intrinsic volumes of convex bodies.

Schneider [703] connects these two fields (namely integral geometry and stochastic geometry) in the following sense: he presents essential techniques coming from integral geometry and being essential for stochastic geometry. He deduces *integral geometric formulae* from Hadwiger's characterization theorem for the quermassintegrals of convex bodies (see [356]), namely (among others) the Crofton formula and the principal kinematic formula. The author calculates integrals with respect to a motion-invariant or Haar measure of the intrinsic volumes of sections of convex bodies, or of intersections of a convex body with congruent copies of a second convex body. In the last chapter of [703], *generalizations of Crofton formulae* are studied. E.g., these concern *extensions to normed spaces*, or to *sections of intermediate dimensions*.

A nice lecture (or "quick course") treating integral geometry, still in the spirit of problems from convex geometry, is given by Zhang [830]. This work shows nicely how fundamental ideas of Crofton, Poincaré, Blaschke, Santaló and others were fruitfully extended in modern times. Kiderlen [452] presents a nice and mathematically rigorous *introduction into integral geometry and stereology*, starting with a nice collection of the most important integral-geometric theorems. He also proves most of them, and he derives, coming from Hadwiger's characterization of intrinsic volumes, the Crofton formula together with other basic results. Applications of such

results in *modern stereology* are presented in the second part. In Section 4 of the exposition [532], Crofton's formula (like also Santaló's chord theorem) are derived in a convenient way. Here we also mention once more the two comprehensive expository papers [489] and [490] of Langevin.

Some Selected, Recent Research Papers

Using sections with random planes or projections in random directions, Goodey et al. [310] study random sets generated from a deterministic n-dimensional convex body K. The kth *Blaschke section body* of K with $2 \leq k \leq n-1$ is a convex body whose surface area measure is derived by integrating the surface area measure of the sections of K over all k-dimensional flats. Analogously, the authors introduce the notion of kth *mean projection body* of K, whose support function is the integral of the support functions of the projections K over all k-dimensional flats. They continue with establishing relations between these associated bodies; in particular, a *Crofton-type formula for Blaschke sums* is obtained.

Curvature measures of sets with positive reach are suitable extensions of classical indefinite integrals of elementary symmetric functions of principal curvatures from smooth hypersurfaces to sets with singularities. In this field, Rother and Zähle [679] derive a kinematic Crofton-type formula relating the kinematic mean of the absolute curvature measure of a flat section of a set X of positive reach to another absolute curvature measure of X. Continuing this, Rataj establishes in [659] a more general related translative and Crofton-type integral formula.

It is well known that *Hilbert's fourth problem* asks for the description of all geometries in which usual line segments are the shortest connections of their endpoints. E.g., there are the classical examples of *Hilbert geometries*, and of the metrics induced by *norms*. To generalize this concept, Busemann naturally proposed to study notions of k-dimensional areas in n-dimensional affine spaces in which k-flats minimize area. Via an integral geometric approach, Crofton measures came into that framework, and in Schneider's exposition [701] this topic is presented in a comprehensive, insightful way. A continuation is the paper [702], where also the settings of *projective Finsler metrics* and k-dimensional *Holmes-Thompson area* (important in normed spaces) are considered. It turns out that there is a unique positive measure on the *space of lines*, yielding a Crofton-type formula for the $(n-1)$-dimensional Holmes-Thompson area. Schneider [702] presents an explicit construction of this measure on the space of lines in a polytopal Hilbert geometry. This yields an integral geometric *Crofton-type representation of the Holmes-Thompson area* of hypersurfaces.

The classical Crofton formula determines an integrated intrinsic volume of the intersection of a given convex body with an invariantly moving k-flat. One way of generalizing this concept is given by the replacement of intrinsic volumes either by local counterparts, such as curvature measures, or by certain *tensor-valued valuations*, the Minkowski tensors. Hug and Schneider [415] derive a complete set of integral geometric formulas of Crofton type for the Minkowski tensors of

convex bodies, involving integrations over Grassmannians. The principal kinematic formulae for Minkowski tensors can be obtained from Crofton formulae. Among other tools, for their proofs they use a translative Crofton formula for curvature measures. In the paper [413], Hug and Weis study versions of the Crofton formula with a tensor valuation, i.e., a valuation with values in the space of symmetric tensors. They present an interesting framework, expressing integrals of tensorial curvature measures via linear combinations of tensorial curvature measures of the considered convex body. The paper is a continuation of [415], generalizing and simplifying the Crofton formulas for tensor valuations given there in various ways.

Further papers in this direction are [413] and [414], also presenting a collection of extensions of the classical Crofton formula for generalized tensorial curvature measures of convex bodies. The authors prove that the considered system of Crofton formulas for such generalized measures is complete and cannot be simplified further. And it is also shown that the generalized tensorial curvature measures, multiplied with powers of the metric tensor, are linearly independent.

Goodey et al. [311] derive principal kinematic and *Crofton-type formulae for surface area measures* of convex bodies. These formulae are related to a localization of Hadwiger's integral geometric theorem, and they are successfully applied to get an extension of Koldobsky's orthogonality relation for *Fourier operators* on the sphere.

Chapter 9
Intersection Graphs of Spherical Caps



You cannot apply mathematics as long as words becloud reality.

— Hermann Weyl

For a family $\mathcal{C} = \{C_1, C_2, \ldots, C_N\}$ of spherical caps on $\mathbb{S}$, its intersection graph $\Omega(\mathcal{C})$ is defined as the graph whose vertices are the centers x_i of the caps C_i, $i = 1, 2, \ldots, N$, and two vertices x_i, x_j are adjacent if and only if $C_i \cap C_j \neq \emptyset$. We present a sufficient condition for the connectedness of $\Omega(\mathcal{C})$. Concerning the caps of the same radius, we specify the family of complete bipartite graphs that can be represented as the intersection graphs of caps, and also determine the range of radius necessary for the representation. In the last section, we consider the asymptotic probability that $\Omega(\mathcal{C})$ becomes connected for a family of random spherical caps of the same radius $r = r(N)$ that is a function of the number N of caps.

9.1 Connectivity

Let $\mathcal{C}$ be a family of spherical caps on the unit sphere $\mathbb{S}$ not necessarily of the same size. We always assume that their centers are all distinct. Let V denote the set of the centers of the caps in $\mathcal{C}$. The cap in $\mathcal{C}$ with center $v \in V$ is denoted by $C(v)$. The *intersection graph* $\Omega(\mathcal{C})$ of $\mathcal{C}$ is a graph with vertex set V in which two vertices $u, v \in V$ are adjacent (u is said to be a *neighbor* of v) whenever $C(u) \cap C(v) \neq \emptyset$. For a pair of adjacent vertices $u, v \in V$ of $\Omega(\mathcal{C})$, we regard the geodesic segment uv on $\mathbb{S}$ as the edge connecting u and v.

Lemma 9.1 *For the intersection graph $\Omega(\mathcal{C})$ of a family $\mathcal{C}$ of spherical caps the following holds: if two vertices x, y are connected by a path, then they can be connected by a path that forms a simple polygonal curve on $\mathbb{S}$.*

Proof Among the paths connecting x, y in $\Omega(\mathcal{C})$, let $v_1 v_2 \ldots v_k$ ($v_1 = x, v_k = y$) be the one that contains the minimum number of vertices. Suppose that $v_i v_{i+1}$

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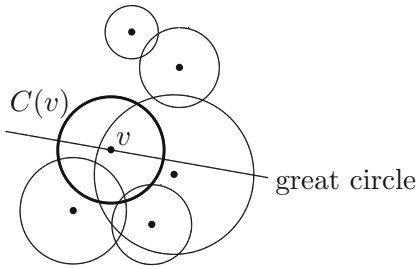


Fig. 9.1 An extremal cap $C(v)$

and $v_j v_{j+1}$ cross each other, where $j > i+1$. Then $(C(v_i) \cup C(v_{i+1})) \cap \big(C(v_j) \cap C(v_{j+1})\big) \neq \emptyset$ since it contains the intersection point of $v_i v_{i+1}$ and $v_j v_{j+1}$. If $C(v_i) \cap \big(C(v_j) \cup C(v_{j+1})\big) \neq \emptyset$, say $C(v_i) \cap C(v_j) \neq \emptyset$, then we have the path $v_1 \dots v_i v_j v_{j+1} \dots v_k$ containing less vertices, a contradiction. If $C(v_i) \cap \big(C(v_j) \cup C(v_{j+1})\big) = \emptyset$, then we have $C(v_{i+1}) \cap \big(C(v_j) \cup C(v_{j+1})\big) \neq \emptyset$, and similarly we can get a path connecting x, y and containing less vertices, a contradiction. Hence the path connecting x, y that contains the smallest number of vertices forms a simple polygonal curve on $\mathbb{S}$. □

A cap $C(u) \in \mathcal{C}$ is called *isolated* if $C(u) \cap C(v) = \emptyset$ for all $v \in V \setminus \{u\}$. It is obvious that if $\mathcal{C}$ contains an isolated cap, then $\Omega(\mathcal{C})$ is disconnected.

A cap $C(v) \in \mathcal{C}$ is called an *extremal cap* if there is a great circle passing through the center v of $C(v)$ for which all the neighbors of v lie in the same side of the great circle, allowing that some of them lie on the great circle, see Fig. 9.1. Note that an isolated cap is also an extremal cap. Note also that if a family $\mathcal{C}$ of caps contains at most 3 caps, then all caps in $\mathcal{C}$ are extremal caps.

Remark 9.1 If all centers of caps in $\mathcal{C}$ lie in a hemisphere, then $\mathcal{C}$ has an extremal cap. This can be seen as follows: Let D be the spherical cap of minimum angular radius θ that contains all centers of the caps of $\mathcal{C}$. Since the centers of the caps in $\mathcal{C}$ are all contained in a hemisphere, we have $\theta \leq \pi/2$, and at least one center v of a cap in $\mathcal{C}$ lies on the boundary of the cap D. Then the cap $C(v)$ is an extremal cap of $\mathcal{C}$.

Theorem 9.1 *Let $\mathcal{C} = \{C(v) : v \in V\}$ be a family of more than 3 spherical caps on $\mathbb{S}$, and let $G = \Omega(\mathcal{C})$ be their intersection graph. If $\Omega(\mathcal{C})$ is disconnected, then $\mathcal{C}$ has an extremal cap.*

Proof Let F, H be two connected components of G, and let U, W be their vertex sets, respectively. Then there is a simple closed curve Γ on $\mathbb{S}$ that separates F from H, such that $\Gamma \cap C(v) = \emptyset$ for all $v \in U \cup W$. (For example, such a curve can be obtained by modifying a certain component of the boundary curve of the region $X_U := \bigcup_{u \in U} C(u)$.) Let x be a point on Γ. We may suppose that the antipodal point x^* of x lies in the same side of Γ as H, allowing $x^* \in \Gamma$. Thus, $\{x, x^*\} \cap X_U = \emptyset$. Note that then we can connect x and x^* by a curve on $\mathbb{S} \setminus X_U$.

Now, let us regard x as the *north-pole*, and x^* as the *south-pole* of $\mathbb{S}$. Then, since $\{x, x^*\} \cap X_U = \emptyset$, for every vertex in U, its *longitude* is well defined. If

$u, u' \in U$ are adjacent, then the difference of their longitude is less than π. (This can be seen as follows: Suppose that the difference of their longitudes is equal to π. Then u, x, u', x^* lie on a great circle, and since $C(u) \cap C(u') \neq \emptyset$, we must have either $x \in C(u) \cup C(u')$ or $x^* \in C(u) \cup C(u')$, which contradicts the equality $\{x, x^*\} \cap X_U = \emptyset$.) Therefore, for every pair of adjacent vertices of U we can tell which vertex lies in the *east* (or *west*) of the other, unless they have the same longitude. Let us call a vertex of U *easternmost* if it has no neighbor that lies east of the vertex. Note that if $v \in U$ is the easternmost vertex, then the cap $C(v)$ is an extremal cap.

Now, starting from an arbitrary vertex $u \in U$, we try to go eastward along the edges of F. If u has a neighbor u' lying east of u, then we go to u'. If u' has a neighbor u'' lying east of u', then we go to u''. In this way we go east and east. Eventually, we either return to a vertex that once we visited, or we reach an eastermost vertex and cannot go further. If the former happens, we have a cycle of F that forms a simple closed curve separating the north-pole x and the south-pole x^*. This is, however, impossible, because there is a curve on $\mathbb{S} \setminus X_U$ that connects x and x^*, as already noted. Therefore, we finally reach an easternmost vertex, and hence, there is an extremal cap. □

Thus, for a family $\mathcal{C}$ of more than 3 spherical caps the following holds:

$$\exists \text{ an isolated cap} \Rightarrow \Omega(\mathcal{C}) \text{ is disconnected} \Rightarrow \exists \text{ an extremal cap.}$$

The idea to use "nonexistence of extremal caps" to show the connectedness of the intersection graph appeared first in [521].

9.2 On the Intersection of a Few Equal Caps

A spherical cap of radius d with center $v \in \mathbb{S}$ is denoted by $C_d(v)$, and its boundary circle is denoted by $\partial C_d(v)$. For a set $\mathcal{X}$ on $\mathbb{S}$, the maximum value of the length uv for $u, v \in \mathcal{X}$ is called the *diameter* of $\mathcal{X}$. A point-set $\mathcal{X}$ on $\mathbb{S}$ is called *d-independent* if the caps $C_d(x), x \in \mathcal{X}$, are all disjoint. Thus, for any two points u, v in a d-independent set, the spherical distance uv is greater than $2d$. For any three points $x, y, z \in \mathbb{S}$, we have $xy + yz + zx \leq 2\pi$. Hence, if $d \geq \pi/3$, then there is no d-independent set of three points.

Let $\breve{\Delta}xyz$ be a spherical triangle such that $yz \geq d, zx \geq d$ and the intersection $C_d(x) \cap C_d(y) \cap C_d(z)$ has interior points. Let $w, \bar{w}$ be the intersection points of $\partial C_d(x)$ and $\partial C_d(y)$. Then, $w \neq \bar{w}$. (If $w = \bar{w}$, then $\partial C_d(x)$ and $\partial C_d(y)$ are tangent and x, y, w lie on a great circle. Since $C_d(x) \cap C_d(y)$ has interior points, we can deduce that $C_d(x) \cup C_d(y) = \mathbb{S}$. Since z is neither an interior point of $C_d(x)$, nor an interior point of $C_d(y)$, we must have $z = w$. But this contradicts the fact that x, y, z are vertices of a spherical triangle.) Consulting Fig. 9.2, we can see that $C_d(x) \cap C_d(y) \cap C_d(z)$ is disconnected if and only if $w, \bar{w} \in C_d(z)$. If $C_d(x) \cap$

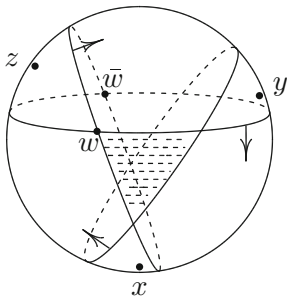


Fig. 9.2
$C_d(x) \cap C_d(y) \cap C_d(z)$ is disconnected

$C_d(y) \cap C_d(z)$ is disconnected, it has two components, consisting of two *curvilinear triangles* (possibly one is degenerated to a point), and they are separated by the boundary circle of the circumscribed cap of $\breve{\triangle}xyz$.

For an equilateral triangle $\breve{\triangle}xyz$ of edge length d, the connected component of $C_d(x) \cap C_d(y) \cap C_d(z)$ that is contained in the circumscribed cap of $\breve{\triangle}xyz$ is called the (spherical) *Reuleaux d-triangle* with vertices x, y, z, and it is denoted by $T_d(xyz)$.

Let $\breve{\triangle}xyz$ be an equilateral triangle of edge length d, and p be the center of the circumscribed cap of $\breve{\triangle}xyz$, p^* be the antipodal point of p. Then $C_d(x) \cap C_d(y) \cap C_d(z)$ is disconnected if and only if $xp^* \leq d$. If $xp^* = d$, then x, y, z, p^* are the vertices of a regular tetrahedron inscribed in $\mathbb{S}$. In this case, $d = xy = xp^*$ is computed as $\cos^{-1}(-1/3)$ and the intersection $C_d(x) \cap C_d(y) \cap C_d(z)$ has two connected components: $\{p^*\}$ and the Reulaeux d-triangle $T_d(xyz)$. Thus the next lemma follows.

Lemma 9.2 *For an equilateral triangle $\breve{\triangle}xyz$ of edge-length d, the intersection $C_d(x) \cap C_d(y) \cap C_d(z)$ is disconnected if and only if*

$$\cos^{-1}(-1/3) \leq d < \cos^{-1}(-1/2) = 2\pi/3.$$

□

Lemma 9.3 *The diameter of the Reuleaux d-triangle is d.*

Proof Let $T_d(xyz)$ be a Reuleaux d-triangle. We show that for any two points $u, v \in T_d(xyz)$, the length uv is at most d. If one of u, v coincides with a vertex of $T_d(xyz)$, then $uv \leq d$ is obvious. So, we may consider the case that none of u, v is a vertex of $T_d(xyz)$, and u lies on the boundary arc $\widehat{xy}$ of $T_d(xyz)$. Then $C_d(u)$ contains x, y, z, and x, y lie in the interior of $C_d(u)$. Let y' be the mirror image of y with respect to the plane uxO. Then the boundary circles $\partial C_d(u)$ and $\partial C_d(x)$ also intersect at y'. Hence the boundary arc $\widehat{yz}$ of $C_d(x)$ lies in the interior of $C_d(u)$ except for the endpoint y. Similarly, the boundary arc $\widehat{xy}$ is contained in $C_d(u)$. Therefore, v lies in $C_d(u)$, and $uv \leq d$. □

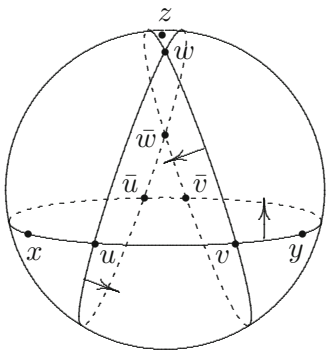


Fig. 9.3 Curvilinear triangles wuv and $\bar{w}\bar{u}\bar{v}$

Corollary 9.1 *Let $d < 2\pi/3$ and $\breve{\triangle}xyz$ be a spherical triangle such that $d \le zx, zy, xy < 2d$. Then the diameter of each component of the intersection $C_d(x) \cap C_d(y) \cap C_d(z)$ is at most d.*

Proof Let $w, \bar{w}$ denote the intersection points of $\partial C_d(x)$ and $\partial C_d(y)$ such that w and z lie in the same side of the plane Oxy, see Fig. 9.3. Then $C_d(z) \cap C_d(x) \cap C_d(y)$ is disconnected if and only if $\bar{w} \in C_d(z)$. Let us consider the case that $C_d(z) \cap C_d(x) \cap C_d(y)$ is disconnected.

Denote the intersection points of $\partial C_d(x)$, $\partial C_d(y)$, $\partial C_d(z)$ by $u, v, \bar{u}, \bar{v}$, as shown in Fig. 9.3. Then the connected components of $C_d(x) \cap C_d(y) \cap C_d(z)$ are curvilinear triangles wuv and $\bar{w}\bar{u}\bar{v}$. Note that $uw < uz = d$. Let p be a point on the arc $\overset{\frown}{w\bar{w}\bar{u}}$ such that $up = d$. Then $\breve{\triangle}puy$ is an equilateral triangle with edge length d, and the curvilinear triangle wuv is contained in the Reuleaux d-triangle $T_d(puy)$. Hence the diameter of the curvilinear triangle is at most d.

Similarly, the curvilinear triangle $\bar{w}\bar{u}\bar{v}$ is contained in a Reuleaux d-triangle $T_d(y\bar{u}q)$, where q is a point on the arc $\overset{\frown}{\bar{w}wu}$ such that $\bar{u}q = d$. Hence its diameter is at most d. □

Corollary 9.2 *Let $d < \pi/2$ and $x, y \in \mathbb{S}$ be two points such that $d \le xy < 2d$. Then $C_d(x) \cap C_d(y)$ cannot contain $(d/2)$-independent three points.*

Proof Since $C_d(x) \cap C_d(y)$ is covered by the union of two Reuleaux d-triangles, the corollary follows. □

The name "Reuleaux triangle" comes from convexity describing the most elementary "curves of constant width" different to the circle. These are special *bodies of constant width*, for which we refer to the survey [153] and to the monograph [558].

9.3 Intersection Graphs of Equal Caps

Recall that a bipartite graph is a graph in which the vertices can be classified into two groups so that no pair of vertices in the same group is adjacent. A *complete bipartite graph* is a bipartite graph in which every vertex in one group is adjacent to all vertices in the other group. A complete bipartite graph in which one group has m vertices, and the other group has n vertices, is denoted by $K(m, n)$. Figure 9.4 shows the complete bipartite graphs $K(2, 3)$, $K(3, 3)$, $K(3, 4)$. For $n \geq 3$, $K(1, n)$ is also called the *n-star*. Figure 9.5 shows the graphs called 4-star, 5-star, 6-star.

A graph G is said to be a *cap graph* if it is the intersection graph of a set of spherical caps of the same radius. If, furthermore, the radius of the caps is d, then G is called a *d-cap graph*. For example, every complete graph K_n is a cap graph, and a d-cap graph for every $0 < d < \pi$. The complete bipartite graph $K(2, 3)$ is also a cap graph. (It is not difficult to find a set of five caps of the same radius whose intersection graph is $K(2, 3)$.)

If the vertex set V of a d-cap graph G contains an *independent* set of k vertices (that is, no pair in these k vertices is adjacent), then the centers of the caps corresponding to these k vertices form a d-independent set on $\mathbb{S}$. If the n-star $K(1, n)$ is a d-cap graph, then since $K(1, n)$ contains n independent vertices, the cap of radius $2d$ whose center corresponds to the vertex of degree n must contain a d-independent set of n points. Since no spherical cap of spherical radius $2d$ can contain a d-independent set of *six* points, $K(1, n)$ is not a cap graph for $n \geq 6$. However, for $n \leq 5$, each n-star $K(1, n)$ is a cap graph. Indeed, for small $d > 0$, we can easily find the set of six caps of radius d whose intersection graph is $K(1, 5)$. On the other hand, if d is large, say, $d \geq \pi/3$, then it is impossible to obtain $K(1, 5)$ as a d-cap graph. In the following, we present, for each $n = 2, 3, 4, 5$, the bound of d such that $K(1, n)$ is a d-cap graph. We also show that among the complete bipartite graphs $K(m, n)$, $n \geq m \geq 2$, only $K(2, 2)$ and $K(2, 3)$ are cap graphs.

Theorem 9.2 *The n-star $K(1, n)$, $n \leq 5$, is a d-cap graph if and only if $d < \rho_n$, where $\rho_n = \pi/n$ for $2 \leq n \leq 4$, and*

$$\rho_5 = \frac{1}{2}\cos^{-1}\left(1/\sqrt{5}\right) \approx 0.553575.$$

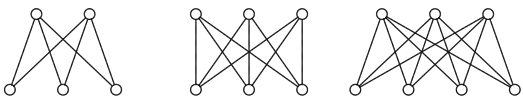


Fig. 9.4 $K(2, 3)$, $K(3, 3)$ and $K(3, 4)$

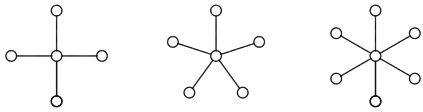


Fig. 9.5 The stars $K(1, 4)$, $K(1, 5)$, $K(1, 6)$

Proof

(i) Since any two hemispheres have nonempty intersection, $K(1,2)$ is not a d-cap graph for $d \geq \pi/2$. If $0 < d < \pi/2$, then a $2d$-cap contains d-independent two points. Hence $\rho_2 = \pi/2$.

(ii) For any three points on $\mathbb{S}$, the minimum distance among the three points is less than or equal to $2\pi/3$. Hence, $\rho_3 \leq \pi/3$. If $d < \pi/3$, we can take a d-independent set of three points on the boundary of a cap of radius $2d$. Hence $K(1,3)$ is the intersection graph of caps of radius d. Therefore, $\rho_3 = \pi/3$.

(iii) For $\pi/4 \leq d < \pi/3$ and $z \in \mathbb{S}$, the cap $C_{2d}(z)$ cannot contain d-independent four points. To see this, divide $C_{2d}(z)$ into four equal "sectors" by four geodesic segments l_1, l_2, l_3, l_4 of length $2d$ emanating from z. Let $l_1 = zx, l_2 = zy$, and let $S(z, xy)$ denote the *sector* bounded by $l_1 = zx, l_2 = zy$ and a minor circular arc $\widehat{xy}$ of $\partial C_{2d}(z)$.

First, we show that the diameter of $S(z, xy)$ is $2d$. If $d = \pi/4$, then $S(z, xy)$ is a Reuleaux $2d$-triangle, and its diameter is $2d$. So, we suppose that $2d > \pi/2$. Let u be the point on xz such that $zu = \pi/2$ and let G be the great circle with a pole u. Then l_2 is an arc of G. Note that G and $\partial C_{2d}(x)$ are tangent at z. Since the area of $\breve{\triangle} zxy$ is greater than the area of $\breve{\triangle}(\pi/2, \pi/2, \pi/2)$, we must have $\angle zxy > \pi/2 = \angle xzy$. Therefore, $xy < zy = 2d$ (cf. Exercise 5.2*). Thus y is an interior point of $C_{2d}(x)$. Since G and $\partial C_{2d}(x)$ are tangent at z, it follows that $l_2 (= yz)$ is contained in $C_{2d}(x)$. Let v be the point on $\partial C_{2d}(z)$ such that $xv = 2d$ and $\widehat{xv}$ contains y. Then the Reuleaux $2d$-triangle $T_{2d}(zxv)$ contains $S(z, xy)$. Thus the diameter of $S(z, xy) = 2d$. Now suppose that $C_{2d}(z)$ contains d-independent four points w_1, w_2, w_3, w_4. We may assume that w_1 lies on $l_1 = zx$. Then the two sectors having $l_1 (= zx)$ in common cannot contain w_2, w_3, w_4. Hence these three points lie on the union of remaining two sectors, which implies that some two points of w_2, w_3, w_4 are contained in a sector, contradicting the fact that the diameter of each sector is $2d$. Thus, $C_{2d}(z)$ cannot contain d-independent four points. Therefore, $\rho_4 \leq \pi/4$.

For $0 < d < \pi/4$ and a point $z \in \mathbb{S}$, let u_1, u_2, u_3, u_4 be four points (in this cyclic order) on the boundary circle of $C_{2d}(z)$ such that $u_1u_2 = u_2u_3 = u_3u_4 = u_4u_1$. Since the isosceles triangle $\breve{\triangle} zu_1u_2$ is a part of an equilateral triangle of side-length $\pi/2$ (and interior angle $\pi/2$), its area is smaller than the area $\pi/2$ of an equilateral triangle with interior angle $\pi/2$. Then, since $\angle u_1zu_2 = \pi/2$, we must have $\angle zu_1u_2 = \angle zu_2u_1 < \pi/2 = \angle u_1zu_2$. This implies that $u_1u_2 > zu_1 = 2d$. Thus, $u_1u_2 = u_2u_3 = u_3u_4 = u_4u_1 > 2d$. Therefore, the four caps $C_d(u_i), i = 1, 2, 3, 4$, are disjoint, and the intersection graph of these four caps and $C_d(z)$ is the 4-star $K(1,4)$. Therefore, $\rho_4 = \pi/4$.

(iv) Since $\rho_4 = \pi/4$, we have $\rho_5 \leq \pi/4$. For a point $z \in \mathbb{S}$ and a radius d, $0 < d < \pi/4$, let v_1, v_2, v_3, v_4, v_5 be five points in this cyclic order on the boundary of $C_{2d}(z)$ that divide the boundary circle into five equal arcs. Then $\angle v_1zv_2 = 2\pi/5$.

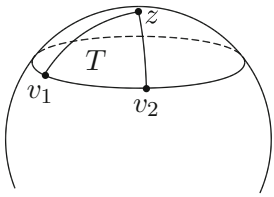


Fig. 9.6 The sector $T = zv_1v_2$

First, we show that $C_{2d}(z)$ contains a d-independent set of five points if and only if $v_1v_2 > zv_1 = 2d$. Let T be the sector obtained as the intersection of $C_{2d}(z)$ and the lune with the interior angle $\angle v_1zv_2$, see Fig. 9.6. Note that the boundary arc $\widehat{v_1v_2}$ of the sector T is not a geodesic segment v_1v_2. Since $\breve{\triangle}zv_1v_2 \subset T$, the interior angle of $\breve{\triangle}zv_1v_2$ at v_1 (and at v_2) is less than $\pi/2$. Note that the diameter of the sector T is $s := \max\{2d, v_1v_2\}$. (To see this, let p, q be two points on the boundary of T. If both p, q lie on the boundary curve $\widehat{v_1v_2}$, then $pq \le v_1v_2$. If p lies on zv_1, then $C_s(p)$ contains T, and hence $pq \le s$.) Suppose that the cap $C_{2d}(z)$ contains a d-independent set $\mathcal{X}$ consisting of five points. We may assume that one point $y_0 \in \mathcal{X}$ lies on the segment zv_1. Then one of the five sectors $zv_1v_2, zv_2v_3, zv_3v_4, zv_4v_5, zv_5v_1$ must contain two points of $\mathcal{X}$. (If neither of the sectors zv_1v_2, zv_5v_1 contains two points of $\mathcal{X}$, then the four points in $\mathcal{X} \setminus \{y_0\}$ must lie in the union of the three sectors $zv_2v_3, zv_3v_4, zv_4v_5$, which implies that one of the three sectors contains some two points of $\mathcal{X} \setminus \{y_0\}$.) Hence the diameter of each sector must be greater than $2d$. Since the diameter of T is $\max\{2d, v_1v_2\}$, we have $v_1v_2 > 2d$. Thus, $v_1v_2 = v_2v_3 = \cdots = v_5v_1 > 2d$. On the other hand, if $v_1v_2 > 2d$, then $\{v_1, v_2, v_3, v_4, v_5\}$ is a d-independent set in $C_{2d}(z)$. Thus, $C_{2d}(z)$ contains a d-independent set of five points if and only if $v_1v_2 > 2d$.

Now, we could ask: For what value δ of d does $v_1v_2 = 2d$ hold? If $zv_1 = v_1v_2 = 2\delta$, then, by the spherical cosine law, we have

$$\cos(2\delta) = \cos^2(2\delta) + \sin^2(2\delta)\cos(2\pi/5),$$

from which we get

$$\cos(2\delta) = \frac{\cos(2\pi/5)}{1 - \cos(2\pi/5)}.$$

Since $\cos(2\pi/5) = \frac{1}{4}(\sqrt{5} - 1)$, we have

$$\cos(2\delta) = \frac{\cos(2\pi/5)}{1 - \cos(2\pi/5)} = \frac{\sqrt{5} - 1}{5 - \sqrt{5}} = \frac{1}{\sqrt{5}},$$

and hence $\delta = \frac{1}{2}\cos^{-1}\left(1/\sqrt{5}\right)$. If $d = \delta$, then $\breve{\triangle}zv_1v_2$ is an equilateral triangle with interior angle $2\pi/5$. If $d < \delta$, then $\breve{\triangle}zv_1v_2$ is an isosceles triangle

whose interior angle at the vertex z is $2\pi/5$, contained in an equilateral triangle with interior angle $2\pi/5$. Hence it has smaller area than the equilateral triangle, and hence $\angle zv_1v_2 < 2\pi/5$, which implies $v_1v_2 > zv_1$. In this case, the intersection graph of the family $\{C_d(z)\} \cup \{C_d(v_i), i = 1, 2, 3, 4, 5\}$ is $K(1, 5)$. Therefore, $K(1, 5)$ is a d-cap graph if and only if $d < \delta$. Thus $\rho_5 = \delta$.

□

Theorem 9.3

(1) $K(2, 2)$ *is a* d*-cap graph if and only if* $0 < d < \pi/2$.
(2) $K(2, 3)$ *is a* d*-cap graph if and only if* $\pi/4 \leq d < \pi/3$.

Proof

(1) The proof is easy and omitted.
(2) If $K(2, 3)$ is a d-cap graph, then $K(1, 3)$ is also a d-cap graph. Hence $K(2, 3)$ is not a d-cap graph for $d \geq \pi/3$.

Suppose that $\pi/4 \leq d < \pi/3$. Let G be a great circle on $\mathbb{S}$, and x, x^* be the poles of G. On the great circle G, we can take three points u, v, w such that the three caps $C_d(u), C_d(v), C_d(w)$ are disjoint. Then these three caps and the caps $C_d(x), C_d(x^*)$ constitute together a family of r-caps whose intersection graph is $K(2, 3)$.

Next, suppose that $d < \pi/4$. Let let $x, y \in \mathbb{S}$ be a pair of points such that $C_d(x) \cap C_d(y) = \emptyset$. Then $2d < \pi/2$, and hence $C_{2d}(x) \cap C_{2d}(y)$ cannot contain a d-independent set of three points by Cororally 9.2. Hence $K(2, 3)$ is not a d-cap graph.

Therefore, $K(2, 3)$ is a d-cap graph if and only if $\pi/4 \leq d < \pi/3$. □

Theorem 9.4

(1) $K(2, 4)$ *is not a cap graph.*
(2) *The complete bipartite graph* $K(3, 3)$ *is not a cap graph.*

Proof

(1) The proof is easy and omitted.
(2) If $K(3, 3)$ is a d-cap graph, then $K(2, 3)$ is also a d-cap graph. So, it is enough to show that for $\pi/4 \leq d < \pi/3$, $K(3, 3)$ is not a d-cap graph. Let $\{x, y, z\} \subset \mathbb{S}$ be a d-independent set. Then $C_{2d}(x) \cap C_{2d}(y) \cap C_{2d}(z)$ consists of at most two components, and each component has diameter at most $2d$ by Cororally 9.1. Hence it cannot contain a d-independent set of three points. Therefore, $K(3, 3)$ is not a cap graph.

□

Remark 9.2 The results in this section are obtained in [540].

9.4 Random Spherical Caps

If we put N spherical caps of the same angular radius $\theta < \pi/2$ *at random* on $\mathbb{S}$ (that is, the centers of the N caps are random points on $\mathbb{S}$ chosen independently), what is the probability that the union of these N caps is connected? Let us estimate this probability using Theorem 9.1.

Lemma 9.4 *Let $C_1, C_2, \ldots, C_N$ be N spherical caps of angular radius $\theta < \pi/2$ put at random on $\mathbb{S}$. Under the condition that C_N intersects only the k caps $C_1, C_2, \ldots, C_k$, the conditional probability that C_N is an extremal cap of the family of N caps is equal to $2k/2^k$.*

Proof Let v_i denote the center of C_i, $i = 1, 2, \ldots, N$. We may suppose that v_N is a fixed point on $\mathbb{S}$. Let D denote the cap of angular radius 2θ centered at v_N. Then, $C_i \cap C_N \neq \emptyset$ is equivalent to $v_i \in D$. Hence, under the condition that C_N intersects $C_1, C_2, \ldots, C_k$, the centers $v_1, v_2, \ldots, v_k$ are independent random points uniformly distributed in D. Since the probability that some of v_i, $i = 1, 2, \ldots k$, coincide with v_N is 0, we may assume that $v_1, \ldots, v_k$ are distributed uniformly in $D \setminus \{v_N\}$. For $i = 1, 2, \ldots, k$, let v_i' denote the intersection point of the boundary circle ∂D of D and the extension of the geodesic segment $v_N v_i$ beyond v_i. Then v_i', $i = 1, 2, \ldots v_k'$ are independently and uniformly distributed random points on the circle ∂D. Under the condition that C_N intersects only the k caps $C_1, C_2, \ldots, C_k$, the following two events (1) and (2) are equivalent:

(1) C_N is an extremal cap,
(2) $v_1', v_2', \ldots, v_k'$ lie on a semicircle of ∂D.

The probability that the event (2) occurs is already computed as $2k/2^k$ in Theorem 8.5.

□

Lemma 9.5 *Let $\{C_1, C_2, \ldots, C_N\}$ be a family of N spherical caps of the same angular radius $\theta < \pi/2$ put at random on $\mathbb{S}$. Then the probability that C_i is an extremal cap of the family is given by*

$$(\cos^2\theta)^{N-1} + (N-1)\sin^2\theta\left(1 - \tfrac{1}{2}\sin^2\theta\right)^{N-2}.$$

Proof Let v_i denote the center of C_i, and let D_i denote the spherical cap of angular radius 2θ with center v_i. The area of D_i is

$$2\pi(1 - \cos 2\theta) = 4\pi \sin^2\theta.$$

Hence, for a random point v on $\mathbb{S}$, $\Pr(v \in D_i) = \sin^2\theta$. The probability that a certain k-tuple of points from $v_1, v_2, \ldots, v_N$ (other than v_i) lies in D_i and the remaining $N - 1 - k$ points lie outside D_i is

$$(\sin^2\theta)^k(1 - \sin^2\theta)^{N-1-k} = (\sin^2\theta)^k(\cos^2\theta)^{N-1-k}.$$

Hence, by Lemma 9.4,

$$\begin{aligned}
&\Pr(C_i \text{ is an extremal cap}) \\
&\quad = (\cos^2\theta)^{N-1} + \sum_{k=1}^{N-1} \binom{N-1}{k} (\sin^2\theta)^k (\cos^2\theta)^{N-1-k} \frac{2k}{2^k} \\
&\quad = (\cos^2\theta)^{N-1} + 2\sum_{k=0}^{N-1} k \binom{N-1}{k} \left(\frac{\sin^2\theta}{2}\right)^k (\cos^2\theta)^{N-1-k}.
\end{aligned}$$

Since $\sum_{k=0}^{n} \binom{n}{k} k x^k y^{n-k} = x \cdot \frac{\partial}{\partial x}(x+y)^n = nx(x+y)^{n-1}$, we have

$$\begin{aligned}
\Pr(C_i \text{ is extremal}) &= (\cos^2\theta)^{N-1} + (N-1)\sin^2\theta \left(\tfrac{1}{2}\sin^2\theta + \cos^2\theta\right)^{N-2} \\
&= (\cos^2\theta)^{N-1} + (N-1)\sin^2\theta \left(1 - \tfrac{1}{2}\sin^2\theta\right)^{N-2}.
\end{aligned}$$

□

Theorem 9.5 *Let $P(\theta, N)$ denote the probability that the intersection graph of the family $\mathcal{C}$ of N spherical caps of the same angular radius $\theta < \pi/2$ put at random on $\mathbb{S}$ is connected. We have*

$$P(\theta, N) \geq 1 - N(\cos^2\theta)^{N-1} - N(N-1)(\sin^2\theta)(1 - \tfrac{1}{2}\sin 2\theta)^{N-2}.$$

Proof By Lemma 9.5, the probability that one of the caps in $\mathcal{C}$ is an extremal cap is at most

$$N(\cos^2\theta)^{N-1} + N(N-1)\sin^2\theta(1 - \frac{1}{2}\sin^2\theta)^{N-2}.$$

Hence the probability that $\mathcal{C}$ contains no extremal cap is at least

$$1 - N(\cos^2\theta)^{N-1} - N(N-1)\sin^2\theta(1 - \frac{1}{2}\sin^2\theta)^{N-2}.$$

By Theorem 9.1, the property that "there is no extremal cap" implies that "the intersection graph of the family is connected". Hence $P(\theta, N)$ is greater than or equal to the probability that $\mathcal{C}$ contains no extremal cap. Therefore,

$$P(\theta, N) \geq 1 - N(\cos^2\theta)^{N-1} - N(N-1)\sin^2\theta(1 - \frac{1}{2}\sin^2\theta)^{N-2}.$$

□

9.5 Chebyshev's Inequality and a Lemma

From Theorem 9.5 we can derive some non-trivial lower bounds of $P(\theta, N)$ for some θ, N. For example,

$$P(\tfrac{\pi}{3}, 15) > 0.65, P(\tfrac{\pi}{6}, 60) > 0.61, P(\tfrac{\pi}{10}, 200) > 0.76.$$

Although these lower bounds look very rough, Theorem 9.5 can be still applied to derive a reasonably good asymptotic result. Preceding to derive this asymptotic result, let us recall here Chebyshev's inequality and prepare a useful lemma.

Let X be a nonnegative random variable, and $f(x)$ be its *density function*. Thus, $\Pr(X \le a) = \int_0^a f(x)dx$. Let $\mu = E(X) > 0$ be the *mean* of X. Since

$$\mu = \int_0^\infty xf(x)dx \ge \int_{t\mu}^\infty xf(x)dx \ge \int_{t\mu}^\infty t\mu f(x)dx = t\mu \Pr(X \ge t\mu),$$

for every $t > 0$, we have

$$\Pr(X \ge tE(X)) \le \frac{1}{t}.$$

This is known as *Markov's inequality*. Let Y be a random variable with mean μ_Y and variance σ^2. Let $X = (Y - \mu_Y)^2$. Then X is nonnegative, and $E(X) = \sigma^2$. In Markov's inequality, put $t = k^2/\sigma^2$. Then $\Pr(X \ge k^2) \le \sigma^2/k^2$. Hence

$$\Pr(|Y - \mu_Y| \ge k) \le \frac{\sigma^2}{k^2}.$$

This is *Chebyshev's inequality*.

Lemma 9.6 *Let $f = f(N), g = g(N)$ be two positive functions of N. If $f \to 0$ as $N \to \infty$, then for sufficiently large N we have $(1 - f)^g < e^{-f \cdot g}$. If, furthermore, $f^2 g \to 0$ as $N \to \infty$, then*

$$(1 - f)^g = e^{-f \cdot g}(1 + o(1)),$$

where $o(1)$ denotes some function of N that tends to 0 as $N \to \infty$.

Proof Put $h(x) = \log(1 - x)$. Then $h'(x) = \frac{-1}{1-x}, h''(x) = \frac{-1}{(1-x)^2}$. Hence, for $0 < t < 1$, we have $h(t) = h(0) + h'(0)t + \frac{1}{2}h''(\xi t)t^2$ for some $0 < \xi < 1$, by Taylor's theorem. Therefore

$$\log(1 - t) = -t + \frac{-t^2}{2(1 - \xi t)^2}.$$

Hence, for large N (such that $f = f(N) < 1$), we have

$$g \cdot \log(1-f) = -f \cdot g - \frac{f^2 \cdot g}{2(1-\xi f)^2} < -f \cdot g.$$

Thus, $(1-f)^g < e^{-f \cdot g}$. If, furthermore, $f^2 \cdot g \to 0$ as $N \to \infty$, then

$$\frac{f^2 \cdot g}{2(1-\xi f)^2} = o(1),$$

and therefore we have

$$(1-f)^g = e^{-f \cdot g + o(1)} = e^{-f \cdot g} \cdot e^{o(1)} = e^{-f \cdot g}(1 + o(1)).$$

□

9.6 Asymptotic Probability

Theorem 9.6 *Let* $\theta = c\sqrt{\frac{1}{N}\log N}$, $c > 0$. *Then*

$$\lim_{N \to \infty} P(\theta, N) = \begin{cases} 0 & (\text{if } c < 1 \quad) \\ 1 & (\text{if } c > \sqrt{2}). \end{cases}$$

Proof First, suppose that $c > \sqrt{2}$. To show $P(\theta, N) \to 1$ as $N \to \infty$, it is enough to verify, by Theorem 9.5, that

$$N(\cos^2\theta)^{N-1} + N(N-1)(\sin^2\theta)(1 - \tfrac{1}{2}\sin 2\theta)^{N-2} \to 0$$

as $N \to \infty$. Since $\theta = c\sqrt{\frac{1}{N}\log N} \to 0$ as $N \to \infty$, we have $\sin^2\theta = \theta^2(1 + o(1)) = (1 + o(1))\frac{c^2}{N}\log N$. Hence, applying Lemma 9.6,

$$\begin{aligned} &N(1-\sin^2\theta)^{N-1} + N(N-1)(\sin^2\theta)(1 - \tfrac{1}{2}\sin 2\theta)^{N-2} \\ &\quad < Ne^{-(N-1)\sin^2\theta} + N(N-1)(\sin^2\theta)(1 - \tfrac{1}{2}\sin^2\theta)^{N-2} \\ &\quad \sim Ne^{-c^2 \log N} + N(c^2 \log N)e^{-(c^2/2)\log N} \sim \frac{1}{N^{c^2-1}} + \frac{c^2 \log N}{N^{c^2/2-1}}, \end{aligned}$$

where the notation $f \sim g$ implies $f = (1 + o(1))g$. Since $c^2 > 2$, we have

$$\frac{1}{N^{c^2-1}} + \frac{c^2 \log N}{N^{c^2/2-1}} \to 0$$

as $N \to \infty$.

Now, suppose that $c < 1$. We have to show that $P(\theta, N) \to 0$ as $N \to \infty$. Let $\mathcal{C} = \{C_1, C_2, \ldots, C_N\}$, and let v_i be the center of C_i. Then $v_1, v_2, \ldots, v_N$ are independent and uniformly distributed random points on $\mathbb{S}$. Denote by D_i the spherical cap of angular radius 2θ with center v_i. The area of D_i is $4\pi \sin^2 \theta$. Since $C_i \cap C_j \neq \emptyset \Leftrightarrow v_i \in D_i$, we have

$$\Pr(C_i \cap C_j \neq \emptyset) = \sin^2 \theta \sim \theta^2.$$

Similarly,

$$\Pr(D_i \cap D_j \neq \emptyset) = \sin^2(2\theta) \sim 4\theta^2.$$

Recall that C_i is an isolated cap of $\mathcal{C}$ if $C_i \cap C_j = \emptyset$ for all $j \neq i$. If $\mathcal{C}$ has an isolated cap, then $\Omega(\mathcal{C})$ is disconnected. For each i define the random variable Z_i by

$$Z_i = \begin{cases} 1 & \text{if } C_i \text{ is an isolated cap} \\ 0 & \text{otherwise.} \end{cases}$$

Then the sum $Z = Z_1 + Z_2 + \cdots + Z_N$ counts the number of isolated caps in $\mathcal{C}$. Since $N(\theta^2)^2 = o(1)$, we have by Lemma 9.6 that the expected value of Z_i is

$$E(Z_i) = \Pr(Z_i = 1) \sim (1 - \theta^2)^{N-1} \sim e^{-N\theta^2} \sim e^{-c^2 \log N} = N^{-c^2}.$$

Hence the expected value of Z is

$$E(Z) = NE(Z_1) \sim N^{1-c^2}.$$

Next, let us find the expected value $E(Z_i Z_j)$ for $i \neq j$. Since $E(Z_i Z_j) = \Pr(Z_i Z_j = 1)$, we have

$$\begin{aligned} E(Z_i Z_j) &< \Pr(D_i \cap D_j = \emptyset) \Pr(v_k \notin D_i \cup D_j \text{ for } k \neq i, j \mid D_i \cap D_j = \emptyset) \\ &\quad + \Pr(D_i \cap D_j \neq \emptyset) \Pr(v_k \notin D_i \cup D_j \text{ for } k \neq i, j \mid D_i \cap D_j \neq \emptyset) \\ &< (1 - 4\theta^2)(1 - 2\theta^2)^{N-2} + 4\theta^2(1 - \theta^2)^{N-2} \\ &\sim (1 - 4\theta^2)e^{-2(N-2)\theta^2} + 4\theta^2 e^{-(N-2)\theta^2} \end{aligned}$$

$$\sim (1-4\theta^2)N^{-2c^2} + (4c^2 \log N)N^{-(1+c^2)}$$
$$= N^{-2c^2}(1-4\theta^2 + (4c^2 \log N)N^{c^2-1}) \sim N^{-2c^2}.$$

Hence

$$\begin{aligned} E(Z^2) &= \sum_{i,j} E(Z_i Z_j) = \sum_i E(Z_i^2) + \sum_{i \neq j} E(Z_i Z_j) \\ &= N \cdot E(Z_1) + N(N-1) \cdot E(Z_1 Z_2) \\ &< N \cdot N^{-c^2} + N^2 \cdot N^{-2c^2} = N^{2(1-c^2)} \left(\frac{1}{N^{1-c^2}} + 1 \right) \\ &\sim N^{2(1-c^2)} \sim E(Z)^2. \end{aligned}$$

Since $E(Z^2) \geq E(Z)^2$, we have $E(Z^2) \sim E(Z)^2$. Note that $Z = 0$ implies that $|Z - E(Z)| \geq E(Z)$ and the variance σ_Z^2 of Z is equal to $E(Z^2) - E(Z)^2$. Now, applying Chebyshev's inequality, we have

$$\Pr(Z = 0) \leq \Pr(|Z - E(Z)| \geq E(Z)) < \frac{E(Z^2) - E(Z)^2}{E(Z)^2} \to 0.$$

Thus, $\Pr(Z \geq 1) \to 1$ $(N \to \infty)$, that is, the probability that there is an isolated cap tends to 1 as $N \to \infty$. Therefore, $P(\theta, N) \to 0$ as $N \to \infty$. □

For more results on intersection graphs of spherical caps see [522].

9.7 Exercises

9.1 Prove that for a regular tetrahedron $wxyz$, the angle $\angle xOy$ is given by $\cos^{-1}(-1/3)$, where O is the center of the circumsphere of the regular tetrahedron.

9.2* Prove that $K(2, 2)$ is a d-cap graph if and only if $0 < d < \pi/2$.

9.3* Prove that $K(2, 4)$ is not a cap graph.

9.4 Let X be a random variable with mean $\mu = 2$ and variance $\sigma^2 = 3$. Using Chebyshev's inequality, estimate the lower bound of

$$\Pr(-2 < X < 6).$$

9.5 Find the probability that two random caps of angular radius $r < \pi/2$ put independently on the unit sphere $\mathbb{S}$ have nonempty intersection.

9.6* Find the expected value of the area of the intersection of two random caps of angular radius $\theta < \pi/2$ put independently on the unit sphere $\mathbb{S}$.

9.7* For n spherical caps of area $4\pi p$ put at random on $\mathbb{S}$, let $\alpha(n, p)$ denote the area of the part of $\mathbb{S}$ that is not covered by any of these n spherical caps. Put

$$U(n, p) = \frac{\alpha(n, p)}{4\pi}.$$

Then $U(n, p)$ is a random variable. Prove that the expected value $E(U(n, p))$ of $U(n, p)$ is given by

$$E(U(n, p)) = (1 - p)^n.$$

9.8 Notes

The most substantial lists of references related thematically to our Chap. 9 refer to intersection graphs as well as Markov's and Chebyshev's inequality. Hence these themes are chosen as corresponding topics of the two subsections presented here.

9.8.1 Notes on Intersection Graphs

For our purpose here we want to define the *intersection graph of a set family* F as the graph whose vertices are presented by the sets from F, and adjacency holds if and only if the corresponding sets intersect. E.g., a well known result of Tietze from 1905 says that any finite graph is the intersection graph of convex 3-polytopes; this is no longer true (in any fixed dimension) if the polytopes have only a bounded number of faces, or if instead of them simple geometric figures are taken which can be described in terms of a bounded number of real parameters. It turns out that intersection graphs of various classes of geometric objects have, even in the plane, interesting structural and extremal properties. We want to reflect some essential survey-like publications from this lively subfield of graph theory, and we note that the graph-theoretic methods discussed here are also interesting in view of other chapters in our book (see, e.g., Chap. 3). Since there are too many related research articles, we refrain from quoting such papers.

Books and Book Parts

In his book [670], Roberts describes how useful graph theory (also combined with other fields) can be to solve complex problems in our daily life, e.g. referring to urban services, energy, pollution problems and many other branches. In Chap. 3 the particular importance of *intersection graphs* within this framework is established.

A book which is completely dedicated to *intersection graphs of set families* is the monograph [564] of McKee and McMorris. The authors treat general intersection representations as well as chordal, interval and competition graphs, clearly showing the common underlying theory and offering also the role that properties of hypergraphs can play within this setting. Chapter 5 is particularly devoted to threshold graphs which have additional properties. Although presenting about 500 references, the authors state that this list is far from being complete.

The comprehensive, encyclopedic work [123] of Brandstädt et al. contains also a part on intersection graphs, namely the whole Chap. 4 treating such graphs with respect to combinatorial objects (e.g., as *line graphs, interval graphs* and *visibility graphs of polygons*). A forerunner of [123] is the book [308] of Golumbic whose second edition we cite here. Also this book contains a lot of material on *intersection graphs and subcases thereof.*

Surveys

In Sect. 8.1 of his important handbook article [233], Eckhoff discusses *intersection graphs in combinatorial convex geometry*. So the studied set families consist mainly of (special types of) convex sets, and the subcases of *interval graphs and circular-arc graphs* or properties like *boxicity and sphericity* occur there. Kostochka [476] surveys the state of the art regarding bounds on the *chromatic number of the intersection graphs of types of geometric figures* (such as, e.g., boxes, translates of convex compacta, or circle chords) with given clique number or girth. A few new results in that direction are also derived.

Circular-arc graphs (i.e., intersection graphs of open arcs on a circle) and *circle graphs* (the intersection graphs of chords on a circle) were comprehensively studied by many authors, and there are various interesting subclasses. Dúran et al. [231] give a structural overview and pose interesting problems. Pach [609] gives an overview to results and problems around *intersection graphs* and the more general notion of *intersection patterns*. Many of the discussed problems go back to famous mathematicians from convexity, graph theory, combinatorics and topology (e.g., treating Helly-type theorems or Ramsey theory). This survey takes the rapid development of computational geometry and algorithms in graph theory into account. In particular, the author presents also applications of results on intersection graphs and patterns in the theory of *topological graphs*. Based on a broad spectrum of mathematical structures and methods, Ellis [242] surveys *intersection problems in extremal combinatorics*. These structures include families of vector subspaces, subsets of finite group actions and also *graphs and hypergraphs*, combined with stronger or weaker intersection conditions. And the used methods, which are also presented, can be purely combinatorial, algebraic, analytic, and probabilistic in nature. The author treats old and new problems and methods, and he gives elegant proofs of several central results. This interesting survey is useful for graduates and researchers.

More algebraic than the definitions of intersection graphs above, Chakrabarty and Kureethara [155] denote by $L(R)$ the set of all non-trivial left ideals of a ring R and define the *intersection graph of ideals of a ring* R as an undirected simple graph $G(R)$ whose vertices are in one-to-one correspondence with $L(R)$, and whose edges come to be iff the corresponding left ideals of R have a non-zero intersection. The comprehensive research on this notion and on properties of $G(R)$ gives rise to this survey.

9.8.2 Notes on Chebyshev's Inequality

Chebyshev's inequality in probability theory (also called Bienaymé-Chebyshev inequality) gives an upper bound on the probability of deviation of a random variable (of finite variance) from its mean. It can be used to prove the *weak law of large numbers*, and we use it here in Sect. 9.5. Also the name "Chebyshev's theorem" is common, and this can also refer to the closely related *Markov inequality*, which is perhaps more used in analysis. Below we collect some basic references related to Chebyshev's famous inequality.

Book Parts

Due to its high applicability, Chebyshev's inequality can be found in various monographs on probability theory and statistics, but also from other fields. We present a little collection, mainly in chronological order.

Johnson and Leone [431] present the Chebyshev inequality in their introductory part of Volume I, treating as groundwork the themes *probability and distribution*. Tuckwell discusses in [783] *applications of probability*. Chapter 6 is dedicated to sequences of random variables and develops also convergence in distribution and convergence in probability. As main tools for this, characteristic functions and Chebyshev's inequality are treated. In Jiang's monograph [429] one can find the Chebyshev inequality in Chapter 5, which collects many numerical and probability inequalities. In Example 5.6 an *exponential probability inequality for sums of independent random variables* is nicely derived from the Chebyshev inequality.

Epps [243, Chapter 3] treats the topics of *integration in abstract spaces and mathematical expectation*, leading to classical inequalities like the Chebyshev one. McKean [563] gives a broad view onto *classical limit theorems in probability* and their applications. As one of the cornerstones in this framework, Chebyshev's inequality is presented in the opening chapter. And in Chap. 2 the *Markov property* and extensions thereof are discussed.

In his basic monograph [50] on *measure theory, integration and real analysis*, Axler dedicates the last chapter to probability theory, introducing a random variable as a measurable function on a probability space, and its expected value as its Lebesgue integral with respect to the probability measure. It is shown that the

expected value of a product of independent random variables is the product of their expected values. And also the Chebyshev inequality is presented, leading to the weak law of large numbers. The book [438] of Kale and Muralidharan, treating the *classical theory of statistical inference*, is an example for the following situation: the author assumes that the reader is familiar with various basic notions and tools, among them also Chebyshev's inequality.

Surveys and Expositions

Here we start with the nice classroom notes [450] of Khatri presenting several elementary inequalities of multivariate Chebyshev type. They are derived in an elementary way, in the spirit of Exercise 6 (p. 256) of the classical monograph [187] of Cramer.

Tong [778] treats the fact that notions like association and dependence of random variables, rearrangements, and heterogeneity via majorization ordering are (directly or indirectly) fundamental for developments around *stochastic inequalities*. It turns out that these notions are strongly related to Chebyshev's inequality, the Hardy-Littlewood-Pólya rearrangement inequality, and Schur functions from analysis. So, in particular, Chebyshev's inequality says that the integral of the product of two increasing functions is larger than or equal to the product of their integrals. This can be viewed as confirmation of the positive dependence of random variables which are monotone functions of a common random variable. In this nice expository paper, Tong wants to emphasize how classical mathematical inequalities of the above described types play a central role for stochastic inequalities.

And in the very interesting and detailed paper [726] of Seneta, the *history of the law of large numbers* is presented. In a successful way, the Chebyshev inequality is characterized as meeting point of French and Russian research efforts. Especially interesting is the presentation of the contributions from the Russian side (besides Chebyshev meaning particularly also Markov and Bernstein).

Selected Research Papers

Huber [407] studies the *equality cases* of the classical Markov and Chebyshev inequalities, which sharply bound the tail probabilities of random variables. Using a simple smoothing via auxiliary randomness, he shows that each of the obtained bounds can be halved, and that in many cases this halving procedure can be reached without using such auxiliary randomness.

Ogasawara [600] derives *multivariate versions of the sharp Markov and Chebyshev inequality*, when associated probabilities are extended to segments of the supports of non-negative random variables. The probabilities take echelon forms, and when some positive lower bounds of these probabilities are reached, then improvements without the echelon forms are attained. Saw et al. [695] study the

necessary (and finally simple) modifications of the Chebyshev inequality for the case that the studied population mean and variance are estimated from a sample.

Based on operations called pseudo-addition and pseudo-multiplication, Agahi et al. [9] define so-called *pseudo-integrals* which are more general than Lebesgue integrals. Generalizing the classical Chebyshev inequality in probability theory, they establish Chebyshev-type inequalities combined with two classes of such pseudo-integrals. Interested in the multivariate version of Chebyshev's in non-additive cases, Agahi [8] shows that the (multivariate) Chebyshev inequality still holds in a generalized probability theory based on the notion of *Choquet integral*.

And in the three short papers [645] of Prakasa Rao, [831] of Zhou and Hu, and [132] of Budny, Chebyshev's inequality is suitably extended to *Hilbert-space-valued random and Banach-space-valued random elements*.

Chapter 10
Quartets on a Sphere



Doubt is the origin of wisdom.

— Rene Descartes

For any region on a sphere, it is impossible to construct a (plane) map of the region in which distances are faithfully represented. This fact can be proved by showing that any given region on a sphere contains a quartet (i.e., a four-point space) that is not isometrically embeddable in the plane. Actually, any region on a sphere contains a quartet whose six distances determine the radius of the sphere uniquely. Moreover, we present an equation to judge if a given quartet can be a quartet on a sphere of a suitable radius.

10.1 Metric Spaces

A nonempty set $\mathcal{X}$ is called a *metric space* if for any two elements $P, Q \in \mathcal{X}$ a real number $d(P, Q)$ is assigned so that the following three conditions hold:

(i) $d(P, Q) = d(Q, P) \geq 0$,
(ii) $d(P, Q) = 0 \Leftrightarrow P = Q$,
(iii) $d(P, Q) + d(Q, R) \geq d(R, P)$ (triangle inequality).

The real number $d(P, Q)$ is called the *distance* between P and Q. Elements of a metric space are called *points*.

For example, the plane $\mathbb{R}^2$ and the 3-dimensional space $\mathbb{R}^3$ are metric spaces by the usual distances. A sphere is also a metric space by the spherical distance. A metric space consisting of three points is called a *trio*, and a metric space consisting of four points is called a *quartet*. Every nonempty subset of a metric space $\mathcal{X}$ is a metric space by the distance defined in $\mathcal{X}$, and it is called the *subspace* of $\mathcal{X}$. Two metric spaces $\mathcal{X}$ and $\mathcal{Y}$ are called *isometric*, if there is a one-to-one correspondence between them such that for any two points in $\mathcal{X}$, the distance between them is equal

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H. Maehara, H. Martini, *Circles, Spheres and Spherical Geometry*,
Birkhäuser Advanced Texts Basler Lehrbücher,
https://doi.org/10.1007/978-3-031-62776-7_10

to the distance between the corresponding two points in $\mathcal{Y}$. If a metric space $\mathcal{X}$ is isometric to a subspace of another metric space $\mathcal{Z}$, then $\mathcal{X}$ is said to be *isometrically embeddable* in $\mathcal{Z}$. A metric space is called *linear* (resp. *planar*), if the metric space is isometrically embeddable in a line (resp. in a plane). For example, any trio (a metric space consisting of three points) is planar. Indeed, if the distances a, b, c between the three points satisfy the triangle inequality, we can take three points in the plane so that the distances between them become a, b, c. If, further on, a, b, c satisfy $a + b = c$ (or $b + c = a$, or $c + a = b$), then the trio is linear.

If a metric space is linear, then it is also planar, and a planar metric space is also isometrically embeddable in $\mathbb{R}^3$.

So far, mostly we take the radius of a sphere as unit of length and regard the sphere as a unit sphere. In this chapter, however, we deal with spheres of different radii at the same time. So we regard unit of length defined in another way. Suppose that a spherical triangle $\breve{\triangle}ABC$ on a sphere with radius r has edge-lengths a, b, c. If we take r as unit of length, then the edge-lengths of $\breve{\triangle}ABC$ become $a/r, b/r, c/r$. Since the size of an angle is invariant under a change of unit of length, the cosine law concerning a spherical triangle $\breve{\triangle}ABC$ with edge-lengths a, b, c on a sphere of radius r becomes

$$\cos(a/r) = \cos(b/r)\cos(c/r) + \sin(b/r)\sin(c/r)\cos\angle A.$$

A metric space that is embeddable in a sphere of some radius is called *spherical*. Any trio is obviously spherical. If A, B, C are three points lying on a sphere $\mathbb{S}(r)$ of radius r, then A, B, C lie on a circle on $\mathbb{S}(r)$, and hence $AB + BC + CA \leq 2\pi r$. Thus, we can see that a trio with mutual distances x, y, z is isometrically embeddable in $\mathbb{S}(r)$ if and only if $x + y + z \leq 2\pi r$. A linear metric space consisting of a finite number of points is also spherical. Indeed, it can be isometrically embeddable in a great circle of a sphere with large radius.

Exercise 10.1 Prove that the quartet consisting of the four vertices of a rhombus in the plane shown in Fig. 10.1 is spherical.

Solution On a sphere $\mathbb{S}(r)$ of radius $r \geq 3d$, take four points A, B, C, D in such a way that

$$AB = AD = BC = BA = BD = d,$$

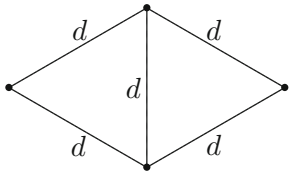


Fig. 10.1 The length of the longest diagonal is $\sqrt{3}d$

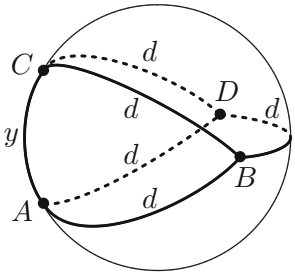


Fig. 10.2 Four points on $\mathbb{S}(r)$

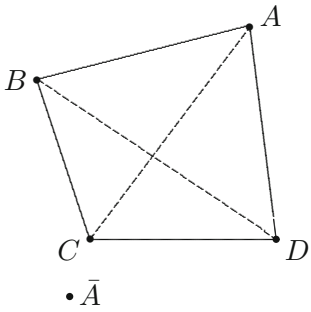


Fig. 10.3 A convex quadrilateral in the plane

as shown in Fig. 10.2, and put $y = AC$. Then $y = y(r)$ is a continuous function of the radius r. When $r = 3d/(2\pi)$, A, B, D lie on a great circle of $\mathbb{S}(r)$, and C, B, D also lie on a great circle. Hence A and C coincide, and $y(r) = 0$. When $r = 4d/(2\pi)$ (that is, $d/r = \pi/2$), the interior angle of $\breve{\triangle}ABD$ is $\pi/2$ by the spherical cosine law. Hence $y(r) = AC = AB + BC = 2d$. Since $0 < \sqrt{3}d < 2d$, there is an r_0 between $3d/(2\pi)$ and $4\pi/(2\pi)$ such that $y(r_0) = \sqrt{3}d$ by the intermediate value theorem. Therefore, the quartet consisting of the vertices of the rhombus is isometrically embeddable in $\mathbb{S}(r_0)$ □

Remark 10.1 By a computation using Theorem 10.4 in Sect. 10.4, we have $r_0/d \approx 0.602289$

10.2 The Vertices of a Convex Quadrilateral

More generally, the following theorem holds.

Theorem 10.1 *For any convex quadrilateral in the plane, the metric space consisting of its four vertices is spherical.*

Proof Let $ABCD$ be a convex quadrilateral in the plane (Fig. 10.3). Among the four triangles $\triangle ABC$, $\triangle BCD$, $\triangle ACD$, $\triangle ABD$, suppose that $\triangle ABD$ has the largest perimeter $2s$. Put $d = AC$. Let $\mathbb{S}(r)$ denote the sphere of radius r centered at the origin O of $\mathbb{R}^3$.

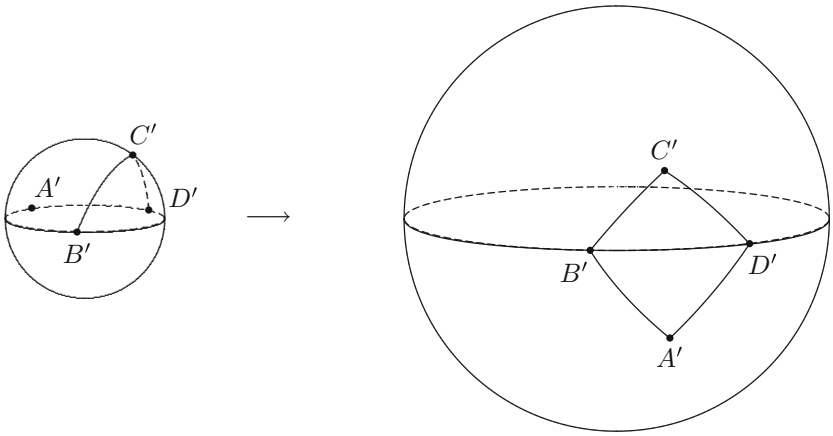


Fig. 10.4 Increasing the radius r

First, put $r = r_0 := s/\pi$, and take four points A', B', C', D' on $\mathbb{S}(r)$ so that

- $B' = (r_0, 0, 0)$, $D' = (x_0, y_0, 0)$, $y_0 > 0$,
- A' lies on the great circle $z = 0$, and C' lies on the hemisphere $z \geq 0$,
- the spherical distances other than $A'C'$ are equal to the corresponding distances in $\{A, B, C, D\}$.

This is clearly possible, see Fig. 10.4 left. If $A'C' = d$ for $r = r_0$, then $\{A', B', C', D'\}$ is an isometric embedding of $\{A, B, C, D\}$ in $\mathbb{S}(r_0)$. If $A'C' \neq d$, then we divide the argument into two cases.

Case (i). $A'C' < d$ at $r = r_0$:

Let us increase the radius r of $\mathbb{S}(r)$ keeping four points A', B', C', D' on it satisfying the conditions

- $B' = (r, 0, 0)$, $D' = (x, y, 0)$, $y > 0$,
- A' lies on the hemisphere $z \leq 0$, and C' lies on the hemisphere $z \geq 0$,
- the spherical distances other than $A'C'$ are the same as in the case $r = r_0$.

This is also possible. When r increases continuously from r_0, four points A', B', C', D' move continuously, and the spherical distance $A'C'$ varies continuously. At $r = r_0$, $B'D'$ is less than the half length of the great circle of $\mathbb{S}(r_0)$, that is, $B'D' < \pi r_0$, which implies that A' and D' do not lie in the same side of the great circle $B'C'$. On the other hand, when r becomes very large, $A'B'C'D'$ becomes closer to a convex quadrilateral in a plane. We can see that A' and D' come to lie in the same side of the great circle $B'C'$, see Fig. 10.4 right. Therefore, at some $r = r_1 > r_0$, A' lies on the great circle $B'C'$. In this case, $A'C' = A'B' + B'C'$. Thus, at $r = r_1$, $A'C' = A'B' + B'C' > d$, and hence, when r increases from r_0 to r_1, there is a moment when $A'C'$ becomes d.

Case (ii). $A'C' > d$ at $r = r_0$:

Let us increase the radius r of $\mathbb{S}(r)$ with keeping the four points A', B', C', D' on it satisfying the conditions

- $B' = (r, 0, 0)$, $D' = (x, y, 0)$, $y > 0$,
- both A' and C' lie on the hemisphere $z \geq 0$,
- spherical distances other than $A'C'$ are the same as in the case $r = r_0$.

Of course this is possible, too. In the convex quadrilateral $ABCD$, let $\bar{A}$ denote the mirror image of A with respect to the line BD, and put $\bar{d} = \bar{A}C$, see Fig. 10.3. Clearly $\bar{d} < d$. When r becomes very large, the distance $A'C'$ comes closer to $\bar{A}C = \bar{d}$. Hence for sufficiently large r we have $A'C' < d$. Thus, when r increases, there is a moment with $A'C' = d$.

□

Example 10.1 [A Nonplanar Quartet on a Sphere] Regarding $\mathbb{S}(r)$ as the surface of the Earth, let N be the north-pole, and A, B, C be the points on the equator with longitudes 10°, 20°, 30°, respectively. Then $AB = BC$, $AC = 2AB$, $AN = BN = CN$, and the trio $\{A, B, C\}$ is linear. Suppose that there is a quartet $\{A', B', C', N'\}$ in the plane that is isometric to $\{A, B, C, N\} \subset \mathbb{S}(r)$. Then, since $A'C' = 2A'B' = A'B' + B'C'$, B' is the midpoint of the line segment $A'C'$. Since $A'N' = B'N' = C'N'$, we have $\triangle A'B'N' \equiv \triangle C'B'N'$. Hence $\angle A'B'N' = 90°$, and $A'N'$ is a hypotenuse of the right triangle $A'B'N'$. This means $A'N' > A'B'$, contradicting that the two quartets are isometric. Therefore, $\{A, B, C, N\}$ is not planar.

Remark 10.2 P. L. Robinson [671] proved that any region on a sphere contains a quartet that cannot be isometrically embedded in the plane. Most results in the following are obtained by H. Maehara in [528].

10.3 Unispherical Quartets

A spherical metric space is called *unispherical* if the radius of the sphere in which the space is isometrically embeddable is unique. Any trio is not unispherical. Indeed, if a, b, c are the mutual distances in the trio, then for any $r > (a + b + c)/(2\pi)$ the trio is isometrically embeddable in a sphere of radius r. Thus, any trio is not unisherical.

Similarly to the proof of Theorem 10.1, the following theorem can be proved.

Theorem 10.2 *A quartet consisting of the four vertices of a convex quadrilateral on a sphere is not unispherical.* □

Is there a quartet on a sphere that is unispherical? Yes. Any (small) region on a sphere contains such a quartet.

Let us call a quartet on a sphere a *three-pointed star* if it consists of the three vertices of an equilateral triangle and the center of the circumscribed cap of the

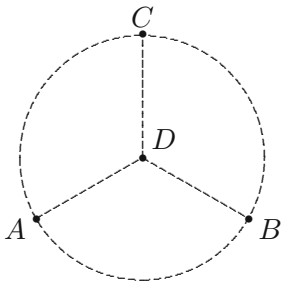


Fig. 10.5 A three-pointed star

triangle, see Fig. 10.5. In a three-pointed star (consulting Fig. 10.5) the angles $\angle ADB, \angle BDC, \angle CDA$ are called the *center-angles* of the three-pointed star, which are all equal to $2\pi/3$. It is clear that if a quartet on a sphere of radius R is isometric to a three-pointed star on a sphere of radius r, then the quartet is also a three-pointed star, and hence its center angle must be equal to $2\pi/3$. Then the triangle comparison theorem (Theorem 5.6) implies that $R = r$. Hence any three-pointed star is unispherical.

Theorem 10.3 *Since any region on a sphere contains a three-pointed star, any region on a sphere contains a quartet whose six distances determine the radius of the sphere uniquely.* □

Corollary 10.1 *For any small region on a sphere, it is impossible to make a (plane) map of the region in which distances are faithfully represented.*

10.4 An Equation for the Radius

For $X \in \mathbb{R}^3$, we denote the vector $\overrightarrow{OX}$ simply by $\vec{X}$. For a quartet

$$\mathcal{X} = \{P_1, P_2, P_3, P_4\},$$

denote by D_{ij} the distance $d(P_i, P_j)$, and define the matrix $M_{\mathcal{X}}(x)$ in the following way:

$$M_{\mathcal{X}}(x) = \begin{pmatrix} 1 & \cos(D_{12}/x) & \cos(D_{13}/x) & \cos(D_{14}/x) \\ \cos(D_{21}/x) & 1 & \cos(D_{23}/x) & \cos(D_{24}/x) \\ \cos(D_{31}/x) & \cos(D_{32}/x) & 1 & \cos(D_{34}/x) \\ \cos(D_{41}/x) & \cos(D_{42}/x) & \cos(D_{43}/x) & 1 \end{pmatrix}$$

If the four points of $\mathcal{X}$ lie on the sphere $\mathbb{S}(x)$ of radius x centered at the origin O of $\mathbb{R}^3$, then the (i, j)-entry of the matrix $x^2 M_{\mathcal{X}}(x)$ is the inner product $\vec{P}_i \cdot \vec{P}_j$ of the vectors $\vec{P}_i$ and $\vec{P}_j$.

In a trio Δ with mutual distances a, b, c, the value $s(\Delta) = (a+b+c)/2$ is called the *semi-perimeter* of the trio Δ. Recall that a trio Δ is isometrically embeddable in a sphere of radius r if and only if $s(\Delta) \leq \pi r$. For a quartet $\mathcal{X}$, we define $s(\mathcal{X})$ by

$$s(\mathcal{X}) = \max\{s(\Delta) : \Delta \subset \mathcal{X}\}.$$

Theorem 10.4 *For a quartet $\mathcal{X} = \{P_1, P_2, P_3, P_4\}$ that is not linear, the following statements are equivalent.*

(1) $\mathcal{X}$ is spherical.
(2) The equation $\det M_{\mathcal{X}}(x) = 0$ *has a solution $x = r$ that satisfies $\pi r \geq s(\mathcal{X})$. (Such a solution is the radius of the sphere in which $\mathcal{X}$ is isometrically embeddable.)*

Proof

(1) $\Rightarrow$ (2): Suppose that a sphere $\mathbb{S}(r)$ of radius r centered at O in $\mathbb{R}^3$ contains $\{P_1', P_2', P_3', P_4'\}$ that is isometric to $\mathcal{X}$. Then $\pi r \geq s(\mathcal{X})$. Since $\angle P_i' O P_j' = D_{ij}/r$, the inner product of two vectors $\vec{P}_i'$ and $\vec{P}_j'$ in $\mathbb{R}^3$ is equal to $r^2 \cos(D_{ij}/r)$. Let U be the (4×3)-matrix with ith row $\vec{P}_i'$ for $i = 1, 2, 3, 4$:

$$U = \begin{pmatrix} \vec{P}_1' \\ \vec{P}_2' \\ \vec{P}_3' \\ \vec{P}_4' \end{pmatrix}.$$

Then $r^2 M_{\mathcal{X}}(r) = UU^t$, where U^t is the transpose of U. Let $\tilde{U}$ be the matrix obtained from U by adding $(0, 0, 0, 0)^t$ as the 4th column:

$$\tilde{U} = \begin{pmatrix} \vec{P}_1' & 0 \\ \vec{P}_2' & 0 \\ \vec{P}_3' & 0 \\ \vec{P}_4' & 0 \end{pmatrix}.$$

Then $\tilde{U}\tilde{U}^t = UU^t$. Since $\det(\tilde{U}) = 0$,

$$\det(r^2 M_{\mathcal{X}}(r)) = \det(UU^t) = \det(\tilde{U}\tilde{U}^t) = 0,$$

and hence $\det M_{\mathcal{X}}(r) = 0$. That is, $\det M_{\mathcal{X}}(x) = 0$ has a solution $x = r$ that satisfies $\pi r \geq s(\mathcal{X})$.

(2) $\Rightarrow$ (1): Suppose that $x = r$ is a solution of $\det M_{\mathcal{X}}(x) = 0$ that satisfies $\pi r \geq s(\mathcal{X})$. Then every trio of $\mathcal{X}$ is isometrically embeddable in the sphere $\mathbb{S}(r)$ of radius r. Let us divide the argument into two cases.

Case (i). The rank of $M_{\mathcal{X}}(r)$ is equal to 2:

Suppose that the first row and the second row of $M_{\mathcal{X}}(r)$ are linearly independent. Then we can eliminate the third row and the fourth row by subtracting suitable linear combinations of the first row and the second row. Since $M_{\mathcal{X}}(r)$ is a symmetric matrix, we can further eliminate the third and the fourth columns by subtracting suitable linear combinations of the first and the second columns. Since the rank of a matrix is invariant under such operations, we have

$$\det \begin{pmatrix} 1 & \cos(D_{12}/r) \\ \cos(D_{21}/r) & 1 \end{pmatrix} \neq 0.$$

Now, suppose that the fourth row of $M_{\mathcal{X}}(r)$ is represented as

$$(\text{the fourth row}) = \lambda_1(\text{the first row}) + \lambda_2(\text{the second row}).$$

Let $\{P_1', P_2', P_3'\}$ be the isometric embedding of the trio $\{P_1, P_2, P_3\}$ of $\mathcal{X}$ into $\mathbb{S}(r)$. Define $P_4' \in R^3$ by

$$\vec{P}_4' = \lambda_1 \vec{P}_1' + \lambda_2 \vec{P}_2'.$$

Since $\{P_1', P_2', P_3'\} \subset \mathbb{S}(r)$ is isometric to $\{P_1, P_2, P_3\}$, we have

$$r^2 \begin{pmatrix} 1 & \cos(D_{12}/r) & \cos(D_{13}(r) \\ \cos(D_{21}/r) & 1 & \cos(D_{23}/r) \\ \cos(D_{31}/r) & \cos(D_{32}/r) & 1 \end{pmatrix} = \begin{pmatrix} \vec{P}_1' \\ \vec{P}_2' \\ \vec{P}_3' \end{pmatrix} \begin{pmatrix} \vec{P}_1' \\ \vec{P}_2' \\ \vec{P}_3' \end{pmatrix}^t.$$

And for $i = 1, 2$ we have

$$\begin{aligned} \vec{P}_4' \cdot \vec{P}_i' &= (\lambda_1 \vec{P}_1' + \lambda_2 \vec{P}_2') \cdot \vec{P}_i' = \lambda_1 \vec{P}_1' \cdot \vec{P}_i' + \lambda_2 \vec{P}_2' \cdot \vec{P}_i' \\ &= r^2\big(\lambda_1 \cos(D_{1i}/r) + \lambda_2 \cos(D_{2i}/r)\big) = r^2 \cos(D_{4i}/r). \end{aligned}$$

Therefore,

$$\begin{pmatrix} \vec{P}_1' \\ \vec{P}_2' \\ \vec{P}_4' \end{pmatrix} \begin{pmatrix} \vec{P}_1' \\ \vec{P}_2' \\ \vec{P}_4' \end{pmatrix}^t = r^2 \begin{pmatrix} 1 & \cos(D_{12}/r) & \cos(D_{14}/r) \\ \cos(D_{21}/r) & 1 & \cos(D_{24}/r) \\ \cos(D_{41}/r) & \cos(D_{42}/r) & |\vec{P}_4'|^2/r^2 \end{pmatrix}.$$

Since the three vectors $\vec{P}_1', \vec{P}_2', \vec{P}_4'$ are linearly dependent, the determinant of this matrix is 0. On the other hand, the first row, the second row, and the fourth row of $M_{\mathcal{X}}(r)$ are linearly dependent,

$$\det \begin{pmatrix} 1 & \cos(D_{12}/r) & \cos(D_{14}/r) \\ \cos(D_{21}/r) & 1 & \cos(D_{24}/r) \\ \cos(D_{41}/r) & \cos(D_{42}/r) & 1 \end{pmatrix} = 0.$$

Hence we must have $|\vec{P}_4'|^2/r^2 = 1$ (and hence $P_4' \in \mathbb{S}(r)$). Thus,

$$\begin{pmatrix} \vec{P}_1' \vec{P}_2' \\ \vec{P}_3' \\ \vec{P}_4' \end{pmatrix} \begin{pmatrix} \vec{P}_1' \\ \vec{P}_2' \\ \vec{P}_3' \\ \vec{P}_4' \end{pmatrix}^t = r^2 \begin{pmatrix} 1 & \cos(D_{12}/r) & \cos(D_{13}/r) & \cos(D_{14}/r) \\ \cos(D_{21}/r) & 1 & \cos(D_{23}/r) & \cos(D_{24}/r) \\ \cos(D_{31}/r) & \cos(D_{32}/r) & 1 & (\vec{P}_3' \cdot \vec{P}_4')/r^2 \\ \cos(D_{41}/r) & \cos(D_{42}/r) & (\vec{P}_3' \cdot \vec{P}_4')/r^2 & 1 \end{pmatrix}.$$

The determinant of the matrix on the left hand side is 0. In the matrix of the right hand side, by subtracting suitable linear combinations of the first row and the second row from the third row and the fourth row, we have the matrix

$$\begin{pmatrix} 1 & \cos(D_{12}/r) & \cos(D_{13}/r) & \cos(D_{14}/r) \\ \cos(D_{21}/r) & 1 & \cos(D_{23}/r) & \cos(D_{24}/r) \\ 0 & 0 & 0 & \frac{\vec{P}_3' \cdot \vec{P}_4'}{r^2} - \cos(D_{34}/r) \\ 0 & 0 & \frac{\vec{P}_3' \cdot \vec{P}_4'}{r^2} - \cos(D_{43}/r) & 0 \end{pmatrix}.$$

Since the determinant of this matrix is 0, we have $(\vec{P}_3' \cdot \vec{P}_4')/r^2 = \cos(D_{43}/r)$. Therefore, $\{P_1', P_2', P_3', P_4'\}$ is an isometric embedding of $\{P_1, P_2, P_3, P_4\}$ in $\mathbb{S}(r)$.

Case (ii). The rank of $M_{\mathcal{X}}(r)$ is equal to 3:

Suppose the first row, the second row, and the third row of $M_{\mathcal{X}}(r)$ are linearly independent. Then we can represent the fourth row by

$$\lambda_1(\text{the first row}) + \lambda_2(\text{the second row}) + \lambda_3(\text{the third row}).$$

First, we take an isometric embedding $\{P_1', P_2', P_3'\}$ of $\{P_1, P_2, P_3\}$ in $\mathbb{S}(r)$, and define $P_4' \in \mathbb{R}^3$ by $\vec{P}_4' = \lambda_1 \vec{P}_1' + \lambda_2 \vec{P}_2' + \lambda_3 \vec{P}_3'$. Then, similarly to the case (i), we can prove that $P_4' \in \mathbb{S}(r)$ and $\{P_1', P_2', P_3', P_4'\} \subset \mathbb{S}(r)$ is isometric to $\mathcal{X}$.

□

Example 10.2 Let $\mathcal{X} = \{P_1, P_2, P_3, P_4\}$ be a metric space in which distances $D_{ij} = d(P_i, P_j)$ are given by

$$(D_{ij}) = \begin{pmatrix} 0 & 5 & 5 & 3 \\ 5 & 0 & 5 & 3 \\ 5 & 5 & 0 & 6 \\ 3 & 3 & 6 & 0 \end{pmatrix},$$

see also the diagram given in Fig. 10.6. Then $s(\mathcal{X})/\pi = 7.5/\pi = 2.38732$. By numerical computation using, say, Mathematica, we have

$$r_1 = 2.4014, \ r_2 = 23.2921$$

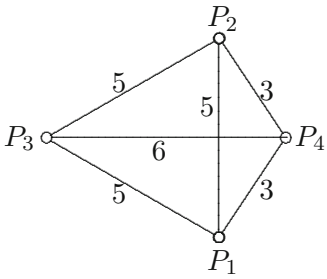


Fig. 10.6 Distances between four points

as solutions of $\det M_{\mathcal{X}}(x) = 0$ that satisfy $\pi r \geq s(\mathcal{X})$. Therefore, $\mathcal{X}$ is spherical, but not unispherical.

10.5 A Planar Quartet That Is also Spherical

By using the triangle comparison theorem for spheres (Theorem 5.6 and Corollary 5.2), the following theorem can be proved easily.

Theorem 10.5 *For a planar triangle $\triangle ABC$ and a point D contained in $\triangle ABC$, the quartet $\{A, B, C, D\}$ is not spherical.*

Proof Suppose that there is a quartet $\{A', B', C', D'\}$ of a sphere that is isometric to $\{A, B, C, D\}$.

Case (i). D is an interior point of $\triangle ABC$: By Corollary 5.2, we have $\angle A'D'B' > \angle ADB$, $\angle B'D'C' > \angle BDC$, $\angle C'D'A' > \angle CDA$. Hence $\angle A'D'B' + \angle B'D'C' + \angle C'D'A' > 2\pi$. If D' lies in the interior of $\breve{\triangle} A'B'C'$, then $\angle A'D'B' + \angle B'D'C' + \angle C'D'A' = 2\pi$, and if D' does not lie in the interior of $\breve{\triangle} A'B'C'$, then $\angle A'D'B' + \angle B'D'C' + \angle C'D'A' < 2\pi$. Thus we have a contradiction.

Case (ii). D lies on the line segment AB: In this case, D' lies on $A'B'$. Hence, $\angle A'D'C' + \angle B'D'C' = \pi$. However, by Corollary 5.2 we have $\angle A'D'C' > \angle ADC$, $\angle B'D'C' > \angle BDC$, and hence $\pi = \angle A'D'C' + \angle B'D'C' > \angle ADC + \angle BDC = \pi$, a contradiction.

□

Corollary 10.2 *A quartet of the plane $\mathbb{R}^2$ is spherical if and only if either the four points are collinear, or the four points are the vertices of a convex quadrilateral in the plane.* □

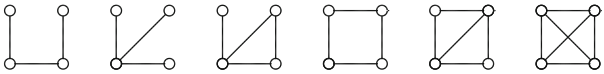


Fig. 10.7 Six connected graphs

10.6 Exercises

10.1* Prove that the quartet shown in Fig. 10.6 is not isometrically embeddable in $\mathbb{R}^3$.

10.2 Let $\mathcal{X}$ be a quartet with six distances 1, 1, 1, 1, 1, 2.

(i) Is $\mathcal{X}$ a planar quartet?
(ii) Is $\mathcal{X}$ a unispherical? If so, find the radius of the sphere that contains a quartet isometric to $\mathcal{X}$.

10.3* Let $ABCD$ be convex quadrilateral on the unit sphere $\mathbb{S}$ such that

$$AB = AC = BC = \pi/2,\ BD = CD = \pi/3.$$

Find the spherical distance AD.

10.4* Let $ABCD$ be the convex quadrilateral on the unit sphere $\mathbb{S}$ given in 10.3*. Find the radius $r \neq 1$ of a sphere into which the quartet $\{A, B, C, D\}$ can be isometrically embedded.

10.5 Every connected graph can be regarded as a metric space by the *path-metric*, that is, the distance $d(x, y)$ between two vertices x, y is defined as the minimum number of edges contained in a path connecting x and y. Classify the six quartets given by the six connected graphs in Fig. 10.7 into the following three types of quartets:

$$\{\textit{not spherical}\},\ \{\textit{spherical but not unispherical}\},\ \{\textit{unispherical}\}.$$

10.7 Notes

The following notes are very short, since the theme of Chap. 10 is (compared with other chapters) very specific in nature. However, we try to give shortly some background.

10.7.1 Notes on Four-Point Systems

Based on the notions of *metric and semi-metric spaces*, Menger and Blumenthal presented with [570] and [96] trend-setting publications giving insights into embeddings and corresponding characterizations of Euclidean, Hilbert, spherical, hyperbolic, and elliptic spaces. Within this framework, so-called *four-point properties* (referring to metrically defined quadruples of points) play an essential role. These quadruples yield, roughly speaking and for example, characterizations of generalized Euclidean spaces obtained via their metrical four-point properties. These say that each complete metric space containing a metric line that connects every pair of its points is a generalized Euclidean space iff each quadruple (from a certain class of quadruples of this space) is congruent with a quadruple of points in a Euclidean space. Clearly, this is closely related to generalizations of the triangle inequality. The book [96] shows various types of such characterizations, also and in particular referring to spherical spaces (see, e.g., Chapter VII and VIII).

Later contributions to four-point properties which characterize different types of spaces are, for example, contained in the following publications: Day [202], Freese [278], Valentine [792], Valentine [793], Freese et al. [280] as well as Freese and Andalafte [279]. For instance, many characterizations of real inner product spaces among metric spaces have been proven based on Euclidean four-point embedding properties, and many results in this direction interestingly refer also to Banach spaces.

Chapter 11
Higher Dimensions



Once we accept our limits, we go beyond them.

— Albert Einstein

First, we compute the volume of an n-dimensional ball $B^n(r)$ of radius r in $\mathbb{R}^n$, and we prove that for every fixed $r > 0$, the volume of the n-dimensional ball tends to 0 as n tends to ∞. How many unit balls in $\mathbb{R}^n$ can touch $B^n(r)$ simultaneously? The maximum number of such unit balls is called the kissing number of $B^n(r)$. We already showed in Chap. 8 that the kissing number of $B^3(1)$ is twelve. It is generally difficult to find the kissing number even for $r = 1$ for large n. For small radius, however, it is possible to determine the kissing number of $B^n(r)$. In this chapter, we determine the kissing number of $B^n(r)$ when $r \leq \sqrt{2} - 1$ for every $n > 0$.

11.1 Figures in High Dimensions

The n-dimensional Euclidean space $\mathbb{R}^n$ is a vector space over $\mathbb{R}$ consisting of all n-tuple row vectors $\boldsymbol{x} = (x_1, x_2, \ldots, x_n)$ where $x_i \in \mathbb{R}$ for $i = 1, 2, \ldots, n$, in which *addition, subtraction, scalar multiple*, and *inner product* are defined as follows: for $\boldsymbol{x} = (x_1, x_2, \ldots, x_n)$, $\boldsymbol{y} = (y_1, y_2, \ldots, y_n)$ and $\lambda \in \mathbb{R}$:

$$\begin{aligned} \boldsymbol{x} \pm \boldsymbol{y} &= (x_1 \pm y_1, x_2 \pm y_2, \ldots, x_n \pm y_n) \\ \lambda \boldsymbol{x} &= (\lambda x_1, \lambda x_2, \ldots, \lambda x_n) \\ \boldsymbol{x} \cdot \boldsymbol{y} &= x_1 y_1 + x_2 y_2 + \cdots + x_n y_n. \end{aligned}$$

Elements of $\mathbb{R}^n$ are called points. For $\boldsymbol{x} \in \mathbb{R}^n$, its *norm* $\|\boldsymbol{x}\|$ is defined as

$$\|\boldsymbol{x}\| = \sqrt{\boldsymbol{x} \cdot \boldsymbol{x}} = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}.$$

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H. Maehara, H. Martini, *Circles, Spheres and Spherical Geometry*,
Birkhäuser Advanced Texts Basler Lehrbücher,
https://doi.org/10.1007/978-3-031-62776-7_11

The distance $d(\boldsymbol{x}, \boldsymbol{y})$ between two points $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$ is defined as $d(\boldsymbol{x}, \boldsymbol{y}) = \|\boldsymbol{x}-\boldsymbol{y}\|$. For $r > 0$, the subset

$$B^n(r) = \{\boldsymbol{x} \in \mathbb{R}^n : \|\boldsymbol{x} - \boldsymbol{p}\| \leq r\}$$

of $\mathbb{R}^n$ is called the n-dimensional *ball* of radius r centered at $\boldsymbol{p}$. Its "boundary"

$$S_n(r) = \{\boldsymbol{x} \in \mathbb{R}^n : \|\boldsymbol{x} - \boldsymbol{p}\| = r\}$$

is a *hypersphere* (simply, a sphere) in $\mathbb{R}^n$ with radius r and center $\boldsymbol{p}$. For $a > 0$, the subset

$$\{(x_1, x_2, \ldots, x_n) \in \mathbb{R}^n : 0 \leq x_i \leq a, i = 1, 2, \ldots, n\}$$

of $\mathbb{R}^n$ is called an n-dimensional *cube* of edge-length a. An n-dimensional cube with edge-length 1 is called an n-dimensional *unit cube*.

For $k + 1$ points $\boldsymbol{x}_i \in \mathbb{R}^n$, $i = 0, 1, 2, \ldots, k$, such that the k vectors

$$\overrightarrow{\boldsymbol{x}_0\boldsymbol{x}_1}, \overrightarrow{\boldsymbol{x}_0\boldsymbol{x}_2}, \ldots, \overrightarrow{\boldsymbol{x}_0\boldsymbol{x}_k}$$

are linearly independent, the set

$$\{\lambda_0\boldsymbol{x}_0 + \lambda_1\boldsymbol{x}_1 + \cdots + \lambda_k\boldsymbol{x}_k \in \mathbb{R}^n : \lambda_0 + \lambda_1 + \cdots + \lambda_k = 1, \lambda_i \geq 0\}$$

is called the k-dimensional *simplex* (simply, k-simplex) spanned by $\boldsymbol{x}_0, \boldsymbol{x}_1, \ldots, \boldsymbol{x}_k$, and each $\boldsymbol{x}_i$ is called a *vertex* of that simplex. Of course, $k \leq n$. A 1-simplex is a line segment, a 2-simplex is a triangle, and a 3-simplex is a terahedron. In an n-simplex, the line segment connecting a pair of vertices is called an *edge* of the simplex. The number of edges of an n-simplex is $\binom{n+1}{2} = n(n+1)/2$. In an n-simplex, any n vertices span an $(n-1)$-simplex, which is called a *facet* of the n-simplex. An n-simplex has $n + 1$ facets.

For an n-simplex in $\mathbb{R}^n$, the (hyper)sphere that passes through all vertices of the simplex is called the *circumscribed sphere* (or circumsphere) of the simplex. This sphere is uniquely determined by the n-simplex. An n-simplex whose edges are all of the same length is called a *regular n-simplex*.

Example 11.1 For $t = \dfrac{1+\sqrt{n+1}}{n}$, or $t = \dfrac{1-\sqrt{n+1}}{n}$, the $n + 1$ points

$$\begin{cases} \boldsymbol{x}_0 &= (t, t, t, \ldots, t, t) \\ \boldsymbol{x}_1 &= (1, 0, 0, \ldots, 0, 0) \\ \boldsymbol{x}_2 &= (0, 1, 0, \ldots, 0, 0) \\ \cdots & \quad \cdots \\ \boldsymbol{x}_n &= (0, 0, 0, \ldots, 0, 1) \end{cases}$$

in $\mathbb{R}^n$ span a regular n-simplex with edge-length $\sqrt{2}$.

Theorem 11.1 *The circumscribed sphere of a regular n-simplex in $\mathbb{R}^n$ with unit edge-length has radius*

$$\sqrt{\frac{n}{2(n+1)}}.$$

Remark 11.1 $\sqrt{\frac{n}{2(n+1)}}$ is a monotonically increasing function of n.

Proof Let $\boldsymbol{x}_0, \boldsymbol{x}_1, \ldots, \boldsymbol{x}_n$ be the vertices of a regular n-simplex in $\mathbb{R}^n$. We may suppose that $\boldsymbol{x}_0$ is the origin O of $\mathbb{R}^n$. Since

$$1 = \|\boldsymbol{x}_1 - \boldsymbol{x}_2\|^2 = \|\boldsymbol{x}_1\|^2 + \|\boldsymbol{x}_2\|^2 - 2\boldsymbol{x}_1 \cdot \boldsymbol{x}_2 = 2 - 2\boldsymbol{x}_1 \cdot \boldsymbol{x}_2,$$

we have $\boldsymbol{x}_1 \cdot \boldsymbol{x}_2 = 1/2$. Hence we have

$$\boldsymbol{x}_i \cdot \boldsymbol{x}_j = \begin{cases} 1 & (\text{for } 1 \le i = j \le n) \\ 1/2 & (\text{for } 1 \le i \ne j \le n). \end{cases}$$

Put $\boldsymbol{p} = \frac{1}{n+1}(\boldsymbol{x}_0 + \boldsymbol{x}_1 + \cdots + \boldsymbol{x}_n)$. Since $\boldsymbol{x}_0 = O$,

$$\begin{aligned} \|\boldsymbol{p} - \boldsymbol{x}_0\|^2 = \boldsymbol{p} \cdot \boldsymbol{p} &= \frac{1}{(n+1)^2}\Big(n + \sum_{1 \le i < j \le n} 2\boldsymbol{x}_i \cdot \boldsymbol{x}_j\Big) \\ &= \frac{1}{(n+1)^2}\big(n + n(n-1)/2\big) = \frac{n}{2(n+1)}. \end{aligned}$$

Hence $\|\boldsymbol{p} - \boldsymbol{x}_0\| = \sqrt{\frac{n}{2(n+1)}}$. Since

$$\begin{aligned} \|\boldsymbol{p} - \boldsymbol{x}_1\|^2 &= \boldsymbol{p} \cdot \boldsymbol{p} + \boldsymbol{x}_1 \cdot \boldsymbol{x}_1 - 2\boldsymbol{p} \cdot \boldsymbol{x}_1 \\ &= \frac{n}{2(n+1)} + 1 - \frac{2}{n+1}\Big(1 + \frac{n-1}{2}\Big) = \frac{n}{2(n+1)}, \end{aligned}$$

we have $|\boldsymbol{p} - \boldsymbol{x}_1| = \sqrt{\frac{n}{2(n+1)}}$. Similarly, for $1 \le i \le n$, we have $\|\boldsymbol{p} - \boldsymbol{x}_i\| = \sqrt{\frac{n}{2(n+1)}}$. Therefore, the sphere with center $\boldsymbol{p}$ and radius $\sqrt{\frac{n}{2(n+1)}}$ passes through the vertices $\boldsymbol{x}_0, \boldsymbol{x}_1, \ldots, \boldsymbol{x}_n$. □

11.2 Moser's Paradox

In a square with edge-length 4 in $\mathbb{R}^2$, we can pack four unit disks as shown in Fig. 11.1. Then, in the room enclosed by these four unit disks at the center of the square, we can pack a disk of radius $\sqrt{2} - 1$. (The centers of the four unit disks form

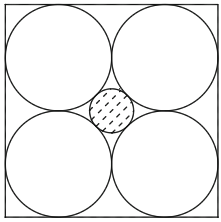


Fig. 11.1 A small disk enclosed by four unit disks

a square of edge-length 2, and its diagonal has length $2\sqrt{2}$. Hence the diameter of the disk packed in the room at the center is $2\sqrt{2} - 2$, and therefore the radius is $\sqrt{2} - 1$.)

In a 3-dimensional cube of edge length 4 in $\mathbb{R}^3$, we can pack eight unit balls at the eight corners of the cube. Then in the room at the center of the cube, we can pack a ball of radius $\sqrt{3} - 1$. In the case of the 4-dimensional cube with edge length 4, we can pack 16 unit balls at the 16 corners of the cube. Then, at the center of the cube, we can pack a ball of radius $\sqrt{4} - 1$. That is, a unit ball can lie in the room. Similarly, in the n-dimensional cube of edge-length 4, we can pack 2^n unit balls at the 2^n corners of the cube, and in the room at the center of the cube, we can pack a ball of radius $\sqrt{n} - 1$. If, for example, $n = 9$, then the radius of the ball that can be packed in the room at the center is $\sqrt{9} - 1 = 2$, and the ball is tangent to the boundary of the cube. If $n > 9$, then the central ball protrudes from the cube!

Remark 11.2 M. Gardner introduced this observation as paradox by Leo Moser in [294, Chapter 3], see also the related notes to this chapter here.

11.3 The Volume of a Ball

Let us compute the volume (n-dimensional volume) of the ball $B^n(r)$ of radius r. Let $V(n)$ denote the volume of an n-dimensional ball of unit radius. Then the volume of the n-dimensional ball of radius r becomes $V(n)r^n$. Since the volume of the "sphere-shell" between two concentric hyperspheres of radii $r + h$ and r is approximated by (surface area of $B^n(r)$) $\times\, h$, and

$$\lim_{h\to 0} \frac{V(n)(r+h)^n - V(n)r^n}{h} = nV(n)r^{n-1},$$

the area of a hypersphere $S_n(r)$ of radius r is given by $nV(n)r^{n-1}$.

Preceding the computation of the volume $V(n)$ of the n-dimensional unit ball, we try an exercise.

Exercise 11.1 Find $\alpha := \int_{-\infty}^{\infty} e^{-x^2} dx$.

Solution We have

$$\alpha^2 = \Big(\int_{-\infty}^{\infty} e^{-x^2} dx\Big)\Big(\int_{-\infty}^{\infty} e^{-y^2} dy\Big) = \int_{\mathbb{R}^2} e^{-(x^2+y^2)} dxdy.$$

By converting cartesian coordinates (x, y) to polar coordinates (r, θ), we get

$$\alpha^2 = \int_0^{\infty} \Big(\int_0^{2\pi} e^{-r^2} d\theta\Big) r dr = 2\pi \int_0^{\infty} r e^{-r^2} dr = 2\pi\big[\tfrac{-1}{2} e^{-r^2}\big]_0^{\infty} = \pi.$$

Therefore, $\alpha = \sqrt{\pi}$. □

To represent $V(n)$, we need the *gamma function* $\Gamma(s)$ defined by a convergent improper integral:

$$\Gamma(s) = \int_0^{\infty} z^{s-1} e^{-z} dz \qquad (s > 0).$$

Concerning the gamma function, the following equalities hold:

- $\Gamma(s+1) = s\Gamma(s)$.
- For an integer $m \geq 0$,

$$\Gamma(m+1) = m!, \quad \Gamma\big(m + \tfrac{1}{2}\big) = \frac{(2m)!}{2^{2m} m!}\sqrt{\pi}.$$

Remark 11.3 For these equalities and the gamma function see, e.g., the book [41] of E. Artin.

Theorem 11.2 *The volume $V(n)$ of an n-dimensional unit ball is given by*

$$V(n) = \frac{\pi^{n/2}}{\Gamma(n/2+1)}.$$

Proof To find the volume $V(n)$, let us consider the integral of the function $e^{-(x_1^2+x_2^2+\cdots+x_n^2)}$ in the whole $\mathbb{R}^n$:

$$I = \int_{\mathbb{R}^n} e^{-(x_1^2+x_2^2+\cdots+x_n^2)} dx_1 dx_2 \ldots dx_n.$$

From Exercise 11.1, we have $I = \big(\sqrt{\pi}\big)^n = \pi^{n/2}$. Let us compute I in a different way: First, integrate the function over the surface of the hypersphere $S_n(r)$ of radius r centered at the origin of $\mathbb{R}^n$, and then integrate the result by r from $r = 0$ to $r = \infty$. Since the value of the integrand is constant ($= e^{-r^2}$) on the surface of

the hypersphere $S_n(r)$, and the area of $S_n(r)$ is $nV(n)r^{n-1}$, its integral over $S_n(r)$ becomes

$$e^{-r^2} nV(n)r^{n-1}.$$

Therefore, we have

$$I = \int_0^\infty e^{-r^2} nV(n)r^{n-1} dr = nV(n) \int_0^\infty r^{n-1} e^{-r^2} dr.$$

By changing variables as $z = r^2$, and noting that $dz = 2rdr$, this yields

$$I = \frac{nV(n)}{2} \int_0^\infty z^{n/2-1} e^{-z} dz.$$

Since the improper integral is equal to $\Gamma(n/2)$,

$$I = \frac{nV(n)}{2} \Gamma(n/2) = V(n)(n/2)\Gamma(n/2) = V(n)\Gamma(n/2+1).$$

Then, from $I = \pi^{n/2}$, we have

$$V(n) = \frac{\pi^{n/2}}{\Gamma(n/2+1)}.$$

□

Figure 11.2 shows the graph of $y = \frac{\pi^{x/2}}{\Gamma(x/2+1)}$ for $0 \le x \le 20$. Using the second equality of the gamma function, we have

$$V(1) = 2,\ V(2) = \pi,\ V(3) = \tfrac{4\pi}{3} \approx 4.19,\ V(4) = \tfrac{\pi^2}{2} \approx 4.93,$$
$$V(5) = \tfrac{8\pi^2}{15} \approx 5.26,\ V(6) = \tfrac{\pi^3}{6} \approx 5.17,\ V(7) = \tfrac{16\pi^3}{105} \approx 4.72.$$

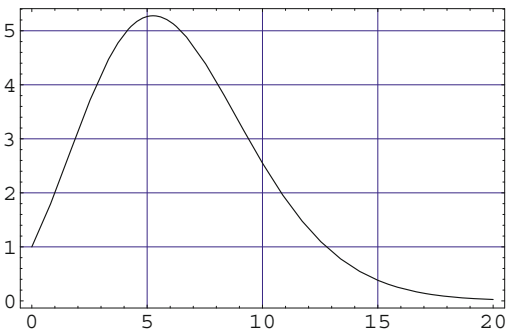


Fig. 11.2 $y = \frac{\pi^{x/2}}{\Gamma(x/2+1)}$

In Fig. 11.2, one sees that the maximum value of $V(n)$ is $V(5)$ and that $V(n) \to 0$ as $n \to \infty$.

Exercise 11.2 Prove that $V(5) > V(n)$ for $n > 5$.

Solution From the approximated value of $V(n)$ for $n \leq 7$, we see that $V(5)$ is the maximum in $\{V(n) : 1 \leq n \leq 7\}$. Since $\Gamma(n/2+1) = (n/2)\Gamma(n/2)$, and $2\pi/n < 1$ for $n \geq 7$, we have

$$V(n)/V(n-2) = \frac{\pi^{n/2}/\Gamma(n/2+1)}{\pi^{n/2-1}/\Gamma(n/2)} = \frac{2\pi}{n} < 1.$$

Thus, $V(n) < V(n-2)$ for $n \geq 7$, and hence $V(n) < V(5)$ follows for $n > 7$. □

Theorem 11.3 *For any $r > 0$, the volume of $B^n(r)$ tends to 0 as n tends to ∞.*

Proof From the equality $\Gamma(s+1) = s\Gamma(s)$, we have $\Gamma(s)/\Gamma(s+1) = 1/s$. Hence, for $n \geq 3$,

$$\frac{r^n V(n)}{r^{n-2}V(n-2)} = r^2 \frac{\pi^{n/2}/\Gamma(n/2+1)}{\pi^{n/2-1}/\Gamma(n/2)} = \frac{\pi r^2}{n/2} = \frac{2\pi r^2}{n},$$

and

$$r^n V(n) = r^{n-2}V(n-2)\Big(\frac{2\pi r^2}{n}\Big).$$

Let $N > 0$ be an integer such that $2\pi r^2/N < 1/2$. Then, for $n \geq N$, we have $r^n V(n) < r^{n-2}V(n-2)/2$. From this we get

$$\begin{aligned} r^{2m+N}V(2m+N) &< r^{2m-2+N}V(2m-2+N)/2 \\ &< r^{2m-4+N}V(2m-4+N)/2^2 \\ &< \cdots \\ &< r^2 V(N)/2^m. \end{aligned}$$

Therefore,

$$\lim_{m\to\infty} r^{2m+N}V(2m+N) = 0,$$

that is, $r^n V(n) \to 0$ $(n \to \infty)$. □

It follows from Theorem 11.3 that for any $\varepsilon > 0$ and any $r > 0$, there is an n_0 such that $r^n V(n) < \varepsilon$ holds for $n > n_0$. Thus, when $n > n_0$, a subset of $\mathbb{R}^n$ with volume $> \varepsilon$ cannot be covered by a ball of radius r. Hence, we have the following corollary.

Corollary 11.1 *For any $\varepsilon > 0$ and for any $r > 0$, there is an $n_0 > 0$ such that if $n > n_0$, then a subset of $\mathbb{R}^n$ with volume $> \varepsilon$ contains a pair of points at least distance r apart.* □

For example, a subset of $\mathbb{R}^n$ with volume 1cm^n can contain, if n is very large, a pair of points more than 100 kilometers apart.

11.4 Two Lemmas

Lemma 11.1 *Let $\{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_m\}$ be a set of m points lying on the sphere $S_n(r)$ of radius r in $\mathbb{R}^n$. Then*

$$m^2 r^2 \geq \sum_{i<j} \|\boldsymbol{x}_i - \boldsymbol{x}_j\|^2$$

holds. Equality holds if and only if the barycenter

$$\boldsymbol{p} := (\boldsymbol{x}_1 + \boldsymbol{x}_2 + \cdots + \boldsymbol{x}_m)/m$$

of $\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_m$ coincides with the center of $S_n(r)$.

Proof Let $\boldsymbol{q}$ be the center of the sphere, and compute $\|\boldsymbol{x}_i - \boldsymbol{q}\|^2$ by using inner products:

$$\begin{aligned}\|\boldsymbol{x}_i - \boldsymbol{q}\|^2 &= (\boldsymbol{x}_i - \boldsymbol{q}) \cdot (\boldsymbol{x}_i - \boldsymbol{q}) \\ &= (\boldsymbol{x}_i - \boldsymbol{p} + \boldsymbol{p} - \boldsymbol{q}) \cdot (\boldsymbol{x}_i - \boldsymbol{p} + \boldsymbol{p} - \boldsymbol{q}) \\ &= \|\boldsymbol{x}_i - \boldsymbol{p}\|^2 + \|\boldsymbol{p} - \boldsymbol{q}\|^2 + 2(\boldsymbol{x}_i - \boldsymbol{p}) \cdot (\boldsymbol{p} - \boldsymbol{q}).\end{aligned}$$

Since $\boldsymbol{p}$ is the barycenter of $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_m$, we have

$$\sum_{i=1}^{m} 2(\boldsymbol{x}_i - \boldsymbol{p}) \cdot (\boldsymbol{p} - \boldsymbol{q}) = 0.$$

Hence

$$\sum_{i=1}^{m} \|\boldsymbol{x}_i - \boldsymbol{q}\|^2 = \sum_{i=1}^{m} \|\boldsymbol{x}_i - \boldsymbol{p}\|^2 + m\|\boldsymbol{p} - \boldsymbol{q}\|^2,$$

and therefore

$$\sum_{i=1}^{m} \|\boldsymbol{x}_i - \boldsymbol{p}\|^2 = \sum_{i=1}^{m} \|\boldsymbol{x}_i - \boldsymbol{q}\|^2 - m\|\boldsymbol{p} - \boldsymbol{q}\|^2. \tag{11.1}$$

On the other hand,

$$\begin{aligned}
2\sum_{i<j}\|\boldsymbol{x}_i-\boldsymbol{x}_j\|^2 &= \sum_{i,j=1}^{m}\|\boldsymbol{x}_i-\boldsymbol{x}_j\|^2\\
&= \sum_{i,j=1}^{m}(\boldsymbol{x}_i-\boldsymbol{x}_j)\cdot\boldsymbol{x}_i-\sum_{i,j=1}^{m}(\boldsymbol{x}_i-\boldsymbol{x}_j)\cdot\boldsymbol{x}_j\\
&= \sum_{i,j=1}^{m}(\boldsymbol{x}_i-\boldsymbol{x}_j)\cdot\boldsymbol{x}_i+\sum_{i,j=1}^{m}(\boldsymbol{x}_j-\boldsymbol{x}_i)\cdot\boldsymbol{x}_j\\
&= 2\sum_{i,j=1}^{m}(\boldsymbol{x}_i-\boldsymbol{x}_j)\cdot\boldsymbol{x}_i.
\end{aligned}$$

Since $\displaystyle\sum_{j=1}^{m}(\boldsymbol{x}_i-\boldsymbol{x}_j)\cdot\boldsymbol{x}_i = m\boldsymbol{x}_i\cdot\boldsymbol{x}_i-m\boldsymbol{p}\cdot\boldsymbol{x}_i = m(\boldsymbol{x}_i-\boldsymbol{p})\cdot\boldsymbol{x}_i$, this equals

$$2m\sum_{i=1}^{m}(\boldsymbol{x}_i-\boldsymbol{p})\cdot\boldsymbol{x}_i,$$

and since $\displaystyle\sum_{i=1}^{m}(\boldsymbol{x}_i-\boldsymbol{p})\cdot\boldsymbol{p}=0$, it becomes

$$2m\sum_{i=1}^{m}(\boldsymbol{x}_i-\boldsymbol{p})\cdot(\boldsymbol{x}_i-\boldsymbol{p}) = 2m\sum_{i=1}^{m}\|\boldsymbol{x}_i-\boldsymbol{p}\|^2.$$

Therefore

$$\sum_{i<j}\|\boldsymbol{x}_i-\boldsymbol{x}_j\|^2 = m\sum_{i=1}^{m}\|\boldsymbol{x}_i-\boldsymbol{p}\|^2.$$

From this and (11.1), we have

$$\begin{aligned}
\sum_{i<j}\|\boldsymbol{x}_i-\boldsymbol{x}_j\|^2 &= m\sum_{i=1}^{m}\|\boldsymbol{x}_i-\boldsymbol{q}\|^2-m^2\|\boldsymbol{p}-\boldsymbol{q}\|^2\\
&\le m\sum_{i=1}^{m}\|\boldsymbol{x}_i-\boldsymbol{q}\|^2\\
&\le m^2r^2.
\end{aligned}$$

Clearly, equality holds only when $\boldsymbol{p}=\boldsymbol{q}$. □

Lemma 11.2 *For m points $\boldsymbol{x}_i, i = 1, 2, \ldots, m$, in $\mathbb{R}^n \setminus \{O\}$, the following statements (i) and (ii) hold:*

(i) If $\boldsymbol{x}_i \cdot \boldsymbol{x}_j < 0$ for every $i \neq j$, then $m \leq n + 1$.
(ii) If $\boldsymbol{x}_i \cdot \boldsymbol{x}_j \leq 0$ for every $i \neq j$, then $m \leq 2n$.

Proof

(i) The assertion is clearly true for $n = 1$. Suppose the assertion is true for $n - 1$ dimensions, and let us consider the n-dimensional case. Let

$$\boldsymbol{x}_i = (x_1^{(i)}, x_2^{(i)}, \ldots, x_n^{(i)}),\ i = 1, 2, \ldots, m,$$

be m points in $\mathbb{R}^n$ such that $\boldsymbol{x}_i \cdot \boldsymbol{x}_j < 0$ for $i \neq j$. By a suitable transformation of coordinates of $\mathbb{R}^n$, we may suppose that the first coordinate of $\boldsymbol{x}_1$ is positive and all other coordinates are 0, namely,

$$\boldsymbol{x}_1 = (x_1^{(1)}, 0, 0, \ldots, 0),\ x_1^{(1)} > 0.$$

Then, since $\boldsymbol{x}_1 \cdot \boldsymbol{x}_j < 0,\ j = 2, 3, \ldots, m$, the first coordinates $x_1^{(j)}$ of $\boldsymbol{x}_j$ ($j = 2, 3, \ldots, m$) are all negative, that is, $x_j^{(1)} < 0$, $j = 2, 3, \ldots, m$. Put

$$\boldsymbol{y}_j = (x_2^{(j)}, x_3^{(j)}, \ldots, x_n^{(j)}),\ j = 2, 3, \ldots, m.$$

Then, $\boldsymbol{y}_2, \boldsymbol{y}_3, \ldots, \boldsymbol{y}_m$ are $m - 1$ points in $\mathbb{R}^{n-1}$. Since $x_1^{(i)} x_1^{(j)} > 0$ and $x_1^{(i)} x_1^{(j)} + \boldsymbol{y}_i \cdot \boldsymbol{y}_j = \boldsymbol{x}_i \cdot \boldsymbol{x}_j < 0$ for $2 \leq i < j \leq m$, we must have $\boldsymbol{y}_i \cdot \boldsymbol{y}_j < 0$. (Hence $\boldsymbol{y}_j \neq O$ for $j = 2, 3, \ldots, m$.) Then by the inductive assumption, we have $m - 1 \leq (n - 1) + 1 = n$, and hence $m \leq n + 1$.

(ii) The assertion is clearly true for $n = 1$. Assuming that the assertion is true for $n - 1$ dimensions, let us consider the n-dimensional case. Let

$$\boldsymbol{x}_i = (x_1^{(i)}, x_2^{(i)}, \ldots, x_n^{(i)}),\ i = 1, 2, \ldots, m,$$

be m points in $\mathbb{R}^n$ such that $\boldsymbol{x}_i \cdot \boldsymbol{x}_j \leq 0$ for $i \neq j$. We divide the argument into two cases.

Case (1). The origin O and some two points $\boldsymbol{x}_i, \boldsymbol{x}_j$ are collinear:
Suppose O and $\boldsymbol{x}_1, \boldsymbol{x}_2$ are collinear. Since $\boldsymbol{x}_1 \cdot \boldsymbol{x}_2 \leq 0$, O must lie between $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$. So, by a suitable coordinate-transformation of $\mathbb{R}^n$, we may assume

$$\boldsymbol{x}_1 = (x_1^{(1)}, 0, 0, \ldots, 0),\ x_1^{(1)} > 0,$$
$$\boldsymbol{x}_2 = (x_1^{(2)}, 0, 0, \ldots, 0),\ x_1^{(2)} < 0.$$

Then, since $\boldsymbol{x}_1 \cdot \boldsymbol{x}_j \leq 0, \boldsymbol{x}_2 \cdot \boldsymbol{x}_j \leq 0,\ j = 3, 4, \ldots, m$, the first coordinate $x_1^{(j)}$ of $\boldsymbol{x}_j$ must be 0 for $j = 3, 4, \ldots, m$. Put

$$\boldsymbol{y}_j = (x_2^{(j)}, x_3^{(j)}, \ldots, x_n^{(j)}),\ j = 3, 4, \ldots, m.$$

Then, $\boldsymbol{y}_3, \boldsymbol{y}_4, \ldots, \boldsymbol{y}_m$ are $m-2$ points in $\mathbb{R}^{n-1} \setminus \{O\}$, and $\boldsymbol{y}_i \cdot \boldsymbol{y}_j = \boldsymbol{x}_i \cdot \boldsymbol{x}_j \leq 0$ for $3 \leq i < j \leq m$. Then, by the inductive assumption, we have $m - 2 \leq 2(n-1)$. Hence, $m \leq 2n$.

Case (2). For any two points $\boldsymbol{x}_i, \boldsymbol{x}_j$, the three points $O, \boldsymbol{x}_1, \boldsymbol{x}_2$ are not collinear: We may assume that the first coordinate of $\boldsymbol{x}_1$ is positive, and all other coordinates of $\boldsymbol{x}_1$ are 0. That is,

$$\boldsymbol{x}_1 = (x_1^{(1)}, 0, 0, \ldots, 0),\ x_1^{(1)} > 0.$$

Since $\boldsymbol{x}_1 \cdot \boldsymbol{x}_j \leq 0,\ j = 2, 3, \ldots, m$, the first coordinate $x_j^{(1)}$ is negative or 0 for $j = 2, 3, \ldots, m$. Put

$$\boldsymbol{y}_j = (x_2^{(j)}, x_3^{(j)}, \ldots, x_n^{(j)}),\ j = 2, 3, \ldots, m.$$

Then, $\boldsymbol{y}_2, \boldsymbol{y}_3, \ldots, \boldsymbol{y}_m$ are $m-1$ points in $\mathbb{R}^{n-1}$. Since $O, \boldsymbol{x}_1, \boldsymbol{x}_j$ are not collinear for $j \geq 2$, $\boldsymbol{y}_j \neq (0, 0, \ldots, 0)$. Moreover, since $x_1^{(i)} x_1^{(j)} \geq 0$ and $x_1^{(i)} x_1^{(j)} + \boldsymbol{y}_i \cdot \boldsymbol{y}_j = \boldsymbol{x}_i \cdot \boldsymbol{x}_j \leq 0$ for $2 \leq i < j \leq m$, we have $\boldsymbol{y}_i \cdot \boldsymbol{y}_j \leq 0$ for $i \neq j$. Then by the inductive assumption, we have $m - 1 \leq 2(n-1)$, and hence $m \leq 2n - 1 \leq 2n$.

□

11.5 Kissing a Small Ball

A point-set $\mathcal{X}$ in $\mathbb{R}^n$ is called *dispersed* if $\|\boldsymbol{x} - \boldsymbol{y}\| \geq 1$ holds for every distinct pair $\boldsymbol{x}, \boldsymbol{y} \in \mathcal{X}$. As we saw in the problem of thirteen balls in Chap. 6, there is a dispersed set of twelve points on the unit sphere in $\mathbb{R}^3$. Let us denote by $\nu[S_n(r)]$ the maximum number of points that a dispersed set on $S_n(r)$ can contain.

Exercise 11.3 Prove that if $r = 1/\sqrt{2}$, then $\nu[S_n(r)] = 2n$ for every $n > 0$.

Proof *(Solution)* For $r = 1/\sqrt{2}$, the set of $2n$ points

$$\begin{array}{cc} (r, 0, 0, \ldots, 0) & (-r, 0, 0, \ldots, 0) \\ (0, r, 0, \ldots, 0) & (0, -r, 0, \ldots, 0) \\ \cdots & \cdots \\ (0, 0, 0 \ldots, r) & (0, 0, 0 \ldots, -r) \end{array}$$

in $\mathbb{R}^n$ is a dispersed set on $S_n(r)$. Hence $\nu[S_n(r)] \geq 2n$.

On the other hand, if $\{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_m\}$ is a dispersed set on $S_n(r)$, then, since

$$1 \leq \|\boldsymbol{x}_i - \boldsymbol{x}_j\|^2 = (\boldsymbol{x}_i - \boldsymbol{x}_j) \cdot (\boldsymbol{x}_i - \boldsymbol{x}_j) = 1/2 + 1/2 - 2\boldsymbol{x}_i \cdot \boldsymbol{x}_j$$

for $i \neq j$, we have $\boldsymbol{x}_i \cdot \boldsymbol{x}_j \leq 0$. This implies, by Lemma 11.2(ii), $m \leq 2n$, and thus $\nu[S_n(r)] \leq 2n$. □

From Exercise 11.3, we see that when $r \geq 1/\sqrt{2}$, $\nu[S_n(r)]$ increases as n increases. To find the exact value of $\nu[S_n(r)]$ for $r > 1/\sqrt{2}$ is difficult. However, in the case $r < 1/\sqrt{2}$, we can derive the exact value of $\nu[S_n(r)]$ for every n.

Theorem 11.4 *For* $0 < r < 1/\sqrt{2}$,

$$\nu[S_n(r)] = \begin{cases} n+1 & \text{for } n < n_r - 1 \\ n_r & \text{for } n \geq n_r - 1 \end{cases},$$

where n_r *denotes the integer part of* $\frac{1}{1-2r^2}$.

For example, when $r = 0.65$, we have $\frac{1}{1-2\times(0.65)^2} = 6.4516$, and hence $n_r = 6$. Thus, $\nu[S_n(0.65)] = n + 1$ for $n < 5$, and $\nu[S_n(0.65)] = 6$ for $n \geq 5$.

Proof

(1) The case $n < n_r - 1$:
Let $\{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_m\}$ be a dispersed set of m points on $S_n(r)$. Since

$$1 \leq \|\boldsymbol{x}_i - \boldsymbol{x}_j\|^2 = 2r^2 - 2\boldsymbol{x}_i \cdot \boldsymbol{x}_j,$$

we have $\boldsymbol{x}_i \cdot \boldsymbol{x}_j < 0$. Then, by Lemma 11.2(i), $m \leq n+1$. On the other hand, the radius of the circumscribed sphere of a regular n-simplex of unit edge-length is $\sqrt{\frac{n}{2(n+1)}}$ by Theorem 11.1. Since $n < n_r - 1$, by the remark after Theorem 11.1 we have

$$\sqrt{\frac{n}{2(n+1)}} \leq \sqrt{\frac{n_r - 1}{2n_r}}.$$

Since $n_r \leq \frac{1}{1-2r^2}$ implies $r \geq \sqrt{\frac{n_r-1}{2n_r}}$, we have $r \geq \sqrt{\frac{n}{2(n+1)}}$. Hence, $S_n(r)$ is a circumscribed sphere of a regular n-simplex with edge length ≥ 1. Thus, $\nu[S_n(r)] \geq n + 1$, and hence $\nu[S_n(r)] = n + 1$

(2) The case $n \geq n_r - 1$:
Let $\{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_m\}$ be independent m points on $S_n(r)$. By Lemma 11.1,

$$m^2 r^2 \geq \sum_{i<j} \|\boldsymbol{x}_i - \boldsymbol{x}_j\|^2 \geq \frac{m(m-1)}{2},$$

from which we have $m \le \frac{1}{1-2r^2}$. Hence $m \le n_r$. On the other hand, the radius of the circumscribed sphere of a regular $(n_r - 1)$-simplex with unit edge length is $\sqrt{\frac{n_r-1}{2n_r}}$ by Theorem 11.1. Since $n_r \le \frac{1}{1-2r^2}$ implies $r \ge \sqrt{\frac{n_r-1}{2n_r}}$, a sphere of radius r in $\mathbb{R}^{n_r-1}$ is the circumscribed sphere of a regular $(n_r - 1)$-simplex with edge length ≥ 1, and hence there is a dispersed set of n_r points. Therefore, $\nu[S_n(r)] = n_r$.

□

Remark 11.4 Theorem 11.4 was essentially obtained by R. A. Rankin in [655].

For an n-dimensional ball $B^n(r)$ of radius r in $\mathbb{R}^n$, the maximum number of unit balls in $\mathbb{R}^n$ that can touch $B^n(r)$ simultaneously (without overlapping) is called the *kissing number* of $B^n(r)$.

Lemma 11.3 *For $r > 1/2$, the kissing number of an n-dimensional ball of radius $2r - 1$ is equal to $\nu[S^n(r)]$.*

Proof Let O be the center of $S_n(r)$ and $\boldsymbol{x}, \boldsymbol{y} \in S_n(r)$ be two points such that $\|\boldsymbol{x} - \boldsymbol{y}\| = 1$. Let $\boldsymbol{x}', \boldsymbol{y}'$ be the points on the rays $\overrightarrow{O\boldsymbol{x}}, \overrightarrow{O\boldsymbol{y}}$, respectively, such that $\|\boldsymbol{x}' - O\| = \|\boldsymbol{y}' - O\| = 2r$. Then $\|\boldsymbol{x}' - \boldsymbol{y}'\| = 2$, and the unit balls with centers $\boldsymbol{x}', \boldsymbol{y}'$ touch the ball $B^n(2r-1)$, see Fig. 11.3. From this we can see that the kissing number of $B^n(2r - 1)$ is at least $\nu[S_n(r)]$. Similarly, we can see that the kissing number of $B^n(2r - 1)$ is at most $\nu[S_n(r)]$. □

Theorem 11.5 *For $0 < \rho < \sqrt{2} - 1$, the kissing number of $B^n(\rho)$ is*

$$\begin{cases} n+1 & \text{for } n < n(\rho) - 1, \\ n(\rho) & \text{for } n \ge n(\rho) - 1, \end{cases}$$

where $n(\rho)$ is the integer part of $\dfrac{2}{2 - (\rho+1)^2}$.

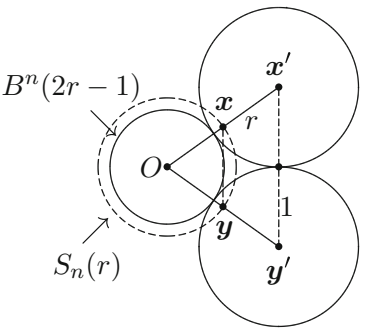


Fig. 11.3 Two unit balls touch $S_n(2r - 1)$

Proof Let $\rho = 2r-1$. Then $r = (\rho+1)/2$, and the kissing number of $B^n(\rho)$ is equal to $\nu[S_n(\frac{\rho+1}{2})]$ by Lemma 11.3. Since $n_r = \frac{1}{1-(\rho+1)^2/2} = 2/(2-(\rho+1)^2) = n(\rho)$, the theorem follows from Theorem 11.4. □

11.6 Exercises

11.1* Prove that among the volumes $V(n)$ of the n-dimensional unit ball, $n = 1, 2, 3, \ldots$, the one that has the maximum value is $V(5)$.

11.2 Show that $\sqrt{\frac{n}{2(n+1)}}$ is monotonically increasing in n.

11.3* Prove that $\Gamma(s+1) = s\Gamma(s)$.

11.4 Prove that $\Gamma(m+1) = m!$ for every integer $m > 0$.

11.5 Using Lemma 11.1, prove Theorem 11.1.

11.6 In R^3, find the maximum number of balls of radius 5 that can simultaneously touch one and the same ball of radius 2.

11.7 Notes

In the above part of Chap. 11, the following geometric figures (in high dimensions) were studied or used: the (unit) ball, the simplex, and the cube. Therefore in the following three subsections (of the notes) precisely these three types of figures are discussed, in n-dimensional version and (due to technical reasons) in reverse order.

11.7.1 Notes on Cubes

In the following we give a selection of nice overviews and results on the geometry of the cube. We see this (like also the next subsection on simplices) as a good "exercise" in high-dimensional geometry, then needed for high-dimensional geometry of balls. In particular, here we lay emphasize on *sections and projections* of n-dimensional cubes.

More elementary results on and properties of *cubes*, mainly in dimension 3, are discussed in the booklets Ranucci and Rollins [656] as well as Ehrenfeucht [240], treating curiosities on dissections, slices, colorings, Pohlke's theorem from descriptive geometry, and geodesics. Similarly, the exposition [804] of Weiss collects also results from 3-space, e.g. referring to *geodesics* (applying Snellius' refraction law) and *billiards*.

A broader and deeper view is given in Zong's booklet [841] which is somehow prepared by the excellent survey [840]. The author treats cross-sections and projections of n-cubes, inscribed simplices and optimal (regarding the number of

needed simplices) triangulations of cubes, also in view of hyperbolic geometry, cube tilings (e.g., Keller's conjecture on twins in cube tilings) and further topics, mainly from *discrete and convex geometry*.

Sections of Cubes

In Ball's lecture [55, Lecture 6] the extremum property of the cube in the symmetric case of the *reverse isoperimetric problem* is nicely discussed. For new contributions extending Ball's well known result on *maximal hyperplanar cube slices* (namely, that the sharp upper bound for the volume of hyperplane sections of a unit cube is $\sqrt{2}$) we refer to the recent paper [422] of Ivanov and Tsiutsiurupa, containing the important references. And we mention here the lecture notes [680] in which also Mahler's conjecture (the *volume product regarding polar bodies* is, in the symmetric case, minimal at cubes) is studied. (Maximal) cube slicing is also related to *section functions* of convex bodies or polytopes, and therefore to *geometric tomography*, i.e., also to Gardner's important monograph [295, Chapters 7 and 8] (in particular, to Note 7.8) and to Martini's survey [547, Section 2.3]. We repeat here Ball's essential result on the largest hyperplanar cube slice in view of a prominent application, namely the famous *Busemann-Petty problem*. And Melbourne and Roberto [567] derive a quantitative form of this result of Ball. Frank and Riede [275] succeed in deducing an elementary formula for the *volume of arbitrary hyperplane slices* of the n-cube.

Chakerian and Logothetti [154] deal with the *geometry and combinatorics of hyperplanar cube-slicing*, studying interesting sequences of cube slices and, related to this, binomial coefficients and properties of Pascal's triangle. The volumes of slices and related slabs are studied, and probabilistic questions coming from these cube slices are also discussed. Also Ball's result and its "Busemann-Petty application" (see above) is discussed. The survey [685] by Saks refers to *combinatorial problems related to sections, partitions and coverings of the n-cube*. E.g., the maximal number of edges that can be intersected by a single hyperplane is precisely determined, and also bounds for the minimal number of hyperplanes needed to meet each edge are discussed. Bartha et al. [64] study *volumes of central slices of unit n-cubes orthogonal to their main diagonals* for increasing n, showing that they give a strictly monotonically increasing function of the dimension.

The volume of central slices of unit cubes with *slabs of fixed width* are studied by Barthe and Koldobsky [66]. They show that if the width is at most $3/4$, then slabs parallel to facets yield the minimum, whereas for large width, the slabs orthogonal to the main diagonal become asymptotically minimal as n goes to infinity. Moody et al. [582] investigate *extremal volumes of non-central slices of the unit n-cube*, meaning that the cutting hyperplanes have to be at a certain distance $d > 0$ from the center. This can also be extended to non-central slices of the unit n-cube with slabs. In [582] the authors solve this completely for one-dimensional slices, and they give a partial solution for extremal hyperplane slices with $d > \sqrt{n-1}/2$. Slices of minimal volume by slabs of width $2d$, with the same bound for d, are

also discussed. Further papers treating slab slices of unit n-cubes are [464] of König and Koldobsky and [463] of König. And in [465] these authors investigate the $(n-2)$-dimensional surface area of central hyperplane slices of the unit n-cube. For instance, they show that the maximum is attained perpendicularly to the direction vector $(1, 1, 0, ..., 0)$. This is applied to give a negative answer to an analogue of the *Busemann-Petty problem for surface area*, namely for dimensions $n \geq 14$. And Ivanov and Tsiutsiurupa [422] discuss *maximal volumes of k-dimensional sections of the unit n-cube*, for $n \geq 2$ presenting a necessary and sufficient condition for a k-subspace to be a local maximizer of this section function.

Projections of Cubes

It is known already for a long time that k-dimensional parallel and, in particular, orthogonal projections of n-cubes are k-dimensional *zonotopes*, i.e., k-dimensional finite sums of line segments. E.g., Naumann [594] uses vector stars and properties of zonotopes *to characterize orthogonal projections* of n-cubes among arbitrary parallel projections. This is practically a result from *descriptive geometry*, more precisely a generalization of *Pohlke's theorem from the theory of axonometry*. Also Bergold [80] slightly generalizes Pohlke's theorem: He shows that two arbitrary parallelepipeds in 3-space can always be turned in such a way that they appear identical, i.e., that their parallel projections onto a 2-plane perpendicular to the direction of sight are similar. Maehara [527] proves that for any $k \geq 2$ and $n \geq k$, there is an n-dimensional unit cube which is projected to a regular $2k$-gon using an orthogonal projection from n-space onto a 2-dimensional subspace. The author studies these projections for increasing dimension n.

McMullen [565] proves that the *orthogonal projections of the unit n-cube onto totally orthogonal subspaces of dimensions k and $n-k$* of Euclidean n-space have the property that their k-volume and $(n-k)$-volume are, respectively, equal. He himself and Schnell [708] extended this result suitably to affine cubes and finite sums of segments (= zonotopes). Chakerian and Filliman [152] obtain several *inequalities* for the k-dimensional Lebesgue measure of the orthogonal projection of the unit n-cube into a k-subspace. It is shown that the studied problem is symmetric in k and $n-k$, and for $k = 2$ and $k = n-2$ the authors give a sharp upper bound and also related results for general k. Filliman [268] adds results on zonotopes, among them also some about special zonotopes which are even *orthogonal k-dimensional projections of n-cubes* with $1 \leq k \leq n$. Also a necessary condition for k-zonotopes to have *largest k-volume* among all k-dimensional orthogonal projections of the n-cube is given. (For the subcase of maximal orthogonal $(n-1)$-dimensional projections of n-cubes we refer to the paper [553].) In this spirit, a nice concrete example is derived: the classical *rhombic triacontahedron* has largest 3-volume among all orthogonal projections of a 6-cube into 3-space.

For unit n-cubes, Ostrovskii [607] investigates *orthogonal projections having minimal k-volume* in k-dimensional linear subspaces. He proves that a k-zonotope is *linearly equivalent* to a minimal-volume projection of a unit n-cube if and only if

it is linearly equivalent to the zonotope spanned by the multiples of rows of a totally unimodular $n \times r$ matrix of rank k. And for every subspace L, the set of minimal-volume projections of unit n-cubes in L contains a parallelepiped, see [606]. On the other hand, for some subspaces there are minimal-volume projections being far away from parallelepipeds (meant with respect to the Banach-Mazur distance).

Paouris et al. [615] deal with the *volume of random projections of the unit n-cube*. They derive a *central limit theorem* for the volume of the orthogonal projection of an n-cube onto a random k-dimensional subspace, distributed according to the appropriate Haar probability measure as $n \to \infty$.

Further Topics

We continue now with some further aspects of the geometry of the n-cube, mentioning for each topic only a few essential and representative publications. For more details we refer to subsection 2.9.1.2 of the recent book [53].

We start here with *vertex embedding of simplices in cubes*, particularly interesting since it is equivalent to the question for the *existence of Hadamard matrices*, for *Jung constants* of the form $n/(n+1)$ in l_1^n, and for a known problem from *coding theory*. From the viewpoint of *computational convexity*, Gritzmann and Klee study this topic in [329, subsection 9.6], discussing also its relations to the problem of *largest k-simplices in n-cubes*. For $k = n$, this is equivalent to the *Hadamard determinant problem* subsuming the problem above on Hadamard matrices. Largest simplices in cubes are also studied in the expositions [330] of Gritzmann, Klee and Larman as well as [408], due to Hudelson, Klee, and Larman. Interesting conjectures are stated, and also other problems regarding simplices and cubes are posed. More recent references on Hadamard matrices are treated in the monograph [722] of Seberry.

The next theme is *triangulations of cubes*, i.e., minimal (regarding number) dissections of the n-cube into n-simplices all whose vertices are vertices of the given n-cube. Good surveys on this topic are given in the books [210, Section 6.3] of De Loera, Rambau, and Santos, and [841, Chapters 3 and 4] of Zong; see also the excellent survey [840] of Zong. For more recent references we refer also to the book [53], more precisely to its subsection 2.9.1.2.

Our next topic is the famous *Keller conjecture* from 1930: If the Euclidean n-space is tiled by congruent parallel unit n-cubes, then there exists some pair of cubes sharing a complete facet. This conjecture was refuted by Lagarias and Shor [486] for $n \geq 10$; see also the corresponding Mathematical Review MR1155280. For further reading regarding the time after the appearance of [486] the references [840] and [841] are again recommended. After a longer period, finally Brakensiek et al. [122] succeeded (excluding the existence of certain types of graphs) to solve Keller's conjecture for all dimensions. And Horak and Kim [402] could show that extensions of Keller's conjecture to tiles having a more complex shape cannot be verified.

Further interesting topics and themes treating the geometry of the n-cube that one might study with the help of MathSciNet, Zentralblatt or problem books like [190] are: *decomposing the cube into cubes, cube colorings, different types of separations of the vertex sets of cubes, Hamming distances, cubes and graphs, cube coverings, dissections of cubes, packings of or in cubes*, the *Banach-Mazur distance*, and many further subjects. Also suitably modified cubes (like, e.g., the *Klee-Minty cube*, see Gale's excellent survey [288]) and their applications present interesting fields.

11.7.2 Notes on Simplices

Simplices of dimension n (shortly called n-simplices) are those convex n-polytopes which have the smallest possible vertex (or facet) number, namely $n + 1$. They are obtained as convex hulls of $n + 1$ affinely independent points, and all their k-dimensional faces are k-simplices, $0 \leq k \leq n - 1$. For $k = 0$ and $k = n - 1$ we call these faces *vertices* and *facets*, respectively. They are n-pyramids over each of their facets (and characterized by this property), and the two- and three-dimensional subcases are called *triangles* and *tetrahedra*, respectively. They have many applications, e.g. in view of triangulations, simplicial complexes, as simple and simplicial polytopes, and as extremal polytopes or even extremal convex bodies with respect to many problems in discrete and convex geometry as well as in polytope theory.

We survey here selected results on simplices which are geometrically intriguing. Some of these results are not directly connected with the content goals of this book, but they are hopefully a welcome completion of the high-dimensional geometry discussed in this chapter.

Books, Book Chapters and Surveys

A comprehensive *survey of results on simplices* is given in the recent book [53], where the following subsections contain the most important parts: 2.9.1.1 on simplices in general, 2.9.1.2 on themes like vertex embeddings or triangulations of simplices or by simplices in cubes, and 4.7.3 on volumes (of sections) of simplices. And there are many further survey-like references where simplices are comprehensively studied or applied. E.g., the following books contain parts dedicated to the geometry of tetrahedra or simplices, or concentrate on problems from fields like elementary geometry, geometric inequalities, non-Euclidean geometries, or convexity (results from Banach-space theory we ignore here): Simon [733], Roché and Comberousse [672], Court [180], Couderc and Ballicioni [178], Thébault [774], Satyanarayana [694], Baston [67], Benson [72], Grünbaum [340], Boltyanski and Yaglom [101], Kelly and Weiss [446], Lay [493], Mitrinović et al. [578], Part I and Part II of Berger's monograph [75], and Matoušek [560]. In several of these references one has (due to the variety of properties treated) to deal with the index of

terms, in others partial chapters on simplices are contained. Clearly, this list is not complete, there are too many further publications and results.

An excellent reference on results and open problems about simplices as extremal bodies in view of *convex geometry* are Schneider's educational talks [700]. The author treats results and problems mainly in the light of themes like *measures of symmetry* with applications, *stability results*, and various *affine inequalities*. Some discussed applications show relations to *stochastic geometry*, and the affine inequalities there refer to *difference bodies, polars of projection bodies, the Blaschke-Santaló inequality, the L^p-Busemann-Petty centroid inequality* and the *Brascamp-Lieb inequality and its reverse*. A survey on special types of convex bodies, in its second section also treating simplices in convex geometry, is the handbook article [391] of Heil and Martini.

An n-simplex whose $n+1$ altitudes have a point O in common is said to be *orthocentric*, O being its *orthocenter*. The introductory part of Edmonds et al. [237] contains a comprehensive survey on the rich geometry of orthocentric simplices (studied by many mathematicians). Based on this overview and also the (partially new) results obtained in [237], the following observation is made: many theorems from planar triangle geometry have natural extensions/analogues in higher dimensions if the considered simplices are orthocentric! For general simplices many theorems do not have such analogues; see the exposition [363], where this viewpoint is discussed.

We should emphasize the monograph [266] of Fiedler, which nicely combines *simplex geometry* with *matrix theory* and *graph theory*. Many scattered results on n-simplices are usefully collected, and used methods and notions are Gram matrices, methods for solving large systems of linear equations, eigenvalue problems, notions from simplex geometry (e.g., circumcenters, dihedral angles etc.), and tools like signed graphs. Special types of simplices (e.g., orthocentric ones) are presented, and in addition the author shows that his methods can also be used for studying other geometric objects like finite point sets, simplicial cones, spherical cones, and also interesting types of degenerate simplices. The author completes this book by the later paper [267], where with similar methods various problems on *pairs of simplices* are solved.

Selected Properties of General Simplices

Any *dihedral angle* of an n-simplex is determined by the other $(n^2+n-2)/2$ dihedral angles (for the analogous problem regarding edge-lengths, also in non-Euclidean geometries, see [207]). Maehara and Martini [533] show that for all dimensions $n \geq 2$ infinitely many similarity classes of n-simplices exist all whose dihedral angles are rational multiples of π, where Hill simplices are essential in this structure. (Recall that a Hill simplex (see, e.g., [354] or [526]) is defined as follows: Let $x_1, \ldots, x_n$ be vectors with the same norm such that the angle between any two of them is γ. An n-dimensional *Hill simplex of angle* γ is the convex hull of the origin together with $x_1, x_1+x_2, \ldots, x_1+x_2+\cdots+x_n$. Further on, recall

that a finite set of real numbers is π-*independent* if the numbers and π are linearly independent over the rational field.) Based on the latter notion, it is verified in [533] that for $n \geq 3$ there are uncountably many, mutually non-similar n-simplices whose dihedral angles and π are linearly independent in that sense, even forming a dense set (in the Hausdorff sense) within the family of all n-simplices.

Napoleon's theorem (see the survey [549]) says that if one erects (to the exterior) equilateral triangles on the sides of an arbitrary triangle, then their three circumcenters form the vertex set of an *equilateral triangle*. A direct extension to n-space is clearly not possible, but if a triangle abc has its Fermat-Torricelli point f (here meaning to have minimal distance sum to the vertices) in its interior, then (an analogue of) Napoleon's theorem can be formulated without these erected equilateral triangles and extendable to higher dimensions. Namely, the vertices a', b', c' of the finally obtained regular triangle are also the intersection points of the lines fa, fb, fc with the circumcircles of the triangles fbc, fac, fab, or the points on the three rays from a, b, c to f for which the distance equalities $|aa'| = |bb'| = |cc'| = |fa| + |fb| + |fc|$ hold, respectively. The n-dimensional version ($n \geq 3$) of the first viewpoint is obtained by Martini and Weissbach [554], that of the second one by Hajja et al. [364].

Let K be an n-dimensional convex body of mean width $M(K)$, and $M(S_K)$ be the *minimal mean width* of all simplices circumscribed about K. Let S denote a regular simplex circumscribed about the unit n-ball. Böröczky and Schneider [110] prove the inequality $2M(S_K) \leq M(K)M(S)$, where equality holds iff K is a ball. The dual problem (to find, among all simplices in a given ball, those of maximal mean width) remains open.

Assuming constant density, Krantz et al. [477] consider *centroids of different k-skeletons of simplices* in Euclidean n-space. They start with the well-known fact that the centroids of the solid simplex and of its vertex set coincide. Then it is shown that the centroids of the other values for k, namely $1 \leq k \leq n-1$, are generically different. The authors present types of simplices which, for all values of k, have this coincidence. E.g., for any $n = pq - 1$, where p and q are integers larger than 2, there exists a non-regular n-simplex for which the centroids of all k-skeletons, with $1 \leq k \leq n$, coincide. The existence of a non-regular 4-simplex all whose centroids coincide is also shown, and various further related results are obtained. Here is also the right place to cite the (relatively unknown) book [54] of Balk and Boltyanski, also dedicated to types of centroids.

The intersections of lines from the vertices to the points where the incircle and excircles of a triangle touch the sides are called *Gergonne and Nagel centers*, respectively. There is no direct higher dimensional extension since the corresponding lines for n-simplices, $n \geq 3$, are not necessarily concurrent. However, Hajja [359] presents equivalent definitions that make such higher dimensional analogues possible. Corresponding nice results are then derived.

Based on earlier results (e.g., due to Butler and Schneider), Martini [545] derives some characterizations of simplices as special convex bodies in an elementary way. Let K be an n-dimensional convex body, and xy be an affine diameter of K (i.e., a chord of K having maximal length in its own direction u). Then K is said to be

an *oblique double cone with respect to u* if every 2-dimensional half-plane whose bounding line is the affine hull of x and y cuts out of K a (possibly degenerate) triangle. The following (and further) properties are equivalent to the fact that K is an n-simplex: (1) For each direction, K is an oblique double cone. (2) For every translation vector t creating nonempty intersection $K \cap (K + t)$, this intersection is either a point or a positive homothet of K. (3) Every *Steiner symmetral* of K has exactly two extreme points outside the corresponding symmetrization hyperplane. (4) The *projection body* and the *difference body* of K are polar to each other regarding the sphere (centered at the origin) of radius $\sqrt{n \cdot V(K)}$. The second characteristic property is essential in view of the notion of *Choquet simplices* (see the basic survey [751]), and V. Soltan [750] derives characterizations of homothetic simplices by intersections of the type $K_1 \cap (K_2 + t)$, where the convex bodies K_1, K_2 belong to at most countably many homothety equivalence classes.

Let again K be an n-dimensional convex body. A finite set X of exterior points of K is said to *see* the whole boundary of K if for any boundary point y there is some $x \in X$ such that K has empty intersection with the *open* segment xy. It is proved in [550] that for any exterior point x_1 of K there are n further exterior points $x_2, \ldots, x_{n+1}$ such that the point set $x_1, x_2, \ldots, x_{n+1}$ sees the whole boundary of K. Further on, it is shown that a convex body K is an n-simplex iff, for every exterior point, there is a second exterior point such that this two-points set sees the whole boundary of K.

An asymptotic estimate on the *volume ratio* $V(S)/V(K)$ for K an n-dimensional convex body and S a simplex containing K is obtained by Galicer et al. [289]. If both have the same centroid, then $(V(S)/V(K))^{1/n} \leq cn^{1/2}$; here $c > 0$ is an absolute constant. The dual situation (regarding simplices contained in K) is also discussed, and a reason for the coincidence of centroids is also given; it has to do with the *Mahler product* of the volumes of S and its polar.

Let S be a *minimal area n-simplex* circumscribed about an n-dimensional strictly convex body K, and h represents the homothety with ratio $1/(1-n)$ whose center is the centroid of a facet of S. Extending results for two and three dimensions, Miernowski et al. [572] confirm that the touching points of K with S are precisely the tangency points of spheres inscribed in images $h(S)$ of S.

Selected Results on Special Types of Simplices

Denoting by S_I the *convex hull of the incenters of the facets* of an n-simplex S, Ma et al. [515] show that the volume ratio of S and S_I satisfies $n^n \leq V(S)/V(S_I)$. Equality holds iff S is a *regular simplex*. They obtain analogous results for convex hulls of the touching points of the insphere, and for the convex hull of the circumcenters of the facets of S.

The following deep result is due to Barthe [65]: each convex body in Euclidean n-space has an affine image whose *mean width* is not larger than that of a regular simplex of the same volume.

Stability versions of Ball's reverse isoperimetric inequality and of the corresponding inequality for *isotropic measures* are obtained by K. J. Böröczky and Hug [109]. They consider affine invariant concepts of distance between convex bodies (based on the Banach-Mazur and on the volume difference distance) and obtain characterizations of regular simplices.

The following is due to Frankl and Maehara [276]: Any finite point set F of cardinality at least 5 has the property that if all triangles formed by triples from F have the same positive area, then F is the vertex set of a *regular simplex*. A result on *k-distance sets* constructed from such point sets is also given, namely in terms of different triangle areas. The characterization of regular simplices is extended by Martini [548] to spherical and hyperbolic space. McMullen [566] calls an n-simplex *k-equiareal* if all its k-faces have the same k-volume (thus, due to [276] any 2-equiareal n-simplex for $n \geq 4$ is regular). Various properties of k-equiareal simplices are proved in [566]. E.g., an n-simplex is $(n-1)$-equiareal iff *its incenter and centroid coincide*. Also, the author constructs $(n-2)$-equiareal n-simplices not being regular and gives examples of $(n-2)$-equiareal n-simplices which are not $(n-1)$-equiareal. One important question for $3 \leq k \leq n-2$ and $n \geq 5$ is whether an $(n-2)$-equiareal n-simplex is necessarily $(n-1)$-equiareal. Although all constructed examples have this property, this is not proved. There exist such non-regular simplices for $n = 5$ and $n \geq 7$, and one for $n = 11$ even having no symmetry. The author confirms also that if the important question above would have a positive answer, then every k-equiareal n-simplex would be *a regular simplex* for $1 \leq k \leq n-3$.

Martini and Wenzel [555] replace "equiareality" by *congruence*. They show that an n-simplex is *regular* if for some k with $1 \leq k \leq n-2$ all its k-faces are congruent, and that this does not hold for $k = n-1 > 2$. Thus, the simplices with congruent facets, called *equifacetal simplices*, remain interesting in this setting. Edmonds shows in [234] that the isometry group of every equifacetal simplex acts *vertex-transitively* (and facet-transitively). It follows that each isometry of an equifacetal simplex has only one fixed point: the centroid (i.e., the barycenter of the vertices). He conjectures that any simplex whose isometry group acts vertex-transitively is equifacetal. There are interesting further results and problems in [234]. And in [235] it is shown that the *set of centers* of a simplex is a subset of the fixed point set of its group of isometries. It is a single point iff this group acts transitively on the vertices, implying that an equifacetal simplex has a unique center. This yields the known *center conjecture*: If a simplex has a unique center, then it is equifacetal. For the dimensions at most six Edmonds proves a strong version of this conjecture. Associated with any equifacetal n-simplex is also a well-defined partition of n that counts the number of edges of each possible length incident at a given vertex. The related *partition problem* asks for a characterization of the partitions coming from equifacetal simplices. In [236] this problem is solved, and in this setting *extremal equifacetal simplices* are characterized.

An n-simplex S is called *replicating* if one can partition it into at least two simplices such that all simplex tiles are mutually congruent and similar to S. The notion of *Hill-simplex* is defined above and goes (for n dimensions) back to

Hadwiger [354]. Every Hill n-simplex is replicating, and hence all such simplices tile n-spce. Hertel (see [393] and [394]) obtained related results and conjectured that a tetrahedron is replicating iff it is a Hill tetrahedron. Maehara [525] constructs counterexamples to this conjecture and shows even that for any $n \geq 3$ there are replicating n-simplices which are not Hill simplices. In [526] he calls a simplex *reflection-symmetric* if it is invariant under some orientation reversing isometry. He characterizes all reflection-symmetric Hill simplices, and for this purpose he explicitly provides all isometries that preserve any given Hill simplex, finding also the orientation reversing cases.

It is well-known that a triangle is *regular* or *equilateral* if two of its classical centers (namely the *centroid, incenter, circumcenter*, and *orthocenter*) coincide. Edmonds et al. [238] aim to extend this concept to higher dimensions, asking for consequences in the facial structure of simplices if two or more centers coincide. They add also the *Fermat-Torricelli point*, having minimum distance sum to the vertices, and replace the orthocenter, in general not existing for $n \geq 3$, by the *Monge point*. It turns out that in 3-space one gets characterizations of isosceles tetrahedra, and that for $n \geq 4$ a more complicated machinery is necessary to get analogous results. If two of the centers coincide, one can geometrically describe consequences for their facial structure, but one cannot usually infer much about other centers. Nevertheless, the authors give a modern and unifying presentation, including many new and also known results in the plane and in 3-space. They continue these investigations in [237], getting much more for the subfamily of *orthocentric simplices* (i.e., simplices which really have an orthocenter). It is confirmed that orthocentric simplices, either with congruent faces, or with any pair from (centroid, circumcentre, incentre, orthocentre) coinciding, have to be *regular*. The used tools are mainly Gram matrices and a characterization of the barycentric coordinates of the orthocenter. Hajja [360] studies *degrees of regularity of simplices* by using coincidences of two or three centers. An n-simplex is called *biregular* if the set of fixed points of its motion group is a line (or, in degenerate cases, a point, yielding the subcase or regular simplices). Solving an old open problem at least for orthocentric simplices of arbitrary dimension, Edmonds et al. [239] prove that the *incenter* of an orthocentric simplex lies on its *Euler line* if and only if this simplex is biregular.

In [365] and [366] the following essential types of n-simplices are studied: *orthocentric simplices* (having concurrent altitudes), *circumscriptible simplices* (for which there exists an $(n-1)$-sphere that touches all edges internally), *isodynamic simplices* (for which the cevians that join the vertices to the incenters of the opposite facets are concurrent), and *isogonic simplices* (a higher-dimensional generalization of isogonic tetrahedra, for which the cevians that join the four vertices to the points where the insphere touches the opposite faces are concurrent). In the first paper [365] it is shown that the intersection between any two such families yields precisely the class of n-kites (these are simplices with n of their vertices forming a regular $(n-1)$-simplex whose vertices are equidistant from the remaining vertex), and in [366] several characterizations of these families are derived or reproved.

Bédaride and Rao [69] investigate the *billiard flow inside a regular n-simplex*, $n > 3$. It is shown that there are two types of *periodic trajectories*, the first of which has period $n+1$ and hits each facet once. The second one, with period $2n$, hits one of the facets n times, and thus any other facet once. In both cases, for the points where the trajectory hits the simplex boundary, the exact coordinates are determined. The authors also study the convex hull of the periodic orbit of length $n + 1$, and it is shown that the orbits obtained for $n = 3$ are stable with respect to perturbations of the respective tetrahedron.

Pompeiu's theorem says that the distances from the vertices of an equilateral triangle to any point in its plane yield the side lengths of some triangle. Al-Afifi et al. [14] generalize this to given regular n-simplices. The authors start with the classical Möbius-Pompeiu theorem and an extension to regular n-simplices obtained by Fiedler. Presenting their new generalization, they show that for a regular n-simplex, $n \geq 2$, and a point x in its affine hull, the distances from x to the vertices of the given simplex can be taken as the volumes of the facets of some n-simplex precisely if x is not a vertex of the considered regular simplex. For this purpose, they prove a *higher-dimensional analogue of the triangle inequality*.

11.7.3 Notes on the Volume of (Unit) Balls

High Dimensional Phenomena

In Sect. 11.3 above we consider the *volume of a unit ball*. It is well-known that if $n \to \infty$, this quantity goes down to zero and, at the same time, concentrates more and more on the thin shell which is close to its spherical boundary. This can be verified in different ways (e.g., by methods from *functional analysis, probability theory, Euclidean geometry, convex geometry*, and even *combinatorics*). An interesting further and recent way is due to Li [505]. Namely, he uses *regularity theory of elliptic partial differential equations and calculus of variations*. And Hayes [387] (in one of the best mathematical writings of 2012) describes nicely the known behaviour of an n-cube C of edge-length 2 and an inscribed n-ball B of radius 1 if the dimension increases (see also the Mathematical Review MR2976645). The author shows the maximality of dimension 5 with respect to the volume of B (and of dimension 7 regarding its surface area). Since B remains inscribed, already for a dimension number like $n = 100$, say, the part of C inside B is close to zero, although this ball further on remains in touching position with each facet of C, and no larger ball is inside C. Thus, the ball somehow "disappears"!

Another nice representation of such phenomena can be found in Artin's classical monograph [41] and Section 13.1 of Matoušek's book [560], and Rabiei [652] presents related *mass concentration phenomena in balls*. There also exact formulas for the mass content of *ball caps* and upper bounds for related ratios of volumes are given, cf. also again Chapter 13 in [560]. In addition we refer here to Gipple [301], where a recursion formula for the volume of the ball is given, and in [746] different approaches, for n even and n odd, to the volume of the n-dimensional ball of radius $r > 0$ are shown. Clearly, there are various further such derivations in the literature.

Inequalities

It might be interesting to add some results on *inequalities* giving bounds on the volume of the unit ball in Euclidean space. Based on earlier results, Alzer [30] derives sharp upper and lower bounds on this volume, and quotients of it using the values for different dimensions are given, too. His tools are the *gamma function* and *its logarithmic derivative*. He continues in [31], presenting there also a related interesting double inequality. Further results in this direction are presented by Chen and Lin [160] and Yin [826], namely sharp upper and lower bounds on ratios of suitably combined volumes of balls of different dimensions. Anderson et al. [32] expose analytical inequalities involving special functions from classical analysis and geometric function theory. Among them the gamma function is discussed, with extensions and again its role regarding the volume of the unit ball.

On Ball Volumes (Mainly) in Three Dimensions

It is well-known that Archimedes [39] discovered the following result for three dimensions: *The volume of a ball is two thirds of the volume of the smallest right cylinder containing it, and the surface area of the ball is also two thirds of the total surface area of this cylinder*. Thus, the ratio "volume/surface area" of this cylinder equals the corresponding ratio of the considered ball. (As third convex body, sometimes a double cone suitably inscribed to the cylinder and, clearly, with volume a third of that of the cylinder, is added.) This nice result is discussed in the book sections [250, Section 1.5, Problem 8] of Eves and [103, Section 4.3] of Borceux. We refer also to the books of Stein [754, Chapter 10] and Aumann [47], the latter nicely collecting and presenting almost all famous contributions of Archimedes to mathematics on a good level for high school students and undergraduates.

A result somehow converse to that of Archimedes is due to Knothe [459], again holding in 3-space: *If the volume of each right cylinder circumscribed about a convex body K is one and a half times that of K, then K is a ball*. Firey [269] extends this slightly characterizing *bodies of constant width* (see the book [558] for this class of convex bodies): Let $C(u)$ be the cylinder whose generators have direction u and which circumscribes a convex body K (thus, the generators are supporting lines of K). Denote by $S(u)$ the lateral area of the part of $C(u)$ lying between the two supporting planes of K orthogonal to u. If $S(u)$ is constant over all directions, then K is a *body of constant width*. This yields an easy proof of Knothe's theorem. If K is the ball inscribed to $C(u)$, and $V(C(u))$, $V(K)$, $A(C(u))$ and $A(K)$ denote the respective volumes and areas, then the ratio $V(K)/V(C(u))$ equals $A(S)/SA(C(u))$. This is confirmed by Du Val and Saban [232] and also suitably generalized to n dimensions. Groemer [332] generalizes Knothe's result to n dimensions and derives interesting *stability versions*. These can be formulated, e.g. in dimension 3, as follows: if the lateral boundary of any cylinder has area close to the surface area of K, then K is close to a ball. Furthermore, the author gives

upper and lower bounds for the average of the second intrinsic volumes of all right cylinders circumscribed to K.

The authors of [215] study the analogues of the above ratios of Archimedes for all small *geodesic spheres of Riemannian manifolds*. Then they continue in proving that these ratios determine the geometry of any two-point homogeneous space inside the family of all Riemannian manifolds. Archimedes' balancing methods led them to the volumes of balls and cylindrical wedges. Apostol and Mnatsakanian [36] introduce new balancing methods and derive interesting relations involving volumes and surface areas of certain types of solid bodies. This helps also to extend Archimedes' classical results on spheres and cylinders to higher dimensions. Besides all this, new related results on centroids of hemispheres are obtained in [36]. The same authors continue in [37]. Namely, inspired by Archimedes' equality of ratios, they introduce classes of solids called globes and having the property that *the ratio of volume to surface area is invariant*.

The two papers [151] and [699] are slightly related to our presentation here. Replacing the circumscribed cylinders in our argumentation by suitably *circumscribed boxes*, their *mean volume* is considered. It is minimal when K is a ball (see Chakerian [151]), and Schneider [699] obtains a similar result for *averaged surface areas*.

Richter and Schicker [668] extend the interesting notion of *ball number*, combining the volume and the surface area of a ball of radius $r > 0$ with extensions to *normed spaces*. They consider the three-dimensional case with the *regular polyhedra as unit balls* (including also the non-symmetric tetrahedral gauge). In the introduction a survey on related results in this direction is presented.

Further Results

Some *monotonicity and concavity properties* of several functions, strongly related to the volume of the unit ball, are generalized by Guo and Qi [348], and properties of corresponding sequences are also concluded. Further related papers are [347] and [651]. Also Mortici [586] derives new bounds for ratios involving the volume of the unit ball. In addition, he shows related monotonicity properties and discusses problems around the *surface area of the unit ball*, too.

In a detailed manner, Dodd and Coll [216] describe hypersurfaces in n-dimensional Euclidean space satisfying higher dimensional analogues of the so-called *equal area zones property* of the sphere, saying that the surface area of a zone between two parallel hyperplanes depends only on the distance between these hyperplanes. These studied objects are called *equizonal n-ovaloids*, and they are hypersurfaces of revolution. Interesting further problems are posed and discussed in [216], e.g. regarding the volume of ovaloids. E.g., the authors prove that the ratio of this volume for radius 1 to that of the unit ball approaches π as $n \to \infty$. Further on, equizonal n-ovaloids are of class C^∞ for even dimension, and only of class $C^{(n-1)/2}$ for the odd case. This is improved in [175], where it is verified that, for even n, equizonal n-ovaloids are analytic, and that for odd dimensions they are everywhere analytic except at their two poles (being there of class C^{n-1}).

Chapter 12
The Cayley-Menger Determinant



Poetry is the art of giving different names to the same thing; Mathematics is the art of giving the same name to different things.

— Henri Poincaré

In this chapter, the Cayley-Menger determinant for a finite semi-metric space is introduced. Using Cayley-Menger determinants, Soddy's formula for mutually tangent spheres and Ptolemy's classical theorem are generalized to n dimensions.

12.1 The Cayley-Menger Matrix

A *semi-metric space* $\mathcal{X}$ is a set in which the *distance* $d(x, y)$ is defined for every $x, y \in \mathcal{X}$ so that

$$d(x, y) = d(y, x) \geq 0 \text{ and } d(x, y) = 0 \Leftrightarrow x = y$$

hold. If, furthermore, the distance $d(x, y)$ satisfies the triangle inequality, then $\mathcal{X}$ is a metric space. Every subset of a semi-metric space is a semi-metric space by the same distance. Every subset of $\mathbb{R}^n$ is a (semi-)metric space by the distance $d(x, y) = \|x - y\|$. Two semi- metric spaces $\mathcal{X}, \mathcal{Y}$ are said to be *isometric* if there is a bijection $f : \mathcal{X} \to \mathcal{Y}$ such that $d(x, y) = d(f(x), f(y))$ holds for every $x, y \in \mathcal{X}$. If $\mathcal{X}$ is isometric to a subset of $\mathbb{R}^n$, then $\mathcal{X}$ is said to be *isometrically embeddable* in $\mathbb{R}^n$. For a semi-metric space $\mathcal{X} = \{x_1, x_2, \ldots, x_k\}$ with k elements, the $k \times k$-matrix whose (i, j)-entry is the squared distance $d(x_i, x_j)^2$ for $i, j = 1, 2, \ldots k$, is called

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the *squared-distance matrix* of $\mathcal{X}$, and denoted by $SD(\mathcal{X})$. And the $(k+1)\times(k+1)$-matrix

$$\Gamma(\mathcal{X}) = \left(\begin{array}{ccc|c} & SD(\mathcal{X}) & & \begin{matrix}1\\ \vdots \\ 1\end{matrix} \\ \hline 1 & \cdots & 1 & 0 \end{array}\right)$$

is called the *Cayley-Menger matrix* of $\mathcal{X}$. Cayley-Menger matrices are real symmetric matrices. For example, for the semi-metric space $\mathcal{X} = \{x_1, x_2, x_3, x_4\}$ with distances $d_{ij} = d(x_i, x_j)$, $1 \le i, j \le 4$, we have

$$SD(\mathcal{X}) = \begin{pmatrix} 0 & d_{12}^2 & d_{13}^2 & d_{14}^2 \\ d_{21}^2 & 0 & d_{23}^2 & d_{24}^2 \\ d_{31}^2 & d_{32}^2 & 0 & d_{34}^2 \\ d_{41}^2 & d_{42}^2 & d_{43}^2 & 0 \end{pmatrix}, \ \Gamma(\mathcal{X}) = \begin{pmatrix} 0 & d_{12}^2 & d_{13}^2 & d_{14}^2 & 1 \\ d_{21}^2 & 0 & d_{23}^2 & d_{24}^2 & 1 \\ d_{31}^2 & d_{32}^2 & 0 & d_{34}^2 & 1 \\ d_{41}^2 & d_{42}^2 & d_{43}^2 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{pmatrix}.$$

The determinant $\det\Gamma(\mathcal{X})$ is called the corresponding *Cayley-Menger determinant*. If we permute the order of the points $x_1, \ldots, x_k$, then the order of the rows and columns of $\Gamma(\mathcal{X})$ are permuted, but there arises no essential difference.

Lemma 12.1 *Let $\mathcal{X} = \{x_1, x_2, x_3\}$ be a semi-metric space with distances $a = d(x_1, x_2)$, $b = d(x_1, x_3)$, $c = d(x_2, x_3)$. If $\det\Gamma(\mathcal{X}) = 0$, then $\mathcal{X}$ is isometrically embeddable in $\mathbb{R}^1$.*

Proof The determinant of $\Gamma(\mathcal{X})$ is calculated as follows.

$$\begin{vmatrix} 0 & a^2 & b^2 & 1 \\ a^2 & 0 & c^2 & 1 \\ b^2 & c^2 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{vmatrix} = \begin{vmatrix} 0 & a^2 & b^2 & 1 \\ a^2 & -a^2 & c^2-b^2 & 0 \\ b^2 & c^2-a^2 & -b^2 & 0 \\ 1 & 1 & 1 & 0 \end{vmatrix} = -\begin{vmatrix} a^2 & -a^2 & c^2-b^2 \\ b^2 & c^2-a^2 & -b^2 \\ 1 & 1 & 1 \end{vmatrix}$$

$$= -\begin{vmatrix} a^2 & -2a^2 & c^2-a^2-b^2 \\ b^2 & c^2-a^2-b^2 & -2b^2 \\ 1 & 0 & 0 \end{vmatrix} = -4a^2b^2 + (c^2-a^2-b^2)^2$$

$$= -(2ab - c^2 + a^2 + b^2)(2ab + c^2 - a^2 - b^2)$$

$$= -((a+b)^2 - c^2)(c^2 - (a-b)^2)$$

$$= -(a+b+c)(a+b-c)(c-a+b)(c+a-b).$$

Hence, $\det\Gamma(\mathcal{X}) = 0$ implies that one of $a+b-c, c-a+b, c+a-b$ is 0. In the case $a+b-c=0$, the correspondence $x_1 \to a$, $x_2 \to 0$, $x_3 \to a+b$ gives an isometric embedding of $\mathcal{X}$ into $\mathbb{R}^1$. Other cases are treated similarly. □

Remark 12.1 Cayley-Menger determinants can be used to make it clear whether a given finite semi-mertic space can be isometrically embedded in a Euclidean space. Chapter 4 of L. M. Blumenthal's book [96] is devoted to the study of relevant related theory.

Theorem 12.1 *For any finite subset $\mathcal{X} \subset \mathbb{R}^n$, the rank of the Cayley-Menger matrix $\Gamma(\mathcal{X})$ is at most $n+2$. Hence, if $\mathcal{X}$ contains more than $n+1$ points, then* $\det \Gamma(\mathcal{X}) = 0$.

Proof Let $\mathcal{X} = \{x_1, x_2, \ldots, x_k\}$. Let X be the $(k-1) \times n$-matrix whose $(i-1)$th row is $\overrightarrow{x_1x_i}$ for $i = 2, 3, \ldots, k$:

$$X = \begin{pmatrix} \overrightarrow{x_1x_2} \\ \overrightarrow{x_1x_3} \\ \vdots \\ \overrightarrow{x_1x_k} \end{pmatrix}.$$

Since each $\overrightarrow{x_1x_i}$ is an n-dimensional vector, it is obvious that the rank of X is at most n. Therefore, the $(k-1) \times (k-1)$-square matrix XX^t (where X^t is the transpose of X) has rank at most n. The (i, j)-entry of XX^t is $\overrightarrow{x_1x_{i+1}} \cdot \overrightarrow{x_1x_{j+1}}$ (inner product of two vectors). Since

$$\begin{aligned} d_{ij}^2 &= \overrightarrow{x_ix_j} \cdot \overrightarrow{x_ix_j} = (\overrightarrow{x_ix_1} + \overrightarrow{x_1x_j}) \cdot (\overrightarrow{x_ix_1} + \overrightarrow{x_1x_j}) \\ &= (-\overrightarrow{x_1x_i} + \overrightarrow{x_1x_j}) \cdot (-\overrightarrow{x_1x_i} + \overrightarrow{x_1x_j}) = d_{1i}^2 + d_{1j}^2 - 2(\overrightarrow{x_1x_i} \cdot \overrightarrow{x_1x_j}), \end{aligned}$$

we have $2(\overrightarrow{x_1x_i} \cdot \overrightarrow{x_1x_j}) = d_{1i}^2 + d_{1j}^2 - d_{ij}^2$. Put

$$M = \begin{pmatrix} 0 & 0 & \ldots & 0 & 1 \\ 0 & & & & 0 \\ \vdots & & 2(XX^t) & & \vdots \\ 0 & & & & 0 \\ 1 & 0 & \ldots & 0 & 0 \end{pmatrix}.$$

This matrix M is a square matrix of degree $k+1$, and since the rank of XX^t is at most n, the rank of M is at most $n+2$. By adding the first row to the ith row, $i = 2, 3, \ldots, k$, and then by adding the first column to the jth column for $j = 2, 3, \ldots, k$, M is changed to

$$M_1 = \begin{pmatrix} 0 & 0 & \ldots & 0 & 1 \\ 0 & & & & 1 \\ \vdots & & 2(XX^t) & & \vdots \\ 0 & & & & 1 \\ 1 & 1 & \ldots & 1 & 0 \end{pmatrix}.$$

Since the (i, j)-entry of $2(XX^t)$ (the $(i+1, j+1)$-entry of M_1) is $d_{i1}^2 + d_{1j}^2 - d_{ij}^2$, by subtracting d_{i1}^2 times the last row from the ith row of M_1 for $i = 2, \ldots, k$, and then by subtracting d_{j1}^2 times the last column from the jth column for $j = 2, \ldots, k$, M_1 becomes

$$M_2 = \begin{pmatrix} 0 & -d_{12}^2 & -d_{13}^2 & \cdots & -d_{1k}^2 & 1 \\ -d_{21}^2 & 0 & -d_{23}^2 & \cdots & -d_{2k}^2 & 1 \\ -d_{31}^2 & -d_{32}^2 & 0 & \cdots & -d_{3k}^2 & 1 \\ \cdots & \cdots & \cdots & \ddots & \cdots & \vdots \\ -d_{k1}^2 & -d_{k2}^2 & \cdots & -d_{k(k-1)}^2 & 0 & 1 \\ 1 & 1 & \cdots & 1 & 1 & 0 \end{pmatrix}.$$

Finally, by mutiplying all rows of M_2 with -1, and then the last row and the last column with -1, M_2 becomes $\Gamma(\mathcal{X})$. Since the rank is invariant under such changes, the rank of $\Gamma(\mathcal{X})$ is equal to (the rank of XX^t)+2. Hence the rank of $\Gamma(\mathcal{X})$ is at most $n + 2$. □

The matrix XX^t used in the above proof is called the *Gram matrix* of the $k-1$ vectors $\overrightarrow{x_1x_i}$, $i = 2, 3, \ldots, k$. The above proof shows that

$$\text{the rank of } \Gamma(\mathcal{X}) = (\text{the rank of } XX^t) + 2.$$

12.2 Odd Integral Distances

As an application of Theorem 12.1, let us show here the following theorem.

Theorem 12.2 ([317]) *There is no four-point-set in the plane in which the distances between the points are all odd integers*

Proof The proof works by contradiction. Suppose that there is a four-point-set $\mathcal{X} = \{x_0, x_1, x_2, x_3\}$ in the plane $\mathbb{R}^2$ in which all distances between points are odd integers. Its Cayley-Menger matrix $\Gamma(\mathcal{X})$ is of degree 5, and its rank is at most $n + 2 = 2 + 2 = 4$ by Theorem 12.1. Thus, $\det \Gamma(\mathcal{X}) = 0$. Hence, the determinant must be 0 when it is computed modulo 8. However, since $(2k+1)^2 = 4k^2+4k+1 = 4k(k+1)+1 \equiv 1 \pmod 8$, if we compute the determinant modulo 8,

$$\det \Gamma(\mathcal{X}) = \begin{vmatrix} 0&1&1&1&1 \\ 1&0&1&1&1 \\ 1&1&0&1&1 \\ 1&1&1&0&1 \\ 1&1&1&1&0 \end{vmatrix} = \begin{vmatrix} 0&1&1&1&1&0 \\ 1&0&1&1&1&0 \\ 1&1&0&1&1&0 \\ 1&1&1&0&1&0 \\ 1&1&1&1&0&0 \\ 0&0&0&0&0&1 \end{vmatrix}$$

$$= \begin{vmatrix} 0&1&1&1&1&0 \\ 1&0&1&1&1&0 \\ 1&1&0&1&1&0 \\ 1&1&1&0&1&0 \\ 1&1&1&1&0&0 \\ 1&1&1&1&1&1 \end{vmatrix} = \begin{vmatrix} -1&0&0&0&0&-1 \\ 0&-1&0&0&0&-1 \\ 0&0&-1&0&0&-1 \\ 0&0&0&-1&0&-1 \\ 0&0&0&0&-1&-1 \\ 1&1&1&1&1&1 \end{vmatrix}$$

$$= \begin{vmatrix} -1&0&0&0&0&-1 \\ 0&-1&0&0&0&-1 \\ 0&0&-1&0&0&-1 \\ 0&0&0&-1&0&-1 \\ 0&0&0&0&-1&-1 \\ 0&0&0&0&0&-4 \end{vmatrix} = 4 \neq 0,$$

yielding a contradiction. □

Similarly, it can be proved that there is no five-point-set in $\mathbb{R}^3$ with all integral distances being odd, see the paper [317] of R. L. Graham, B. L. Rothschild, and E. G. Straus. They proved that there is an $(n+2)$-point-set in $\mathbb{R}^n$ with all integral distances being odd if and only if $n \equiv 14 \pmod{16}$.

12.3 Orthogonal-Sphere Systems

Two spheres σ_1, σ_2 in $\mathbb{R}^n$ with centers p_1, p_2 and radii r_1, r_2 are called *orthogonally crossing* if

$$\|p_1 - p_2\|^2 = r_1^2 + r_2^2. \tag{12.1}$$

Thus, for any intersection point x of σ_1 and σ_2, $\angle p_1 x p_2 = 90°$ holds. A family of spheres is called an *orthogonal-sphere system* if every pair of spheres in the family is orthogonally crossing.

Example 12.1 In $\mathbb{R}^n$, n unit spheres with centers

$$(1, 0, 0, \ldots, 0), (0, 1, 0, \ldots, 0), \ldots, (0, 0, \ldots, 0, 1)$$

and a sphere with center $(-1, -1, \ldots, -1)$ and radius $\sqrt{n+2}$ form together an orthogonal-sphere system. This can be seen by checking (12.1) for every pair of spheres.

Theorem 12.3 *The maximum number of spheres in an orthogonal-sphere system in* $\mathbb{R}^n$ *is* $n + 1$.

Proof As seen in Example 12.1, there is an orthogonal-sphere system in $\mathbb{R}^n$ consisting of $n+1$ spheres.

Let $\{\sigma_1, \sigma_2, \ldots, \sigma_k\}$ be an orthogonal-sphere system of k spheres in $\mathbb{R}^n$. Let p_i, r_i be the center and the radius of σ_i for $i = 1, 2, \ldots, k$. Put $\mathcal{X} = \{p_1, p_2, \ldots, p_k\}$ and let d_{ij} be the distance between p_i and p_j. Then, by (12.1), $d_{ij}^2 = r_i^2 + r_j^2$. Hence, for $i, j \leq k$ the (i, j)-entry of the Cayley-Menger matrix $\Gamma(\mathcal{X})$ is $r_i^2 + r_j^2$. For $1 \leq i, j \leq k$, by subtracting from the ith row of $\Gamma(\mathcal{X})$, r_j^2 times the last row, and subtracting from the jth column r_j^2 times the last column, we have

$$\det \Gamma(\mathcal{X}) = \begin{vmatrix} -2r_1^2 & 0 & \ldots & 0 & 1 \\ 0 & -2r_1^2 & \ldots & 0 & 1 \\ & \ldots & & \ldots & 1 \\ 0 & \ldots & \ldots & -2r_k^2 & 1 \\ 1 & 1 & \ldots & 1 & 0 \end{vmatrix}$$

$$= \begin{vmatrix} -2r_1^2 & 0 & \ldots & 0 & 1 \\ 0 & -2r_1^2 & \ldots & 0 & 1 \\ & \ldots & & \ldots & 1 \\ 0 & \ldots & \ldots & -2r_k^2 & 1 \\ 0 & 0 & \ldots & 0 & \sum \frac{1}{2r_i^2} \end{vmatrix} \neq 0.$$

Since the rank of $\Gamma(\mathcal{X})$ is at most $n+2$ by Theorem 12.1, we must have $k+1 \leq n+2$. Therefore, $k \leq n+1$. □

12.4 Tangent-Sphere Systems

Two spheres in $\mathbb{R}^n$ (two circles for $n = 2$) are *tangent* if they share exactly one point, and this common point is the contact point of the two spheres. If two spheres have the radii r_1, r_2, then the two spheres are tangent if and only if

$$\text{(the distance between the centers of the sphers)} = |r_1 \pm r_2|,$$

where the signs $+, -$ correspond to “externally tangent" or “internally tangent", respectively.

A family of spheres in $\mathbb{R}^n$ is called a *tangent-sphere system* if the spheres are mutually tangent, and no three spheres are tangent at the same point. Thus, in a tangent-sphere system in $\mathbb{R}^n$, if a sphere σ encloses another sphere (that is, the ball bounded by σ contains another sphere), then σ encloses all other spheres.

Example 12.2 In $\mathbb{R}^n$, $n+1$ unit spheres centered at the vertices of an n-dimensional regular simplex with edge-lengths 2 are mutually tangent. And there is a sphere of suitable radius, centered at the center (= centroid) of the regular simplex that is tangent to all $n+1$ unit spheres. Thus, there are mutually tangent $n+2$ spheres in $\mathbb{R}^n$ that form a tangent-sphere system.

Let $\{\sigma_1, \sigma_2, \ldots, \sigma_k\}$ be a tangent-sphere system in $\mathbb{R}^n$. Let us define the *signed radius* ρ_i of each σ_i as follows:

$$\rho_i = \begin{cases} -(\text{the radius of } \sigma_i) & (\text{if } \sigma_i \text{ encloses another } \sigma_j), \\ +(\text{the radius of } \sigma_i) & (\text{otherwise}). \end{cases}$$

By using the signed radius, the distance between the centers of tangent spheres σ_i and σ_j can be always represented by $|\rho_i + \rho_j|$, no matter whether they are internally tangent or externally tangent.

Descartes' circle theorem and Soddy's formula in Chap. 2 are generalized in the following way.

Theorem 12.4 *The maximum number of spheres in a tangent-sphere system in $\mathbb{R}^n$, $n \geq 2$, is equal to $n+2$. If $\{\sigma_1, \sigma_2, \ldots, \sigma_{n+2}\}$ is a tangent-sphere-ststem in $\mathbb{R}^n$, then*

$$\Big(\sum_{i=1}^{n+2} \frac{1}{\rho_i}\Big)^2 = n \sum_{i=1}^{n+2} \Big(\frac{1}{\rho_i}\Big)^2 \tag{12.2}$$

holds, where ρ_i is the signed radius of σ_i.

Preceding the proof of Theorem 12.4, we present an example to compute a determinant.

Example 12.3 To compute the determinant

$$D = \begin{vmatrix} -1 & 1 & 1 & \ldots & 1 & a_1 \\ 1 & -1 & 1 & \ldots & 1 & a_2 \\ & \ldots & \ddots & & \ldots & \\ & \ldots & & \ddots & \ldots & \\ 1 & 1 & 1 & \ldots & -1 & a_k \\ a_1 & a_2 & a_3 & \ldots & a_k & 0 \end{vmatrix},$$

we use a method to increase the degree of the matrix. Namely,

$$D = \begin{vmatrix} -1 & 1 & 1 & \dots & 1 & a_1 & 1 \\ 1 & -1 & 1 & \dots & 1 & a_2 & 1 \\ & \dots & \ddots & & \dots & & 1 \\ & \dots & & \ddots & \dots & & \\ 1 & 1 & 1 & \dots & -1 & a_k & 1 \\ a_1 & a_2 & a_3 & \dots & a_k & 0 & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 & 1 \end{vmatrix} = \begin{vmatrix} -2 & 0 & 0 & \dots & 0 & a_1 & 1 \\ 0 & -2 & 0 & \dots & 0 & a_2 & 1 \\ & \dots & \ddots & & \dots & & 1 \\ & \dots & & \ddots & \dots & & \\ 0 & 0 & 0 & \dots & -2 & a_k & 1 \\ a_1 & a_2 & a_3 & \dots & a_k & 0 & 0 \\ -1 & -1 & -1 & \dots & -1 & 0 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} -2 & 0 & 0 & \dots & 0 & a_1 & 1 \\ 0 & -2 & 0 & \dots & 0 & a_2 & 1 \\ & \dots & \ddots & & \dots & & 1 \\ & \dots & & \ddots & \dots & & \\ 0 & 0 & 0 & \dots & -2 & a_k & 1 \\ 0 & 0 & 0 & \dots & 0 & \sum(a_i^2/2) & \sum(a_i/2) \\ 0 & 0 & 0 & \dots & 0 & \sum(-a_i/2) & 1-k/2 \end{vmatrix}$$

$$= (-2)^k \begin{vmatrix} \frac{1}{2}\sum a_i^2 & \frac{1}{2}\sum a_i \\ -\frac{1}{2}\sum a_i & \frac{1}{2}(2-k) \end{vmatrix} = \frac{(-2)^k}{4}\Big(\big(\sum_{i=1}^{k} a_i\big)^2 - (k-2)\sum_{i=1}^{k} a_i^2\Big).$$

Remark 12.2 Similarly to the above computation, we can see that the determinant of the square matrix of degree k whose entries in the main diagonal are all -1, and whose remaining entries are all 1, is equal to

$$(-2)^k(1-k/2),$$

and hence, if $k \geq 3$, then such a matrix is non-singular.

Proof of Theorem 12.4 We already showed that there is a tangent-sphere system of $n+2$ spheres in $\mathbb{R}^n$. Now, let $\{\sigma_1, \sigma_2, \dots, \sigma_k\}$ be a tangent-sphere system in $\mathbb{R}^n$. Let $\rho_1, \rho_2, \dots, \rho_k$ be the signed radii of these spheres, and $\mathcal{X} = \{p_1, p_2, \dots, p_k\}$ be the set of their centers. Let us compute the Cayley-Menger determinant of $\mathcal{X}$. Let d_{ij} denote the distance between p_i and p_j. Then $d_{ij} = |\rho_i + \rho_j|$, and hence $d_{ij}^2 = \rho_i^2 + \rho_j^2 + 2\rho_i\rho_j$. For $i, j = 1, 2, \dots, k+1$, by subtracting ρ_i^2 times the

last row from the ith row, and by subtracting ρ_j^2 times the last column from the jth column, we have

$$\det\Gamma(\mathcal{X}) = \begin{vmatrix} -2\rho_1^2 & 2\rho_1\rho_2 & \dots & \dots & 2\rho_1\rho_k & 1 \\ 2\rho_2\rho_1 & -2\rho_2^2 & \dots & \dots & 2\rho_2\rho_k & 1 \\ \dots & \dots & \ddots & \dots & \dots & \\ \dots & \dots & \dots & \ddots & \dots & \\ 2\rho_k\rho_1 & 2\rho_k\rho_2 & \dots & 2\rho_k\rho_{k-1} & -2\rho_k^2 & 1 \\ 1 & 1 & \dots & \dots & 1 & 0 \end{vmatrix}$$

$$= (\rho_1\rho_2\dots\rho_k)^2 \begin{vmatrix} -2 & 2 & \dots\dots & 2 & 1/\rho_1 \\ 2 & -2 & \dots\dots & 2 & 1/\rho_2 \\ \dots & \dots & \ddots\dots & \dots & \\ \dots & \dots & \dots\ddots & \dots & \\ 2 & 2 & \dots\ 2 & -2 & 1/\rho_k \\ 1/\rho_1 & 1/\rho_2 & \dots\dots & 1/\rho_k & 0 \end{vmatrix}$$

$$= 2^{k-1}(\rho_1\rho_2\dots\rho_k)^2 \begin{vmatrix} -1 & 1 & \dots\dots & 1 & 1/\rho_1 \\ 1 & -1 & \dots\dots & 1 & 1/\rho_2 \\ \dots & \dots & \ddots\dots & \dots & \\ \dots & \dots & \dots\ddots & \dots & \\ 1 & 1 & \dots\ 1 & -1 & 1/\rho_k \\ 1/\rho_1 & 1/\rho_2 & \dots\dots & 1/\rho_k & 0 \end{vmatrix}.$$

In this determinant, the $(k \times k)$-square matrix obtained from it by removing the last row and the last column (in which the entries in the main diagonal are all -1 and the remaining entries are all 1) is nonsingular for $k \geq 3$ by Remark 12.2. Hence the rank of $\Gamma(\mathcal{X})$ is at least k. Since the rank of $\Gamma(\mathcal{X})$ is at most $n+2$ by Theorem 12.1, we can deduce that $k \leq n+2$. Thus, the number of mutually tangent spheres with distinct contact points is at most $n+2$.

The last determinant can be computed as in Example 12.3:

$$\det\Gamma(\mathcal{X}) = 2^{k-1}(\rho_1\rho_2\dots\rho_k)^2 \frac{(-2)^k}{4}\Big(\big(\sum_{i=1}^{k}\frac{1}{\rho_i}\big)^2 - (k-2)\sum_{i=1}^{k}\frac{1}{\rho_i^2}\Big).$$

Therefore, if $k = n + 2$ (that is, $\{\sigma_1, \sigma_2, \ldots, \sigma_{n+2}\}$ is a family of mutually tangent spheres with distinct contact points), then $\Gamma(\mathcal{X})$ is a square matrix of degree $n + 3$, and hence $\det \Gamma(\mathcal{X}) = 0$ by Theorem 12.1. Therefore,

$$\Big(\sum_{i=1}^{n+2} \frac{1}{\rho_i}\Big)^2 - n \sum_{i=1}^{n+2} \frac{1}{\rho_i^2} = 0$$

holds. □

12.5 A Generalization of Ptolemy's Theorem

The inversion of $\mathbb{R}^n$ with respect to a sphere with center p and radius r is the transformation of $\mathbb{R}^n \setminus \{p\}$ that sends $x \in \mathbb{R}^n \setminus \{p\}$ to a point x' on the ray $\overrightarrow{px}$ such that $|px| \cdot |px'| = r^2$. Similarly to Theorem 1.8 on inversions of $\mathbb{R}^3$, it can be easily proved that an inversion of $\mathbb{R}^n$ maps a sphere to a sphere or a hyperplane. (Instead of "hypersphere" we use also here the word "sphere".) More precisely,

- a sphere that does not pass through the center p of the inversion is transformed to a sphere that does not pass through p.
- a sphere that passes through p is transformed to a hyperplane that does not pass through p and *vice versa*, and
- a hyperplane that passes through p is transformed to itself.

Lemma 12.2

$$\begin{vmatrix} 0 & a^2 & b^2 & c^2 \\ a^2 & 0 & d^2 & e^2 \\ b^2 & d^2 & 0 & f^2 \\ c^2 & e^2 & f^2 & 0 \end{vmatrix} = (af + be + cd)(af + be - cd)(af - be + cd)(af - be - ce).$$

Proof Expanding the left-hand side determinant along the first column, we get $(af)^4 + (be)^4 + (cd)^4 - 2(afbe)^2 - 2(afcd)^2 - 2(becd)^2$. On the other hand, the right-hand side can be expanded in the following way:

$$\begin{aligned}
&(af + be + cd)(af + be - cd)(af - be + cd)(-af + be + cd) \\
&= [(af + be)^2 - (cd)^2][(af - be)^2 - (cd)^2] \\
&= [(af)^2 + (be)^2 - (cd)^2 + 2(afbe)][(af)^2 + (be)^2 - (cd)^2 - 2(afbe)] \\
&= [(af)^2 + (be)^2 - (cd)^2]^2 - 4(afbe)^2 \\
&= (af)^4 + (be)^4 + (cd)^4 - 2(afbe)^2 - 2(afcd)^2 - 2(becd)^2.
\end{aligned}$$

□

Theorem 12.5 *An $(n+2)$-point-set $\mathcal{X}$ in $\mathbb{R}^n$ lies on a sphere or a hyperplane in $\mathbb{R}^n$ if and only if* $\det SD(\mathcal{X}) = 0$.

Remark 12.3 For a 4-point-set $\mathcal{X} = \{x_1, x_2, x_3, x_4\}$ in $\mathbb{R}^2$, the determinant $\det SD(\mathcal{X})$ of the 4×4-matrix $SD(\mathcal{X})$ can be factorized by Lemma 12.2, as

$$\begin{aligned}&(d_{12}d_{34} + d_{13}d_{24} + d_{23}d_{14})(d_{12}d_{34} - d_{13}d_{24} + d_{23}d_{14})\\ &\times (d_{12}d_{34} + d_{13}d_{24} - d_{23}d_{14})(d_{12}d_{34} - d_{13}d_{24} - d_{23}d_{14}),\end{aligned}$$

where $d_{ij} = d(x_i, x_j)$. Hence, $SD(\mathcal{X}) = 0$ if and only if (just) one of

$$\begin{aligned}d_{12}d_{34} - d_{13}d_{24} + d_{23}d_{14},\\ d_{12}d_{34} + d_{13}d_{24} - d_{23}d_{14},\\ -d_{12}d_{34} + d_{13}d_{24} + d_{23}d_{14}\end{aligned}$$

is 0. Thus, the case $n = 2$ of Theorem 12.5 implies Ptolemy's classical theorem.

Proof of Theorem 12.5 Let $\mathcal{X} = \{x_1, x_2, \ldots, x_{n+1}, x_{n+2}\}$, and denote x_{n+2} by z. Let φ be the inversion of $\mathbb{R}^n$ with respect to the unit sphere centered at z. Put $y_i = \varphi(x_i)$, $i = 1, 2, \ldots, n+1$, and put $\mathcal{Y} = \{y_1, y_2, \ldots, y_{n+1}\}$. Then $\|y_i - z\| = 1/\|x_i - z\|$ for $i = 1, 2, \ldots, n+1$. Since $\|y_i - z\|\|x_i - z\| = 1 = \|y_j - z\|\|x_j - z\|$, we have

$$\|y_j - z\| : \|y_i - z\| = \|x_i - z\| : \|x_j - z\|,$$

and hence $\triangle z x_i x_j$ is similar to $\triangle z y_j y_i$. Therefore,

$$\|y_i - y_j\| = \frac{\|x_j - x_i\|\|y_i - z\|}{\|x_j - z\|} = \frac{\|x_i - x_j\|}{\|x_i - z\|\|x_j - z\|}.$$

Then, by multiplying the ith row of the Cayley-Menger matrix $\Gamma(\mathcal{Y})$, for $i = 1, 2, \ldots, n+1$, with $\|x_i - z\|^2$ and the jth column of the resulting matrix, for $j = 1, 2, \ldots, n+1$, with $\|x_j - z\|^2$, we get the matrix $SD(\mathcal{X})$. Therefore

$$\left(\prod_{i=1}^{n+1} \|x_i - x_{n+2}\|^4\right) \cdot \det \Gamma(\mathcal{Y}) = \det SD(\mathcal{X}). \tag{12.3}$$

Now, $x_1, x_2, \ldots, x_{n+2}$ lie on a sphere or a hyperplane

$$\begin{aligned}&\Leftrightarrow y_1, y_2, \ldots, y_{n+1} \text{ lie on a hyperplane}\\ &\Leftrightarrow \det \Gamma(\mathcal{Y}) = 0 \text{ (by Theorem 12.1)}\\ &\Leftrightarrow \det SD(\mathcal{X}) = 0 \text{ (by (12.3)).}\end{aligned}$$

□

12.6 Exercises

12.1 Confirm that in $\mathbb{R}^n$, n unit spheres with centers

$$(1, 0, 0, \ldots, 0), (0, 1, 0, \ldots, 0), \ldots, (0, \ldots, 0, 1)$$

and the sphere with center $(-1, -1, \ldots, -1)$ and radius $\sqrt{n+2}$ form together an orthogonal-sphere system.

12.2* Compute the determinant of the square matrix of degree k whose entries in the main diagonal are all -1, and whose remaining entries are all 1.

12.3 Prove that there is no five-point set in $\mathbb{R}^3$ with all integral distances being odd.

12.4 Confirm that in $\mathbb{R}^n$, $n+1$ spheres of radius $1/\sqrt{2}$ with centers

$$(1, 0, \ldots, 0), (0, 1, 0, \ldots, 0), \ldots, (0, \ldots, 0, 1), (a, a, \ldots, a),$$

where $a = (1 - \sqrt{1+n})/n$, and the sphere centered at the barycenter of these $n+1$ points having radius

$$\sqrt{\frac{n}{n+1}} - \frac{1}{\sqrt{2}}$$

form a tangent-sphere system.

12.5 In $\mathbb{R}^4$, five unit spheres and a sphere of unknown radius r form a tangent-sphere system. Find r.

12.6* Let $\boldsymbol{x}_i$, $i = 1, 2, 3, 4$ be four points in the plane $\mathbb{R}^2$ and

$$d(\boldsymbol{x}_1, \boldsymbol{x}_2) = 1, d(\boldsymbol{x}_1, \boldsymbol{x}_3) = 3, d(\boldsymbol{x}_1, \boldsymbol{x}_4) = 2, d(\boldsymbol{x}_2, \boldsymbol{x}_3) = 3,$$
$$d(\boldsymbol{x}_3, \boldsymbol{x}_4) = 3.$$

Find the distance $d(\boldsymbol{x}_2, \boldsymbol{x}_4)$.

12.7 Notes

In these notes, we bring together knowledge only about Cayley-Menger determinants and related topics. To the other subjects we could not locate too many references, except for Ptolemy's famous theorem. But this is discussed in the notes to the next chapter, since Casey's theorem directly generalizes it.

12.7.1 Notes on Cayley-Menger Determinants

As helpful tools and for their own sake, Cayley-Menger matrices and Cayley-Menger determinants are introduced in Sect. 12.1.

Assignment and Some History

Volume computation is an important subfield of pure and applied mathematics. Just referring only to *geometric aspects and applications* and strongly related notions (like, e.g., that of *mixed volumes* in convexity) one gets already a large and interesting area. A comprehensive variety of approaches to volumes (especially from the viewpoint of *computational convexity*, where notions from convexity are studied in normed vector spaces of finite, but generally *non-restricted dimension*) is collected in the important exposition [329] of Gritzmann and Klee. Basic notions and concepts studied there are representations of polytopes, triangulations, lattice-point enumeration, sweeping-plane methods, approximation theory, randomized approaches and also special types of convex bodies (like simplices or zonotopes). For *simplices*, Section 3.6 of [329] is particularly interesting. There the authors discuss volumes of simplices in terms of the *Gram matrix* of inner products of their vertices, in terms of the coefficients appearing in the affine functionals defining the facets, and (last, but not least) in terms of their (squared) edge-lengths, thus using Cayley-Menger determinants.

We start now with some historical remarks treating Cayley-Menger determinants. In the 1930s there was a strong tendency to look for axiomatization possibilities in several mathematical fields, also influenced by Hilbert's formalism. This led Menger [570] to present in 1928 a new axiomatic approach to metric spaces based on the concept of distance and on the congruence relation. Menger understood that an algebraic machinery, developed by Cayley, is very useful in this framework, and so the Cayley-Menger determinants came to be. With the help of this notion, Menger extended also *Heron's theorem* to the computation of the volume of n-simplices.

More precisely, the *Cayley-Menger determinant*, essential mainly in *linear algebra, geometry, and trigonometry*, is a determinant yielding the volume of an n-dimensional simplex S in terms of all *squared distances* between vertices of S (edge-lengths). Besides the proper sources [147] due to Cayley and [570, second investigation] of Menger, recommendable older references treating this notion are due to Dörrie [218, pp. 285-289] (discussing the *three-dimensional case*), Blumenthal [96, Part III] (presenting it as a cornerstone of *distance geometry*, also in spherical, elliptic, and hyperbolic spaces as well as in semi-metric ones; see also Menger's booklet [571]), Blumenthal and Gillam [97] (treating the geometrical interpretation of the signs of the cofactors of the elements of a Cayley-Menger determinant), and Sommerville [752, Chapter VIII] (for *volume calculations*). A good expository article on Cayley-Menger determinants from those days was written by Bottema [116].

For more recent expositions and overviews we refer to the book parts and surveys discussed below, and to the web page [174].

So Cayley-Menger determinants became mainly a useful tool for several partial fields of and concepts in mathematics, and in applications. Among these mathematical fields and concepts we might emphasize: volume computation, distance geometry, non-Euclidean geometries, embeddibility and rigidity questions, k-distance sets, (semi-)metric spaces, graph theory etc.; and essential application fields are e.g. robotics, network theory, physical chemistry and molecular structures.

Overviews in Books, and Surveys

Here we bring book- and survey-like publications treating different aspects of Cayley-Menger determinants in chronological order.

The essential monograph [96] of Blumenthal, which is one of the most fundamental references treating *distance geometry*, uses Cayley-Menger determinants and generalizations of them as important tools. This can be clearly seen in the Parts II, III, and IV of the book. Various embedding problems and derived characterizations (e.g., of Euclidean or Hilbert spaces) are given in terms of these determinants. And one can also see that some properties of Cayley-Menger determinants, hardly derivable in a purely algebraic way, get a better approach when they are presented in the language of the metric relations studied in [96].

In his nice monograph [266], Fiedler summarizes many (mainly own) theorems from the *geometry of simplices*, going back to the early fifties, but containing also modern results (and various parts are even new). All these results are obtained by the successful use of the interplay between Euclidean geometry, graph theory, matrices and determinants corresponding with simplices. Also the Cayley-Menger determinant (introduced in Sect. 1.4) is studied, and Fiedler presents many applications (e.g., in electrical network theory).

Hampton [378] gives a nice survey on the *geometry of triangles, tetrahedra and higher-dimensional simplices*. Old and new formulas and identities are presented, in particular referring to distances, areas, and angles. The author also treats Cayley-Menger determinants within that framework.

With more than 200 references, Liberti et al. [509] give a comprehensive survey on the Euclidean geometry based on the concept of distance. The authors emphasize information for applications mainly in the following direction: the input data are given in a non-complete system of distances, and the output is a point set in Euclidean n-space where the given distances are realized. After an introductory part, the main mathematical results are given in Sects. 12.2, and 12.3 presents the applications of distance geometry to *molecular conformation problems*. In the final fourth section applications to wireless sensor networks, statics, and robotics are discussed. We continue with a nice exposition of problems around distance geometry (and in some cases combinable with Cayley-Menger determinants), namely the paper [508] of Liberti and Lavor. It contains, nicely presented, a beautiful

discussion of Heron's formula, Cauchy's rigidity theorem on polyhedra, Cayley's extension of Heron's formula to higher dimensions, Menger's characterization of abstract semi-metric spaces, Gödel's result on metric spaces on the sphere, and Schoenberg's equivalence of distance and positive semidefinite matrices, giving the fundament of multidimensional scaling.

Ungar encoded in 1988 the Einstein velocity addition law and got an algebraic structure that he called *gyroalgebra*. Based on this, Ungar's book [788] introduces for the first time the concept of *hyperbolic n-simplex* as an important tool and subject of research in this direction of *n-dimensional hyperbolic geometry*. The basic notions of gyroalgebra and related geometric notions are excellently introduced in Chapter 1 of [788], and in Part IV (Chapter 10) Ungar discusses hyperbolic hyperplanes, hyperspheres and simplices in this context, getting, e.g., notions like gyrofaces, gyrocenters, and gyoradii for gyrosimplices. Successfully using Cayley-Menger determinants, this concept enables him to study an interesting portion of related hyperbolic simplex geometry, e.g. treating in- and circumcenters, in- and circumradii, altitudes, and Lemoine points.

The book [20] of Alfakih covers knowledge about *Euclidean distance matrices* (EDMs) and their applications mainly in *rigidity theory*. After introducing mathematical notions from matrix theory, graph theory, convexity, and semidefinite programming, Chaps. 3 and 4 deal with basics from the theory of EDMs themselves and various special types, such as Manhattan distance matrices on grids, distance matrices of hypercubes and trees, and in particular also the *Cayley-Menger matrices*. Several interesting topics are discussed in later chapters, such as spherical and nonspherical EDMs, connections among EDMs, adjacency matrices of graphs and combinatorial designs, and the completion of incomplete EDMs. In the last two chapters notions like *rigidity, local rigidity, infinitesimal rigidity, and universal rigidity* are introduced and studied. In a balanced and detailed manner, the author nicely presents classical results and various recent developments. It makes sense to present directly here, close to the book [20], the nicely written exposition [18] treating also results on EDMs regarding the old question: how to find out whether a real, symmetric, zero-diagonal quadratic matrix is an EDM? The authors show the correctness of a related algorithm which is faster than the classical algorithms and based on sphere intersections. The complexity of this algorithm is linear in the number of rows and columns of the input matrix, and quadratically in the minimum embedding dimension.

Havel and Li [384] give a survey on *distance geometry, conformal geometric algebra and its applications to physical chemistry* against the background of metric spaces on which the group of isometries acts transitively. The authors start with Cayley-Menger determinants, Schoenberg's quadratic forms and further related and needed tools, and they describe then conformal molecular chemistry as an application field of distance geometry. More general Euclidean invariants are discussed, too. Summarizing the discussed subjects one can say that this survey is mainly focused on applications of basic concepts and classical results of metric geometry in physical chemistry.

In the recent exposition [86] a computational technique to determine the volume of n-dimensional polytopes is presented. Initially, the volume is computed for an n-simplex which is obtained by signed simplex decomposition of the polytope. A recursive algorithm makes it possible to compute then the volume of the whole polytope. In this procedure, Cayley-Menger determinants are used to get the needed simplex volumes.

Simplex Geometry

Blumenthal and Sen [98] deal with *curvature in metric spaces* (where the curvature of three and the torsion of four points can be introduced via Cayley-Menger determinants). The latter yields two *extensions of the sine law* of tetrahedra, the first referring to dihedral angles, and the second to solid angles. Using their tools, they extend this also to Euclidean n-simplices.

Alexander [19] defines so-called r-acute n-simplices with the help of equal squared and non-squared interpoint distances and proves a metric simplex result implying a *lower bound for some Cayley-Menger determinants*.

Yang and Zhang [824] extend the *Neuberg-Pedoe inequality* on pairs of triangles in the plane, whose case of equality characterizes pairs of directly similar triangles, to Euclidean n-space and give also some related non-Euclidean results. Namely, the generalized inequalities for pairs of n-simplices contain, besides the volumes and edge-lengths of the two participating n-simplices, also *cofactors of the Cayley-Menger determinants* for one squared volume. Regarding non-Euclidean analogues it is shown that some analogues do not hold in the hyperbolic case, but at least in the 2-dimensional elliptic case. Further interesting results of these authors in the same direction are cited.

Gregorac [324] solves the following problem: Given non-negative numbers $a_1, ..., a_{n+1}$ and a regular n-simplex S with vertices v_i. Under what circumstances is there a point p and a scaling factor s such that $pv_i = a_i/s$? If one solution exists, inversion in the circumsphere of S yields another one. Using the Cayley-Menger determinant regarding the distances, the author solves the problem and applies the result to other problems.

Kokkendorff [467] studies finite metric spaces and creates Gram matrices as tool for studying the *geometry of simplices in space forms*. He gives a formula relating the determinant of a normalized Gram matrix (here defined via cosines of interpoint distances) to the geometry of the considered simplex. A machinery is constructed that allows to work with Gram matrices in a very general context, including also a formula for the determinant being a *generalization of the Cayley-Menger determinant formula* and an extension of *Heron's formula*. Based on these results, the author gives conditions for a metric space to be realizable as an *isometric embedding into Riemannian and semi-Riemannian space forms*, and also into *hyperbolic spaces*.

The starting point of D'Andrea and Sombra [196] is the classical Cayley-Menger determinant regarding an n-simplex. The main result says that the Cayley-Menger

determinant is absolutely irreducible over $\mathbf{C}[d_{ij}]$ for any $n \geq 3$, with d_{ij} as set of distances between all vertices. Moreover, the authors prove that a further polynomial related to the problem of computing the radius of the circumsphere of a simplex is also absolutely irreducible in all dimensions, but the smallest one (this has to do with Ptolemy's theorem). Hajja et al. [368] deal with the equality, holding for regular n-simplices ($n \geq 2$), which combines the total edge-length and the $n + 1$ vertex distances from a point in its affine hull. As a by-product the authors get new approaches to the *irreducibility of Cayley-Menger determinants*.

With the help of the Cayley-Menger determinant one can write the n-volume of an n-simplex with integer edge-lengths of the form $q\sqrt{k}$, where q is a rational number and k denotes a square-free positive integer which is called the *characteristic of the simplex*. Any triangle spanned by three non-collinear points being from this family is known to have the same characteristic. By Kurz [484] this is extended to n dimensions, and he also shows that for any n-dimensional integral point set all non-degenerate n-simplices have the same characteristic. The author continues with generating finite *integral point sets*, e.g. of minimal diameter.

Inspired by the fact that the sum of the angles of all Euclidean triangles is equal for all triangles, Hajja and Martini raised the corresponding question for higher dimensions in [363], namely whether something similar holds for n-simplices with $n \geq 3$. Some related property (via corner angles) holds for the *orthocentric case*, but for various further important classes of simplices (e.g., circumscriptible ones) some property like this is not holding, as shown by Hajja and Hayajneh [361]. Here characterizations of special Cayley-Menger determinants corresponding with special simplex classes form an important tool, like also embeddability properties.

Gaifullin [287] asks whether there are relations between *k-volumes of k-faces* of an n-simplex S similar to those between edge-lengths and the volume of S (the latter ones leading to Cayley-Menger determinants). He has special interest in the case $k = 2$ and shows that, for $n \geq 4$, the square volume of S satisfies a polynomial relation with coefficients dependent on the squares of the areas of its 2-faces. The respective minimal degree is computed, and the existence of a monic polynomial relation between the volume of S with coefficients depending on the areas of the 2-faces of S is confirmed if and only if n is even and at least 6. The leading coefficients of polynomial relations satisfied by the volume for other dimensions are also studied.

Schwartz [717] proves the following two non-trivial results: for any tetrahedron T there exists a tetrahedron T' such that every edge of T' is one unit longer than the corresponding edge of T. Furthermore, the ratio of the volume of T' to that of T has the sharp lower bound $(1 + 6/s)^3$, where s denotes the sum of edge-lengths of T. An important tool in proving these results is given by Cayley-Menger determinants. And so also results for higher dimensions are obtained.

In [367], Hajja et al. prove many nice results on *pre-kites*, i.e., on n-simplices having one regular $(n - 1)$-face. Their proofs use Cayley-Menger determinants specified for pre-kites. E.g., they prove the following result: for $n \geq 3$ an arbitrary non-regular n-dimensional pre-kite cannot have more than two regular facets. The metric structure of pre-kites with two regular facets is completely clarified. An n-

dimensional pre-kite P is a regular n-simplex if one of the following items holds true: (i) the circumcenter and the centroid of P coincide; (ii) the circumcenter and the incenter of P coincide; (iii) the incenter and the centroid of P coincide and $n \leq 5$. We add that, if $n \geq 6$, then an arbitrary pre-kite with coinciding incenter and centroid is not necessarily a regular simplex.

A *circumscriptible tetrahedron* has a sphere touching all its edges. Clearly, a general tetrahedron is not uniquely determined by its face areas (even not, if additionally the volume and circumradius are known). Using a characterization of the tetrahedron by the Cayley-Menger determinant, Hajja and Krasopoulos [362] prove that a circumscriptible tetrahedron is determined by its face areas.

Aomoto and Machida (see [34]) present a new Schläfli-type formula for the volume of pseudo-simplices which are spherically faced in Euclidean space. This formula is described in terms of Cayley-Menger determinants and their differentials involved with hypersphere arrangements in [35]. This is based on hypergeometric integrals and goes back to a variational formula also coming from hypergeometric integrals, but associated with hyperplane arrangements. Also using Cayley-Menger determinants, this was first studied by them (see [34]).

Taking Heron's formula (giving the area of a triangle in terms of its side-lengths) as starting point, Kock [460] presents the corresponding Cayley-Menger determinant with the possibility to extend this way everything to higher dimensions. He applies the formula to infinitesimal simplices in Riemannian manifolds, yielding a geometric description of the volume form of the respective manifold.

It is a natural question whether there can be several non-congruent tetrahedra with the same volume, circumradius and facet areas. Dehbi et al. proved (in an older paper) that for these given metric invariants at most eight real tetrahedra types exist (except that three facet areas are equal). Now, in [205], they reprove this more elegantly, and they show (using Cayley-Menger determinants) that the determination of tetrahedra from given metric invariants can be reduced to solving two polynomial equations having two unknowns. Further simplifications are discussed, and finally numerical experiments show that the number of real solutions might be not larger than 6.

At the end of this subsection here we mention once more Fiedler's monograph [266] which contains results from various research papers of the author dedicated to simplex geometry.

Extensions and Modifications

Some extensions and modifications of Cayley-Menger determinants and their applications were already mentioned above, and we will not repeat them here.

Seidel [724] presents a distance-geometric theory of *planar Euclidean geometry* based on properties of Cayley-Menger determinants, and he continues in [725] giving the fundaments for metrical geometry in *spherical and hyperbolic planes* using Cayley-Menger determinants based on cosines or hyperbolic cosines of the mutual distances in finite sets.

Havel [383] gives an algebraic approach to prove theorems on Euclidean point configurations by using the *interpoint distances instead of the coordinates of the given points*. Thus Cayley-Menger determinants and *Cayley-Menger bideterminants* (see below) are taken to present distances and further measures in an algebraic way. The author states four geometric theorems (e.g., Simson's theorem is one of them), but gives on this way algebraic proofs, and further applications (e.g., to get the topological structure of a simple linkage mechanism) are also presented.

Basically using *Cayley-Menger bideterminants* whose entries are 0, 1 and squared distances, Dalbec [195] derives the first and the second fundamental theorem for vector invariants of the Euclidean group over the field of characteristic $\neq 2$. Also a presented related algorithm is based on these determinants.

Rylov [681] presents a new method for investigating a metric space X which is based on a classification of finite subspaces of X. Due to this one gets informations on X and can remove constraints imposed usually on the metric. So one can use X for the description of the space-time geometry and also further geometries, with indefinite metrics. Interestingly, as a measure for a finite subspace in this framework (called length or volume), the author introduces a *determinant of metric functions* which is similar to the Cayley-Menger determinant.

Li and Leng present in [506] a *discrete version of mixed volumes*, e.g. of two finite vector sets, deriving also the corresponding analogue to the Minkowski inequality for mixed volumes. They continue by defining the concepts of *mixed Cayley-Menger determinants* and *mixed vertex angles of simplices*, thus extending the concept of classical Cayley-Menger determinants. The relations between all these notions are studied, and based on this, interesting inequalities and equalities for simplices containing also the mixed Cayley-Merger determinant and the mixed vertex angles are derived.

Soland [748] elegantly shows that one can characterize the Euclidean plane by using the common axioms for metric spaces together with three further simple axioms involving solely the Cayley-Menger determinants.

Dordovskii [221] shows that the metric spaces which are pretangent to a Euclidean or unitary space (finite-dimensional) are isometric to this space. Based on this property, he describes the *metric pretangent spaces* at the non-singular points of smooth surfaces. The proof of the main result is strongly based on the fact that the Cayley-Menger determinant of $n + 2$ points in n-space is zero. The author proves also that there exist spaces pretangent to Hilbert spaces without this isometry property. Also Bilet and Dovgoshey [88] deal with pretangent spaces, more precisely with conditions for the isometric embeddability of pretangent metric spaces in the real Euclidean space. This means that they extend results on isometric embeddability due to Menger, Schoenberg and Blumenthal to this more general case.

Thomas and Porta [775] use a known identity of Clifford to extend Euclidean distance problems regarding point sets to *families of hyperspheres* (taking the so-called power between hyperspheres as distance). Choosing the radii suitably, the Clifford identity comes conveniently down to *generalized Cayley-Menger determinants*, simplifying the approaches to the extended distance problems.

Applications

Schoenberg [709] uses Cayley-Menger determinants to describe the mobility of rigid segment structures (like, e.g., the edge system of the regular octahedron) which lose their rigidity and become movable linkages with increasing dimension of the spaces in which they are embedded. His motivation is to study general classes of such linkages.

Szenthe deals in [767] with the concepts of *curvature in the metric theory of curves* and the *standard differential geometry of curves* in Euclidean spaces. Using Cayley-Menger determinants, the relations between these concepts are studied.

Graham et al. [317] give an exact criterion that $n+2$ points in Euclidean n-space have only odd integers as their pairwise distances, and they prove this (and further similar results) with the help of a factorization of the respective Cayley-Menger determinant.

Sippl and Scheraga [735] expose results on the problem of calculating Cartesian coordinates from a matrix of interpoint distances (thus yielding the so-called *embedding problem*). They present an efficient algorithm for the transformation of distances to coordinates and show this way that the embedding problem is related to the theory of symmetric matrices. Using methods from distance geometry, they continue in [736] with attacking the embedding problem regarding *nuclear magnetic resonance for macromolecules* (determining a proper subset of the complete system of distances between given points, which then has to be filled with a minimum set of errors).

Crippen [189] proposes to apply methods from Blumenthal's distance geometry (see [96]) for the calculation of *conformation of molecules*, i.e., to use direct calculation of the interpoint distances and results on Cayley-Menger determinants and embeddability from distance geometry. His concept is to use all known distances at their right places, and to approximate or modify the unknown distances so that they satisfy geometric constraints. Following the proposals from [189], Klapper and DeBrota [454] provide an algorithm to compute the occurring distances in a *molecular structure* directly, and they present an interesting example.

To make some useful results from distance geometry accessible to applied scientists, Gower [312] nicely discusses properties of the interpoint distance matrix of finite point sets in Euclidean n-space. And he studies under what circumstances a symmetric matrix can be such a *distance matrix*.

By gradual *contraction of the dimensionality*, Purisima and Scheraga [650] succeed to get a low-energy three-dimensional structure from a very-low-energy high-dimensional conformation. For this dimension contraction they use Cayley-Menger determinants, and interesting examples (e.g., full-atom representations of certain terminally blocked amino acids) are given.

Using a Cayley-Menger determinant, Volenec [797] proves that if all distances occurring between five points in the Euclidean plane take only two different values, then they form the vertex set of a regular pentagon. (Note that here a "correct"

labelling of the distances is not assumed; otherwise the result is well known and true even without assuming 2-dimensionality.)

Serré et al. [728] deal with *conceptual design*, where designers have to verify the consistency of engineering specifications; more precisely, here the analysis of the consistency of a 3D geometric specification is described. Since the usual way via Cartesian modelling does not give easy analysis of consistency, the authors present two non-Cartesian modellings: the first by using points and Cayley-Menger determinants regarding their mutual distances, and the second based on Gram determinants regarding points and vectors. Finally, the advantages due to these two ways of modelling are shown. Namely, distance geometry is often used in molecular geometry, and vectorial modelling is used in fields like kinematics, dynamics and tolerancing.

For Cayley-Menger determinants used regarding *k-distance sets* (i.e., point sets having only k different values of interpoint distances) we refer to Section 5.4 of the problem book [125] of Brass, Moser and Pach.

In *geometric constraint solving* Cayley-Menger determinants are preferably used in *robotics and molecular chemistry*. Distances are considered as coordinates yielding systems where the unknowns are interpoint distances. This can reduce the expense of them to solve, but these systems can also be hard to solve. Mathis and Schreck [559] describe two algorithms making it possible to get such systems with a minimal number of equations.

Giving proofs of two recent conjectures concerning the *m-body matrix*, the paper [327] mainly refers to the problem of m particles under a potential field that is only a function of the distances among the pairs of the particles. Whenever the positions of the considered masses are in a nonsingular configuration (i.e., they do not lie in an affine subspace of dimension at most $m-2$), this matrix defines a Riemannian metric on the space coordinatized by their interpoint distances. The corresponding determinant can be factored into the product of the participating Cayley-Menger determinant (of order m) and a factor which is mass dependent and of one sign on all nonsingular mass configurations.

Based on information from *protein geometry and distance data*, obtained via *nuclear magnetic resonance technique*, the authors of the paper [595] treat a branch-and-prune algorithm to search graph realizations in *molecular distance geometry* and, with the help of its dual, to look for completions of corresponding non-complete distance matrices. The authors use a *global optimization method* which makes it possible that also the dual branch-and-prune algorithm can efficiently attack problems from nuclear magnetic resonance experiments.

Inversive distance circle packings were introduced as a generalization of *Thurston's circle packings* on surfaces, allowing adjacent circles to separate. They live on triangulated surfaces with given weights on all undirected edges and circle centers at the vertices, clearly satisfying certain conditions regarding the inversive distance of neighboring circles. A central question refers to their *rigidity*, and for the proof of that a variational principle has to be applied. With new proof steps, Xu [819] follows that way, and in the first step he uses Cayley-Menger determinants allowing also to extend at least this proof part to higher dimensions.

Chapter 13
Casey's Theorem



The more you know, the more you know you don't know.

— Aristotle

Casey's theorem is an extension of Ptolemy's theorem: four points in Ptolemy's theorem are replaced by four circles, and distances between points are replaced by common tangent distances between circles. The proof of Casey's theorem is elaborated and requires a few new techniques. Casey's theorem can be also extended to a set of circles consisting of more than four circles.

13.1 Bicolored Sets of Circles

To state Casey's theorem, let us introduce *bicolored sets of circles*. A set $\mathcal{F}$ of circles is called a bicolored set of circles if each circle in $\mathcal{F}$ has one of two colores (red and blue, say) and the following properties (1) and (2) hold.

(1) No circle in $\mathcal{F}$ is contained in the closed disk bounded by another circle in $\mathcal{F}$.
(2) Circles in $\mathcal{F}$ of different colors are disjoint to each other.

In a bicolored set of circles, one color class may be empty. We also allow that some colored circles have radius 0 (i.e., are colored points, called *null circles*).

Let $\mathcal{F} = \{c_1, c_2, \ldots, c_n\}$ be a bicolored set of circles in $\mathbb{R}^2$. We define the distance $t(c_i, c_j)$ (called the *common tangent distance*) between two circles c_i, c_j in $\mathcal{F}$ by the distance between the two contact points of a common tangent, where we take an external tangent if they have the same color. Otherwise, we take an internal tangent. Let us state this more precisely. For a circle c_i in $\mathcal{F}$, let z_i be its center, and

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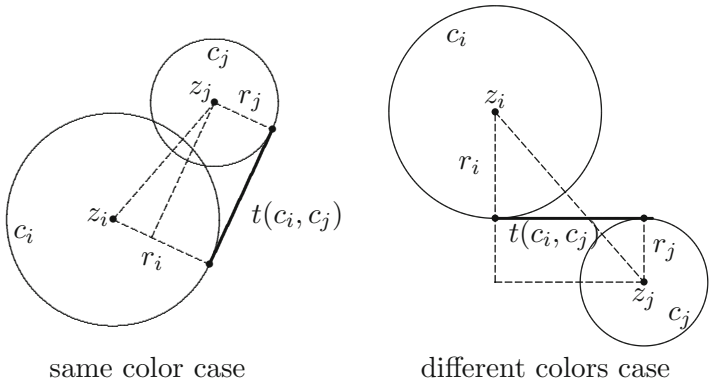


Fig. 13.1 Common tangent distances

r_i denote its radius. For a pair of circles c_i, c_j in $\mathcal{F}$, their common tangent distance $t(c_i, c_j)$ is defined by

$$t(c_i, c_j) = \begin{cases} \sqrt{|z_i z_j|^2 - (r_i - r_j)^2} & \text{if } c_i, c_j \text{ have the same color,} \\ \sqrt{|z_i z_j|^2 - (r_i + r_j)^2} & \text{if } c_i, c_j \text{ have different colors,} \end{cases}$$

see Fig. 13.1. If c_i, c_j are null circles, then $t(c_i, c_j) = \|z_i - z_j\|$. Due to the common tangent distances, a bicolored set of circles becomes a semi-metric space.

A circle or a line is called *properly tangent* to the circles in a bicolored set of circles, if circles of the same color are tangent to the circle (or line) from the same side, and the circles of different colors are tangent to the circle (or the line) from the opposite side.

Now, Casey's theorem can be stated as follows.

Theorem 13.1 (Casey's Theorem) *For a bicolored set $\mathcal{F} = \{c_1, c_2, c_3, c_4\}$ of four circles, the following two statements are equivalent.*

(i) There is a circle or a line that is properly tangent to the circles in $\mathcal{F}$.
(ii) $\det SD(\mathcal{F}) = 0$.

Remark 13.1 By Lemma 12.2, the statement (ii) implies that one of

$$t_{12}t_{34} - t_{13}t_{24} + t_{23}t_{14},$$
$$t_{12}t_{34} + t_{13}t_{24} - t_{23}t_{14},$$
$$-t_{12}t_{34} + t_{13}t_{24} + t_{23}t_{14}$$

is equal to 0, where $t_{ij} = t(c_i, c_j)$ are the common tangent distances between c_i and c_j. Thus, if all circles in $\mathcal{F}$ are null circles, then Casey's theorem becomes Ptolemy's classical theorem.

To prove Casey's theorem, we prepare several lemmas.

13.2 Inversion Invariants

Lemma 13.1 *Let c_1, c_2 be two circles in a bicolored set of circles, and let $x \in c_1$ be a point that lies outside of c_2. Then a circle (or a line) that passes through x and orthogonally crosses both c_1, c_2 is uniquely determined.*

Proof Let us consider the case that x does not lie on the line through the centers of c_1, c_2. Then the center of a circle that orthogonally crosses both c_1, c_2 and passes through x is the intersection point of the tangent line of c_1 at x and the radical axis of c_1, c_2, see Lemma 3.5. Hence such a circle is uniquely determined. □

Lemma 13.2 *Let φ be the inversion of the plane with respect to a circle with center p and radius r. Let us denote $\varphi(x)$ by x'. For any two points x, y (both different from p) we have*

$$|x'y'| = \frac{r^2|xy|}{|px| \cdot |py|}.$$

Proof Since $|px| \cdot |px'| = r^2 = |py| \cdot |py'|$, the triangles $\triangle pxy$ and $\triangle py'x'$ are similar. Hence

$$\frac{|x'y'|}{|yx|} = \frac{|px'|}{|py|} = \frac{r^2}{|px| \cdot |py|}.$$

From this the lemma follows. □

For four points w, x, y, z in the plane, let us define $[w, x, y, z]$ by

$$[w, x, y, z] = \frac{|wx| \cdot |yz|}{|wy| \cdot |xz|},$$

which is called a *Möbius invariant* of $\{w, x, y, z\}$. The next corollary is obtained by straightforward calculation using the above lemma.

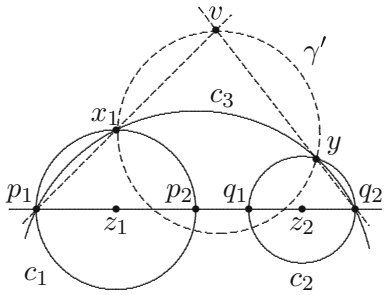
Corollary 13.1 *The Möbius invariant $[w, x, y, z]$ is invariant under any inversion φ whose center p is different from w, x, y, z, namely,*

$$[\varphi(w), \varphi(x), \varphi(y), \varphi(z)] = [w, x, y, z].$$

□

Lemma 13.3 *Let c_1, c_2 be two circles in a bicolored set of circles, and let z_i, r_i be the center and the radius of c_i for $i = 1, 2$. Let γ be a circle that orthogonally cuts both c_1, c_2. Let $x_1 \in \gamma \cap c_1$ be a point that lies outside c_2. When we trace γ from x_1 in the direction towards the interior of c_1, let x_2 be the point where we meet c_1*

Fig. 13.2 $c_1 \perp \gamma' \perp c_2$



again, y_1 be the point where we first meet c_2, and y_2 be the point where we meet c_2 the second time. Then we have

$$[x_1, y_1, x_2, y_2] = \frac{|z_1z_2|^2 - (r_1 - r_2)^2}{(2r_1)(2r_2)}. \tag{13.1}$$

If $|z_1z_2| > r_1 + r_2$, then we also have

$$[x_1, y_2, x_2, y_1] = \frac{|z_1z_2|^2 - (r_1 + r_2)^2}{(2r_1)(2r_2)}. \tag{13.2}$$

Proof Let us consider the case that $|z_1z_2| > r_1 + r_2$. Let p_1, p_2 and q_1, q_2 be the points where the line z_1z_2 cuts c_1 and c_2 so that p_2, q_1 lie between p_1, q_2. Let c_3 be the circle determined by the three points p_1, q_2, x_1, and let y be the intersection of c_2 and c_3 different to q_2, see Fig. 13.2. Let v be the intersection of the lines p_1x_1 and q_2y. Let γ' be the circle determined by v, x_1, y. Then $|vx_1| \cdot |vp_1| = |vy| \cdot |vq_2|$. If we denote by ψ the inversion with center v and power $|vx_1| \cdot |vp_1|$, then $\psi(c_i) = c_i$, for $i = 1, 2, 3$, where $\psi(\gamma')$ is the line z_1z_2. Since the inversion preserves the size of angles, we can deduce that γ' orthogonally cuts c_1, c_2. Hence $\gamma = \gamma'$ by Lemma 13.1, and therefore $v \in \gamma$ and $y = y_2$. Thus,

$$\psi(p_1) = x_1, \psi(p_2) = x_2, \psi(q_1) = y_1, \psi(q_2) = y_2.$$

Since the Möbius invariant is invariant under inversions, we have

$$[p_1, q_1, p_2, q_2] = [x_1, y_1, x_2, y_2] \text{ and } [p_1, q_2, p_2, q_1] = [x_1, y_2, x_2, y_1].$$

Now, since $|p_1q_1| = |z_1z_2| + r_1 - r_2$, $|p_2q_2| = |z_1z_2| - r_1 - r_2$, $|p_1p_2| = 2r_1$, and $|q_1q_2| = 2r_2$, we have

$$\begin{aligned}[x_1, y_1, x_2, y_2] &= [p_1, q_1, p_2, q_2] \\ &= \frac{(|z_1z_2| + r_1 - r_2)(|z_1z_2| - r_1 - r_2)}{(2r_1)(2r_2)} \\ &= \frac{|z_1z_2|^2 - (r_1 - r_2)^2}{(2r_1)(2r_2)}.\end{aligned}$$

Similarly, (13.2) can be obtained. □

Let $\mathcal{F}$ be a bicolored set of circles, and let p be a point exterior to all circles in $\mathcal{F}$. Then by the inversion with respect to the unit circle centered at p, circles in $\mathcal{F}$ are transformed to circles. Since p is exterior to all circles in $\mathcal{F}$, no transformed circle is enclosed in another transformed circle. Hence, by assuming that every transformed circle bears the same color as its predecessor (and we always assume this), we get another bicolored set of circles.

For a circle c, let us denote its center by $z(c)$, and its radius by $r(c)$.

Theorem 13.2 *For a pair of circles c_1, c_2 in a bicolored set of circles,*

$$\frac{t(c_1, c_2)^2}{r(c_1)r(c_2)}$$

is invariant under any inversion with center lying outside of both c_1, c_2.

Proof Let us consider the case where c_1, c_2 have the same color. Let φ be an inversion with center p lying outside of both c_1, c_2. Let $\sigma_i = \varphi(c_i)$, $i = 1, 2$, and γ be a circle such that $c_1 \perp \gamma \perp c_2$ and $p \notin \gamma$. Let $x_1 \in c_1 \cap \gamma$ be a point lying outside c_1. Define x_2, y_1, y_2 as in Lemma 13.3. Then,

$$4[x_1, y_1, x_2, y_2] = \frac{t(c_1, c_2)^2}{r(c_1)r(c_2)}$$

by Lemma 13.3. Let $\eta = \varphi(\gamma)$. Then $\sigma_1 \perp \eta \perp \sigma_2$. Put $x_i' = \varphi(x_i)$, $y_i' = \varphi(y_i)$, $i = 1, 2$. Then $x_1' \in \sigma_1 \cap \eta$, which lies outside σ_2, and

$$4[x_1', y_1', x_2', y_2'] = \frac{t(\sigma_1, \sigma_2)^2}{r(\sigma_1)r(\sigma_2)}$$

holds. Since the Möbius invariant is invariant under inversions, we have the equality $[x_1, y_1, x_2, y_2] = [x_1', y_1', x_2', y_2']$. Hence the theorem follows. □

13.3 Isometric Radii Change

Recall that for a pair of circles c_i, c_j in a bicolored set $\mathcal{F}$ of circles,

$$t(c_i, c_j)^2 = \begin{cases} |z_i z_j|^2 - (r_i - r_j)^2 & \text{if } c_i, c_j \text{ have the same color} \\ |z_i z_j|^2 - (r_i + r_j)^2 & \text{otherwise,} \end{cases}$$

where z_i, r_i denote the center and the radius of c_i. Hence, if we simultaneously increase the radii of the circles in $\mathcal{F}$ of the same color by the same amount, say ε, and decrease the radii of circles of the other color by ε, then the values of $t(c_i, c_j)$ are kept unchanged. More precisely, if ε_r denote the radius of the smallest red circle in $\mathcal{F}$, then for any $0 < \varepsilon \leq \varepsilon_r$, it is possible to decrease the radii of all red circles by ε, and to increase the radii of all blue circles by ε without changing the common tangent distances. Similarly, if ε_b denotes the radius of the smallest blue circle in $\mathcal{F}$, then for any $0 < \varepsilon \leq \varepsilon_b$, it is possible to decrease the radii of all blue circles by ε and to increase the radii of all red circles by ε without changing the common tangent distances. Thus, by assuming that every modified circle bears the same color as its predecessor (and we always assume this), the bicolored set of modified circles is isometric to the original bicolored set of circles as semi-metric space. Let us call such a simultaneous radii change an *isometric radii change*. If there is a circle that is properly tangent to the circles in $\mathcal{F}$, then by increasing or decreasing the radius of the circle by ε, we get a circle that is properly tangent to the modified circles. If there is a line that is properly tangent to the circles in $\mathcal{F}$, then, by translating the line in a direction through distance ε, we get a line that is properly tangent to the modified circles. Hence we have the following lemma.

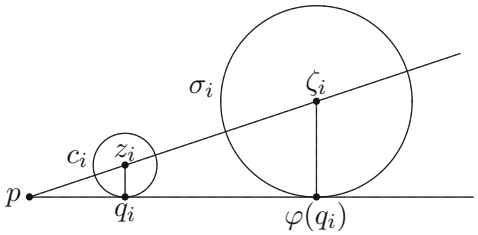
Lemma 13.4 *The circles in a bicolored set have a circle or a line that is properly tangent to them if and only if the circles modified by an isometric radii change have a circle or a line that is properly tangent to them.* □

Example 13.1 For any bicolored set of three circles c_1, c_2, c_3, there is a circle (or a line) that is properly tangent to the three circles. This can be seen as follows. By Lemma 13.4, we may assume that one of them, say c_3, is a null circle $\{p\}$. The point p is exterior to both c_1, c_2. Let φ be the inversion with respect to the unit circle with center p. Then $\{\varphi(c_1), \varphi(c_2)\}$ is a bicolored set of circles. Obviously, there is a line l properly tangent to $\varphi(c_1), \varphi(c_2)$. Then $\varphi(l)$ is a circle (or a line) that is properly tangent to c_1, c_2, c_3.

Lemma 13.5 *Let $\mathcal{F} = \{c_1, c_2, c_3, c_4\}$ be a bicolored set of four circles, where $c_4 = \{p\}$, a null circle. Let φ be the inversion of $\mathbb{R}^2$ with respect to the unit circle with center p, and let $\sigma_i = \varphi(c_i)$ for $i = 1, 2, 3$. Then*

$$t(\sigma_i, \sigma_j)^2 = \frac{t(c_i, c_j)^2}{t(c_i, c_4)^2 t(c_j, c_4)^2} \quad \text{for } 1 \leq i < j \leq 3. \tag{13.3}$$

Fig. 13.3
$\triangle pq_iz_i \sim \triangle p\varphi(q_i)\zeta_i$



Proof Let z_i, r_i be the center and the radius of c_i, and let ζ_i, ρ_i be the center and the radius of σ_i for $i = 1, 2, 3$. First suppose that $r_i > 0$, $\rho_i > 0$. Let q_i be the point on c_i such that $\angle pq_iz_i = \pi/2$. Then $\angle p\varphi(q_i)\zeta_i = \pi/2$, and $\triangle pq_iz_i \sim \triangle p\varphi(q_i)\zeta_i$, see Fig. 13.3. Hence $|p\varphi(q_i)|/\rho_i = |pq_i|/r_i$, and therefore $|p\varphi(q_i)| = |pq_i|\rho_i/r_i$. Since $|pq_i||p\varphi(q_i)| = 1$, we have $|pq_i|^2\rho_i/r_i = 1$, and hence $t(c_i, c_4)^2 = |pq_i|^2 = r_i/\rho_i$. Similarly, $t(c_j, c_4)^2 = r_j/\rho_j$. Now, by Theorem 13.2, we have

$$\frac{t(\sigma_i, \sigma_j)^2}{\rho_i\rho_j} = \frac{t(c_i, c_j)^2}{r_ir_j}.$$

Hence

$$t(\sigma_i, \sigma_j)^2 = \frac{t(c_i, c_j)^2}{(r_i/\rho_i)(r_j/\rho_j)} = \frac{t(c_i, c_j)^2}{t(c_i, c_4)^2t(c_j, c_4)^2}.$$

Now, for fixed p, z_i, z_j,

$$t(c_i, c_j), t(\sigma_i, \sigma_j), t(c_i, c_4), t(\sigma_i, \sigma_4), \rho_i, \rho_j$$

are continuous functions of r_i, r_j. For $k \in \{i, j\}$,

$$\lim_{r_k\to 0}\rho_k = 0, \lim_{\rho_k\to 0}\frac{r_k}{\rho_k} = \lim_{r_k\to 0}|pq_k|^2 = |pz_k|^2,$$

and when $r_k = 0$, we have $|pz_k|^2 = t(c_4, c_k)^2$. Hence (13.3) holds even in the case where one of c_i, c_j or both are null circles. □

Lemma 13.6 *Let $\mathcal{F} = \{c_1, c_2, c_3, c_4\}$ be a bicolored set of four circles, where $c_4 = \{p\}$, a null circle. Let φ be the inversion of $\mathbb{R}^2$ with respect to the unit circle with center p, let $\sigma_i = \varphi(c_i)$, $i = 1, 2, 3$, and put $\mathcal{E} = \{\sigma_1, \sigma_2, \sigma_3\}$. Then*

$$\prod_{i=1}^{3} t(c_i, c_4)^4 \det\Gamma(\mathcal{E}) = \det SD(\mathcal{F}).$$

Proof For $1 \leq i, j \leq 3$, the (i, j)-entry of $SD(\mathcal{E})$ is given by (13.3). Multiplying the ith row of $\Gamma(\mathcal{E})$, for $i = 1, 2, 3$, with $t(c_i, c_4)^2$ and then the jth column, for $j = 1, 2, 3$, with $t(c_j, c_4)^2$, we have $SD(\mathcal{F})$. Hence the theorem follows. □

Lemma 13.7 *For a bicolored set of three circles $\mathcal{E} = \{\sigma_1, \sigma_2, \sigma_3\}$, the following two statements are equivalent.*

(i) There is a line that is properly tangent to all circles in $\mathcal{E}$.
(ii) As a semi-metric space, $\mathcal{E}$ is isometrically embeddable in $\mathbb{R}^1$.

Proof

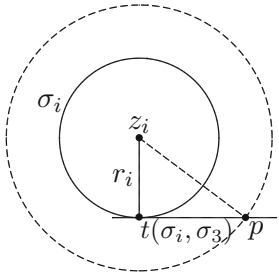
(i) ⇒ (ii). Suppose that there is a line L that is properly tangent to all circles in $\mathcal{E}$. Then the set of contact points $L \cap \sigma_i$, $i = 1, 2, 3$, is a subset of L that is isometric to $\mathcal{E}$. Hence $\mathcal{E}$ is isometrically embeddable in $\mathbb{R}^1$.

(ii) ⇒ (i). Noting that the common tangent distance is preserved by isometric radii change, and by Lemma 13.4, we may suppose that one circle in $\mathcal{E}$ is a null circle. Let $\sigma_3 = \{q\}$, a null circle. Obviously, there is a line L that is properly tangent to σ_1, σ_2. Let q_i be the contact points $L \cap \sigma_i$, $i = 1, 2$. Since $\mathcal{E}$ is isometrically embeddable in a line, there is a point u on L such that $\{q_1, q_2, u\}$ is isometric to $\mathcal{E}$. If $p = u$, then we are done. Suppose $p \neq u$. Let z_i, r_i be the center and the radius of σ_i, for $i = 1, 2$. Then $\|z_i - p\|^2 = r_i^2 + t(\sigma_i, \sigma_3)^2$, for $i = 1, 2$. Hence, p lies on the circles with center z_i and radius $\sqrt{r_i^2 + t(\sigma_i, \sigma_3)^2}$, $i = 1, 2$, see Fig. 13.4.

The point u also lies on these circles. Hence $\triangle z_1 z_2 u$ and $\triangle z_1 z_2 p$ are congruent. Let ψ be the reflection of $\mathbb{R}^2$ with respect to the line $z_1 z_2$. Since $p \neq u$, we must have $\psi(u) = p$. Then, since $p \in \psi(L)$, $\psi(L)$ is a line that is properly tangent to σ_1, σ_2 and passes through p.

□

Fig. 13.4 $\sigma_3 = \{p\}$



13.4 Proof of Casey's Theorem

By Lemma 13.4, we may consider the case that one of the circles in $\mathcal{F}$ is a null circle. Let $c_4 = \{p\}$, a null circle. Note that p is exterior to all other circles in $\mathcal{F}$. Let φ be the inversion of $\mathbb{R}^2$ with respect to the unit circle with center p, let $\sigma_i = \varphi(c_i)$, $i = 1, 2, 3$, and $\mathcal{E} = \{\sigma_1, \sigma_2, \sigma_3\}$. By Lemma 13.6,

$$\det \Gamma(\mathcal{E}) = 0 \Leftrightarrow \det SD(\mathcal{F}) = 0. \tag{13.4}$$

(i) $\Rightarrow$ (ii): Suppose that there is a circle γ that is properly tangent to all circles in $\mathcal{F}$. Then $\varphi(\gamma)$ is a line that is properly tangent to the circles in $\mathcal{E}$. Then, by Lemma 13.7, $\mathcal{E}$ is isometric to a 3-point-subset of $\mathbb{R}^1$. Hence $\det \Gamma(\mathcal{E}) = 0$ by Theorem 12.1. Therefore, $\det SD(\mathcal{F}) = 0$ by (13.4).

(ii) $\Rightarrow$ (i): Suppose that $\det SD(\mathcal{F}) = 0$. Then $\det \Gamma(\mathcal{E}) = 0$, by (12.7). Hence $\mathcal{E}$ is isometrically embeddable in $\mathbb{R}^1$, and hence there is a line L that is properly tangent to the circles in $\mathcal{E}$, by Lemma 13.7. Then $\varphi(L)$ is a circle that is properly tangent to the circles in $\mathcal{F}$. □

Remark 13.2 Casey's theorem can be also generalized to *bicolored sets of spheres* in $\mathbb{R}^n$, see [535].

13.5 A Theorem on Five Circles

With similar methods, we can derive a related theorem referring to a quintuple of circles.

Theorem 13.3 *Let $\mathcal{F} = \{c_1, c_2, c_3, c_4, c_5\}$ be a bicolored set of five circles in $\mathbb{R}^2$. Suppose that, for each $i = 1, 2, 3, 4, 5$, there is a circle (or a line) γ_i that is properly tangent to the circles in $\mathcal{F} \setminus \{c_i\}$. Then some two of $\gamma_i, i = 1, 2, \ldots, 5$, coincide, and hence there is a circle or a line that is properly tangent to all five circles in $\mathcal{F}$.*

Proof We may suppose that at least three circles in $\mathcal{F}$ are red circles. Let us diminish the radius of the smallest blue circle, to zero if $\mathcal{F}$ has a blue circle. Otherwise, diminish the radius of the smallest red circle to zero by isometric radii change. So, we may suppose, from the first, $c_5 \in \mathcal{F}$ is a null circle $\{p\}$. Let φ be the inversion of $\mathbb{R}^2$ with respect to the unit circle with center p and let $\mathcal{E} = \{\sigma_1, \sigma_2, \sigma_3, \sigma_4\}$ be the bicolored set, where $\sigma_i = \varphi(c_i)$, $i = 1, 2, 3, 4$. Let $l_i = \varphi(\gamma_i)$ for $i = 1, 2, 3, 4$, and $\tau = \varphi(\gamma_5)$. Then l_i is a line that is properly tangent to the circles in $\mathcal{E} \setminus \{\sigma_i\}$ for $i = 1, 2, 3, 4$, and τ is a circle or a line that is properly tangent to the four circles in $\mathcal{E}$. Since at least three circles in $\mathcal{E}$ are red circles, we may suppose that $\sigma_1, \sigma_2, \sigma_3$ are red circles, and σ_4 is either a blue circle, or the smallest red circle. By applying isometric radii change, we may furthermore suppose that σ_4 is a null circle $\{q\}$. (Of course, the lines l_i and τ (if τ is a line) are then translated accordingly, and the

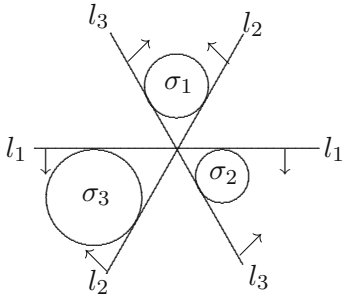


Fig. 13.5 The three angular regions are disjoint

radius of τ (if τ is a circle) is changed accordingly.) Note that three lines l_1, l_2, l_3 all pass through q, and l_4 is properly tangent to $\sigma_1, \sigma_2, \sigma_3$.

To prove the theorem, it is sufficient to show that $l_i = l_j$ for some $1 \le i \ne j \le 4$.

If one of $\sigma_1, \sigma_2, \sigma_3$ is a null circle, say $\sigma_3 = \{q'\}$, then since both l_1, l_2 pass through the two points q, q', we have $l_1 = l_2$. So, we may consider the case that none of $\sigma_1, \sigma_2, \sigma_3$ is a null circle.

Suppose that l_1, l_2, l_3, l_4 are all different lines. Since $\sigma_1, \sigma_2, \sigma_3$ are all red circles and $\sigma_4 = \{q\}$, the circles in $\mathcal{E} \setminus \{\sigma_i\}$ are tangent to the line l_i from the same side of l_i, $i \le 3$. For $i = 1, 2, 3, 4$, let us denote by $[l_i]$ the side of l_i containing the circles in $\mathcal{E} \setminus \{\sigma_i\}$. Then

$$\sigma_1 \subset [l_2] \cap [l_3],\ \sigma_2 \subset [l_1] \cap [l_3],\ \sigma_3 \subset [l_1] \cap [l_2].$$

Since three lines l_1, l_2, l_3 all pass through the point q, either one of the three angular regions $[l_1] \cap [l_2]$, $[l_1] \cap [l_3]$, $[l_2] \cap [l_3]$ is contained in another angular region, or the interiors of the three angular regions are disjoint. If the interiors of three angular regions are disjoint, then it is impossible to cut simultaneously all three angular regions by a single line, and hence it is impossible that $\sigma_1, \sigma_2, \sigma_3$ are tangent to l_4 from the same side (see Fig. 13.5), which is a contradiction. Hence we may suppose that one of the three angular regions is contained in another angular region, say

$$[l_1] \cap [l_2] \subset [l_1] \cap [l_3].$$

In this case, $[l_1] \cap [l_2]$ is also contained in $[l_2] \cap [l_3]$, see Fig. 13.6. Let x be the contact point of σ_1 and l_3, and y be the contact point of σ_1 and l_2.

Since τ passes through q and τ is tangent to $\sigma_1, \sigma_2, \sigma_3$ from the same side, τ must be a circle externally tangent to them. Then the contact point of σ_1 and τ must lie on the minor arc $\overset{\frown}{xy}$ of σ_1. Since σ_3 ($\subset [l_1] \cap [l_2]$) is also externally tangent to τ, the contact point of σ_3 and l_2 must lie on the line segment qy (otherwise, τ cannot be tangent to σ_3). Then σ_3 is contained in the union of the disk bounded by σ_1 and $\triangle xyq$. Since the contact point of σ_2 ($\subset [l_1] \cap [l_3]$) and l_3 lies on the extension of the line segment xq beyond q, no line can be properly tangent to all three circles

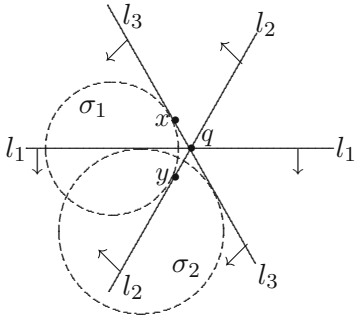


Fig. 13.6
$[l_1] \cap [l_2] \subset [l_1] \cap [l_3]$

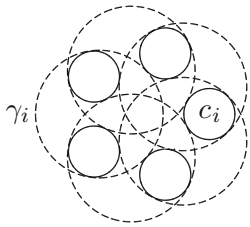


Fig. 13.7 $i \neq j \Rightarrow \gamma_i$ and c_j are tangent

$\sigma_1, \sigma_2, \sigma_3$, contradicting the property that l_4 is properly tangent to these three circles. Thus, if we assume that l_1, l_2, l_3, l_4 are all different, then we have a contradiction. □

Remark 13.3 Theorem 13.3 implies the *non-existence of double 5-systems of circles*. For details, see [536].

Remark 13.4 It is impossible to replace "properly tangent" simply by "tangent" in Theorem 13.3, see Fig. 13.7.

13.6 An Extension of Casey's Theorem

For a semi-metric space $\mathcal{X}$ and $k > 2$, we write

$$\det SD(\mathcal{X} : k) \equiv 0$$

if $\det SD(\mathcal{Y}) = 0$ holds for every subset $\mathcal{Y} \subset \mathcal{X}$ with k elements. Using this notation, Ptolemy's theorem can be extended as follows: For a subset $\mathcal{X} \subset \mathbb{R}^2$ consisting of n points, $n \geq 5$, $\mathcal{X}$ lies on a circle or a line if and only if $\det SD(\mathcal{X} : 4) \equiv 0$. However, since three points determine a circle uniquely, this is trivial. On the other hand, a similar extension of Casey's theorem seems to be not obvious.

Theorem 13.4 *For a bicolored set $\mathcal{F}$ of n (≥ 4) circles in $\mathbb{R}^2$, the following two statements are equivalent.*

(i) There is a circle or a line that is properly tangent to the circles in $\mathcal{F}$.
(ii) $\det SD(\mathcal{F} : 4) \equiv 0$.

Lemma 13.8 *For a bicolored set of three circles $\mathcal{F} = \{c_1, c_2, c_3\}$, the number of circles or lines that are properly tangent to the circles in $\mathcal{F}$ is at most two.*

Proof By Lemma 13.4, we may suppose that a circle in $\mathcal{F}$, say c_3, is a null circle $\{p\}$. The point p lies outside both c_1, c_2. Let φ be the inversion of the plane with respect to the unit circle with center p. Let $\sigma_1 = \varphi(c_1), \sigma_2 = \varphi(c_2)$. If γ is a circle or a line that is properly tangent to c_1, c_2, c_3, then $\varphi(\gamma)$ is a *line* that is properly tangent to σ_1, σ_2. The number of lines that are properly tangent to σ_1, σ_2 is at most two. (If one of σ_1, σ_2 is not a null circle, then there are exactly two lines that are properly tangent to σ_1, σ_2.) Hence the number of circles or lines that are properly tangent to c_1, c_2, c_3 is at most two. □

Proof of Theorem 13.4 The part (i) $\Rightarrow$ (ii) follows from Casey's theorem. So we show the part (ii) $\Rightarrow$ (i). Suppose that (ii) holds. If $n = 4$, (i) follows from Casey's theorem. If $n = 5$, then (i) follows from Theorem 13.3. Assume that (i) holds for an $m \geq 5$, and consider the case $n = m + 1$. Let $c_1, c_2, \ldots, c_{m+1}$ be colored circles of $\mathcal{F}$. For each $i = 1, 2, \ldots, m + 1$, there is a circle or a line γ_i that is properly tangent to the circles in $\mathcal{F} \setminus \{c_i\}$ by the inductive assumption. By Lemma 13.8, there are at most two circles (or lines) that are properly tangent to three circles c_1, c_2, c_3. Since $\gamma_4, \gamma_5, \gamma_6$ are all properly tangent to c_1, c_2, c_3, some two of $\gamma_4, \gamma_5, \gamma_6$ coincide. Suppose $\gamma_4 = \gamma_5$. Then $\gamma_4 (= \gamma_5)$ is properly tangent to all circles in $\mathcal{F}$, and hence (i) holds. □

13.7 Exercises

13.1 Give an example of a bicolored set in $\mathbb{R}^2$ in which the common-tangent distance does not satisfy the triangle inequality.

13.2 Using Lemma 13.2, prove Corollary 13.1.

13.3 Let c_1 be a red circle of unit radius centered at $(-2, 0)$, and c_2 be the blue circle of unit radius centered at $(2, 0)$. Let φ be the inversion of the plane with respect to the unit circle centered at $(3, 3)$. What is the common tangent distance $t(\varphi(c_1), \varphi(c_2))$?

13.4* Prove the following assertion: If all circles in a bicolored set $\mathcal{F}$ of four circles, each having positive radius, intersect at a common point, then $\det SD(\mathcal{F}) = 0$.

13.5* Suppose that two circles c_1, c_2 of the same color intersect with angle θ, see Fig. 13.8. Prove that

$$\frac{t(c_1, c_2)^2}{r_1 r_2} = 2(1 + \cos\theta),$$

where r_1, r_2 are radii of c_1, c_2, respectively.

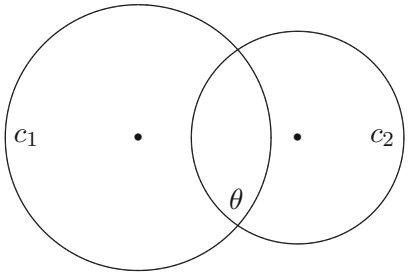


Fig. 13.8 Two circles intersecting with angle θ

13.8 Notes

Although Ptolemy's theorem was already discussed in Sect. 12.5, we present the literature around it here in Chap. 13. We do this due to its proximity to Casey's theorem, which gets also a broad representation in the notes here.

13.8.1 Notes on Ptolemy's Theorem

Ptolemy's theorem from planar Euclidean geometry, going back to the first book of Claudius Ptolemy's *Almagest* (see also our Sect. 12.5 here), and its converse can be formulated together as follows: *a quadrilateral is cyclic* (i.e., inscribed to a circle) *iff the product of the lengths of its diagonals equals the sum of the products of the lengths of the opposite pairs of sides*. Directly related is *Ptolemy's inequality* which is an equality exactly when the four considered vertices of a quadrilateral form a cyclic quadrilateral (otherwise the product of the diagonal lengths is smaller). Ptolemy's result can be found at many places in the literature. As already said, we discuss notes on Ptolemy's theorem in this chapter here since it is a special case of and a good starting point for *Casey's theorem*, which is the main topic of Chap. 13.

Some History

Due to its popularity, Ptolemy's theorem is mentioned in various books and surveys on *history of mathematics*. For a more detailed discussion, one might look at the chapter on trigonometry in the monograph [309] of González-Velasco, where also applications and further work of Ptolemy are studied. Another interesting historical reference is the recent expository paper [418] of Ibragimov and Suceava, discussing *Ptolemy's theorem and Ptolemy's inequality*. Also applications are given there, and Ptolemy's inequality is discussed in the context of several *metrics* (such as, e.g., the taxicab one). Furthermore, the authors discuss in an informative way Ptolemy's work *Almagest*.

Due to Tanaka et al. [772], Ptolemy's theorem can also be found in the *Japanese temple geometry*. And Gupta [350] presents extensions of Ptolemy's theorem (with simple proofs) due to *Indian mathematicians*.

Book Chapters and Surveys

We will mention first the following books where Ptolemy's theorem is nicely presented (the list is certainly not complete): Altshiller-Court [28, pp. 127–134], Johnson [430, Chapter IV], Perfect [630, Chapter 3], Coxeter and Greitzer [186, pp. 42–43], Eves [250, Section 3.8], Berger [75, 10.9.2], Pedoe [625, Section 24.1], Bottema [117, Section 22], Borceux [103, Section 4.13], Aumann [48, Section 6.1], as well as Posamentier and Geretschläger [644, Chapter 2].

Further books containing Ptolemy's theorem and related topics are [358] due to Hahn, based on *complex numbers*, and the book [326] of Grigorieva collecting *elementary geometric problems of competition type* for high school or college students. And also in the relatively recent books [25, Chapter 2] and [26, Chapter 1] by Alsina and Nelsen, showing the rich geometry of *quadrilaterals and polygons*, Ptolemy's theorem is studied.

Extensions of Ptolemy's Theorem

The motivation for Bowman's paper [119] goes back to Möbius who studied the problem of finding, with given side-lengths, all cyclic configurations of polygons. To get progress in this question, Bowman uses repeated application of Ptolemy's theorem. Various *elliptic analogues* of theorems from elementary geometry are proved in [342], among them Ptolemy's theorem. Valentine (see [790] and [791]) presents *spherical and hyperbolic variants of Ptolemy's theorem*. Bilinski [89] develops a general concept containing, e.g., *Ptolemy functions and Ptolemy matrices*, to unify approaches to many (at first glance seemingly independent) theorems from elementary geometry. These are, e.g., the theorems of Monge, Möbius, Kolmogorov, Ptolemy, and Casey.

Using *systems of forces being in equilibrium*, Jacob [426] derives elegantly the theorems of Ptolemy, Möbius, Neuberg and Pompeiu getting cross-connections to further theorems. His starting point is a theorem due to Mitrinović, Pécarić and Volenec which expresses the cyclic position of four points in terms of a suitable accompanying triangle; it is degenerate iff the four points lie on a circle. Smith [744] extends the triangle inequality to an alternating inequality for each cyclic polygon with an odd vertex number, where equality holds for an even number. Via *inversion* the triangle inequality is transformed into Ptolemy's inequality, and this is extended to *cyclic polygons*, yielding a generalization of Ptolemy's theorem. The author studies the analogous results in Minkowski planes, and he continues these investigations in [745] showing the connections between Euclidean and Minkowski geometry based on *Ptolemy's theorem with a Minkowski metric*.

Alsina et al. [27] give a characterization of those bijections that preserve the property of quadrilaterals to be cyclic, and they extend this to *real inner product spaces*. Popescu [642] gives several extensions of Ptolemy's theorem to sets consisting of *more than four points*. Gregorac [323] studies a *Feuerbach and a Möbius analogue of Ptolemy's theorem*, and he extends this (using simplex volumes instead of triangle areas) to *n dimensions*. Here the proof is based on *inversion* in $(n-1)$-spheres and a determinant rule for the n-dimensional version of *Heron's formula*. A determinant is used which disappears iff $n+2$ considered points lie in an $(n-1)$-sphere or a hyperplane; several interesting special cases are discussed, too.

Askey [42] uses *complex numbers* to extend Ptolemy's theorem from cyclic quadrilaterals to general ones. The same author discusses in [43] *Brahmagupta's extension* of Ptolemy's theorem to find (in the cyclic quadrilaterals) the lengths of the diagonals, of the segments generated by the intersection points of the diagonals, and the sides. This is extended also to the figures obtained by prolongation of the opposite sides until they meet, etc.; the author gives nice proofs.

Interesting Proof Methods

The papers [642] and [323] discussed already above elegantly use *inversion arguments* to prove their theorems. And Askey [42] successfully uses *complex numbers*.

Pech [622] proves several theorems around Ptolemy's theorem via computations based on the theory of *automated geometry theorem proving*, using *Gröbner bases*. The novelty is to use this method for proving *geometric inequalities*.

It is known that cyclic quadrilaterals have many geometric properties which have the character of necessary conditions. The fact that most of them are also sufficient is not so much taken care of. Josefsson (see [432] and [433]) presents altogether 34 *characterizations* with proofs based on very different tools. The paper [274] of Fraivert et al. is similar in nature.

Applications

Motivated by Ptolemy's theorem and the Brahmagupta formula, Khimshiashvili [451] studies *cyclic polygons* in relation to *polygonal linkages*, also in view of mathematical models of mechanical and chemical systems. The investigations in [451] are based on a new aspect, namely to interpret cyclic polygons as critical points of certain functions on the moduli space of the respective polygonal linkage.

Percy and Rogers [629] nicely discuss the relations between the *Fermat-Torricelli problem* and Ptolemy's theorem, including also a historical survey.

Schwarz and Smith [720] study *cyclic quadrilaterals* having correspondingly the same four side lengths a, b, c, and d, but different order of the corresponding sides. Namely, given such a quadrilateral, there are in general two other ones with the same side lengths in different order. Their circumradius R and area remain, and

the lengths of their diagonals can reach altogether only three values: p, q, and r. It is proved that for any of the three configurations the following relations hold: (1) $(p+q+r)-(a+b+c+d) = (abcd)/tR^2$, where $t := a+b+c+d+p+q+r$; (2) $a+b+c+d < p+q+r$; (3) $(p^2+q^2+r^2)-(a^2+b^2+c^2+d^2) = (abcd)/R^2$, implying $a^2+b^2+c^2+d^2 < p^2+q^2+r^2$. E.g., for proving (1) they use three applications of Ptolemy's theorem.

13.8.2 Notes on Casey's Theorem

Ptolemy's theorem on cyclic quadrilaterals is a natural starting point for considering Casey's theorem.

The Theorem Itself and Some History

Ptolemy's theorem was generalized by John Casey (cf. [142] and [143]) who replaced the vertices of Q by circles tangent (outside or inside) to C at the vertices of Q (see also Taylor [773]). More precisely, Casey proved the following statement, usually called *Casey's theorem*: *Four circles are tangent to one circle if and only if the sum of the products of the two pairs of opposite tangential distances of these four circles equals the product of the diagonal tangential distances*. For example, one can take in all cases (sides and diagonals) outer tangents throughout. Thus, the property "four points lie on a circle" (Ptolemy) is analogous to "four circles are tangent to a common circle" (Casey) if we replace points by circles and the usual distance of points suitably by the notion of tangential distance of circles.

Like, e.g., the Malfatti problem and various further theorems and problems, Casey's discovery is also an example from the set of geometric theorems which were found, independently and earlier than by Europeans, in the Japanese Temple Geometry of the Edo period. For interesting comments, proofs, applications, and connections to related theorems we refer to Hayashi [386], Mikami[573], as well as Fukagawa and Pedoe [284, Section 3.3]. More precisely, the theorem was known and used already in 1874, as a Sangaku problem from the Gunma prefecture. Moreover, the Japanese mathematician Chochu Siraishi knew it already in 1820.

Meanwhile Casey's theorem is well-established and discussed in various monographs. E.g., one can find it (differently proved and applied) in the following books: Coolidge [177, pp. 37–38], Johnson [430, p. 124], Eves [249, p. 129], Bottema [117, Chapter 22], Yaglom [822, pp. 69–71], and Aumann [48, pp. 121–123]. It is also cited in the encyclopedia [805], under "Casey's theorem", and it even found access to classroom geometry, as the two short and nicely written articles [739] and [821] show. In the first paper six problems are treated which are based on tangential distances of circles inscribed into another circle *in various configurations*, and in the second article historical background and applications of the theorems of Ptolemy, Casey, and Feuerbach are given.

Extensions, Modifications, and Applications

In view of the tangency notion defined for *oriented circles* (cycles), Zacharias [829] presents Casey's theorem in a more rigorous form. Casey's theorem has also many *applications*, e.g. with respect to *Feuerbach circles* (cf. [664] and also [343], where many relations between Ptolemy's, Casey's and further basic theorems are derived). Also Duparc [229] lists interesting applications of Casey's theorem. Finding it among other nice theorems from elementary geometry, the same author shows in [230] how this theorem is intertwined with the others.

Bilinski [89] introduces certain incidence configurations and matrices in projective n-space to study comprehensively their subcases, which include an impressive list of classical theorems from elementary geometry, among them theorems of Euler, Monge, Möbius, Ptolemy, and Casey. *Higher-dimensional versions* of Casey's theorem are given by Abrosimov and Aseev [3] and by Maehara and Martini [535].

There exist also numerous extensions to non-Euclidean geometries: Kubota [478] gives a *hyperbolic version of Casey's theorem*, and an *elliptic analogue* is derived by van Gruting [342]. Kurnik and Volenec (see [482] and [483]) obtain non-Euclidean versions extended to an arbitrary number of circles; these can be also horo- or hypercycles in the hyperbolic case. *Hyperbolic analogues* are also due to Abroshimov and Mikaiylova [4] as well as Kostin and Kostina [475], in the first paper additionally considering *spherical geometry*, and in the second *pseudo-Euclidean and pseudo-hyperbolic geometry*. Abrosimov and Mednykh [5] present various interesting results referring to area and volume in spherical and hyperbolic geometries. Among them, they also present *spherical and hyperbolic analogues of Casey's theorem*.

A *bipartite set of spheres in n-space* is a set of colored $(n-1)$-spheres, where two colors are used, no sphere is contained in the closed ball bounded by another sphere in the set, and spheres of different colors are disjoint. For any two such spheres, their *common-tangent distance* is the distance between two tangent points in a common tangent hyperplane, where an outer common tangent hyperplane is taken if the two spheres have the same color (otherwise, a common inner tangent hyperplane is taken). Due to this distance type, a bipartite set becomes a *semi-metric space*. E.g., Casey's theorem presents a condition for a bipartite set of four circles in the plane having a circle which is suitably tangent to all these circles of a bipartite set. Based on Ptolemy configurations in n-space obtained via Cayley-Menger determinants and among various further results, Maehara and Martini [535] derive a Casey-type extension of the n-dimensional Ptolemy configuration, referring to bipartite families of $n+2$ spheres in n-space. And this is then further extended to *bipartite sphere families of arbitrarily large cardinality*.

Chapter 14
Solutions to the Selected Exercises

Obvious is the most dangerous word in mathematics.

— Eric Temple Bell

1.4* Let $\boldsymbol{a}, \boldsymbol{b}$ be the centers and r_A, r_B be the radii of the disjoint circles A, B, respectively. Since B lies in the interior of A, we have $r_A > r_B$. Let C be a circle that is internally tangent to A and externally tangent to B. Let $\boldsymbol{x}$ be the center of C and r be the radius of C. Then

$$|\boldsymbol{ax}| = r_A - r,\ |\boldsymbol{bx}| = r_B + r.$$

Hence $|\boldsymbol{ax}| + |\boldsymbol{bx}| = r_A + r_B = const$. Therefore the trace of $\boldsymbol{x}$ is an ellipse with foci $\boldsymbol{a}, \boldsymbol{b}$. □

1.7* Let Π be a regular icosahedron inscribed in a sphere $\mathbb{S}$, and let V be the vertex set of Π. Note that each vertex of Π has degree five. For each $x \in V$, let $C(x, r)$ be the spherical cap on $\mathbb{S}$ with center x and angular radius r. If r is very small, then the spherical caps $C(x, r), x \in V$, are disjoint. When we increase r continuously, there is a moment that each spherical cap $C(x, r)$ touches its *neighboring* caps. At this moment, the boundary circles $\partial C(x, r), x \in V$, form a family of circles on $\mathbb{S}$ such that each circle is tangent to exactly five other circles in the family. Let $p \in \mathbb{S}$ be a point that is not covered by any spherical cap, and let φ be the stereographic projection of $\mathbb{S}$ from p to the plane that is tangent to $\mathbb{S}$ at p^*. Then, $\{\varphi(\partial C(x, r)) : x \in V\}$ is a family of circles in which each circle is tangent to exactly five other circles in the family. □

2.4* By Soddy's formula, $b(\tau_1)$ and $b(\tau_2)$ are the solutions of the quadratic equation in X:

$$\left(X + \sum_{i=1}^{4} b(\sigma_i)\right)^2 = 3\left(X^2 + \sum_{i=1}^{4} b(\sigma_i)^2\right).$$

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H. Maehara, H. Martini, *Circles, Spheres and Spherical Geometry*,
Birkhäuser Advanced Texts Basler Lehrbücher,
https://doi.org/10.1007/978-3-031-62776-7_14

Rewriting this equation, we have

$$2X^2 - 2\left(\sum_{i=1}^{4} b(\sigma_i)\right) X + 3\sum_{i=1}^{4} b(\sigma_i) - \left(\sum_{i=1}^{4} b(\sigma_i)\right)^2 = 0.$$

Then, by Vieta's formula, we obtain

$$b(\tau_1) + b(\tau_2) = b(\sigma_1) + b(\sigma_2) + b(\sigma_3) + b(\sigma_4).$$

□

2.5* Since $\sigma_1, \sigma_2, \alpha, \beta$ are mutually tangent circles and $\sigma_3, \sigma_2, \alpha, \beta$ are also mutually tangent circles, we can see, by Descartes' formula, that $b(\sigma_1)$ and $b(\sigma_3)$ are the solutions of the following quadratic equation in X:

$$(X + b(\sigma_2) + b(\alpha) + b(\beta))^2 = 2(X^2 + b(\sigma_2)^2 + b(\alpha)^2 + b(\beta)^2).$$

Rewriting this equation we have

$$\begin{aligned} &X^2 - 2\left(b(\sigma_2) + b(\alpha) + b(\beta)\right) X \\ &\quad + 2b(\sigma)^2 + 2b(\alpha)^2 + 2b(\beta)^2 - \left(b(\sigma_2) + b(\alpha) + b(\beta)\right)^2 = 0. \end{aligned}$$

Then, by Vieta's formula, we get

$$b(\sigma_1) + b(\sigma_3) = 2\left(b(\alpha) + b(\beta) + b(\sigma_2)\right).$$

Similarly, from the mutually tangent four circles $\sigma_i, \sigma_3, \alpha, \beta$ for $i = 2, 4$, we have

$$b(\sigma_2) + b(\sigma_4) = 2\left(b(\alpha) + b(\beta) + b(\sigma_3)\right).$$

By eliminating $b(\alpha) + b(\beta)$ from these two, we have $b(\sigma_1) + 3b(\sigma_3) = b(\sigma_4) + 3b(\sigma_2)$. □

3.5* Let v, e, f denote the numbers of vertices, edges, and faces, respectively, of a quadrangulation G. Noting that the minimum degree of G is at least 3, we have

$$3v \leq 2e = 4f.$$

Hence $e = 2f$ and $3v \leq 4f$. By Euler's formula, one obtains

$$2 = v - e + f = v - 2f + f = v - f,$$

and hence $v = f + 2$. Thus $3v = 3f + 6 \leq 4f$, from which we have $6 \leq f$. Therefore, $v = f + 2 \geq 6 + 2 = 8$. □

3.7* Since G is a 3-connected plane graph, its radial graph $R(G)$ is an irreducible quadrangulation. Let $\mathcal{F}$ be an OCR of $R(G)$. For four circles of $\mathcal{F}$ that meet at a point, connect the centers of each pair of tangent circles by a line-segment. (If one circle is a special circle C_ω, then take, instead of a line segment, a half line emanating from the center of the circle that is internally tangent to C_ω in the direction to the contact point.) Then each pair of intersecting line-segments (or a line segment and a half line) crosses orthogonally. Such a pair of line-segments corresponds to an edge of G and its dual edge of the dual graph G^*. Thus, we can get a desired drawing of G and G^*. □

Remark This question on an *orthogonal drawing* of G and G^* was first asked by W. T. Tutte in 1963 ([785, §13]), and it was answered by G. R. Brightwell and E. R. Sheinerman [128].

4.4* Let α, β, γ be the interior angles of the spherical triangles $\breve{\triangle}ABC$ at A, B, C respectively. Then the interior angles of $\breve{\triangle}A^*B^*C$ at A^*, B^*, C are $\pi-\alpha, \pi-\beta, \gamma$ respectively. Now

$$\begin{aligned}[ABC] > \pi &\Leftrightarrow \alpha + \beta + \gamma - \pi > \pi \\ &\Leftrightarrow \gamma > (\pi - \alpha) + (\pi - \beta) \\ &\Leftrightarrow \overset{\frown}{A^*CB^*} \text{ is a minor arc.}\end{aligned}$$

□

4.5* If $P \in X^\circ$, then, since $X^\circ = \bigcap_{Q \in X} H(Q)$, we have $PQ \leq \pi/2$ for all $Q \in X$. Hence $X \subset H(P)$. Then $conv(X) \subset H(P)$, and hence $conv(X) \subset X^{\circ\circ}$.

If $X \subset H(P)$, then $X^\circ \supset H(P)^\circ = \{P\}$, and hence $X^{\circ\circ} \subset H(P)$. Therefore, $X^{\circ\circ} \subset conv(X)$. □

4.6* Let $r > \rho$ be the radius of the sphere, O the center of the sphere, Z the point on the sphere where the needle of the compass is fixed, and P be a point on the circle. Let $\theta = \angle POZ$. Then, by Example 4.1, the area of the cap is equal to $2\pi r^2(1 - \cos\theta)$. Since $1 - \cos\theta = 2\sin^2(\theta/2)$, and $\rho = 2r\sin(\theta/2)$, the area of the cap is

$$2\pi r^2(1 - \cos\theta) = 2\pi r^2 2\sin^2(\theta/2) = \pi\rho^2,$$

which is independent of r. □

5.1* Consider the spherical triangle ABN, where N is the north-pole. Then $AN = BN = \pi/2 - \pi/4 = \pi/4$ and $\angle ANB = \pi/2$. Applying the spherical cosine law we have

$$\cos AB = \cos(\pi/4)\cos(\pi/4) - 0 = 1/2.$$

Hence $AB = \pi/3$. □

5.2* By the spherical cosine law,

$$\cos d = \cos^2 d + \sin^2 d \cos\theta = \cos^2 d + (1 - \cos^2 d)\cos\theta.$$

From this we have

$$(\cos d - 1)((1 - \cos\theta)\cos d - \cos\theta) = 0.$$

Since $0 < d$, we have $\cos d = \frac{\cos\theta}{1-\cos\theta}$, and $\cos\theta = \frac{\cos d}{1+\cos d}$. □

5.3* Let $\beta = \angle B$, $\gamma = \angle C$. Let φ be the stereographic projection of $\mathbb{S}$ from A^* to the plane Π that is tangent to $\mathbb{S}$ at A.

(i) Let Λ be the spherical cap with center A and radius AC. Then B lies in the interior of Λ. The image $\varphi(\Lambda)$ is a disk in the plane Π with center $\bar{A} = A$, $\bar{B}$ lies in the interior of the disk $\varphi(\Lambda)$, and $\bar{C}$ lies on the boundary of this disk. Hence $|\bar{A}\bar{C}| > |\bar{A}\bar{B}|$ and so $\angle\bar{B} > \angle\bar{C}$ by the Proposition 18 of Euclid's Elements. Since the angles of the Cesàro triangle $\bar{A}\bar{B}\bar{C}$ at $\bar{B}, \bar{C}$ are $\beta - \varepsilon, \gamma - \varepsilon$, respectively, it follows that $\beta - \varepsilon > \gamma - \gamma$. Hence $\beta > \gamma$.

(ii) can be proved similarly, and then (iii) becomes obvious. □

5.4* It is sufficient to consider the case that X, Y lie on the boundary of $\breve{\triangle}ABC$. Suppose X lies on AB and Y lies on AC, see Fig. 14.1. Note that one of the angles $\angle AXY$, $\angle BXY$ is not less than $\pi/2$. If $\angle BXA \geq \pi/2$, then in $\breve{\triangle}AXY$, $\angle A \leq \angle X$ implies $XY \leq AY$ by Exercise 5.3*. And since $AY < AC$, we have $XY < AC$. Suppose that $\angle BXY \geq \pi/2$. In the triangle $\breve{\triangle}BXY$, we have $\angle BXY > \angle XBY$. Hence $XY < BY$. By a similar argument, we have $AY < \max\{AB, BC\}$. Hence $XY < \max\{AB, BC, CA\}$. □

5.6* Let $ABCD$ be a convex quadrilateral on the sphere $\mathbb{S}$ with $AB = AD = a$, $BC = DC = b$. Then $[ABCD] = [ABC] + [ADC] = 2[ABC]$. Hence

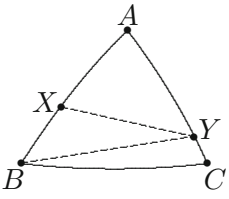


Fig. 14.1 Exercise 5.4*

$[ABCD]$ becomes the maximum when $[ABC]$ becomes the maximum. By Theorem 5.7 (2), $[ABC]$ becomes the maximum when $\overset{\frown}{ABC}$ is a semicircle. Hence $[ABCD]$ becomes the maximum when it is inscribed in a spherical cap. □

6.2* Suppose that there is a quadrangulation G with 9 vertices. Let e, f be the numbers of edges, faces, respectively. Then $2e = 4f$, and hence $e = 2f$. By Euler's formula, $2 = 9 - e + f = 9 - f$, and $f = 7$. Therefore, $e = 14$. Since the degree of each vertex is at least 3, we can deduce, by considering the hand-shaking lemma, that just one vertex has degree 4, and the remainig 8 vertices have degree 3. Let us try to draw such a quadrangulation G on the plane. Let A be the vertex with degree 4 and $ABCD$ be a quadrangular face of G. We may assume that $ABCD$ is not the outer face of G. Then, from A two edges must emanate, and from each of B, C, D, one edge must emanate. Let AA_1, AA_2, BB', CC', DD' be such edges in the counter-clockwise order. It is obvious that none of A_1, A_2, B', C', D' belongs to $\{A, B, C, D\}$. It is also not difficult to see that A_1, A_2, B', C', D' are all different. Since all faces are quadrangles, A_1B', $B'C'$, $C'D'$, $D'A_2$ must be edges of G, see Fig. 14.2. Then, since 13 edges appeared so far, there remains just one edge we can add. Since A_1, A_2 are not adjacent, we need (to make their degrees 3) two more edges, which is impossible. Therefore, it is impossible to draw a quadrangulation with nine vertices. □

6.4* Suppose that two balls of radius 2 touch one unit ball with center O at P, Q. Then, from

$$4^2 \geq 3^2 + 3^2 - 2 \times 3^2 \cos \angle POQ,$$

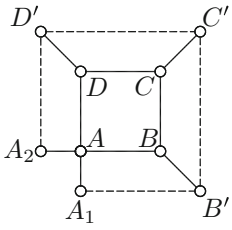
we have $\angle POQ \geq d := \cos^{-1}(1/9)$. Hence the spherical distance between any two contact points on the boundary sphere $\mathbb{S}$ of the unit ball is at least

$$d := \cos^{-1}(1/9) \approx 1.45945.$$

By Exercise 5.2*, the interior angle θ of a $\breve{\triangle}(d, d, d)$ is given by

$$\theta = \cos^{-1}\left(\frac{\cos d}{1 + \cos d}\right) = \cos^{-1}(1/19) \approx 1.47063.$$

Fig. 14.2 Exercise 6.2*



Hence the area of an equilateral spherical triangle with edge-length d is

$$3\theta - \pi \approx 1.270.$$

Let n be the maximum value of the number of balls of radius 2 that can simultaneously touch one and the same unit ball. Let $\mathcal{X}$ be the set of contact points of n balls with the boundary $\mathbb{S}$ of the unit ball. Then $\mathcal{X}$ is not contained in a hemisphere. (Otherwise, another ball of radius 2 can touch the unit ball, contradicing that n is the maximum number. Hence the convex hull $conv(\mathcal{X})$ of $\mathcal{X}$ is a convex polyhedron containing the center of $\mathbb{S}$. Triangulate each face of $conv(\mathcal{X})$ by adding diagonals, if necessary, and project each triangle, from the center O onto $\mathbb{S}$. Then $\mathbb{S}$ is divided into spherical triangles. Let f be the number of spherical triangles and e be the number of *edges* in this triangulation of $\mathbb{S}$. Note that each spherical triangle is inscribed in a spherical cap with angular radius less than d. (Otherwise, another ball of radius 2 can touch at the center of the circumscribed spherical cap, contradicting the maximality of n.) Then, by Fejes Tóth's Lemma, each spherical triangle has area $\geq [\breve{\triangle}(d, d, d)] \approx 1.27$. Since the sum of the areas of f spherical triangles is equal to 4π, and each spherical triangle has area at least $[\breve{\triangle}(d, d, d)]$, we have $f \leq 4\pi/1.27 \approx 9.894$. Thus $f \leq 9$. If $f = 9$, then $2e = 3f$ which contradicts the fact that e is an integer. Hence $f \leq 8$. If $f = 8$, then from Euler's formula we have $4 = 2n - 2e + 2f = 2n - f$, and $n = (8 + 4)/2 = 6$. Thus, $n \leq 6$. On the other hand, six balls of radius 2 with centers at

$$(\pm 3, 0, 0), (0, \pm 3, 0), (0, 0 \pm 3)$$

are disjoint balls that touch the unit ball with center $O = (0, 0, 0)$. Hence $n = 6$. □

7.2* Consider the spherical quadrilaterals $ABCP, ABQC$. It is clear that $[ABP] = \frac{1}{2}[ABCP] = [ABC]$. Hence $[ABP] = [ABC]$. Similarly, we have $[ABQ] = [ABC]$. Thus, P, C, Q lie on the Lexell arc $\overset{\frown}{A^*CB^*}$. Therefore, $\overset{\frown}{PCQ}$ is a subarc of $\overset{\frown}{A^*CB^*}$. □

7.3* Let $\breve{\triangle}ABX$ be a spherical triangle such that $AB = a, BX = b, XA = c$. Let K_A, K_B be the circles on $\mathbb{S}$ of spherical radius a with centers A and B, respectively. Let $C \in \mathbb{S}$ be the intersection point of K_A and K_B lying in the same side of AB as X. Let K_C be the circle on $\mathbb{S}$ with spherical radius a and center C. Let P be the intersection point of K_A, K_C different to B, and Q be the intersection point of K_C, K_B different to A. Since the circumradius of $\breve{\triangle}(a, b, c)$ is smaller than a, the vertex X lies in the interior of K_C (that is, the same side of K_C as C), and since $a \leq b \leq c$, X lies in the exterior of K_A and in the exterior of K_B. Since $b \leq c$, the edge AX must intersect the minor

arc $\widehat{PC}$ of K_A. Therefore, AX intersects the arc $\widehat{PCQ}$. Then, by Exercise 7.1, we have $[ABX] \geq [ABC] = [\breve{\triangle}(a, a, a)]$. □

7.6* Among the spherical regions with a given perimeter s, let Ω be one that has the maximum area. We may suppose that Ω is a convex region. (Otherwise, it would be possible to increase the area by deforming Ω with preservation of the perimeter.) Suppose that Ω is not a spherical cap. Then there are four points A, B, C, D (in this cyclic order) on the boundary of Ω such that the spherical quadrilateral $ABCD$ is not cyclic. The four segments AB, BC, CD, DA divide Ω into five regions: the quadrilateral $ABCD$ and four "lune-like" regions $\alpha, \beta, \gamma, \delta$ (some of them could be empty). Since the convex quadrilateral $ABCD$ is not cyclic, it has a proper diagonal. By increasing the length of this proper diagonal a little bit keeping four edge-lengths fixed, we can get a quadrilateral with a larger area by the proper-diagonal lemma. By attaching four regions congruent to $\alpha, \beta, \gamma, \delta$ to this new quadrilateral from the outside of this new quadrilateral, we have a region whose area is greater than Ω and whose perimeter is s. This contradicts the fact that the spherical region Ω is one that has the maximum area. □

8.2* Let $X = \{\boldsymbol{x}_i : i = 1, 2, 3, 4, 5\}$ be the set of five random points on $\mathbb{S}$ chosen independently. By Exercise 4.5*, we have $conv(X) = X^{\circ\circ}$. Then

$$
\begin{aligned}
&X \text{ is the vertex set of a convex pentagon} \\
&\quad \Leftrightarrow X^\circ \text{ is a spherical pentagon (by Theorem 4.8)} \\
&\quad \Leftrightarrow H(\boldsymbol{x}_1) \cap H(\boldsymbol{x}_2) \cap H(\boldsymbol{x}_3) \cap H(\boldsymbol{x}_4) \cap H(\boldsymbol{x}_5) \text{ is a pentagon.}
\end{aligned}
$$

Now let us take five random points in the following two-steps way: First choose five random great circles $\boldsymbol{G}_1, \ldots, \boldsymbol{G}_5$ independently. Next, from the pair of poles of each great circle $\boldsymbol{G}_i$, choose either one with equal probability. In this two-steps way, we can take five random points $\boldsymbol{x}_i, i = 1, 2, \ldots, 5$. Five great circles $\boldsymbol{G}_i, i = 1, 2, \ldots, 5$, divide $\mathbb{S}$ into 22 regions. To choose a pole of $\boldsymbol{G}_i$ corresponds to choosing one side of $\boldsymbol{G}_i$, and $X^\circ = H(\boldsymbol{x}_1) \cap \cdots \cap H(\boldsymbol{x}_5)$ is nonempty if and only if X° is one of the 22 regions divided by $\boldsymbol{G}_i, i = 1, \ldots, 5$. Since there are $2^5 = 32$ ways to choose poles, and among 22 regions, only two are pentagons, we can deduce that

$$\Pr(X^\circ \text{ is a pentagon}) = 2/2^5 = 1/16.$$

□

8.6* Let Γ be a semicircle of spherical radius ρ, and A, B be the endpoints of Γ. Then $AB = 2\rho$ and $conv(\Gamma)$ is bounded by $\Gamma \cup AB$. The length of Γ is $\pi \sin\rho$, and the perimeter of $conv(\Gamma)$ is $\pi \sin\rho + 2\rho$. Note that the probability that a random great circle $\boldsymbol{G}$ intersects $conv(\Gamma)$ in only one point is zero. A random great circle $\boldsymbol{G}$ cuts $conv(\Gamma)$ if and only if (i) $\boldsymbol{G}$ intersects Γ twice or (ii) $\boldsymbol{G}$ crosses AB. Since the events (i) and (ii) are exclusive, we

have

$$\Pr(\boldsymbol{G} \cap conv(X) \neq \emptyset) = \Pr((i) \text{ occurs}) + \Pr((ii) \text{ occurs}).$$

Therefore,

$$\begin{aligned}&\Pr(\boldsymbol{G} \text{ intersects } \Gamma \text{ in two points})\\ &\quad = \Pr(\boldsymbol{G} \cap conv(\Gamma) \neq \emptyset) - \Pr(\boldsymbol{G} \cap AB \neq \emptyset).\end{aligned}$$

By Crofton's formula we have

$$\Pr(\boldsymbol{G} \cap conv(\Gamma) \neq \emptyset) = \frac{\pi \sin\rho + 2\rho}{2\pi},\ \Pr(\boldsymbol{G} \cap AB \neq \emptyset) = \frac{2\rho}{\pi},$$

and hence

$$\Pr(\boldsymbol{G} \text{ intersects } \Gamma \text{ in two points}) = \frac{\pi \sin\rho - 2\rho}{2\pi}.$$

□

9.2* If $d \geq \pi/2$, then any two caps of radius d intersect. Hence, if $d \geq \pi/2$, then $K(2,2)$ is not a d-cap graph. Suppose $\pi/4 \leq d < \pi/2$. Let $w, x, y, z \in \mathbb{S}$ be four points lying on a great circle in this cyclic order and satisfying $wx = xy = yz = zw = \pi/2$. Then, the intersection graph of the four caps $C_d(w), C_d(x), C_d(y), C_d(z)$ is $K(2,2)$. Next, suppose that $0 < d < \pi/4$. Let w, x, y, z be four points lying on a circle K in this cyclic order such that $wx = xy = yz = zw = 2d$. Then the angular radius of K is greater than d. Hence the intersection graph of the four caps $C_d(w), C_d(x), C_d(y), C_d(z)$ is $K(2,2)$. Thus, for any $0 < d < \pi/2$, $K(2,2)$ is a d-cap graph. □

9.3* If $d \geq \pi/4$, then since $K(1,4)$ is not a d-cap graph, $K(2,4)$ is not a d-cap graph. Suppose $d < \pi/4$. Then for any d-independent two points x, y, $C_{2d}(x) \cap C_{2d}(y)$ cannot contain d-independent three points by Cororally 9.2. Hence $K(2,4)$ is not a d-cap graph. □

9.6* Let $\boldsymbol{x}, \boldsymbol{y}$ be two random points on the unit sphere $\mathbb{S}$ chosen independently, and $C(\boldsymbol{x}), C(\boldsymbol{y})$ be the caps of angular radius θ and with centers $\boldsymbol{x}, \boldsymbol{y}$, respectively. Then $C(\boldsymbol{x}), C(\boldsymbol{y})$ are two random caps. Let W be the area of $C(\boldsymbol{x}) \cap C(\boldsymbol{y})$. Then W is a random variable. Let $g(w)$ denote the density function of W. Then $E(W) = \int wg(w)dw$. Let $\boldsymbol{z}$ be a random point on $\mathbb{S}$. By the law of total probability, we have

$$\begin{aligned}\Pr(\boldsymbol{z} \in C(\boldsymbol{x}) \cap C(\boldsymbol{y})) &= \int \Pr(\boldsymbol{z} \in C(\boldsymbol{x}) \cap C(\boldsymbol{y}) | W = w) g(w) dw\\ &= \int \frac{w}{4\pi} g(w) dw = \frac{E(W)}{4\pi}.\end{aligned}$$

Let $C(\boldsymbol{z})$ be the spherical cap of angular radius θ with center $\boldsymbol{z}$. Then

$$\begin{aligned}\Pr(\boldsymbol{z} \in C(\boldsymbol{x}) \cap C(\boldsymbol{y})) &= \Pr(\boldsymbol{x}, \boldsymbol{y} \in C(\boldsymbol{z})) \\ &= \left(\frac{2\pi(1-\cos\theta)}{4\pi}\right)^2 = \frac{(1-\cos\theta)^2}{4}.\end{aligned}$$

Therefore, $E(W) = \pi(1-\cos\theta)^2$. □

9.7* Let $V(n, p) = \mathbb{S} \setminus$ (the union of n random caps). (Thus, $\alpha(n, p)$ is the area of $V(n, p)$.) For a random point $\boldsymbol{x}$ and n random spherical caps of area $4\pi p$ chosen independently on $\mathbb{S}$, let us compute the probability $\Pr(\boldsymbol{x} \in V(n, p))$ in two different ways.

First, let us regard that n random caps are chosen before we choose a random point $\boldsymbol{x}$. Since

$$\Pr(\boldsymbol{x} \in V(n, p) | U(n, p) = u) = u,$$

we have

$$\begin{aligned}\Pr(\boldsymbol{x} \in V(n, p)) &= \int \Pr(\boldsymbol{x} \in V(n, p) | U(n, p) = u) g(u) du \\ &= \int u g(u) du = E(U(n, p)).\end{aligned}$$

Next, let us regard that n random caps are chosen after a random point $\boldsymbol{x}$ is chosen. Then we may regard $\boldsymbol{x}$ as a fixed point, and $\boldsymbol{x} \in V(n, p)$ is equivalent to the event that the center of each random cap lies outside of the spherical cap of area $4\pi p$ with center $\boldsymbol{x}$. Hence

$$\Pr(\boldsymbol{x} \in V(n, p)) = (1-p)^n.$$

Therefore, $E(U(n, p)) = (1-p)^n$. □

10.1* Let $\{P_1', P_2', P_3'\} \subset \mathbb{R}^3$ be a trio that is isometric to $\{P_1, P_2, P_3\}$. Then every point at the same distance 3 from P_1' and P_2' lies on the plane E that perpendicularly bisects the line segment $P_1'P_2'$. Among the points on E at distance 3 from P_1', P_2', the one that is farthest from P_3' is at distance $\sqrt{5^2-(2.5)^2} + \sqrt{3^2-(2.5)^2} \approx 5.988$ from P_3'. Since $5.988 < 6$, there is no point P_4' on E such that $P_4'P_3' = 6$. Hence the quartet $\{P_1, P_2, P_3, P_4\}$ is not isometrically embeddable in $\mathbb{R}^3$. □

10.3* Let $y = \cos AD$. Since $\mathcal{X} := \{A, B, C, D\}$ lies on the unit sphere $\mathbb{S}$, by putting $x = 1$ in the equation $M_{\mathcal{X}}(x) = 0$, we have

$$\begin{vmatrix} 1 & 0 & 0 & y \\ 0 & 1 & 0 & 1/2 \\ 0 & 0 & 1 & 1/2 \\ y & 1/2 & 1/2 & 1 \end{vmatrix} = 0,$$

from which we get $y^2 = 1/2$. Hence $y = \pm 1/\sqrt{2}$, and $AD = \pi/4$ or $AD = 3\pi/4$. Since $ABCD$ is a convex quadrilateral, we have $AD = 3\pi/4$. □

10.4* The value of $s(\mathcal{X})$ is

$$\frac{1}{2}(\pi/2 + 3\pi/4 + \pi/3) = 19\pi/24 \approx 2.48709.$$

Hence $s(\mathcal{X})/\pi = 19/24 \approx 0.791667$. The equation $M_{\mathcal{X}}(x) = 0$ is given by

$$\begin{vmatrix} 1 & \cos(\pi/2x) & \cos(\pi/2x) & \cos(3\pi/4x) \\ \cos(\pi/2x) & 1 & \cos(\pi/2x) & \cos(\pi/3x) \\ \cos(\pi/2x) & \cos(\pi/2x) & 1 & \cos(\pi/3x) \\ \cos(3\pi/4x) & \cos(\pi/3x) & \cos(\pi/3x) & 1 \end{vmatrix} = 0.$$

By numerical computation, we have $r = 1$ and $r \approx 0.80608$, as solutions of $\det M_{\mathcal{X}}(x) = 0$ that satisfy $r \geq s(\mathcal{X})/\pi = 0.791667$. Hence $r \approx 0.80608$. □

11.2* From the approximated value of $V(n)$ for $n \leq 7$, we see that $V(5)$ is the maximum in $\{V(n) : 1 \leq n \leq 7\}$. Since $\Gamma(n/2 + 1) = (n/2)\Gamma(n/2)$, and $2\pi/n < 1$ for $n \geq 7$, we have

$$V(n)/V(n-2) = \frac{\pi^{n/2}/\Gamma(n/2+1)}{\pi^{n/2-1}/\Gamma(n/2)} = \frac{2\pi}{n} < 1.$$

Thus, $V(n) < V(n-2)$ for $n \geq 7$, and hence $V(n) < V(5)$ follows for $n > 7$. □

11.3*

$$\begin{aligned} s\Gamma(s) &= s\int_0^\infty z^{s-1}e^{-z}dz = \int_0^\infty (z^s)'e^{-z}dz \\ &= [z^s e^{-z}]_0^\infty - \int_0^\infty z^s(e^{-z})'dz = \int_0^\infty z^s e^{-z}dz = \Gamma(s+1). \end{aligned}$$

□

12.2*

$$\begin{vmatrix} -1 & 1 & 1 & \dots & 1 \\ 1 & -1 & 1 & \dots & 1 \\ 1 & 1 & -1 & \dots & 1 \\ \vdots & \vdots & & \ddots & \vdots \\ 1 & 1 & 1 & \dots & -1 \end{vmatrix} = \begin{vmatrix} -1 & 1 & 1 & \dots & 1 & 1 \\ 1 & -1 & 1 & \dots & 1 & 1 \\ 1 & 1 & -1 & \dots & 1 & 1 \\ \vdots & \vdots & & \ddots & \vdots & 1 \\ 1 & 1 & 1 & \dots & -1 & 1 \\ 0 & 0 & 0 & \dots & 0 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} -2 & 0 & 0 & \dots & 0 & 1 \\ 0 & -2 & 0 & \dots & 0 & 1 \\ 0 & 0 & -2 & \dots & 0 & 1 \\ \vdots & \vdots & & \ddots & \vdots & 1 \\ 0 & 0 & 0 & \dots & -2 & 1 \\ -1 & -1 & -1 & \dots & -1 & 1 \end{vmatrix} = \begin{vmatrix} -2 & 0 & 0 & \dots & 0 & 1 \\ 0 & -2 & 0 & \dots & 0 & 1 \\ 0 & 0 & -2 & \dots & 0 & 1 \\ \vdots & \vdots & & \ddots & \vdots & 1 \\ 0 & 0 & 0 & \dots & -2 & 1 \\ 0 & 0 & 0 & \dots & 0 & 1-\frac{k}{2} \end{vmatrix}$$

$$= (-2)^k(1-\tfrac{k}{2}).$$

□

12.6* Put $x = d(\boldsymbol{x}_2, \boldsymbol{x}_4)$. Since the Cayley-Menger determinant of $\{\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3, \boldsymbol{x}_4\}$ is zero, we have

$$0 = \begin{vmatrix} 0 & 1 & 9 & 4 & 1 \\ 1 & 0 & 9 & x^2 & 1 \\ 9 & 9 & 0 & 9 & 1 \\ 4 & x^2 & 9 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{vmatrix} = \begin{vmatrix} 0 & 1 & 9 & 4 & 1 \\ 0 & -1 & 8 & x^2-1 & 1 \\ 0 & 0 & -9 & 0 & 1 \\ 0 & x^2-4 & 5 & -4 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 9 & 4 & 1 \\ -1 & 8 & x^2-1 & 1 \\ 0 & -9 & 0 & 1 \\ x^2-4 & 5 & -4 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 9 & 4 & 1 \\ -2 & 7 & x^2-2 & 0 \\ -1 & -10 & -1 & 0 \\ x^2-5 & 4 & -5 & 0 \end{vmatrix}$$

$$= -\begin{vmatrix} 1 & 9 & 4 \\ -2 & 7 & x^2-2 \\ -1 & -10 & -1 \end{vmatrix} = x^2 - 85.$$

Therefore, $x = \sqrt{85}$. □

13.4* Let c_1, c_2, c_3, c_4 be four circles intersecting at a point p. Then these four circles have the same color, say red. Let us regard p as a colored point, say a blue null circle. Then, by performing isometric radii change to $\{c_1, c_2, c_3, c_4, \{p\}\}$, we can decrease the radii of $c_i, i = 1, 2, 3, 4$, by some small ε, and we get circles $c_i', i = 1, 2, 3, 4$. These circles are properly

tangent to a circle of radius ε with center p. Hence, by Theorem 13.1 (Casey's theorem) we have $SD(\{c_1', c_2', c_3', c_4'\})) = 0$. Since the isometric radii change does not change the common tangent distance, we have $SD(\mathcal{F}) = 0$. □

13.5* Let z_i be the center of c_i for $i = 1, 2$, and let p be an intersection point of c_1, c_2. Let l_i be the tangent lines of c_i at p. Then the intersection angle of l_1, l_2 is θ. Since the line $z_i p$ and l_i cross orthogonally, it follows that $\angle z_1 p z_2 = \pi - \theta$. Hence

$$|z_1 z_2|^2 = r_1^2 + r_2^2 - 2r_1 r_2 \cos(\pi - \theta).$$

Therefore,

$$\begin{aligned} t(c_1, c_2)^2 &= |z_1 z_2|^2 - (r_1 - r_2)^2 = -2r_1 r_2 \cos(\pi - \theta) + 2r_1 r_2 \\ &= 2r_1 r_2 (1 + \cos\theta). \end{aligned}$$

Thus

$$\frac{t(c_1, c_2)^2}{r_1 r_2} = 2(1 + \cos\theta).$$

□

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Index

Symbols

$B^n(r)$ (n-dimensional ball of radius r), 214
$C(v)$ (cap of center v), 181
$C_d(v)$ (cap of radius d and center v), 183
$H(P)$ (hemisphere with pole P), 85
$K(m, n)$ (complete bipartite graph), 186
$K_1 \perp K_2$ (orthogonally crossing), 3
K_n (complete graph with n vertices), 50
$M_{\mathcal{X}}(x)$ (a special matrix), 206
$N(v)$ (neighborhood of v), 50
$SD(\mathcal{X})$ (squared-distance matrix of $\mathcal{X}$), 240
$S_n(r)$ (boundary of $B^n(r)$), 214
$T_d(xyz)$ (Reuleaux d-triangle), 184
$V(n)$ (volume of $B^n(1)$), 216
X^* (antipodal point of X), 75
$[w, x, y, z]$ (Möbius invariant), 263
$\breve{\triangle}ABC$ (spherical triangle ABC), 80
$\breve{\mathbb{S}}$ (a modification of the 2-sphere), 77
$\det SD(\mathcal{X} : k)$, 271
$v[S_n(r)]$ (maximum cardinality of a dispersed set on $S_n(r)$), 223
$\partial C_d(v)$ (boundary of $C_d(v)$), 183
ρ_i (signed radius), 245
$\Gamma(s)$ (gamma function), 217
$\Gamma(\mathcal{X})$ (Cayley-Menger matrix of $\mathcal{X}$), 240
$\Omega(\mathcal{C})$ (intersection graph of $\mathcal{C}$), 181
$|\boldsymbol{xy}|$ (the length of the line segment connecting x to y), 1
$b(\sigma)$ (bend of σ), 31
d-cap graph, 186
d-independent, 183
$e(G)$ (number of edges of G), 53
$f(G)$ (number of faces of G), 53
n-star $K(1, n)$, 186
$r(\sigma)$ (radius of σ), 31
$s(\Delta)$ (semi-perimeter of Δ), 207
$s(\mathcal{X})$, 207
$t(c_i, c_j)$ (common tangent distance), 261
$v(G)$ (number of vertices of G), 53
$\mathbb{S}$, $\mathbb{S}(r)$ (sphere of radius r), 75
$\mathbb{R}^n$ (n-dimensional. Euclidean space), 213
$\mathcal{X}^\circ$ (polar set of $\mathcal{X}$), 85

A

Abresch, U., 120
Abroshimov, N.V., 277
Adachi, T., 121
Addition of a bypass, 52
Addition of a pointed edge, 52
Adjacency relation, 49
Aeppli, A., 43
Agahi, H., 200
Agaoka, Y., 96
Agarwal, P.K., 66, 69
Agricoly, I., 92
Aigner, M., 136
Akama, Y., 96
Akopyan, A., 19
Al-Afifi, C., 236
Aleksandrov, A.D., 118, 119
Aleksandrov, P.S., 17
Aleksandrov-Toponogov theorem, 118
Alencar, J., 253
Alexander, R., 254
Alfakih, A., 70, 253
Almohammad, S.M., 73

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Alonso, J., 7
Alsina, C., 18, 274, 275
Altshiller-Court, N., 16, 18, 22, 274
Amann, A., 26
Andalafte, E.Z., 212
Andreev, E, 66
Angular radius, 79
Anstreicher, K.M., 134, 136
Antipodal point, 75
Aomoto, K., 256
Apollonian circle packings, 44
Apollonian group, 45
Apollonius, 16, 21
Apollonius circle, 25
Apollonius problem, 25, 46
Arbeiter, E., 173
Archimedes, 21, 78
Armitage, D.H., 97
Aseev, V.V., 277
Askey, R., 275
Asplund, E., 6
Asymptotic probability, 193
Atanasian, L.S., 17
Athen. H., 116
Atzema, E.J., 157
Aumann, G., 17, 18, 23, 46, 274, 276
Avelino, C.P., 96
Average kissing number, 140
Axler, S., 198

B
Bachoc, C., 138
Bakelman, I.J., 16
Balestro, V., 229, 230
Balk, M.B., 232
Ball, K., 227
Ballicioni, A., 230
Baragar, A., 48
Baralić, D., 19
Bárány, I., 174, 175
Barden, D., 171, 173
Barnes, J., 18, 20
Barnette, D., 68, 71, 72
Barrett, A.N., 171
Bartha, F.A., 227
Barthe, F., 227, 233
Baston, V.J.D., 230
Bayer, M.M., 72
Beecroft, P., 43
Bédaride, N., 236
Belotti, M., 73
Bend, 31
Benson, R.V., 230
Benz, W., 25
Berger, A., 48
Berger, M., 7, 16, 23, 27, 44, 46, 90, 92, 94, 117, 119, 141, 158, 230, 274
Berggren, J.L., 22
Bergold, H., 228
Besau, F., 175
Bezdek, K., 137, 141
Bhattacharya, A., 254
Bicolored set of circles, 261, 266
Bigalke, H.-G., 23, 92, 94, 117
Bilet, V., 257
Bilinski, S., 274, 277
Billiard knots, 28
Billiards, 27, 28, 236
Bipartite graph, 55, 186
Birational geometry, 25
Björnbo, A.A., 115
Blair, D.E., 16
Bland, J.R., 116
Blaschke, W., 22, 175
Blocking number, 137
Blumenthal, L.M., 89, 212, 241, 251, 252, 254, 258
Bodies of constant width, 90, 98, 185, 237
Bokowski, J., 72
Boltyanski, V., 116, 230
Boltyanski, V.G., 232
Bonnet, G., 175
Borceux, F., 17, 23, 89, 117, 274
Boreland, B., 17
Böröczky, K., 134, 136, 140, 232, 234
Böröczky, K.J., 173
Borsuk, K., 22
Bos, H.J.M., 26, 43
Bottema, O., 251, 274, 276
Boundary edge, 52
Boundary graph of a face, 53
Bourgain, J., 46
Bowman, F., 274
Boyvalenkov, P., 138
Brakensiek, J., 229
Brandstädt, A., 197
Brass, P., 69, 139, 259
Brauchart, S., 174
Breda, A.M.R., 96
Brightwell, G.R., 63, 69
Brocard porism, 29
Bruen, A.A., 47
Budny, K., 200
Burago, D., 118
Burago, Yu., 118, 122
Burkholder, E.F., 117
Byer, O.D., 19

C
Caddeo, R., 93
Calinger, R., 157, 158
Campos, M., 139
Cap graph, 186
Carne, T.K., 171, 173
Cartography, 24, 93
Casey, J., 157, 276
Casey's theorem, 262, 276
Casselman, B., 97, 134, 138
Catalan, E.C., 157
Cayley, A., 251
Cayley-Menger determinant, 240
Cayley-Menger matrix, 240, 251
Cecil, T.E., 24
Celestial mechanics, 117
Center-angle, 206
Cerioli, M.R., 70
Cesàro, G., 99
Cesàro triangle, 99, 158
Chakerian, G.D., 185, 227, 228
Chakrabarty, I., 198
Chang, Y., 96
Chapple, W., 18, 26
Charitos, C., 93
Chavel, I., 118–120
Chebyshev's inequality, 192, 198
Cheeger, J., 119
Chern, S.-S., 176, 177
Chilakamarri, K.B., 155
Chintshin, A.J., 17
Cieślak, W., 28
Circumcircle, 6, 7
Circumscribed cap, 83, 112
Circumscribed sphere, 214
Clancy, C., 17
Classical geometry, 25
Claudius Ptolemy, 273
Clemens, C.H., 94
Cohen-Macaulay simplicial complexes, 73
Cohn, H., 138, 139
Cohn-Vossen, S., 5, 17, 22
Coin graph, 50, 65
Coin graph theorem, 66, 69
Coin representation, 65
Colding, T.H., 121
Colin de Verdière, Y., 70
Colins, K.D., 252
Common tangent distance, 261
Complete bipartite graph, 186
Complete graph, 50, 54, 186
Complete set, 90, 98
Complex numbers, 16, 274, 275
Component, connected component, 52
Congruent, 84
Connected, 51
Convex, 80, 82
Convex hull, 88, 162
Coolidge, J.L., 16, 18, 276
Couderc, P., 230
Courant, R., 16
Court, N.A., 230
Covering number, 137
Coxeter, H.S.M., 1, 16–18, 20, 22, 43, 47, 136, 158, 274
Cramer, H., 199
Crawley, E., 116
Crippen, G.M., 258
Croft, H.T., 139, 171, 230
Crofton, M.W., 175
Crofton's formula, 165, 167, 175
Crossing angle, 3
Crystallography, 23
Cube, 214
Cube slicing, 227
Cundy, H.M., 17
Cut set, 63
Cycle, 51
Cyclic, 111
Cylindrical projection, 77, 90

D
d'Aguilon, 22
Dai, X., 121
Dalbec, J.P., 257
D'Andrea, K., 254
Datta, B.B., 115
Davis, D.J., 6
Dawson, R.J.MaxG., 95
Day, M.M., 212
Deák, I., 171
Debarnot, M.-T., 115
DeBrota, D., 258
de Comberousse, C., 230
Degree, 50
Dehbi, L., 256
Dekster, B.V., 231
Del Centina, A., 26
De Loera, J., 229
Delsarte, P., 138
Deltheil, R., 175
Density function, 192
de Oliveira Filho, F.M., 140
Derived triangle, 99
de Salles Neto, L.L., 259
Descartes' circle theorem, 37, 43
Diameter, 183

Dietrich, U., 115
Differential geometry, 23, 24, 91–93, 117, 258
Diminnie, C., 212
Dirnböck, H., 155
Disconnected, 51
Dispersed, 223
Distance, 201
Distance geometry, 89
Dodecahedral conjecture, 141
Dodunekov, S., 138
Donnay, J.D.H., 22, 101
Dörrie, H., 46, 116, 117, 251
Dordovskii, D., 257
Dos Santos, A.M.F, 96
Dos Santos, J.M., 96
Dostert, M., 140
Dovgoshey, O., 257
Doyle, B., 95
Dragović, V., 27, 28
Drechsler. K., 47
Dual edge, 68
Dual graph, 68
Dubey, K.K., 254
Dunajski, M., 90
Dunham, W., 18
Duparc, H.J.A., 277
Dupin cyclide, 20
Dúran, G., 197

E
Ebin, D.G., 119
Eckhoff, J., 197
Edge, 80
Edge of a graph, 50
Edmonds, A.L., 231, 234, 235
Ehrenfeucht, A., 226
Elerath, D.E., 120
Elkies, N., 138
Ellis, D.C., 197
Embeddable, 52
Epps, T.W., 198
Eppstein, D., 70
Equiareal simplices, 234
Equilateral spherical triangle, 146
Euler, L., 23, 156, 157
Euler inequality, 17
Euler's formula, 53
Euler's formula for spherical excess, 105
Euler's polyhedral formula, 82
Euler's triangle theorem, 6, 17
Even-cycle, 51
Eves, H., 17, 18, 20, 274
Ewald, G., 16, 71, 72
Exterior of ellipse, 144
Extremal cap, 182

F
Fabricius-Bjerre, F., 155
Face of a plane graph, 52
Face-point, 65
Facet, 214
Falconer, K.J., 139, 171, 230
Faria. L., 70
Feeman, T.G., 24, 92
Fejes Tóth, G., ix, 135, 140, 157, 158
Fejes Tóth, L., ix, 95, 104, 135, 140, 141, 146, 157, 158
Fejes Tóth's lemma, 135, 154
Felsner, S., 69, 71
Fenn, R., 94
Fenyvesi, K., 158
Ferguson, S.P., 140
Ferreira, T.O., 70
Fiala, F., 92
Fiedler, M., 231, 252
Filliman, P., 228
Fisher, J.C., 47
Fitz-Gerald, J.M., 47
Flatto, L., 27
Florian, A., 95
Foci, 143
Fodor, F., 98, 227
Four-point properties, 126, 130, 162, 203, 205, 210, 211, 275
Fractals, 17
Fraivert, D., 275
Frame, M, 17
Frank, R., 227
Frankl, P, 234
Freeden, W., 91
Freese, R.W., 212
Frenkel, E,, 158
Friedrich, T., 92
Fuchs, D., 27
Fuchs, E., 46
Fukagawa, H., vii, 19, 20, 37, 43, 276

G
Gaifullin, A.A., 255
Gale, D,, 230
Galicer, D., 233
Gamma function, 217
Gao, F., 90, 95, 173
Garcia, R., 28

Garcia-Roig, J.-L., 275
Gardner, M., 216
Gardner, R.J., 227
Gauss-Bonnet theorem, 23
Gdawiec, K., 17
Gdowski, B., 92
Geiges, H., 26
General position, 162
Geodesic segment, 76
Geodesy, 24
Geometric probability, 170, 172
Ger, R., 275
Geretschläger, R., 17, 18, 27, 44, 46, 274
Giering, O., 16, 23, 27
Gillam, B.E,, 251
Girard's formula, 81
Girstmair, K., 115
Glaeser, G., 23
Glasauer, S., 171
Glazyrin, A., 97, 136
Goethals, J.M., 138
Goldberg, M., 17, 98
Golumbic, M.C., 197
González Merino, B., 227
González-Velasco, E., 273
Goodey, P.R., 178, 179
Gower, J.C., 258
Gowers, T., 90
Grünbaum, B., 6, 68, 69, 71, 95, 139, 230
Grace, J.H., 7
Grace-Danielsson inequality, 7
Graf, U., 116
Grafarend, E., 91
Graham, R.L., 45, 243, 258
Gram matrix, 242
Graph, 49
Graph of convex polyhedron, 50
Graph of polyhedron, 49
Great circle, 75
Greenberg, M.J., 16
Gregorac, R.J., 254, 275
Gregory-Newton problem, 123, 128, 135, 138
Greinke, W., 93
Greitzer, S.L., 16–18, 22, 274
Grigoriev, A., 48
Grigorieva, E., 274
Grinberg, D., 259
Grippo, L., 197
Gritzmann, P., 229, 251
Groemer, H., 171, 185
Gromoll, D., 119
Gromov, M., 120, 122
Grove, K., 119
Gruber, P.M., 69, 71, 138
Gueron, S., 277
Guggenheimer, H.W., 16
Guo, Q., 90
Guo, R., 97
Gupta, R.C., 274
Guy, R.K., 139, 171, 230
Gyarmathi, L., 47

H
Ha, J.-S., 98
Hadamard, J., 16, 158
Hadwiger number, 137
Hadwiger, H., 176, 177, 231, 235
Hahn, L.-S., 274
Hajós, G., 16
Hajja, M., 231, 232, 235, 236, 255
Halbeisen, L., 28
Hales, T.C., vii, 140, 141
Halley, 22
Hamada, T., 274
Hamburger, P., 155
Hamdan, A., 236
Hammond, J.R., 116
Hammoudeh, I., 235, 255
Hampton, M., 252
Hand-shaking lemma, 50
Harriot, 22
Harris, S.G., 120
Hartl, J., 25
Hartsfield, N., 69
Hartshorne, R., 16, 46, 89, 94
Havel, T., 253, 257
Havil, J., 115
Havlicek, H., 93
Hayajneh, M., 235, 255
Hayashi, T., 276
Hebda, J.J., 122
Heidari, F., 20
Heil, E., 231
Hemisphere, 75
Henk, M., 140, 173
Hersh, R., 6
Hertel, E., 235
Hessenberg, G., 116
Heule, M., 229
Hexlet, 10
Hexlet of Soddy, 20
Heyda, J.F., 117
Hibi, T, 73
Hilbert, D., 5, 17, 22
Hill simplex, 231

Hinks, A.R., 91
Hipparchus, 21
Hlinený, P., 70
Holshouser, A.L., 44
Honari, B., 20
Honsberger, R., 18, 20
Horak, P., 229
Horizontal line, 77
Horizontal plane, 77
Horváth, À., 155, 156
Hoschek, J., 92
Hsiang, W.-Y., 136
Hu, S., 122
Hu, Z-C., 200
Huber, M., 199
Hudelson, H., 229
Hug, D., 90, 172–174, 177, 178, 234
Hui, M.-T., 158
Hull, T., 117
Hungerbühler, N., 19, 28
Hypersphere, 214

I
Ibragimov, Z.S., 273
Ikeda, Y., 122
Incircle, 5, 7
Inscribed, 112
Inscribed angle theorem, 83
Integral geometry, 176, 177
Interior of ellipse, 144
Intersection graphs, 181, 186, 196, 197
Invariance of angles, 12
Invariant of graphs, 70
Inversion, 1, 36
Inversion in circles, 15
Inversor of Hart, 17
Invisibility, 23
Ionascu, E.J., 25
Irreducible, 55
Isolated, 182
Isometric, 201, 239
Isometrically embeddable, 202, 239, 257, 268
Isometric radii change, 266
Isoperimetric theorem for spherical quadrilaterals, 112
Isoperimetric-type results, 96, 97, 111, 146, 150, 158
Itokawa, Y., 120
Ivanov, G., 227, 228
Ivanov, S., 118
Ivory's theorem, 156
Iwata, S., 19, 47
Izmestiev, I., 155

J
Jacob, C., 274
Japanese temple geometry, 19, 20, 43, 274, 276
Jeger, M., 118
Jennings, G.A., 89, 104, 117
Jenssen, M., 139
Jiang, J., 198
Johnson, N.L., 198
Johnson, R.A., 5, 16–18, 22, 46, 274
Jokanović, D.S., 19
Josefsson, M., 275
Joswig, M., 73, 140

K
Kabluchko, Z., 174
Kakulashvili, G., 97
Kalai, G., 72
Kale, B.K., 199
Kambly, R., 116
Kamiyama, Y., 97
Kang, M.-H., 155
Karcher, H., 119
Kautzleben, H., 93
k-connected, 63
Keller's conjecture, 229
Kells, L.M., 116
Kelly, P.J., 230
Kendall, D.G., 171, 173
Kendall, M.G., 175
Kepler conjecture, 140
Kepler problem, 26
Kern, W.F., 116
Kers, C., 26
Khatri, C.G., 199
Khimshiashvili, G., 275
Kiderlen, M., 177, 178
Kim, D., 229
Kissing number, 137, 225
Kitrick, C.J., 90, 94
Klain, D, 176
Klapper, M.H., 258
Klee, V., 229, 251
Klein, F., 22
Kleinschmidt, P., 72
Klingenberg, W., 119
Klotzek, B., 92
Knörrer, H., 23
Knight, R.D., 47
Kock, A., 256
Koebe, P., 66, 69, 73
Koecher, M., 92
Kokkendorff, S.L., 254
Koldobsky, A., 227, 228

Kollár, J., 25
Kolpakov, A., 140
Komori, Y., 98
Kondo, K., 121, 122
König, H., 228
König, R., 24, 91
Kontorovich, A., 44, 48
Kostin, A.V., 277
Kostina, N.N., 277
Kostochka, A., 197
Krantz, S.G., 232
Krasopoulos, P., 256
Kratochvíl, J., 70
Krieg, A., 92
Krumm, F., 91
Kubota, T., 277
Kuklin, N.A., 138
Kumar, A., 139
Kunze, H., 17
Kuperberg, G., 139
Kuperberg, W., ix, 135, 140, 157, 158
Kureethara, J.V., 198
Kurnik, Z., 277
Kurosaki, K., 274
Kürzel, M., 121
Kusejko, K., 19

L
Lagarias, J.C., 44, 45, 140, 229
Lambert, J.H., 78
Lángi, Z.L., 73
Lang, U., 121
Langevin, R., 175, 178
Langlet, G.A., 47
Larcher, H., 120
Larman, D., 229
Lassak, M., 90, 95, 98
Lavor, C., 252, 253, 259
Law of total probability, 167
Lay, S.R., 230
Le, H., 171, 173
Le, H.-L., 173
Le, V.B., 197
Lebesgue, H.L., 157, 175
Lee, C.W., 72
Lee, T.-U., 96
Leech, J., 134, 136
Legendre, A.-M., 82, 157, 158
Leng, G., 257
Length of a cycle, 51
Length of a path, 51
Leone, F.C., 198
Leonhardt, U., 24
Levi, S, 28
Levrie, P., 43
Lexell, A.J., 156, 157
Lexell's theorem, 25, 145, 156
Li, H., 253
Li, X., 257
Liberti, L., 138, 252, 253
Lie contact geometry, 47
Lietzmann, W., 116, 117
Lin, W.-H., 155
Linear, 202
Linked 4-cycle pair, 41
Liu, C., 96
Liu, Y., 96
Location problem, 48
Lodwick, W., 259
Logothetti, D., 227
Lovász, L., 69, 71
Lune, 81
Lysenko, V.I., 158
Lyuter, I.O., 46

M
Ma, T., 233
Macaulay, W.H., 116
Machida, Y., 256
Machigashira, Y., 120
Mackay, A.L., 171
Mackey, J., 229
Maculan, N., 252
Maeda, Y., 174
Maehara, H., 5, 19, 21, 68, 97, 101, 106, 111, 119, 134, 136, 140, 146, 156, 157, 164, 172–174, 178, 228, 231, 234, 235, 271, 277
Magnus, L.J., 1
Major triangle, 125
Mallows, C.L., 44
Mandelbrot, B.B., 17
Marenich, V.B., 121
Markov's inequality, 192, 198
Markushevich, A.I., 17
Martini, H., 5–7, 26, 28, 97, 101, 140, 146, 157, 164, 172, 174, 178, 185, 227–232, 234, 235, 255, 271, 277
Mathematical cartography, 90
Mathis, P., 259
Matoušek, J., ix, 69, 71, 230
Maximal plane graph, 54
Maximum degree, 50
Maximum independent set, 70
McCarthy, J.E., 232
McCleary, J., 118

McKean, H., 198
McKee, T.A., 197
McLaughlin, S., 141
McMorris, F.R., 197
McMullen, P., 228, 234
Mean width, 233
Mednykh, A., 277
Melbourne, J., 227
Melluish, R.K., 91
Melzak, Z.A., 90
Menger, K., 212, 251
Mercator, 22
Merzbacher, M., 233
Mesiar, R., 200
Metric space, 201
Meyer, W., 119
Michelen, M., 139
Michiwaki, Y., 274
Miernowski, A., 233
Mikaiylova, L.A., 277
Mikami, Y., 276
Miles, R.E., 171
Milićević, M., 19
Miller, S.D., 139
Milman, R.S., 22
Minimal area simplices, 233
Minimal cut set, 63
Minimal enclosing ellipse, 156
Minimum degree, 50
Mirman, B., 28
Mitchell, L., 70
Mitrinović, D.S., 230
Mittelmann, H., 138
Mnëv, N.E., 72
Mo, T.C., 199
Möbius geometry, 16, 19, 22, 23, 25, 275
Möbius invariant, 263
Mohseni Takaloo, S., 25, 48
Molchanov, S., 44
Molnár, J., 90
Mondal, B., 254
Montejano, L., 185
Moody, J., 227
Moran, P.A.P., 175
Morgan, J., 120
Morita, K., 27, 47
Moritsugu, S., 44
Moser's paradox, 215
Moser, L., 216
Moser, W., 69, 139, 259
Mozgawa, W., 28, 233
Mucherino, A., 252
Multisection lasers, 26
Muralidharan, K., 199
Musin, O.R., 19, 21, 138, 140, 141

N
Namikawa, Y., 155
Napoleon's theorem, 232
Narváez, D., 229
Naszódi, M., 73
Naumann, H., 228
Neighbor, 50, 181
Neighborhood, 50
Nelsen, R., 18, 274
Neuberg-Pedoe inequality, 118
Neutsch, W., 117
Newton number, 137
Nguyen, B., 255
Non-Euclidean geometries, 16, 25
Non-facial 3-cycle, 126
Norm, 214
Normal family, 31
Northshield, S., 44, 45
Null circle, 261
Nushizumi, Y., 43

O
O'Connor, C.P.J., 26
OCR theorem, 58
Odd-cycle, 51
Odehnal, B., 28
Ogasawara, H,, 199
Ogilvy, C.S., 18, 20, 46
Oh, H., 46
Ohta, S., 122
Olivares, J.G., 115
Oliveros, D., 185
Olver, P.J., 259
Onn, S., 72
Oort, F., 26
O'Reilly, E., 175
Oriented matroid, 72
Origami, 117
Orthocentric simplices, 231, 235
Orthogonal-circle-representation, 58
Orthogonally crossing, 3, 243
Orthogonal-sphere system, 243
Ortuzar, A., 259
Oshiro, A, 21
Ostermann, A., 117
Ostrovskii, M.I., 228
Ouyang, Y., 200
Oyama, M., 274

P
Pach, J., 66, 69, 139, 197, 259
Pachner, U., 72
Paluszny, M., 47
Pambuccian, V., 18, 106, 119, 158
Panin, A., 48
Panizzut, M., 73
Paouris, G., 229
Papadopoulos, A., vii, 89, 93, 115, 146, 157, 158
Parameter, 143
Parker, G.D., 22
Parks, H., 232
Path, 51
Peaucellier linkage, 5
Peaucellier's inversor, 17
Pecarić, J.E., 230
Pech, P., 275
Pecker, D., 28
Pedoe, D., vii, 16, 18–20, 23, 37, 43, 46, 47, 274, 276
Pekonen, O., 158
Peng, Y., 90
Penny graph, 69
Percy, A., 275
Perel'man, G.Y., 122
Perfect, H., 16, 274
Perimeter deviation, 98
Persson, U., 158
Petersen, P., 119
Petrera, M., 118
Pfender, F., 97, 138
Pfiefer, R.E., 174
Philbin, T., 24
Pinasco, D., 233
Pippert, R.E., 155
Pisanski, T., 69
Pivovarov, P., 174, 229
Pizzetti, P., 119
Planar, 202
Planar graph, 52
Plane graph, 52
Plofker, K., 115
Polar set, 85
Polar triangle, 87
Pole, 79
Polya, G., 6
Polyhedral graphs, 71
Pompeiu property, 97
Pompeiu's theorem, 236
Poncelet polygon, 26
Poncelet's porism, 7, 12, 26, 28
Poncelet triangle, 28
Popescu, C., 275
Popko, E.S., 25, 90, 94
Porta, J.M., 257
Posamentier, A.S., 17, 18, 27, 44, 46, 274
Power, 1
Prakasa Rao, B.S.L., 200
Prasolov, V.V., 22, 27, 89
Problem of Apollonius, 43, 46
Projections of cubes, 228
Proper diagonal, 124
Properly tangent, 262
Proper subgraph, 50
Propp, J.G., 139
Proskurowski, W., 93
Protasov, V.Yu., 19
Protti, F., 70
Ptolemy, 21
Ptolemy's theorem, 249, 273, 274
Purisima, E.O., 258

Q
Quadrangulation, 54
Quadrics, 23
Quartet, 201

R
Rademacher, H., 5, 71
Radial edges, 65
Radial graph, 65
Radical axis, 56
Radnović, M., 27, 28
Rambau, J., 229
Ramírez, G.N., 17
Randić, M., 69
Random diameter, 160
Random great circle, 160
Random point, 159, 171
Random polygons, 170
Random polyhedral cones, 172
Random polytopes, 170, 172, 174
Random spherical cap, 173, 190
Random spherical triangle, 160
Ranucci, E.R., 226
Rao, M., 236
Rashed, R., 115
Rataj, J., 178
Rauch, H.E., 119
Raven, D.W., 26
Realization, 50
Realization spaces for polytopes, 72
Reduced bodies, 90, 98
Reed, N., 17
Regular n-simplex, 214

Regular spherical polygon, 97, 98, 150
Reiter, H., 44
Reitzner, M., 172
Ren, D.L., 176
Replicating simplices, 234
Representation of a graph, 51
Reuleaux d-triangle, 184
Reuschel, A., 277
Reventós Tarrida, A., 17, 19
Reznik, D., 28
Reznikov, A.B., 174
Richter-Gebert, J., 25, 71, 72
Riede, H., 227
Riemann's number ball, 23
Rigidity, 253, 259
Ringel, G., 69
Rivière, A., 259
Roanes-Lozano, E., 19, 21
Robbins, H., 16
Roberto, C., 227
Roberts, F.S., 196
Roché, E., 230
Rogers, D.G., 275
Rollett, A.P., 17
Rollins, W.E., 226
Ronda, J., 28
Rosen, D.A., 175
Rosenfeld, B.A., 15, 22, 89, 95, 117
Roshdi, R., vii
Rota, G.-C., 176
Rother, W., 178
Rothman, T., vii
Rothschild, B.L., 243, 258
Ryabogin, D., 227
Ryan, P.J., 24, 70
Rylov, Yu.A., 257
Rzymowski,W., 233

S
Saban, G., 175
Sachs, H., 69–71
Safe, M., 197
Saff, E.B., 174
Sahasrabudhe, J., 139
Sakano, Y., 96
Saks, M.E., 227
Santaló, L.A., 170, 175, 176
Santaló's chord theorem, 169
Santos, F., 229
Sarnak, P., 46
Satyanarayana, K., 230
Saw, J.G., 199
Schacht , C., 18
Scheffers, G., 24
Scheinerman, E.R., 63, 69
Scheraga, H.A., 258
Scherer, K., 117
Schilling, F., 155
Schmidt, H., 17, 22, 46
Schneider, R., 90, 171, 173, 174, 176–178, 231, 232
Schnell, U., 228
Schoenberg, I.J., 7, 27, 258
Schramm, O., 70, 139
Schreck, P., 259
Schreiner, M., 91
Schröcker, H.-P., 155
Schröder, E., 22, 24, 92, 93
Schroeder, V., 121
Schroth, A.E., 47
Schulz, C., 72
Schuster, R., 178
Schütte, K., 97, 134, 136
Schwartz, R.E., 20, 26, 44, 255
Schwarz, D., 275
Schwerdtfeger, H., 16, 23
Seberry, J., 229
Seidel, J.J., 138, 256
Semialgebraic sets, 72
Semi-metric space, 239
Semi-perimeter, 207
Sen, H., 254
Seneta, E., 199
Separating 4-cycle, 55
Serré, P., 259
Shaqaqha, S., 255
Sharygin, I.F., 44
Shephard, G.C., 95
Shi, N., 95
Shiga, K., 27
Shin, S.-Y., 98
Shiohama, K., 120, 122
Shor, P/ W., 229
Shtogrin, M.I., 44
Sigl, R., 116
Sigler, A., 275
Signed radius, 245
Silverberg, J.S., 115
Simon, M., 230
Simonić, A., 25, 146, 157
Simple polygonal curve, 80
Simplex, 214, 231, 235, 253, 254
Simplex centers, 232, 235
Singh, A.N., 115
Sippl, M.J., 258
Skau, C., 27
Skopets, Z.A., 46, 276

Skoruppa, N.-P., 138
Sloan, I.H., 174
Sloane, N.J.A., 137
Slutskiy, D., 118
Small circle, 75
Smart, W.M., 89
Smeltzer, D.L., 19
Smith, G.C., 275
Smith, J.D., 274
Soddy, Frederick, 20, 37
Soddy's formula, 39, 43
Soddy's hexlet, 10, 21, 35
Soland, C., 257
Solomon, H., 176
Soltan, V., 233
Sombra, M., 254
Sönmez, N., 97
Sphere packing problems, 137
Spherical, 202
Spherical n-gon, 80
Spherical astronomy, 89
Spherical cap, 75
Spherical codes, 19, 138
Spherical cosine law, 89, 103, 114
Spherical Delaunay triangulations, 136
Spherical ellipse, 143, 151, 155
Spherical excess, 81
Spherical polygon, 80, 94
Spherical rotors, 98
Spherical sine law, 89, 104, 114
Spherical tilings, 94–96
Spherical triangle, 80
Spherical trigonometry, 90, 114
Sphero-conics, 155
Spinrad, J.P., 197
Spirova, M., 6, 7
Springborn, B.A,, 73
Squared-distance matrix, 240
Stén, J., 157, 158
Steiner chain of circles, 18, 20, 21
Steiner cycle, 8, 18, 31
Steiner's porism, 8, 18
Steiner, J., 157
Steinert, K.-G., 116
Steinitz' theorem, 67, 71
Steinitz, E., 67, 71
Stereographic projection, 11, 21, 99, 118, 145
Sterz, U., 47
Stoka, M., 176
Stone, C., 227
Straus, E.G., 243, 258
Strong dodecahedral conjecture, 141
Stupel, M., 275
Sturmfels, B., 72
Su, W., 158
Su, X., 122
Subgraph, 50
Subspace, 201
Suceava, B., 273
Sugimoto, T., 173
Sulanke, R., 176
Super-Apollonian group, 45
Surface area, 178, 237, 238
Suris, Y.B., 118
Svrtan, D., 18
Swanepoel, K.J., 7, 137
Swart, D., 25
Sylvester's four-point-problem, 163, 174
Szabó, L., 140
Szenthe, J., 258
Szpiro, G.D., 140

T

Tabachnikov, S., 20, 26–28, 44
Taherian, S.Gh., 25, 48
Takayama, Y, 73
Tammes problem, 140
Tanaka, M., 121, 122, 274
Tanemura, M., 173
Tangent, 3
Tangent-sphere system, 244
Tarasov, A.S., 140
Taylor, J.H., 276
Teixeira, R., 229, 230
Thäle, C., 172, 175
Thébault, V., 230
Thomas, F., 257
Three-circles theorem, 5
Three-pointed star, 205
Thurston, W.P., 66, 69
Tian, G., 120
Tikhomirov, V.M., 22, 27, 89
Titeica, G. (Ţiţeica, G.), 5
Titeica-Johnson theorem, 5
Toeplitz, O., 5
Tong, Y.L., 199
Toponogov's theorem, 118
Toponogov's triangle comparison theorem, 106
Toponogov, V.A., 106, 118, 119
Torrence, B., 25
Trapp, K., 92
Triangulations of cubes, 229
Triangle comparison theorem for spheres, 105, 118
Triangle free penny graphs, 70
Triangular face, triangle, 54
Triangulation of a quadrilateral, 126

Triangulations of cubes, 229
Trio, 201
Tropfke, J., 116
Tsiutsiurupa, I., 227, 228
Tuckwell, H.C., 198
Tupan, A., 44
Two ears theorem, 147

U
Ueno, K., 27
Ueno, Y., 96
Umezawa, R., 98
Ungar, A.A,, 253
Unispherical, 205
Unit-distance graph, 49
Urmaev, N.A., 92

V
Valdés, A., 28
Valentine, J.E., 212, 274
Vallès, J., 28
Vallentin, F., 138
Van Brummelen, G., vii, 23, 90, 94, 101, 116, 117
Van der Waerden, B.L., 97, 134, 136
Van Gruting, C.J., 274, 277
Veljan, D., 18
Vertex, 49, 80
Vertex embeddings of simplices in cubes, 229
Viazovska, M.S., 139
Volenec, V., 230, 258, 277
Volume, 213, 216, 236, 237
Volume of a ball, 216, 237
Vu, T.Khoi, 97

W
Wagner, K., 92
Wang, E., 96
Wang, Y., 122
Wang, Y.G., 174
Wanner, G., 117
Wasan, 19, 44, 276
Weber, M.J., 156
Wei, G., 121
Weil, W., 171, 176–178
Weis, J.A., 179
Weise, K.H., 24, 91
Weiss, A.I., 44
Weiss, G., 7, 226
Weiss, M.L., 230
Weissbach, B., 228, 232
Weisstein, E.W., 276
Wells, D., 5, 18, 20
Wendel's theorem, 164
Wendel, J.G., 164
Wenninger, M.J., 94
Wenzel, W., 234
Whittlesey, M.A., vii, 89, 94
Wieacker, J.A., 177
Wieczorek, S., 26
Wiegert, P.A., 158
Wilker, J.B., 17, 21, 25, 43, 47, 231
Wilks, A.R., 44, 45
Wilson, P.M.H., 23, 89, 94, 117
Wimmer, L., 96
Winkler, P., 111
Wintgen, P., 176
Womersley, R.S., 174
Wu, S., 6
Wünsch, V., 92

X
Xie, Y.M., 96
Xu, X., 259

Y
Yaglom, I.M., 16–18, 22, 46, 116, 117, 230, 276
Yan, C.H., 45
Yan, M., 95, 96
Yang, L., 118, 254, 256
Yang, M.C., 199
Yasui, T., 98
Ye. Q.-Z., 158
Yiu, P., 19
Yuan, J., 233

Z
Zach, D., 227
Zacharias, M., 116, 277
Zähle, M., 173, 178
Zamfirescu, T., 106, 119
Zeng, Z., 256
Zhang, G., 177
Zhang, J.Z., 118, 254
Zhao, L., 233
Zhou, L., 200
Zhukova, A., 158
Ziegler, G.M., 68, 69, 71, 72, 97, 136, 138, 140
Zinn, J., 229
Zong, C., 136–139, 226, 229
Zvavitch, A., 227